Solutions Manual

Theory of Finite Automata with an Introduction to Formal Languages

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Contents

Chapter 0

Preliminaries — Solutions

0.1. **Construct truth tables for:**

(a) $\neg r \vee (\neg p \downarrow \neg q)$

Recall that $p \downarrow q$ (NOR) is $\neg(p \vee q)$.

p	q	r	$\neg p$	$\neg q$	$\neg p \downarrow \neg q$	$\neg r$	$\neg r \vee (\neg p \downarrow \neg q)$
0	0	0	1	1	0	1	1
0	0	1	1	1	0	0	0
0	1	0	1	0	0	1	1
0	1	1	1	0	0	0	0
1	0	0	0	1	0	1	1
1	0	1	0	1	0	0	0
1	1	0	0	0	1	1	1
1	1	1	0	0	1	0	1

(b) $(p \wedge \neg q) \vee \neg(p \uparrow q)$

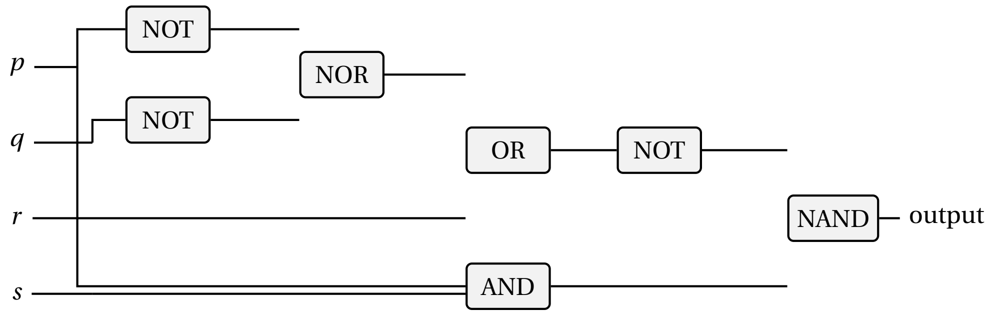
Recall that $p \uparrow q$ (NAND) is $\neg(p \wedge q)$.

p	q	$p \wedge \neg q$	$p \uparrow q$	$\neg(p \uparrow q)$	$(p \wedge \neg q) \vee \neg(p \uparrow q)$
0	0	0	1	0	0
0	1	0	1	0	0
1	0	1	1	0	1
1	1	0	0	1	1

0.2. **Draw circuit diagrams for:**

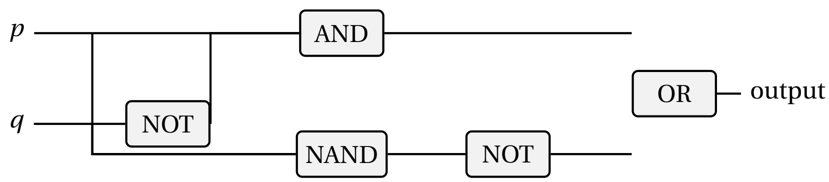
(a) $\neg(r \vee (\neg p \downarrow \neg q)) \uparrow (s \wedge p)$

One corresponding circuit is:



(b) $(p \wedge \neg q) \vee \neg(p \uparrow q)$

One corresponding circuit is:



0.3. **Show that** $\{1,2\} \times \{a,b\}$ **and** $\{a,b\} \times \{1,2\}$ **are not equal.**

$$\{1,2\} \times \{a,b\} = \{\langle 1,a\rangle, \langle 1,b\rangle, \langle 2,a\rangle, \langle 2,b\rangle\}$$

$$\{a,b\} \times \{1,2\} = \{\langle a,1\rangle, \langle a,2\rangle, \langle b,1\rangle, \langle b,2\rangle\}$$

Since $\langle 1,a\rangle \in \{1,2\}\times\{a,b\}$ but $\langle 1,a\rangle \notin \{a,b\}\times\{1,2\}$ (the first component of any element of $\{a,b\}\times\{1,2\}$ is a letter, not a number), the two sets are not equal.

0.4. **Let** $X = \{1,2,3,4\}$**.**

(a) $<= \{\langle 1,2\rangle, \langle 1,3\rangle, \langle 1,4\rangle, \langle 2,3\rangle, \langle 2,4\rangle, \langle 3,4\rangle\}$

(b) $== \{\langle 1,1\rangle, \langle 2,2\rangle, \langle 3,3\rangle, \langle 4,4\rangle\}$

(c) $= \cup <= \{\langle 1,1\rangle, \langle 2,2\rangle, \langle 3,3\rangle, \langle 4,4\rangle, \langle 1,2\rangle, \langle 1,3\rangle, \langle 1,4\rangle, \langle 2,3\rangle, \langle 2,4\rangle, \langle 3,4\rangle\}$

(d) $\leq == \cup <$, which is the set computed in part (c).

0.5. **Show that** $\equiv_n$ **is an equivalence relation.**

We must show reflexivity, symmetry, and transitivity.

Reflexivity: $a - a = 0 = 0 \cdot n$, so $a \equiv_n a$.

Symmetry: If $a \equiv_n b$, then $n \mid (a-b)$, so $a - b = kn$ for some $k \in \mathbb{Z}$. Then $b - a = (-k)n$, so $n \mid (b-a)$, giving $b \equiv_n a$.

Transitivity: If $a \equiv_n b$ and $b \equiv_n c$, then $a - b = kn$ and $b - c = mn$ for integers k, m. Then $a - c = (a-b) + (b-c) = (k+m)n$, so $a \equiv_n c$.

0.6. **Equivalence classes of** $\equiv_0$ **on** $\mathbb{N}$**.**

$a \equiv_0 b$ means $0 \mid (a-b)$, i.e., $a - b = 0$, i.e., $a = b$. So the equivalence classes are all singletons: $\{0\}, \{1\}, \{2\}, \dots$

0.7. **Equivalence classes of $\equiv_1$ on $\mathbb{N}$.**

$a \equiv_1 b$ means $1 \mid (a-b)$, which is always true. So there is exactly one equivalence class: $\mathbb{N}$ itself.

0.8. **Equivalence classes of $\equiv_1$ on $\mathbb{R}$.**

Here $x \equiv_1 y$ means $1 \mid (x-y)$, i.e., $x - y \in \mathbb{Z}$ (the difference is an integer). Two reals are equivalent iff they have the same fractional part $\{x\} = x - \lfloor x \rfloor$. The equivalence classes are the cosets of $\mathbb{Z}$ in $(\mathbb{R},+)$:

$$[r]_{\equiv_1} = \{r+n \mid n \in \mathbb{Z}\}, \qquad r \in [0,1).$$

There are uncountably many such classes, one for each $r \in [0,1)$.

0.9. **Prove the distinct equivalence classes of R form a partition of X.**

Let $[x]_R = \{y \in X \mid yRx\}$.

Coverage: For any $x \in X$, reflexivity gives $x \in [x]_R$, so every element of X belongs to at least one class.

Disjointness: Suppose $[x]_R \cap [y]_R \neq \emptyset$; pick z with zRx and zRy. By symmetry xRz, and by transitivity xRy. For any $w \in [x]_R$ we have $wRxRy$, so $w \in [y]_R$; thus $[x]_R \subseteq [y]_R$. By symmetry $[y]_R \subseteq [x]_R$, so $[x]_R = [y]_R$.

Hence the distinct classes are disjoint and cover X, forming a partition.

0.10. **Prove X equals the union of the sets in P.**

Let $P = \{A_1, \ldots, A_n\}$ be a partition of X.

By definition of partition, each $A_i \subseteq X$, so $\bigcup_i A_i \subseteq X$.

Also by definition, every $x \in X$ belongs to some A_i, so $X \subseteq \bigcup_i A_i$.

Therefore $X = \bigcup_{i=1}^{n} A_i$.

0.11. **Prove $R(P)$ is an equivalence relation.**

Define $xR(P)y$ iff x and y belong to the same block of P.

Reflexivity: x is in the same block as itself.

Symmetry: If x and y are in the same block, so are y and x.

Transitivity: If x, y are in block A_i and y, z are in block A_j, then since $y \in A_i \cap A_j$ and blocks are disjoint, $A_i = A_j$. So x and z are in the same block.

0.12. **Let $X = \{1,2,3,4\}$.**

(a) Example where $R(P)$ is a function: $P = \{\{1\},\{2\},\{3\},\{4\}\}$. Then $R(P) = \{\langle x, x\rangle \mid x \in X\}$, which is the identity function.

(b) Example where $R(P)$ is not a function: $P = \{\{1,2\},\{3,4\}\}$. Then $\langle 1,1\rangle, \langle 1,2\rangle \in R(P)$, so 1 maps to two elements; it is not a function.

0.13. **Find the flaw in the "proof" that symmetric and transitive implies reflexive.**

The flaw is in the first step. Symmetry says: if xRy then yRx. This requires the hypothesis that xRy. If x is not related to any y at all, we cannot conclude xRx.

Counterexample: Let $X = \{1,2,3\}$ and $R = \{\langle 1,1\rangle, \langle 2,2\rangle, \langle 1,2\rangle, \langle 2,1\rangle\}$. Then R is symmetric and transitive but $3 \not R\, 3$, so R is not reflexive.

0.14. **Prove equality refines R.**

We need: $x = y \Rightarrow xRy$. If $x = y$, then by reflexivity of R, $xRx = xRy$. $\square$

0.15. **The "function" $t : \mathbb{R} \to \mathbb{R}$, $x \mapsto \arccos(x)$.**

(a) t is not well defined. A function must assign *exactly one* output to each input in its domain. But if $x \in [-1,1]$, there are generally many real numbers whose cosine is x. For example,

$$\cos\left(\frac{\pi}{3}\right) = \frac{1}{2} = \cos\left(\frac{5\pi}{3}\right),$$

so the input $x = \frac{1}{2}$ would have more than one output. Also, if $x \notin [-1,1]$, there is no real number whose cosine is x, so the proposed rule is not even defined everywhere on $\mathbb{R}$.

(b) Restrict domain to $[-1,1]$ and range to $[0,\pi]$: $t : [-1,1] \to [0,\pi]$, $t(x) = \arccos(x)$, is a valid function.

0.16. **Show the converse of $s' : \mathbb{R} \to \mathbb{R}$, $s'(x) = x^2$, is not a function.**

s'^{-1} (converse) is $\{\langle y,x\rangle \mid x^2 = y\}$. For $y = 1$, both $x = 1$ and $x = -1$ satisfy $x^2 = 1$, so $\langle 1,1\rangle$ and $\langle 1,-1\rangle$ are both in the converse, which is therefore not a function (1 maps to two values).

0.17. **Show s^{-1} exists for $s : \mathbb{P} \to \mathbb{P}$, $s(x) = x^2$, where $\mathbb{P}$ is nonneg. reals.**

s is injective: if $x^2 = y^2$ and $x, y \geq 0$ then $x = y$. s is surjective onto $\mathbb{P}$: for any $y \geq 0$, $x = \sqrt{y} \geq 0$ satisfies $s(x) = y$. Since s is a bijection, $s^{-1}(y) = \sqrt{y}$ exists.

0.18. **Prove: converse of $f : X \to Y$ is a function iff f is a bijection.**

($\Rightarrow$) Suppose $\tilde{f} = \{\langle y,x\rangle \mid \langle x,y\rangle \in f\}$ is a function (from Y to X). Then every $y \in Y$ has a unique pre-image under f. *Injectivity*: if $f(x_1) = f(x_2) = y$, then $\tilde{f}(y)$ must equal both x_1 and x_2; since $\tilde{f}$ is a function, $x_1 = x_2$. *Surjectivity*: since $\tilde{f}$ is a function *with domain* Y, every $y \in Y$ is in the domain of $\tilde{f}$, meaning there exists x with $f(x) = y$. Hence f is surjective. (Note: totality of f does not by itself imply surjectivity; surjectivity follows here from $\tilde{f}$ being defined on *all* of Y.)

($\Leftarrow$) If f is injective, each y has at most one pre-image, so $\tilde{f}$ assigns at most one value. If f is surjective, each $y \in Y$ has at least one pre-image, so $\tilde{f}$ is defined everywhere on Y. Together, $\tilde{f}$ is a function.

0.19. (a) $\sim(\sim A) = A$: By definition, $\sim A = \{x \mid x \notin A\}$. Then $\sim(\sim A) = \{x \mid x \notin \sim A\} = \{x \mid x \in A\} = A$.

(b) $\sim(\sim R) = R$ for a relation R (where $\sim R$ is the converse): $\sim R = \{\langle y,x\rangle \mid \langle x,y\rangle \in R\}$. Then $\sim(\sim R) = \{\langle x,y\rangle \mid \langle y,x\rangle \in \sim R\} = \{\langle x,y\rangle \mid \langle x,y\rangle \in R\} = R$.

0.20. **If $f : X \to Y$ and $g : Y \to Z$ are one-to-one, prove $g \circ f$ is one-to-one.**

Suppose $(g \circ f)(x_1) = (g \circ f)(x_2)$. Then $g(f(x_1)) = g(f(x_2))$. Since g is injective, $f(x_1) = f(x_2)$. Since f is injective, $x_1 = x_2$.

0.21. **If $f : X \to Y$ and $g : Y \to Z$ are onto, prove $g \circ f$ is onto.**

Let $z \in Z$. Since g is surjective, $\exists y \in Y$ with $g(y) = z$. Since f is surjective, $\exists x \in X$ with $f(x) = y$. Then $(g \circ f)(x) = g(f(x)) = g(y) = z$.

0.22. **Define two functions for which $f \circ g = g \circ f$.**

Let $f, g : \mathbb{R} \to \mathbb{R}$, $f(x) = x+1$, $g(x) = x+2$. Then $(f \circ g)(x) = f(x+2) = x+3 = g(x+1) = (g \circ f)(x)$.

0.23. **Define bijections.**

(a) $\mathbb{N}$ and $\mathbb{Z}$: $f : \mathbb{N} \to \mathbb{Z}$ defined by

$$f(n) = \begin{cases} n/2 & \text{if } n \text{ is even} \\ -(n+1)/2 & \text{if } n \text{ is odd} \end{cases}$$

maps $0 \mapsto 0, 1 \mapsto -1, 2 \mapsto 1, 3 \mapsto -2, 4 \mapsto 2, \ldots$

(b) $\mathbb{N}$ and $\mathbb{N} \times \mathbb{N}$: Use the Cantor pairing function $f(m,n) = \frac{(m+n)(m+n+1)}{2} + n$. Its inverse gives a bijection $\mathbb{N} \to \mathbb{N} \times \mathbb{N}$.

(c) $\mathbb{N}$ and $\mathbb{Q}$: The positive rationals can be arranged in a grid and enumerated by diagonals (Cantor's enumeration). Combined with the bijection $\mathbb{N} \to \mathbb{Z}$ from part (a), one obtains a bijection $\mathbb{N} \to \mathbb{Q}$.

(d) $\mathbb{N}$ and $\{a, b, c\}$: No bijection exists, since $\mathbb{N}$ is infinite and $\{a, b, c\}$ has only 3 elements.

0.24. **Use induction to prove $\sum_{k=1}^{n} k^3 = \frac{n^2(n+1)^2}{4}$.**

Base case ($n = 1$): $1^3 = 1 = \frac{1 \cdot 4}{4}$. ✓

Inductive step: Assume $\sum_{k=1}^{n} k^3 = \frac{n^2(n+1)^2}{4}$. Then

$$\begin{aligned} \sum_{k=1}^{n+1} k^3 &= \frac{n^2(n+1)^2}{4} + (n+1)^3 \\ &= \frac{n^2(n+1)^2 + 4(n+1)^3}{4} \\ &= \frac{(n+1)^2(n^2 + 4(n+1))}{4} \\ &= \frac{(n+1)^2(n^2 + 4n + 4)}{4} \\ &= \frac{(n+1)^2(n+2)^2}{4}. \end{aligned}$$

This is the formula with n replaced by $n+1$. $\square$

0.25. **Prove $\sum_{k=1}^{n} k < n^2$ for $n > 1$.**

Base case ($n = 2$): $1 + 2 = 3 < 4 = 2^2$. ✓

Inductive step: Assume $\sum_{k=1}^{n} k < n^2$. Then

$$\sum_{k=1}^{n+1} k = \sum_{k=1}^{n} k + (n+1) < n^2 + n + 1.$$

We need $n^2 + n + 1 \le (n+1)^2 = n^2 + 2n + 1$, which holds since $n \ge 1$. $\square$

0.26. **Prove $n! > n^2$ for $n > 3$.**

Base case ($n = 4$): $4! = 24 > 16 = 4^2$. ✓

Inductive step: Assume $n! > n^2$ for some $n \geq 4$. Then

$$(n+1)! = (n+1)\cdot n! > (n+1)\cdot n^2.$$

We need $(n+1)n^2 \geq (n+1)^2$, i.e., $n^2 \geq n+1$, i.e., $n^2 - n - 1 \geq 0$. For $n \geq 4$, $n^2 \geq 16 > 5 > n+1$. □

0.27. **Prove $n! > 2^n$ for $n > 3$.**

Base case ($n = 4$): $24 > 16$. ✓

Inductive step: Assume $n! > 2^n$ for $n \geq 4$. Then

$$(n+1)! = (n+1)\cdot n! > (n+1)\cdot 2^n \geq 5\cdot 2^n > 2\cdot 2^n = 2^{n+1}. \quad \square$$

0.28. **Prove $1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$.**

Base case ($n = 1$): $1 = \frac{1\cdot 2\cdot 3}{6} = 1$. ✓

Inductive step: Assume the formula holds for n. Then

$$\begin{aligned}\sum_{k=1}^{n+1} k^2 &= \frac{n(n+1)(2n+1)}{6} + (n+1)^2 \\ &= \frac{n(n+1)(2n+1) + 6(n+1)^2}{6} \\ &= \frac{(n+1)[n(2n+1) + 6(n+1)]}{6} \\ &= \frac{(n+1)(2n^2+7n+6)}{6} \\ &= \frac{(n+1)(n+2)(2n+3)}{6}. \quad \square\end{aligned}$$

0.29. **Prove $X \cap (X_1 \cup \cdots \cup X_n) = (X \cap X_1) \cup \cdots \cup (X \cap X_n)$ by induction.**

Base case ($n = 1$): $X \cap X_1 = X \cap X_1$. ✓

Inductive step: Assume the result for n. Then

$$\begin{aligned}X \cap \left(\bigcup_{i=1}^{n+1} X_i\right) &= X \cap \left(\left(\bigcup_{i=1}^{n} X_i\right) \cup X_{n+1}\right) \\ &= \left(X \cap \bigcup_{i=1}^{n} X_i\right) \cup (X \cap X_{n+1}) \quad \text{(distributivity for } n = 2\text{)} \\ &= \left(\bigcup_{i=1}^{n} (X \cap X_i)\right) \cup (X \cap X_{n+1}) \quad \text{(inductive hypothesis)} \\ &= \bigcup_{i=1}^{n+1} (X \cap X_i). \quad \square\end{aligned}$$

0.30. **Prove $\sim(X_1 \cup \cdots \cup X_n) = (\sim X_1) \cap \cdots \cap (\sim X_n)$ by induction.**

Base case ($n = 2$): De Morgan's law: $\sim(X_1 \cup X_2) = (\sim X_1) \cap (\sim X_2)$. ✓

Inductive step: Assume for n. Then

$$\sim\left(\bigcup_{i=1}^{n+1} X_i\right) = \sim\left(\bigcup_{i=1}^{n} X_i \cup X_{n+1}\right) = \sim\left(\bigcup_{i=1}^{n} X_i\right) \cap (\sim X_{n+1}) = \left(\bigcap_{i=1}^{n} \sim X_i\right) \cap (\sim X_{n+1}) = \bigcap_{i=1}^{n+1} \sim X_i. \quad \square$$

0.31. **Prove $|\wp(A)| = 2^{|A|}$ by induction.**

Base case ($|A| = 0$): $\wp(\emptyset) = \{\emptyset\}$, so $|\wp(\emptyset)| = 1 = 2^0$. ✓

Inductive step: Assume $|\wp(A)| = 2^n$ for all sets with $|A| = n$. Let $|B| = n+1$ and pick any $b \in B$. Set $A = B \setminus \{b\}$. Every subset of B either contains b or not. Those not containing b are exactly $\wp(A)$ (2^n of them). Those containing b are in bijection with $\wp(A)$ via $S \mapsto S \setminus \{b\}$ (2^n more). Total: $2^n + 2^n = 2^{n+1}$. $\square$

0.32. **Prove strong induction is equivalent to ordinary induction.**

Let $P(n)$ be a statement and $Q(n) = (\forall i \le n)P(i)$.

($\Rightarrow$) Assume ordinary induction. The strong induction hypotheses are $P(0)$ and $(\forall m)((\forall i \le m)P(i) \Rightarrow P(m+1))$. Define $Q(n) = (\forall i \le n)P(i)$. Clearly $Q(0) \Leftrightarrow P(0)$. The strong hypothesis says $Q(m) \Rightarrow P(m+1)$, which means $Q(m) \Rightarrow Q(m+1)$ (since $Q(m+1) = Q(m) \wedge P(m+1)$). By ordinary induction on Q, $(\forall n)Q(n)$, hence $(\forall n)P(n)$.

($\Leftarrow$) Ordinary induction is the special case of strong induction where the hypothesis uses only $P(m)$ instead of $(\forall i \le m)P(i)$.

0.33. **Determine what strings the BNF defines.**

The grammar defines exactly the strings of the following four forms:

$$\langle\text{integer}\rangle, \qquad \langle\text{integer}\rangle., \qquad \langle\text{integer}\rangle.\langle\text{natural}\rangle, \qquad \langle\text{integer}\rangle.\langle\text{natural}\rangle E\langle\text{integer}\rangle.$$

Here $\langle$natural$\rangle$ is any nonempty string of decimal digits, and $\langle$integer$\rangle$ is either a natural number or a sign followed by a natural number. Thus the BNF defines signed or unsigned decimal numerals, with an optional fractional part and an optional exponent part.

0.34. **Cofinite sets with universal set $\mathbb{Z}$.**

(a) A finite set: $\{-5, 0, 3, 7\}$.

(b) A cofinite set: $\mathbb{Z} \setminus \{0\} = \{n \in \mathbb{Z} \mid n \ne 0\}$ (complement is $\{0\}$, which is finite).

(c) Neither: $2\mathbb{Z} = \{\ldots, -4, -2, 0, 2, 4, \ldots\}$, the even integers. Its complement is the set of odd integers, which is infinite; so it is not cofinite. It is infinite, so not finite.

0.35. **Prove the equipotent relation is an equivalence relation.**

Recall: $A \sim B$ iff there exists a bijection $f : A \to B$.

(a) *Reflexive*: $\mathrm{id}_A : A \to A$ is a bijection, so $A \sim A$.

(b) *Symmetric*: If $f : A \to B$ is a bijection, then $f^{-1} : B \to A$ is also a bijection.

(c) *Transitive*: If $f : A \to B$ and $g : B \to C$ are bijections, then $g \circ f : A \to C$ is a bijection.

0.36. **Define a function showing $|\mathbb{N}| = |\mathbb{N} \times \mathbb{N}|$.**

Let $T_k = \frac{k(k+1)}{2}$ be the kth triangular number. For each $n \in \mathbb{N}$ there are unique integers k, m with

$$n = T_k + m, \qquad 0 \le m \le k.$$

Define $f : \mathbb{N} \to \mathbb{N} \times \mathbb{N}$ by

$$f(n) = \langle k - m, m \rangle.$$

Thus

$$f(0) = \langle 0,0 \rangle, \quad f(1) = \langle 1,0 \rangle, \quad f(2) = \langle 0,1 \rangle, \quad f(3) = \langle 2,0 \rangle, \quad f(4) = \langle 1,1 \rangle, \quad f(5) = \langle 0,2 \rangle, \ \ldots$$

so f lists the pairs by diagonals. Every pair $\langle a, b \rangle$ occurs exactly once, namely when $k = a + b$ and $m = b$, so f is a bijection.

0.37. **Show $\mathbb{N}$ is equipotent to $\mathbb{Z}$.**

One bijection $f : \mathbb{N} \to \mathbb{Z}$ is

$$f(n) = \begin{cases} n/2, & \text{if } n \text{ is even,} \\ -(n+1)/2, & \text{if } n \text{ is odd.} \end{cases}$$

Thus

$$0 \mapsto 0, \quad 1 \mapsto -1, \quad 2 \mapsto 1, \quad 3 \mapsto -2, \quad 4 \mapsto 2, \quad 5 \mapsto -3, \ \ldots$$

so every integer appears exactly once. Hence $\mathbb{N}$ and $\mathbb{Z}$ are equipotent.

0.38. **Show $\mathbb{N}$ is equipotent to $\mathbb{Q}$.**

Let

$$S = \{\langle a, b \rangle \mid a, b \in \{1,2,3,\ldots\} \text{ and } \gcd(a,b) = 1\}.$$

Order the pairs in S first by increasing $a + b$, and among pairs with the same sum, by increasing a. This gives a list

$$\langle a_1, b_1 \rangle, \ \langle a_2, b_2 \rangle, \ \langle a_3, b_3 \rangle, \ \ldots$$

Define $h : \{1,2,3,\ldots\} \to \mathbb{Q}^+$ by

$$h(n) = \frac{a_n}{b_n}.$$

Every positive rational number has a unique reduced form a/b with $\gcd(a,b) = 1$, so h is a bijection from the positive integers onto $\mathbb{Q}^+$.

Now define $f : \mathbb{N} \to \mathbb{Q}$ by

$$f(0) = 0, \qquad f(2n-1) = h(n), \qquad f(2n) = -h(n) \quad (n \ge 1).$$

Then f lists 0, then the positive rationals, then their negatives:

$$0, \ h(1), \ -h(1), \ h(2), \ -h(2), \ldots$$

Hence f is a bijection from $\mathbb{N}$ onto $\mathbb{Q}$, so $\mathbb{N}$ and $\mathbb{Q}$ are equipotent.

0.39. **Show $\wp(\mathbb{N})$ is equipotent to $\{f : \mathbb{N} \to \{\text{Yes,No}\}\}$.**

Define $\phi : \wp(\mathbb{N}) \to \{f : \mathbb{N} \to \{\text{Yes,No}\}\}$ by $\phi(A) = \chi_A$, where $\chi_A(n) = \text{Yes}$ if $n \in A$ and No otherwise. This is clearly a bijection (its inverse maps f to $\{n \mid f(n) = \text{Yes}\}$).

0.40. **Show $\wp(\mathbb{N})$ is equipotent to $\mathbb{R}$.**

One explicit bijection is as follows.

By Exercise 0.39, $\wp(\mathbb{N})$ is equipotent to the set

$$B = \{(a_0, a_1, a_2, \ldots) \mid a_i \in \{0,1\}\}$$

of infinite binary sequences. It therefore suffices to show that B is equipotent to $\mathbb{R}$.

Let $E \subseteq B$ be the set of sequences that are eventually all 1s. This set is countable, because such a sequence is determined by a finite initial segment and the position after which only 1s occur. Let

$$D = \left\{ \frac{k}{2^n} \;\middle|\; n \in \mathbb{N},\ 0 \le k \le 2^n \right\}$$

be the set of dyadic rationals in $[0,1]$; D is also countable. Choose a bijection $\eta : E \to D$.

Now define $\Phi : B \to [0,1]$ by

$$\Phi(a_0, a_1, a_2, \ldots) = \begin{cases} \displaystyle\sum_{i=0}^{\infty} \frac{a_i}{2^{i+1}}, & \text{if } (a_0, a_1, a_2, \ldots) \notin E, \\ \eta(a_0, a_1, a_2, \ldots), & \text{if } (a_0, a_1, a_2, \ldots) \in E. \end{cases}$$

If a sequence is not eventually all 1s, its binary expansion is the unique binary expansion of a nondyadic real in $[0,1]$. Thus the first clause is a bijection from $B \setminus E$ onto $[0,1] \setminus D$. The second clause is a bijection from E onto D. Therefore Φ is a bijection from B onto $[0,1]$.

It remains to show that $[0,1]$ is equipotent to $\mathbb{R}$. First define a bijection $j : [0,1] \to (0,1)$ by

$$j(x) = \begin{cases} \frac{1}{2}, & x = 0, \\ \frac{1}{4}, & x = 1, \\ \frac{1}{2^{n+2}}, & x = \frac{1}{2^n} \text{ for } n \ge 1, \\ x, & \text{otherwise.} \end{cases}$$

This simply shifts the countable set $\{1, 1/2, 1/4, 1/8, \ldots\}$ inward and fixes all other points. Next define

$$g : (0,1) \to \mathbb{R}, \qquad g(x) = \begin{cases} 2 - \frac{1}{x}, & 0 < x < \frac{1}{2}, \\ 0, & x = \frac{1}{2}, \\ \frac{1}{1-x} - 2, & \frac{1}{2} < x < 1. \end{cases}$$

The first branch maps $(0, \frac{1}{2})$ onto $(-\infty, 0)$, the middle point maps to 0, and the third branch maps $(\frac{1}{2}, 1)$ onto $(0, \infty)$. Hence g is a bijection from $(0,1)$ onto $\mathbb{R}$, and $g \circ j$ is a bijection from $[0,1]$ onto $\mathbb{R}$.

Since $\wp(\mathbb{N}) \sim B$, $B \sim [0,1]$, and $[0,1] \sim \mathbb{R}$, the equipotent relation is transitive, so $\wp(\mathbb{N}) \sim \mathbb{R}$.

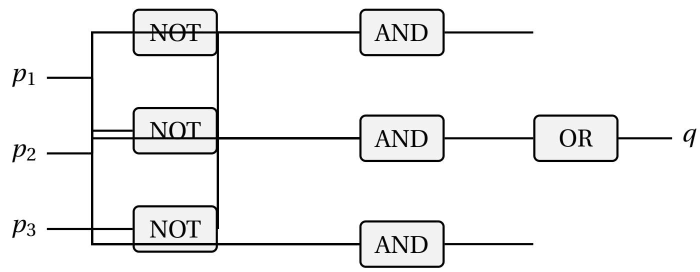
0.41. **Draw a circuit for q (Figure 0.8).**

The truth table shows $q = 1$ when $(p_1, p_2, p_3) \in \{(0,1,0), (1,0,0), (1,1,1)\}$.

Using the rows with $q = 1$, the PDNF is

$$q = (\bar{p}_1 p_2 \bar{p}_3) \vee (p_1 \bar{p}_2 \bar{p}_3) \vee (p_1 p_2 p_3).$$

The corresponding AND-OR circuit is:



0.42. **Draw circuits for q_1, q_2, q_3 (Figure 0.9).**

Reading the truth table row by row (rows indexed 0–15, with $p_1 p_2 p_3 p_4$ in binary):

$q_1 = 1$ at rows 0,2,3,5,6,8,9,12 (i.e., minterms $m_0, m_2, m_3, m_5, m_6, m_8, m_9, m_{12}$): (0000), (0010), (0011), (0101), (0110), (

$q_2 = 1$ at rows 0,1,3,4,6,10,12,13,14 (minterms $m_0, m_1, m_3, m_4, m_6, m_{10}, m_{12}, m_{13}, m_{14}$): (0000), (0001), (0011), (0100),

$q_3 = 1$ at rows 0,4,8,12,14 (minterms $m_0, m_4, m_8, m_{12}, m_{14}$): (0000), (0100), (1000), (1100), (1110).

Using the PDNF construction from the text, the required formulas are:

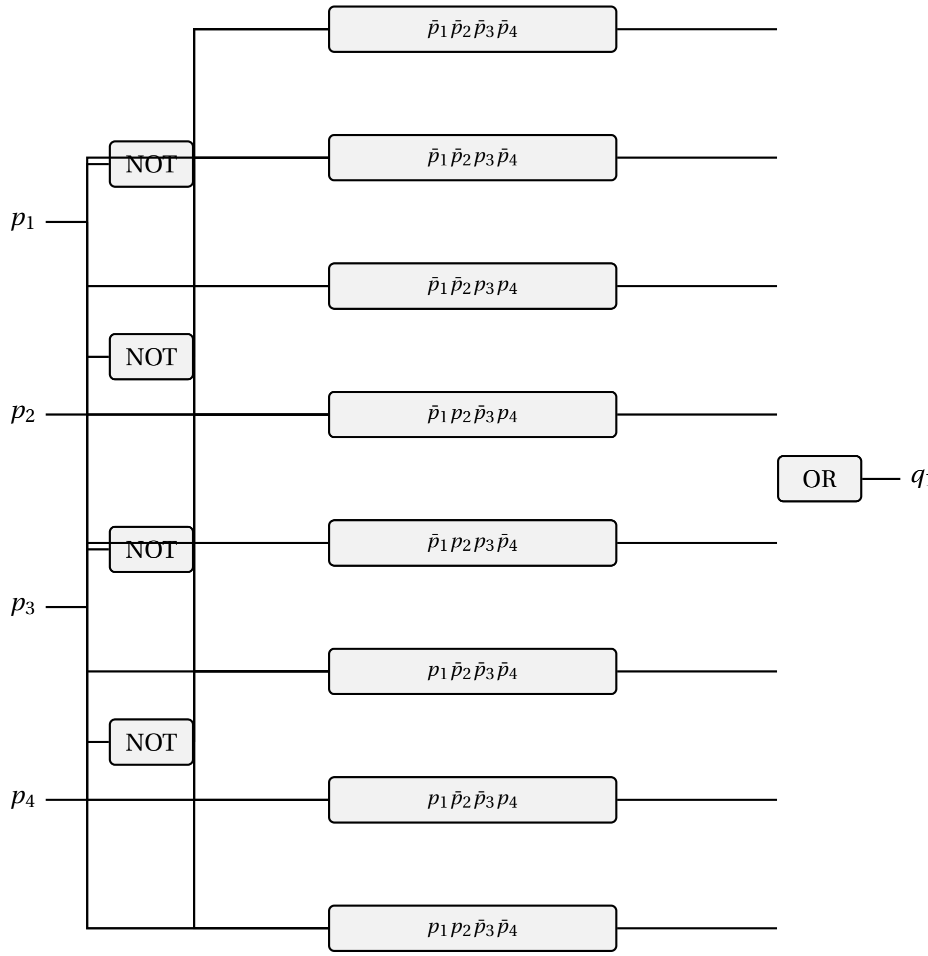
$$\begin{aligned} q_1 = {} & \bar{p}_1\bar{p}_2\bar{p}_3\bar{p}_4 \vee \bar{p}_1\bar{p}_2 p_3\bar{p}_4 \vee \bar{p}_1\bar{p}_2 p_3 p_4 \vee \bar{p}_1 p_2\bar{p}_3 p_4 \\ & \vee \bar{p}_1 p_2 p_3\bar{p}_4 \vee p_1\bar{p}_2\bar{p}_3\bar{p}_4 \vee p_1\bar{p}_2\bar{p}_3 p_4 \vee p_1 p_2\bar{p}_3\bar{p}_4 \end{aligned}$$

$$\begin{aligned} q_2 = {} & \bar{p}_1\bar{p}_2\bar{p}_3\bar{p}_4 \vee \bar{p}_1\bar{p}_2\bar{p}_3 p_4 \vee \bar{p}_1\bar{p}_2 p_3 p_4 \vee \bar{p}_1 p_2\bar{p}_3\bar{p}_4 \\ & \vee \bar{p}_1 p_2 p_3\bar{p}_4 \vee p_1\bar{p}_2 p_3\bar{p}_4 \vee p_1 p_2\bar{p}_3\bar{p}_4 \vee p_1 p_2\bar{p}_3 p_4 \vee p_1 p_2 p_3\bar{p}_4 \end{aligned}$$

$$\begin{aligned} q_3 = {} & \bar{p}_1\bar{p}_2\bar{p}_3\bar{p}_4 \vee \bar{p}_1 p_2\bar{p}_3\bar{p}_4 \vee p_1\bar{p}_2\bar{p}_3\bar{p}_4 \\ & \vee p_1 p_2\bar{p}_3\bar{p}_4 \vee p_1 p_2 p_3\bar{p}_4 \end{aligned}$$

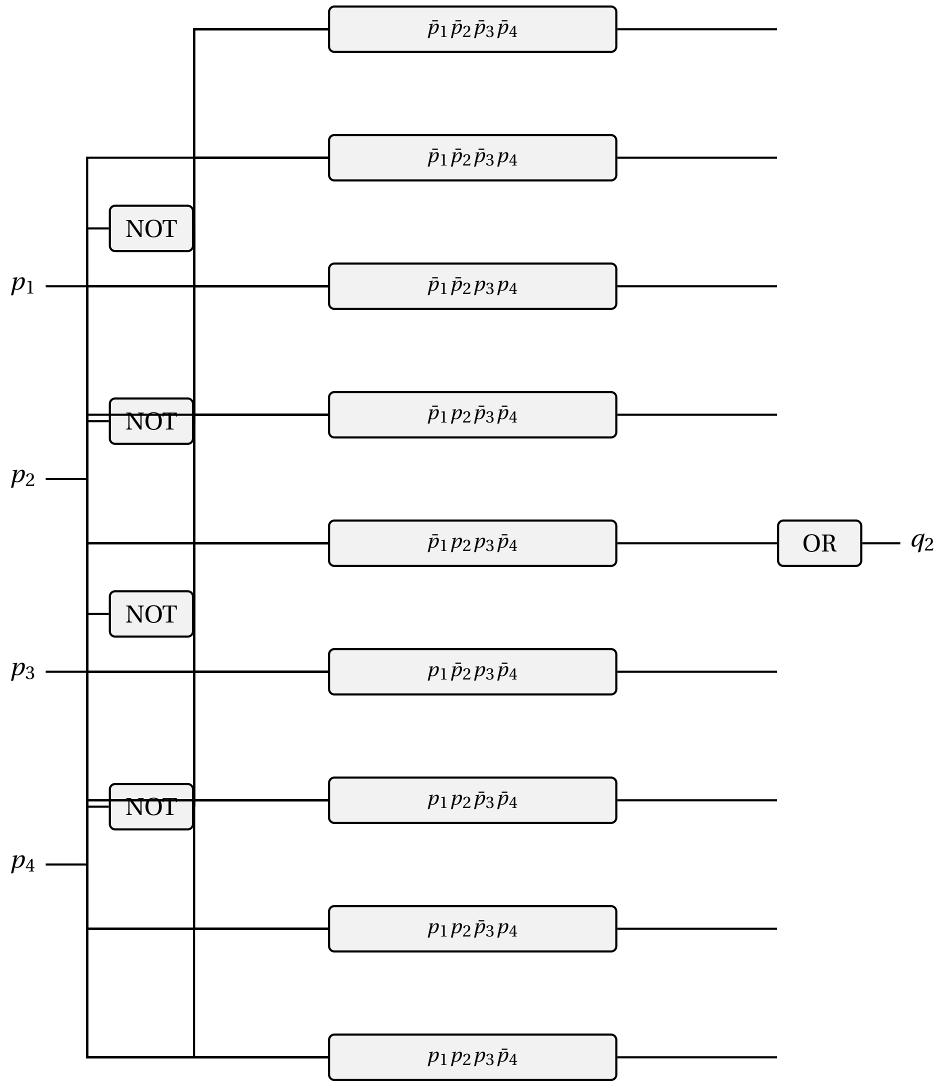
Each circuit below is the corresponding two-level PDNF circuit: NOT gates first, then one AND subcircuit for each product term, feeding a final OR gate.

(a) For q_1:

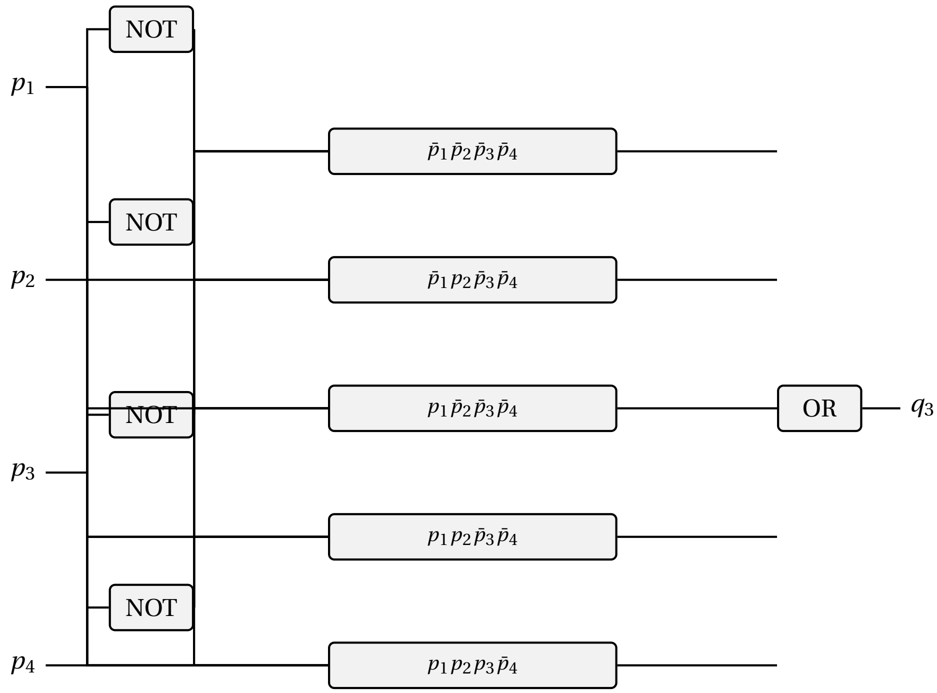


$\bar{p}_1\bar{p}_2\bar{p}_3\bar{p}_4$
NOT
$\bar{p}_1\bar{p}_2 p_3\bar{p}_4$
p_1
$\bar{p}_1\bar{p}_2 p_3 p_4$
NOT
p_2
$\bar{p}_1 p_2\bar{p}_3 p_4$
OR
q_1
NOT
$\bar{p}_1 p_2 p_3\bar{p}_4$
p_3
$p_1\bar{p}_2\bar{p}_3\bar{p}_4$
NOT
p_4
$p_1\bar{p}_2\bar{p}_3 p_4$
$p_1 p_2\bar{p}_3\bar{p}_4$

(b) For q_2:


$\bar{p}_1\bar{p}_2\bar{p}_3\bar{p}_4$
$\bar{p}_1\bar{p}_2\bar{p}_3 p_4$
NOT
p_1
$\bar{p}_1\bar{p}_2 p_3 p_4$
NOT
$\bar{p}_1 p_2\bar{p}_3\bar{p}_4$
p_2
$\bar{p}_1 p_2 p_3\bar{p}_4$
OR
q_2
NOT
p_3
$p_1\bar{p}_2 p_3\bar{p}_4$
NOT
$p_1 p_2\bar{p}_3\bar{p}_4$
p_4
$p_1 p_2\bar{p}_3 p_4$
$p_1 p_2 p_3\bar{p}_4$

(c) For q_3:



NOT
NOT
NOT
NOT
p_1
p_2
p_3
p_4
$\bar{p}_1\bar{p}_2\bar{p}_3\bar{p}_4$
$\bar{p}_1 p_2\bar{p}_3\bar{p}_4$
$p_1\bar{p}_2\bar{p}_3\bar{p}_4$
$p_1 p_2\bar{p}_3\bar{p}_4$
$p_1 p_2 p_3\bar{p}_4$
OR
q_3

Chapter 1

Introduction and Basic Definitions — Solutions

1.1. **Show $\overline{\delta}_t = \overline{\delta}_h$.**

Let $P(n)$: $(\forall s \in S)(\forall x \in \Sigma^n)(\overline{\delta}_t(s,x) = \overline{\delta}_h(s,x))$.

Base: $n = 0$: $\overline{\delta}_t(s,\lambda) = s = \overline{\delta}_h(s,\lambda)$. For $n = 1$: $\overline{\delta}_t(s,\mathbf{a}) = \delta(s,\mathbf{a}) = \overline{\delta}_h(s,\mathbf{a})$.

Inductive step: Let $x = ya$ where $|y| = n$ and $a \in \Sigma$. Assume $\overline{\delta}_t(s,y) = \overline{\delta}_h(s,y)$ for all s, y with $|y| \le n$.

We prove $\overline{\delta}_t(s,wa) = \delta(\overline{\delta}_t(s,w),a)$ for all $w \in \Sigma^*$, $a \in \Sigma$, $s \in S$, by induction on $|w|$.

If $|w| = 0$: $\overline{\delta}_t(s,a) = \delta(s,a)$. ✓

If $|w| = n+1$, write $w = bv$ with $b \in \Sigma$: $\overline{\delta}_t(s,bva) = \overline{\delta}_t(\delta(s,b),va) = \delta(\overline{\delta}_t(\delta(s,b),v),a)$ [IH] $= \delta(\overline{\delta}_t(s,bv),a)$. This establishes the claim. Now:

$$\overline{\delta}_h(s,wa) = \delta(\overline{\delta}_h(s,w),a) = \delta(\overline{\delta}_t(s,w),a) \text{ [IH]} = \overline{\delta}_t(s,wa). \quad \square$$

1.2. **Modify vending machine to include coin-return r.**

Add a new input symbol **r** (coin return). When **r** is pressed, transition back to the initial state s_0 (0 cents deposited) from any state. All other transitions remain unchanged. Formally, for every state s_i (representing i cents), $\delta(s_i,\mathbf{r}) = s_0$.

1.3. **The DFA of Figure 1.26.**

(a) The quintuple is

$$A = \langle\{\mathbf{0},\mathbf{1}\},\{q_0,q_1\},q_0,\delta,\{q_1\}\rangle,$$

where

$$\delta(q_0,\mathbf{0}) = q_1, \quad \delta(q_0,\mathbf{1}) = q_1, \quad \delta(q_1,\mathbf{0}) = q_1, \quad \delta(q_1,\mathbf{1}) = q_0.$$

(b) The language is

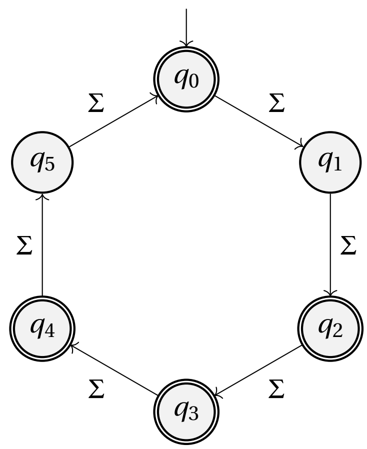
$$L(A) = \Sigma^*\mathbf{0}(\mathbf{11})^* \cup \mathbf{1}(\mathbf{11})^*, \qquad \Sigma = \{\mathbf{0},\mathbf{1}\}.$$

Equivalently, a word is accepted iff it either ends in **0** followed by an even number of trailing **1**s, or consists entirely of an odd number of **1**s.

1.4. **Build a DFA accepting $L = \{x \mid |x|$ is a multiple of 2 or 3$\}$ over $\{\mathbf{a}, \mathbf{b}, \mathbf{c}\}$.**

We need $|x| \equiv 0 \pmod 2$ or $|x| \equiv 0 \pmod 3$, i.e., $|x| \in \{0,2,3,4,6,8,9,\ldots\}$. The complement is $\{|x| \equiv 1 \pmod 6$ or $|x| \equiv 5 \pmod 6\}$. Use a 6-state DFA tracking $|x|$ mod 6, with final states $\{0,2,3,4\}$ (i.e., everything except 1 and 5 mod 6).

$S = \{q_0, q_1, q_2, q_3, q_4, q_5\}$, $s_0 = q_0$, $F = \{q_0, q_2, q_3, q_4\}$, $\delta(q_i, a) = q_{(i+1) \bmod 6}$ for all $a \in \Sigma$.



1.5. **DFAs for L_1, L_2, L_3 over $\{0,1\}$.**

(a) $L_1 = \{x \mid |x| \bmod 7 = 4\}$: 7-state DFA with states $q_0, \ldots, q_6$, $F = \{q_4\}$, $\delta(q_i, a) = q_{(i+1) \bmod 7}$.

(b) $L_2 = \Sigma^* \setminus \{w \mid w$ ends in $1\}$: The complement is all words ending in $\mathbf{1}$. So L_2 accepts words not ending in $\mathbf{1}$: either λ, or words ending in $\mathbf{0}$. Two-state DFA: s_0 (start, final – not ending in 1, or empty), s_1 (ending in 1, nonfinal). $\delta(s_0, \mathbf{0}) = s_0$, $\delta(s_0, \mathbf{1}) = s_1$, $\delta(s_1, \mathbf{0}) = s_0$, $\delta(s_1, \mathbf{1}) = s_1$.

(c) $L_3 = \{y \mid |y|_0 = |y|_1\}$: not FAD. Suppose some DFA with n states accepted it. Consider the $n+1$ strings

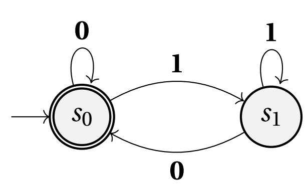
$$\lambda,\ \mathbf{0},\ \mathbf{00},\ \ldots,\ \mathbf{0}^n.$$

Two of them, say $\mathbf{0}^i$ and $\mathbf{0}^j$ with $i < j$, must lead from the start state to the same state. Then appending $\mathbf{1}^i$ would give the same final outcome for both strings. But

$$\mathbf{0}^i\mathbf{1}^i \in L_3 \quad \text{while} \quad \mathbf{0}^j\mathbf{1}^i \notin L_3,$$

since the second string has more $\mathbf{0}$s than $\mathbf{1}$s. This contradiction shows that no DFA can accept L_3.

The 2-state DFA for L_2:

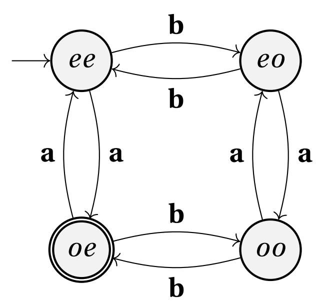


1.6. **DFAs A_1, A_2, A_3, A_4 over $\{\mathbf{a}, \mathbf{b}\}$.**

(a) Use a 4-state DFA with states $\{ee, eo, oe, oo\}$ (parity of #a, parity of #b). A_1: $F = \{oe\}$ (#a odd, #b even). A_2: $F = \{eo, oe, oo\}$ (not ee). A_3: $F = \{oe, eo\}$ (exactly one count even). A_4: $F = \{ee, oe\}$ (#a even).

(b) Each machine is a quotient or sub-machine of the 4-state product machine from Example 1.10, which tracks parities of #a and #b.

The common 4-state machine (shown here for A_1 with $F = \{oe\}$; change F for the other sub-machines):



1.7. **Modify M to accept $L(M) - \{\lambda\}$.**

If $s_0 \notin F$ then $\lambda \notin L(M)$ and M already accepts $L(M) \setminus \{\lambda\} = L(M)$; nothing to do.

If $s_0 \in F$, add a fresh nonfinal start state s_0' and set

$$\delta'(s_0', a) = \delta(s_0, a) \quad \text{for all } a \in \Sigma,$$

leaving all other transitions and the final-state set F unchanged. The new machine $M' = (\Sigma, S \cup \{s_0'\}, s_0', \delta', F)$ accepts exactly $L(M) \setminus \{\lambda\}$.

Correctness: λ is rejected because $s_0' \notin F$. For any nonempty $x = a_1 \cdots a_k$, the computation in M' follows $s_0' \xrightarrow{a_1} \delta(s_0, a_1) \xrightarrow{a_2} \cdots$, which mirrors the computation of M from s_0 on x, so $x \in L(M')$ iff $x \in L(M)$.

Why not remove s_0 from F? That would reject nonempty strings x with $\bar{\delta}(s_0, x) = s_0$, which belong to $L(M)$ and should remain accepted.

1.8. **General procedure to modify M to accept $L(M) \cup \{\lambda\}$.**

If $s_0 \in F$ already, M already accepts λ. Otherwise, add s_0 to F: $F' = F \cup \{s_0\}$. All transitions stay the same. This accepts $L(M) \cup \{\lambda\}$.

1.9. **DFA for $\Psi = \{x \mid (x$ begins with $\mathbf{d}) \vee (x$ contains $\mathbf{bb})\}$.**

The machine must distinguish "first character is **d**" (which makes the string accepted regardless of what follows) from "**d** appears later". A 3-state machine conflates these; the minimal DFA has **4 states**:

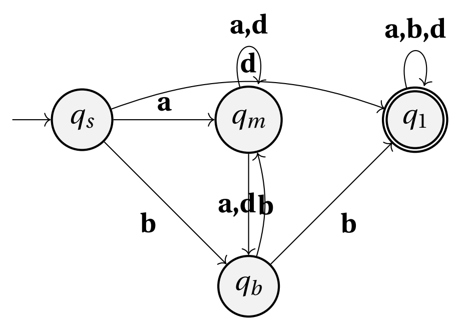
- q_s: start (no characters read yet).

- q_m: at least one character read; string does not start with **d** and no **bb** seen; last char was not **b**.
- q_b: same as q_m but last char was **b** (pending **bb**).
- q_1: accepting sink (string starts with **d**, or **bb** has been seen).

$F = \{q_1\}$.

	a	**b**	**d**
q_s	q_m	q_b	q_1
q_m	q_m	q_b	q_m
q_b	q_m	q_1	q_m
q_1	q_1	q_1	q_1

Spot-check: **bd** — $q_s \xrightarrow{b} q_b \xrightarrow{d} q_m$, rejected. ✓ **bb** — $q_s \xrightarrow{b} q_b \xrightarrow{b} q_1$, accepted. ✓ **d** — $q_s \xrightarrow{d} q_1$, accepted. ✓



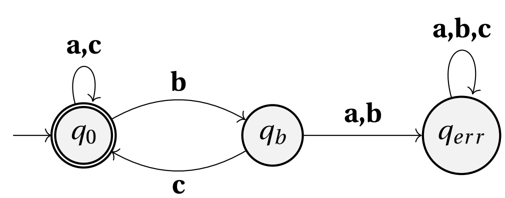
1.10. **DFA for $\Phi = \{x \mid$ every b in x is immediately followed by c$\}$.**

States: q_0 (ok, last char not **b**), q_b (last char was **b**), q_{err} (dead state). $F = \{q_0\}$.

$$\delta(q_0, \mathbf{a}) = q_0, \quad \delta(q_0, \mathbf{b}) = q_b, \quad \delta(q_0, \mathbf{c}) = q_0,$$

$$\delta(q_b, \mathbf{c}) = q_0, \quad \delta(q_b, \mathbf{a}) = q_{err}, \quad \delta(q_b, \mathbf{b}) = q_{err},$$

$$\delta(q_{err}, \mathbf{a}) = \delta(q_{err}, \mathbf{b}) = \delta(q_{err}, \mathbf{c}) = q_{err}.$$



1.11. **DFA for Ω (numbers divisible by 7).**

The key insight: if the current state represents $n \bmod 7$ (the value of the number read so far), then reading digit d gives the new value $(10n + d) \bmod 7$.

States $q_0, \ldots, q_6$ (representing $0, \ldots, 6 \bmod 7$). $s_0 = q_0$, $F = \{q_0\}$.

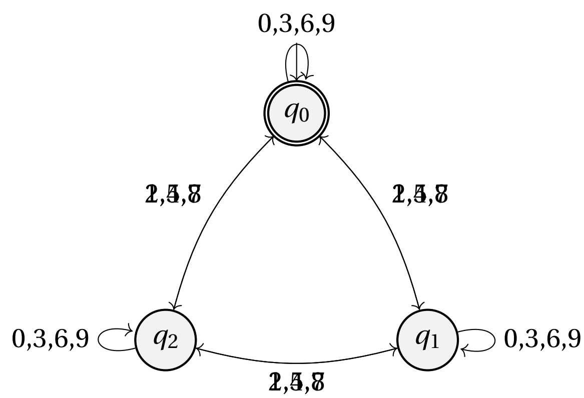
$$\delta(q_i, d) = q_{(10i+d) \bmod 7}.$$

1.12. **DFA for Γ (divisible by 3).**

Three states q_0, q_1, q_2 (representing $0, 1, 2 \bmod 3$). $F = \{q_0\}$.

$$\delta(q_i, d) = q_{(i+d) \bmod 3}$$

since $(10i + d) \bmod 3 = (i + d) \bmod 3$ (because $10 \equiv 1 \pmod 3$).



1.13. **DFA for K (divisible by 5).**

Since $10 \equiv 0 \pmod 5$, we have $(10i + d) \bmod 5 = d \bmod 5$. So only the last digit matters.

Five-state DFA: $\delta(q_i, d) = q_{d \bmod 5}$; $F = \{q_0\}$.

Two-state DFA: q_0 (final, last digit in $\{0, 5\}$), q_1 (nonfinal, last digit not in $\{0, 5\}$). $\delta(q_i, d) = q_0$ if $d \in \{0, 5\}$, else q_1.

1.14. **DFA for the first eight primes:** $\{2, 3, 5, 7, 11, 13, 17, 19\}$**.**

This language is finite, so it is FAD. A concrete DFA is the prefix-tree machine for the set

$$P = \{\lambda, \mathbf{1}, \mathbf{11}, \mathbf{13}, \mathbf{17}, \mathbf{19}, \mathbf{2}, \mathbf{3}, \mathbf{5}, \mathbf{7}\}.$$

Let the states be $\{q_x \mid x \in P\} \cup \{q_{dead}\}$, with start state q_λ and final states

$$F = \{q_{\mathbf{2}}, q_{\mathbf{3}}, q_{\mathbf{5}}, q_{\mathbf{7}}, q_{\mathbf{11}}, q_{\mathbf{13}}, q_{\mathbf{17}}, q_{\mathbf{19}}\}.$$

For any state q_x and any digit $\boldsymbol{d}$, define

$$\delta(q_x, \boldsymbol{d}) = \begin{cases} q_{x\boldsymbol{d}} & \text{if } x\boldsymbol{d} \in P, \\ q_{dead} & \text{otherwise.} \end{cases}$$

Also $\delta(q_{dead}, \boldsymbol{d}) = q_{dead}$ for every digit $\boldsymbol{d}$. Thus q_λ branches to $q_{\mathbf{2}}, q_{\mathbf{3}}, q_{\mathbf{5}}, q_{\mathbf{7}}$, and to $q_{\mathbf{1}}$ on input $\mathbf{1}$; then $q_{\mathbf{1}}$ branches to $q_{\mathbf{11}}, q_{\mathbf{13}}, q_{\mathbf{17}}, q_{\mathbf{19}}$ on inputs $\mathbf{1}, \mathbf{3}, \mathbf{7}, \mathbf{9}$, respectively. Every other transition goes to q_{dead}, so the machine accepts exactly the first eight primes.

1.15. **Combinations of u, v, w with $uvw = $ cab.**

(a) The string **cab** has length 3. We must split it as uvw with $u, v, w \in \Sigma^*$. The split points are $i = |u|$, $j = |u| + |v|$ with $0 \le i \le j \le 3$. There are $\binom{n+2}{2} = \binom{5}{2} = 10$ ways for length 3:
(u, v, w): $(\lambda, \lambda, \mathbf{cab})$, $(\lambda, \mathbf{c}, \mathbf{ab})$, $(\lambda, \mathbf{ca}, \mathbf{b})$, $(\lambda, \mathbf{cab}, \lambda)$, $(\mathbf{c}, \lambda, \mathbf{ab})$, $(\mathbf{c}, \mathbf{a}, \mathbf{b})$, $(\mathbf{c}, \mathbf{ab}, \lambda)$, $(\mathbf{ca}, \lambda, \mathbf{b})$, $(\mathbf{ca}, \mathbf{b}, \lambda)$, $(\mathbf{cab}, \lambda, \lambda)$.

(b) For a string of length n: the number of ways to choose split points $0 \le i \le j \le n$ is $\binom{n+2}{2} = \frac{(n+1)(n+2)}{2}$.

1.16. **DFA for $\Xi = \{x \mid x$ contains ≥ 2 consecutive bs and does not contain aa$\}$.**

The DFA tracks two conditions simultaneously: whether **aa** has ever appeared, and whether **bb** has ever appeared. Once **aa** appears the string is rejected regardless; once **bb** appears we continue watching for **aa**.

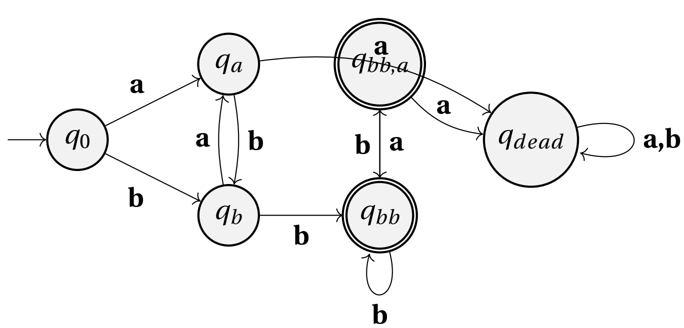
States (6 total):

- q_0: start; no **bb**, no **aa**, last char none.
- q_a: no **bb**, no **aa**, last char = **a**.
- q_b: no **bb**, no **aa**, last char = **b**.
- q_{bb}: seen **bb**, no **aa**, last char = **b**.
- $q_{bb,a}$: seen **bb**, no **aa**, last char = **a**.
- q_{dead}: seen **aa** (trap).

$F = \{q_{bb}, q_{bb,a}\}$.

Transitions:

	a	**b**
q_0	q_a	q_b
q_a	q_{dead}	q_b
q_b	q_a	q_{bb}
q_{bb}	$q_{bb,a}$	q_{bb}
$q_{bb,a}$	q_{dead}	q_{bb}
q_{dead}	q_{dead}	q_{dead}



1.17. **Modify Figure 1.11 to reject DO and DATA.**

Keep the structure of Example 1.9, but split the ordinary identifier states so that the machine can remember the exceptional prefixes ***D***, ***DO***, ***DA***, ***DAT***, and ***DATA*** until it is clear whether the identifier is exactly one of the reserved words.

One convenient design uses the states

$$s_0,\ d_1,\ do_2,\ da_2,\ dat_3,\ data_4,\ g_1,\ g_2,\ g_3,\ g_4,\ g_5,\ g_6,\ s_7,$$

where g_i means "a legal identifier of length i whose prefix is no longer one of the exceptional words," and s_7 is the dead state. Embedded blanks behave exactly as in Example 1.9: they loop in every nondead state.

The key transitions are:

- From s_0, the letter ***D*** goes to d_1, every other letter goes to g_1, digits and characters from Γ go to s_7, and $\diamond$ loops to s_0.
- From d_1, the letter ***O*** goes to do_2, ***A*** goes to da_2, and every other letter or digit goes to g_2.
- From do_2, any further letter or digit goes to g_3; if the identifier ends while the machine is still in do_2, the word is rejected.
- From da_2, the letter ***T*** goes to dat_3; every other letter or digit goes to g_3.
- From dat_3, the letter ***A*** goes to $data_4$; every other letter or digit goes to g_4.
- From $data_4$, any further letter or digit goes to g_5; if the identifier ends while the machine is still in $data_4$, the word is rejected.
- The states $g_1, \ldots, g_6$ behave exactly like the length-counting states $s_1, \ldots, s_6$ of Example 1.9.

The final states are

$$\{d_1, da_2, dat_3, g_1, g_2, g_3, g_4, g_5, g_6\}.$$

So identifiers such as ***D***, ***DA***, ***DAT***, ***DOG***, and ***DATA*1** are accepted, while exactly ***DO*** and exactly ***DATA*** are rejected.

1.18. **Modify machine for real-number constants (extended BNF).**

The new BNF allows leading decimal points like `.5` and `+.5`. Modify the DFA from Figure 1.13 to add transitions that accept the input immediately after a sign symbol or at the start by going to a decimal-point state, from which digits are read as the fractional part.

(a) A concrete modified DFA is obtained by using the states

$$s_0 = \text{start}, \quad s_1 = \text{sign}, \quad s_2 = \text{integer part}, \quad s_3 = \text{dot after integer},$$

$$s_4 = \text{E just read}, \quad s_5 = \text{sign after E}, \quad s_6 = \text{exponent digits},$$

$$s_8 = \text{fractional digits}, \quad s_9 = \text{leading dot before any digit}, \quad s_7 = \text{dead}.$$

The accepting states are $\{s_2, s_3, s_8, s_6\}$. The important transitions are:

$$
\begin{array}{llll}
\delta(s_0, \text{digit}) = s_2, & \delta(s_0, \boldsymbol{+}) = s_1, & \delta(s_0, \boldsymbol{-}) = s_1, & \delta(s_0, \boldsymbol{.}) = s_9, \\
\delta(s_1, \text{digit}) = s_2, & \delta(s_1, \boldsymbol{.}) = s_9, & & \\
\delta(s_2, \text{digit}) = s_2, & \delta(s_2, \boldsymbol{.}) = s_3, & & \\
\delta(s_3, \text{digit}) = s_8, & & & \\
\delta(s_8, \text{digit}) = s_8, & \delta(s_8, \boldsymbol{E}) = s_4, & & \\
\delta(s_9, \text{digit}) = s_8, & & & \\
\delta(s_4, \text{digit}) = s_6, & \delta(s_4, \boldsymbol{+}) = s_5, & \delta(s_4, \boldsymbol{-}) = s_5, & \\
\delta(s_5, \text{digit}) = s_6, & & & \\
\delta(s_6, \text{digit}) = s_6. & & &
\end{array}
$$

All unspecified transitions go to s_7, and s_7 loops to itself. This accepts strings such as `.5`, `+.5`, and `-.5E7`, while still rejecting strings such as `8.E-10`, because there must be at least one digit after the decimal point if an exponent is present.

(b) Here is a Pascal implementation in the same style as Figures 1.4–1.8.

```
program RealConst(input, output);
type
   Sigma = (digit, plus, minus, dot, emark, other);
   State = (start, sign, intpart, dotafterint, expmark,
            expsign, expdigits, dead, fracpart, leadingdot);
var
   TransitionTable : array [State, Sigma] of State;
   FinalState : set of State;

function ClassOf(c : char) : Sigma;
begin
   if c in ['0'..'9'] then
      ClassOf := digit
   else if c = '+' then
      ClassOf := plus
   else if c = '-' then
      ClassOf := minus
   else if c = '.' then
      ClassOf := dot
   else if c = 'E' then
      ClassOf := emark
   else
      ClassOf := other
end; { ClassOf }

function Delta(s : State; a : Sigma) : State;
begin
   Delta := TransitionTable[s, a]
```

```
end; { Delta }

function DeltaBar(s : State) : State;
var
   t : State;
begin
   t := s;
   while not eoln(input) do
      begin
         t := Delta(t, ClassOf(input^));
         get(input)
      end;
   DeltaBar := t
end; { DeltaBar }

function Accept : boolean;
begin
   Accept := DeltaBar(start) in FinalState
end; { Accept }

procedure Initialize;
var
   s : State;
   a : Sigma;
begin
   FinalState := [intpart, dotafterint, fracpart, expdigits];

   for s := start to leadingdot do
      for a := digit to other do
         TransitionTable[s, a] := dead;

   TransitionTable[start, digit] := intpart;
   TransitionTable[start, plus] := sign;
   TransitionTable[start, minus] := sign;
   TransitionTable[start, dot] := leadingdot;

   TransitionTable[sign, digit] := intpart;
   TransitionTable[sign, dot] := leadingdot;

   TransitionTable[intpart, digit] := intpart;
   TransitionTable[intpart, dot] := dotafterint;

   TransitionTable[dotafterint, digit] := fracpart;
   TransitionTable[leadingdot, digit] := fracpart;
```

```
    TransitionTable[fracpart, digit] := fracpart;
    TransitionTable[fracpart, emark] := expmark;

    TransitionTable[expmark, digit] := expdigits;
    TransitionTable[expmark, plus] := expsign;
    TransitionTable[expmark, minus] := expsign;

    TransitionTable[expsign, digit] := expdigits;

    TransitionTable[expdigits, digit] := expdigits
 end; { Initialize }

 begin
    Initialize;
    if Accept then
       writeln(output, 'Accepted')
    else
       writeln(output, 'Rejected')
 end.
```

1.19. **Show Definition 1.11(i) is implied by (ii) and (iii).**

In Definition 1.11, part (i) is the statement

$$(\forall s \in S)(\forall \boldsymbol{a} \in \Sigma)\, \overline{\delta}(s,\boldsymbol{a}) = \delta(s,\boldsymbol{a}).$$

Assume only parts (ii) and (iii). Then for any state s and any letter $\boldsymbol{a}$,

$$\overline{\delta}(s,\boldsymbol{a}) = \overline{\delta}(s,\boldsymbol{a}\lambda)$$

because $\boldsymbol{a}\lambda = \boldsymbol{a}$. Now apply part (iii) with $x = \lambda$:

$$\overline{\delta}(s,\boldsymbol{a}\lambda) = \overline{\delta}(\delta(s,\boldsymbol{a}),\lambda).$$

Finally, by part (ii),

$$\overline{\delta}(\delta(s,\boldsymbol{a}),\lambda) = \delta(s,\boldsymbol{a}).$$

Therefore $\overline{\delta}(s,\boldsymbol{a}) = \delta(s,\boldsymbol{a})$ for every $s \in S$ and every $\boldsymbol{a} \in \Sigma$, which is exactly part (i).

1.20. **Develop a succinct description of the transition function from Example 1.9.**

Let Γ be the set defined in Example 1.9, so that the alphabet is partitioned into letters, digits, blanks $\diamond$, and the remaining symbols Γ. Then the transition function can be described uniformly as follows:

$$\delta(s_0,\text{letter}) = s_1, \qquad \delta(s_0,\text{digit}) = s_7, \quad \delta(s_0,\diamond) = s_0, \quad \delta(s_0,\gamma) = s_7 \quad (\gamma \in \Gamma),$$

$$\delta(s_i,\text{letter or digit}) = s_{i+1} \qquad (1 \le i \le 5),$$

$$\delta(s_i,\diamond) = s_i \qquad (1 \le i \le 6),$$

$$\delta(s_i,\gamma) = s_7 \qquad (1 \le i \le 6,\ \gamma \in \Gamma),$$

$$\delta(s_6,\text{letter or digit}) = s_7,$$

$$\delta(s_7,\sigma) = s_7 \qquad (\sigma \in \Sigma).$$

So s_i means "a legal identifier of length i has been seen," blanks do not change the current length, and any illegal character or seventh nonblank character forces the machine into the dead state s_7.

1.21. **Finite, cofinite, neither over $\{\mathbf{a}, \mathbf{b}\}^*$.**

(a) Finite: $\{\mathbf{a}, \mathbf{ab}\}$.

(b) Cofinite: $\{\mathbf{a}, \mathbf{b}\}^* \setminus \{\mathbf{a}\}$ (complement is $\{\mathbf{a}\}$, finite).

(c) Neither: $\{\mathbf{a}^n \mid n \text{ even}\}$. Infinite, and its complement $\{\mathbf{a}^n \mid n \text{ odd}\} \cup \{x \mid x \text{ contains } \mathbf{b}\}$ is also infinite.

1.22. **DFA of Figure 1.27.**

(a) The quintuple is

$$\langle\{\boldsymbol{a}, \boldsymbol{b}\}, \{q_0, q_1, q_2, q_3, q_4\}, q_0, \delta, \{q_2, q_3\}\rangle,$$

where

$$\begin{aligned} \delta(q_0, \boldsymbol{a}) &= q_1, & \delta(q_0, \boldsymbol{b}) &= q_0, \\ \delta(q_1, \boldsymbol{a}) &= q_2, & \delta(q_1, \boldsymbol{b}) &= q_1, \\ \delta(q_2, \boldsymbol{a}) &= q_2, & \delta(q_2, \boldsymbol{b}) &= q_4, \\ \delta(q_3, \boldsymbol{a}) &= q_3, & \delta(q_3, \boldsymbol{b}) &= q_2, \\ \delta(q_4, \boldsymbol{a}) &= q_4, & \delta(q_4, \boldsymbol{b}) &= q_3. \end{aligned}$$

(b) The machine first waits for two occurrences of $\boldsymbol{a}$. The state q_2 is reached as soon as the second $\boldsymbol{a}$ is seen. After that, only the number of $\boldsymbol{b}$s modulo 3 matters:

$$q_2 \equiv 0, \qquad q_4 \equiv 1, \qquad q_3 \equiv 2 \pmod 3.$$

Since q_2 and q_3 are final and q_4 is not, the language is

$$\{\boldsymbol{b}^i \boldsymbol{a} \boldsymbol{b}^j \boldsymbol{a} x \mid i, j \geq 0,\ x \in \{\boldsymbol{a}, \boldsymbol{b}\}^*,\ |x|_{\boldsymbol{b}} \equiv 0 \text{ or } 2 \pmod 3\}.$$

Equivalently: the string must contain at least two $\boldsymbol{a}$s, and after the second $\boldsymbol{a}$ the number of $\boldsymbol{b}$s must not be congruent to $1 \pmod 3$.

1.23. **Names of everyone in China – FAD?**

The set is finite (China has a finite population), hence regular (every finite language is FAD). Yes.

1.24. **Legal infix arithmetic expressions – FAD?**

Yes. Without parentheses, normal precedence rules affect evaluation, not syntactic well-formedness. A DFA only has to remember whether the next symbol should be an operand or an operator. The symbol **–** is special because it can serve as unary negation while the machine is expecting an operand, and as subtraction while the machine is expecting an operator.

A suitable DFA has three states:

- q_E: expecting an operand,
- q_O: expecting an operator or the end of the string,

- q_D: dead state.

The start state is q_E, the only final state is q_O, and the transitions are

	$\boldsymbol{A}$	$\boldsymbol{B}$	$\boldsymbol{+}$	$\boldsymbol{-}$	$\boldsymbol{\times}$	$\boldsymbol{/}$
q_E	q_O	q_O	q_D	q_E	q_D	q_D
q_O	q_D	q_D	q_E	q_E	q_E	q_E
q_D	q_D	q_D	q_D	q_D	q_D	q_D

Thus $\boldsymbol{A}$, $\boldsymbol{-A}$, $\boldsymbol{A-B}$, and $\boldsymbol{A\times -B}$ are accepted, while strings such as $\boldsymbol{+A}$, $\boldsymbol{AB}$, and $\boldsymbol{A/}$ are rejected.

1.25. **Properties of a DFA M.**

(a) $\lambda \in L(M)$ iff $s_0 \in F$.

(b) $L(M) = \Sigma^*$ iff every *reachable* state is a final state. (Unreachable non-final states do not affect the language: a DFA with one reachable final start state and one unreachable non-final state still accepts Σ^*.)

(c) $L(M) = \emptyset$ iff no final state is reachable from s_0.

1.26. **DFAs for substring recognition.**

(a) For "$\boldsymbol{abc}$ is a substring," use the states q_0, q_1, q_2, q_3, where q_i means that the longest suffix seen so far that is also a prefix of $\boldsymbol{abc}$ has length i. Let q_3 be an accepting sink. The transition table is

	$\boldsymbol{a}$	$\boldsymbol{b}$	$\boldsymbol{c}$
q_0	q_1	q_0	q_0
q_1	q_1	q_2	q_0
q_2	q_1	q_0	q_3
q_3	q_3	q_3	q_3

(b) For "$\boldsymbol{acaba}$ is a substring," use the states $p_0, p_1, p_2, p_3, p_4, p_5$, where p_5 is an accepting sink. The transition table is

	$\boldsymbol{a}$	$\boldsymbol{b}$	$\boldsymbol{c}$
p_0	p_1	p_0	p_0
p_1	p_1	p_0	p_2
p_2	p_3	p_0	p_0
p_3	p_1	p_4	p_2
p_4	p_5	p_0	p_0
p_5	p_5	p_5	p_5

Again, p_i records the longest suffix that is also a prefix of $\boldsymbol{acaba}$.

1.27. **Add pennies to vending machine.**

The original machine tracks deposited amounts in multiples of 5¢, needing 7 states (0, 5, 10, 15, 20, 25, 30¢). Adding pennies requires tracking every cent from 0 to 30, so the minimum number of states is **31**: states $q_0, q_1, \ldots, q_{30}$ representing $0, 1, \ldots, 30$ cents deposited. State q_{30} is the accepting (dispensing) state; from q_{30} all additional input is ignored (or loops back). Each extra penny input on state q_i (for $i < 30$) transitions to q_{i+1}.

1.28. **The DFA (t_0, t_1) of Exercise 1.29.**

(a) The language: $L = \{x \in \{\mathbf{a},\mathbf{b}\}^* \mid |x|_{\mathbf{b}} \text{ is odd}\}$, i.e., strings with an odd number of **b**s. (t_0 = even #b, t_1 = odd #b, $F = \{t_1\}$.)

(b) *Proof*: By induction on $|x| = n$. Let $P(n)$: $(\forall x \in \Sigma^n)(\overline{\delta}(t_0, x) = t_0 \Leftrightarrow |x|_{\mathbf{b}} \text{ even}) \wedge (\overline{\delta}(t_0, x) = t_1 \Leftrightarrow |x|_{\mathbf{b}} \text{ odd})$.

Base ($n = 0$): $\overline{\delta}(t_0, \lambda) = t_0$, $|\lambda|_{\mathbf{b}} = 0$ (even). ✓

Step: Let $x = ya$, $|y| = n$. Case $a = \mathbf{a}$: $\overline{\delta}(t_0, ya) = \delta(\overline{\delta}(t_0, y), \mathbf{a})$. Since $\delta(t_0, \mathbf{a}) = t_0$ and $\delta(t_1, \mathbf{a}) = t_1$, the state after ya equals the state after y, and $|ya|_{\mathbf{b}} = |y|_{\mathbf{b}}$. Case $a = \mathbf{b}$: δ flips the state; $|ya|_{\mathbf{b}} = |y|_{\mathbf{b}} + 1$ changes parity. Both cases follow from the IH. □

1.29. **Vending machine: pennies, nickels, dimes, quarters, 10¢ bars.**

(a) States $q_0, q_1, \ldots, q_{10}$ where $q_i = i$ cents deposited so far, q_{10} = accepting (dispensing) sink. $\delta(q_i, p) = q_{\min(i+1,10)}$, $\delta(q_i, n) = q_{\min(i+5,10)}$, $\delta(q_i, d) = q_{\min(i+10,10)}$, $\delta(q_i, q) = q_{\min(i+25,10)} \to q_{10}$. $F = \{q_{10}\}$.

(b) 11 states.

(c) The minimum is 11 states (must track exact amount from 0 to 10 cents).

1.30. **Vending machine: nickels, dimes, quarters, 10¢ bars.**

(a) States q_0, q_5, q_{10} (representing 0, 5, 10+ cents). $\delta(q_0, n) = q_5$, $\delta(q_0, d) = \delta(q_5, n) = q_{10}$, $\delta(q_0, q) = \delta(q_5, d) = \delta(q_5, q) = q_{10}$, $\delta(q_{10}, -) = q_{10}$. $F = \{q_{10}\}$.

(b) Minimum: 3 states (0, 5, ≥10 cents).

(c) Encode the states by

$$q_0 = \mathbf{00}, \qquad q_5 = \mathbf{01}, \qquad q_{10} = \mathbf{10},$$

with **11** unused. Encode the input alphabet $\{\langle EOS\rangle, \boldsymbol{n}, \boldsymbol{d}, \boldsymbol{q}\}$ by

$$\langle EOS\rangle = \mathbf{00}, \qquad \boldsymbol{n} = \mathbf{01}, \qquad \boldsymbol{d} = \mathbf{10}, \qquad \boldsymbol{q} = \mathbf{11}.$$

Let the state bits be t_1, t_2, and the input bits be $\boldsymbol{a}_1, \boldsymbol{a}_2$. One corresponding circuit uses two D flip-flops with inputs

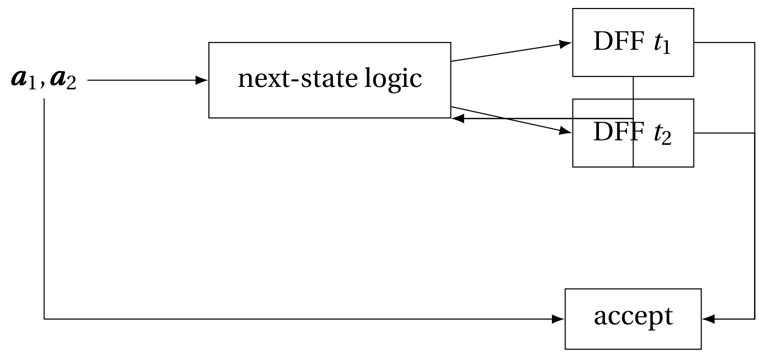
$$t_1' = (\neg \boldsymbol{a}_1 \wedge \neg \boldsymbol{a}_2 \wedge t_1) \vee (\neg \boldsymbol{a}_1 \wedge \boldsymbol{a}_2 \wedge (t_1 \vee t_2)) \vee \boldsymbol{a}_1,$$

$$t_2' = \neg \boldsymbol{a}_1 \wedge \big((\neg \boldsymbol{a}_2 \wedge t_2) \vee (\boldsymbol{a}_2 \wedge \neg t_1 \wedge \neg t_2)\big).$$

The accept light should turn on exactly when <EOS> is scanned in state q_{10}, so

$$\textbf{accept} = t_1 \wedge \neg t_2 \wedge \neg \boldsymbol{a}_1 \wedge \neg \boldsymbol{a}_2.$$

One corresponding circuit diagram is:



1.31. **Circuit for Exercise 1.29 DFA.**

(a) Use one state bit t_1, with $t_1 = \mathbf{0}$ for state t_0 and $t_1 = \mathbf{1}$ for state t_1. Encode

$$<EOS> = \mathbf{00}, \qquad \boldsymbol{a} = \mathbf{01}, \qquad \boldsymbol{b} = \mathbf{10}, \qquad <SOS> = \mathbf{11}.$$

Then <EOS> and $\boldsymbol{a}$ leave the parity unchanged, $\boldsymbol{b}$ toggles it, and <SOS> resets to the start state. Thus one D flip-flop with input

$$t_1' = (t_1 \wedge \neg \boldsymbol{a}_1) \vee (\neg t_1 \wedge \boldsymbol{a}_1 \wedge \neg \boldsymbol{a}_2)$$

implements the next-state logic. The accept circuitry is

$$\textbf{accept} = t_1 \wedge \neg \boldsymbol{a}_1 \wedge \neg \boldsymbol{a}_2,$$

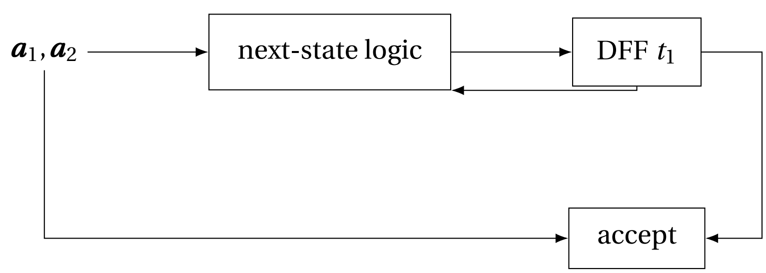
since the string is accepted exactly when <EOS> is scanned in the odd-$\boldsymbol{b}$ state.

(b) Without <SOS> and <EOS>, only one input bit is needed; let $x = \mathbf{0}$ represent $\boldsymbol{a}$ and $x = \mathbf{1}$ represent $\boldsymbol{b}$. The next state is just

$$t_1' = t_1 \oplus x.$$

So the circuit is a single D flip-flop whose input is the exclusive-or of the current state bit and the input bit.

A corresponding circuit diagram for the version with <SOS> and <EOS> is:



1.32. **Circuit for Exercise 1.7 DFA.**

It is convenient to implement the language of Exercise 1.7 by keeping the parity bits from Example 1.10 together with one extra bit u that records whether at least one input symbol has been read. Let

$$p = \text{parity of } |x|_{\boldsymbol{a}}, \qquad q = \text{parity of } |x|_{\boldsymbol{b}}, \qquad u = \text{“have seen input”}.$$

Then the machine accepts exactly when $p = q = \mathbf{0}$ and $u = \mathbf{1}$.

(a) With <SOS> and <EOS>, use the encoding

$$<EOS> = \mathbf{00}, \qquad \boldsymbol{a} = \mathbf{01}, \qquad \boldsymbol{b} = \mathbf{10}, \qquad <SOS> = \mathbf{11}.$$

One corresponding circuit uses three D flip-flops with inputs

$$p' = (\neg \boldsymbol{a}_1 \wedge \neg \boldsymbol{a}_2 \wedge p) \vee (\neg \boldsymbol{a}_1 \wedge \boldsymbol{a}_2 \wedge \neg p) \vee (\boldsymbol{a}_1 \wedge \neg \boldsymbol{a}_2 \wedge p),$$

$$q' = (\neg \boldsymbol{a}_1 \wedge \neg \boldsymbol{a}_2 \wedge q) \vee (\neg \boldsymbol{a}_1 \wedge \boldsymbol{a}_2 \wedge q) \vee (\boldsymbol{a}_1 \wedge \neg \boldsymbol{a}_2 \wedge \neg q),$$

$$u' = (\neg \boldsymbol{a}_1 \wedge \neg \boldsymbol{a}_2 \wedge u) \vee (\neg \boldsymbol{a}_1 \wedge \boldsymbol{a}_2) \vee (\boldsymbol{a}_1 \wedge \neg \boldsymbol{a}_2).$$

The accept light is

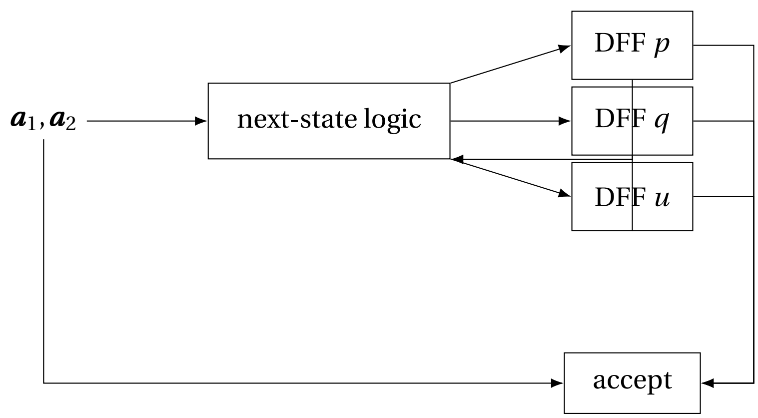
$$\mathbf{accept} = \neg p \wedge \neg q \wedge u \wedge \neg \boldsymbol{a}_1 \wedge \neg \boldsymbol{a}_2.$$

(b) Without <SOS> and <EOS>, use one input bit x with $x = \mathbf{0}$ for $\boldsymbol{a}$ and $x = \mathbf{1}$ for $\boldsymbol{b}$. Then

$$p' = p \oplus \neg x, \qquad q' = q \oplus x, \qquad u' = \mathbf{1}.$$

After the last clock pulse, the machine accepts iff $\neg p \wedge \neg q \wedge u$.

One corresponding circuit diagram for the <SOS>/<EOS> version is:



1.33. **Modify Example 1.12 to correctly handle <SOS>.**

Use the same one-bit state encoding as Example 1.12: $s_0 = \mathbf{0}$ and $s_1 = \mathbf{1}$. Keep the book's input convention and encode

$$<EOS> = \mathbf{00}, \qquad \boldsymbol{b} = \mathbf{01}, \qquad \boldsymbol{c} = \mathbf{10}, \qquad <SOS> = \mathbf{11}.$$

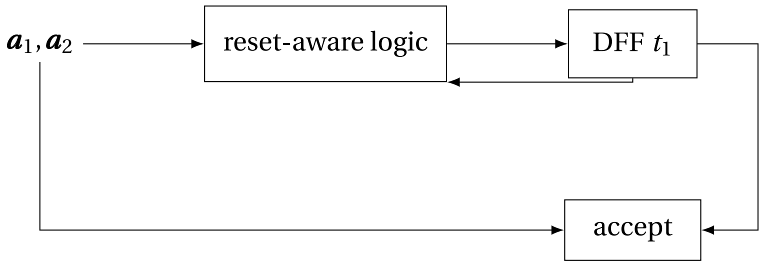
The only extra requirement is that scanning <SOS> from either state must send the machine to s_0. The resulting next-state function is

$$t_1' = (\neg \boldsymbol{a}_1 \wedge \neg \boldsymbol{a}_2 \wedge t_1) \vee (\boldsymbol{a}_1 \wedge \neg \boldsymbol{a}_2 \wedge \neg t_1).$$

This agrees with Example 1.12 on $\boldsymbol{b}$, $\boldsymbol{c}$, and <EOS>, and forces a reset to the start state on <SOS>. The accept circuitry is unchanged:

$$\mathbf{accept} = t_1 \wedge \neg \boldsymbol{a}_1 \wedge \neg \boldsymbol{a}_2.$$

So the corrected circuit is exactly the Example 1.12 circuit with the input combination **11** wired to force the next state to **0**.



1.34. **Circuit for Example 1.6 machine.**

(a) Encode the states by

$$s_0 = \mathbf{00}, \qquad s_1 = \mathbf{01}, \qquad s_2 = \mathbf{10},$$

with **11** unused. Use the standard input encoding

$$<EOS> = \mathbf{00}, \qquad \boldsymbol{a} = \mathbf{01}, \qquad \boldsymbol{b} = \mathbf{10}, \qquad <SOS> = \mathbf{11}.$$

If t_1, t_2 are the state bits, then a corresponding pair of D inputs is

$$t_1' = (\neg \boldsymbol{a}_1 \wedge \neg \boldsymbol{a}_2 \wedge t_1) \vee (\boldsymbol{a}_1 \wedge \neg \boldsymbol{a}_2 \wedge \neg t_1),$$

$$t_2' = (\neg \boldsymbol{a}_1 \wedge \neg \boldsymbol{a}_2 \wedge t_2) \vee (\neg \boldsymbol{a}_1 \wedge \boldsymbol{a}_2).$$

The machine accepts in state s_1, so the accept circuitry is

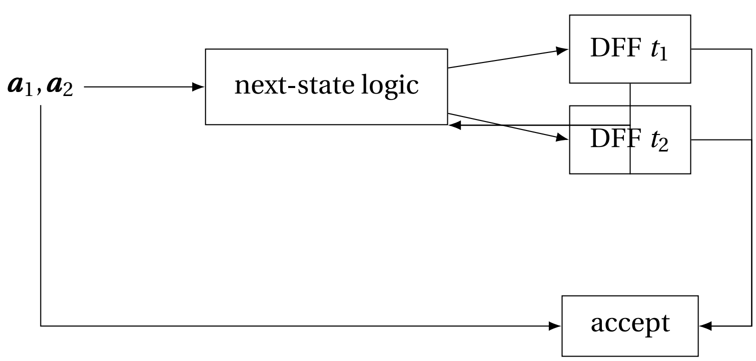
$$\mathbf{accept} = \neg t_1 \wedge t_2 \wedge \neg \boldsymbol{a}_1 \wedge \neg \boldsymbol{a}_2.$$

(b) Without <SOS> and <EOS>, only one input bit x is needed, with $x = \mathbf{0}$ for $\boldsymbol{a}$ and $x = \mathbf{1}$ for $\boldsymbol{b}$. Then

$$t_1' = x \wedge \neg t_1, \qquad t_2' = \neg x.$$

Thus two D flip-flops together with the above combinational logic implement the machine.

One corresponding circuit diagram for the <SOS>/<EOS> version is:



1.35. **Circuit for Example 1.10 machine.**

Let p and q be the parity bits for $|x|_{\boldsymbol{a}}$ and $|x|_{\boldsymbol{b}}$, respectively. Then the four states of Example 1.10 are exactly the four combinations of (p, q).

(a) With <SOS> and <EOS>, use

$$<EOS> = \mathbf{00}, \qquad \boldsymbol{a} = \mathbf{01}, \qquad \boldsymbol{b} = \mathbf{10}, \qquad <SOS> = \mathbf{11}.$$

Then <EOS> leaves the state unchanged, $\boldsymbol{a}$ flips p, $\boldsymbol{b}$ flips q, and <SOS> resets both bits to $\mathbf{0}$. One corresponding circuit uses two D flip-flops with

$$p' = (\neg\boldsymbol{a}_1 \wedge \neg\boldsymbol{a}_2 \wedge p) \vee (\neg\boldsymbol{a}_1 \wedge \boldsymbol{a}_2 \wedge \neg p) \vee (\boldsymbol{a}_1 \wedge \neg\boldsymbol{a}_2 \wedge p),$$

$$q' = (\neg\boldsymbol{a}_1 \wedge \neg\boldsymbol{a}_2 \wedge q) \vee (\neg\boldsymbol{a}_1 \wedge \boldsymbol{a}_2 \wedge q) \vee (\boldsymbol{a}_1 \wedge \neg\boldsymbol{a}_2 \wedge \neg q).$$

The accept light turns on exactly at <EOS> when $p = q = \mathbf{0}$:

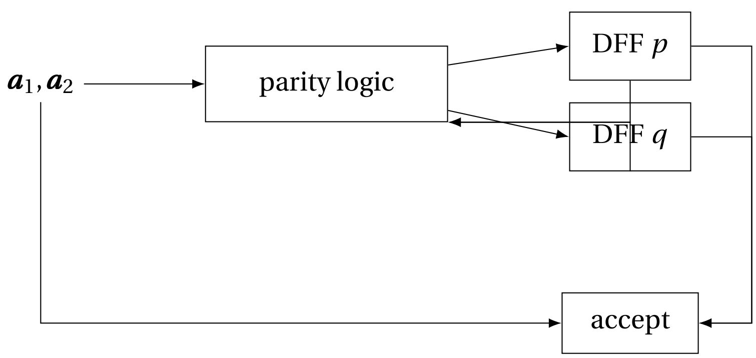
$$\mathbf{accept} = \neg p \wedge \neg q \wedge \neg\boldsymbol{a}_1 \wedge \neg\boldsymbol{a}_2.$$

(b) Without <SOS> and <EOS>, let $x = \mathbf{0}$ encode $\boldsymbol{a}$ and $x = \mathbf{1}$ encode $\boldsymbol{b}$. Then

$$p' = p \oplus \neg x, \qquad q' = q \oplus x.$$

So the circuit is just two D flip-flops fed by these two exclusive-or subcircuits.

One corresponding circuit diagram for the <SOS>/<EOS> version is:



1.36. **Circuit for Example 1.14 (vending machine), with EOS.**

Encode the seven states of Example 1.14 by the binary codes for the numbers 0 through 6:

$$s_0 = \mathbf{000},\ s_5 = \mathbf{001},\ s_{10} = \mathbf{010},\ s_{15} = \mathbf{011},\ s_{20} = \mathbf{100},\ s_{25} = \mathbf{101},\ s_{30} = \mathbf{110},$$

leaving **111** unused. Encode the input alphabet by

$$<EOS> = \mathbf{00}, \qquad \boldsymbol{n} = \mathbf{01}, \qquad \boldsymbol{d} = \mathbf{10}, \qquad \boldsymbol{q} = \mathbf{11}.$$

One clean circuit design uses three D flip-flops together with a combinational subcircuit that does the following:

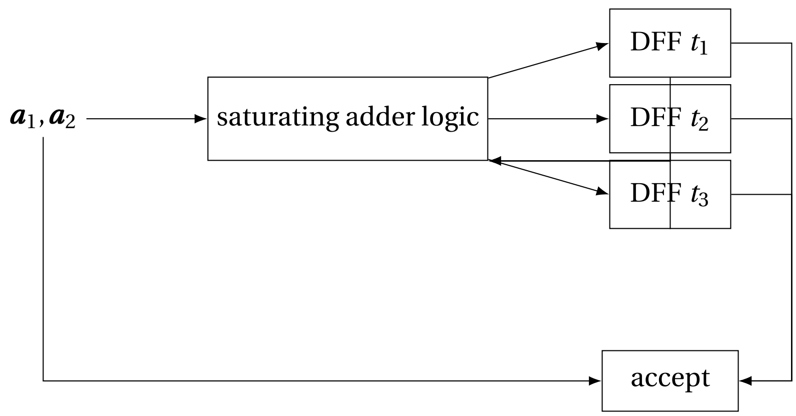
- decode the current state bits as the number $i \in \{0, 1, \ldots, 6\}$,
- decode the input bits as the increment c, where $c = 0$ for <EOS>, $c = 1$ for $\boldsymbol{n}$, $c = 2$ for $\boldsymbol{d}$, and $c = 5$ for $\boldsymbol{q}$,

- compute $j = \min\{6, i + c\}$,
- write the three-bit encoding of j into the D inputs.

The accept light is on exactly when <EOS> is scanned in state s_{30}:

$$\textbf{accept} = t_1 \wedge t_2 \wedge \neg t_3 \wedge \neg \boldsymbol{a}_1 \wedge \neg \boldsymbol{a}_2,$$

if $t_1 t_2 t_3$ is the chosen encoding of the current state and $s_{30} = \mathbf{110}$.



1.37. **Circuit for Example 1.16, with SOS and EOS.**

The protocol machine has four states, so two state bits suffice. For example, use

$$A = \mathbf{00}, \qquad R = \mathbf{01}, \qquad RF = \mathbf{10}, \qquad RD = \mathbf{11}.$$

The input alphabet is $\{B, D, Z, H, S, <EOS>, <SOS>\}$, so use three input bits and encode

$$<EOS> = \mathbf{000}, \quad B = \mathbf{001}, \quad D = \mathbf{010}, \quad H = \mathbf{011},$$

$$S = \mathbf{100}, \quad Z = \mathbf{101}, \quad <SOS> = \mathbf{111},$$

leaving **110** unused.

The next-state combinational block should implement exactly the protocol described in Example 1.16:

- scanning <SOS> from any state sends the machine to R,
- scanning <EOS> leaves the current state unchanged,
- from R, only S is legal and sends the machine to RF,
- from RF, only H is legal and sends the machine to RD,
- from RD, D stays in RD, Z goes to RF, and B goes to R,
- every unexpected packet sends the machine to A,
- from A, every packet except <SOS> leaves the machine in A.

Thus the circuit consists of two D flip-flops whose inputs are the Boolean functions defined by this complete transition table. Since Example 1.16 is being used as a control automaton rather than as a language acceptor, there is no separate accept light; the outputs of interest are simply the state bits themselves.

1.38. **DFA for** $L = \{x \in \{\mathbf{a}, \mathbf{b}, \mathbf{c}\}^* \mid |x|_{\mathbf{b}} = 2\}$**.**

(a) Three states q_0, q_1, q_2 (counting #b seen so far, capped at 2), plus a dead state q_3. $F = \{q_2\}$. $\delta(q_i, \mathbf{a}) = \delta(q_i, \mathbf{c}) = q_i$; $\delta(q_0, \mathbf{b}) = q_1$; $\delta(q_1, \mathbf{b}) = q_2$; $\delta(q_2, \mathbf{b}) = q_3$; $\delta(q_3, -) = q_3$.

(b) The quintuple is

$$\langle\{\mathbf{a}, \mathbf{b}, \mathbf{c}\}, \{q_0, q_1, q_2, q_3\}, q_0, \delta, \{q_2\}\rangle,$$

where

$$\delta(q_0, \mathbf{a}) = q_0, \quad \delta(q_0, \mathbf{b}) = q_1, \quad \delta(q_0, \mathbf{c}) = q_0,$$

$$\delta(q_1, \mathbf{a}) = q_1, \quad \delta(q_1, \mathbf{b}) = q_2, \quad \delta(q_1, \mathbf{c}) = q_1,$$

$$\delta(q_2, \mathbf{a}) = q_2, \quad \delta(q_2, \mathbf{b}) = q_3, \quad \delta(q_2, \mathbf{c}) = q_2,$$

$$\delta(q_3, \mathbf{a}) = q_3, \quad \delta(q_3, \mathbf{b}) = q_3, \quad \delta(q_3, \mathbf{c}) = q_3.$$

1.39. **DFA for every b eventually followed by c.**

Two states track whether there is an outstanding **b** with no subsequent **c**: q_0 (no unresolved **b**), q_b (at least one unresolved **b**). $F = \{q_0\}$.

$$\delta(q_0, \mathbf{a}) = q_0, \quad \delta(q_0, \mathbf{b}) = q_b, \quad \delta(q_0, \mathbf{c}) = q_0,$$

$$\delta(q_b, \mathbf{a}) = q_b, \quad \delta(q_b, \mathbf{b}) = q_b, \quad \delta(q_b, \mathbf{c}) = q_0.$$

Reading **c** in q_b resolves all pending **b**s; **a** or **b** keeps them pending. Accept iff input ends in q_0.

1.40. **DFA for no consecutive as and no consecutive bs.**

(a) States: q_0 (start/even alternation), q_a (last char **a**), q_b (last char **b**), q_{dead}. $F = \{q_0, q_a, q_b\}$. $\delta(q_0, \mathbf{a}) = q_a$, $\delta(q_0, \mathbf{b}) = q_b$, $\delta(q_a, \mathbf{a}) = q_{dead}$, $\delta(q_a, \mathbf{b}) = q_b$, $\delta(q_b, \mathbf{b}) = q_{dead}$, $\delta(q_b, \mathbf{a}) = q_a$, $\delta(q_{dead}, -) = q_{dead}$.

(b) The quintuple is

$$\langle\{\mathbf{a}, \mathbf{b}\}, \{q_0, q_a, q_b, q_{dead}\}, q_0, \delta, \{q_0, q_a, q_b\}\rangle,$$

where

$$\delta(q_0, \mathbf{a}) = q_a, \quad \delta(q_0, \mathbf{b}) = q_b,$$

$$\delta(q_a, \mathbf{a}) = q_{dead}, \quad \delta(q_a, \mathbf{b}) = q_b,$$

$$\delta(q_b, \mathbf{a}) = q_a, \quad \delta(q_b, \mathbf{b}) = q_{dead},$$

$$\delta(q_{dead}, \mathbf{a}) = q_{dead}, \quad \delta(q_{dead}, \mathbf{b}) = q_{dead}.$$

1.41. **DFA for** $L = \{x \in \{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}\}^* \mid |x|_{\boldsymbol{a}} \equiv 0 \bmod 3\}$**.**

(a) Use three states r_0, r_1, r_2, where r_i means that the number of $\boldsymbol{a}$s seen so far is congruent to i (mod 3). The start state is r_0, the only final state is r_0, and

$$\delta(r_i,\boldsymbol{a}) = r_{i+1 \bmod 3}, \qquad \delta(r_i,\boldsymbol{b}) = r_i, \qquad \delta(r_i,\boldsymbol{c}) = r_i.$$

(b) The quintuple is

$$\langle\{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}\},\{r_0,r_1,r_2\},r_0,\delta,\{r_0\}\rangle,$$

where

$$\delta(r_0,\boldsymbol{a}) = r_1, \quad \delta(r_0,\boldsymbol{b}) = r_0, \quad \delta(r_0,\boldsymbol{c}) = r_0,$$

$$\delta(r_1,\boldsymbol{a}) = r_2, \quad \delta(r_1,\boldsymbol{b}) = r_1, \quad \delta(r_1,\boldsymbol{c}) = r_1,$$

$$\delta(r_2,\boldsymbol{a}) = r_0, \quad \delta(r_2,\boldsymbol{b}) = r_2, \quad \delta(r_2,\boldsymbol{c}) = r_2.$$

1.42. **DFAs for Pascal comments.**

(a) For "exactly one valid Pascal comment and no illegal comment fragments," use

$$A = \langle\Sigma, S, P_0, \delta, F\rangle,$$

where

$$\Sigma = \{\boldsymbol{a},\boldsymbol{b},\boldsymbol{(},\boldsymbol{*},\boldsymbol{)}\},$$

$$S = \{P_0, P_(, P_*, I_0, I_*, Q_0, Q_(, Q_*, D\}, \qquad F = \{Q_0, Q_(, Q_*\}.$$

Here P means "before the unique comment," I means "inside the comment," Q means "after the unique comment," and D is dead. The full transition function is:

	$\boldsymbol{a}$	$\boldsymbol{b}$	**(**	$\boldsymbol{*}$	**)**
P_0	P_0	P_0	$P_($	P_*	P_0
$P_($	P_0	P_0	$P_($	I_0	P_0
P_*	P_0	P_0	$P_($	P_*	D
I_0	I_0	I_0	I_0	I_*	I_0
I_*	I_0	I_0	I_0	I_*	Q_0
Q_0	Q_0	Q_0	$Q_($	Q_*	Q_0
$Q_($	Q_0	Q_0	$Q_($	D	Q_0
Q_*	Q_0	Q_0	$Q_($	Q_*	D
D	D	D	D	D	D

(b) For "zero or more valid unnested Pascal comments," use

$$B = \langle\Sigma, T, O_0, \delta, F'\rangle,$$

where

$$T = \{O_0, O_(, O_*, I_0, I_*, D\}, \qquad F' = \{O_0, O_(, O_*\}.$$

Here O means "outside any comment." The full transition function is:

	$\boldsymbol{a}$	$\boldsymbol{b}$	**(**	$\boldsymbol{*}$	**)**
O_0	O_0	O_0	$O_($	O_*	O_0
$O_($	O_0	O_0	$O_($	I_0	O_0
O_*	O_0	O_0	$O_($	O_*	D
I_0	I_0	I_0	I_0	I_*	I_0
I_*	I_0	I_0	I_0	I_*	O_0
D	D	D	D	D	D

This machine accepts exactly the strings that end outside a comment and never produce an illegal delimiter fragment.

1.43. **Postfix expressions.**

(a) Yes. A convenient DFA for the case "at most 2 operators" uses states

$$q_{d,m} \qquad (0 \le d \le 3,\ 0 \le m \le 2),$$

together with a dead state q_{dead}. Here d is the current stack surplus

$$d = (\#\text{operands read}) - (\#\text{operators read}),$$

and m is the number of operators used so far. The start state is $q_{0,0}$. On an operand $\boldsymbol{A}$ or $\boldsymbol{B}$,

$$\delta(q_{d,m}, \boldsymbol{A}) = \delta(q_{d,m}, \boldsymbol{B}) = \begin{cases} q_{d+1,m} & d < 3, \\ q_{dead} & d = 3, \end{cases}$$

and on any operator $\odot \in \{\boldsymbol{+}, \boldsymbol{-}, \boldsymbol{\times}, \boldsymbol{/}\}$,

$$\delta(q_{d,m}, \odot) = \begin{cases} q_{d-1,m+1} & d \ge 2 \text{ and } m < 2, \\ q_{dead} & \text{otherwise.} \end{cases}$$

The final states are the states $q_{1,0}, q_{1,1}, q_{1,2}$, because a complete postfix expression leaves exactly one value on the stack. This is the required machine.

(b) For four or fewer operators, use exactly the same construction with

$$q_{d,m} \qquad (0 \le d \le 5,\ 0 \le m \le 4),$$

again plus a dead state. The same transition rules apply, with the bounds 3 and 2 replaced by 5 and 4, respectively. The final states are all $q_{1,m}$ with $0 \le m \le 4$.

(c) For eight or fewer operators, use the same DFA schema once more, now with

$$q_{d,m} \qquad (0 \le d \le 9,\ 0 \le m \le 8),$$

plus a dead state. On operands the machine increments d; on operators it checks that $d \ge 2$ and then decrements d while incrementing m. The final states are all $q_{1,m}$ with $0 \le m \le 8$.

(d) The set of *all* postfix expressions is not FAD. Intuitively, a recognizer must keep track of how many partial results are currently on the stack, and that number is unbounded. More formally, the strings

$$\boldsymbol{A},\ \boldsymbol{AA},\ \boldsymbol{AAA},\ \ldots,\ \boldsymbol{A}^n,\ \ldots$$

have pairwise different futures: from $\boldsymbol{A}^n$, the continuation $\boldsymbol{+}^{n-1}$ completes a legal postfix expression, but the same continuation does not complete a legal expression from $\boldsymbol{A}^m$ when $m \neq n$. Therefore infinitely many distinct states would be required.

1.44. **DFA for words that begin and end with different letters.**

(a) Use one start state s, and for each ordered pair $(u, v) \in \{\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}\}^2$ a state q_{uv} meaning "the first letter seen was u and the current last letter is v." From the start state,

$$\delta(s, \boldsymbol{a}) = q_{aa}, \qquad \delta(s, \boldsymbol{b}) = q_{bb}, \qquad \delta(s, \boldsymbol{c}) = q_{cc}.$$

Thereafter,

$$\delta(q_{uv}, w) = q_{uw} \qquad \text{for all } u, v, w \in \{\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}\}.$$

The accepting states are precisely the six states q_{uv} with $u \neq v$.

(b) The quintuple is

$$\langle \{\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}\}, S, s, \delta, F \rangle,$$

where

$$S = \{s\} \cup \{q_{uv} \mid u, v \in \{\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}\}\},$$

and

$$F = \{q_{uv} \mid u \neq v\}.$$

The transition function is

$$\delta(s, \boldsymbol{a}) = q_{aa}, \qquad \delta(s, \boldsymbol{b}) = q_{bb}, \qquad \delta(s, \boldsymbol{c}) = q_{cc},$$

and for every $u, v \in \{\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}\}$,

$$\delta(q_{uv}, \boldsymbol{a}) = q_{ua}, \qquad \delta(q_{uv}, \boldsymbol{b}) = q_{ub}, \qquad \delta(q_{uv}, \boldsymbol{c}) = q_{uc}.$$

1.45. **DFA that rejects words whose last two letters match.**

(a) A convenient DFA uses the seven states

$$s,\ a,\ b,\ c,\ aa,\ bb,\ cc.$$

The start state is s. The states a, b, c mean that the current last letter is $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}$ and the last two letters do *not* match; the states aa, bb, cc mean that the last two letters are equal. The transitions are:

$$\delta(s, \boldsymbol{a}) = a, \quad \delta(s, \boldsymbol{b}) = b, \quad \delta(s, \boldsymbol{c}) = c,$$

$$\delta(a, \boldsymbol{a}) = aa,\ \delta(a, \boldsymbol{b}) = b,\ \delta(a, \boldsymbol{c}) = c,$$

$$\delta(b, \boldsymbol{a}) = a,\ \delta(b, \boldsymbol{b}) = bb,\ \delta(b, \boldsymbol{c}) = c,$$

$$\delta(c, \boldsymbol{a}) = a,\ \delta(c, \boldsymbol{b}) = b,\ \delta(c, \boldsymbol{c}) = cc,$$

and from aa, bb, cc the same rule applies: a repeated last letter stays in the corresponding double-letter state, while a different last letter moves to $a, b,$ or c. The accepting states are s, a, b, c.

(b) The quintuple is

$$\langle \{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}\}, \{s,a,b,c,aa,bb,cc\}, s, \delta, \{s,a,b,c\}\rangle,$$

where

$$\delta(s,\boldsymbol{a}) = a, \quad \delta(s,\boldsymbol{b}) = b, \quad \delta(s,\boldsymbol{c}) = c,$$

$$\delta(a,\boldsymbol{a}) = aa, \quad \delta(a,\boldsymbol{b}) = b, \quad \delta(a,\boldsymbol{c}) = c,$$

$$\delta(b,\boldsymbol{a}) = a, \quad \delta(b,\boldsymbol{b}) = bb, \quad \delta(b,\boldsymbol{c}) = c,$$

$$\delta(c,\boldsymbol{a}) = a, \quad \delta(c,\boldsymbol{b}) = b, \quad \delta(c,\boldsymbol{c}) = cc,$$

$$\delta(aa,\boldsymbol{a}) = aa, \quad \delta(aa,\boldsymbol{b}) = b, \quad \delta(aa,\boldsymbol{c}) = c,$$

$$\delta(bb,\boldsymbol{a}) = a, \quad \delta(bb,\boldsymbol{b}) = bb, \quad \delta(bb,\boldsymbol{c}) = c,$$

$$\delta(cc,\boldsymbol{a}) = a, \quad \delta(cc,\boldsymbol{b}) = b, \quad \delta(cc,\boldsymbol{c}) = cc.$$

1.46. **DFA that rejects words whose first two letters match.**

(a) Use the six states

$$s,\ a,\ b,\ c,\ q_{acc},\ q_{rej}.$$

After the first input symbol, the machine remembers that symbol in one of the states a, b, c. On the second symbol it decides permanently:

$$\delta(a,\boldsymbol{a}) = q_{rej}, \quad \delta(a,\boldsymbol{b}) = \delta(a,\boldsymbol{c}) = q_{acc},$$

and similarly for b and c. Both q_{acc} and q_{rej} are sink states. The accepting states are s, a, b, c, q_{acc}, since words of length 0 or 1 do not have matching first two letters.

(b) The quintuple is

$$\langle \{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}\}, \{s,a,b,c,q_{acc},q_{rej}\}, s, \delta, \{s,a,b,c,q_{acc}\}\rangle,$$

where

$$\delta(s,\boldsymbol{a}) = a, \qquad \delta(s,\boldsymbol{b}) = b, \qquad \delta(s,\boldsymbol{c}) = c,$$

$$\delta(a,\boldsymbol{a}) = q_{rej}, \qquad \delta(a,\boldsymbol{b}) = q_{acc}, \qquad \delta(a,\boldsymbol{c}) = q_{acc},$$

$$\delta(b,\boldsymbol{a}) = q_{acc}, \qquad \delta(b,\boldsymbol{b}) = q_{rej}, \qquad \delta(b,\boldsymbol{c}) = q_{acc},$$

$$\delta(c,\boldsymbol{a}) = q_{acc}, \qquad \delta(c,\boldsymbol{b}) = q_{acc}, \qquad \delta(c,\boldsymbol{c}) = q_{rej},$$

and

$$\delta(q_{acc}, x) = q_{acc}, \qquad \delta(q_{rej}, x) = q_{rej}$$

for every $x \in \{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}\}$.

1.47. **Prove that the empty word is unique.**

Suppose x and y are both empty words. By definition, this means $|x| = 0$ and $|y| = 0$. The definition of equality of strings says that two strings are equal iff they have the same length and agree in every position. Here the lengths are equal, and there are no positions at all to check because the common length is 0. Thus the positional condition is vacuously satisfied, so $x = y$.

1.48. **Show that** $|xy| = |x| + |y|$**.**

We prove this by induction on $|y|$.

Base case: If $|y| = 0$, then $y = \lambda$, so

$$|xy| = |x\lambda| = |x| = |x| + 0 = |x| + |y|.$$

Inductive step: Assume that $|xz| = |x| + |z|$ whenever $|z| = n$. Let $|y| = n+1$, so $y = \boldsymbol{a}z$ for some letter $\boldsymbol{a}$ and some string z of length n. Then

$$|xy| = |x\boldsymbol{a}z| = |(x\boldsymbol{a})z|.$$

By the induction hypothesis,

$$|(x\boldsymbol{a})z| = |x\boldsymbol{a}| + |z| = (|x| + 1) + |z| = |x| + (1 + |z|) = |x| + |y|.$$

Hence $|xy| = |x| + |y|$ for all strings x and y.

1.49. **The DFA** C**.**

(a) The state transition diagram has two states, t_0 and t_1, with t_0 as start state and t_1 final. The transitions are

$$\delta(t_0, \boldsymbol{a}) = t_0, \quad \delta(t_0, \boldsymbol{b}) = t_1, \quad \delta(t_0, \boldsymbol{c}) = t_1,$$

$$\delta(t_1, \boldsymbol{a}) = t_0, \quad \delta(t_1, \boldsymbol{b}) = t_1, \quad \delta(t_1, \boldsymbol{c}) = t_0.$$

(b) The letter $\boldsymbol{a}$ always resets the machine to t_0. After the last $\boldsymbol{a}$, the state depends only on the remaining suffix over $\{\boldsymbol{b}, \boldsymbol{c}\}$. Starting from t_0, a $\boldsymbol{b}$ moves to t_1 and then further $\boldsymbol{b}$s leave the machine there, while each $\boldsymbol{c}$ toggles the state. Therefore $L(C)$ consists exactly of those strings for which, after the last $\boldsymbol{a}$, either

- there is no $\boldsymbol{b}$ and the number of $\boldsymbol{c}$s is odd, or
- there is a last $\boldsymbol{b}$ and it is followed by an even number of $\boldsymbol{c}$s.

(c) Use one state bit t, with $t = \mathbf{0}$ for t_0 and $t = \mathbf{1}$ for t_1. Encode

$$\text{<}EOS\text{>} = \mathbf{00}, \qquad \boldsymbol{a} = \mathbf{01}, \qquad \boldsymbol{b} = \mathbf{10}, \qquad \boldsymbol{c} = \mathbf{11}.$$

Then <EOS> leaves the state unchanged, $\boldsymbol{a}$ sends the machine to t_0, $\boldsymbol{b}$ sends it to t_1, and $\boldsymbol{c}$ toggles the state. So one D flip-flop with input

$$t' = (\neg t \wedge \boldsymbol{a}_1) \vee (t \wedge \neg \boldsymbol{a}_2)$$

implements the next-state logic. The accept circuitry is

$$\textbf{accept} = t \wedge \neg \boldsymbol{a}_1 \wedge \neg \boldsymbol{a}_2.$$

1.50. **Roman numerals.**

(a) For strict-order Roman numerals, a DFA is

$$\langle\{\boldsymbol{I},\boldsymbol{V},\boldsymbol{X},\boldsymbol{L},\boldsymbol{C},\boldsymbol{D},\boldsymbol{M}\},\{q_s,q_M,q_D,q_C,q_L,q_X,q_V,q_I,q_{dead}\},q_s,\delta,F\rangle,$$

where

$$F=\{q_M,q_D,q_C,q_L,q_X,q_V,q_I\}.$$

The states represent the highest-valued symbol class that may still appear. The transitions are:

$$\begin{array}{lllll}
\delta(q_s,\boldsymbol{M})=q_M, & \delta(q_s,\boldsymbol{D})=q_D, & \delta(q_s,\boldsymbol{C})=q_C, & & \\
\delta(q_s,\boldsymbol{L})=q_L, & \delta(q_s,\boldsymbol{X})=q_X, & \delta(q_s,\boldsymbol{V})=q_V, & \delta(q_s,\boldsymbol{I})=q_I, & \\
\delta(q_M,\boldsymbol{M})=q_M, & \delta(q_M,\boldsymbol{D})=q_D, & \delta(q_M,\boldsymbol{C})=q_C, & & \\
\delta(q_M,\boldsymbol{L})=q_L, & \delta(q_M,\boldsymbol{X})=q_X, & \delta(q_M,\boldsymbol{V})=q_V, & \delta(q_M,\boldsymbol{I})=q_I, & \\
\delta(q_D,\boldsymbol{C})=q_C, & \delta(q_D,\boldsymbol{L})=q_L, & \delta(q_D,\boldsymbol{X})=q_X, & \delta(q_D,\boldsymbol{V})=q_V, & \delta(q_D,\boldsymbol{I})=q_I, \\
\delta(q_C,\boldsymbol{C})=q_C, & \delta(q_C,\boldsymbol{L})=q_L, & \delta(q_C,\boldsymbol{X})=q_X, & \delta(q_C,\boldsymbol{V})=q_V, & \delta(q_C,\boldsymbol{I})=q_I, \\
\delta(q_L,\boldsymbol{X})=q_X, & \delta(q_L,\boldsymbol{V})=q_V, & \delta(q_L,\boldsymbol{I})=q_I, & & \\
\delta(q_X,\boldsymbol{X})=q_X, & \delta(q_X,\boldsymbol{V})=q_V, & \delta(q_X,\boldsymbol{I})=q_I, & & \\
\delta(q_V,\boldsymbol{I})=q_I, \quad \delta(q_I,\boldsymbol{I})=q_I. & & & &
\end{array}$$

Every unspecified transition goes to q_{dead}, and q_{dead} loops to itself. This accepts exactly the strings of the form

$$\boldsymbol{M}^*\boldsymbol{D}^{0,1}\boldsymbol{C}^*\boldsymbol{L}^{0,1}\boldsymbol{X}^*\boldsymbol{V}^{0,1}\boldsymbol{I}^*,$$

excluding the empty word.

(b) For ordinary Roman numerals, one explicit DFA is

$$A=\langle\Sigma,S,q_s,\delta,F\rangle,$$

where

$$\Sigma=\{\boldsymbol{I},\boldsymbol{V},\boldsymbol{X},\boldsymbol{L},\boldsymbol{C},\boldsymbol{D},\boldsymbol{M}\},$$

$$S=\{q_s,q_M,q_T,q_{HC},q_{HCC},q_{HCCC},q_{HD},q_{HDC},q_{HDCC},q_{HDCCC},$$

$$q_{TX},q_{TXX},q_{TXXX},q_{TL},q_{TLX},q_{TLXX},q_{TLXXX},q_O,q_{OI},q_{OII},q_{OIII},q_{OV},q_{OVI},q_{OVII},q_{OVIII},q_{done},q_{dead}\},$$

$$F=S-\{q_s,q_{dead}\}.$$

The state names indicate the phase that has been reached: q_T means that the thousands and hundreds phases are complete and the tens phase has not yet begun, while q_O means that the tens phase is complete and the ones phase has not yet begun. The transition function is:

$$
\begin{aligned}
&\delta(q_s,\boldsymbol{M}) = q_M, && \delta(q_s,\boldsymbol{C}) = q_{HC},\\
&\delta(q_s,\boldsymbol{X}) = q_{TX}, && \delta(q_s,\boldsymbol{L}) = q_{TL},\\
&\delta(q_M,\boldsymbol{M}) = q_M, && \delta(q_M,\boldsymbol{C}) = q_{HC},\\
&\delta(q_M,\boldsymbol{X}) = q_{TX}, && \delta(q_M,\boldsymbol{L}) = q_{TL},\\
&\delta(q_{HC},\boldsymbol{M}) = q_T, && \delta(q_{HC},\boldsymbol{D}) = q_T,\\
&\delta(q_{HC},\boldsymbol{X}) = q_{TX}, && \delta(q_{HC},\boldsymbol{L}) = q_{TL},\\
&\delta(q_{HCC},\boldsymbol{C}) = q_{HCCC}, && \delta(r,\boldsymbol{X}) = q_{TX},\\
&\delta(r,\boldsymbol{I}) = q_{OI}, && \delta(r,\boldsymbol{V}) = q_{OV}\\
&\delta(q_{HD},\boldsymbol{C}) = q_{HDC}, \quad \delta(q_{HDC},\boldsymbol{C}) = q_{HDCC}, \quad \delta(q_{HDCC},\boldsymbol{C}) = q_{HDCCC}, &&\\
&\delta(q_T,\boldsymbol{X}) = q_{TX}, && \delta(q_T,\boldsymbol{L}) = q_{TL},\\
&\delta(q_{TX},\boldsymbol{C}) = q_O, && \delta(q_{TX},\boldsymbol{L}) = q_O,\\
&\delta(q_{TX},\boldsymbol{I}) = q_{OI}, && \delta(q_{TX},\boldsymbol{V}) = q_{OV},\\
&\delta(q_{TXX},\boldsymbol{X}) = q_{TXXX}, && \delta(q_{TXX},\boldsymbol{I}) = q_{OI},\\
&\delta(q_{TXXX},\boldsymbol{I}) = q_{OI}, && \delta(q_{TXXX},\boldsymbol{V}) = q_{OV},\\
&\delta(q_{TL},\boldsymbol{X}) = q_{TLX}, && \delta(q_{TL},\boldsymbol{I}) = q_{OI},\\
&\delta(q_{TLX},\boldsymbol{X}) = q_{TLXX}, && \delta(q_{TLX},\boldsymbol{I}) = q_{OI},\\
&\delta(q_{TLXX},\boldsymbol{X}) = q_{TLXXX}, && \delta(q_{TLXX},\boldsymbol{I}) = q_{OI},\\
&\delta(q_{TLXXX},\boldsymbol{I}) = q_{OI}, && \delta(q_{TLXXX},\boldsymbol{V}) = q_{OV},\\
&\delta(q_O,\boldsymbol{I}) = q_{OI}, && \delta(q_O,\boldsymbol{V}) = q_{OV},\\
&\delta(q_{OI},\boldsymbol{I}) = q_{OII}, && \delta(q_{OI},\boldsymbol{V}) = q_{done},\\
&\delta(q_{OII},\boldsymbol{I}) = q_{OIII}, &&\\
&\delta(q_{OV},\boldsymbol{I}) = q_{OVI}, \quad \delta(q_{OVI},\boldsymbol{I}) = q_{OVII}, \quad \delta(q_{OVII},\boldsymbol{I}) = q_{OVIII}. &&
\end{aligned}
$$

For every state $q \in S$ and every input letter not covered by the equations above,

$$\delta(q,\sigma) = q_{dead},$$

and q_{dead} loops to itself on every input letter. This machine accepts exactly the standard Roman numerals.

(c) The following Pascal program recognizes exactly the language of part (b), using a transition table and a clean δ function.

```
program Roman(input, output);
type
   Sigma = (symI, symV, symX, symL, symC, symD, symM, other);
   State = (qs, qM, qT, qHC, qHCC, qHCCC, qHD, qHDC, qHDCC, qHDCCC,
```

```
            qTX, qTXX, qTXXX, qTL, qTLX, qTLXX, qTLXXX,
            qOI, qOII, qOIII, qOV, qOVI, qOVII, qOVIII, qO, qdone, qdead);
var
   TransitionTable : array [State, Sigma] of State;
   FinalState : set of State;

function ClassOf(c : char) : Sigma;
begin
   case c of
      'I': ClassOf := symI;
      'V': ClassOf := symV;
      'X': ClassOf := symX;
      'L': ClassOf := symL;
      'C': ClassOf := symC;
      'D': ClassOf := symD;
      'M': ClassOf := symM
   else
      ClassOf := other
   end
end; { ClassOf }

function Delta(s : State; a : Sigma) : State;
begin
   Delta := TransitionTable[s, a]
end; { Delta }

function DeltaBar(s : State) : State;
var
   t : State;
begin
   t := s;
   while not eoln(input) do
      begin
         t := Delta(t, ClassOf(input^));
         get(input)
      end;
   DeltaBar := t
end; { DeltaBar }

function Accept : boolean;
begin
   Accept := DeltaBar(qs) in FinalState
end; { Accept }

procedure Initialize;
```

```
var
   s : State;
   a : Sigma;
begin
   FinalState := [qM, qT, qHC, qHCC, qHCCC, qHD, qHDC, qHDCC, qHDCCC,
                  qTX, qTXX, qTXXX, qTL, qTLX, qTLXX, qTLXXX,
                  qOI, qOII, qOIII, qOV, qOVI, qOVII, qOVIII, qO, qdone];

   for s := qs to qdead do
      for a := symI to other do
         TransitionTable[s, a] := qdead;

   for a := symI to other do
      TransitionTable[qdead, a] := qdead;

   TransitionTable[qs, symM] := qM;
   TransitionTable[qM, symM] := qM;

   TransitionTable[qs, symC] := qHC;
   TransitionTable[qM, symC] := qHC;
   TransitionTable[qs, symD] := qHD;
   TransitionTable[qM, symD] := qHD;
   TransitionTable[qs, symX] := qTX;
   TransitionTable[qM, symX] := qTX;
   TransitionTable[qs, symL] := qTL;
   TransitionTable[qM, symL] := qTL;
   TransitionTable[qs, symI] := qOI;
   TransitionTable[qM, symI] := qOI;
   TransitionTable[qs, symV] := qOV;
   TransitionTable[qM, symV] := qOV;

   TransitionTable[qHC, symM] := qT;
   TransitionTable[qHC, symD] := qT;
   TransitionTable[qHC, symC] := qHCC;
   TransitionTable[qHC, symX] := qTX;
   TransitionTable[qHC, symL] := qTL;
   TransitionTable[qHC, symI] := qOI;
   TransitionTable[qHC, symV] := qOV;

   TransitionTable[qHCC, symC] := qHCCC;
   TransitionTable[qHD, symC] := qHDC;
   TransitionTable[qHDC, symC] := qHDCC;
   TransitionTable[qHDCC, symC] := qHDCCC;

   TransitionTable[qHCC, symX] := qTX;
```

```
TransitionTable[qHCC, symL] := qTL;
TransitionTable[qHCC, symI] := qOI;
TransitionTable[qHCC, symV] := qOV;
TransitionTable[qHCCC, symX] := qTX;
TransitionTable[qHCCC, symL] := qTL;
TransitionTable[qHCCC, symI] := qOI;
TransitionTable[qHCCC, symV] := qOV;
TransitionTable[qHD, symX] := qTX;
TransitionTable[qHD, symL] := qTL;
TransitionTable[qHD, symI] := qOI;
TransitionTable[qHD, symV] := qOV;
TransitionTable[qHDC, symX] := qTX;
TransitionTable[qHDC, symL] := qTL;
TransitionTable[qHDC, symI] := qOI;
TransitionTable[qHDC, symV] := qOV;
TransitionTable[qHDCC, symX] := qTX;
TransitionTable[qHDCC, symL] := qTL;
TransitionTable[qHDCC, symI] := qOI;
TransitionTable[qHDCC, symV] := qOV;
TransitionTable[qHDCCC, symX] := qTX;
TransitionTable[qHDCCC, symL] := qTL;
TransitionTable[qHDCCC, symI] := qOI;
TransitionTable[qHDCCC, symV] := qOV;

TransitionTable[qT, symX] := qTX;
TransitionTable[qT, symL] := qTL;
TransitionTable[qT, symI] := qOI;
TransitionTable[qT, symV] := qOV;

TransitionTable[qTX, symC] := qO;
TransitionTable[qTX, symL] := qO;
TransitionTable[qTX, symX] := qTXX;
TransitionTable[qTXX, symX] := qTXXX;
TransitionTable[qTX, symI] := qOI;
TransitionTable[qTXX, symI] := qOI;
TransitionTable[qTXXX, symI] := qOI;
TransitionTable[qTX, symV] := qOV;
TransitionTable[qTXX, symV] := qOV;
TransitionTable[qTXXX, symV] := qOV;

TransitionTable[qTL, symX] := qTLX;
TransitionTable[qTLX, symX] := qTLXX;
TransitionTable[qTLXX, symX] := qTLXXX;
TransitionTable[qTL, symI] := qOI;
TransitionTable[qTLX, symI] := qOI;
```

```
   TransitionTable[qTLXX, symI] := qOI;
   TransitionTable[qTLXXX, symI] := qOI;
   TransitionTable[qTL, symV] := qOV;
   TransitionTable[qTLX, symV] := qOV;
   TransitionTable[qTLXX, symV] := qOV;
   TransitionTable[qTLXXX, symV] := qOV;

   TransitionTable[qO, symI] := qOI;
   TransitionTable[qO, symV] := qOV;
   TransitionTable[qOI, symX] := qdone;
   TransitionTable[qOI, symV] := qdone;
   TransitionTable[qOI, symI] := qOII;
   TransitionTable[qOII, symI] := qOIII;
   TransitionTable[qOV, symI] := qOVI;
   TransitionTable[qOVI, symI] := qOVII;
   TransitionTable[qOVII, symI] := qOVIII
end; { Initialize }

begin
   Initialize;
   if Accept then
      writeln(output, 'Accepted')
   else
      writeln(output, 'Rejected')
end.
```

1.51. **Describe the set of words accepted by the DFA in Figure 1.9.**

Let y be the suffix of the input after the last occurrence of $\boldsymbol{b}$; if no $\boldsymbol{b}$ occurs, let y be the entire string. The DFA accepts exactly those words for which

$$|y|_{\boldsymbol{a}} + 2|y|_{\boldsymbol{c}} \equiv 2 \pmod 3.$$

Equivalently, $\boldsymbol{b}$ resets the machine to state s_0, while $\boldsymbol{a}$ adds 1 modulo 3 and $\boldsymbol{c}$ adds 2 modulo 3, so acceptance depends only on the contribution of the suffix after the last $\boldsymbol{b}$.

1.52. **The languages L_7, L_3, and L_n for digit sums.**

(a) For L_7, use seven states $q_0, \ldots, q_6$, where q_i means that the sum of the digits seen so far is congruent to $i \pmod 7$. The start state is q_0, the only final state is q_0, and

$$\delta(q_i, \boldsymbol{d}) = q_{i+d \bmod 7}.$$

(b) The corresponding quintuple is

$$\langle\{\mathbf{0}, \mathbf{1}, \ldots, \mathbf{9}\}, \{q_0, \ldots, q_6\}, q_0, \delta, \{q_0\}\rangle,$$

with $\delta(q_i, \boldsymbol{d}) = q_{i+d \bmod 7}$.

(c) For L_3, use three states r_0, r_1, r_2, where r_i records the digit sum modulo 3. Again r_0 is both the start state and the only final state, and

$$\delta(r_i, \boldsymbol{d}) = r_{i+d \bmod 3}.$$

(d) The corresponding quintuple is

$$\langle\{\mathbf{0}, \mathbf{1}, \ldots, \mathbf{9}\}, \{r_0, r_1, r_2\}, r_0, \delta, \{r_0\}\rangle,$$

with $\delta(r_i, \boldsymbol{d}) = r_{i+d \bmod 3}$.

(e) In general,

$$\langle\{\mathbf{0}, \mathbf{1}, \ldots, \mathbf{9}\}, \{s_0, \ldots, s_{n-1}\}, s_0, \delta, \{s_0\}\rangle$$

recognizes L_n if

$$\delta(s_i, \boldsymbol{d}) = s_{i+d \bmod n}$$

for every residue class i and every digit $\boldsymbol{d}$. The state always records the current digit sum modulo n.

1.53. **Explain the last row of Table 1.3.**

The last row corresponds to the case where the circuitry has somehow been initialized to the unused state **11** and then scans <SOS>, which is also encoded by **11**. Unlike the other rows containing unused state/input combinations, this one is *not* a don't-care, because the text explicitly requires <SOS> to reset the machine to the real start state. Therefore the outputs in that row must be fixed so that the circuit moves from the bogus state back to s_0 rather than behaving arbitrarily.

Chapter 2

Nerode's Theorem — Solutions

2.1. **Show Ψ is not a right congruence.**

$x\Psi y \Leftrightarrow |x|-|y|$ is odd. Take $x=\mathbf{a}$, $y=\mathbf{b}$ (so $|x|-|y|=0$, not odd, thus $x\not\Psi y$). Also check: is Ψ even an equivalence relation? $|x|-|x|=0$ (even), so $x\not\Psi x$ (not reflexive). Hence Ψ is not an equivalence relation and a fortiori not a right congruence.

2.2. Q **defined by** $|x|_{\mathbf{a}}-|y|_{\mathbf{a}}\equiv 0 \pmod 3$**.**

(a) *Equivalence*: Reflexive: $|x|_{\mathbf{a}}-|x|_{\mathbf{a}}=0\equiv 0$. Symmetric: if $3\mid(|x|_{\mathbf{a}}-|y|_{\mathbf{a}})$ then $3\mid(|y|_{\mathbf{a}}-|x|_{\mathbf{a}})$. Transitive: if $3\mid(a-b)$ and $3\mid(b-c)$ then $3\mid(a-c)$.

(b) *Right congruence*: Suppose xQy, i.e., $|x|_{\mathbf{a}}\equiv|y|_{\mathbf{a}} \pmod 3$. For any $z\in\Sigma^*$: $|xz|_{\mathbf{a}}=|x|_{\mathbf{a}}+|z|_{\mathbf{a}}$ and $|yz|_{\mathbf{a}}=|y|_{\mathbf{a}}+|z|_{\mathbf{a}}$. So $|xz|_{\mathbf{a}}-|yz|_{\mathbf{a}}=|x|_{\mathbf{a}}-|y|_{\mathbf{a}}\equiv 0 \pmod 3$, i.e., $xzQyz$.

2.3. **All languages L over $\{\mathbf{a},\mathbf{b},\mathbf{c}\}$ with $\operatorname{rk}(R_L)=1$.**

$\operatorname{rk}(R_L)=1$ means R_L has exactly one equivalence class, namely all of Σ^*. This occurs iff $R_L=\Sigma^*\times\Sigma^*$, i.e., xR_Ly for all x,y. This happens iff $(\forall z)(xz\in L\Leftrightarrow yz\in L)$ for all x,y,z, which holds iff either $L=\emptyset$ or $L=\Sigma^*$.

Justification: If $L\neq\emptyset$ and $L\neq\Sigma^*$, pick $w\in L$ and $u\notin L$. Then wR_Lu fails (take $z=\lambda$), so R_L has at least 2 classes.

2.4. **Show P (with 3 equivalence classes as given) is a right congruence.**

The classes are $[\lambda]_P=\{\mathbf{0},\mathbf{1}\}^0$, $[\mathbf{1}]_P=\{\mathbf{0},\mathbf{1}\}^1$, $[\mathbf{00}]_P=\bigcup_{n\geq 2}\{\mathbf{0},\mathbf{1}\}^n$.

We must check: if xPy then $xaPya$ for all $a\in\{0,1\}$.

Case $x,y\in[\lambda]_P$: $|x|=|y|=0$, so $|xa|=|ya|=1$, thus $xa,ya\in[\mathbf{1}]_P$.

Case $x,y\in[\mathbf{1}]_P$: $|x|=|y|=1$, so $|xa|=|ya|=2$, thus $xa,ya\in[\mathbf{00}]_P$.

Case $x,y\in[\mathbf{00}]_P$: $|x|,|y|\geq 2$, so $|xa|,|ya|\geq 3\geq 2$, thus $xa,ya\in[\mathbf{00}]_P$.

In every case $xaPya$, so P is a right congruence. □

2.5. **All languages L with $R_L=P$ (from Exercise 2.4).**

Recall $A_0=[\lambda]_P=\{\lambda\}$, $A_1=[\mathbf{1}]_P=\Sigma^1$, $A_2=[\mathbf{00}]_P=\Sigma^{\geq 2}$. For $R_L=P$ we need L to be a union of P-classes *and* no two distinct P-classes to be merged by R_L. Two classes A_i,A_j are merged (identified) by R_L iff for every $z \in \Sigma^*$: $A_i \cdot z \subseteq L \Leftrightarrow A_j \cdot z \subseteq L$ (more precisely, for representative elements $u \in A_i$, $v \in A_j$: $uz \in L \Leftrightarrow vz \in L$ for all z).

Check each of the 6 nonempty proper unions:

- $L = A_0 = \{\lambda\}$: For any $z \neq \lambda$, both $\mathbf{0}z \not i nL$ (length$\geq$ 2) and $\mathbf{00}z \not i nL$; for $z = \lambda$, $\mathbf{0} \not i nL$ and $\mathbf{00} \not i nL$. So A_1 and A_2 are *not* distinguished: $R_L \neq P$.
- $L = A_1 = \Sigma^1$: $z = \lambda$: $\lambda \not i nL$ but $\mathbf{0} \in L$ (distinguishes A_0 from A_1). $\mathbf{0} \in L$ but $\mathbf{00} \not i nL$ (distinguishes A_1 from A_2). All three classes remain distinct: $R_L = P$. ✓
- $L = A_2 = \Sigma^{\geq 2}$: $z = \mathbf{0}$: $\lambda\mathbf{0} = \mathbf{0} \not i nL$ but $\mathbf{0} \cdot \mathbf{0} = \mathbf{00} \in L$ (distinguishes A_0 from A_1). $z = \lambda$: $\mathbf{0} \not i nL$ but $\mathbf{00} \in L$ (distinguishes A_1 from A_2). So $R_L = P$. ✓
- $L = A_0 \cup A_1 = \Sigma^{\leq 1}$: $z = \mathbf{0}$: $\lambda \cdot \mathbf{0} = \mathbf{0} \in L$ but $\mathbf{0} \cdot \mathbf{0} = \mathbf{00} \not i nL$ (distinguishes A_0 from A_1). $z = \lambda$: $\mathbf{0} \in L$ but $\mathbf{00} \not i nL$ (distinguishes A_1 from A_2). So $R_L = P$. ✓
- $L = A_0 \cup A_2$: $z = \lambda$: $\lambda \in L$ but $\mathbf{0} \not i nL$ (distinguishes A_0, A_1). $z = \lambda$: $\mathbf{0} \not i nL$ but $\mathbf{00} \in L$ (distinguishes A_1, A_2). So $R_L = P$. ✓
- $L = A_1 \cup A_2 = \Sigma^+$: For any z, $\mathbf{0}z \in \Sigma^+$ and $\mathbf{00}z \in \Sigma^+$, both in L. So A_1 and A_2 are not distinguished: $R_L \neq P$.

Therefore exactly **4** languages satisfy $R_L = P$: A_1, A_2, $A_0 \cup A_1$, and $A_0 \cup A_2$.

2.6. **All languages L with $R_L = Q$ (from Exercise 2.2).**

Q has three equivalence classes: $[x]_Q$ depends on $|x|_{\mathbf{a}}$ mod 3. So Q-classes are $C_0 = \{x : |x|_{\mathbf{a}} \equiv 0\}$, $C_1 = \{x : |x|_{\mathbf{a}} \equiv 1\}$, $C_2 = \{x : |x|_{\mathbf{a}} \equiv 2\}$. Any nonempty proper union of these classes gives $R_L = Q$.

2.7. **Analysis of Q on $\{\mathbf{a}, \mathbf{b}\}^*$.**

(a) *Right congruence, classes*: $\lambda Q \lambda$ by definition; for non-empty strings, xQy iff $(|x|, |y|$ have the same parity. Classes: $[\lambda]_Q$ (just λ) and $C_e = \{x \neq \lambda : |x| \text{ even}\}$ and $C_o = \{x : |x| \text{ odd}\}$. This is a right congruence because appending any a flips parity of length for non-empty strings, consistently within each class.

(b) $L = [\lambda]_Q \cup [\mathbf{aa}]_Q = \{\lambda\} \cup C_e$. Simple description: $L = \{x : |x| \text{ is even}\}$. R_L has two classes: even-length strings and odd-length strings.

(c) A_Q: 3 states ($[\lambda]_Q$, C_e, C_o). Final states for L: $\{[\lambda]_Q, C_e\}$.

(d) A_{R_L}: 2 states (even, odd), final: even.

(e) A_Q with relabeled final states: Yes. $C_e \cup [\lambda]_Q$ vs. C_o – this collapses to A_{R_L}.

(f) The eight languages are the $2^3 = 8$ unions of subsets of $\{[\lambda]_Q, C_e, C_o\}$. A language K satisfies the criterion of part (e) iff $R_K = Q$, i.e., A_Q (relabeled with new final states for K) is already minimal.

Checking all eight: $\emptyset$ and Σ^* give $\mathrm{rk}(R_K) = 1$; $\{\lambda\}$, all-even ($\{\lambda\} \cup C_e$), C_o, and Σ^+ ($= C_e \cup C_o$) give $\mathrm{rk}(R_K) = 2$ (two of the three Q-classes merge in each case). Only two choices keep all three Q-classes distinct:

- $K = C_e$: $z = \lambda$ distinguishes $[\lambda]$ from C_e (since $\lambda \not i nK$ but $aa \in K$); $z = \lambda$ distinguishes C_e from C_o (since $aa \in K$ but $a \not i nK$). So $R_K = Q$.
- $K = \{\lambda\} \cup C_o$: $z = aa$ distinguishes $[\lambda]$ from C_o ($\lambda \cdot aa = aa \not i nK$ but $a \cdot aa = aaa \in K$); $z = \lambda$ distinguishes C_o from C_e ($a \in K$ but $aa \not i nK$). So $R_K = Q$.

These are the only two of the eight languages satisfying part (e).

2.8. **I = identity on $\{\mathbf{a}\}^*$.**

(a) $xIy \Leftrightarrow x = y$. Right congruence: if $x = y$, then $xa = ya$ for any a. Yes.

(b) Each equivalence class is a singleton: $[\mathbf{a}^n]_I = \{\mathbf{a}^n\}$.

2.9. **$L = \{\lambda\} \cup \{\mathbf{a}\} \cup \{\mathbf{aa}\}$, I has infinite rank. Is L FAD?**

L is a finite set, hence FAD. Nerode's theorem says L is FAD iff R_L has finite rank, not iff I (or any other right congruence) has finite rank. Here R_L has rank equal to the number of R_L-classes, which is finite (since L is finite and FAD). The fact that I has infinite rank is irrelevant – we need I to *refine* R_L (which it does, trivially, since I is the finest relation), but Nerode's theorem requires R_L itself to have finite rank. $\square$

2.10. **Machine M with $\mathrm{rk}(R_{L(M)}) \neq |S_M|$.**

Let M be the two-state DFA: $\Sigma = \{\mathbf{a}\}$, $S = \{q_0, q_1\}$, $s_0 = q_0$, $F = \{q_0, q_1\}$ (all states final), $\delta(q_0, \mathbf{a}) = q_1$, $\delta(q_1, \mathbf{a}) = q_0$. Then $L(M) = \Sigma^*$, so $R_{L(M)}$ has rank 1, but $|S_M| = 2$.

2.11. **Show F_{R_L} is well defined.**

$F_{R_L} = \{[x]_{R_L} \mid x \in L\}$. We must show: if $[x]_{R_L} = [y]_{R_L}$, then $x \in L \Leftrightarrow y \in L$ (so each R_L-class is either entirely inside L or entirely outside L, making the assignment of classes to F_{R_L} unambiguous). Since $[x]_{R_L} = [y]_{R_L}$, we have xR_Ly, meaning $(\forall z \in \Sigma^*)(xz \in L \Leftrightarrow yz \in L)$. Taking $z = \lambda$: $x \in L \Leftrightarrow y \in L$. $\square$

2.12. **Show δ_{R_L} is well defined.**

$\delta_{R_L}([x]_{R_L}, a) = [xa]_{R_L}$. We must show: if $[x]_{R_L} = [y]_{R_L}$, then $[xa]_{R_L} = [ya]_{R_L}$. Since xR_Ly, for any z: $xaz \in L \Leftrightarrow x(az) \in L \Leftrightarrow y(az) \in L \Leftrightarrow yaz \in L$. Thus xaR_Lya. $\square$

2.13. **Prove $\overline{\delta}_{R_L}([x]_{R_L}, y) = [xy]_{R_L}$ by induction on $|y|$.**

Base $|y| = 0$: $\overline{\delta}_{R_L}([x], \lambda) = [x] = [x\lambda]$. ✓

Step: $y = wa$, $|w| = n$. $\overline{\delta}_{R_L}([x], wa) = \delta_{R_L}(\overline{\delta}_{R_L}([x], w), a) = \delta_{R_L}([xw], a) = [xwa] = [xy]$. $\square$

2.14. **Find R^P for automaton P in Example 2.6, check $R^P = R_L$.**

R^P is defined by $xR^Py \Leftrightarrow \overline{\delta}(s_0, x) = \overline{\delta}(s_0, y)$. Since the machine P in Example 2.6 is the minimal machine for L, each state is reachable and distinguishable. The relation R^P has the same classes as R_L (each state corresponds to one R_L-class). Hence $R^P = R_L$.

2.15. **Find R^A for machines from Chapter 1 exercises.**

For each DFA A built in Chapter 1: R^A is defined by $xR^Ay \Leftrightarrow \overline{\delta}(s_0, x) = \overline{\delta}(s_0, y)$ (same state reached). Compare to $R_{L(A)}$: by definition of R^A, it always refines $R_{L(A)}$ (since if same state, same future acceptance). For minimal machines they coincide; for non-minimal ones R^A may be strictly finer.

2.16. **Prove the pumping lemma statement by induction.**

Let M be a DFA with states S, $|S| = n$. For a word $w = uvw'$ with $|uv| \leq n$, $|v| \geq 1$, and $uvw' \in L$. Let $s = \overline{\delta}(s_0, u)$, so $\overline{\delta}(s, v) = s$ (since the path through v loops back to s).

Claim: $(\forall i \in \mathbb{N})(\overline{\delta}(s_0, uv^i w') = \overline{\delta}(s_0, uvw') = f \in F)$.

Proof by induction on i:

$i = 0$: $\overline{\delta}(s_0, uw') = \overline{\delta}(\overline{\delta}(s_0, u), w') = \overline{\delta}(s, w')$. And $\overline{\delta}(s_0, uvw') = \overline{\delta}(\overline{\delta}(s_0, uv), w') = \overline{\delta}(s, w')$ (since $\overline{\delta}(s, v) = s$). Equal. ✓

$i \to i+1$: By the inner induction, $\overline{\delta}(s_0, uv^i) = s$ (base: $\overline{\delta}(s_0, u) = s$; step: $\overline{\delta}(s_0, uv^{i+1}) = \overline{\delta}(\overline{\delta}(s_0, uv^i), v) = \overline{\delta}(s, v) = s$). Therefore

$$\overline{\delta}(s_0, uv^{i+1} w') = \overline{\delta}(\overline{\delta}(s_0, uv^i), vw') = \overline{\delta}(s, vw') = \overline{\delta}(\overline{\delta}(s, v), w') = \overline{\delta}(s, w').$$

So $\overline{\delta}(s_0, uv^i w') = \overline{\delta}(s, w') = f$ for all i. □

2.17. **Prove Theorem 2.1.**

Theorem 2.1 states: L is FAD iff R_L has finite rank.

(⇒) If $L = L(M)$ for a DFA M with n states, define $\mu : \Sigma^* \to S$ by $\mu(x) = \overline{\delta}(s_0, x)$. Then $xR^M y \Leftrightarrow \mu(x) = \mu(y)$, so R^M has at most n classes. Since R^M refines R_L and R_L is coarser, $\mathrm{rk}(R_L) \le \mathrm{rk}(R^M) \le n < \infty$.

(⇐) If R_L has finite rank k, construct the minimal machine $A_{R_L} = \langle \Sigma, [\cdot]_{R_L}, [\lambda]_{R_L}, \delta_{R_L}, F_{R_L} \rangle$ with k states (shown well-defined in Exercises 2.11–2.12). Show $L(A_{R_L}) = L$: $x \in L(A_{R_L}) \Leftrightarrow \overline{\delta}_{R_L}([\lambda], x) \in F_{R_L} \Leftrightarrow [x] \in F_{R_L} \Leftrightarrow x \in L$.

2.18. **Find a language giving rise to I (identity on $\{\mathbf{a}\}^*$); show it cannot be FAD.**

(a) Take
$$L = \{\mathbf{a}^{2^m} \mid m \ge 0\} \subseteq \{\mathbf{a}\}^*.$$
We claim $R_L = I$. Let $i < j$ and $d = j - i \ge 1$. Choose m so that $2^m > d$, and set $z = \mathbf{a}^{2^m - i}$. Then
$$\mathbf{a}^i z = \mathbf{a}^{2^m} \in L,$$
but
$$\mathbf{a}^j z = \mathbf{a}^{2^m + d}$$
with
$$2^m < 2^m + d < 2^{m+1},$$
so $2^m + d$ is not a power of 2. Hence $\mathbf{a}^j z \notin L$. Thus $\mathbf{a}^i \not R_L \mathbf{a}^j$ for all $i \ne j$, so every class is a singleton and $R_L = I$.

(b) Such a language cannot be FAD: $R_L = I$ has infinite rank (one class per string in $\{\mathbf{a}\}^*$), so by Nerode's theorem L is not FAD. □

2.19. **Prove Theorem 2.4 from Theorem 2.3.**

Theorem 2.3: If L is FAD then R_L has finite rank. Theorem 2.4 (Nerode): L is FAD $\Leftrightarrow$ R_L has finite rank.

We need the converse: finite rank ⇒ FAD. Given R_L has k classes, the machine A_{R_L} has k states and accepts L (by Exercises 2.11–2.13). Hence L is FAD.

2.20. **Prove Theorem 2.5.**

Theorem 2.5: L is FAD $\Leftrightarrow$ there exists a right congruence R of finite rank such that L is a union of R-classes.

($\Rightarrow$) Take $R = R^M$ for a DFA M accepting L: finite rank (at most $|S|$), and $L = \bigcup_{s \in F} [s]_{R^M}$.

($\Leftarrow$) Given such R, build A_R with states = R-classes, F_R = classes $\subseteq L$. Then $L = L(A_R)$, so L is FAD.

2.21. **Prove Theorem 2.6.**

Theorem 2.6 states:

$$(\exists n)(\forall x \not{i}nL,\ |x| \geq n)(\exists u, v, w)\big[x = uvw,\ |uv| \leq n,\ |v| \geq 1,\ (\forall i \geq 0)\ uv^i w \not{i}nL\big].$$

Assume L is FAD, accepted by DFA $M = \langle \Sigma, S, s_0, \delta, F \rangle$, with $|S| = n$. Take any $x \not{i}nL$ with $|x| \geq n$. Look at the first $n+1$ states on the run of x:

$$s_0,\ \overline{\delta}(s_0, a_1),\ \overline{\delta}(s_0, a_1 a_2),\ \ldots,\ \overline{\delta}(s_0, a_1 \cdots a_n).$$

By pigeonhole, two are equal. So for some decomposition $x = uvw$ with $|uv| \leq n$, $|v| \geq 1$,

$$\overline{\delta}(s_0, u) = \overline{\delta}(s_0, uv) = q.$$

Hence v is a loop at q, so for every $i \geq 0$,

$$\overline{\delta}(s_0, uv^i w) = \overline{\delta}(q, w) = \overline{\delta}(s_0, uvw) = \overline{\delta}(s_0, x).$$

Since $x \not{i}nL$, $\overline{\delta}(s_0, x) \not{i}nF$. Therefore $\overline{\delta}(s_0, uv^i w) \not{i}nF$, i.e., $uv^i w \not{i}nL$ for all $i \geq 0$. This is exactly Theorem 2.6. $\square$

2.22. **Prove Theorem 2.7.**

Theorem 2.7 states: if M is an n-state DFA accepting L, then every $x = a_1 a_2 \cdots a_m \in L$ with $m \geq n$ has a subsequence $a_{i_1} a_{i_2} \cdots a_{i_j} \in L$ with $j < n$.

Let $x \in L$, $|x| = m \geq n$. By Theorem 2.3 (or Exercise 2.16), we can write

$$x = u_1 v_1 w_1, \quad |u_1 v_1| \leq n,\ |v_1| \geq 1, \text{ and } u_1 v_1^i w_1 \in L\ \forall i \geq 0.$$

In particular, with $i = 0$, $x_1 = u_1 w_1 \in L$ and $|x_1| < |x|$.

If $|x_1| < n$, stop. Otherwise apply the same argument to x_1. Each step removes at least one symbol and keeps membership in L, so after finitely many steps we obtain

$$y \in L, \quad |y| < n,$$

and y is obtained from x by deleting some blocks. Therefore y is a subsequence of x:

$$y = a_{i_1} a_{i_2} \cdots a_{i_j}$$

for an increasing index sequence $i_1 < i_2 < \cdots < i_j$, with $j = |y| < n$. This is exactly Theorem 2.7. $\square$

2.23. **Show $L = \{x \mid |x|_{\mathbf{a}} < |x|_{\mathbf{b}}\}$ is not FAD.**

For any n, take $w = \mathbf{a}^n\mathbf{b}^{n+1} \in L$, $|w| = 2n+1 \geq n$. Any decomposition $w = uvw'$ with $|uv| \leq n$ and $|v| \geq 1$ has $v = \mathbf{a}^k$ for some $k \geq 1$ (since $|uv| \leq n$). Then $uv^0w' = uw' = \mathbf{a}^{n-k}\mathbf{b}^{n+1}$ which has $n-k$ **a**s and $n+1$ **b**s. Since $n-k < n+1$, this is in L. And $uv^2w' = \mathbf{a}^{n+k}\mathbf{b}^{n+1}$: need $n+k < n+1$, i.e., $k < 1$, impossible since $k \geq 1$. So $uv^2w' \not{i}nL$. The pumping lemma fails, so L is not FAD. □

2.24. **Show $G = \{x \mid |x|_{\mathbf{a}} \geq |x|_{\mathbf{b}}\}$ is not FAD.**

For any n, take $w = \mathbf{a}^n\mathbf{b}^n \in G$. Any $v = \mathbf{a}^k$ ($k \geq 1$, since $|uv| \leq n$). Then $uv^0w' = \mathbf{a}^{n-k}\mathbf{b}^n$: need $n-k \geq n$, i.e., $k \leq 0$. Contradiction. So pumping down takes us out of G. Not FAD. □

2.25. **Show $P = \{\mathbf{d}^p \mid p$ prime$\}$ is not FAD.**

Suppose P is FAD with n states. Take a prime $p > n$ and $w = \mathbf{d}^p \in P$. By pumping: $w = uvw'$ with $|uv| \leq n$, $|v| = k \geq 1$, so $v = \mathbf{d}^k$ and $u = \mathbf{d}^j$, $w' = \mathbf{d}^{p-j-k}$. Then $uv^iw' = \mathbf{d}^{j+ik+(p-j-k)} = \mathbf{d}^{p+(i-1)k}$ must be in P for all $i \geq 0$. So $p+(i-1)k$ must be prime for all $i \geq 0$. Take $i = p+1$: $p + pk = p(1+k)$, which is composite (both factors > 1 since $p \geq 2$ and $k \geq 1$). Contradiction. □

2.26. **Show $\Gamma = \{w \cdot \mathbf{2} \cdot w \mid w \in \{\mathbf{0}, \mathbf{1}\}^*\}$ is not FAD.**

We use Theorem 2.6. Suppose Γ were FAD, and let n be the number supplied by that theorem.

Take

$$x = \mathbf{0}^{n!}\mathbf{2}\mathbf{0}^{2n!}.$$

This string is *not* in Γ, because the block to the left of $\mathbf{2}$ has length $n!$ while the block to the right has length $2n!$, so the two sides are unequal.

By Theorem 2.6 there should be a decomposition

$$x = uvw, \qquad |uv| \leq n, \qquad |v| \geq 1,$$

such that

$$uv^iw \not{i}n\Gamma \qquad \text{for every } i \geq 0.$$

Since the first n symbols of x are all $\mathbf{0}$'s, the substring v must be of the form $\mathbf{0}^k$ for some $1 \leq k \leq n$. Because $k \leq n$, the integer k divides $n!$. Choose

$$i = \frac{n!}{k} + 1.$$

Then pumping v changes only the block of $\mathbf{0}$'s to the left of $\mathbf{2}$, and its new length is

$$n! + (i-1)k = n! + \frac{n!}{k}k = 2n!.$$

Therefore

$$uv^iw = \mathbf{0}^{2n!}\mathbf{2}\mathbf{0}^{2n!} \in \Gamma.$$

This contradicts Theorem 2.6.

Hence Γ is not FAD. □

2.27. **Show $\Psi = \{ww \mid w \in \{\mathbf{0}, \mathbf{1}\}^*\}$ is not FAD.**

For each $n \geq 0$, let $x_n = \mathbf{0}^n\mathbf{1}$. We show the x_n are pairwise in distinct R_Ψ-classes.

Fix $i \neq j$ and take suffix $z_i = \mathbf{0}^i\mathbf{1}$. Then

$$x_i z_i = \mathbf{0}^i\mathbf{1}\mathbf{0}^i\mathbf{1} = ww \in \Psi$$

with $w = \mathbf{0}^i\mathbf{1}$.

Now consider

$$x_j z_i = \mathbf{0}^j\mathbf{1}\mathbf{0}^i\mathbf{1}.$$

Suppose this were in Ψ, so $x_j z_i = tt$ for some t. The string has exactly two $\mathbf{1}$s, so each half must contain exactly one $\mathbf{1}$. Hence the position of the second $\mathbf{1}$ must be "half-length" after the first. Its first $\mathbf{1}$ is at position $j+1$, second at $j+i+2$, so the half-length must be $i+1$. But half-length is also $\frac{i+j+2}{2}$, giving $\frac{i+j+2}{2} = i+1$, hence $j = i$, contradiction. Therefore $x_j z_i \not in \Psi$.

So $x_i \not R_\Psi x_j$ for $i \neq j$, giving infinitely many Nerode classes. Thus Ψ is not FAD. □

2.28. **Show $K = \{w \mid w = w^r\}$ (palindromes) is not FAD.**

For any n, the strings $\mathbf{0}, \mathbf{0}^2, \ldots, \mathbf{0}^n$ are in distinct R_K-classes: $\mathbf{0}^i R_K \mathbf{0}^j$ requires $(\forall z)(\mathbf{0}^i z \in K \Leftrightarrow \mathbf{0}^j z \in K)$. Take $z = \mathbf{0}^i$: $\mathbf{0}^{2i}$ is a palindrome (yes). $\mathbf{0}^{j+i}$: palindrome (yes, any string of one letter is a palindrome). So we need a subtler argument. Take $z = \mathbf{1}\mathbf{0}^i$: $\mathbf{0}^i\mathbf{1}\mathbf{0}^i \in K$ (palindrome), but $\mathbf{0}^j\mathbf{1}\mathbf{0}^i \in K$ iff $\mathbf{0}^j\mathbf{1}\mathbf{0}^i = \mathbf{0}^i\mathbf{1}\mathbf{0}^j$, i.e., $i = j$. So distinct $i \neq j$ give distinct classes. Infinite rank. Not FAD. □

2.29. **Show $\Phi = \{\mathbf{a}^j\mathbf{b}^k\mathbf{c}^m \mid j \geq 3, k = m\}$ is not FAD (second pumping lemma).**

We use Theorem 2.5. Suppose, for contradiction, that Φ is FAD. Then there is a right congruence R of finite rank such that Φ is a union of R-classes.

For each $i \in \mathbb{N}$, let

$$x_i = \mathbf{a}^3\mathbf{b}^i.$$

We claim that the x_i lie in distinct R-classes. If $i \neq j$ and $x_i R x_j$, then by the right-congruence property,

$$x_i\mathbf{c}^i R x_j\mathbf{c}^i.$$

But

$$x_i\mathbf{c}^i = \mathbf{a}^3\mathbf{b}^i\mathbf{c}^i \in \Phi,$$

while

$$x_j\mathbf{c}^i = \mathbf{a}^3\mathbf{b}^j\mathbf{c}^i \not in \Phi$$

because $j \neq i$. Since Φ is a union of R-classes, two R-equivalent strings cannot disagree about membership in Φ. Contradiction.

So $x_0, x_1, x_2, \ldots$ belong to infinitely many distinct R-classes, contradicting the finite rank of R. Therefore Φ is not FAD. □

Why first pumping lemma is hard: Any long word in Φ starts with ≥ 3 **a**s. Pumping within the **a**-block may still satisfy $j \geq 3$; pumping within **b** or **c** block destroys $k = m$.

2.30. **Show $C = \{\mathbf{d}^q \mid q$ non-prime$\}$ is not FAD.**

We use Theorem 2.6. Suppose C were FAD, and let n be the number supplied by that theorem. Choose a prime $p > n$ and let

$$x = \mathbf{d}^p.$$

Since p is prime, $x \not inC$. By Theorem 2.6 there should be a decomposition

$$x = uvw, \qquad |uv| \le n, \qquad |v| \ge 1,$$

such that

$$uv^i w \not inC \qquad \text{for every } i \ge 0.$$

Because x is unary, $v = \mathbf{d}^k$ for some $k \ge 1$. Then

$$uv^i w = \mathbf{d}^{p+(i-1)k}.$$

Now choose $i = p + 1$. Its length is

$$p + pk = p(1+k),$$

which is composite. Hence

$$uv^{p+1}w \in C,$$

contradicting Theorem 2.6.

Therefore C is not FAD. □

2.31. **R_L with 3 classes: $\{\lambda\}$, odd-length, even-length$\setminus\{\lambda\}$.**

(a) Why $L = \{x \mid |x| \text{ odd}\}$ can't give this R_L: Compute $R_{\{x:|x| \text{ odd}\}}$. We need $xR_L y \Leftrightarrow (\forall z)(|xz| \text{ odd} \Leftrightarrow |yz| \text{ odd})$, i.e., $|x| \equiv |y| \pmod 2$. So R_L has only 2 classes (even-length, odd-length), not 3. λ is not distinguished from other even-length strings.

(b) Let

$$A = \{\lambda\}, \quad B = \{x : |x| \text{ odd}\}, \quad C = \{x : |x| \text{ even and } x \ne \lambda\}.$$

Checking all 8 unions of A, B, C, exactly two give this same 3-class relation:

$$L = C \quad \text{and} \quad L = A \cup B.$$

For each of these, A, B, C are pairwise distinguishable:

- A vs C: suffix $z = \lambda$.
- A vs B: suffix z of odd length.
- B vs C: suffix $z = \lambda$.

All other unions merge at least one pair of classes.

2.32. **Describe R_Ψ and draw machine for $\Psi = \{x \mid |x|_{\mathbf{a}}$ even, ends with $\mathbf{b}\}$.**

R_Ψ classes: we need to determine when x, y have the same future w.r.t. Ψ. The future of x depends on: (1) parity of $|x|_{\mathbf{a}}$, and (2) whether x ends with $\mathbf{b}$.

Thus R_Ψ has four classes:

- $C_{e,b} = \{x \mid |x|_{\mathbf{a}}$ even and x ends with $\mathbf{b}\}$.
- $C_{e,\neg b} = \{x \mid |x|_{\mathbf{a}}$ even and x does not end with $\mathbf{b}\}$.
- $C_{o,b} = \{x \mid |x|_{\mathbf{a}}$ odd and x ends with $\mathbf{b}\}$.
- $C_{o,\neg b} = \{x \mid |x|_{\mathbf{a}}$ odd and x does not end with $\mathbf{b}\}$.

Here $\lambda \in C_{e,\neg b}$ (even number of **a**'s, and it does not end with **b**).

So the minimal DFA has 4 states, start state $C_{e,\neg b}$, and unique final state $C_{e,b}$.

2.33. **Show $\Xi = \{\lambda, \mathbf{a}, \mathbf{a}^4, \mathbf{a}^9, \ldots\}$ (perfect square lengths) is not FAD.**

We use Theorem 2.3. Suppose Ξ were FAD, and let n be a pumping length for Ξ. Take

$$w = \mathbf{a}^{n^2} \in \Xi.$$

For any decomposition

$$w = uvw, \qquad |uv| \le n, \qquad |v| \ge 1,$$

the string v must be of the form $\mathbf{a}^k$ with $1 \le k \le n$.

Pump once upward:

$$uv^2w = \mathbf{a}^{n^2+k}.$$

Since $1 \le k \le n$, we have

$$n^2 < n^2 + k < n^2 + (2n+1) = (n+1)^2.$$

So $n^2 + k$ is strictly between two consecutive squares, hence is not itself a square. Therefore

$$uv^2w \not{i}n\Xi,$$

contradicting Theorem 2.3.

Thus Ξ is not FAD. □

2.34. **Show $\Phi = \{\mathbf{b}, \mathbf{b}^2, \mathbf{b}^4, \mathbf{b}^8, \ldots\}$ (powers of 2) is not FAD.**

We use Theorem 2.3. Suppose Φ were FAD, and let n be a pumping length for Φ. Choose m so that $2^m > n$, and let

$$w = \mathbf{b}^{2^m} \in \Phi.$$

For any decomposition

$$w = uvw, \qquad |uv| \le n, \qquad |v| \ge 1,$$

we have $v = \mathbf{b}^k$ for some $1 \le k \le n < 2^m$.

Pump once upward:

$$uv^2w = \mathbf{b}^{2^m+k}.$$

Because $0 < k < 2^m$,

$$2^m < 2^m + k < 2^{m+1}.$$

So $2^m + k$ lies strictly between two consecutive powers of 2, and therefore is not itself a power of 2. Hence

$$uv^2w \not{i}n\Phi,$$

contradicting Theorem 2.3.

Therefore Φ is not FAD. □

2.35. **Five-class R_L over $\{\mathbf{a},\mathbf{b}\}$.**

(a) Let
$$C_0 = \{\lambda\}, C_1 = \{\mathbf{a}\}, C_2 = \{\mathbf{aa}\}, C_3 = \{\mathbf{a}^k : k \geq 3\}, C_4 = \{x : x \text{ contains a } \mathbf{b}\}.$$
If L is exactly one class and still has R_L equal to this 5-class partition, the only possibility is
$$L = C_3 = \{\mathbf{a}^3, \mathbf{a}^4, \ldots\}.$$
Choosing any other single class merges some classes under R_L.

(b) The other languages L that still give this same R_L are exactly:
$$C_0 \cup C_3, \quad C_1 \cup C_3, \quad C_0 \cup C_1 \cup C_3,$$
$$C_2 \cup C_4, \quad C_0 \cup C_2 \cup C_4, \quad C_1 \cup C_2 \cup C_4, \quad C_0 \cup C_1 \cup C_2 \cup C_4.$$
Together with part (a), there are 8 such languages.

2.36. **Find R_Ω for $\Omega = \{y \mid (y$ has exactly one $\mathbf{0}) \vee (y$ has even $\#\mathbf{1}$s$)\}$.**

$xz \in \Omega$ iff $(|x|_\mathbf{0} + |z|_\mathbf{0} = 1)$ or $(|x|_\mathbf{1} + |z|_\mathbf{1} \equiv 0 \pmod 2)$. The future of x (i.e., which z satisfy $xz \in \Omega$) depends only on $c_0 = |x|_\mathbf{0}$ (capped at 2) and $p = |x|_\mathbf{1} \bmod 2$. Writing (c_0, p) for these two pieces of information, the six cases yield the following future descriptions:

- $(0,0)$: $xz \in \Omega \Leftrightarrow |z|_\mathbf{0} = 1$ or $|z|_\mathbf{1}$ even.
- $(0,1)$: $xz \in \Omega \Leftrightarrow |z|_\mathbf{0} = 1$ or $|z|_\mathbf{1}$ odd.
- $(1,0)$: $xz \in \Omega \Leftrightarrow |z|_\mathbf{0} = 0$ or $|z|_\mathbf{1}$ even.
- $(1,1)$: $xz \in \Omega \Leftrightarrow |z|_\mathbf{0} = 0$ or $|z|_\mathbf{1}$ odd.
- $(2,0)$: $xz \in \Omega \Leftrightarrow |z|_\mathbf{1}$ even. ($|xz|_\mathbf{0} \geq 2$ so the first disjunct can never hold.)
- $(2,1)$: $xz \in \Omega \Leftrightarrow |z|_\mathbf{1}$ odd.

All six futures are distinct (for example, $(0,0)$ and $(2,0)$ differ on $z = \mathbf{01}$: the first accepts it since $|z|_\mathbf{0} = 1$, the second rejects since $|z|_\mathbf{1} = 1$ is odd; and so on for the other pairs). Hence R_Ω has exactly **6** equivalence classes:
$$C_{c_0,p} = \{x \in \{\mathbf{0},\mathbf{1}\}^* \mid |x|_\mathbf{0} = c_0 \text{ (or } \geq 2 \text{ if } c_0 = 2) \text{ and } |x|_\mathbf{1} \equiv p \pmod 2\},$$
for $(c_0, p) \in \{0,1,2\} \times \{0,1\}$. The minimal DFA for Ω has 6 states. Final states are those in Ω: $(1,0), (1,1), (0,0), (2,0)$ — i.e., strings with exactly one $\mathbf{0}$ (states $(1,\cdot)$) or with even $\#\mathbf{1}$s (states $(\cdot,0)$).

2.37. **Which of $L_1, L_2, L_1 \cap L_2, \sim L_2, L_1 \cup L_2$ are FAD?**

$L_1 = \{x \mid |x|_\mathbf{a} > |x|_\mathbf{b}\}$: **not FAD**. If it were FAD, let n be a pumping length from Theorem 2.3. Take
$$w = \mathbf{a}^{n+1}\mathbf{b}^n \in L_1.$$
Any split $w = uvw'$ with $|uv| \leq n$ and $|v| \geq 1$ has $v = \mathbf{a}^k$ for some $k \geq 1$. Pumping down gives
$$uw' = \mathbf{a}^{n+1-k}\mathbf{b}^n,$$

and now $n+1-k \le n$, so this pumped string is not in L_1. Contradiction.

$L_2 = \{x \mid |x|_{\mathbf{a}} < 3\}$: **FAD**. A DFA only needs to remember whether it has seen 0, 1, 2, or at least 3 copies of **a**.

$L_1 \cap L_2$: **FAD**. In fact this language is finite:

$$L_1 \cap L_2 = \{\mathbf{a}, \mathbf{aa}, \mathbf{aab}, \mathbf{aba}, \mathbf{baa}\}.$$

Every finite language is FAD.

$\sim L_2 = \{x \mid |x|_{\mathbf{a}} \ge 3\}$: **FAD**. The same 4-state machine used for L_2 works here too; we simply make the "at least 3 **a**'s" state final.

$L_1 \cup L_2$: **not FAD**. Suppose it were FAD, and let n be a pumping length. Consider

$$w = \mathbf{a}^{n+4}\mathbf{b}^{n+3} \in L_1 \cup L_2$$

because it has more **a**'s than **b**'s. Again any split $w = uvw'$ with $|uv| \le n$ and $|v| \ge 1$ has $v = \mathbf{a}^k$ for some $k \ge 1$. Pumping down gives

$$uw' = \mathbf{a}^{n+4-k}\mathbf{b}^{n+3}.$$

Here $n+4-k \ge 4$, so the pumped string is not in L_2; and also $n+4-k \le n+3$, so it is not in L_1. Thus $uw' \not in L_1 \cup L_2$, contradicting Theorem 2.3.

2.38. **R_m defined by $|x| - |y|$ is a multiple of m.**

(a) *Equivalence relation and right congruence*: Reflexive, symmetric, and transitive are immediate from divisibility by m. If $m \mid (|x| - |y|)$, then

$$|xz| - |yz| = |x| - |y|,$$

so $m \mid (|xz| - |yz|)$ as well. Thus R_m is a right congruence.

(b) $R_2 \cap R_3 = R_6$: $xR_2 \cap R_3 y \Leftrightarrow 2 \mid |x| - |y|$ and $3 \mid |x| - |y| \Leftrightarrow 6 \mid |x| - |y| \Leftrightarrow xR_6 y$. So every R_6-class is contained in an R_2-class (and also in an R_3-class); in particular, $R_6 \subseteq R_2$.

(c) In general: since both R and S are equivalence relations, $R \cap S$ is also an equivalence relation. And if $xR \cap Sy$, then xRy and xSy. For any z, right congruence of R and S gives $xzRyz$ and $xzSyz$, hence $xz(R \cap S)yz$. So $R \cap S$ is a right congruence. Also $R \cap S \subseteq R$ and $R \cap S \subseteq S$, so its classes refine the classes of both R and S.

(d) $R_2 \cup R_3$: $\lambda R_2 \mathbf{aa}$ and $\mathbf{a} R_3 \mathbf{aaaa}$. But is $\lambda R_2 \cup R_3 \mathbf{a}$? $|\lambda| - |\mathbf{a}| = -1$, not divisible by 2 or 3. No. Is $\lambda R_2 \cup R_3 \mathbf{aaa}$? $0 - 3 = -3$, divisible by 3. Yes. Now $\lambda R_2 \cup R_3 \mathbf{aaa}$ and $\mathbf{aaa} R_2 \cup R_3 \mathbf{aaaaa}$ (length diff=2). But is $\lambda R_2 \cup R_3 \mathbf{aaaaa}$? Length diff=5, not divisible by 2 or 3. No. So transitivity fails: not an equivalence relation.

(e) If $R \cup S$ is an equivalence relation: for any x, y with $xR \cup Sy$, we have xRy or xSy. Either way, for any z: $xzRyz$ or $xzSyz$ (by right congruence of R or S), hence $xzR \cup Syz$.

2.39. **Example of two right congruences whose union is not a right congruence.**

Let $R = R_2$ and $S = R_3$. As shown above, $R_2 \cup R_3$ is not even an equivalence relation, hence not a right congruence.

2.40. **Analysis of** $\Gamma = \{\lambda, \mathbf{a}, \mathbf{ab}, \mathbf{ba}, \mathbf{bb}, \mathbf{bbb}\} \cup \{x : |x| \geq 4\}$**.**

(a) $\mathbf{ab} R_\Gamma \mathbf{ba}$: For all z: $\mathbf{ab}z \in \Gamma \Leftrightarrow \mathbf{ba}z \in \Gamma$. If $|z| = 0$: $\mathbf{ab}, \mathbf{ba} \in \Gamma$ (both listed). $\checkmark$ If $|z| = 1$: $|\mathbf{ab}z| = 3$. Since Γ contains only $\mathbf{bbb}$ at length 3, and $\mathbf{ab}$ cannot be extended to $\mathbf{bbb}$ by appending one letter, $\mathbf{ab}z \notin \Gamma$; likewise $\mathbf{ba}z \notin \Gamma$ (since $\mathbf{baa}, \mathbf{bab} \neq \mathbf{bbb}$). Both outside Γ. $\checkmark$ If $|z| \geq 2$: $|\mathbf{ab}z|, |\mathbf{ba}z| \geq 4$, so both in Γ. $\checkmark$ Hence $\mathbf{ab} R_\Gamma \mathbf{ba}$.

(b) $\mathbf{ab}$ is not related to $\mathbf{bb}$: take $z = \mathbf{b}$. Then

$$\mathbf{ab}z = \mathbf{abb} \not in \Gamma$$

because among length-3 strings only $\mathbf{bbb}$ is in Γ. But

$$\mathbf{bb}z = \mathbf{bbb} \in \Gamma.$$

So $\mathbf{ab} \not R_\Gamma \mathbf{bb}$. $\square$

(c) The 8 classes are exactly:

$$C_0 = \{\lambda\},\ C_1 = \{\mathbf{a}\},\ C_2 = \{\mathbf{b}\},\ C_3 = \{\mathbf{aa}\},\ C_4 = \{\mathbf{bb}\},$$

$$C_5 = \{\mathbf{ab}, \mathbf{ba}\},\ C_6 = \{x : |x| = 3,\ x \neq \mathbf{bbb}\},\ C_7 = \{\mathbf{bbb}\} \cup \{x : |x| \geq 4\}.$$

These are pairwise distinct by suitable suffixes (for instance, C_4 vs C_5 by suffix $\mathbf{b}$: $\mathbf{bbb} = \mathbf{bbb} \in \Gamma$ but $\mathbf{abb} = \mathbf{abb} \not in \Gamma$).

(d) Minimal DFA has these 8 states, start C_0, final states

$$F = \{C_0, C_1, C_4, C_5, C_7\}.$$

Transition function:

$$\delta(C_0, \mathbf{a}) = C_1,\ \delta(C_0, \mathbf{b}) = C_2,$$
$$\delta(C_1, \mathbf{a}) = C_3,\ \delta(C_1, \mathbf{b}) = C_5,$$
$$\delta(C_2, \mathbf{a}) = C_5,\ \delta(C_2, \mathbf{b}) = C_4,$$
$$\delta(C_3, \mathbf{a}) = C_6,\ \delta(C_3, \mathbf{b}) = C_6,$$
$$\delta(C_4, \mathbf{a}) = C_6,\ \delta(C_4, \mathbf{b}) = C_7,$$
$$\delta(C_5, \mathbf{a}) = C_6,\ \delta(C_5, \mathbf{b}) = C_6,$$
$$\delta(C_6, \mathbf{a}) = C_7,\ \delta(C_6, \mathbf{b}) = C_7,\ \delta(C_7, \mathbf{a}) = C_7,\ \delta(C_7, \mathbf{b}) = C_7.$$

2.41. **Prove** R^M **is a right congruence.**

R^M: $xR^M y \Leftrightarrow \overline{\delta}(s_0, x) = \overline{\delta}(s_0, y)$.

Since this relation is defined by equality of destination state, it is immediately an equivalence relation: reflexive, symmetric, and transitive all follow from equality.

To check the right-congruence property, assume $xR^M y$, so $\overline{\delta}(s_0, x) = \overline{\delta}(s_0, y) = s$. Then for any $z \in \Sigma^*$,

$$\overline{\delta}(s_0, xz) = \overline{\delta}(s, z) = \overline{\delta}(s_0, yz),$$

so $xzR^M yz$. Therefore R^M is a right congruence. $\square$

2.42. **Is Pascal (set of all valid Pascal programs) FAD?**

No. We use a Chapter 2 distinct-futures argument: matching nested `begin/end` blocks requires arbitrarily many different continuations.

For each $n \geq 1$, let x_n be the ASCII string

```
program p; begin begin ...  begin
```

with exactly n occurrences of the keyword `begin`, and let z_n be the ASCII string

```
end end ...  end.
```

with exactly n occurrences of the keyword `end`.

Then $x_n z_n$ is a valid Pascal program: after the header `program p;`, the body is a nested compound statement with n occurrences of `begin` followed by the matching n occurrences of `end`, and then the final period.

If $m \neq n$, then $x_m z_n$ is not a valid Pascal program. If $m < n$, there are too many `end`s; if $m > n$, there are unmatched `begin`s left over before the final period.

Therefore x_n and x_m have different futures with respect to the language Pascal, since

$$x_n z_n \in \text{Pascal} \qquad \text{but} \qquad x_m z_n \not{i}n\text{Pascal} \;\; (m \neq n).$$

So the strings $x_1, x_2, x_3, \ldots$ lie in distinct R_{Pascal}-classes. Hence R_{Pascal} has infinitely many equivalence classes, and Pascal is not FAD.

2.43. **Is Short Pascal (programs with $< 10^6$ characters) FAD?**

Yes. Short Pascal is a finite language (finitely many programs of length $< 10^6$ over the ASCII alphabet). Every finite language is FAD. The DFA would have a huge but finite number of states (roughly exponential in 10^6, but finite).

2.44. $L = \{\mathbf{a}^n\mathbf{b}^k\mathbf{c}^j \mid n < 3$ **or** $(n \geq 3$ **and** $k = j)\}$**.**

(a) **Hypothesis false, conclusion true.**
First, the hypothesis "L is FAD" is false. For each $r \in \mathbb{N}$, let

$$x_r = \mathbf{a}^3\mathbf{b}^r.$$

If $r \neq s$, append the suffix $z = \mathbf{c}^r$. Then

$$x_r z = \mathbf{a}^3\mathbf{b}^r\mathbf{c}^r \in L,$$

but

$$x_s z = \mathbf{a}^3\mathbf{b}^s\mathbf{c}^r \not{i}nL$$

because the **b**- and **c**-counts are unequal. So the strings $x_0, x_1, x_2, \ldots$ lie in distinct R_L-classes. Hence R_L has infinite rank, and by Theorem 2.1 the language L is not FAD.
Now show the pumping *conclusion* of Theorem 2.3 does hold for L. Take pumping length $N = 3$. Let

$$x = \mathbf{a}^n\mathbf{b}^k\mathbf{c}^j \in L, \quad |x| \geq 3.$$

We choose a decomposition $x = uvw$ with $|uv| \le 3$, $|v| \ge 1$, and show $uv^m w \in L$ for all $m \ge 0$.
Case 1: $n \ge 3$ (so $k = j$). Choose

$$u = \lambda,\ v = \mathbf{a},\ w = \mathbf{a}^{n-1}\mathbf{b}^k\mathbf{c}^k.$$

Then $uv^m w = \mathbf{a}^{n-1+m}\mathbf{b}^k\mathbf{c}^k$. If $n-1+m < 3$, it is in L by the $n < 3$ clause. If $n-1+m \ge 3$, it is in L because $k = j$ still holds.

Case 2: $n < 3$. Since $|x| \ge 3$, there is at least one non-**a** symbol, and its position is at most 3. Let v be that first non-**a** symbol (either **b** or **c**), with u the preceding prefix. Then $|uv| \le 3$, $|v| = 1$. Pumping changes only the number of **b**'s or **c**'s, not the number of **a**'s, so $n < 3$ remains true for all m. Therefore $uv^m w \in L$ for all $m \ge 0$.

So the pumping conclusion holds even though L is not FAD.

(b) The contrapositive of Theorem 2.3: "if pumping fails then L is not FAD." This is trivially true (it's logically equivalent to Theorem 2.3). But here pumping doesn't fail, so the contrapositive doesn't give us information. The converse ("if pumping holds then L is FAD") is *false*, as this example shows.

2.45. **Show F_Q in Definition 2.5 is well defined.**

$F_Q = \{[x]_Q \mid x \in L\}$.

First, this is defined everywhere it needs to be: if $x \in L$, then certainly $x \in \Sigma^*$, so its equivalence class $[x]_Q$ exists and is an element of

$$S_Q = \{[y]_Q \mid y \in \Sigma^*\}.$$

Second, it is not multiply defined. Suppose $[x]_Q = [y]_Q$ and $x \in L$. Then x and y lie in the same Q-class. Since L is assumed to be a union of Q-classes, the whole class $[x]_Q = [y]_Q$ is either contained in L or disjoint from L. Because $x \in L$, that class must be contained in L, so $y \in L$ as well.

Thus the choice of representative does not matter, and F_Q is a well-defined subset of S_Q. □

2.46. **Show δ_Q in Definition 2.5 is well defined.**

$\delta_Q([x]_Q, a) = [xa]_Q$.

First, it is defined for every state and input letter: if $[x]_Q \in S_Q$ and $a \in \Sigma$, then $xa \in \Sigma^*$, so $[xa]_Q \in S_Q$.

Second, it is independent of the chosen representative of the class. If $[x]_Q = [y]_Q$, then xQy. Since Q is a right congruence, $xaQya$, and therefore

$$[xa]_Q = [ya]_Q.$$

So δ_Q is well defined. □

2.47. **Prove $\overline{\delta}_Q([x]_Q, y) = [xy]_Q$ by induction.**

Base: $\overline{\delta}_Q([x]_Q, \lambda) = [x]_Q = [x\lambda]_Q$. ✓

Step: $y = wa$. $\overline{\delta}_Q([x]_Q, wa) = \delta_Q(\overline{\delta}_Q([x]_Q, w), a) = \delta_Q([xw]_Q, a) = [xwa]_Q = [xy]_Q$. □

2.48. **Prove $L(A_Q) = L$.**

$x \in L(A_Q) \Leftrightarrow \overline{\delta}_Q([\lambda]_Q, x) \in F_Q \Leftrightarrow [x]_Q \in F_Q \Leftrightarrow x \in L$ (by definition of F_Q). □

2.49. **Prove or disprove $Q = R^{A_Q}$.**

R^{A_Q}: $xR^{A_Q}y \Leftrightarrow \overline{\delta}_Q([\lambda], x) = \overline{\delta}_Q([\lambda], y) \Leftrightarrow [x]_Q = [y]_Q \Leftrightarrow xQy$. So $R^{A_Q} = Q$. □

2.50. **Prove $R_L = R^{A_{R_L}}$.**

By Exercise 2.43 applied with $Q = R_L$: $R^{A_{R_L}} = R_L$. □

2.51. **Show the converse of Theorem 2.3 is false.**

The converse would say: if a language satisfies the conclusion of Theorem 2.3, then it must be FAD. The immediately preceding exercise already provides a counterexample:

$$L = \{\mathbf{a}^n\mathbf{b}^k\mathbf{c}^j \mid n < 3 \text{ or } (n \ge 3 \text{ and } k = j)\}.$$

Part (a) showed that the pumping conclusion of Theorem 2.3 does hold for this language, while the hypothesis fails because L is not FAD.

So a language can satisfy the pumping conclusion without being FAD. Therefore the converse of Theorem 2.3 is false. □

2.52. **Show $L = \{x : |x|_{\mathbf{a}} = 2|x|_{\mathbf{b}}\}$ is not FAD.**

For any n: strings $\mathbf{a}^0, \mathbf{a}^2, \mathbf{a}^4, \dots, \mathbf{a}^{2n}$ are in distinct R_L-classes. Indeed: $\mathbf{a}^{2i}R_L\mathbf{a}^{2j}$ requires for all z: $\mathbf{a}^{2i}z \in L \Leftrightarrow \mathbf{a}^{2j}z \in L$. Take $z = \mathbf{b}^i$: $\mathbf{a}^{2i}\mathbf{b}^i \in L$ (yes). $\mathbf{a}^{2j}\mathbf{b}^i \in L$ iff $2j = 2i$, i.e., $j = i$. So distinct $i \ne j$ give distinct classes. □

2.53. K **palindromes: $[\mathbf{110}]_{R_K}$ and description of R_K.**

Let $K = \{w \mid w = w^r\}$.

To describe R_K, suppose xR_Ky. Take suffixes

$$z_0 = \mathbf{0}x^r, \qquad z_1 = \mathbf{1}x^r.$$

Both $xz_0 = x\mathbf{0}x^r$ and $xz_1 = x\mathbf{1}x^r$ are palindromes, so by xR_Ky, both yz_0 and yz_1 must be palindromes.

Now:

- If a palindrome ends with $\mathbf{0}x^r$, it begins with $x\mathbf{0}$.
- If a palindrome ends with $\mathbf{1}x^r$, it begins with $x\mathbf{1}$.

Hence if $|y| \ge |x| + 1$, the first $|x| + 1$ symbols of y would have to be both $x\mathbf{0}$ and $x\mathbf{1}$, impossible. So $|y| \le |x|$. By symmetry (since equivalence relations are symmetric), also $|x| \le |y|$. Therefore $|x| = |y|$. With equal lengths, from yz_0 palindrome we get that the first $|x|$ symbols of yz_0 equal x; those first $|x|$ symbols are exactly y. So $y = x$.

Thus $xR_Ky \Rightarrow x = y$, and the converse is trivial. Therefore R_K is the identity relation on $\{\mathbf{0}, \mathbf{1}\}^*$.

So

$$[\mathbf{110}]_{R_K} = \{\mathbf{110}\}.$$

2.54. **Prove or disprove:** $R_{L_1} \cap R_{L_2} = R_{L_1 \cap L_2}$.

This is false in general.

However, $R_{L_1} \cap R_{L_2}$ always refines $R_{L_1 \cap L_2}$. Indeed, if $x R_{L_1} y$ and $x R_{L_2} y$, then for every z,

$$xz \in L_1 \cap L_2 \Leftrightarrow (xz \in L_1 \text{ and } xz \in L_2) \Leftrightarrow (yz \in L_1 \text{ and } yz \in L_2) \Leftrightarrow yz \in L_1 \cap L_2,$$

so $x R_{L_1 \cap L_2} y$.

Sometimes equality does happen. For example, over $\{\mathbf{a}, \mathbf{b}\}$ let

$$L_1 = \{x : |x| \text{ even}\}, \qquad L_2 = \{x : |x|_{\mathbf{a}} \text{ even}\}.$$

Then both $R_{L_1} \cap R_{L_2}$ and $R_{L_1 \cap L_2}$ have the same four classes, determined by the parity of $|x|$ and the parity of $|x|_{\mathbf{a}}$.

But in general the equality can fail. Let $L_1 = \{\lambda\}$ and $L_2 = \{\mathbf{a}\}$ over $\{\mathbf{a}\}^*$. Then R_{L_1} has 2 classes, R_{L_2} has 3 classes, and so $R_{L_1} \cap R_{L_2}$ has 3 classes. On the other hand,

$$L_1 \cap L_2 = \emptyset,$$

so $R_{L_1 \cap L_2} = R_\emptyset$ has just one class, namely all of Σ^*.

Therefore $R_{L_1} \cap R_{L_2} = R_{L_1 \cap L_2}$ is false in general. $\square$

2.55. **Prove: each R-class has a representative of length $< n$.**

R is a right congruence of rank n. Consider the DFA A_R with n states. By the pigeonhole principle, the BFS tree of a DFA with n states reaches every state by a path of length at most $n-1$: any path of length $\geq n$ must revisit a state. Hence every reachable state has a witness string of length $\leq n-1 < n$, so every R-class has a representative of length $< n$. $\square$

2.56. **Find all L with $R_L = R^N$ for R^N from Example 2.6.**

Let the four R^N-classes from Example 2.6 be

$$A_0 = [\lambda], \quad A_1 = [\mathbf{1}], \quad A_2 = [\mathbf{11}], \quad A_3 = [\mathbf{000}].$$

Not every union of these classes preserves $R_L = R^N$; many choices collapse states. The unions that keep all four classes pairwise Nerode-distinct are exactly:

$$L = A_0 \cup A_1, \quad L = A_1 \cup A_2, \quad L = A_0 \cup A_3, \quad L = A_2 \cup A_3.$$

These are precisely the languages with $R_L = R^N$.

2.57. **Right congruences for languages of Exercise 1.6.**

In Exercise 1.6(a), let

$$p(x) = \big(|x|_{\mathbf{a}} \bmod 2,\ |x|_{\mathbf{b}} \bmod 2\big) \in \{0,1\}^2.$$

Appending a suffix adds parities mod 2, so each R_L is determined by which parity information must be remembered.

(a) $L_1 = \{x : (|x|_{\mathbf{a}} \text{ odd}) \wedge (|x|_{\mathbf{b}} \text{ even})\}$.

$$xR_{L_1}y \iff p(x) = p(y).$$

So there are 4 classes (one per parity pair).

(b) $L_2 = \{x : (|x|_{\mathbf{a}} \text{ even}) \vee (|x|_{\mathbf{b}} \text{ odd})\}$. Again all 4 parity pairs are distinguishable, so

$$xR_{L_2}y \iff p(x) = p(y).$$

(c) $L_3 = \{x : (|x|_{\mathbf{a}} \text{ even})\overline{\vee}(|x|_{\mathbf{b}} \text{ even})\}$. This depends only on whether the two parities are equal or different:

$$xR_{L_3}y \iff \big(|x|_{\mathbf{a}} + |x|_{\mathbf{b}} \bmod 2\big) = \big(|y|_{\mathbf{a}} + |y|_{\mathbf{b}} \bmod 2\big).$$

So there are 2 classes: "same parity" and "different parity".

(d) $L_4 = \{x : |x|_{\mathbf{a}} \text{ even}\}$. Only parity of **a** matters:

$$xR_{L_4}y \iff |x|_{\mathbf{a}} \equiv |y|_{\mathbf{a}} \pmod 2.$$

So there are 2 classes.

2.58. **$L \subseteq \{\mathbf{a}\}^*$, $\lambda R_L \mathbf{a}$.**

$\lambda R_L \mathbf{a}$ means: for all $z \in \{\mathbf{a}\}^*$, $z \in L \Leftrightarrow \mathbf{a}z \in L$. This means L is closed under left-prepending **a** and left-removing **a** (when applicable). Combined, $\mathbf{a}^n \in L \Leftrightarrow \mathbf{a}^{n+1} \in L$ for all n. So either $L = \emptyset$ or $L = \{\mathbf{a}\}^*$.

2.59. **$\mathbf{a}R_L\mathbf{aa}$ over $\{\mathbf{a}\}^*$.**

$\mathbf{a}R_L\mathbf{aa}$: $(\forall z)(\mathbf{a}z \in L \Leftrightarrow \mathbf{aa}z \in L)$. So $\mathbf{a}^{n+1} \in L \Leftrightarrow \mathbf{a}^{n+2} \in L$. This means membership in L is the same at all lengths ≥ 1 except possibly at length 0 (where λ may or may not be in L). Also $\mathbf{a}R_L\mathbf{aa}$ means $[\mathbf{a}] = [\mathbf{aa}]$, so $L \cap \{\mathbf{a}^n : n \geq 1\}$ is either $\emptyset$ or $\{\mathbf{a}^n : n \geq 1\}$. With the possible inclusion of λ: 4 possibilities: $\emptyset$, $\{\lambda\}$, $\{\mathbf{a}^n : n \geq 1\}$, $\{\mathbf{a}\}^*$.

2.60. **$\lambda R_L\mathbf{aa}$ over $\{\mathbf{a}\}^*$.**

$(\forall z)(z \in L \Leftrightarrow \mathbf{aa}z \in L)$. So $\mathbf{a}^n \in L \Leftrightarrow \mathbf{a}^{n+2} \in L$: membership depends on parity of length. Possibilities: $\emptyset$, $\{\mathbf{a}^{2k} : k \geq 0\}$ (even lengths), $\{\mathbf{a}^{2k+1} : k \geq 0\}$ (odd lengths), $\{\mathbf{a}\}^*$.

2.61. **Examples with $R^M = R_{L(M)}$ or $R^M \neq R_{L(M)}$.**

(a) $R^M = R_{L(M)}$: any minimal DFA. E.g., the 2-state DFA for $\{\mathbf{a}^{2k} : k \geq 0\}$.

(b) $R^M \neq R_{L(M)}$: any non-minimal DFA. E.g., a 3-state DFA for the same language, where two states are equivalent. R^M has 3 classes (one per state) but $R_{L(M)}$ has 2.

(c) For every DFA M, R^M refines $R_{L(M)}$: if $\overline{\delta}(s_0, x) = \overline{\delta}(s_0, y) = s$, then for any z: $xz \in L(M) \Leftrightarrow \overline{\delta}(s_0, xz) \in F \Leftrightarrow \overline{\delta}(s, z) \in F \Leftrightarrow \overline{\delta}(s_0, yz) \in F \Leftrightarrow yz \in L(M)$.

2.62. **Find R^M and $R_{L(M)}$ for machine of Example 1.5.**

In Example 1.5 (vending machine), states track deposited amount in

$$T = \{0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50\}$$

(capped at 50), and final states are amounts ≥ 30.

R^M**.**

$$xR^M y \Longleftrightarrow \overline{\delta}(s_0, x) = \overline{\delta}(s_0, y).$$

So R^M has one class for each amount in T (11 classes).

$R_{L(M)}$**.** All classes for amounts < 30 remain distinct, while all amounts ≥ 30 merge into one Nerode class.

Why amounts < 30 are distinct: if $t_1 < t_2 < 30$, append coins totaling $30 - t_2$ (possible with $\mathbf{n}, \mathbf{d}, \mathbf{q}$); then from t_2 the result is accepted (reaches 30), while from t_1 it is still < 30 and rejected.

Why amounts ≥ 30 are equivalent: once at least 30 cents has been deposited, any further input keeps the machine in accepting territory.

Hence $R_{L(M)}$ has 7 classes:

$$[0], [5], [10], [15], [20], [25], \ [\geq 30].$$

So R^M is strictly finer than $R_{L(M)}$.

2.63. **Is $A_{R_{L(A)}}$ always equivalent to A?**

Yes. $A_{R_{L(A)}}$ is the minimal DFA for $L(A)$, hence $L(A_{R_{L(A)}}) = L(A)$. So they are always equivalent (accept the same language). They need not be isomorphic (if A is not minimal).

2.64. $L_4 = \{y : |y|_{\mathbf{0}} = |y|_{\mathbf{1}}\}$**: does the example u, v, w show L_4 is FAD?**

No. The pumping lemma says: for each sufficiently long word in a regular language, there exists *some* decomposition that pumps. Finding one convenient decomposition for one word does not prove regularity. To prove non-regularity, we choose one long word and show that *every* allowed decomposition fails. For L_4, a standard choice is $y = \mathbf{0}^n\mathbf{1}^n$: any v with $|uv| \leq n$ lies in the $\mathbf{0}$-block, and pumping changes the $\mathbf{0}/\mathbf{1}$ balance, leaving L_4.

2.65. **Find R_{L_4}.**

$L_4 = \{y : |y|_{\mathbf{0}} = |y|_{\mathbf{1}}\}$. R_{L_4} is determined by: $xR_{L_4}y \Leftrightarrow |x|_{\mathbf{0}} - |x|_{\mathbf{1}} = |y|_{\mathbf{0}} - |y|_{\mathbf{1}}$ (the "balance" of 0s over 1s must be equal). This gives infinitely many classes (one per integer balance $\in \mathbb{Z}$), confirming L_4 is not FAD.

2.66. **Show postfix expressions over $\{\mathbf{A}, \mathbf{B}, +, -\}$ are not FAD.**

For any n: $\mathbf{A}^n$ (n copies of A) is a prefix of postfix expressions. The strings $\mathbf{A}^i$ for $i = 0, 1, \ldots, n$ are in distinct R_L-classes: $\mathbf{A}^i$ needs i more operands consumed (by operators) to complete a valid postfix expression. Specifically: $\mathbf{A}^i z \in L$ iff z is a sequence of $i - 1$ binary operators (plus the right structure). So the future of $\mathbf{A}^i$ differs from $\mathbf{A}^j$ for $i \neq j$. Infinite rank. Not FAD. □

2.67. **Show parenthesized infix expressions are not FAD.**

Similar: $(^n$ is a prefix; each n gives a distinct future (requiring n closing parens eventually). Infinite rank. Not FAD. □

2.68. R_L **vs.** $R_{\sim L}$**.**

$R_L = R_{\sim L}$. Proof: $xR_L y \Leftrightarrow (\forall z)(xz \in L \Leftrightarrow yz \in L) \Leftrightarrow (\forall z)(xz \notin L \Leftrightarrow yz \notin L) \Leftrightarrow (\forall z)(xz \in \sim L \Leftrightarrow yz \in \sim L) \Leftrightarrow xR_{\sim L}y$. □

2.69. **Show no L has $R_L = Q$ where Q has classes $\{\lambda\}, \{\mathbf{a}\}, \{\mathbf{b}\}, \{x : |x| \geq 2\}$.**

Q is a right congruence: every class maps consistently under each letter (e.g. $\{\mathbf{a}\} \cdot \sigma$ and $\{\mathbf{b}\} \cdot \sigma$ both land in $\{|x| \geq 2\}$ for any σ). Nevertheless, no language L satisfies $R_L = Q$.

Proof. If $R_L = Q$, then $\mathbf{a}$ and $\mathbf{b}$ must lie in different R_L-classes. For $R_L = Q$ to hold, any candidate L must be a union of Q-classes. Consider $L = \{x : |x| \geq 2\}$. Then $\mathbf{a} R_L \mathbf{b}$: for all z, $\mathbf{a}z \in L \Leftrightarrow |\mathbf{a}z| \geq 2 \Leftrightarrow |z| \geq 1$, and $\mathbf{b}z \in L \Leftrightarrow |z| \geq 1$; the futures are identical. So R_L merges $\{\mathbf{a}\}$ and $\{\mathbf{b}\}$ into one class — contradicting Q. A case analysis over all unions of Q-classes shows the same merging occurs in every case. Hence $R_L \neq Q$ for any L. □

2.70. **Q with classes $\{\lambda, \mathbf{a}, \mathbf{aa}\}$ and $\{\mathbf{a}^3, \mathbf{a}^4, \ldots\}$ is not a right congruence.**

(a) $\mathbf{aa} Q \lambda$ (same class). Append $\mathbf{a}$: $\mathbf{aaa} \in$ second class, but $\lambda \cdot \mathbf{a} = \mathbf{a} \in$ first class. So $\mathbf{aaa} \not{Q} \mathbf{a}$, violating right congruence.

(b) Building A_Q: states are the two Q-classes. $\delta([\lambda]_Q, \mathbf{a}) = [?]$. For λ: $\lambda \cdot \mathbf{a} = \mathbf{a} \in [\lambda]_Q$. For $\mathbf{a}$: $\mathbf{a} \cdot \mathbf{a} = \mathbf{aa} \in [\lambda]_Q$. For $\mathbf{aa}$: $\mathbf{aa} \cdot \mathbf{a} = \mathbf{aaa} \in [\mathbf{a}^3]_Q$. So from the same class $[\lambda]_Q$, different elements lead to different successor classes. δ is not well-defined. This is the failure.

(c) Part (a) says Q is not a right congruence. Part (b) shows δ fails to be a function, which is precisely because right congruence is the condition needed for δ to be well-defined.

2.71. **Show $\{\mathbf{a}^i \mathbf{b}^j \mathbf{c}^k \mid i, j, k \in \mathbb{N}, i + j = k\}$ is not FAD.**

For any n: consider $\mathbf{a}^m$ for $m = 0, 1, \ldots, n$. $\mathbf{a}^m R_L \mathbf{a}^{m'}$: need $(\forall z) \mathbf{a}^m z \in L \Leftrightarrow \mathbf{a}^{m'} z \in L$. Take $z = \mathbf{b}^0 \mathbf{c}^m = \mathbf{c}^m$: $\mathbf{a}^m \mathbf{c}^m \in L \Leftrightarrow m + 0 = m$ (yes). $\mathbf{a}^{m'} \mathbf{c}^m \in L \Leftrightarrow m' + 0 = m \Leftrightarrow m' = m$. So for $m \neq m'$, $\mathbf{a}^m \not{R}_L \mathbf{a}^{m'}$. Infinite rank. Not FAD. □

2.72. **Show Q with classes $[\lambda] = \{\lambda\}$, $[\mathbf{a}] = \{\mathbf{a}\}$, $[\mathbf{aa}] = \{\mathbf{a}\}^{\geq 2}$ is a right congruence on $\{\mathbf{a}\}^*$.**

Case $x, y \in [\lambda]$: $x = y = \lambda$, $xa = ya = \mathbf{a} \in [\mathbf{a}]$.

Case $x, y \in [\mathbf{a}]$: $x = y = \mathbf{a}$, $xa = ya = \mathbf{aa} \in [\mathbf{aa}]$.

Case $x, y \in [\mathbf{aa}]$: $|x|, |y| \geq 2$, so $|xa|, |ya| \geq 3 \geq 2$, thus $xa, ya \in [\mathbf{aa}]$.

In each case $xaQya$. □

2.73. **R_2 on $\{\mathbf{a}, \mathbf{b}, \mathbf{c}\}^*$: same last letter.**

(a) Equivalence: Reflexive (x ends with same letter as x). Symmetric (same). Transitive (same). But for λ: λ has no last letter. By convention, λ is in its own class (or we restrict to Σ^+). Assuming λ is its own class: $\lambda R_2 \lambda$, and for $x, y \in \Sigma^+$, $x R_2 y \Leftrightarrow \text{last}(x) = \text{last}(y)$.

(b) Right congruence: If $x R_2 y$ (same last letter, say c), then for any $z \in \Sigma^+$: $\text{last}(xz) = \text{last}(z) = \text{last}(yz)$, so $xz R_2 yz$. For $z = \lambda$: $x R_2 y$. □

2.74. **R_3 on $\{\mathbf{a}, \mathbf{b}, \mathbf{c}\}^*$: same first letter.**

(a) Equivalence: reflexive, symmetric, transitive by same argument (λ its own class).

(b) Right congruence: Suppose $x R_3 y$, i.e., x and y have the same first letter (or both are λ). For any $z \in \Sigma^*$: if $z = \lambda$ then $xz = x$ and $yz = y$, so $xz R_3 yz$ trivially. If $z \neq \lambda$ then $\text{first}(xz) = \text{first}(x) = \text{first}(y) = \text{first}(yz)$, so $xz R_3 yz$. Hence R_3 is a right congruence. □

2.75. **Which of $L_1, \ldots, L_5$ are FAD?**

(a) L_1 = last letter matches first letter: FAD. Use 5 states: q_ε (no input yet), and $(a,a), (a,b), (b,a), (b,b)$ where the first component is the first symbol seen and the second is the current last symbol. Final states are (a,a) and (b,b) (and include/exclude q_ε depending on the convention for λ).

(b) L_2 = odd-length words whose first letter equals the center letter: Not FAD. Suppose L_2 were FAD, and let p be a pumping length. Take

$$w = \mathbf{a}^{p+1}\mathbf{b}^p.$$

This word has odd length $2p+1$, its first letter is **a**, and its center letter is also **a**, so $w \in L_2$.
For any split $w = uvw'$ with $|uv| \le p$ and $|v| \ge 1$, the substring v lies entirely in the initial **a**-block. So $v = \mathbf{a}^k$ for some $k \ge 1$. Pumping down gives

$$uw' = \mathbf{a}^{p+1-k}\mathbf{b}^p.$$

If k is odd, then $|uw'| = 2p+1-k$ is even, so $uw' \not{i}nL_2$. If k is even, then $|uw'|$ is odd, but its center position is

$$\frac{|uw'|+1}{2} = p+1-\frac{k}{2},$$

which is strictly larger than the length $p+1-k$ of the remaining **a**-block. So the center letter lies in the **b**-block, while the first letter is still **a**. Hence again $uw' \not{i}nL_2$.
This contradicts Theorem 2.3. Therefore L_2 is not FAD.

(c) L_3 = last letter does not appear in any earlier position: **FAD**. The DFA tracks (i) the set of characters seen in positions $1, \ldots, n-1$ and (ii) the current last character. Since Σ is fixed and finite, there are at most $2^{|\Sigma|} \times |\Sigma|$ states (plus a start state for λ). Accept when the last character is not in the set of characters seen before it. Both components update in one step: reading a sets last-char = a and adds the previous last-char to the "seen" set. This is a finite machine.

(d) L_4 = even-length, two center letters match: Not FAD. Suppose L_4 were FAD, and let p be a pumping length. Consider

$$w = \mathbf{a}^{2p-1}\mathbf{b}(\mathbf{ba})^p.$$

This word has length $4p$, so it is even. Its two center letters are the two consecutive **b**'s, so $w \in L_4$.
For any split $w = uvw'$ with $|uv| \le p$ and $|v| \ge 1$, we have $v = \mathbf{a}^k$ for some $k \ge 1$, because the first p symbols are all **a**'s. Pumping down gives

$$uw' = \mathbf{a}^{2p-1-k}\mathbf{b}(\mathbf{ba})^p.$$

If k is odd, then $|uw'| = 4p-k$ is odd, so $uw' \not{i}nL_4$. If $k = 2r$ is even, then the center positions of uw' are

$$2p-r \quad \text{and} \quad 2p-r+1.$$

After the initial block $\mathbf{a}^{2p-1-2r}$, the remaining suffix is

$$\mathbf{b}(\mathbf{ba})^p = \mathbf{bbaba}\cdots.$$

Those two center positions become positions $r+1$ and $r+2$ within this suffix, and these consecutive symbols are always different. So the two center letters of uw' do not match, and $uw' \not{i}nL_4$.
This contradicts Theorem 2.3. Therefore L_4 is not FAD.

(e) L_5 = odd-length, center letter matches none of the others: Not FAD. Suppose L_5 were FAD, and let p be a pumping length. Take

$$w = \mathbf{a}^p\mathbf{b}\mathbf{a}^p.$$

This word has odd length, its center letter is **b**, and that **b** appears nowhere else, so $w \in L_5$. For any split $w = uvw'$ with $|uv| \le p$ and $|v| \ge 1$, again $v = \mathbf{a}^k$ for some $k \ge 1$. Pumping down gives

$$uw' = \mathbf{a}^{p-k}\mathbf{b}\mathbf{a}^p.$$

If k is odd, then $|uw'| = 2p+1-k$ is even, so $uw' \not{i}nL_5$. If $k = 2r$ is even, then the center position of uw' is

$$\frac{|uw'|+1}{2} = p+1-r.$$

But the unique **b** is at position $p-2r+1$, so the center lies strictly to the right of that **b**, inside the final **a**-block. Thus the center letter is **a**, which certainly matches other letters of the word. Hence $uw' \not{i}nL_5$.

This contradicts Theorem 2.3. Therefore L_5 is not FAD.

2.76. **Complete proof of case 1 in (2)⇒(3) of Nerode's theorem.**

(a) Case 1: $x = \lambda$. We need $\lambda R_L y$ (for some y with $\overline{\delta}(s_0, y) = \overline{\delta}(s_0, \lambda) = s_0$). Since $\overline{\delta}(s_0, \lambda) = s_0 = \overline{\delta}(s_0, y)$, for all z: $\overline{\delta}(s_0, \lambda z) = \overline{\delta}(s_0, z) = \overline{\delta}(s_0, yz)$ (using $\overline{\delta}(s_0, y) = s_0$ and the homomorphism property of $\overline{\delta}$). So $\lambda z \in L \Leftrightarrow yz \in L$, i.e., $\lambda R_L y$.

(b) Case 1 *can* be included under case 2 if we allow x to be a prefix of y with $|x| = 0$: the proof of case 2 (induction on $|x|$) can start at $|x| = 0$ (the λ case), so yes.

2.77. **Right congruence property can use ⇔ instead of ⇒.**

The original right congruence (RC): $xRy \Rightarrow xuRyu$.

Claim: $xRy \Leftrightarrow xuRyu$ (for all u).

The ⇒ direction is given. For ⇐: assume $xuRyu$ for all u; take $u = \lambda$: $x\lambda Ry\lambda$, i.e., xRy (since $x\lambda = x$ and $y\lambda = y$). □

2.78. **Show $\overline{\delta}(s, uv^i) = q$ by induction.**

Given $\overline{\delta}(s, u) = q$ and $\overline{\delta}(q, v) = q$.

Base ($i = 0$): $\overline{\delta}(s, uv^0) = \overline{\delta}(s, u) = q$. ✓

Step: $\overline{\delta}(s, uv^{i+1}) = \overline{\delta}(\overline{\delta}(s, uv^i), v) = \overline{\delta}(q, v) = q$ (by IH). □

2.79. L **= strings agreeing with $\mathbf{0}^1\mathbf{1}\mathbf{0}^2\mathbf{1}\mathbf{0}^3\mathbf{1}\cdots$ is not FAD.**

The strings in L are all prefixes of this infinite sequence. Consider prefix $\mathbf{0}^1\mathbf{1}\mathbf{0}^2\mathbf{1}\cdots\mathbf{0}^n\mathbf{1}$ (length $n(n+3)/2$) and the prefix $\mathbf{0}^1\mathbf{1}\mathbf{0}^2\mathbf{1}\cdots\mathbf{0}^n$ (length $n(n+3)/2 - n = n(n+1)/2$). The futures of these two strings are different (one expects a **1** next, the other expects $n+1$ zeros followed by **1**). So different-length prefixes that cut at different points in the pattern have different futures. An unbounded number of distinct futures exist. Not FAD. □

2.80. **Show the "simple" BNF language is not FAD.**

⟨simple⟩ ::= **a** | (⟨simple⟩). The language is $L = \{(^n\mathbf{a})^n \mid n \geq 0\}$. By Myhill-Nerode: the strings $(^0, (^1, \ldots, (^n$ are pairwise in distinct R_L-classes. The distinguishing suffix for $(^j$ vs. $(^k$ (with $j \neq k$) is $z = \mathbf{a})^j$: $(^j\mathbf{a})^j \in L$ but $(^k\mathbf{a})^j \not\in L$ (since $k \neq j$). Hence R_L has infinite rank and L is not FAD. □

2.81. L_2 **(binary powers of 2) is FAD;** L_{10} **(decimal powers of 2) is not.**

(a) $L_2 = \{\mathbf{1}, \mathbf{10}, \mathbf{100}, \mathbf{1000}, \ldots\}$: these are exactly the strings whose first symbol is **1** and whose remaining symbols, if any, are all **0**. This is FAD: use a 2-state DFA with $q_0 \xrightarrow{\mathbf{1}} q_1$, $q_1 \xrightarrow{\mathbf{0}} q_1$, $F = \{q_1\}$, and a dead state for all other transitions.

(b) $L_{10} = \{1, 2, 4, 8, 16, 32, \ldots\}$ (decimal powers of 2) is not FAD. *Proof (by pumping lemma and modular arithmetic).* Assume L_{10} is FAD with pumping length p. Choose n so large that if w is the decimal representation of 2^n, then

$$|w| = d > p + 2$$

and

$$4 \cdot 5^{d-p-1} > (p+1)\log_2 10.$$

Write $w = xyz$ with $|xy| \leq p$, $|y| = t \geq 1$, and let $|z| = r$. Then $r = d - |xy| \geq d - p$.
By pumping, $xy^2z \in L_{10}$, so for some $m > n$,

$$\mathrm{val}(xy^2z) = 2^m, \qquad \mathrm{val}(xyz) = 2^n.$$

Because pumping duplicates a block before the final z, both numbers end with the same r digits z. Hence

$$2^m \equiv 2^n \pmod{10^r},$$

so

(*) $$2^{m-n} \equiv 1 \pmod{5^r}.$$

Lemma. If $2^q \equiv 1 \pmod{5^r}$, then $4 \cdot 5^{r-1} \mid q$.
Why: the order of 2 modulo 5^r is $4 \cdot 5^{r-1}$. Base $r = 1$: order mod 5 is 4. Induction step: if the order mod 5^r is $4 \cdot 5^{r-1}$, then

$$2^{4\cdot 5^{r-1}} = 1 + u5^r, \quad 5 \nmid u,$$

so

$$2^{4\cdot 5^r} = (1 + u5^r)^5 \equiv 1 + u5^{r+1} \not\equiv 1 \pmod{5^{r+2}},$$

which shows the order at the next level is multiplied by 5. Thus order(2 mod 5^r) $= 4 \cdot 5^{r-1}$ for all $r \geq 1$.
Applying the lemma to $(*)$ gives

(1) $$m - n \geq 4 \cdot 5^{r-1} \geq 4 \cdot 5^{d-p-1}.$$

Now get an upper bound for $m - n$ from decimal length. The pumped string xy^2z has length $d + t \leq d + p$, so

$$2^m = \mathrm{val}(xy^2z) < 10^{d+t} \leq 10^{d+p}.$$

Also $2^n = \mathrm{val}(w) \geq 10^{d-1}$. Therefore

$$2^{m-n} < 10^{p+1},$$

so

$$m - n < (p+1)\log_2 10. \tag{2}$$

(1) and (2) contradict our choice of n. Hence L_{10} is not FAD. □

Chapter 3

Minimization — Solutions

3.1. **Prove $\mu(\overline{\delta}(s,x)) = \overline{\delta}_{R_L}(\mu(s),x)$ by induction for the mapping μ in Theorem 3.1.**

$\mu(s) = [s]_{R^M}$ (the R^M-class of state s, identified with an R_L-class via the homomorphism). Let $P(n)$: $(\forall s \in S)(\forall x \in \Sigma^n)(\mu(\overline{\delta}(s,x)) = \overline{\delta}_{R_L}(\mu(s),x))$.

Base ($n = 0$): $\mu(\overline{\delta}(s,\lambda)) = \mu(s) = \overline{\delta}_{R_L}(\mu(s),\lambda)$. ✓

Step: $x = ya$, $|y| = n$. Using the homomorphism property $\mu(\delta(t,a)) = \delta_{R_L}(\mu(t),a)$:

$$\mu(\overline{\delta}(s,ya)) = \mu(\delta(\overline{\delta}(s,y),a)) = \delta_{R_L}(\mu(\overline{\delta}(s,y)),a) = \delta_{R_L}(\overline{\delta}_{R_L}(\mu(s),y),a) = \overline{\delta}_{R_L}(\mu(s),ya). \quad \square$$

3.2. **Show $\overline{\delta}_{E_A}([s]_{E_A},x) = [\overline{\delta}(s,x)]_{E_A}$ by induction.**

Base ($x = \lambda$): $\overline{\delta}_{E_A}([s]_{E_A},\lambda) = [s]_{E_A} = [\overline{\delta}(s,\lambda)]_{E_A}$. ✓

Step: $x = ya$.

$$\overline{\delta}_{E_A}([s]_{E_A},ya) = \delta_{E_A}(\overline{\delta}_{E_A}([s]_{E_A},y),a) = \delta_{E_A}([\overline{\delta}(s,y)]_{E_A},a) = [\delta(\overline{\delta}(s,y),a)]_{E_A} = [\overline{\delta}(s,ya)]_{E_A}. \quad \square$$

3.3. **Prove $E_A = \bigcap_{j=0}^{\infty} E_{jA}$.**

Define E_{0A}: $sE_{0A}t \Leftrightarrow (s \in F \Leftrightarrow t \in F)$.

Define $E_{i+1,A}$: $sE_{i+1,A}t \Leftrightarrow sE_{iA}t \wedge (\forall a \in \Sigma)(\delta(s,a)E_{iA}\delta(t,a))$.

E_A (state equivalence): $sE_At \Leftrightarrow L(A^s) = L(A^t)$ where A^s is A with start state s.

($\subseteq$): We show $E_A \subseteq E_{jA}$ for every j by induction on j.

Base ($j = 0$): if sE_At, then $L(A^s) = L(A^t)$. In particular,

$$\lambda \in L(A^s) \Leftrightarrow \lambda \in L(A^t),$$

so

$$s \in F \Leftrightarrow t \in F.$$

Hence $sE_{0A}t$.

Step: assume $E_A \subseteq E_{jA}$ and let sE_At. Then $L(A^s) = L(A^t)$. For every $a \in \Sigma$,

$$L(A^{\delta(s,a)}) = \{x \in \Sigma^* : ax \in L(A^s)\} = \{x \in \Sigma^* : ax \in L(A^t)\} = L(A^{\delta(t,a)}),$$

so $\delta(s,a) E_A \delta(t,a)$. By the induction hypothesis,

$$sE_{jA}t \qquad \text{and} \qquad \delta(s,a)E_{jA}\delta(t,a) \quad (\forall a \in \Sigma).$$

Therefore $sE_{j+1,A}t$. Thus $E_A \subseteq E_{jA}$ for every j, and so

$$E_A \subseteq \bigcap_{j=0}^{\infty} E_{jA}.$$

($\supseteq$): Suppose $sE_{jA}t$ for every j. We claim that for every $n \geq 0$ and every string x with $|x| \leq n$,

$$\overline{\delta}(s,x) \in F \Leftrightarrow \overline{\delta}(t,x) \in F.$$

We prove this by induction on n.

Base ($n = 0$): the only string of length at most 0 is λ. Since $sE_{0A}t$, we have

$$s \in F \Leftrightarrow t \in F,$$

that is,

$$\overline{\delta}(s,\lambda) \in F \Leftrightarrow \overline{\delta}(t,\lambda) \in F.$$

Step: assume the claim holds for n. Let $|x| \leq n+1$. If $x = \lambda$, then $sE_{n+1,A}t$ implies $sE_{nA}t$, so the induction hypothesis already gives the result. If $x = ay$ with $a \in \Sigma$ and $|y| \leq n$, then $sE_{n+1,A}t$ implies

$$\delta(s,a)E_{nA}\delta(t,a).$$

Applying the induction hypothesis to the pair $\delta(s,a), \delta(t,a)$ and the string y, we get

$$\overline{\delta}(\delta(s,a),y) \in F \Leftrightarrow \overline{\delta}(\delta(t,a),y) \in F.$$

Since

$$\overline{\delta}(\delta(s,a),y) = \overline{\delta}(s,ay) \qquad \text{and} \qquad \overline{\delta}(\delta(t,a),y) = \overline{\delta}(t,ay),$$

the claim follows.

Now let $x \in \Sigma^*$ be arbitrary and take $n = |x|$. Then

$$\overline{\delta}(s,x) \in F \Leftrightarrow \overline{\delta}(t,x) \in F,$$

which means

$$x \in L(A^s) \Leftrightarrow x \in L(A^t).$$

Therefore $L(A^s) = L(A^t)$, i.e., sE_At. So

$$\bigcap_{j=0}^{\infty} E_{jA} \subseteq E_A.$$

Hence $E_A = \bigcap_{j=0}^{\infty} E_{jA}$. □

3.4. **Show δ_{E_A} in Definition 3.8 is well defined.**

$\delta_{E_A}([s]_{E_A}, a) = [\delta(s,a)]_{E_A}$. If $[s]_{E_A} = [t]_{E_A}$, i.e., sE_At, then $\delta(s,a)E_A\delta(t,a)$ (proved in Exercise 3.3, $\supseteq$ direction): $L(A^{\delta(s,a)}) = \{x : ax \in L(A^s)\} = \{x : ax \in L(A^t)\} = L(A^{\delta(t,a)})$. So $[\delta(s,a)]_{E_A} = [\delta(t,a)]_{E_A}$. □

3.5. **Show F_{E_A} is well defined.**

$F_{E_A} = \{[s]_{E_A} \mid s \in F\}$. If $[s]_{E_A} = [t]_{E_A}$ and $s \in F$: $sE_A t \Rightarrow L(A^s) = L(A^t) \Rightarrow \lambda \in L(A^s) \Leftrightarrow \lambda \in L(A^t) \Rightarrow s \in F \Leftrightarrow t \in F$. So $t \in F$. $\square$

3.6. **Show δ^c maps into S^c.**

S^c = connected states = $\{\overline{\delta}(s_0, x) \mid x \in \Sigma^*\}$. δ^c is δ restricted to S^c. For any $s \in S^c$ and $a \in \Sigma$: $s = \overline{\delta}(s_0, x)$ for some x, so $\delta(s, a) = \delta(\overline{\delta}(s_0, x), a) = \overline{\delta}(s_0, xa) \in S^c$. $\square$

3.7. **Prove Lemma 3.1.**

Suppose $sE_{m+1A}t$. By definition, this means

$$(\forall x \in \Sigma^* \ni |x| \le m+1)(\overline{\delta}(s,x) \in F \Leftrightarrow \overline{\delta}(t,x) \in F).$$

In particular, the same equivalence holds for every x with $|x| \le m$. Hence

$$(\forall x \in \Sigma^* \ni |x| \le m)(\overline{\delta}(s,x) \in F \Leftrightarrow \overline{\delta}(t,x) \in F),$$

which is exactly $sE_{mA}t$. Therefore $E_{m+1A} \subseteq E_{mA}$. $\square$

3.8. **Prove Lemma 3.3.**

By definition,

$$sE_{0A}t \Leftrightarrow (\forall x \in \Sigma^* \ni |x| \le 0)(\overline{\delta}(s,x) \in F \Leftrightarrow \overline{\delta}(t,x) \in F).$$

The only string of length 0 is λ, so

$$sE_{0A}t \Leftrightarrow \overline{\delta}(s,\lambda) \in F \Leftrightarrow \overline{\delta}(t,\lambda) \in F \Leftrightarrow s \in F \Leftrightarrow t \in F.$$

Therefore the equivalence classes of E_{0A} are exactly F and $S - F$, unless one of these sets is empty, in which case there is only the single class S. $\square$

3.9. **Prove Theorem 3.9.**

Assume $E_{mA} = E_{m+1A}$. We prove by induction on k that $E_{m+k,A} = E_{mA}$.

Base case ($k=0$): $E_{m+0,A} = E_{mA}$ trivially.

Inductive step: assume $E_{m+k,A} = E_{mA}$. Let $s, t \in S$. By Theorem 3.8,

$$\begin{aligned} sE_{m+k+1,A}t &\Leftrightarrow sE_{m+k,A}t \wedge (\forall a \in \Sigma)(\delta(s,a)E_{m+k,A}\delta(t,a)) \\ &\Leftrightarrow sE_{mA}t \wedge (\forall a \in \Sigma)(\delta(s,a)E_{mA}\delta(t,a)) \end{aligned}$$

by the induction hypothesis. Applying Theorem 3.8 again,

$$sE_{m+k+1,A}t \Leftrightarrow sE_{m+1,A}t.$$

Since $E_{m+1A} = E_{mA}$ by hypothesis, this gives

$$sE_{m+k+1,A}t \Leftrightarrow sE_{mA}t.$$

Therefore $E_{m+k+1,A} = E_{mA}$. By induction, $(\forall k \in \mathbb{N})(E_{m+k,A} = E_{mA})$. $\square$

3.10. **Prove $\mu(\overline{\delta}_A(s,x)) = \overline{\delta}_B(\mu(s),x)$ by induction for a homomorphism μ.**

Base: $\mu(\overline{\delta}_A(s,\lambda)) = \mu(s) = \overline{\delta}_B(\mu(s),\lambda)$. ✓

Step: $x = ya$. $\mu(\overline{\delta}_A(s,ya)) = \mu(\delta_A(\overline{\delta}_A(s,y),a)) = \delta_B(\mu(\overline{\delta}_A(s,y)),a)$ (homomorphism property) $=$ $\delta_B(\overline{\delta}_B(\mu(s),y),a)$ (IH) $= \overline{\delta}_B(\mu(s),ya)$. □

3.11. **Prove $L(A) = L(B)$ for a homomorphism $\mu : A \to B$.**

For any $x \in \Sigma^*$,

$$\begin{aligned}
x \in L(A) &\Leftrightarrow \overline{\delta}_A(s_{0A},x) \in F_A \\
&\Leftrightarrow \mu(\overline{\delta}_A(s_{0A},x)) \in F_B \qquad \text{(because } \mu(s) \in F_B \Leftrightarrow s \in F_A\text{)} \\
&\Leftrightarrow \overline{\delta}_B(\mu(s_{0A}),x) \in F_B \qquad \text{(by Exercise 3.10)} \\
&\Leftrightarrow \overline{\delta}_B(s_{0B},x) \in F_B \qquad \text{(since } \mu(s_{0A}) = s_{0B}\text{)} \\
&\Leftrightarrow x \in L(B).
\end{aligned}$$

Therefore $L(A) = L(B)$. □

3.12. **Examples with connectivity and $A/_{E_A}$.**

(a) A not connected, $A/_{E_A}$ not connected: take

$$A = \langle \{\boldsymbol{a}\}, \{s_0,s_1,s_2\}, s_0, \delta, \{s_0,s_1\}\rangle,$$

where

$$\delta(s_0,\boldsymbol{a}) = s_1, \qquad \delta(s_1,\boldsymbol{a}) = s_1, \qquad \delta(s_2,\boldsymbol{a}) = s_2.$$

The state s_2 is unreachable, so A is not connected. Also $s_0 E_A s_1$, because from either state every continuation is accepted, while s_2 is not equivalent to them because from s_2 every continuation is rejected. Thus $A/_{E_A}$ has exactly the two states $[s_0] = [s_1]$ and $[s_2]$, and $[s_2]$ is still unreachable. Hence $A/_{E_A}$ is not connected.

(b) A not connected, $A/_{E_A}$ connected: take

$$A = \langle \{\boldsymbol{a}\}, \{t_0,t_1,t_2\}, t_0, \delta, \{t_1,t_2\}\rangle,$$

where

$$\delta(t_0,\boldsymbol{a}) = t_1, \qquad \delta(t_1,\boldsymbol{a}) = t_1, \qquad \delta(t_2,\boldsymbol{a}) = t_2.$$

Again t_2 is unreachable, so A is not connected. But now $t_1 E_A t_2$, because from either state every continuation is accepted. Therefore $A/_{E_A}$ has just the two states $[t_0]$ and $[t_1] = [t_2]$, and both are reachable from the start class: $[t_0]$ is the start state and

$$\delta([t_0],\boldsymbol{a}) = [t_1].$$

Hence $A/_{E_A}$ is connected.

3.13. **Show there exists a homomorphism from A to $A/_{E_A}$.**

Define $\mu : S_A \to S_{E_A}$ by $\mu(s) = [s]_{E_A}$. Then:

- $\mu(s_{0A}) = [s_{0A}]_{E_A} = s_{0,E_A}$. ✓
- $\mu(\delta_A(s,a)) = [\delta_A(s,a)]_{E_A} = \delta_{E_A}([s]_{E_A}, a) = \delta_{E_A}(\mu(s), a)$. ✓
- $s \in F_A \Leftrightarrow [s]_{E_A} \in F_{E_A}$, i.e., $\mu(s) \in F_{E_A} \Leftrightarrow s \in F_A$. ✓

So μ is a homomorphism. □

3.14. **Show there exists a homomorphism from connected A to $A_{R_{L(A)}}$.**

(a) Define $\psi : S \to \{R_{L(A)}\text{-classes}\}$ by $\psi(s) = [x_s]_{R_{L(A)}}$ where x_s is any string with $\overline{\delta}(s_{0A}, x_s) = s$ (well-defined by part (b)).

More precisely: since A is connected, for each $s \in S$ there exists $x_s \in \Sigma^*$ with $\overline{\delta}(s_{0A}, x_s) = s$. Define $\psi(s) = [x_s]_{R_{L(A)}}$.

(b) Well-definedness: if $\overline{\delta}(s_0, x) = \overline{\delta}(s_0, y) = s$, then $xR^A y$ (same state), and R^A refines $R_{L(A)}$, so $xR_{L(A)}y$, giving $[x]_{R_{L(A)}} = [y]_{R_{L(A)}}$. □

(c) Homomorphism:

- $\psi(s_{0A}) = [\lambda]_{R_{L(A)}} = s_{0,R_{L(A)}}$. ✓
- $\psi(\delta(s,a)) = [x_s a]_{R_{L(A)}} = \delta_{R_{L(A)}}([x_s]_{R_{L(A)}}, a) = \delta_{R_{L(A)}}(\psi(s), a)$. ✓
- $s \in F \Leftrightarrow \lambda \in L(A^s) \Leftrightarrow x_s \in L(A) \Leftrightarrow [x_s]_{R_{L(A)}} \in F_{R_{L(A)}} \Leftrightarrow \psi(s) \in F_{R_{L(A)}}$. ✓

3.15. **Show there may not be a homomorphism $A \to A_{R_{L(A)}}$ if A is not connected.**

Take $\Sigma = \{\mathbf{0}\}$ and let

$$A = \langle \{\mathbf{0}\}, \{s_0, s_1\}, s_0, \delta, \{s_1\} \rangle$$

where $\delta(s_0, \mathbf{0}) = s_0$ and $\delta(s_1, \mathbf{0}) = s_1$. The state s_1 is unreachable, so A is not connected. Since the reachable part is just the nonfinal loop at s_0, we have $L(A) = \emptyset$. Therefore $A_{R_{L(A)}}$ is the one-state DFA for $\emptyset$, with a single nonfinal state q and $\delta(q, \mathbf{0}) = q$.

If there were a homomorphism $\psi : A \to A_{R_{L(A)}}$, then $\psi(s_0) = q$ because s_0 is the start state. But a homomorphism must also preserve finality:

$$s_1 \in F_A \Leftrightarrow \psi(s_1) \in F_{R_{L(A)}}.$$

This is impossible, because q is the only state of $A_{R_{L(A)}}$ and q is not final. Hence no such homomorphism exists.

3.16. **Show there may still be a homomorphism $A \to A_{R_{L(A)}}$ if A is not connected.**

Take $\Sigma = \{\mathbf{0}\}$ and let

$$A = \langle \{\mathbf{0}\}, \{s_0, s_1\}, s_0, \delta, \{s_0, s_1\} \rangle$$

where $\delta(s_0, \mathbf{0}) = s_0$ and $\delta(s_1, \mathbf{0}) = s_1$. The state s_1 is unreachable, so A is not connected, but $L(A) = \Sigma^*$ because every string keeps the start state s_0 in a final state.

Therefore $A_{R_{L(A)}}$ is the one-state DFA

$$\langle \{\mathbf{0}\}, \{q\}, q, \delta', \{q\} \rangle$$

with $\delta'(q, \mathbf{0}) = q$. Define $\psi : A \to A_{R_{L(A)}}$ by

$$\psi(s_0) = q, \qquad \psi(s_1) = q.$$

This is a homomorphism: the start state is preserved, finality is preserved because all three states involved are final, and

$$\psi(\delta(s_i,\mathbf{0})) = \psi(s_i) = q = \delta'(q,\mathbf{0}) = \delta'(\psi(s_i),\mathbf{0})$$

for $i = 0, 1$.

3.17. **Show there need not be a homomorphism from A_{R_L} to A_R.**

Let $\Sigma = \{\boldsymbol{a}\}$ and let Q be the right congruence with two classes

$$[\lambda]_Q = \{\lambda\}, \qquad [\boldsymbol{a}]_Q = \{x \in \Sigma^* : |x| \geq 1\}.$$

Let R refine Q by splitting the nonempty strings according to parity:

$$[\lambda]_R = \{\lambda\}, \qquad [\boldsymbol{a}]_R = \{x : |x| \text{ is odd}\}, \qquad [\boldsymbol{aa}]_R = \{x : |x| \text{ is even and } |x| \geq 2\}.$$

Take $L = \{\lambda\}$, which is a union of Q-classes. Then A_Q has two states,

$$q_0 = [\lambda]_Q \quad \text{(start and final)}, \qquad q_1 = [\boldsymbol{a}]_Q$$

with $\delta_Q(q_0, \boldsymbol{a}) = q_1$ and $\delta_Q(q_1, \boldsymbol{a}) = q_1$.

The DFA A_R has three states,

$$r_0 = [\lambda]_R \quad \text{(start and final)}, \qquad r_1 = [\boldsymbol{a}]_R, \qquad r_2 = [\boldsymbol{aa}]_R,$$

with transitions

$$\delta_R(r_0, \boldsymbol{a}) = r_1, \qquad \delta_R(r_1, \boldsymbol{a}) = r_2, \qquad \delta_R(r_2, \boldsymbol{a}) = r_1.$$

Assume there is a homomorphism $\psi : A_Q \to A_R$. Since start states must map to start states, $\psi(q_0) = r_0$. Then

$$\psi(q_1) = \psi(\delta_Q(q_0, \boldsymbol{a})) = \delta_R(\psi(q_0), \boldsymbol{a}) = \delta_R(r_0, \boldsymbol{a}) = r_1.$$

But now

$$\psi(q_1) = \psi(\delta_Q(q_1, \boldsymbol{a})) = \delta_R(\psi(q_1), \boldsymbol{a}) = \delta_R(r_1, \boldsymbol{a}) = r_2,$$

which says $r_1 = r_2$, a contradiction. Therefore there need not be a homomorphism from A_Q to A_R.

3.18. **Is $\cong$ appropriate to ask if it's a right congruence?**

$\cong$ is a relation on DFAs, not on Σ^*. Right congruences are relations on Σ^*. So asking whether $\cong$ (between DFAs) is a right congruence (on strings) is a category error – it makes no sense. One could ask whether the relation induced on strings by $\cong$ is a right congruence, but the question itself is ill-formed in the original sense.

3.19. **Is E_A a right congruence?**

E_A is a relation on states S, not on Σ^*. So strictly speaking, E_A is not a right congruence (which is a relation on Σ^*). However, R^A (induced on strings) is a right congruence, and E_A induces R^A via $xR^A y \Leftrightarrow \overline{\delta}(s_0, x) E_A \overline{\delta}(s_0, y)$.

3.20. **If A is reduced then A^c is reduced.**

A^c = connected part of A. States of A^c are $S^c \subseteq S$. If A is reduced, no two distinct states in S are equivalent; in particular, no two distinct states in $S^c \subseteq S$ are equivalent. So A^c is reduced. $\square$

3.21. **For a homomorphism $\mu : S_A \to S_B$, prove $\mu(s) E_B \mu(t) \Leftrightarrow s E_A t$.**

($\Rightarrow$) Suppose $\mu(s) E_B \mu(t)$, i.e., $L(B^{\mu(s)}) = L(B^{\mu(t)})$. By Exercise 3.10: $L(A^s) = L(B^{\mu(s)})$ and $L(A^t) = L(B^{\mu(t)})$ (since μ is a homomorphism from A^s to $B^{\mu(s)}$). So $L(A^s) = L(A^t)$, i.e., $s E_A t$.

($\Leftarrow$) Suppose $s E_A t$. Then $L(A^s) = L(A^t)$. By the homomorphism property: $L(B^{\mu(s)}) = L(A^s) = L(A^t) = L(B^{\mu(t)})$, so $\mu(s) E_B \mu(t)$. $\square$

3.22. **Define $\psi : M^c \to A_{R^M}$ and prove it is an isomorphism.**

(a) For each state $s \in S^c$, choose a string $x_s \in \Sigma^*$ with

$$\overline{\delta}(s_0, x_s) = s,$$

and define

$$\psi(s) = [x_s]_{R^M}.$$

(b) *Well-defined*: if x and y both satisfy

$$\overline{\delta}(s_0, x) = \overline{\delta}(s_0, y) = s,$$

then by definition of R^M we have $x R^M y$. Hence

$$[x]_{R^M} = [y]_{R^M},$$

so $\psi(s)$ does not depend on which string reaching s was chosen.

(c) *Homomorphism*: first,

$$\psi(s_0) = [\lambda]_{R^M},$$

because λ reaches s_0. Now let $s \in S^c$ and $a \in \Sigma$. Since

$$\overline{\delta}(s_0, x_s a) = \delta(\overline{\delta}(s_0, x_s), a) = \delta(s, a),$$

the string $x_s a$ reaches $\delta(s, a)$, so

$$\psi(\delta(s, a)) = [x_s a]_{R^M} = \delta_{R^M}([x_s]_{R^M}, a) = \delta_{R^M}(\psi(s), a).$$

Also,

$$s \in F \Leftrightarrow \overline{\delta}(s_0, x_s) \in F \Leftrightarrow x_s \in L(M) \Leftrightarrow [x_s]_{R^M} \in F_{R^M} \Leftrightarrow \psi(s) \in F_{R^M}.$$

Thus ψ preserves start state, transitions, and finality.

(d) *Bijection*: to show injectivity, suppose $\psi(s) = \psi(t)$. Then

$$[x_s]_{R^M} = [x_t]_{R^M},$$

so $x_s R^M x_t$. By definition of R^M,

$$\overline{\delta}(s_0, x_s) = \overline{\delta}(s_0, x_t),$$

hence $s = t$. For surjectivity, let $[x]_{R^M}$ be any state of A_{R^M}. Then

$$s = \overline{\delta}(s_0, x) \in S^c,$$

and by definition of ψ,

$$\psi(s) = [x]_{R^M}.$$

(e) $M^c \cong A_{R^M}$. $\square$

3.23. **For the machine A in Figure 3.10(a), find E_A, $L(A)$, $A_{R_{L(A)}}$, $R_{L(A)}$, and $A/_{E_A}$.**

(a)
$$E_{0A} = \{\{q_0, q_1, q_3, q_6\}, \{q_2, q_4, q_5\}\}.$$

Refining once gives
$$E_{1A} = \{\{q_0, q_1\}, \{q_3, q_6\}, \{q_2\}, \{q_4, q_5\}\}.$$

Refining again gives
$$E_{2A} = \{\{q_0\}, \{q_1\}, \{q_2\}, \{q_3, q_6\}, \{q_4, q_5\}\}.$$

This is already stable, so $E_A = E_{2A}$.

(b) The language is
$$L(A) = \{\lambda, 1\} \cup 00\Sigma^* \cup 11\Sigma^*.$$

Indeed, λ is accepted at q_0, the string 1 is accepted at q_1, and every string of length at least 2 is accepted exactly when its first two symbols are equal.

(c) $A_{R_{L(A)}}$ has five states, represented by the classes of λ, 1, 0, 00, and 01. Writing these as $[\lambda], [1], [0], [00], [01]$, the transitions are

	0	1
$[\lambda]$	$[0]$	$[1]$
$[1]$	$[01]$	$[00]$
$[0]$	$[00]$	$[01]$
$[00]$	$[00]$	$[00]$
$[01]$	$[01]$	$[01]$

with final states $[\lambda], [1], [00]$. Here $[11] = [00]$ and $[10] = [01]$.

(d) The right-congruence classes of $R_{L(A)}$ may be written as
$$[\lambda], \qquad [1], \qquad [0], \qquad [00] = [11], \qquad [01] = [10].$$

Equivalently, they are
$$\{\lambda\}, \quad \{1\}, \quad \{0\}, \quad \{x : |x| \geq 2 \text{ and the first two symbols are equal}\},$$

and
$$\{x : |x| \geq 2 \text{ and the first two symbols are different}\}.$$

(e) $A/_{E_A}$ has states
$$[q_0],\ [q_1],\ [q_2],\ [q_3] = [q_6],\ [q_4] = [q_5],$$

with $[q_0]$ initial and final states $[q_0], [q_1], [q_3] = [q_6]$. Its transition table is

	0	1
$[q_0]$	$[q_2]$	$[q_1]$
$[q_1]$	$[q_4] = [q_5]$	$[q_3] = [q_6]$
$[q_2]$	$[q_3] = [q_6]$	$[q_4] = [q_5]$
$[q_3] = [q_6]$	$[q_3] = [q_6]$	$[q_3] = [q_6]$
$[q_4] = [q_5]$	$[q_4] = [q_5]$	$[q_4] = [q_5]$

so $A/_{E_A} \cong A_{R_{L(A)}}$.

3.24. **Exercises 3.24–3.26.**

(a) **Machine B of Figure 3.10(b).**

i.

$$E_{0B} = \{\{t_0\}, \{t_1, t_2, t_3\}\}.$$

All three nonfinal states have the same 0- and 1-transitions into the nonfinal class, so $E_{1B} = E_{0B}$. Hence $E_B = E_{0B}$.

ii. $L(B) = \{\lambda\}$.

iii. $A_{R_{L(B)}}$ is the two-state DFA with states $[\lambda]$ and $[\mathbf{0}]$ (equivalently, $[\mathbf{1}]$), where $[\lambda]$ is initial and final, $[\mathbf{0}]$ is nonfinal, and both letters send $[\mathbf{0}]$ to itself.

iv. $R_{L(B)}$ has exactly two classes:

$$[\lambda] = \{\lambda\}, \qquad [\mathbf{0}] = \Sigma^+.$$

v. $B/_{E_B}$ has states $[t_0]$ and $[t_1] = [t_2] = [t_3]$, with $[t_0]$ initial and final, and

$$\delta([t_0], 0) = \delta([t_0], 1) = [t_1] = [t_2] = [t_3],$$

while both letters fix $[t_1] = [t_2] = [t_3]$. Thus $B/_{E_B} \cong A_{R_{L(B)}}$.

(b) **Machine C of Figure 3.10(c).**

i.

$$E_{0C} = \{\{q_1, q_4\}, \{q_0, q_2, q_3, q_5\}\}.$$

Refining once gives

$$E_{1C} = \{\{q_1\}, \{q_4\}, \{q_0, q_2\}, \{q_3, q_5\}\}.$$

This is already stable, so $E_C = E_{1C}$.

ii. The language is

$$L(C) = 1(1 \cup 01)^*,$$

that is, the strings that begin with 1, end with 1, and never contain the substring 00.

iii. $A_{R_{L(C)}}$ has three states, represented by the classes of λ, 1, and 0. Writing them as $[\lambda], [1], [0]$, the transition table is

	0	1
$[\lambda]$	$[0]$	$[1]$
$[1]$	$[\lambda]$	$[1]$
$[0]$	$[0]$	$[0]$

with only $[1]$ final. Here $[10] = [\lambda]$, while $[11] = [1]$ and $[00] = [01] = [0]$.

iv. The classes of $R_{L(C)}$ can be represented by $[\lambda], [1], [0]$. Concretely,

$$[1] = 1(1 \cup 01)^*,$$

$$[\lambda] = \{\lambda\} \cup 1(1 \cup 01)^* 0,$$

and

$$[0] = \Sigma^* - \big([\lambda] \cup [1]\big).$$

v. $C/_{E_C}$ has the four states

$$[q_1],\ [q_4],\ [q_0]=[q_2],\ [q_3]=[q_5].$$

The initial state is $[q_0]=[q_2]$, the final states are $[q_1]$ and $[q_4]$, and the transition table is

	0	1
$[q_0]=[q_2]$	$[q_3]=[q_5]$	$[q_1]$
$[q_1]$	$[q_0]=[q_2]$	$[q_1]$
$[q_3]=[q_5]$	$[q_3]=[q_5]$	$[q_3]=[q_5]$
$[q_4]$	$[q_3]=[q_5]$	$[q_3]=[q_5]$

The state $[q_4]$ is disconnected, so $C/_{E_C}$ is not connected; its connected part is isomorphic to $A_{R_{L(C)}}$.

(c) **Machine D of Figure 3.10(d).**

i.

$$E_{0D}=\{\{S_1,S_4,S_7\},\{S_0,S_2,S_3,S_5,S_6,S_8,S_9\}\}.$$

Refining once gives

$$E_{1D}=\{\{S_1,S_4\},\{S_7\},\{S_0,S_3\},\{S_2,S_5,S_6\},\{S_8,S_9\}\}.$$

Refining again gives

$$E_{2D}=\{\{S_1,S_4\},\{S_7\},\{S_0,S_3\},\{S_2,S_5\},\{S_6\},\{S_8\},\{S_9\}\}.$$

This is already stable, so $E_D=E_{2D}$.

ii. From the initial state S_0, the reachable classes are $[S_0]=[S_3]$, $[S_1]=[S_4]$, and $[S_2]=[S_5]$. Therefore $L(D)$ consists of exactly those strings over $\{a,b\}$ that contain at least one a and do not end with two or more consecutive a's.

iii. $A_{R_{L(D)}}$ has three states represented by $[\lambda]$, $[a]$, and $[aa]$, with transition table

	a	b
$[\lambda]$	$[a]$	$[\lambda]$
$[a]$	$[aa]$	$[a]$
$[aa]$	$[aa]$	$[a]$

and only $[a]$ final.

iv. The classes of $R_{L(D)}$ may be described as

$$[\lambda]=b^*,$$

$$[a]=L(D),$$

and

$$[aa]=\{x\in\{a,b\}^* : x \text{ ends with at least two consecutive } a\text{'s}\}.$$

v. $D/_{E_D}$ has the seven states

$$[S_0] = [S_3],\ [S_1] = [S_4],\ [S_2] = [S_5],\ [S_6],\ [S_7],\ [S_8],\ [S_9].$$

Its initial state is $[S_0] = [S_3]$, its final states are $[S_1] = [S_4]$ and $[S_7]$, and its transition table is

	a	b
$[S_0] = [S_3]$	$[S_1] = [S_4]$	$[S_0] = [S_3]$
$[S_1] = [S_4]$	$[S_2] = [S_5]$	$[S_1] = [S_4]$
$[S_2] = [S_5]$	$[S_2] = [S_5]$	$[S_1] = [S_4]$
$[S_6]$	$[S_6]$	$[S_7]$
$[S_7]$	$[S_7]$	$[S_7]$
$[S_8]$	$[S_0] = [S_3]$	$[S_9]$
$[S_9]$	$[S_2] = [S_5]$	$[S_2] = [S_5]$

The connected part of this quotient is the three-state machine $A_{R_{L(D)}}$ above.

3.25. **Prove: if A, B are reduced, connected, equivalent DFAs then $A \cong B$.**

(a) Since A is connected, for each $s \in S_A$ choose a string x_s with $\overline{\delta}_A(s_{0A}, x_s) = s$. Define

$$\psi(s) = \overline{\delta}_B(s_{0B}, x_s).$$

(b) To show this is well defined, suppose x and y both satisfy $\overline{\delta}_A(s_{0A}, x) = \overline{\delta}_A(s_{0A}, y) = s$. Then for every $z \in \Sigma^*$,

$$xz \in L(A) \Leftrightarrow yz \in L(A),$$

because xz and yz continue from the same state s in A. Since $L(A) = L(B)$, it follows that

$$xz \in L(B) \Leftrightarrow yz \in L(B)$$

for every z. Therefore the states $\overline{\delta}_B(s_{0B}, x)$ and $\overline{\delta}_B(s_{0B}, y)$ have the same future language in B. Because B is reduced, these states must be equal. Hence ψ is well defined.

(c) First,

$$\psi(s_{0A}) = \overline{\delta}_B(s_{0B}, \lambda) = s_{0B}.$$

Now let $s \in S_A$ and $a \in \Sigma$. Since

$$\overline{\delta}_A(s_{0A}, x_s a) = \delta_A(\overline{\delta}_A(s_{0A}, x_s), a) = \delta_A(s, a),$$

we may use $x_s a$ as a string reaching $\delta_A(s, a)$. Therefore

$$\psi(\delta_A(s, a)) = \overline{\delta}_B(s_{0B}, x_s a) = \delta_B(\overline{\delta}_B(s_{0B}, x_s), a) = \delta_B(\psi(s), a).$$

Also,

$$s \in F_A \Leftrightarrow x_s \in L(A) \Leftrightarrow x_s \in L(B) \Leftrightarrow \psi(s) \in F_B.$$

So ψ is a homomorphism.

(d) *Injective*: suppose $\psi(s) = \psi(t)$. Then for every $z \in \Sigma^*$,

$$x_s z \in L(B) \Leftrightarrow x_t z \in L(B),$$

because x_s and x_t reach the same state in B. Since $L(A) = L(B)$, we also have

$$x_s z \in L(A) \Leftrightarrow x_t z \in L(A)$$

for every z. Thus s and t have the same future language in A. Because A is reduced, $s = t$.
Surjective: let $u \in S_B$. Since B is connected, there exists $y \in \Sigma^*$ such that $\overline{\delta}_B(s_{0B}, y) = u$. Let $s = \overline{\delta}_A(s_{0A}, y)$. Then y is one of the strings reaching s, so by the definition of ψ and well-definedness,

$$\psi(s) = \overline{\delta}_B(s_{0B}, y) = u.$$

Hence ψ is surjective.

3.26. **Show the transition from line 3 to 4 in Theorem 3.1(6) is not $\Leftrightarrow$.**

In that step, we used $\overline{\delta}_A(s,x) \in F_A \Rightarrow \mu(\overline{\delta}_A(s,x)) \in F_B$. The converse would require $\mu(s) \in F_B \Rightarrow s \in F_A$, which is the definition of a homomorphism (both directions). If μ is only required to satisfy $s \in F_A \Rightarrow \mu(s) \in F_B$ (not $\Leftarrow$), then we only get $L(A) \subseteq L(B)$, not equality. Example: A has states $\{q_0, q_1\}$, $F_A = \{q_1\}$; B has $\{p_0\}$, $F_B = \{p_0\}$ (all final); $\mu(q_i) = p_0$. Then $L(A) \subset L(B) = \Sigma^*$.

3.27. **Supply reasons for equivalences in proof of Theorem 3.8.**

Reading down the proof of Theorem 3.8:

(a)
$$sE_{i+1A}t \Leftrightarrow (\forall x \in \Sigma^* \ni |x| \le i+1)(\overline{\delta}(s,x) \in F \Leftrightarrow \overline{\delta}(t,x) \in F)$$

is just the definition of E_{i+1A}.

(b) Splitting this into the strings of length at most i and the strings of length exactly $i+1$ is valid because those two cases exhaust the condition $|x| \le i+1$.

(c) Replacing "$|y| = i+1$" by "$1 \le |y| \le i+1$" is valid because the strings of lengths $1, \ldots, i$ are already handled by the first conjunct.

(d) Passing from nonempty y with $|y| \le i+1$ to strings of the form $\mathbf{a}x$ with $\mathbf{a} \in \Sigma$ and $|x| \le i$ uses the fact that every nonempty string can be written uniquely in that form.

(e)
$$(\forall x \in \Sigma^* \ni |x| \le i)(\overline{\delta}(s,x) \in F \Leftrightarrow \overline{\delta}(t,x) \in F) \Leftrightarrow sE_{iA}t$$

is the definition of E_{iA}.

(f)
$$\overline{\delta}(s, \mathbf{a}x) = \overline{\delta}(\delta(s,\mathbf{a}), x)$$

and similarly for t is the definition of the extended transition function $\overline{\delta}$.

(g) Finally,

$$(\forall x \in \Sigma^* \ni |x| \le i)(\overline{\delta}(\delta(s,\mathbf{a}),x) \in F \Leftrightarrow \overline{\delta}(\delta(t,\mathbf{a}),x) \in F) \Leftrightarrow \delta(s,\mathbf{a})E_{iA}\delta(t,\mathbf{a})$$

is again the definition of E_{iA}.

3.28. **Minimize the machine of Figure 3.3.**

Let

$$E_{0A} = \{\{S_0, S_5\}, \{S_1, S_2, S_3, S_4\}\}.$$

Refining once gives

$$E_{1A} = \{\{S_0, S_5\}, \{S_1, S_2\}, \{S_3, S_4\}\},$$

and this is already stable. Thus the minimal machine has the three states

$$A = [S_0] = [S_5], \qquad B = [S_1] = [S_2], \qquad C = [S_3] = [S_4],$$

with A initial and final, and transition table

	a	b
A	B	B
B	C	C
C	A	A

So the machine minimizes to a 3-state DFA in which both letters move one step forward around the cycle $A \to B \to C \to A$. □

3.29. **Examples where A not reduced, A^c not/is reduced.**

(a) A not reduced, A^c not reduced: take

$$A = \langle\{\boldsymbol{a}\}, \{q_0, q_1\}, q_0, \delta, \{q_0, q_1\}\rangle,$$

where

$$\delta(q_0, \boldsymbol{a}) = q_1, \qquad \delta(q_1, \boldsymbol{a}) = q_1.$$

Both states are reachable, so $A^c = A$. Also $q_0 E_A q_1$, because from either state every continuation is accepted. Thus A is not reduced, and neither is A^c.

(b) A not reduced, A^c is reduced: take

$$A = \langle\{\boldsymbol{a}\}, \{q_0, q_1, q_2\}, q_0, \delta, \{q_1, q_2\}\rangle,$$

where

$$\delta(q_0, \boldsymbol{a}) = q_1, \qquad \delta(q_1, \boldsymbol{a}) = q_1, \qquad \delta(q_2, \boldsymbol{a}) = q_2.$$

The state q_2 is unreachable, and $q_1 E_A q_2$ because from either state every continuation is accepted. So A is not reduced. However,

$$A^c = \langle\{\boldsymbol{a}\}, \{q_0, q_1\}, q_0, \delta|_{S^c \times \Sigma}, \{q_1\}\rangle,$$

and q_0 and q_1 are not equivalent in A^c because λ is rejected from q_0 but accepted from q_1. Hence A^c is reduced.

3.30. **Prove $\cong$ is an equivalence relation on DFAs.**

(a) If $g : A \to B$ and $f : B \to C$ are isomorphisms, then $f \circ g : A \to C$ satisfies all isomorphism conditions (since both are bijective homomorphisms and composition of bijections is bijective, composition of homomorphisms is a homomorphism).

(b) Symmetry: if $f : A \to B$ is an isomorphism (bijective homomorphism), then $f^{-1} : B \to A$ is also an isomorphism: $f^{-1}(s_{0B}) = s_{0A}$ (since $f(s_{0A}) = s_{0B}$); $f^{-1}(\delta_B(t,a)) = f^{-1}(\delta_B(f(s),a)) = f^{-1}(f(\delta_A(s,a))) = \delta_A(s,a) = \delta_A(f^{-1}(t),a)$; $t \in F_B \Leftrightarrow f^{-1}(t) \in F_A$.

(c) Reflexivity: $\mathrm{id} : A \to A$ is an isomorphism.

(d) From (a),(b),(c): $\cong$ is an equivalence relation. □

3.31. **Homomorphism is not an equivalence relation on DFAs.**

Homomorphism is not symmetric. Let $\Sigma = \{a\}$, and let A be the one-state DFA with state p (initial, nonfinal) and $\delta_A(p,a) = p$, so $L(A) = \emptyset$. Let B have two states q_0, q_1, with q_0 initial and nonfinal, q_1 final and unreachable, and both states looping on a.

There is a homomorphism $\mu : A \to B$ given by $\mu(p) = q_0$: initials match, transitions are preserved, and $p \in F_A \Leftrightarrow q_0 \in F_B$.

However, there is no homomorphism $\nu : B \to A$, because $q_1 \in F_B$ would have to imply $\nu(q_1) \in F_A$, and A has no final state. Therefore homomorphism is not an equivalence relation on DFAs. □

3.32. **For R and R_L from Theorem 2.2, show homomorphism from A_R to A_{R_L}.**

R refines R_L: each R-class is contained in some R_L-class. Define

$$\mu : S_{A_R} \to S_{A_{R_L}} \qquad \text{by} \qquad \mu([x]_R) = [x]_{R_L}.$$

This is well defined: if $[x]_R = [y]_R$, then xRy, and since R refines R_L, also xR_Ly, so $[x]_{R_L} = [y]_{R_L}$.

Now

$$\mu(s_{0,R}) = \mu([\lambda]_R) = [\lambda]_{R_L} = s_{0,R_L}.$$

Also, for any $a \in \Sigma$,

$$\mu(\delta_R([x]_R, a)) = \mu([xa]_R) = [xa]_{R_L} = \delta_{R_L}([x]_{R_L}, a) = \delta_{R_L}(\mu([x]_R), a).$$

Finally,

$$[x]_R \in F_R \Leftrightarrow x \in L \Leftrightarrow [x]_{R_L} \in F_{R_L} \Leftrightarrow \mu([x]_R) \in F_{R_L}.$$

Therefore μ is a homomorphism from A_R to A_{R_L}. □

3.33. **Prove: if there is a homomorphism $A \to B$ then R^A refines R^B.**

$\mu : A \to B$ homomorphism. If xR^Ay: $\overline{\delta}_A(s_0,x) = \overline{\delta}_A(s_0,y)$. Then $\overline{\delta}_B(s_0,x) = \mu(\overline{\delta}_A(s_0,x)) = \mu(\overline{\delta}_A(s_0,y)) = \overline{\delta}_B(s_0,y)$, so xR^By. □

3.34. **Prove: if $A \cong B$ then $R^A = R^B$.**

(a) Via Exercise 3.35: an isomorphism $\mu : A \to B$ is in particular a homomorphism, so R^A refines R^B (Exercise 3.35). By symmetry ($\mu^{-1} : B \to A$ is also a homomorphism), R^B refines R^A. So $R^A = R^B$.

(b) Directly: $xR^Ay \Leftrightarrow \overline{\delta}_A(s_0,x) = \overline{\delta}_A(s_0,y) \Leftrightarrow \mu(\overline{\delta}_A(s_0,x)) = \mu(\overline{\delta}_A(s_0,y))$ (μ injective) $\Leftrightarrow \overline{\delta}_B(s_0,x) = \overline{\delta}_B(s_0,y)$ (Exercise 3.10) $\Leftrightarrow xR^By$.

3.35. **A not homomorphic to B, $R^A = R^B$.**

(a) $L(A) = L(B)$: let $\Sigma = \{a\}$. Let B be the minimal 2-state DFA for even-length strings: q_0 is initial and final, q_1 is nonfinal, and $\delta(q_0, a) = q_1$, $\delta(q_1, a) = q_0$. Let A have the same reachable part, plus an unreachable final state r with $\delta(r, a) = r$.

Then $L(A) = L(B)$ and $R^A = R^B$, because R^A only depends on the states reached from the initial state, and A and B have the same reachable part. However, there is no homomorphism $A \to B$: since r is final, its image would have to be the final state q_0, but then

$$\mu(\delta_A(r, a)) = \mu(r) = q_0$$

while

$$\delta_B(\mu(r), a) = \delta_B(q_0, a) = q_1,$$

a contradiction.

(b) $L(A) \neq L(B)$: let A again be the even-length machine, and let B be the same transition graph but with final state q_1 instead of q_0, so $L(B)$ is the odd-length strings. Then $R^A = R^B$ (both relations simply partition Σ^* into even and odd lengths), but A is not homomorphic to B because the initial state of A is final while the initial state of B is not.

(c) No. Suppose A and B are connected, $L(A) = L(B)$, and $R^A = R^B$. Define

$$\psi(\overline{\delta}_A(s_{0A}, x)) = \overline{\delta}_B(s_{0B}, x).$$

Because $R^A = R^B$, this is well defined: if $\overline{\delta}_A(s_{0A}, x) = \overline{\delta}_A(s_{0A}, y)$, then $xR^A y$, hence $xR^B y$, and therefore $\overline{\delta}_B(s_{0B}, x) = \overline{\delta}_B(s_{0B}, y)$. Since both machines are connected, the map ψ is onto: given $u \in S_B$, choose x with $\overline{\delta}_B(s_{0B}, x) = u$, and then

$$\psi(\overline{\delta}_A(s_{0A}, x)) = u.$$

It is also one-to-one: if $\psi(s) = \psi(t)$, choose strings x_s, x_t with

$$\overline{\delta}_A(s_{0A}, x_s) = s, \qquad \overline{\delta}_A(s_{0A}, x_t) = t.$$

Then

$$\overline{\delta}_B(s_{0B}, x_s) = \psi(s) = \psi(t) = \overline{\delta}_B(s_{0B}, x_t),$$

so $x_s R^B x_t$. Since $R^A = R^B$, also $x_s R^A x_t$, hence

$$s = \overline{\delta}_A(s_{0A}, x_s) = \overline{\delta}_A(s_{0A}, x_t) = t.$$

For transitions,

$$\psi(\delta_A(\overline{\delta}_A(s_{0A}, x), a)) = \psi(\overline{\delta}_A(s_{0A}, xa)) = \overline{\delta}_B(s_{0B}, xa) = \delta_B(\overline{\delta}_B(s_{0B}, x), a) = \delta_B(\psi(\overline{\delta}_A(s_{0A}, x)), a).$$

Finally, for any $x \in \Sigma^*$,

$$\overline{\delta}_A(s_{0A}, x) \in F_A \Leftrightarrow x \in L(A) \Leftrightarrow x \in L(B) \Leftrightarrow \overline{\delta}_B(s_{0B}, x) \in F_B,$$

so ψ preserves finality. Thus ψ is an isomorphism $A \cong B$, so in particular there is a homomorphism from A to B. Such examples therefore cannot exist.

(d) Yes. The example from part (a) already works if we note that both machines are reduced: B is minimal and therefore reduced, and the extra state r in A has language Σ^*, which is different from the state languages of q_0 and q_1. So A and B are both reduced, $L(A) = L(B)$, and $R^A = R^B$, but A is still not homomorphic to B.

3.36. **Disprove: if A homomorphic to B then $R^A = R^B$.**

Counterexample: A = 2-state DFA for $\{a^{2k}\}$, B = 1-state DFA for Σ^* (all states final). There is a homomorphism $\mu : A \to B$ mapping both states to the one state of B. R^A has 2 classes (even/odd), R^B has 1 class. $R^A \neq R^B$. □

3.37. **Prove or give counterexamples (homomorphisms involving A_{R^M}).**

(a) Homomorphism $\psi : A_{R^M} \to M$: Yes. Define

$$\psi([x]_{R^M}) = \overline{\delta}_M(s_0, x).$$

This is well defined: if $[x]_{R^M} = [y]_{R^M}$, then $xR^M y$, so by definition of R^M,

$$\overline{\delta}_M(s_0, x) = \overline{\delta}_M(s_0, y).$$

It is a homomorphism because

$$\psi([\lambda]_{R^M}) = \overline{\delta}_M(s_0, \lambda) = s_0,$$

$$\psi(\delta_{R^M}([x]_{R^M}, a)) = \psi([xa]_{R^M}) = \overline{\delta}_M(s_0, xa) = \delta_M(\overline{\delta}_M(s_0, x), a) = \delta_M(\psi([x]_{R^M}), a),$$

and

$$[x]_{R^M} \in F_{R^M} \Leftrightarrow x \in L(M) \Leftrightarrow \overline{\delta}_M(s_0, x) \in F_M \Leftrightarrow \psi([x]_{R^M}) \in F_M.$$

(b) Isomorphism $\psi : A_{R^M} \to M$: Not always. Only if M is connected (since $A_{R^M} \cong M^c$, isomorphism only if $M = M^c$, i.e., M connected).

(c) Homomorphism $\psi : M \to A_{R^M}$: Not always. The reachable part of M maps naturally to A_{R^M} (send each reachable state s to $[x_s]_{R^M}$ where x_s is any string reaching s), but unreachable states may obstruct extension.

Counterexample. Let $\Sigma = \{\mathbf{a}\}$, $S = \{s_0, s_1, s_d\}$, s_0 initial, $F = \{s_1\}$, with

$$\delta(s_0, \mathbf{a}) = s_1, \quad \delta(s_1, \mathbf{a}) = s_1, \quad \delta(s_d, \mathbf{a}) = s_0.$$

State s_d is unreachable. The accessible quotient A_{R^M} has two states q_0, q_1 with

$$\delta'(q_0, \mathbf{a}) = q_1, \quad \delta'(q_1, \mathbf{a}) = q_1,$$

q_0 initial, $F' = \{q_1\}$.

A homomorphism $\psi : M \to A_{R^M}$ must satisfy $\psi(s_0) = q_0$ (initial states correspond) and $\delta'(\psi(s), \mathbf{a}) = \psi(\delta(s, \mathbf{a}))$ for every s. From s_d: $\delta'(\psi(s_d), \mathbf{a}) = \psi(\delta(s_d, \mathbf{a})) = \psi(s_0) = q_0$. But $\delta'(q_0, \mathbf{a}) = q_1 \neq q_0$ and $\delta'(q_1, \mathbf{a}) = q_1 \neq q_0$, so no state in A_{R^M} can serve as $\psi(s_d)$. The homomorphism does not exist. □

3.38. **Prove: if A minimal then $R^A = R_{L(A)}$.**

Let

$$A = \langle \Sigma, S, s_0, \delta, F \rangle$$

be minimal.

First, A must be connected. If not, then its connected part A^c accepts the same language as A but has fewer states, contradicting minimality.

Second, A must be reduced. If not, then two distinct states are E_A-equivalent, so the quotient $A/_{E_A}$ accepts the same language as A and has fewer states, again contradicting minimality.

Because A is connected, every state is reached by some string, so the R^A-classes are in one-to-one correspondence with the states of A. Hence

$$rk(R^A) = |S|.$$

Also, Exercise 2.57(c) says that R^A refines $R_{L(A)}$, so

$$rk(R^A) \geq rk(R_{L(A)}).$$

On the other hand, Corollary 3.2 says that $A_{R_{L(A)}}$ is minimal for $L(A)$. Since A is also minimal for $L(A)$, the two minimal DFAs have the same number of states:

$$|S| = rk(R_{L(A)}).$$

Therefore

$$rk(R^A) = |S| = rk(R_{L(A)}).$$

A refinement of finite rank cannot be proper if the two ranks are equal. Hence $R^A = R_{L(A)}$. □

3.39. **Example where Exercise 3.40 fails if A not minimal.**

Take $\Sigma = \{a\}$ and let

$$A = \langle \{a\}, \{q_0, q_1, q_2\}, q_0, \delta, \{q_0\} \rangle,$$

where

$$\delta(q_0, a) = q_1, \qquad \delta(q_1, a) = q_0, \qquad \delta(q_2, a) = q_2.$$

The state q_2 is unreachable, so A is not minimal. From the initial state, the machine accepts exactly the even-length strings, so

$$L(A) = \{a^{2k} \mid k \geq 0\}.$$

Therefore $R_{L(A)}$ has exactly two classes, represented by λ and a. But R^A is the state congruence of the whole machine, so it has three classes, represented by the three states q_0, q_1, q_2. Hence

$$R^A \neq R_{L(A)}.$$

3.40. **Example where Exercise 3.40 holds even if A not minimal.**

Take the minimal 2-state DFA for the language of even-length strings over $\{a\}$, with states q_0 (initial and final) and q_1 (nonfinal), and add an extra unreachable state q_2 with $\delta(q_2, a) = q_2$. The resulting machine A is not minimal, because it is not connected. However, R^A is unchanged by the presence of q_2, since no string from the initial state ever reaches q_2. Therefore R^A still has exactly the two classes "even length" and "odd length," which are precisely the classes of $R_{L(A)}$. So in this example A is not minimal but $R^A = R_{L(A)}$. □

3.41. **Quotient by arbitrary relation R.**

(a) $R = E_{0A}$: $A/_{E_{0A}}$ partitions into final and nonfinal. $\delta_{E_{0A}}$ maps $[s]_{E_{0A}} \times \Sigma \to S_{E_{0A}}$: need $[s] \ni s'$ with $\delta(s,a)$ and $\delta(s',a)$ in the same E_{0A}-class... not necessarily. So $A/_{E_{0A}}$ is not always well-defined. Example: A = 3-state DFA, $F = \{q_1\}$, $\delta(q_0,a) = q_1$, $\delta(q_2,a) = q_0$. E_{0A}-classes: $\{q_0,q_2\}$ (nonfinal) and $\{q_1\}$ (final). $\delta_{E_{0A}}(\{q_0,q_2\},a)$: $\delta(q_0,a) = q_1 \in \{q_1\}$ but $\delta(q_2,a) = q_0 \in \{q_0,q_2\}$: two different classes! Not well-defined.

(b) If R refines E_A, then $A/_R$ is well defined exactly when R is stable under the transitions:

$$sRt \Rightarrow \delta(s,a)R\delta(t,a)$$

for every $a \in \Sigma$.
For such well-defined refinements R, the analogues of Theorems 3.4, 3.5, and 3.6 behave as follows.
Analogue of Theorem 3.4: false in general. Take R to be the identity relation on the states of a DFA A that is not reduced. Then R certainly refines E_A, and $A/_R = A$, which is not reduced.
Analogue of Theorem 3.5: true. If A is connected and $[s]_R$ is a state of $A/_R$, choose x with $\overline{\delta}(s_0,x) = s$. Since $A/_R$ is well defined,

$$\overline{\delta}_R([s_0]_R,x) = [\overline{\delta}(s_0,x)]_R = [s]_R,$$

so every state class is reachable.
Analogue of Theorem 3.6: true. Because R refines E_A, every R-class is contained in a single E_A-class, so no R-class mixes final and nonfinal states; hence F_R is well defined. Then the proof of Theorem 3.6 goes through verbatim:

$$x \in L(A/_R) \Leftrightarrow \overline{\delta}_R([s_0]_R,x) \in F_R \Leftrightarrow [\overline{\delta}(s_0,x)]_R \in F_R \Leftrightarrow \overline{\delta}(s_0,x) \in F \Leftrightarrow x \in L(A).$$

Thus $L(A/_R) = L(A)$.

3.42. **Homomorphisms between $M/_{E_M}$ and $A_{R_{L(M)}}$.**

(a) Not always. Consider the DFA

$$M = \langle \{a\}, \{s_0,s_1,s_d\}, s_0, \delta, \{s_1\} \rangle$$

with

$$\delta(s_0,a) = s_1, \qquad \delta(s_1,a) = s_1, \qquad \delta(s_d,a) = s_0.$$

The state s_d is unreachable. The three states have distinct future languages:

$$L(M^{s_0}) = a^+, \qquad L(M^{s_1}) = a^*, \qquad L(M^{s_d}) = a^{\geq 2},$$

so E_M is the identity relation and $M/_{E_M} = M$.
Now $L(M) = a^+$, and $A_{R_{L(M)}}$ is the usual 2-state DFA with initial nonfinal state q_0, final state q_1, and transitions $\delta(q_0,a) = q_1$, $\delta(q_1,a) = q_1$.
If there were a homomorphism $\psi : M/_{E_M} \to A_{R_{L(M)}}$, then $\psi(s_0) = q_0$ and $\psi(s_1) = q_1$. Since s_d is nonfinal, $\psi(s_d)$ would have to be q_0. But then

$$\psi(\delta(s_d,a)) = \psi(s_0) = q_0$$

while

$$\delta(\psi(s_d), a) = \delta(q_0, a) = q_1,$$

a contradiction. So the statement is false.

(b) Yes. Define

$$\psi([x]_{R_{L(M)}}) = [\overline{\delta}_M(s_0, x)]_{E_M}.$$

This is well defined: if $xR_{L(M)}y$, then for every $z \in \Sigma^*$,

$$xz \in L(M) \Leftrightarrow yz \in L(M),$$

so the states $\overline{\delta}_M(s_0, x)$ and $\overline{\delta}_M(s_0, y)$ have the same future language and are therefore E_M-equivalent.

Now

$$\psi([\lambda]_{R_{L(M)}}) = [s_0]_{E_M},$$

and for any $a \in \Sigma$,

$$\psi(\delta_{R_{L(M)}}([x]_{R_{L(M)}}, a)) = \psi([xa]_{R_{L(M)}}) = [\overline{\delta}_M(s_0, xa)]_{E_M} = [\delta(\overline{\delta}_M(s_0, x), a)]_{E_M} = \delta_{E_M}(\psi([x]_{R_{L(M)}}), a).$$

Finally,

$$[x]_{R_{L(M)}} \in F_{R_{L(M)}} \Leftrightarrow x \in L(M) \Leftrightarrow \overline{\delta}_M(s_0, x) \in F \Leftrightarrow [\overline{\delta}_M(s_0, x)]_{E_M} \in F_{E_M}.$$

So ψ is a homomorphism from $A_{R_{L(M)}}$ to $M/_{E_M}$. $\square$

3.43. **Prove: $\exists m < \|S\|$ with $E_{mA} = E_{m+1,A}$.**

The sequence $E_{0A} \supseteq E_{1A} \supseteq \cdots$ is decreasing (each refinement removes some equivalences). Since S is finite, the number of equivalence classes is bounded by $|S|$. Each strict refinement step increases the number of classes by at least 1. Starting from E_{0A} (which has at most 2 classes: final/nonfinal), we can have at most $|S| - 1$ strict refinement steps before reaching E_A. So $E_{|S|-1,A} = E_{|S|,A} = E_A$, meaning $m \leq |S| - 1 < |S|$. $\square$

3.44. **Equivalence vs. isomorphism.**

(a) False. Let B be the minimal DFA for the even-length strings over

$$\Sigma = \{\boldsymbol{a}\},$$

with states r_0, r_1, start state r_0, final set $\{r_0\}$, and transitions

$$\delta_B(r_0, \boldsymbol{a}) = r_1, \qquad \delta_B(r_1, \boldsymbol{a}) = r_0.$$

Now let A be the DFA obtained by adjoining an unreachable state r_2:

$$A = \langle \{\boldsymbol{a}\}, \{r_0, r_1, r_2\}, r_0, \delta_A, \{r_0\} \rangle,$$

where

$$\delta_A(r_0, \boldsymbol{a}) = r_1, \qquad \delta_A(r_1, \boldsymbol{a}) = r_0, \qquad \delta_A(r_2, \boldsymbol{a}) = r_2.$$

Then $L(A) = L(B)$, so A and B are equivalent, but they are not isomorphic because one has three states and the other has two.

(b) True: if $A \cong B$ then in particular $L(A) = L(B)$ (by Exercise 3.11).

3.45. **Prove $C_i \subseteq C_{i+1}$.**

$C_0 = \{s_0\}$, $C_{i+1} = C_i \cup \{\delta(s,a) : s \in C_i, a \in \Sigma\}$. Clearly $C_i \subseteq C_{i+1}$ by definition. □

3.46. **Prove C_i = states reachable from s_0 by strings of length $\leq i$.**

Let $R_i = \{\overline{\delta}(s_0, x) : |x| \leq i\}$.

Base: $R_0 = \{s_0\} = C_0$. ✓

Step: $R_{i+1} = \{\overline{\delta}(s_0, x) : |x| \leq i+1\} = R_i \cup \{\delta(\overline{\delta}(s_0, x), a) : |x| \leq i, a \in \Sigma\} = C_i \cup \{\delta(s,a) : s \in C_i, a \in \Sigma\} = C_{i+1}$. □

3.47. **Prove: $C_i = C_{i+1} \Rightarrow (\forall k)(C_i = C_{i+k})$.**

Base $k = 0$: trivial. $k = 1$: given.

Step: Assume $C_i = C_{i+k}$. Then $C_{i+k+1} = C_{i+k} \cup \{\delta(s,a) : s \in C_{i+k}\} = C_i \cup \{\delta(s,a) : s \in C_i\} = C_{i+1} = C_i$. □

3.48. **Prove: for a DFA A, $C_i = C_{i+1} \Rightarrow C_i = S^c$.**

$S^c = \bigcup_{k \geq 0} C_k$. If $C_i = C_{i+1}$, then by Exercise 3.49, $C_i = C_{i+k}$ for all $k \geq 0$. So $S^c = \bigcup_k C_k = C_i$ (since $C_k \subseteq C_i$ for $k \leq i$ by $C_i \supseteq C_{i-1} \supseteq \cdots$, and $C_k = C_i$ for $k \geq i$). □

3.49. **Prove: $\exists i$ with $C_i = C_{i+1}$.**

Since S is finite, the sequence $C_0 \subseteq C_1 \subseteq \cdots$ is non-decreasing and bounded above by S. So it must stabilize: $\exists i$ with $C_i = C_{i+1}$. □

3.50. **Prove: $\exists i \leq \|S\|$ with $C_i = S^c$.**

Each strict inclusion $C_i \subsetneq C_{i+1}$ adds at least one new state. Starting from $|C_0| = 1$, after at most $|S| - 1$ strict inclusions we reach $C_{|S|-1} \supseteq \cdots$, which must equal $C_{|S|}$ (no room to grow beyond $|S|$ states). So $i \leq |S| - 1 \leq |S|$ and $C_i = S^c$. □

3.51. **Argue the C_i algorithm is correct and terminates.**

Correctness: by Exercise 3.48, C_i is exactly the reachable states within depth i. By Exercise 3.51, $C_i = S^c$ when stable. Termination: by Exercise 3.51 and 3.52, stability occurs by step $|S|$. □

3.52. **Give two DFAs with homomorphisms in both directions but no isomorphism.**

Let
$$L = \{a^{2k} : k \geq 0\}.$$
Let
$$A = \langle \{a\}, \{q_0, q_1\}, q_0, \delta_A, \{q_0\} \rangle,$$
where
$$\delta_A(q_0, a) = q_1, \qquad \delta_A(q_1, a) = q_0.$$
This is the minimal 2-state DFA for L.

Let
$$B = \langle \{a\}, \{p_0, p_1, p_2\}, p_0, \delta_B, \{p_0, p_2\} \rangle,$$

where

$$\delta_B(p_0, a) = p_1, \qquad \delta_B(p_1, a) = p_0, \qquad \delta_B(p_2, a) = p_1.$$

Here p_2 is unreachable, and $p_0 E_B p_2$, so B is a non-minimal DFA for L.

There is a homomorphism $\mu : B \to A$ collapsing $p_0, p_2 \mapsto q_0$ and $p_1 \mapsto q_1$: this satisfies $\delta_A(\mu(p), a) = \mu(\delta_B(p, a))$ and $p \in F_B \Leftrightarrow \mu(p) \in F_A$.

There is also a homomorphism $\nu : A \to B$ sending $q_0 \mapsto p_0$ and $q_1 \mapsto p_1$ (the natural inclusion into the connected part of B): $\nu(\delta_A(q, a)) = \delta_B(\nu(q), a)$ since $\delta_A(q_0, a) = q_1$ and $\delta_B(p_0, a) = p_1 = \nu(q_1)$, and $q_0 \in F_A \Leftrightarrow p_0 \in F_B$.

But $A \ncong B$ since they have different numbers of states.

3.53. **R and Q right congruences of finite rank, R refines Q, L union of Q-classes.**

(a) L is a union of R-classes: since R refines Q, each Q-class is a union of R-classes. So $L = \bigcup(\text{some } Q\text{-classes}) = \bigcup(\text{corresponding } R\text{-classes})$.

(b) Homomorphism $A_R \to A_Q$: Define $\mu([x]_R) = [x]_Q$ (well-defined since R refines Q: $[x]_R = [y]_R \Rightarrow xRy \Rightarrow xQy \Rightarrow [x]_Q = [y]_Q$). Check: $\mu(s_{0,R}) = \mu([\lambda]_R) = [\lambda]_Q = s_{0,Q}$. $\mu(\delta_R([x]_R, a)) = \mu([xa]_R) = [xa]_Q = \delta_Q([x]_Q, a) = \delta_Q(\mu([x]_R), a)$. $[x]_R \in F_R \Leftrightarrow x \in L \Leftrightarrow [x]_Q \in F_Q \Leftrightarrow \mu([x]_R) \in F_Q$.

(c) Not in general. Take $\Sigma = \{\boldsymbol{a}\}$, let

$$[\lambda]_Q = \{\lambda\}, \qquad [\boldsymbol{a}]_Q = \{x \in \Sigma^* : |x| \geq 1\},$$

and let R refine Q by splitting the nonempty strings according to parity:

$$[\lambda]_R = \{\lambda\}, \qquad [\boldsymbol{a}]_R = \{x : |x| \text{ is odd}\}, \qquad [\boldsymbol{aa}]_R = \{x : |x| \text{ is even and } |x| \geq 2\}.$$

Let $L = \{\lambda\}$, which is a union of Q-classes. Then A_Q has two states, $q_0 = [\lambda]_Q$ (start and final) and $q_1 = [\boldsymbol{a}]_Q$, with

$$\delta_Q(q_0, \boldsymbol{a}) = q_1, \qquad \delta_Q(q_1, \boldsymbol{a}) = q_1.$$

The DFA A_R has three states, $r_0 = [\lambda]_R$ (start and final), $r_1 = [\boldsymbol{a}]_R$, and $r_2 = [\boldsymbol{aa}]_R$, with

$$\delta_R(r_0, \boldsymbol{a}) = r_1, \qquad \delta_R(r_1, \boldsymbol{a}) = r_2, \qquad \delta_R(r_2, \boldsymbol{a}) = r_1.$$

Assume there is a homomorphism $\psi : A_Q \to A_R$. Since start states must map to start states, $\psi(q_0) = r_0$. Then

$$\psi(q_1) = \psi(\delta_Q(q_0, \boldsymbol{a})) = \delta_R(\psi(q_0), \boldsymbol{a}) = \delta_R(r_0, \boldsymbol{a}) = r_1.$$

But also

$$\psi(q_1) = \psi(\delta_Q(q_1, \boldsymbol{a})) = \delta_R(\psi(q_1), \boldsymbol{a}) = \delta_R(r_1, \boldsymbol{a}) = r_2,$$

which is impossible. Therefore there need not be a homomorphism from A_Q to A_R.

3.54. **Prove A_Q is connected.**

Every state $[x]_Q$ of A_Q is reachable: $\overline{\delta}_Q([\lambda]_Q, x) = [x]_Q$. □

3.55. **Prove: if $A \cong B$ and A connected then B connected.**

Let $\mu : A \to B$ be an isomorphism. μ is surjective (bijection), so every state of B is $\mu(s)$ for some $s \in S_A$. Since A is connected, $s = \overline{\delta}_A(s_0, x)$ for some x. Then $\mu(s) = \mu(\overline{\delta}_A(s_0, x)) = \overline{\delta}_B(\mu(s_0), x) = \overline{\delta}_B(s_{0B}, x)$ (by Exercise 3.10). So every state of B is reachable. □

3.56. **Prove: if $A \cong B$ and B connected then A connected.**

$\mu^{-1} : B \to A$ is also an isomorphism. By Exercise 3.56, A is connected. □

3.57. **Disprove: homomorphism $A \to B$, A connected $\Rightarrow$ B connected.**

Take

$$A = \langle \{\mathbf{0}\}, \{s\}, s, \delta_A, \{s\} \rangle$$

with $\delta_A(s, \mathbf{0}) = s$. This DFA is connected. Let

$$B = \langle \{\mathbf{0}\}, \{t, u\}, t, \delta_B, \{t\} \rangle$$

where $\delta_B(t, \mathbf{0}) = t$ and $\delta_B(u, \mathbf{0}) = u$. The state u is unreachable from t, so B is not connected.

Define $\mu : A \to B$ by $\mu(s) = t$. Then μ preserves the start state, it preserves finality because s and t are both final, and

$$\mu(\delta_A(s, \mathbf{0})) = \mu(s) = t = \delta_B(t, \mathbf{0}) = \delta_B(\mu(s), \mathbf{0}).$$

Hence μ is a homomorphism even though B is not connected.

3.58. **Disprove: homomorphism $A \to B$, B connected $\Rightarrow$ A connected.**

Take

$$A = \langle \{\mathbf{0}\}, \{s_0, s_1\}, s_0, \delta_A, \{s_0, s_1\} \rangle$$

with $\delta_A(s_0, \mathbf{0}) = s_0$ and $\delta_A(s_1, \mathbf{0}) = s_1$. The state s_1 is unreachable, so A is not connected. Let

$$B = \langle \{\mathbf{0}\}, \{t\}, t, \delta_B, \{t\} \rangle$$

with $\delta_B(t, \mathbf{0}) = t$; this DFA is connected.

Define $\mu : A \to B$ by

$$\mu(s_0) = t, \qquad \mu(s_1) = t.$$

Then μ preserves the start state, it preserves finality because all these states are final, and

$$\mu(\delta_A(s_i, \mathbf{0})) = \mu(s_i) = t = \delta_B(t, \mathbf{0}) = \delta_B(\mu(s_i), \mathbf{0})$$

for $i = 0, 1$. So there is a homomorphism from A to connected B even though A is not connected.

3.59. **$A_{R^A} \cong A^c$: define ψ and prove it is an isomorphism.**

(a) $\psi([x]_{R^A}) = \overline{\delta}(s_0, x)$: sends each R^A-class (= set of strings reaching the same state) to that state.

(b) Well-defined: $[x]_{R^A} = [y]_{R^A} \Rightarrow \overline{\delta}(s_0, x) = \overline{\delta}(s_0, y)$. ✓

(c) Homomorphism: $\psi([\lambda]_{R^A}) = s_0$. $\psi([xa]_{R^A}) = \overline{\delta}(s_0, xa) = \delta(\overline{\delta}(s_0, x), a) = \delta(\psi([x]_{R^A}), a)$. $[x] \in F_{R^A} \Leftrightarrow x \in L(A) \Leftrightarrow \overline{\delta}(s_0, x) \in F = F^c$. ✓

(d) Isomorphism: Injective: $\psi([x]) = \psi([y]) \Rightarrow \overline{\delta}(s_0, x) = \overline{\delta}(s_0, y) \Rightarrow [x] = [y]$. Surjective (onto S^c): any $s \in S^c$ has $s = \overline{\delta}(s_0, x_s) = \psi([x_s])$.

3.60. **Prove $\psi = \mu^{-1}$ when A, B connected and $\psi : A \to B$, $\mu : B \to A$ isomorphisms.**

$(\mu \circ \psi)(s) = \mu(\psi(s))$. For connected A: $s = \overline{\delta}_A(s_0, x)$ for some x. $\psi(s) = \overline{\delta}_B(s_{0B}, x)$ (by Exercise 3.10 applied to ψ). $\mu(\psi(s)) = \mu(\overline{\delta}_B(s_{0B}, x)) = \overline{\delta}_A(s_{0A}, x) = s$ (by Exercise 3.10 applied to μ). So $\mu \circ \psi = \mathrm{id}_A$, i.e., $\psi = \mu^{-1}$. □

3.61. **Give example for which $\psi \neq \mu^{-1}$ (not connected).**

Take $A = B$ to be the DFA

$$\langle \{\mathbf{0}\}, \{q_0, q_1, q_2\}, q_0, \delta, \{q_0\} \rangle$$

where

$$\delta(q_0, \mathbf{0}) = q_0, \qquad \delta(q_1, \mathbf{0}) = q_1, \qquad \delta(q_2, \mathbf{0}) = q_2.$$

This machine is not connected because q_1 and q_2 are unreachable.

Let $\psi : A \to B$ be the identity isomorphism:

$$\psi(q_i) = q_i \qquad (i = 0, 1, 2).$$

Let $\mu : B \to A$ swap the two unreachable states:

$$\mu(q_0) = q_0, \qquad \mu(q_1) = q_2, \qquad \mu(q_2) = q_1.$$

Because q_1 and q_2 have exactly the same final/nonfinal status and the same loop transition on $\mathbf{0}$, μ is also an isomorphism. But

$$\mu^{-1} = \mu \neq \psi,$$

since, for example, $\mu^{-1}(q_1) = q_2$ while $\psi(q_1) = q_1$.

3.62. **Three-state DFA with E_{0A} having one class. Can $E_{0A} \neq E_{1A}$?**

E_{0A} has one class iff all states are equivalent under E_{0A}, i.e., all are final or all are nonfinal. Example: $F = S = \{q_0, q_1, q_2\}$ (all final). Then $E_{0A} = S \times S$ (one class). Can $E_{1A} \neq E_{0A}$? E_{1A}: $qE_{1A}r \Leftrightarrow qE_{0A}r \wedge (\forall a)\delta(q, a)E_{0A}\delta(r, a)$. Since $E_{0A} = S \times S$, all δ-targets are in the same class, so $E_{1A} = E_{0A}$. So no, if E_{0A} has one class then $E_{1A} = E_{0A}$.

3.63. **If A, B reduced and connected and $\psi : A \to B$ is a homomorphism, must ψ be an isomorphism?**

Yes. Since ψ is a homomorphism, it already satisfies the start-state, final-state, and transition conditions from Definition 3.5. We only need to show that ψ is one-to-one and onto.

To prove injectivity, assume $\psi(s) = \psi(t)$. Then certainly $\psi(s)E_B\psi(t)$, and Exercise 3.20 gives

$$\psi(s)E_B\psi(t) \Leftrightarrow sE_At.$$

Since A is reduced, sE_At implies $s = t$. Thus ψ is one-to-one.

To prove surjectivity, let $u \in S_B$. Because B is connected, there exists $x \in \Sigma^*$ such that $\overline{\delta}_B(s_{0B}, x) = u$. Let $s = \overline{\delta}_A(s_{0A}, x) \in S_A$. By Exercise 3.10,

$$\psi(s) = \psi(\overline{\delta}_A(s_{0A}, x)) = \overline{\delta}_B(\psi(s_{0A}), x) = \overline{\delta}_B(s_{0B}, x) = u.$$

So every state of B lies in the image of ψ.

Therefore ψ is a bijective homomorphism, hence an isomorphism. □

3.64. **Prove Corollary 3.2.**

By Theorem 2.2, A_{R_L} accepts L. The discussion immediately preceding the corollary shows that if $M = \langle \Sigma, S, s_0, \delta, F\rangle$ is any DFA with $L(M) = L$, then

$$\|S\| \geq rk(R^M) \geq rk(R_L) = \|S_{R_L}\|.$$

Thus every DFA accepting L has at least as many states as A_{R_L}, so A_{R_L} is a minimal DFA for L. □

3.65. **Prove $S^c = \{\overline{\delta}(s_0, x) : |x| \leq \|S\|\}$.**

By Exercise 3.52: $\exists i \leq |S|$ with $C_i = S^c$. So $S^c = C_i = \{\overline{\delta}(s_0, x) : |x| \leq i\} \subseteq \{\overline{\delta}(s_0, x) : |x| \leq |S|\}$. The other inclusion is trivial (all strings of length $\leq |S|$ reach states in S^c). □

3.66. **Prove $sE_A t \Leftrightarrow L(A^s) = L(A^t)$.**

This is the definition of E_A. Formally: $sE_A t$ is defined as $L(A^s) = L(A^t)$ where $A^s = \langle \Sigma, S, s, \delta, F\rangle$. The equivalence is by definition. (The substantive content is showing E_A satisfies the right congruence property and the iterative characterization from Exercises 3.3–3.4.) □

3.67. **Properties of initial and terminal sets.**

(a) $\{I(A, t) \mid t \in S\}$ partitions Σ^*: Each $x \in \Sigma^*$ reaches exactly one state $\overline{\delta}(s_0, x) = t$. So $x \in I(A, t)$ for exactly one t. The sets are disjoint and cover Σ^*. □

(b) Terminal sets need not partition Σ^*. For example, let $A = \langle\{\boldsymbol{a}\}, \{q_0, q_1\}, q_0, \delta, \{q_0, q_1\}\rangle$ with $\delta(q_0, \boldsymbol{a}) = q_0$ and $\delta(q_1, \boldsymbol{a}) = q_1$. Then

$$T(A, q_0) = \Sigma^* = T(A, q_1).$$

Thus the terminal sets are not disjoint, so they do not form a partition.

(c) Terminal sets can partition Σ^*. Let $A = \langle\{\boldsymbol{a}, \boldsymbol{b}\}, \{q_0, q_1\}, q_0, \delta, \{q_0\}\rangle$ where both letters toggle the states:

$$\delta(q_0, \boldsymbol{a}) = \delta(q_0, \boldsymbol{b}) = q_1, \qquad \delta(q_1, \boldsymbol{a}) = \delta(q_1, \boldsymbol{b}) = q_0.$$

Then $T(A, q_0)$ is the set of even-length strings, while $T(A, q_1)$ is the set of odd-length strings. These two sets are disjoint and their union is all of Σ^*, so in this example the terminal sets do form a partition.

Chapter 4

Nondeterministic Finite Automata — Solutions

4.1. Using the local Figure 4.25, the connected deterministic equivalents are as follows. In each table, $\emptyset$ is the dead state and all missing subsets from $\wp(S)$ are disconnected.

(a) Figure 4.25(a): start state $\{t\}$, final state $\{t\}$.

	a	b	c
$\{t\}$	$\{t\}$	$\emptyset$	$\{t\}$
$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$

(b) Figure 4.25(b): start state $\{q_1\}$, final state $\{q_1\}$.

	a	b	c
$\{q_1\}$	$\{q_0\}$	$\{q_1\}$	$\emptyset$
$\{q_0\}$	$\emptyset$	$\{q_1\}$	$\{q_0\}$
$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$

(c) Figure 4.25(c): start state $\{r_0\}$, final states $\{r_0\}$ and $\{r_0, r_1\}$.

	a	b	c
$\{r_0\}$	$\{r_0, r_1\}$	$\emptyset$	$\{r_1\}$
$\{r_0, r_1\}$	$\{r_0, r_1\}$	$\{r_0\}$	$\{r_1\}$
$\{r_1\}$	$\emptyset$	$\{r_0\}$	$\emptyset$
$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$

(d) Figure 4.25(d): start state $\{s_0\}$, final state $\{s_2\}$.

	a	b	c
$\{s_0\}$	$\{s_1\}$	$\emptyset$	$\emptyset$
$\{s_1\}$	$\{s_2\}$	$\{s_3\}$	$\emptyset$
$\{s_2\}$	$\emptyset$	$\emptyset$	$\emptyset$
$\{s_3\}$	$\emptyset$	$\{s_1\}$	$\emptyset$
$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$

(e) Figure 4.25(e): start state $\{s_1\}$, final state $\{s_1, s_2\}$.

	$\boldsymbol{a}$	$\boldsymbol{b}$	$\boldsymbol{c}$
$\{s_1\}$	$\{s_1, s_2\}$	$\emptyset$	$\emptyset$
$\{s_1, s_2\}$	$\{s_1, s_2\}$	$\{s_1\}$	$\emptyset$
$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$

(f) Figure 4.25(f): after taking the λ-closures, the connected DFA has start state $\{s_0\}$ and final state $\{s_0, s_1, s_3\}$.

	$\boldsymbol{a}$	$\boldsymbol{b}$	$\boldsymbol{c}$
$\{s_0\}$	$\{s_0, s_1\}$	$\emptyset$	$\emptyset$
$\{s_0, s_1\}$	$\{s_0, s_1\}$	$\{s_2\}$	$\emptyset$
$\{s_2\}$	$\emptyset$	$\emptyset$	$\{s_0, s_1, s_3\}$
$\{s_0, s_1, s_3\}$	$\{s_0, s_1\}$	$\{s_2\}$	$\emptyset$
$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$

(g) Figure 4.25(g): start state $\{s_0\}$, final state $\{s_0\}$.

	$\boldsymbol{a}$	$\boldsymbol{b}$	$\boldsymbol{c}$
$\{s_0\}$	$\{s_1\}$	$\emptyset$	$\emptyset$
$\{s_1\}$	$\{s_0\}$	$\{s_3\}$	$\{s_2\}$
$\{s_2\}$	$\emptyset$	$\emptyset$	$\{s_1\}$
$\{s_3\}$	$\emptyset$	$\{s_1\}$	$\emptyset$
$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$

4.2. Consider Example 4.17, that is, the automaton E of Figure 4.23. Reading the figure gives

$$E = \langle \{\boldsymbol{a}, \boldsymbol{b}\}, \{s_1, s_2, s_3\}, \{s_1\}, \delta_\lambda, \{s_2\} \rangle$$

with ordinary transitions

$$\delta_\lambda(s_1, \boldsymbol{a}) = \{s_1, s_2\}, \qquad \delta_\lambda(s_1, \boldsymbol{b}) = \{s_3\},$$
$$\delta_\lambda(s_2, \boldsymbol{a}) = \{s_1\}, \qquad \delta_\lambda(s_2, \boldsymbol{b}) = \{s_1\},$$
$$\delta_\lambda(s_3, \boldsymbol{a}) = \emptyset, \qquad \delta_\lambda(s_3, \boldsymbol{b}) = \{s_1\},$$

and one λ-transition $\delta_\lambda(s_1, \lambda) = \{s_2\}$.

(a) *Remove λ-transitions (Definition 4.9).* The λ-closures are:

$$\Lambda(s_1) = \{s_1, s_2\}, \qquad \Lambda(s_2) = \{s_2\}, \qquad \Lambda(s_3) = \{s_3\}.$$

Since $\lambda \in L(E)$, Definition 4.9 adds the isolated start state q_0 and also makes q_0 final. Thus

$$E_\lambda^\sigma = \langle \{\boldsymbol{a}, \boldsymbol{b}\}, \{q_0, s_1, s_2, s_3\}, \{q_0, s_1\}, \delta_\lambda^\sigma, \{q_0, s_2\} \rangle$$

with

$$\delta_\lambda^\sigma(q_0, \boldsymbol{a}) = \delta_\lambda^\sigma(q_0, \boldsymbol{b}) = \emptyset,$$
$$\delta_\lambda^\sigma(s_1, \boldsymbol{a}) = \{s_1, s_2\}, \qquad \delta_\lambda^\sigma(s_1, \boldsymbol{b}) = \{s_1, s_2, s_3\},$$
$$\delta_\lambda^\sigma(s_2, \boldsymbol{a}) = \{s_1, s_2\}, \qquad \delta_\lambda^\sigma(s_2, \boldsymbol{b}) = \{s_1, s_2\},$$
$$\delta_\lambda^\sigma(s_3, \boldsymbol{a}) = \emptyset, \qquad \delta_\lambda^\sigma(s_3, \boldsymbol{b}) = \{s_1, s_2\}.$$

(b) *Convert this NDFA into a DFA (Definition 4.5).* The full deterministic machine has $2^4 = 16$ subset states, but only three are reachable from the start set $\{q_0, s_1\}$:

$$Q_0 = \{q_0, s_1\}, \qquad Q_1 = \{s_1, s_2\}, \qquad Q_2 = \{s_1, s_2, s_3\}.$$

All three are final, because $q_0 \in Q_0$ and $s_2 \in Q_1 \cap Q_2$. Their transitions are

	$\boldsymbol{a}$	$\boldsymbol{b}$
Q_0	Q_1	Q_2
Q_1	Q_1	Q_2
Q_2	Q_1	Q_2

so the connected part of the DFA is a 3-state machine.

4.3. Consider the NDFA A from Example 4.4: $A = \langle\{\boldsymbol{a}, \boldsymbol{b}\}, \{r, s, t\}, \{r, s\}, \delta, \{t\}\rangle$ with

$$\delta(r, \boldsymbol{a}) = \{s\}, \quad \delta(r, \boldsymbol{b}) = \emptyset, \quad \delta(s, \boldsymbol{a}) = \emptyset, \quad \delta(s, \boldsymbol{b}) = \{r, t\},$$

$$\delta(t, \boldsymbol{a}) = \emptyset, \qquad \delta(t, \boldsymbol{b}) = \{t\}.$$

(a) *Circuit diagram (no <SOS> or <EOS>).* Using one activity bit for each state, the next-state equations are

$$r' = s \wedge \boldsymbol{b}, \qquad s' = r \wedge \boldsymbol{a}, \qquad t' = (s \wedge \boldsymbol{b}) \vee (t \wedge \boldsymbol{b}).$$

Initially the active states are r and s, and t is inactive.

(b) *Circuit diagram (with <SOS> and <EOS>).* <SOS> resets the activity bits to $r = s = 1,\ t = 0$. The accept line is $t \wedge$ <EOS>.

(c) *DFA via subset construction (connected portion only).* The start state of A^d is $\{r, s\}$. The reachable states are

$$\{r, s\}, \quad \{s\}, \quad \{r, t\}, \quad \{t\}, \quad \emptyset.$$

Their transition table is

State	$\boldsymbol{a}$	$\boldsymbol{b}$
$\{r, s\}$	$\{s\}$	$\{r, t\}$
$\{s\}$	$\emptyset$	$\{r, t\}$
$\{r, t\}$	$\{s\}$	$\{t\}$
$\{t\}$	$\emptyset$	$\{t\}$
$\emptyset$	$\emptyset$	$\emptyset$

The final states are $\{r, t\}$ and $\{t\}$.

(d) *DFA including disconnected portion.* The full state set is

$$\wp(\{r, s, t\}) = \{\emptyset, \{r\}, \{s\}, \{t\}, \{r, s\}, \{r, t\}, \{s, t\}, \{r, s, t\}\}.$$

Its transition table is

State	$\boldsymbol{a}$	$\boldsymbol{b}$
$\emptyset$	$\emptyset$	$\emptyset$
$\{r\}$	$\{s\}$	$\emptyset$
$\{s\}$	$\emptyset$	$\{r, t\}$
$\{t\}$	$\emptyset$	$\{t\}$
$\{r, s\}$	$\{s\}$	$\{r, t\}$
$\{r, t\}$	$\{s\}$	$\{t\}$
$\{s, t\}$	$\emptyset$	$\{r, t\}$
$\{r, s, t\}$	$\{s\}$	$\{r, t\}$

Final states are exactly those subsets containing t.

4.4. Consider the NDFA from Example 4.2, which has multiple transitions on **0**: from s_0 to both s_1 and s_2.

(a) *Circuit diagram (with <SOS> and <EOS>).* Use one activity flip-flop for each state $s_0, \ldots, s_5$. The next-state equations are

$$s_0' = \text{<SOS>}, \qquad s_1' = (s_0 \wedge \mathbf{0}) \vee (s_1 \wedge \mathbf{0}) \vee (s_5 \wedge \mathbf{1}),$$

$$s_2' = (s_0 \wedge \mathbf{0}) \vee (s_0 \wedge \mathbf{1}) \vee (s_4 \wedge \mathbf{0}) \vee (s_4 \wedge \mathbf{1}),$$

$$s_3' = (s_0 \wedge \mathbf{1}) \vee (s_1 \wedge \mathbf{1}) \vee (s_3 \wedge \mathbf{0}),$$

$$s_4' = (s_2 \wedge \mathbf{0}) \vee (s_2 \wedge \mathbf{1}), \qquad s_5' = (s_3 \wedge \mathbf{1}) \vee (s_5 \wedge \mathbf{0}).$$

Since the final states in Figure 4.3 are s_0, s_1, and s_4, the accept line is

$$(s_0 \vee s_1 \vee s_4) \wedge \text{<EOS>}.$$

(b) *DFA via subset construction.* Starting from the set of start state $\{s_0\}$, the connected portion of the subset DFA has the following reachable states:

$$Q_0 = \{s_0\}, \quad Q_1 = \{s_1, s_2\}, \quad Q_2 = \{s_2, s_3\}, \quad Q_3 = \{s_1, s_4\}, \quad Q_4 = \{s_3, s_4\},$$

$$Q_5 = \{s_4, s_5\}, \quad Q_6 = \{s_2, s_5\}.$$

Their transitions are

	0	**1**
Q_0	Q_1	Q_2
Q_1	Q_3	Q_4
Q_2	Q_4	Q_5
Q_3	Q_1	Q_2
Q_4	Q_2	Q_6
Q_5	Q_6	Q_3
Q_6	Q_5	Q_3

The final states are exactly those subsets containing one of the original final states s_0, s_1, or s_4, namely

$$Q_0,\ Q_1,\ Q_3,\ Q_4,\ Q_5.$$

4.5. Consider the NDFA from Example 4.3, which accepts the same language as Example 4.2 but uses multiple start states instead of multiple transitions.

(a) *Circuit diagram (no <SOS> or <EOS>).* Again use one activity flip-flop per state. For the automaton of Figure 4.4,

$$s_1' = (s_1 \wedge \mathbf{0}) \vee (s_5 \wedge \mathbf{0}), \qquad s_2' = (s_4 \wedge \mathbf{0}) \vee (s_4 \wedge \mathbf{1}),$$

$$s_3' = (s_1 \wedge \mathbf{1}) \vee (s_3 \wedge \mathbf{0}), \qquad s_4' = (s_2 \wedge \mathbf{0}) \vee (s_2 \wedge \mathbf{1}),$$

$$s_5' = s_3 \wedge \mathbf{1}.$$

The initial configuration has $s_1 = s_4 = 1$ and the other activity bits equal to 0. Since the final states are s_1 and s_4, the accept line is $s_1 \vee s_4$.

(b) *Circuit diagram (with <SOS> and <EOS>).* Use the same next-state equations as in part (a), but OR the start pulse into the two start-state inputs:

$$s_1' = \text{<SOS>} \vee (s_1 \wedge \mathbf{0}) \vee (s_5 \wedge \mathbf{0}), \qquad s_4' = \text{<SOS>} \vee (s_2 \wedge \mathbf{0}) \vee (s_2 \wedge \mathbf{1}),$$

with the equations for s_2', s_3', and s_5' unchanged. The accept output is

$$(s_1 \vee s_4) \wedge \text{<EOS>}.$$

(c) *DFA (connected portion only).* The start state of A^d is the set of all start states, namely $\{s_1, s_4\}$. The reachable subsets are

$$P_0 = \{s_1, s_4\}, \quad P_1 = \{s_1, s_2\}, \quad P_2 = \{s_2, s_3\}, \quad P_3 = \{s_3, s_4\}, \quad P_4 = \{s_4, s_5\}, \quad P_5 = \{s_2, s_5\},$$

$$P_6 = \{s_2\}, \qquad P_7 = \{s_4\}.$$

Their transitions are

	0	**1**
P_0	P_1	P_2
P_1	P_0	P_3
P_2	P_3	P_4
P_3	P_2	P_5
P_4	P_1	P_6
P_5	P_0	P_7
P_6	P_7	P_7
P_7	P_6	P_6

The final states are

$$P_0,\ P_1,\ P_3,\ P_4,\ P_7,$$

since these are exactly the reachable subsets containing s_1 or s_4.

(d) *Is this DFA isomorphic to any of those in Exercise 4.4?* No. The DFA from Exercise 4.4(b) has seven reachable states

$$Q_0, \ldots, Q_6,$$

while the DFA from part (c) above has eight reachable states

$$P_0, \ldots, P_7.$$

Since isomorphic automata must have the same number of states, these two subset-construction DFAs are not isomorphic. They do accept the same language, but that only implies that their *minimal* DFAs are isomorphic, not these unreduced machines.

4.6. Consider the λ-NDFA from Example 4.14, with states $\{s_0, s_1, s_2, s_3\}$ and λ-transitions $s_0 \to s_1$ and $s_1 \to s_2$.

(a) *Circuit diagram for the original λ-NDFA (no SOS/EOS).* Figure 4.19 uses four activity bits, one for each state. Ignoring the λ-moves for a moment, the ordinary transitions give

$$s_0' = 0, \qquad s_1' = (s_0 \wedge \boldsymbol{a}) \vee (s_3 \wedge \boldsymbol{b}),$$

$$s_2' = (s_1 \wedge \boldsymbol{c}) \vee (s_2 \wedge \boldsymbol{d}), \qquad s_3' = s_1 \wedge \boldsymbol{b}.$$

The λ-moves $s_0 \to s_1$ and $s_1 \to s_2$ are implemented exactly as in Example 4.17 by OR-ing the source activity line directly into the destination input. Thus the circuit can be specified by

$$D_0 = 0, \qquad D_1 = s_0 \vee (s_0 \wedge \boldsymbol{a}) \vee (s_3 \wedge \boldsymbol{b}),$$

$$D_2 = s_1 \vee (s_1 \wedge \boldsymbol{c}) \vee (s_2 \wedge \boldsymbol{d}), \qquad D_3 = s_1 \wedge \boldsymbol{b}.$$

Initially only s_0 is active; the accept output is simply s_2.

(b) *Circuit diagram for the NDFA from Definition 4.9 (part (b) of Example 4.14, no SOS/EOS).* The corresponding λ-free automaton is the one shown in Figure 4.20. Use one activity bit for each of q_0, s_0, s_1, s_2, s_3; then

$$q_0' = 0, \qquad s_0' = 0,$$

$$s_1' = (s_0 \wedge \boldsymbol{a}) \vee (s_3 \wedge \boldsymbol{b}),$$

$$s_2' = (s_0 \wedge \boldsymbol{a}) \vee (s_0 \wedge \boldsymbol{c}) \vee (s_0 \wedge \boldsymbol{d}) \vee (s_1 \wedge \boldsymbol{c}) \vee (s_1 \wedge \boldsymbol{d}) \vee (s_3 \wedge \boldsymbol{b}) \vee (s_2 \wedge \boldsymbol{d}),$$

$$s_3' = (s_0 \wedge \boldsymbol{b}) \vee (s_1 \wedge \boldsymbol{b}).$$

The initial activity bits are $q_0 = s_0 = 1$ and $s_1 = s_2 = s_3 = 0$, and the accept output is $q_0 \vee s_2$.

4.7. Consider the second NDFA of Example 4.16, which accepts the language $L^+ = \{x_1 x_2 \cdots x_k \mid k \geq 1, x_i \in L\}$ where L is the language of strings over $\{\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}\}$ containing exactly one $\boldsymbol{b}$ immediately followed by $\boldsymbol{c}$.

(a) *Circuit diagram for A_λ^σ (with EOS but no SOS).* For the λ-NDFA of Figure 4.22, use one activity bit for each of s_0, s_1, s_2. The ordinary transitions give

$$s_0' = s_0 \wedge (\boldsymbol{a} \vee \boldsymbol{c}), \qquad s_1' = s_0 \wedge \boldsymbol{b},$$

$$s_2' = (s_1 \wedge \boldsymbol{c}) \vee (s_2 \wedge (\boldsymbol{a} \vee \boldsymbol{c})).$$

Because Figure 4.22 also has the λ-transition $s_2 \to s_0$, OR the activity line for s_2 directly into the input for s_0. Thus

$$D_0 = s_2 \vee (s_0 \wedge (\boldsymbol{a} \vee \boldsymbol{c})), \qquad D_1 = s_0 \wedge \boldsymbol{b},$$

$$D_2 = (s_1 \wedge \boldsymbol{c}) \vee (s_2 \wedge (\boldsymbol{a} \vee \boldsymbol{c})).$$

The accept line is $s_2 \wedge$ <EOS>.

(b) *NDFA without λ-transitions (Definition 4.9).* For Figure 4.22 the λ-closures are

$$\Lambda(s_0) = \{s_0\}, \qquad \Lambda(s_1) = \{s_1\}, \qquad \Lambda(s_2) = \{s_0, s_2\}.$$

Since $\lambda \not{\!in} L(A_\lambda)$, Definition 4.9 adds q_0 as an extra disconnected start state but does not make it final. Thus

$$A_\lambda^\sigma = \langle\{\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}\}, \{q_0, s_0, s_1, s_2\}, \{q_0, s_0\}, \delta_\lambda^\sigma, \{s_2\}\rangle,$$

where

$$\delta_\lambda^\sigma(q_0, \boldsymbol{a}) = \delta_\lambda^\sigma(q_0, \boldsymbol{b}) = \delta_\lambda^\sigma(q_0, \boldsymbol{c}) = \emptyset,$$

$$\delta_\lambda^\sigma(s_0, \boldsymbol{a}) = \{s_0\}, \qquad \delta_\lambda^\sigma(s_0, \boldsymbol{b}) = \{s_1\}, \qquad \delta_\lambda^\sigma(s_0, \boldsymbol{c}) = \{s_0\},$$

$$\delta_\lambda^\sigma(s_1, \boldsymbol{a}) = \emptyset, \qquad \delta_\lambda^\sigma(s_1, \boldsymbol{b}) = \emptyset, \qquad \delta_\lambda^\sigma(s_1, \boldsymbol{c}) = \{s_0, s_2\},$$

$$\delta_\lambda^\sigma(s_2, \boldsymbol{a}) = \{s_0, s_2\}, \qquad \delta_\lambda^\sigma(s_2, \boldsymbol{b}) = \{s_1\}, \qquad \delta_\lambda^\sigma(s_2, \boldsymbol{c}) = \{s_0, s_2\}.$$

(c) *Circuit diagram for the λ-free NDFA (with EOS, no SOS).* Use one activity bit for each of q_0, s_0, s_1, s_2. The next-state equations are

$$q_0' = 0,$$

$$s_0' = (s_0 \wedge \boldsymbol{a}) \vee (s_0 \wedge \boldsymbol{c}) \vee (s_1 \wedge \boldsymbol{c}) \vee (s_2 \wedge \boldsymbol{a}) \vee (s_2 \wedge \boldsymbol{c}),$$

$$s_1' = (s_0 \wedge \boldsymbol{b}) \vee (s_2 \wedge \boldsymbol{b}),$$

$$s_2' = (s_1 \wedge \boldsymbol{c}) \vee (s_2 \wedge \boldsymbol{a}) \vee (s_2 \wedge \boldsymbol{c}).$$

The initial configuration has $q_0 = s_0 = 1$ and $s_1 = s_2 = 0$, and the accept line is $s_2 \wedge$ <EOS>.

(d) *DFA (connected portion only).* Starting from $\{q_0, s_0\}$, the reachable states are

$$Q_0 = \{q_0, s_0\}, \qquad Q_1 = \{s_0\}, \qquad Q_2 = \{s_1\}, \qquad Q_3 = \emptyset, \qquad Q_4 = \{s_0, s_2\}.$$

Their transitions are

	$\boldsymbol{a}$	$\boldsymbol{b}$	$\boldsymbol{c}$
Q_0	Q_1	Q_2	Q_1
Q_1	Q_1	Q_2	Q_1
Q_2	Q_3	Q_3	Q_4
Q_3	Q_3	Q_3	Q_3
Q_4	Q_4	Q_2	Q_4

The only final state is Q_4.

4.8. **Complementing the final states of an NDFA does not yield a machine accepting the complement language.**

For a DFA, $L(\overline{A}) = \Sigma^* \setminus L(A)$ because every word follows exactly one path and is accepted iff it ends in a final state. Flipping final and nonfinal states reverses acceptance on that unique path.

For an NDFA, a word w may follow *multiple* paths, some ending in final states and some not. Under Definition 4.4, $w \in L(A)$ iff *at least one* path ends in a final state. If we complement the final states, a word is accepted by the modified machine iff at least one path ends in a *formerly nonfinal* state. This is *not* the same as $w \not{\!in} L(A)$, because a word can simultaneously have a path to a (formerly) final state and a path to a (formerly) nonfinal state; the complemented machine would accept it, but it was already in $L(A)$.

Example. Let $\Sigma = \{\boldsymbol{a}\}$, $S = \{s_0, s_1\}$, $S_0 = \{s_0\}$, $F = \{s_1\}$, $\delta(s_0, \boldsymbol{a}) = \{s_0, s_1\}$, $\delta(s_1, \boldsymbol{a}) = \emptyset$. Then $\boldsymbol{a} \in L(A)$ because $s_1 \in \delta(s_0, \boldsymbol{a})$. After complementing, $F' = \{s_0\}$, and $\boldsymbol{a}$ is still accepted (via the path $s_0 \to s_0 \in F'$), yet $\boldsymbol{a} \not{i} n\Sigma^* \setminus L(A)$ since $\boldsymbol{a} \in L(A)$. Hence the complement-of-final-states trick fails for NDFAs.

4.9. **Given an NDFA A without λ-transitions, construct a λ-NDFA A' with exactly one start state, exactly one final state, and $L(A') = L(A)$.**

Construction. Let $A = \langle \Sigma, S, S_0, \delta, F \rangle$. Define $A' = \langle \Sigma, S', q_0, \delta', \{q_f\} \rangle$ where:

- $S' = S \cup \{q_0, q_f\}$ (two new states).
- q_0 is the unique start state.
- $\delta'(q_0, \lambda) = S_0$ (a λ-transition from q_0 to every start state of A).
- For all $s \in S$ and $\boldsymbol{a} \in \Sigma$: $\delta'(s, \boldsymbol{a}) = \delta(s, \boldsymbol{a})$.
- $\delta'(f, \lambda) \ni q_f$ for every $f \in F$ (λ-transitions from each old final state to the new unique final state q_f).
- $\delta'(q_f, \boldsymbol{a}) = \emptyset$ for all $\boldsymbol{a}$ (and no transitions out of q_f).

Correctness. A word $w \in \Sigma^+$ is accepted by A iff there is a path $q_0 \xrightarrow{\lambda} s_i \xrightarrow{w} f$ for some $s_i \in S_0$ and $f \in F$; by the λ-transition $f \xrightarrow{\lambda} q_f$, this same path exists in A' ending at q_f. Conversely, any accepting path in A' uses the initial λ-step to some $s_i \in S_0$, processes w within S (by identical transitions), then uses a final λ-step to q_f, which corresponds to an accepting path in A.

If $\lambda \in L(A)$, add the extra λ-transition $\delta'(q_0, \lambda) \ni q_f$. Then λ is accepted while A' still has exactly one start state (q_0) and exactly one final state (q_f). Hence $L(A') = L(A)$ in all cases.

4.10. **Can part (ii) of Definition 4.8 be deduced from parts (i) and (iii)?**

Definition 4.8 defines $\overline{\delta}$ for NDFAs:

(i) $\overline{\delta}(s, \lambda) = \{s\}$,

(ii) $(\forall \boldsymbol{a} \in \Sigma)\overline{\delta}(s, \boldsymbol{a}) = \delta(s, \boldsymbol{a})$,

(iii) $(\forall x \in \Sigma^*)(\forall \boldsymbol{a} \in \Sigma)\overline{\delta}(s, x\boldsymbol{a}) = \bigcup_{t \in \overline{\delta}(s,x)} \delta(t, \boldsymbol{a})$.

Part (ii) concerns single-letter strings. Setting $x = \lambda$ in part (iii):

$$\overline{\delta}(s, \lambda\boldsymbol{a}) = \bigcup_{t \in \overline{\delta}(s,\lambda)} \delta(t, \boldsymbol{a}) = \bigcup_{t \in \{s\}} \delta(t, \boldsymbol{a}) = \delta(s, \boldsymbol{a}),$$

where we used part (i) to obtain $\overline{\delta}(s, \lambda) = \{s\}$ and the fact that $\lambda\boldsymbol{a} = \boldsymbol{a}$. Since $\lambda\boldsymbol{a} = \boldsymbol{a}$, this gives $\overline{\delta}(s, \boldsymbol{a}) = \delta(s, \boldsymbol{a})$, which is exactly part (ii).

Conclusion. Yes, part (ii) can be derived from parts (i) and (iii). It is included explicitly to make the base case of inductive proofs more transparent, but it is logically redundant.

4.11. **Alternative construction for removing λ-transitions.**

Define $S' = S$, $S'_0 = \Lambda(S_0)$, $F' = F$, and $\delta'(s, \boldsymbol{a}) = \overline{\delta}_\lambda(s, \boldsymbol{a})$ for all $s \in S$, $\boldsymbol{a} \in \Sigma$.

This construction does work. Let

$$A' = \langle \Sigma, S, \Lambda(S_0), \delta', F \rangle.$$

We show $L(A') = L(A_\lambda)$.

First, for every $s \in S$ and every nonempty word $x \in \Sigma^+$,

$$\overline{\delta'}(s,x) = \overline{\delta}_\lambda(s,x).$$

This is proved by the same induction as Lemma 4.2: for a single letter $\boldsymbol{a}$, it is just the definition $\delta'(s,\boldsymbol{a}) = \overline{\delta}_\lambda(s,\boldsymbol{a})$, and the inductive step extends in the usual nondeterministic manner.

Case 1: $x = \lambda$**.** Then

$$\lambda \in L(A') \iff \Lambda(S_0) \cap F \neq \emptyset.$$

But this says exactly that some start state of A_λ can reach a final state by λ-moves alone, which is the condition $\lambda \in L(A_\lambda)$.

Case 2: $x \in \Sigma^+$**.** Using the identity above,

$$x \in L(A') \iff \Big(\bigcup_{t \in \Lambda(S_0)} \overline{\delta'}(t,x)\Big) \cap F \neq \emptyset \iff \Big(\bigcup_{t \in \Lambda(S_0)} \overline{\delta}_\lambda(t,x)\Big) \cap F \neq \emptyset.$$

If $t \in \Lambda(S_0)$, then $t \in \Lambda(s_0)$ for some $s_0 \in S_0$, so the initial λ-moves taking s_0 to t may be prefixed to any path counted in $\overline{\delta}_\lambda(t,x)$. Hence

$$\bigcup_{t \in \Lambda(S_0)} \overline{\delta}_\lambda(t,x) = \bigcup_{s_0 \in S_0} \overline{\delta}_\lambda(s_0,x),$$

and therefore

$$x \in L(A') \iff \Big(\bigcup_{s_0 \in S_0} \overline{\delta}_\lambda(s_0,x)\Big) \cap F \neq \emptyset \iff x \in L(A_\lambda).$$

So this is a valid alternative to Definition 4.9: instead of adding q_0, it enlarges the start-state set to $\Lambda(S_0)$.

4.12. **Algorithm for union of two FAD languages using λ-NDFAs.**

Let $L_1 = L(A_1)$ and $L_2 = L(A_2)$ where A_1 and A_2 are λ-NDFAs. By Exercise 4.9 we may assume each A_i has exactly one start state q_0^i and exactly one final state q_f^i. Construct A' as follows:

(1) Introduce a new start state q_0 and a new final state q_f.

(2) Add λ-transitions: $q_0 \xrightarrow{\lambda} q_0^1$ and $q_0 \xrightarrow{\lambda} q_0^2$.

(3) Add λ-transitions: $q_f^1 \xrightarrow{\lambda} q_f$ and $q_f^2 \xrightarrow{\lambda} q_f$.

(4) Keep all existing transitions of A_1 and A_2.

Then $L(A') = L(A_1) \cup L(A_2)$: a word is accepted iff it is accepted by A_1 or by A_2, because the λ-transitions route the computation into one of the two machines and collect the result at q_f.

4.13. **Example of NDFA A such that A^d is not connected, not reduced, but is minimal.**

(a) A^d is not connected. Any NDFA in which some subsets of states are never reachable from the start set under the transition function will produce an A^d with unreachable states. For example, let $\Sigma = \{\boldsymbol{a}\}$, $S = \{s_0, s_1\}$, $S_0 = \{s_0\}$, $F = \{s_1\}$, $\delta(s_0, \boldsymbol{a}) = \{s_1\}$, $\delta(s_1, \boldsymbol{a}) = \emptyset$. Then A^d has states $\{\emptyset, \{s_0\}, \{s_1\}, \{s_0, s_1\}\}$ but $\{s_0, s_1\}$ and $\{s_1\}$ with $\delta^d(\{s_1\}, \boldsymbol{a}) = \emptyset$ are reachable, while $\{s_0, s_1\}$ is not reached from $\{s_0\}$.

(b) A^d is not reduced. Use a different example:

$$A = \langle \{\boldsymbol{a}, \boldsymbol{b}\}, \{p, u, v\}, \{p\}, \delta, \{u, v\} \rangle$$

where

$$\delta(p, \boldsymbol{a}) = \{u\}, \qquad \delta(p, \boldsymbol{b}) = \{v\},$$

and all other transitions are empty. Then the reachable subset states of A^d are

$$\{p\}, \qquad \{u\}, \qquad \{v\}, \qquad \emptyset.$$

Both $\{u\}$ and $\{v\}$ are final, and from either one every input leads to $\emptyset$. Therefore

$$\{u\} \; E_{A^d} \; \{v\},$$

so A^d is not reduced.

(c) A^d is minimal. Here we use a different example. Let $A = \langle \{\boldsymbol{a}, \boldsymbol{b}\}, \{s\}, \{s\}, \delta, \{s\} \rangle$ where

$$\delta(s, \boldsymbol{a}) = \{s\}, \qquad \delta(s, \boldsymbol{b}) = \emptyset.$$

Then $L(A) = \boldsymbol{a}^*$. The deterministic machine A^d has exactly two states, $\{s\}$ and $\emptyset$, with

$$\delta^d(\{s\}, \boldsymbol{a}) = \{s\}, \quad \delta^d(\{s\}, \boldsymbol{b}) = \emptyset, \quad \delta^d(\emptyset, \boldsymbol{a}) = \delta^d(\emptyset, \boldsymbol{b}) = \emptyset.$$

This is the standard 2-state DFA for $\boldsymbol{a}^*$ over the alphabet $\{\boldsymbol{a}, \boldsymbol{b}\}$, so A^d is minimal.

4.14. **Why was the extra state q_0 included in A_λ^σ?**

The role of q_0 is to preserve acceptance of λ while leaving the original start states and all ordinary transitions unchanged. In Definition 4.9 the extra state q_0 is disconnected from the rest of the machine, and

$$q_0 \in F^\sigma \iff \lambda \in L(A_\lambda).$$

Thus q_0 serves as a separate witness for the empty word.

Example. Let

$$A_\lambda = \langle \{\boldsymbol{a}\}, \{s_0, f\}, \{s_0\}, \delta_\lambda, \{f\} \rangle$$

where $\delta_\lambda(s_0, \lambda) = \{f\}$ and all ordinary transitions are empty. Then $\lambda \in L(A_\lambda)$ because $f \in \Lambda(s_0)$.

If we try to remove λ-moves *without* adding q_0 and keep the same start state s_0, then the new machine has no way to accept λ: no input is read, and s_0 is not final. The disconnected state q_0 fixes exactly this problem. We keep s_0 for all nonempty words, and we make q_0 final so that the empty word is still accepted.

4.15. **Union construction using NDFAs *without* λ-transitions.**

(a) **Algorithm.** Let $A_1 = \langle \Sigma, S_1, S_0^1, \delta_1, F_1 \rangle$ and $A_2 = \langle \Sigma, S_2, S_0^2, \delta_2, F_2 \rangle$ (assume $S_1 \cap S_2 = \emptyset$). Construct $A' = \langle \Sigma, S_1 \cup S_2, S_0^1 \cup S_0^2, \delta', F_1 \cup F_2 \rangle$ where $\delta'(s, \boldsymbol{a}) = \delta_i(s, \boldsymbol{a})$ for $s \in S_i$. The machine A' starts in the combined set of initial states and accepts any word accepted by either A_1 or A_2. Thus $L(A') = L(A_1) \cup L(A_2)$.

(b) **Comparison.** This is actually *easier* than the λ-transition approach (Exercise 4.12): no new states are needed at all, since we simply merge the state sets and initial-state sets.

(c) **Can we ensure exactly one start state and one final state?** Not in full generality. The disjoint-union construction in part (a) gives a correct union machine, but it usually has several start states and several final states. If the resulting language does *not* contain λ, then Exercise 4.22 shows how to convert a no-λ NDFA into another no-λ NDFA with exactly one start state and exactly one final state. But Exercise 4.23 shows that this can fail when λ is in the language. So the correct conclusion is: one start state and one final state can be forced under the hypothesis $\lambda \not{i} nL$, but not uniformly in general.

4.16. Consider the λ-NDFA A_λ^σ from Example 4.15 (the machine without λ-transitions corresponding to the automaton of Example 4.14).

(a) *Circuit diagram (no SOS or EOS).* Example 4.15 gives the λ-free machine of Figure 4.20, so we can use one activity bit for each of q_0, s_0, s_1, s_2, s_3. The next-state equations are

$$q_0' = 0, \qquad s_0' = 0,$$

$$s_1' = (s_0 \wedge \boldsymbol{a}) \vee (s_3 \wedge \boldsymbol{b}),$$

$$s_2' = (s_0 \wedge \boldsymbol{a}) \vee (s_0 \wedge \boldsymbol{c}) \vee (s_0 \wedge \boldsymbol{d}) \vee (s_1 \wedge \boldsymbol{c}) \vee (s_1 \wedge \boldsymbol{d}) \vee (s_3 \wedge \boldsymbol{b}) \vee (s_2 \wedge \boldsymbol{d}),$$

$$s_3' = (s_0 \wedge \boldsymbol{b}) \vee (s_1 \wedge \boldsymbol{b}).$$

The initial configuration has $q_0 = s_0 = 1$ and $s_1 = s_2 = s_3 = 0$. Since q_0 and s_2 are final states in Example 4.15, the accept output is $q_0 \vee s_2$.

(b) *Convert to $A_\lambda^{\sigma d}$ (connected portion only).* In Example 4.15, the start states of A_λ^σ are

$$\{q_0, s_0\},$$

and the final states are q_0 and s_2. The connected part of the deterministic equivalent has the reachable states

$$R_0 = \{q_0, s_0\}, \quad R_1 = \{s_1, s_2\}, \quad R_2 = \{s_3\}, \quad R_3 = \{s_2\}, \quad R_4 = \emptyset.$$

Their transitions are

	$\boldsymbol{a}$	$\boldsymbol{b}$	$\boldsymbol{c}$	$\boldsymbol{d}$
R_0	R_1	R_2	R_3	R_3
R_1	R_4	R_2	R_3	R_3
R_2	R_4	R_1	R_4	R_4
R_3	R_4	R_4	R_4	R_3
R_4	R_4	R_4	R_4	R_4

The final states are

$$R_0, \ R_1, \ R_3,$$

because these are exactly the reachable subsets containing q_0 or s_2.

4.17. (a) **Proof that $\overline{\delta}(s,\boldsymbol{a}) = \delta(s,\boldsymbol{a})$ for any NDFA without λ-transitions.**

By Definition 4.8(ii), $\overline{\delta}(s,\boldsymbol{a}) = \delta(s,\boldsymbol{a})$ for all $s \in S$ and $\boldsymbol{a} \in \Sigma$. This is immediate from the definition; alternatively, apply part (iii) with $x = \lambda$:

$$\overline{\delta}(s,\lambda\boldsymbol{a}) = \bigcup_{t\in\overline{\delta}(s,\lambda)} \delta(t,\boldsymbol{a}) = \bigcup_{t\in\{s\}} \delta(t,\boldsymbol{a}) = \delta(s,\boldsymbol{a}),$$

using part (i): $\overline{\delta}(s,\lambda) = \{s\}$. Since $\lambda\boldsymbol{a} = \boldsymbol{a}$, we conclude $\overline{\delta}(s,\boldsymbol{a}) = \delta(s,\boldsymbol{a})$. □

(b) **Counterexample for λ-NDFAs.**

Let $\Sigma = \{\boldsymbol{a}\}$, $S = \{s_0, s_1\}$, $S_0 = \{s_0\}$, $F = \{s_1\}$, $\delta_\lambda(s_0,\lambda) = \{s_1\}$, $\delta_\lambda(s_0,\boldsymbol{a}) = \emptyset$, $\delta_\lambda(s_1,\boldsymbol{a}) = \emptyset$.
Then $\Lambda(s_0) = \{s_0, s_1\}$. So

$$\overline{\delta}_\lambda(s_0,\boldsymbol{a}) = \Lambda\Big(\bigcup_{t\in\Lambda(s_0)} \delta_\lambda(t,\boldsymbol{a})\Big) = \Lambda(\emptyset\cup\emptyset) = \emptyset,$$

but $\delta_\lambda(s_0,\boldsymbol{a}) = \emptyset$ also. Let us try a different example: $\delta_\lambda(s_0,\lambda) = \{s_1\}$, $\delta_\lambda(s_1,\boldsymbol{a}) = \{s_1\}$. Then $\delta_\lambda(s_0,\boldsymbol{a}) = \emptyset$, yet

$$\overline{\delta}_\lambda(s_0,\boldsymbol{a}) = \Lambda\Big(\bigcup_{t\in\{s_0,s_1\}} \delta_\lambda(t,\boldsymbol{a})\Big) = \Lambda(\{s_1\}) = \{s_1\} \neq \emptyset = \delta_\lambda(s_0,\boldsymbol{a}).$$

This demonstrates that $\overline{\delta}_\lambda(s,\boldsymbol{a}) \neq \delta_\lambda(s,\boldsymbol{a})$ is possible for λ-NDFAs.

4.18. Consider the NDFA accepting the original language L from Example 4.10: $L = \{x \in \{\boldsymbol{a},\boldsymbol{b}\}^* \mid x \text{ begins with } \boldsymbol{a} \vee x \text{ contains } \boldsymbol{ba}\}$.

(a) *Circuit diagram (no SOS/EOS).* The NDFA has a start state q_0, two final states q_1, q_3, and the transitions of Figure 4.10. Using one activity bit per state, the next-state equations are

$$q_0' = 0, \qquad q_1' = (q_0 \wedge \boldsymbol{a}) \vee (q_1 \wedge \boldsymbol{a}) \vee (q_1 \wedge \boldsymbol{b}),$$

$$q_2' = (q_0 \wedge \boldsymbol{b}) \vee (q_2 \wedge \boldsymbol{b}), \qquad q_3' = (q_2 \wedge \boldsymbol{a}) \vee (q_3 \wedge \boldsymbol{a}) \vee (q_3 \wedge \boldsymbol{b}).$$

Initially only q_0 is active, and the accept line is $q_1 \vee q_3$.

(b) *DFA (connected portion only).* Apply the subset construction to Figure 4.10. Starting from $\{q_0\}$, the reachable states are

$$Q_0 = \{q_0\}, \qquad Q_1 = \{q_1\}, \qquad Q_2 = \{q_2\}, \qquad Q_3 = \{q_3\}.$$

Their transitions are

	$\boldsymbol{a}$	$\boldsymbol{b}$
Q_0	Q_1	Q_2
Q_1	Q_1	Q_1
Q_2	Q_3	Q_2
Q_3	Q_3	Q_3

The final states are Q_1 and Q_3.

4.19. Consider the NDFA accepting $L^r = \{x \in \{\boldsymbol{a},\boldsymbol{b}\}^* \mid x \text{ ends with } \boldsymbol{a} \vee x \text{ contains } \boldsymbol{ab}\}$ from Example 4.10.

(a) *Circuit diagram (no SOS/EOS).* This NDFA is obtained by reversing the arrows in Figure 4.10 and swapping the start and final states. Using activity bits q_0, q_1, q_2, q_3, the next-state equations are

$$q_0' = (q_1 \wedge \boldsymbol{a}) \vee (q_2 \wedge \boldsymbol{b}),$$

$$q_1' = (q_1 \wedge \boldsymbol{a}) \vee (q_1 \wedge \boldsymbol{b}), \qquad q_2' = (q_3 \wedge \boldsymbol{a}) \vee (q_2 \wedge \boldsymbol{b}),$$

$$q_3' = (q_3 \wedge \boldsymbol{a}) \vee (q_3 \wedge \boldsymbol{b}).$$

The initial configuration has $q_1 = q_3 = 1$ and $q_0 = q_2 = 0$, and the accept line is q_0.

(b) *DFA (connected portion only).* From Figure 4.11, the NDFA has start states $\{q_1, q_3\}$ and only one final state, q_0. Its nonempty transitions are

$$\delta(q_1, \boldsymbol{a}) = \{q_1, q_0\}, \qquad \delta(q_1, \boldsymbol{b}) = \{q_1\},$$

$$\delta(q_2, \boldsymbol{b}) = \{q_2, q_0\}, \qquad \delta(q_3, \boldsymbol{a}) = \{q_3, q_2\}, \qquad \delta(q_3, \boldsymbol{b}) = \{q_3\}.$$

Starting from $Q_0 = \{q_1, q_3\}$, Definition 4.5 gives

$$\delta^d(Q_0, \boldsymbol{a}) = \{q_0, q_1, q_2, q_3\} = Q_1, \qquad \delta^d(Q_0, \boldsymbol{b}) = \{q_1, q_3\} = Q_0.$$

From Q_1 we get

$$\delta^d(Q_1, \boldsymbol{a}) = Q_1, \qquad \delta^d(Q_1, \boldsymbol{b}) = Q_1.$$

Therefore the connected DFA has just two states:

$$Q_0 = \{q_1, q_3\} \quad \text{(nonfinal)}, \qquad Q_1 = \{q_0, q_1, q_2, q_3\} \quad \text{(final)},$$

with transition table

	$\boldsymbol{a}$	$\boldsymbol{b}$
Q_0	Q_1	Q_0
Q_1	Q_1	Q_1

4.20. **Are the pumping lemma arguments still valid for NDFAs?**

Yes. The pumping lemma (Chapter 2) applies to any language accepted by a DFA (or equivalently, any FAD language). Since every NDFA A is equivalent to a DFA A^d (Theorem 4.1), $L(A) = L(A^d)$ is FAD, and the pumping lemma applies to $L(A)$.

More directly: the pumping lemma is a property of the *language*, not the machine representation. If L is FAD (accepted by some NDFA), then all the arguments from Chapter 2 go through unchanged. In particular, if A has n states, then A^d has at most 2^n states; words of length at least 2^n must cause a repeated state in A^d, yielding the pumping decomposition.

4.21. **Does the conclusion of Theorem 2.7 still hold for NDFAs?**

Yes. Theorem 2.7 characterizes FAD languages via the Myhill–Nerode theorem: L is FAD iff $\equiv_L$ has finitely many equivalence classes. This is a property of the language L, not of the particular automaton (DFA or NDFA) accepting it. Since NDFAs and DFAs accept exactly the same class of languages (the FAD languages, by Theorem 4.1), the conclusion holds: a language is accepted by some NDFA iff it is accepted by some DFA iff $\equiv_L$ has finitely many classes iff the minimal DFA is finite.

4.22. **Later no-λ exercise: given an NDFA A without λ-transitions with $\lambda \not{i}nL(A)$, construct A'' with exactly one start state, one final state, and $L(A'') = L(A)$.**

Construction. Let $A = \langle \Sigma, S, S_0, \delta, F \rangle$ with $\lambda \not{i}nL(A)$, so $S_0 \cap F = \emptyset$. Define

$$A'' = \langle \Sigma, S \cup \{q_0, q_f\}, \{q_0\}, \delta'', \{q_f\} \rangle$$

where q_0 and q_f are new states and:

1. For each $\boldsymbol{a} \in \Sigma$, let

$$\delta''(q_0, \boldsymbol{a}) = \Big(\bigcup_{s \in S_0} \delta(s, \boldsymbol{a}) \Big) \cup \begin{cases} \{q_f\} & \text{if } (\bigcup_{s \in S_0} \delta(s, \boldsymbol{a})) \cap F \neq \emptyset, \\ \emptyset & \text{otherwise.} \end{cases}$$

2. For each old state $s \in S$ and each $\boldsymbol{a} \in \Sigma$, let

$$\delta''(s, \boldsymbol{a}) = \delta(s, \boldsymbol{a}) \cup \begin{cases} \{q_f\} & \text{if } \delta(s, \boldsymbol{a}) \cap F \neq \emptyset, \\ \emptyset & \text{otherwise.} \end{cases}$$

3. $\delta''(q_f, \boldsymbol{a}) = \emptyset$ for every $\boldsymbol{a} \in \Sigma$.

This machine has exactly one start state (q_0) and exactly one final state (q_f).

Why this preserves the language. Since $\lambda \not{i}nL(A)$, every accepted word has positive length. Suppose $w = \boldsymbol{a}_1 \cdots \boldsymbol{a}_n \in L(A)$. Then there is a path in A from some start state to some final state that reads all of w. In A'', use the same path for the first $n-1$ letters. On the last letter, when the old path would move into a final state, choose the extra target q_f instead. Thus $w \in L(A'')$.

Conversely, any accepting path in A'' must end in q_f. The only way to enter q_f is on some actual input symbol $\boldsymbol{a}$ from q_0 or from an old state s, and this happens only when the corresponding old transition set contains at least one state of F. Replacing that last step by a move into one of those original final states yields an accepting path in A. Hence $L(A'') = L(A)$.

4.23. **If $\lambda \in L(A)$, the three conditions of Exercise 4.22 may be incompatible (without λ-transitions).**

Example. Let

$$A = \langle \{\boldsymbol{a}\}, \{s_0, s_1\}, \{s_0\}, \delta, \{s_0, s_1\} \rangle$$

where $\delta(s_0, \boldsymbol{a}) = \{s_1\}$ and $\delta(s_1, \boldsymbol{a}) = \emptyset$. Then

$$L(A) = \{\lambda, \boldsymbol{a}\}.$$

We claim there is no NDFA without λ-transitions that has exactly one start state, exactly one final state, and also accepts exactly $\{\lambda, \boldsymbol{a}\}$.

Assume to the contrary that

$$B = \langle \{\boldsymbol{a}\}, S, \{q\}, \delta_B, \{f\} \rangle$$

has those three properties and $L(B) = \{\lambda, \boldsymbol{a}\}$. Since $\lambda \in L(B)$, the unique start state must also be the unique final state:

$$q = f.$$

Since $\boldsymbol{a} \in L(B)$ and there are no λ-transitions, there must be a transition on $\boldsymbol{a}$ from q to the only final state, namely to q itself. Thus $q \in \delta_B(q, \boldsymbol{a})$.

But then the same move can be used repeatedly, so every positive power of $\boldsymbol{a}$ is accepted:

$$\boldsymbol{a}^n \in L(B) \qquad \text{for all } n \geq 1.$$

In particular, $\boldsymbol{aa} \in L(B)$, contradicting $L(B) = \{\lambda, \boldsymbol{a}\}$. Therefore the three conditions may indeed be incompatible.

4.24. **Prove Lemma 4.2.**

Lemma 4.2: For any $s \in S$ and any $x \in \Sigma^+$, $\overline{\delta^\sigma_\lambda}(s,x) = \overline{\delta}_\lambda(s,x)$.

Proof. We proceed by induction on $|x| = k \geq 1$.

Base case ($k = 1$). Let $x = \boldsymbol{a} \in \Sigma$. By definition of δ^σ_λ:

$$\delta^\sigma_\lambda(s,\boldsymbol{a}) = \Lambda\Big(\bigcup_{t \in \Lambda(s)} \delta_\lambda(t,\boldsymbol{a})\Big) = \overline{\delta}_\lambda(s,\boldsymbol{a}).$$

So $\overline{\delta^\sigma_\lambda}(s,\boldsymbol{a}) = \delta^\sigma_\lambda(s,\boldsymbol{a}) = \overline{\delta}_\lambda(s,\boldsymbol{a})$.

Inductive step. Assume the result holds for all words of length k. Let $y = x\boldsymbol{a}$ with $|x| = k$ and $\boldsymbol{a} \in \Sigma$. Then:

$$\begin{aligned}
\overline{\delta^\sigma_\lambda}(s,x\boldsymbol{a}) &= \bigcup_{t \in \overline{\delta^\sigma_\lambda}(s,x)} \delta^\sigma_\lambda(t,\boldsymbol{a}) \\
&= \bigcup_{t \in \overline{\delta}_\lambda(s,x)} \delta^\sigma_\lambda(t,\boldsymbol{a}) \quad \text{(induction hypothesis)} \\
&= \bigcup_{t \in \overline{\delta}_\lambda(s,x)} \overline{\delta}_\lambda(t,\boldsymbol{a}) \quad \text{(base case)} \\
&= \overline{\delta}_\lambda(s,x\boldsymbol{a}). \quad \text{(definition of } \overline{\delta}_\lambda\text{)}
\end{aligned}$$

By induction, the result holds for all $x \in \Sigma^+$. □ □

4.25. **A DFA can be viewed as an NDFA, and $L(A^n) = L(A)$.**

Construction. Given a DFA $A = \langle \Sigma, S, s_0, \delta, F\rangle$ (with $\delta : S \times \Sigma \to S$ a total function), define the NDFA $A^n = \langle \Sigma, S, \{s_0\}, \delta^n, F\rangle$ where $\delta^n : S \times \Sigma \to \wp(S)$ is given by $\delta^n(s,\boldsymbol{a}) = \{\delta(s,\boldsymbol{a})\}$.

A^n is a valid NDFA: S is finite, $S^n_0 = \{s_0\}$ is a nonempty set of start states, and δ^n maps to subsets of S.

Claim: $L(A^n) = L(A)$**.**

Proof. By induction on $|x|$, we show $\overline{\delta}^n(s_0,x) = \{\overline{\delta}(s_0,x)\}$ (i.e., the nondeterministic extended transition function applied to the singleton start always yields the singleton $\{\overline{\delta}(s_0,x)\}$).

Base: $\overline{\delta}^n(s_0,\lambda) = \{s_0\} = \{\overline{\delta}(s_0,\lambda)\}$.

Step: $\overline{\delta}^n(s_0,x\boldsymbol{a}) = \bigcup_{t \in \overline{\delta}^n(s_0,x)} \delta^n(t,\boldsymbol{a}) = \bigcup_{t \in \{\overline{\delta}(s_0,x)\}} \{\delta(t,\boldsymbol{a})\} = \{\delta(\overline{\delta}(s_0,x),\boldsymbol{a})\} = \{\overline{\delta}(s_0,x\boldsymbol{a})\}$.

Hence $x \in L(A^n)$ iff $\overline{\delta}^n(s_0,x) \cap F \neq \emptyset$ iff $\overline{\delta}(s_0,x) \in F$ iff $x \in L(A)$. □ □

4.26. **Show that $\overline{\delta^\sigma_\lambda}(s,\lambda) \neq \overline{\delta}_\lambda(s,\lambda)$ in general.**

By definition:

- $\overline{\delta^\sigma_\lambda}(s,\lambda) = \{s\}$ (definition of the extended transition function for zero-length input).
- $\overline{\delta}_\lambda(s,\lambda) = \Lambda(s)$ (definition of $\overline{\delta}_\lambda$ for λ).

These differ whenever $\Lambda(s) \neq \{s\}$, i.e., whenever there exists at least one λ-transition out of s.

Example. In Example 4.14, $\Lambda(s_0) = \{s_0, s_1, s_2\}$ but $\overline{\delta^\sigma_\lambda}(s_0,\lambda) = \{s_0\}$. Hence $\overline{\delta^\sigma_\lambda}(s_0,\lambda) \neq \overline{\delta}_\lambda(s_0,\lambda)$.

This is why Lemma 4.2 is stated only for $x \in \Sigma^+$ (words of length ≥ 1) and cannot be extended to λ.

4.27. **Prove:** $(\forall A, B \in \wp(S))(\forall \boldsymbol{a} \in \Sigma)\delta^d(A \cup B, \boldsymbol{a}) = \delta^d(A, \boldsymbol{a}) \cup \delta^d(B, \boldsymbol{a})$**.**

Proof. By Definition 4.5, $\delta^d(Q,\boldsymbol{a}) = \bigcup_{q \in Q} \delta(q,\boldsymbol{a})$ for any $Q \in \wp(S)$. Therefore:

$$\begin{aligned}\delta^d(A \cup B, \boldsymbol{a}) &= \bigcup_{q \in A \cup B} \delta(q,\boldsymbol{a}) \\ &= \Big(\bigcup_{q \in A} \delta(q,\boldsymbol{a})\Big) \cup \Big(\bigcup_{q \in B} \delta(q,\boldsymbol{a})\Big) \\ &= \delta^d(A,\boldsymbol{a}) \cup \delta^d(B,\boldsymbol{a}). \quad \square\end{aligned}$$

$\square$

4.28. **Acceptance requiring *every* reachable state to be final.**

(a) **New acceptance definition.** Define $L_\forall(A) = \{w \in \Sigma^* \mid \overline{\delta}(S_0, w) \neq \emptyset \wedge \overline{\delta}(S_0, w) \subseteq F\}$. That is, w is accepted iff (i) at least one state is reachable and (ii) *every* reachable state is final.

(b) **Modified A^d.** Define $A^\forall = A^d$ with the same state set $\wp(S)$ and transition function δ^d as before, but with the new final-state condition:

$$F^\forall = \{Q \in \wp(S) \mid Q \neq \emptyset \wedge Q \subseteq F\}.$$

That is, Q is a final state of $A^\forall$ iff Q is nonempty and all states in Q are final states of A. By the same argument as Theorem 4.1 (modified for this new acceptance criterion), $L_\forall(A) = L(A^\forall)$. Note: this "universal" acceptance is still FAD (the modified $A^\forall$ is a DFA), so $L_\forall(A)$ is always a FAD language.

4.29. **Draw the connected part of T^d, the deterministic equivalent of the NDFA T in Example 4.7.**

Let

$$Q_0 = \{s_0\}, \quad Q_1 = \{s_0, s_1\}, \quad Q_2 = \{s_0, s_2\}, \quad Q_3 = \{s_0, s_1, s_3\},$$

$$Q_4 = \{s_0, s_1, s_4\}, \quad Q_5 = \{s_0, s_2, s_5\}, \quad Q_6 = \{s_0, s_1, s_3, s_6\},$$

$$Q_7 = \{s_0, s_1, s_4, s_6\}, \quad Q_8 = \{s_0, s_2, s_6\}, \quad Q_9 = \{s_0, s_1, s_6\},$$

$$Q_{10} = \{s_0, s_2, s_5, s_6\}, \qquad Q_{11} = \{s_0, s_6\}.$$

These are exactly the reachable subsets from $\{s_0\}$ under Definition 4.5. The accepting states are those containing s_6, namely

$$Q_6, Q_7, Q_8, Q_9, Q_{10}, Q_{11}.$$

Their transitions are:

	0	**1**
Q_0	Q_1	Q_0
Q_1	Q_1	Q_2
Q_2	Q_3	Q_0
Q_3	Q_4	Q_2
Q_4	Q_1	Q_5
Q_5	Q_6	Q_0
Q_6	Q_7	Q_8
Q_7	Q_9	Q_{10}
Q_8	Q_6	Q_{11}
Q_9	Q_9	Q_8
Q_{10}	Q_6	Q_{11}
Q_{11}	Q_9	Q_{11}

This is the connected part of T^d.

4.30. **Modify T to revert to a nonfinal state when 000111 (EOT) is detected.**

Keep the original start-of-transmission arm for **010010**, but replace the single final sink s_6 by six final "recording" states $r_0, \dots, r_5$ that track the longest suffix of the data seen so far that is also a prefix of the EOT string **000111**. The only nonfinal states are the original $s_0, \dots, s_5$.

	0	**1**
s_0	$\{s_0, s_1\}$	$\{s_0\}$
s_1	$\emptyset$	$\{s_2\}$
s_2	$\{s_3\}$	$\emptyset$
s_3	$\{s_4\}$	$\emptyset$
s_4	$\emptyset$	$\{s_5\}$
s_5	$\{r_0\}$	$\emptyset$
r_0	$\{r_1\}$	$\{r_0\}$
r_1	$\{r_2\}$	$\{r_0\}$
r_2	$\{r_3\}$	$\{r_0\}$
r_3	$\{r_3\}$	$\{r_4\}$
r_4	$\{r_1\}$	$\{r_5\}$
r_5	$\{r_1\}$	$\{s_0\}$

Here r_i means that recording is on and the last i scanned bits match the first i bits of **000111**. Thus $r_5 \xrightarrow{\mathbf{1}} s_0$ turns the recorder off exactly when the full EOT signal **000111** has just been seen, while every other input keeps the machine in one of the final recording states.

4.31. Consider the NDFA A from Example 4.14.

(a) **Diagram of A_λ^σ.** For Example 4.14, $\lambda \in L(A_\lambda)$, so Definition 4.9 yields

$$A_\lambda^\sigma = \langle \{\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}, \boldsymbol{d}\}, \{q_0, s_0, s_1, s_2, s_3\}, \{q_0, s_0\}, \delta_\lambda^\sigma, \{q_0, s_2\} \rangle$$

with transition table

	$\boldsymbol{a}$	$\boldsymbol{b}$	$\boldsymbol{c}$	$\boldsymbol{d}$
q_0	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$
s_0	$\{s_1, s_2\}$	$\{s_3\}$	$\{s_2\}$	$\{s_2\}$
s_1	$\emptyset$	$\{s_3\}$	$\{s_2\}$	$\{s_2\}$
s_2	$\emptyset$	$\emptyset$	$\emptyset$	$\{s_2\}$
s_3	$\emptyset$	$\{s_1, s_2\}$	$\emptyset$	$\emptyset$

This table is exactly the state-transition diagram of Figure 4.20.

(b) **Diagram of $A_\lambda^{\sigma d}$ (connected portion).** Starting from the actual start set $\{q_0, s_0\}$, the reachable subsets are

$$R_0 = \{q_0, s_0\}, \quad R_1 = \{s_1, s_2\}, \quad R_2 = \{s_3\}, \quad R_3 = \{s_2\}, \quad R_4 = \emptyset.$$

Their transitions are

	$\boldsymbol{a}$	$\boldsymbol{b}$	$\boldsymbol{c}$	$\boldsymbol{d}$
R_0	R_1	R_2	R_3	R_3
R_1	R_4	R_2	R_3	R_3
R_2	R_4	R_1	R_4	R_4
R_3	R_4	R_4	R_4	R_3
R_4	R_4	R_4	R_4	R_4

The final states are R_0, R_1, and R_3.

4.32. **What is wrong with the given "proof" of Lemma 4.2?**

The flaw is in the inductive step. The "proof" starts the induction at $k = 1$ (base case for single letters), but the inductive step uses $\overline{\delta_\lambda^\sigma}(s, x)$ for $x \in \Sigma^k$ and then applies the "induction hypothesis" $\overline{\delta_\lambda^\sigma}(s, x) = \overline{\delta}_\lambda(s, x)$.

The critical error is on the line:

$$\delta_\lambda^\sigma(\overline{\delta_\lambda^\sigma}(s, x), \boldsymbol{a}) = \overline{\delta_\lambda^\sigma}(\overline{\delta}_\lambda(s, x), \boldsymbol{a}).$$

Here, $\overline{\delta_\lambda^\sigma}(s, x)$ is a *set* of states (since we're in an NDFA), but δ_λ^σ is defined on individual states, not sets. The expression $\delta_\lambda^\sigma(Q, \boldsymbol{a})$ where Q is a set is not directly the same as $\overline{\delta_\lambda^\sigma}(Q, \boldsymbol{a})$ for a set-valued argument: the extended function $\overline{\delta_\lambda^\sigma}$ on a set-argument must take the union over all states in the set.

More specifically, the step

$$\delta_\lambda^\sigma\big(\overline{\delta_\lambda^\sigma}(s, x), \boldsymbol{a}\big) = \overline{\delta_\lambda^\sigma}\big(\overline{\delta}_\lambda(s, x), \boldsymbol{a}\big)$$

conflates applying δ_λ^σ to a *set* with applying the set-union version of $\overline{\delta_\lambda^\sigma}$ to a set argument. These are not the same: the left side applies δ_λ^σ to the entire set at once (which is not how the function is defined), while the right side correctly takes the union over individual states. This makes the step invalid without justification.

4.33. Consider the NDFA T from Example 4.7 (detecting substring **010010**).

(a) *Circuit diagram (no SOS/EOS).* T has 8 states (one start state that loops on all input plus the six states that track the pattern **010010**). Using one activity bit per state, the next-state equations are

$$s_0' = s_0, \qquad s_1' = s_0 \wedge \mathbf{0}, \qquad s_2' = s_1 \wedge \mathbf{1},$$

$$s_3' = s_2 \wedge \mathbf{0}, \qquad s_4' = s_3 \wedge \mathbf{0}, \qquad s_5' = s_4 \wedge \mathbf{1},$$

$$s_6' = (s_5 \wedge \mathbf{0}) \vee (s_6 \wedge \mathbf{0}) \vee (s_6 \wedge \mathbf{1}).$$

Initially only s_0 is active, and the accept line is s_6.

(b) *DFA T^d (connected portion).* See Exercise 4.29 for the subset construction. The connected DFA is exactly the 12-state machine listed there, with states $Q_0, \ldots, Q_{11}$ and the transition table given in Exercise 4.29.

4.34. Consider the NDFA from Example 4.11, which accepts all words containing **10110**, **1010**, or **01101** as substrings.

(a) *Circuit diagram (no SOS/EOS).* The NDFA has a single initial state s_0 that loops on Σ^*, three pattern-matching arms, and the final sink s_{18} from Figure 4.12. Since s_1, s_7, and s_{12} are only shown for clarity in the text, they can be omitted from the actual circuit. With one activity bit per remaining state, the nonzero next-state equations are

$$s_0' = s_0,$$

$$s_2' = s_0 \wedge \mathbf{1}, \quad s_3' = s_2 \wedge \mathbf{0}, \quad s_4' = s_3 \wedge \mathbf{1}, \quad s_5' = s_4 \wedge \mathbf{1}, \quad s_6' = s_5 \wedge \mathbf{0},$$

$$s_8' = s_0 \wedge \mathbf{1}, \quad s_9' = s_8 \wedge \mathbf{0}, \quad s_{10}' = s_9 \wedge \mathbf{1}, \quad s_{11}' = s_{10} \wedge \mathbf{0},$$

$$s_{13}' = s_0 \wedge \mathbf{0}, \quad s_{14}' = s_{13} \wedge \mathbf{1}, \quad s_{15}' = s_{14} \wedge \mathbf{1}, \quad s_{16}' = s_{15} \wedge \mathbf{0}, \quad s_{17}' = s_{16} \wedge \mathbf{1},$$

$$s_{18}' = s_6 \vee s_{11} \vee s_{17} \vee s_{18}.$$

The accept line is $s_6 \vee s_{11} \vee s_{17} \vee s_{18}$.

(b) *DFA (connected portion only).* Since all three patterns are started from the looping state s_0, the subset construction yields the following 23 reachable states:

$Q_0 = \{s_0\}$,	$Q_1 = \{s_0, s_{13}\}$,	$Q_2 = \{s_0, s_2, s_8\}$,
$Q_3 = \{s_0, s_{14}, s_2, s_8\}$,	$Q_4 = \{s_0, s_{13}, s_3, s_9\}$,	$Q_5 = \{s_0, s_{15}, s_2, s_8\}$,
$Q_6 = \{s_0, s_{10}, s_{14}, s_2, s_4, s_8\}$,	$Q_7 = \{s_0, s_{13}, s_{16}, s_3, s_9\}$,	$Q_8 = \{s_0, s_{11}, s_{13}, s_3, s_9\}$,
$Q_9 = \{s_0, s_{15}, s_2, s_5, s_8\}$,	$Q_{10} = \{s_0, s_{10}, s_{14}, s_{17}, s_2, s_4, s_8\}$,	$Q_{11} = \{s_0, s_{13}, s_{18}\}$,
$Q_{12} = \{s_0, s_{10}, s_{14}, s_{18}, s_2, s_4, s_8\}$,	$Q_{13} = \{s_0, s_{13}, s_{16}, s_3, s_6, s_9\}$,	$Q_{14} = \{s_0, s_{11}, s_{13}, s_{18}, s_3, s_9\}$,
$Q_{15} = \{s_0, s_{15}, s_{18}, s_2, s_5, s_8\}$,	$Q_{16} = \{s_0, s_{14}, s_{18}, s_2, s_8\}$,	$Q_{17} = \{s_0, s_{10}, s_{14}, s_{17}, s_{18}, s_2, s_4, s_8\}$,
$Q_{18} = \{s_0, s_{13}, s_{16}, s_{18}, s_3, s_6, s_9\}$,	$Q_{19} = \{s_0, s_{18}, s_2, s_8\}$,	$Q_{20} = \{s_0, s_{13}, s_{18}, s_3, s_9\}$,
$Q_{21} = \{s_0, s_{15}, s_{18}, s_2, s_8\}$,	$Q_{22} = \{s_0, s_{13}, s_{16}, s_{18}, s_3, s_9\}$.	

Their transition table is

	0	**1**
Q_0	Q_1	Q_2
Q_1	Q_1	Q_3
Q_2	Q_4	Q_2
Q_3	Q_4	Q_5
Q_4	Q_1	Q_6
Q_5	Q_7	Q_2
Q_6	Q_8	Q_9
Q_7	Q_1	Q_{10}
Q_8	Q_{11}	Q_{12}
Q_9	Q_{13}	Q_2
Q_{10}	Q_{14}	Q_{15}
Q_{11}	Q_{11}	Q_{16}
Q_{12}	Q_{14}	Q_{15}
Q_{13}	Q_{11}	Q_{17}
Q_{14}	Q_{11}	Q_{12}
Q_{15}	Q_{18}	Q_{19}
Q_{16}	Q_{20}	Q_{21}
Q_{17}	Q_{14}	Q_{15}
Q_{18}	Q_{11}	Q_{17}
Q_{19}	Q_{20}	Q_{19}
Q_{20}	Q_{11}	Q_{12}
Q_{21}	Q_{22}	Q_{19}
Q_{22}	Q_{11}	Q_{17}

The final states are exactly those containing one of s_6, s_{11}, s_{17}, or s_{18}:

$$Q_8, Q_{10}, Q_{11}, Q_{12}, Q_{13}, Q_{14}, Q_{15}, Q_{16}, Q_{17}, Q_{18}, Q_{19}, Q_{20}, Q_{21}, Q_{22}.$$

4.35. Consider the NDFA B from Example 4.8 and its deterministic equivalent B^d.

(a) *Circuit diagram for B (no SOS/EOS).* Encode the states of B using one activity bit for r and one for s. From Figure 4.7,

$$r' = (r \wedge \boldsymbol{a}) \vee (s \wedge \boldsymbol{a}), \qquad s' = (r \wedge \boldsymbol{a}) \vee (r \wedge \boldsymbol{b}),$$

with initial state $r = 1$, $s = 0$. Since s is the only final state, the accept output is s.

(b) *Circuit diagram for B^d (no SOS/EOS).* The states of B^d are

$$\varnothing, \qquad \{r\}, \qquad \{s\}, \qquad \{r, s\}.$$

Using the binary state assignment

$$\varnothing = 00, \qquad \{r\} = 01, \qquad \{s\} = 10, \qquad \{r, s\} = 11,$$

with present-state bits x_1x_0, the transition table is

x_1x_0	$\boldsymbol{a}$	$\boldsymbol{b}$
00	00	00
01	11	10
10	01	00
11	11	10

so the DFA circuit is given by

$$x_1' = x_0, \qquad x_0' = \boldsymbol{a} \wedge (x_1 \vee x_0).$$

The start state is 01 and the accept output is x_1, since the final subset states are $\{s\}$ and $\{r, s\}$.

4.36. **Circuit diagrams for each NDFA in Figure 4.25 (no SOS/EOS).**

For each NDFA A_i in Figure 4.25, with $\Sigma = \{\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}\}$:

1. **(a)** With activity bit t for the single state:

$$t' = t \wedge (\boldsymbol{a} \vee \boldsymbol{c}), \qquad \text{accept} = t.$$

2. **(b)** With activity bits q_1, q_0:

$$q_1' = (q_1 \wedge \boldsymbol{b}) \vee (q_0 \wedge \boldsymbol{b}), \qquad q_0' = (q_1 \wedge \boldsymbol{a}) \vee (q_0 \wedge \boldsymbol{c}),$$

   and the accept output is q_1.

3. **(c)** With activity bits r_0, r_1:

$$r_0' = (r_0 \wedge \boldsymbol{a}) \vee (r_1 \wedge \boldsymbol{b}), \qquad r_1' = (r_0 \wedge \boldsymbol{a}) \vee (r_0 \wedge \boldsymbol{c}),$$

   and the accept output is r_0.

4. **(d)** With activity bits s_0, s_1, s_2, s_3:

$$s_0' = 0, \qquad s_1' = (s_0 \wedge \boldsymbol{a}) \vee (s_3 \wedge \boldsymbol{b}),$$

$$s_2' = s_1 \wedge \boldsymbol{a}, \qquad s_3' = s_1 \wedge \boldsymbol{b},$$

   and the accept output is s_2.

5. **(e)** With activity bits s_1, s_2:

$$s_1' = (s_1 \wedge \boldsymbol{a}) \vee (s_2 \wedge \boldsymbol{a}) \vee (s_2 \wedge \boldsymbol{b}), \qquad s_2' = s_1 \wedge \boldsymbol{a},$$

   and the accept output is s_2.

6. **(f)** The λ-machine has states s_0, s_1, s_2, s_3 with ordinary transitions $s_0 \xrightarrow{\boldsymbol{a}} s_1$, $s_1 \xrightarrow{\boldsymbol{b}} s_2$, $s_2 \xrightarrow{\boldsymbol{c}} s_3$ and λ-moves $s_1 \to s_0$, $s_3 \to s_1$. The standard circuit therefore uses

$$D_0 = s_1, \qquad D_1 = (s_0 \wedge \boldsymbol{a}) \vee s_3, \qquad D_2 = s_1 \wedge \boldsymbol{b}, \qquad D_3 = s_2 \wedge \boldsymbol{c},$$

   with accept output s_3. Adding the transitive λ-connection $s_3 \to s_0$ as well is also consistent with the text's discussion.

7. **(g)** With activity bits s_0, s_1, s_2, s_3:

$$s_0' = s_1 \wedge \boldsymbol{a}, \qquad s_1' = (s_0 \wedge \boldsymbol{a}) \vee (s_2 \wedge \boldsymbol{c}) \vee (s_3 \wedge \boldsymbol{b}),$$

$$s_2' = s_1 \wedge \boldsymbol{c}, \qquad s_3' = s_1 \wedge \boldsymbol{b},$$

and the accept output is s_0.

4.37. **Circuit diagrams for each NDFA in Figure 4.25 (with SOS and EOS).**

Extend each circuit from Exercise 4.36 by forcing the start state(s) active on <SOS> and gating the final-state output with <EOS>:

- **(a)** $t' = \text{<SOS>} \vee (t \wedge (\boldsymbol{a} \vee \boldsymbol{c}))$ and accept $= t \wedge \text{<EOS>}$.
- **(b)** OR <SOS> into the input for q_1:

  $$q_1' = \text{<SOS>} \vee (q_1 \wedge \boldsymbol{b}) \vee (q_0 \wedge \boldsymbol{b}), \qquad q_0' = (q_1 \wedge \boldsymbol{a}) \vee (q_0 \wedge \boldsymbol{c}),$$

  and accept $= q_1 \wedge \text{<EOS>}$.
- **(c)** OR <SOS> into the input for r_0 and gate the output:

  $$r_0' = \text{<SOS>} \vee (r_0 \wedge \boldsymbol{a}) \vee (r_1 \wedge \boldsymbol{b}), \qquad r_1' = (r_0 \wedge \boldsymbol{a}) \vee (r_0 \wedge \boldsymbol{c}),$$

  with accept $= r_0 \wedge \text{<EOS>}$.
- **(d)** OR <SOS> into s_0' and use accept $= s_2 \wedge \text{<EOS>}$.
- **(e)** OR <SOS> into the corrected first equation:

  $$s_1' = \text{<SOS>} \vee (s_1 \wedge \boldsymbol{a}) \vee (s_2 \wedge \boldsymbol{a}) \vee (s_2 \wedge \boldsymbol{b}), \qquad s_2' = s_1 \wedge \boldsymbol{a},$$

  and use accept $= s_2 \wedge \text{<EOS>}$.
- **(f)** OR <SOS> into D_0 and use accept $= s_3 \wedge \text{<EOS>}$.
- **(g)** OR <SOS> into s_0' and use accept $= s_0 \wedge \text{<EOS>}$.

4.38. **Adapting Definition 3.10 for NDFAs and proving that an algorithm exists to find the connected portion.**

(a) **Adapted definition.** The connected portion of an NDFA $A = \langle \Sigma, S, S_0, \delta, F \rangle$ is defined as follows. Let

$$\text{Reach}_0 = S_0, \qquad \text{Reach}_{k+1} = \text{Reach}_k \cup \bigcup_{\boldsymbol{a} \in \Sigma} \bigcup_{s \in \text{Reach}_k} \delta(s, \boldsymbol{a}).$$

The connected portion is $\text{Reach} = \bigcup_{k \geq 0} \text{Reach}_k$, together with all transitions among states in Reach and the restriction of F to Reach.

(b) **Proof that the algorithm terminates.**

Proof. The sets Reach_k form a non-decreasing chain $\text{Reach}_0 \subseteq \text{Reach}_1 \subseteq \cdots \subseteq S$. Since S is finite, the chain must stabilize: there exists k^* such that $\text{Reach}_{k^*} = \text{Reach}_{k^*+1}$. At that point, no new states are added. The algorithm terminates in at most $|S|$ iterations (since at least one new state must be added in each non-stable step, and there are only $|S|$ states).

To compute this algorithmically:

1. Initialize $R := S_0$, New $:= S_0$.
2. While New $\neq \emptyset$:
   (a) Next $:= \bigcup_{\boldsymbol{a} \in \Sigma} \bigcup_{s \in \text{New}} \delta(s, \boldsymbol{a}) \setminus R$.
   (b) $R := R \cup$ Next; New $:=$ Next.
3. Output R as the connected portion.

Each iteration of the while loop adds at least one state to R (otherwise New $= \emptyset$ and we stop). Thus the loop executes at most $|S|$ times, and the algorithm runs in $O(|S| \cdot |\Sigma|)$ time. □ □

Chapter 5

Closure Properties — Solutions

5.1. Let F_Σ be the collection of all finite languages over Σ.

(a) **Complementation: No.** Σ^* is infinite, so for any finite L, $\Sigma^* \setminus L$ is infinite (assuming $|\Sigma| \geq 1$), hence not in F_Σ.

(b) **Union: Yes.** If L_1, L_2 are finite, then $|L_1 \cup L_2| \leq |L_1| + |L_2| < \infty$.

(c) **Intersection: Yes.** $|L_1 \cap L_2| \leq |L_1| < \infty$.

(d) **Concatenation: Yes.** $|L_1 \cdot L_2| \leq |L_1| \cdot |L_2| < \infty$.

(e) **Kleene closure: No.** Let $L = \{\boldsymbol{a}\}$. Then $L^* = \{\boldsymbol{a}\}^* = \{\lambda, \boldsymbol{a}, \boldsymbol{aa}, \ldots\}$ is infinite.

(f) **Relative complement: Yes.** $L_1 \setminus L_2 \subseteq L_1$, so $|L_1 \setminus L_2| \leq |L_1| < \infty$.

5.2. Let C_Σ be the collection of all cofinite languages (complements of finite languages).

(a) **Complementation: No.** Let $L = \Sigma^* \setminus \{\boldsymbol{a}\}$. Then $L \in C_\Sigma$, but $\sim L = \{\boldsymbol{a}\}$ is finite, hence not in C_Σ.

(b) **Union: Yes.** If $L_1 = \Sigma^* \setminus F_1$ and $L_2 = \Sigma^* \setminus F_2$, then $L_1 \cup L_2 = \Sigma^* \setminus (F_1 \cap F_2)$. Since $F_1 \cap F_2$ is finite, $L_1 \cup L_2 \in C_\Sigma$.

(c) **Intersection: Yes.** $L_1 \cap L_2 = \Sigma^* \setminus (F_1 \cup F_2)$. Since F_1 and F_2 are finite, so is $F_1 \cup F_2$, hence $L_1 \cap L_2 \in C_\Sigma$.

(d) **Concatenation: Yes.** If $L_1, L_2 \in C_\Sigma$ (both infinite), then $L_1 \cdot L_2$ contains arbitrarily long words, but let us verify cofiniteness. Since $\Sigma^* \setminus (L_1 \cdot L_2) \subseteq$ the set of words not writable as a product from L_1 and L_2; this set is finite because L_1 and L_2 each omit only finitely many words. More precisely: there exists N_1 such that all words of length $\geq N_1$ are in L_1, and similarly N_2 for L_2. Then any word of length $\geq N_1 + N_2$ can be split so that both parts are in L_1 and L_2, so $L_1 \cdot L_2$ is cofinite.

(e) **Kleene closure: Yes.** Since L is cofinite, there is an N such that every word of length at least N lies in L. Hence every word of length at least N also lies in L^* (just take it as a one-factor product). Therefore $\Sigma^* \setminus L^*$ contains only words of length $< N$, so it is finite and $L^* \in C_\Sigma$.

(f) **Relative complement: No.** $L_1 \setminus L_2 = L_1 \cap \sim L_2$. Here $\sim L_2$ is finite, so $L_1 \setminus L_2 \subseteq \sim L_2$ is finite, not cofinite in general. Counterexample: $L_1 = \Sigma^* \setminus \emptyset = \Sigma^*$ and $L_2 = \Sigma^* \setminus \{\boldsymbol{a}\}$. Then $L_1 \setminus L_2 = \{\boldsymbol{a}\}$, which is finite, not cofinite.

5.3. $B_\Sigma = F_\Sigma \cup C_\Sigma$.

(a) **Complementation: Yes.** The complement of a finite language is cofinite and vice versa.

(b) **Union: Yes.** If both L_1, L_2 are finite, $L_1 \cup L_2$ is finite. If one is cofinite, $L_1 \cup L_2$ is cofinite. If both are cofinite, $L_1 \cup L_2$ is cofinite (by 5.2b). **Closed.**

(c) **Intersection: Yes.** If both finite: finite. If one cofinite and one finite: finite. If both cofinite: cofinite (by 5.2c).

(d) **Concatenation: No.** Let $L_1 = C_\Sigma$ element and $L_2 = C_\Sigma$ element; concatenation is cofinite (by 5.2d). Let $L_1, L_2 \in F_\Sigma$; product is finite. Let $L_1 \in F_\Sigma$ and $L_2 \in C_\Sigma$ or vice versa: if $L_1 = \emptyset$, product is $\emptyset \in F_\Sigma$. If L_1 is finite and nonempty: the product may be neither finite nor cofinite. **Counterexample:** $L_1 = \{\boldsymbol{a}\}$ (finite), $L_2 = \Sigma^* \setminus \{\boldsymbol{aa}\}$ (cofinite). Then $L_1 \cdot L_2$ = all words of the form $\boldsymbol{a}w$ where $w \neq \boldsymbol{aa}$, i.e., all words starting with $\boldsymbol{a}$ except $\boldsymbol{aaa}$. This is neither finite nor cofinite (it's infinite and its complement is infinite). So **not closed**.

(e) **Kleene closure: No.** $\{\boldsymbol{a}\}^*$ is infinite but $\{\boldsymbol{a}\} \in F_\Sigma$; $\{\boldsymbol{a}\}^*$ is not in F_Σ. Is it cofinite? $\{\boldsymbol{a}\}^* = \{\lambda, \boldsymbol{a}, \boldsymbol{aa}, \ldots\}$, and its complement contains all words with any $\boldsymbol{b}$ (if $|\Sigma| \geq 2$), which is infinite, so $\{\boldsymbol{a}\}^*$ is not cofinite. **Not closed.**

(f) **Relative complement: Yes.** If $L_1 \in C_\Sigma$ and $L_2 \in C_\Sigma$, then $L_1 \setminus L_2 = L_1 \cap \sim L_2$ where $\sim L_2$ is finite, so the result is finite and therefore belongs to $F_\Sigma \subseteq B_\Sigma$. If $L_1 \in F_\Sigma$, then $L_1 \setminus L_2 \subseteq L_1$ is finite. Hence B_Σ is closed under relative complement.

5.4. $I_\Sigma = \wp(\Sigma^*) \setminus F_\Sigma$, the infinite languages.

(a) **Complementation: No.** $\Sigma^* \setminus L$ may be finite. E.g., $L = \Sigma^* \setminus \{\lambda\}$ is infinite; $\sim L = \{\lambda\}$ is finite.

(b) **Union: Yes.** Union of two infinite sets is infinite.

(c) **Intersection: No.** $L_1 = \{\boldsymbol{a}^{2n} \mid n \geq 0\}$ and $L_2 = \{\boldsymbol{a}^{2n+1} \mid n \geq 0\}$ are both infinite, but $L_1 \cap L_2 = \emptyset$, which is finite.

(d) **Concatenation: Yes.** If L_1, L_2 are infinite, pick any fixed word $y \in L_2$. Then

$$\{xy \mid x \in L_1\} \subseteq L_1 \cdot L_2.$$

The map $x \mapsto xy$ is injective, so $\{xy \mid x \in L_1\}$ is infinite. Therefore $L_1 \cdot L_2$ is infinite.

(e) **Kleene closure: Yes.** $L^* \supseteq L$ when $\lambda \in L^*$, and L^* is always infinite if L contains any nonempty word.

(f) **Relative complement: No.** $L_1 \setminus L_2$ could be finite even if L_1, L_2 are infinite (e.g., $L_1 = L_2$).

5.5. $J_\Sigma = \wp(\Sigma^*) \setminus C_\Sigma$, languages with infinite complements.

(a) **Complementation: No.** $L \in J_\Sigma$ means $\sim L$ is infinite, i.e., $\sim L \in I_\Sigma$. Then $\sim(\sim L) = L$; is $\sim L \in J_\Sigma$? $\sim(\sim L) = L$ is infinite (since $L \in J_\Sigma$ is not cofinite, its complement is infinite, so L itself need not be infinite). Counterexample: $L = \emptyset \in J_\Sigma$ (complement is Σ^*, infinite), but $\sim L = \Sigma^* \not{i}nJ_\Sigma$ (complement is $\emptyset$, finite). **Not closed.**

(b) **Union: No.** Over $\Sigma = \{\boldsymbol{a}\}$, let $L_1 = \{\boldsymbol{a}^{2n} \mid n \geq 0\}$ and $L_2 = \{\boldsymbol{a}^{2n+1} \mid n \geq 0\}$. Both have infinite complements, so both belong to J_Σ. But $L_1 \cup L_2 = \Sigma^*$, whose complement is finite, so $L_1 \cup L_2 /$ inJ_Σ.

(c) **Intersection: Yes.** If $\sim L_1$ and $\sim L_2$ are both infinite, then $\sim(L_1 \cap L_2) = \sim L_1 \cup \sim L_2 \supseteq \sim L_1$, which is infinite.

(d) **Concatenation: No.** Over $\Sigma = \{\boldsymbol{a}\}$, let $L_1 = \{\boldsymbol{a}^n \mid n \text{ odd or } n = 2\}$ and $L_2 = \{\boldsymbol{a}^{2k} \mid k \geq 1\}$. Then $\sim L_1 = \{\lambda, \boldsymbol{a}^4, \boldsymbol{a}^6, \ldots\}$ and $\sim L_2 = \{\lambda, \boldsymbol{a}, \boldsymbol{a}^3, \boldsymbol{a}^5, \ldots\}$ are both infinite, so $L_1, L_2 \in J_\Sigma$. But $L_1 \cdot L_2 = \{\boldsymbol{a}^n \mid n \geq 3\}$ (any $n \geq 3$ is expressible as odd+even or 2+even), whose complement $\{\lambda, \boldsymbol{a}, \boldsymbol{a}^2\}$ is finite—so $L_1 \cdot L_2 \not in J_\Sigma$.

(e) **Kleene closure: No.** Over $\Sigma = \{\boldsymbol{a}\}$, let $L = \{\boldsymbol{a}^n \mid n \text{ odd}\}$. Then $\sim L = \{\lambda, \boldsymbol{a}^2, \boldsymbol{a}^4, \ldots\}$ is infinite, so $L \in J_\Sigma$. But odd+odd=even, even+odd=odd, so L^k contains all $\boldsymbol{a}$-strings of the right parity at length $\geq k$; inductively $L^* = \{\boldsymbol{a}\}^* = \Sigma^*$. Then $\sim L^* = \emptyset$ is finite, so $L^* \not in J_\Sigma$.

(f) **Relative complement: Yes.** $\sim (L_1 \setminus L_2) = \sim (L_1 \cap \sim L_2) = \sim L_1 \cup L_2 \supseteq \sim L_1$. Since $L_1 \in J_\Sigma$, $\sim L_1$ is infinite, so $\sim (L_1 \setminus L_2)$ is infinite, hence $L_1 \setminus L_2 \in J_\Sigma$. □

5.6. $\boldsymbol{E}$: all languages over $\{\boldsymbol{a}, \boldsymbol{b}\}$ that contain ***abba***.

(a) **Complementation: No.** $\sim L$ contains all words *not* in L; since L contains ***abba***, $\sim L$ does not contain ***abba***, so $\sim L \not in \boldsymbol{E}$.

(b) **Union: Yes.** If ***abba*** $\in L_1$, then ***abba*** $\in L_1 \cup L_2$.

(c) **Intersection: Yes.** If $L_1, L_2 \in \boldsymbol{E}$ then both contain ***abba***, so $L_1 \cap L_2$ also contains ***abba***. **Closed.**

(d) **Concatenation: No.** Here $\boldsymbol{E}$ means languages that have the specific word ***abba*** as an element. Let $L_1 = L_2 = \{$***abba***$\}$. Then $L_1, L_2 \in \boldsymbol{E}$, but $L_1 \cdot L_2 = \{$***abbaabba***$\}$, which does not contain ***abba*** as a member. So $\boldsymbol{E}$ is not closed under concatenation.

(e) **Kleene closure: Yes.** L^* contains $L^1 = L$, so ***abba*** $\in L \subseteq L^*$.

(f) **Relative complement: No.** $L_1 \setminus L_2$ might not contain ***abba*** if ***abba*** $\in L_2$. E.g., $L_1 = L_2 = \{$***abba***$\}$: $L_1 \setminus L_2 = \emptyset \not in \boldsymbol{E}$.

5.7. **Intersection closed does not imply union closed.**

Counterexample. Let $\mathscr{C} = \{\emptyset\} \cup \{\{x\} \mid x \in \Sigma^*\}$ (the empty language together with all singleton languages).

Closed under intersection: $\{x\} \cap \{y\} = \{x\}$ if $x = y$, and $\emptyset$ if $x \neq y$; both are in $\mathscr{C}$.

Not closed under union: $\{x\} \cup \{y\} = \{x, y\}$ for $x \neq y$, which is a two-element set and not in $\mathscr{C}$. □

5.8. **If closed under intersection and complement, then closed under union (by De Morgan).**

If $\mathscr{C}$ is closed under intersection and complement, then for any $L_1, L_2 \in \mathscr{C}$: $\sim L_1, \sim L_2 \in \mathscr{C}$ (by closure under complement), so $\sim L_1 \cap \sim L_2 \in \mathscr{C}$ (by closure under intersection), so $\sim (\sim L_1 \cap \sim L_2) = L_1 \cup L_2 \in \mathscr{C}$ (by De Morgan and closure under complement). Hence $\mathscr{C}$ is also closed under union. □

5.9. **Closed under concatenation does not imply closed under Kleene closure.**

Let $\mathscr{C} = F_\Sigma$ (all finite languages). Closed under concatenation (5.1d). Not closed under Kleene closure: $\{\boldsymbol{a}\}^*$ is infinite (5.1e).

5.10. **Closed under Kleene closure does not imply closed under concatenation.**

Let

$$\mathscr{C} = \{\{w\}^* \mid w \in \Sigma^*\}.$$

This includes $\{\lambda\} = \{\lambda\}^*$ (take $w = \lambda$). Each language in $\mathscr{C}$ is closed under Kleene closure, because

$$(\{w\}^*)^* = \{w\}^*.$$

But $\mathscr{C}$ is not closed under concatenation: over $\Sigma = \{\boldsymbol{a}, \boldsymbol{b}\}$,

$$\{\boldsymbol{a}\}^* \cdot \{\boldsymbol{b}\}^* = \{\boldsymbol{a}^m \boldsymbol{b}^n \mid m, n \geq 0\},$$

and this language is not of the form $\{u\}^*$ for any word u.

5.11. **Closed under complement does not imply closed under relative complement.**

Counterexample. Let

$$K = \{\emptyset, \Sigma^*\} \cup \{\{w\} \mid w \in \Sigma^*\} \cup \{\Sigma^* \setminus \{w\} \mid w \in \Sigma^*\}.$$

This collection is closed under complement by construction. But it is not closed under relative complement: if $u \neq v$, then

$$(\Sigma^* \setminus \{u\}) \setminus \{v\} = \Sigma^* \setminus \{u, v\} \not inK.$$

So closure under complement does not imply closure under relative complement.

5.12. **A finite set of numbers closed under $\sqrt{}$:**

$\{0, 1\}$. $\sqrt{0} = 0 \in \{0, 1\}$ and $\sqrt{1} = 1 \in \{0, 1\}$.

5.13. **An infinite set of numbers closed under $\sqrt{}$:**

$[0, 1] = \{x \in \mathbb{R} \mid 0 \leq x \leq 1\}$. For any $x \in [0, 1]$, $\sqrt{x} \in [0, 1]$ since $0 \leq x \leq 1 \Rightarrow 0 \leq \sqrt{x} \leq 1$.

5.14. **Algorithm for deterministic union machine A^U.**

Given $A_1 = \langle \Sigma, S_1, s_0^1, \delta_1, F_1 \rangle$ and $A_2 = \langle \Sigma, S_2, s_0^2, \delta_2, F_2 \rangle$ (both DFAs, $S_1 \cap S_2 = \emptyset$):

1. Build the NDFA $A^\cup = \langle \Sigma, S_1 \cup S_2, \{s_0^1, s_0^2\}, \delta^\cup, F_1 \cup F_2 \rangle$ (as in Theorem 5.2). 2. Apply Definition 4.5 (subset construction) to $A^\cup$ to produce a DFA A^U. Start state: $\{s_0^1, s_0^2\}$. For each reachable state-set $Q \subseteq S_1 \cup S_2$ and each $\boldsymbol{a} \in \Sigma$: $\delta^U(Q, \boldsymbol{a}) = \delta^d(Q, \boldsymbol{a}) = \bigcup_{q \in Q} \delta^\cup(q, \boldsymbol{a})$. Final states: all Q with $Q \cap (F_1 \cup F_2) \neq \emptyset$.

5.15. (a) **Direct product machine for union.** Define $A^u = \langle \Sigma, S_1 \times S_2, \langle s_0^1, s_0^2 \rangle, \delta^u, F^u \rangle$ where $\delta^u(\langle s, t \rangle, \boldsymbol{a}) = \langle \delta_1(s, \boldsymbol{a}), \delta_2(t, \boldsymbol{a}) \rangle$ and $F^u = \{\langle s, t \rangle \mid s \in F_1 \vee t \in F_2\}$.

(b) **Proof.** By induction, $\overline{\delta^u}(\langle s_0^1, s_0^2 \rangle, x) = \langle \overline{\delta^1}(s_0^1, x), \overline{\delta^2}(s_0^2, x) \rangle$. Then $x \in L(A^u)$ iff $\overline{\delta^u}(\langle s_0^1, s_0^2 \rangle, x) \in F^u$ iff $\overline{\delta^1}(s_0^1, x) \in F_1$ or $\overline{\delta^2}(s_0^2, x) \in F_2$ iff $x \in L(A_1) \cup L(A_2)$.

(c) **State counts (no minimization).** $A^\cup$ (NDFA union): $|S_1| + |S_2|$ states (plus the new start state if added separately, but in Theorem 5.2 no new state is needed). A^U (subset construction of $A^\cup$): up to $2^{|S_1|+|S_2|}$ states, but in practice far fewer. A^u (product): $|S_1| \cdot |S_2| = nm$ states. With $n = |S_1|$ and $m = |S_2|$: $A^\cup$ has $n + m$ states; A^U has up to 2^{n+m} states; A^u has nm states.

5.16. **$\mathfrak{D}_\Sigma$ is closed under the prefix operator P.**

Proof. Let $L \in \mathfrak{D}_\Sigma$ and $A = \langle \Sigma, S, s_0, \delta, F \rangle$ be a DFA with $L(A) = L$. Define $A' = \langle \Sigma, S, s_0, \delta, F' \rangle$ where $F' = \{s \in S \mid \exists y \in \Sigma^* \ni \overline{\delta}(s, y) \in F\}$. (A state is final in A' iff some continuation from it leads to acceptance in A.)

Claim: $L(A') = P(L)$.

$x \in P(L)$ iff $\exists y \in \Sigma^* \ni xy \in L$ iff $\exists y \in \Sigma^* \ni \overline{\delta}(s_0, xy) \in F$ iff $\exists y \in \Sigma^* \ni \overline{\delta}(\overline{\delta}(s_0, x), y) \in F$ iff $\overline{\delta}(s_0, x) \in F'$ iff $x \in L(A')$.

Since F' is computable (for each state s, check if any state in F is reachable from s—a finite graph problem), A' is a valid DFA and $P(L) = L(A') \in \mathfrak{D}_\Sigma$. □ □

5.17. $\mathfrak{D}_\Sigma$ **is closed under the suffix operator** S**.**

(a) **Construction.** Given DFA $A = \langle \Sigma, S, s_0, \delta, F \rangle$, build NDFA $A' = \langle \Sigma, S, S_0', \delta, F \rangle$ where $S_0' = \{\overline{\delta}(s_0, y) \mid y \in \Sigma^*\}$, the set of states reachable from s_0.

(b) **Proof.** $x \in S(L)$ iff $\exists y \in \Sigma^* \ni yx \in L$ iff $\exists y \in \Sigma^* \ni \overline{\delta}(s_0, yx) \in F$ iff $\exists s = \overline{\delta}(s_0, y) \in S_0' \ni \overline{\delta}(s, x) \in F$ iff $\overline{\delta}'(\{s\}, x) \cap F \neq \emptyset$ for some $s \in S_0'$ iff $x \in L(A')$.

(c) Since A' is an NDFA over a finite state set, $L(A') \in \mathfrak{D}_\Sigma$ by Theorem 4.1. Hence $S(L) \in \mathfrak{D}_\Sigma$. □

5.18. $\mathfrak{D}_\Sigma$ **is closed under the center operator** C**.**

(a) **Construction.** Let $R = \{\overline{\delta}(s_0, y) \mid y \in \Sigma^*\}$ be the set of reachable states. Build an NDFA $A' = \langle \Sigma, S, R, \delta, F' \rangle$ where $F' = \{s \in S \mid \exists y \in \Sigma^* \ni \overline{\delta}(s, y) \in F\}$ (states from which F is reachable).

(b) **Proof.** $x \in C(L)$ iff $\exists y, z \ni yxz \in L$ iff $\exists y, z \ni \overline{\delta}(\overline{\delta}(\overline{\delta}(s_0, y), x), z) \in F$ iff $\exists s = \overline{\delta}(s_0, y) \in R$ and $\overline{\delta}(s, x) \in F'$ iff $x \in L(A')$.

(c) Since A' is an NDFA, $C(L) \in \mathfrak{D}_\Sigma$. □

5.19. $\mathfrak{D}_\Sigma$ **is closed under** F **(maximal words not extended in** L**).**

Let $A = \langle \Sigma, S, s_0, \delta, F \rangle$ be a minimal DFA for L. $F(L) = \{x \in L \mid$ no nonempty word y satisfies $xy \in L\}$. Define $A' = \langle \Sigma, S, s_0, \delta, F'' \rangle$ where $F'' = \{s \in F \mid (\forall y \in \Sigma^+)(\overline{\delta}(s, y) \not{i}nF)\}$. Then $x \in L(A')$ iff $\overline{\delta}(s_0, x) \in F''$ iff $x \in L$ and there is no nonempty y with $\overline{\delta}(s_0, xy) \in F$, iff $x \in F(L)$. So $F(L) = L(A') \in \mathfrak{D}_\Sigma$. □

5.20. $\mathfrak{D}_\Sigma$ **is closed under reversal.**

(a) **Construction.** Given DFA $A = \langle \Sigma, S, s_0, \delta, F \rangle$, build NDFA $A^r = \langle \Sigma, S, F, \delta^r, \{s_0\} \rangle$ where $\delta^r(s, \boldsymbol{a}) = \{t \in S \mid \delta(t, \boldsymbol{a}) = s\}$ (reverse all arrows).

(b) **Proof.** We show $x \in L(A^r)$ iff $x^r \in L(A)$.
By induction on $|x|$: a path $f \xrightarrow{a_1} s_1 \xrightarrow{a_2} \cdots \xrightarrow{a_k} s_0$ in A^r (from some $f \in F$ to s_0) corresponds to the path $s_0 \xrightarrow{a_k} \cdots \xrightarrow{a_1} f$ in A. Hence $a_1 a_2 \cdots a_k \in L(A^r)$ iff $a_k \cdots a_1 \in L(A)$, i.e., $x \in L(A^r)$ iff $x^r \in L(A)$.

(c) $L^r = L(A^r) \in \mathfrak{D}_\Sigma$ (since A^r is an NDFA which is equivalent to some DFA). □

5.21. (a) $\mathfrak{D}_\Sigma$ **is not closed under** $G(L) = \{w^r \cdot w \mid w \in L\}$**.**
Let $L = \Sigma^*$ over $\Sigma = \{\boldsymbol{a}, \boldsymbol{b}\}$. Then $G(L) = \{w^r w \mid w \in \Sigma^*\}$, the set of even-length palindromes (of the form $w^r w$). By the pumping lemma, this is not FAD: take $w = \boldsymbol{a}^n \boldsymbol{b}^n$; then $w^r w = \boldsymbol{b}^n \boldsymbol{a}^n \boldsymbol{a}^n \boldsymbol{b}^n \in G(L)$. If $G(L)$ were FAD with n states, pump the string $\boldsymbol{b}^n \boldsymbol{a}^n \boldsymbol{a}^n \boldsymbol{b}^n$ to get a contradiction (pumping the first block of $\boldsymbol{b}$s destroys the palindrome structure).

(b) $\mathfrak{N}_\Sigma$ **is closed under** G**.** Let $L \in \mathfrak{N}_\Sigma$. If $G(L)$ were FAD, then Theorem 5.11 would imply that $G(L)^b$ is FAD. But $G(L)^b = L^r$, and Exercise 5.20 shows that reversal preserves the FAD languages. Hence $L = (L^r)^r$ would be FAD, contradicting $L \in \mathfrak{N}_\Sigma$. Therefore $G(L) \in \mathfrak{N}_\Sigma$.

(c) $\mathfrak{N}_\Sigma$ **is not closed under** b**.** By Lemma 5.6, $L = \{\boldsymbol{a}^n\boldsymbol{b}^n \mid n \geq 0\} \in \mathfrak{N}_\Sigma$, but $L^b = \{\boldsymbol{a}^n \mid n \geq 0\}$ is FAD.

5.22. $\mathfrak{N}_\Sigma$ **is closed under reversal.**

Assume $L \in \mathfrak{N}_\Sigma$ but $L^r \not in\mathfrak{N}_\Sigma$. Then $L^r \in \mathfrak{D}_\Sigma$. By Exercise 5.20, $\mathfrak{D}_\Sigma$ is closed under reversal, so $L = (L^r)^r \in \mathfrak{D}_\Sigma$, a contradiction. Hence $L^r \in \mathfrak{N}_\Sigma$ whenever $L \in \mathfrak{N}_\Sigma$.

5.23. $\mathfrak{N}_\Sigma$ **is *not* closed under** P **(prefix operator).**

Let $K = \{x \in \{\boldsymbol{a},\boldsymbol{b}\}^* \mid |x|_{\boldsymbol{a}} = |x|_{\boldsymbol{b}}\}$. This language belongs to $\mathfrak{N}_\Sigma$ (it is the language used in Lemma 5.4). But $P(K) = \{\boldsymbol{a},\boldsymbol{b}\}^*$: given any word x, if $|x|_{\boldsymbol{a}} - |x|_{\boldsymbol{b}} = d$, append $|d|$ copies of the deficient symbol to obtain a balanced word. Thus $P(K)$ is FAD, so $\mathfrak{N}_\Sigma$ is not closed under P.

5.24. $\mathfrak{N}_\Sigma$ **is not closed under** S **(suffix operator).**

Using the same language K, we have $S(K) = \{\boldsymbol{a},\boldsymbol{b}\}^*$: for any word x, prefix it by enough copies of the deficient symbol to balance the counts. Since $K \in \mathfrak{N}_\Sigma$ but $S(K)$ is FAD, $\mathfrak{N}_\Sigma$ is not closed under S.

5.25. $\mathfrak{N}_\Sigma$ **is not closed under** C **(center operator).**

Again let $K = \{x \mid |x|_{\boldsymbol{a}} = |x|_{\boldsymbol{b}}\}$. Then $C(K) = \{\boldsymbol{a},\boldsymbol{b}\}^*$, because for any word x we can choose $y = \boldsymbol{a}^m$ and $z = \boldsymbol{b}^{m+|x|_{\boldsymbol{a}}-|x|_{\boldsymbol{b}}}$ with m large enough so that yxz has equally many $\boldsymbol{a}$s and $\boldsymbol{b}$s. Hence $C(K)$ is FAD while $K \in \mathfrak{N}_\Sigma$.

5.26. $\mathfrak{N}_\Sigma$ **is not closed under** F**.**

For the same language K, every word in K has a proper extension in K: if $x \in K$, then $x\boldsymbol{ab} \in K$. Therefore $F(K) = \emptyset$, which is FAD. So $\mathfrak{N}_\Sigma$ is not closed under F.

5.27. **Complementing final states of an NDFA does not give complement language.**

From Exercise 4.8: if A is an NDFA, $L(A^\sim) \neq \sim L(A)$ in general.

Example. Let A have $\Sigma = \{\boldsymbol{a}\}$, $S = \{s_0, s_1\}$, $S_0 = \{s_0\}$, $F = \{s_1\}$, $\delta(s_0,\boldsymbol{a}) = \{s_0, s_1\}$, $\delta(s_1,\boldsymbol{a}) = \{s_1\}$. Then $L(A) = \{\boldsymbol{a}\}^+ = \{\boldsymbol{a},\boldsymbol{aa},\ldots\}$. After complementing: $F^\sim = \{s_0\}$. $\boldsymbol{a}$: from s_0, reach $\{s_0, s_1\}$; $s_0 \in F^\sim$, so $\boldsymbol{a} \in L(A^\sim)$. But $\boldsymbol{a} \in L(A)$, so $\boldsymbol{a} \not in \sim L(A)$. Thus $\boldsymbol{a} \in L(A^\sim)$ but $\boldsymbol{a} \not in \sim L(A)$, so $L(A^\sim) \neq \sim L(A)$.

5.28. **Complete the proof of Lemma 5.7.**

Proof. Let $L = \{x \in \{\boldsymbol{a},\boldsymbol{b}\}^* \mid |x|_{\boldsymbol{a}} \neq |x|_{\boldsymbol{b}}\}$, the language used in Lemma 5.7. We show that $L/L = \Sigma^*$.

Take any $x \in \Sigma^*$ and let $d = |x|_{\boldsymbol{a}} - |x|_{\boldsymbol{b}}$. Define $y = \boldsymbol{a}^{|d|+1}$. Then $y \in L$, because it has strictly more $\boldsymbol{a}$s than $\boldsymbol{b}$s. Also

$$|xy|_{\boldsymbol{a}} - |xy|_{\boldsymbol{b}} = d + (|d| + 1) \neq 0,$$

so $xy \in L$ as well. Hence $x \in L/L$. Since x was arbitrary, $L/L = \Sigma^*$. Therefore $\mathfrak{N}_\Sigma$ is not closed under quotient. $\square$

5.29. **Collection closed under union, concatenation, Kleene closure, but not intersection.**

Take $\mathscr{C} = I_\Sigma$, the collection of all infinite languages from Exercise 5.4. Exercise 5.4 shows that I_Σ is closed under union, concatenation, and Kleene closure, but not under intersection. So I_Σ is the required example.

5.30. **Closed under union does not imply closed under intersection.**

Counterexample. Again use I_Σ, the infinite languages. Exercise 5.4(b) shows it is closed under union, while Exercise 5.4(c) shows it is not closed under intersection.

5.31. **Prove $L(A^\bullet) = L_1 \cdot L_2$ (Theorem 5.4 construction).**

Proof. If $x \in L_1 \cdot L_2$, write $x = uv$ with $u \in L_1$ and $v \in L_2$. Run A_1 on u from s_{0_1} to some state of F_1.

If $v = \lambda$, then $\lambda \in L_2$, so Theorem 5.4 makes every state of F_1 final in $A^\bullet$; hence the same path accepts x.

If $v = \boldsymbol{a}w$ is nonempty, then from the final state reached after u the definition of $\delta^\bullet$ allows the next input letter $\boldsymbol{a}$ to be processed exactly as it would be from s_{0_2} in A_2. The rest of v then follows the transitions of A_2, so some path in $A^\bullet$ accepts x. Thus $L_1 \cdot L_2 \subseteq L(A^\bullet)$.

Conversely, let $x \in L(A^\bullet)$. An accepting path either stays entirely in S_1 and ends in a state of F_1; this can happen only when $\lambda \in L_2$, so then $x \in L_1 = L_1 \cdot \lambda \subseteq L_1 \cdot L_2$. Otherwise there is a first transition that enters S_2. Just before that step, the path is at some state $f \in F_1$, so the already-read prefix u lies in L_1. The rest of the input, beginning with the symbol that caused the move into S_2, follows A_2 from s_{0_2} to a final state, so it lies in L_2. Hence $x = uv \in L_1 \cdot L_2$. □

5.32. **Prove $L(A_*) = L^*$ (Theorem 5.5 construction).**

Proof. Since $q_0 \in F_*$, the empty word is accepted, so $L^0 = \{\lambda\} \subseteq L(A_*)$.

Now let $x = x_1 x_2 \cdots x_k$ with each $x_i \in L$. Run the original machine A on x_1 from some start state in S_0 to a final state in F. Whenever one factor ends and the next factor begins with a letter $\boldsymbol{a}$, the definition of δ_* for final states adds the transitions $\bigcup_{t \in S_0} \delta(t, \boldsymbol{a})$, so the same input letter can also begin a new run of A on the next factor. Repeating this for all factors yields an accepting path for x, so $L^* \subseteq L(A_*)$.

Conversely, consider an accepting path of A_* on some nonempty word x. Because $\delta_*(q_0, \boldsymbol{a}) = \emptyset$, the path must begin in some state of S_0. Each time the path uses one of the extra transitions $\bigcup_{t \in S_0} \delta(t, \boldsymbol{a})$ from a final state, it starts a fresh copy of the original machine on the next factor of the input. Thus the input splits into finitely many consecutive blocks, each of which labels a path in A from a start state to a final state; hence each block lies in L. Therefore $x \in L^*$. □

5.33. **Amplify the proof of Theorem 5.2 (union).**

Theorem 5.2: $\mathfrak{D}_\Sigma$ is closed under union. The proof uses $A^\cup$ (NDFA combining A_1 and A_2 with joint start-state set $S_0^1 \cup S_0^2$). The key equivalences:

$x \in L(A^\cup)$ [Definition of $L(A^\cup)$] $\Leftrightarrow \overline{\delta}^\cup(S_0^1 \cup S_0^2, x) \cap (F_1 \cup F_2) \neq \emptyset$ [Def. 4.4] $\Leftrightarrow (\overline{\delta}^\cup(S_0^1, x) \cup \overline{\delta}^\cup(S_0^2, x)) \cap (F_1 \cup F_2) \neq \emptyset$ [Exercise 4.27] $\Leftrightarrow (\overline{\delta}_1(s_0^1, x) \cup \overline{\delta}_2(s_0^2, x)) \cap (F_1 \cup F_2) \neq \emptyset$ [paths stay in S_i] $\Leftrightarrow \overline{\delta}_1(s_0^1, x) \cap F_1 \neq \emptyset$ or $\overline{\delta}_2(s_0^2, x) \cap F_2 \neq \emptyset \Leftrightarrow x \in L_1 \cup L_2$.

5.34. **Given DFA A, define NDFA accepting $L(A)^+$.**

Define $A^+ = \langle \Sigma, S, s_0, \delta^+, F\rangle$ where $\delta^+(s, \boldsymbol{a}) = \{\delta(s, \boldsymbol{a})\} \cup \{s_0 \mid \delta(s, \boldsymbol{a}) \in F\}$. That is, whenever a transition would lead to a final state, simultaneously allow the machine to restart from s_0. The final states remain F. This accepts $L^+ = \bigcup_{k\geq 1} L^k$.

5.35. **Given NDFA A, define NDFA accepting $L(A)^+$.**

Same construction as 5.34 adapted to NDFAs: $A^+ = \langle \Sigma, S, S_0, \delta^+, F\rangle$ where for each $s \in S$ and $\boldsymbol{a} \in \Sigma$: $\delta^+(s, \boldsymbol{a}) = \delta(s, \boldsymbol{a}) \cup (S_0 \text{ if } \delta(s, \boldsymbol{a}) \cap F \neq \emptyset)$. This allows returning to any start state after reaching a final state.

5.36. **If L is FAD, must all subsets of L be FAD?**

No. Let $\Sigma = \{\boldsymbol{a}, \boldsymbol{b}\}$ and $L = \Sigma^*$ (FAD). The subset $\{a^n b^n \mid n \geq 0\}$ is not FAD (by the pumping lemma).

5.37. **If $L \in \mathfrak{D}_\Sigma$, must all supersets of L be in $\mathfrak{D}_\Sigma$?**

No. Let $L = \emptyset \in \mathfrak{D}_\Sigma$. The superset $\{a^n b^n \mid n \geq 0\}$ is not FAD.

5.38. **If $L \in \mathfrak{N}_\Sigma$, must all subsets be in $\mathfrak{N}_\Sigma$?**

No. Let $K = \{x \in \{\boldsymbol{a}, \boldsymbol{b}\}^* \mid |x|_{\boldsymbol{a}} = |x|_{\boldsymbol{b}}\} \in \mathfrak{N}_\Sigma$. Then $\{\lambda\} \subseteq K$, but $\{\lambda\} \in \mathfrak{D}_\Sigma$.

5.39. **If $L \in \mathfrak{N}_\Sigma$, must all supersets be in $\mathfrak{N}_\Sigma$?**

No. With the same language $K \in \mathfrak{N}_\Sigma$, we have $K \subseteq \{\boldsymbol{a}, \boldsymbol{b}\}^*$, but $\{\boldsymbol{a}, \boldsymbol{b}\}^* \in \mathfrak{D}_\Sigma$.

5.40. **Why is q_0 needed in the Kleene closure construction (Theorem 5.5)?**

The extra start state q_0 handles the case $\lambda \in L^*$ (which is always true) independently from the rest of the machine. Without q_0, making s_0 both the start and a final state could create spurious acceptance: words that loop back through the original start states might be accepted even when they are not valid concatenations of words from L. The new state q_0 is needed only to accept λ without changing the behavior of the original start states.

5.41. **Redesign Theorem 5.4 (concatenation) using λ-transitions.**

Given DFAs A_1 and A_2, construct $A^\bullet_\lambda = \langle \Sigma, S_1 \cup S_2, s^1_0, \delta^\bullet_\lambda, F_2\rangle$ where: $\delta^\bullet_\lambda(s, \boldsymbol{a}) = \delta_1(s, \boldsymbol{a})$ for $s \in S_1$, $\delta^\bullet_\lambda(s, \boldsymbol{a}) = \delta_2(s, \boldsymbol{a})$ for $s \in S_2$, and $\delta^\bullet_\lambda(f, \lambda) \ni s^2_0$ for each $f \in F_1$. The λ-transition from each $f \in F_1$ to s^2_0 cleanly separates the two phases.

5.42. **Redesign Theorem 5.5 (Kleene closure) using λ-transitions, without extra q_0.**

In general this cannot be done correctly without changing the original start state's behavior, so the extra q_0 is still necessary.

The tempting construction is to make s_0 final (so that λ is accepted) and add λ-transitions from each $f \in F$ back to s_0. But this is wrong: making s_0 final can cause the new machine to accept words that merely return to s_0, even when they are not in L^*.

For example, let $L = \{\boldsymbol{aa}\}$ over $\Sigma = \{\boldsymbol{a}, \boldsymbol{b}\}$, and let A be a DFA for L in which s_0 loops on $\boldsymbol{b}$. If we simply make s_0 final, then the redesigned machine accepts $\boldsymbol{b}$, because

$$\delta(s_0, \boldsymbol{b}) = s_0 \in F'.$$

But $\boldsymbol{b}\ \not{\lambda} nL^*$, since every nonempty word of L^* is a concatenation of copies of $\boldsymbol{aa}$.

So even with λ-transitions, the fresh start state q_0 is needed to accept λ without altering the behavior of the original start state.

5.43. **Redesign Theorem 5.4 for NDFAs A_1 and A_2.**

Given NDFAs $A_1 = \langle \Sigma, S_1, S_0^1, \delta_1, F_1 \rangle$ and $A_2 = \langle \Sigma, S_2, S_0^2, \delta_2, F_2 \rangle$ with $S_1 \cap S_2 = \emptyset$: Define

$$A^\bullet = \langle \Sigma, S_1 \cup S_2, S_0^1, \delta^\bullet, F^\bullet \rangle,$$

where

$$F^\bullet = \begin{cases} F_2, & \lambda \not{i}nL_2, \\ F_1 \cup F_2, & \lambda \in L_2, \end{cases}$$

and

$$\delta^\bullet(s, \boldsymbol{a}) = \begin{cases} \delta_1(s, \boldsymbol{a}), & s \in S_1 - F_1, \\ \delta_1(s, \boldsymbol{a}) \cup \bigcup\limits_{t \in S_0^2} \delta_2(t, \boldsymbol{a}), & s \in F_1, \\ \delta_2(s, \boldsymbol{a}), & s \in S_2. \end{cases}$$

This is the exact NDFA analogue of Theorem 5.4: when the current state is already in F_1, the same input symbol may also start a run of A_2 from any start state in S_0^2.

5.44. **Redesign Theorem 5.5 for DFA A.**

If $A = \langle \Sigma, S, s_0, \delta, F \rangle$ is deterministic, the redesigned machine is the NDFA

$$A_* = \langle \Sigma, S \cup \{q_0\}, \{q_0, s_0\}, \delta_*, F \cup \{q_0\} \rangle,$$

where

$$\delta_*(s, \boldsymbol{a}) = \begin{cases} \{\delta(s, \boldsymbol{a})\}, & s \in S - F, \\ \{\delta(s, \boldsymbol{a}), \delta(s_0, \boldsymbol{a})\}, & s \in F, \\ \emptyset, & s = q_0. \end{cases}$$

This is the DFA-specialized form of Theorem 5.5, with singleton outputs exactly because the old transition function δ is deterministic.

5.45. **Comparing R_L and $R_{\sim L}$.**

The right congruence $\equiv_L$ on Σ^* is defined by $x \equiv_L y$ iff $\forall z \in \Sigma^*, xz \in L \Leftrightarrow yz \in L$. The right congruence $\equiv_{\sim L}$ satisfies $x \equiv_{\sim L} y$ iff $\forall z, xz \in \sim L \Leftrightarrow yz \in \sim L$, i.e., $xz \not{i}nL \Leftrightarrow yz \not{i}nL$, i.e., $xz \in L \Leftrightarrow yz \in L$. So $\equiv_L = \equiv_{\sim L}$: the two right congruences are identical.

This makes sense: if A is a minimal DFA for L, then $A^\sim$ (with complemented final states) is a minimal DFA for $\sim L$, and both have the same state set and transition function. Thus $R_L = R_{\sim L}$.

5.46. (a) **Examples where $R_{L_1 \cap L_2} = R_{L_1} \cap R_{L_2}$.** Let $L_1 = L_2$ be any FAD language. Then $L_1 \cap L_2 = L_1$, so $R_{L_1 \cap L_2} = R_{L_1} = R_{L_1} \cap R_{L_1} = R_{L_1} \cap R_{L_2}$.

(b) **Examples where $R_{L_1 \cap L_2} \neq R_{L_1} \cap R_{L_2}$.** Let $L_1 = \{\boldsymbol{a}\}$ and $L_2 = \{\boldsymbol{b}\}$ over $\Sigma = \{\boldsymbol{a}, \boldsymbol{b}\}$. Then $L_1 \cap L_2 = \emptyset$, so $R_{L_1 \cap L_2} = R_\emptyset$ is the universal relation. But $\lambda \not{R}_{L_1} \boldsymbol{a}$, because with suffix λ the word $\boldsymbol{a}$ is in L_1 and λ is not. Hence

$$R_{L_1 \cap L_2} \neq R_{L_1} \cap R_{L_2}.$$

5.47. **$\mathfrak{D}_\Sigma$ is closed under relative complement.**

(a) **By theorems.** $L_1 \setminus L_2 = L_1 \cap \sim L_2$. Since $\mathfrak{D}_\Sigma$ is closed under complement (Theorem 5.1) and intersection (Lemma 5.1), $L_1 \setminus L_2 \in \mathfrak{D}_\Sigma$.

(b) **Machine definition.** Define A^- as the product of A_1 and $A_2^\sim$ (complement of A_2), using the intersection machine from Lemma 5.1 with $F^- = F_1 \times (S_2 \setminus F_2)$.

(c) **Proof.** By induction (as in Exercise 5.56/5.57), $\overline{\delta^-}(\langle s_0^1, s_0^2\rangle, x) = \langle \overline{\delta^1}(s_0^1, x), \overline{\delta^2}(s_0^2, x)\rangle$. Then $x \in L(A^-)$ iff $\overline{\delta^1}(s_0^1, x) \in F_1$ and $\overline{\delta^2}(s_0^2, x) \not\in nF_2$ iff $x \in L_1 \setminus L_2$. □

5.48. $\mathfrak{L}_\Sigma$ = languages recognized by λ-NDFAs. By Theorem 4.2, $\mathfrak{L}_\Sigma = \mathfrak{D}_\Sigma$. Hence $\mathfrak{L}_\Sigma$ has exactly the same closure properties as $\mathfrak{D}_\Sigma$ (Theorems 5.1–5.12).

5.49. (a) **Example of L with $\lambda \in L^+$.** Let $L = \{\lambda\}$. Then $L^+ = \bigcup_{k\geq 1} L^k = \{\lambda\}$, and $\lambda \in L^+$.

(b) **Three languages with $L^+ = L$.**

(i) $L = \Sigma^*$: $(\Sigma^*)^+ = \Sigma^*$.

(ii) $L = \{\boldsymbol{a}\}^*$: $(\{\boldsymbol{a}\}^*)^+ = \{\boldsymbol{a}\}^*$.

(iii) $L = \Sigma^+$: $(\Sigma^+)^+ = \Sigma^+$ (since $\Sigma^+ \cdot \Sigma^+ \subseteq \Sigma^+$).

5.50. (a) **Prove $\overline{\delta^\cup}(s, x) = \overline{\delta}_i(s, x)$ for $s \in S_i$.**

Proof. By induction on $|x|$.
Base ($|x| = 0$): $\overline{\delta^\cup}(s, \lambda) = \{s\} = \overline{\delta}_i(s, \lambda)$ (by definition of extended transition functions for single start-states, noting NDFAs).
Step: Assume $\overline{\delta^\cup}(s, x) = \overline{\delta}_i(s, x)$ for $s \in S_i$. For $x\boldsymbol{a}$:

$$\overline{\delta^\cup}(s, x\boldsymbol{a}) = \bigcup_{t\in\overline{\delta^\cup}(s,x)} \delta^\cup(t, \boldsymbol{a}) = \bigcup_{t\in\overline{\delta}_i(s,x)} \delta_i(t, \boldsymbol{a}) = \overline{\delta}_i(s, x\boldsymbol{a}),$$

where we used the fact that $\overline{\delta}_i(s, x) \subseteq S_i$ (paths from $s \in S_i$ stay in S_i, since $\delta_i : S_i \to \wp(S_i)$) and $\delta^\cup(t, \boldsymbol{a}) = \delta_i(t, \boldsymbol{a})$ for $t \in S_i$. □ □

(b) **Was this used in Theorem 5.2?** Yes, implicitly: the proof that $L(A^\cup) = L_1 \cup L_2$ relies on the fact that paths starting in S_1 stay in S_1 and paths starting in S_2 stay in S_2, which is exactly this lemma.

5.51. (a) **$\mathfrak{N}_\Sigma$ is not closed under relative complement.** Let $K = \{x \in \{\boldsymbol{a}, \boldsymbol{b}\}^* \mid |x|_{\boldsymbol{a}} = |x|_{\boldsymbol{b}}\} \in \mathfrak{N}_\Sigma$. Then $K - K = \emptyset \in \mathfrak{D}_\Sigma$.

(b) **$\mathfrak{N}_\Sigma$ is not closed under union.** If $K \in \mathfrak{N}_\Sigma$, then $\sim K \in \mathfrak{N}_\Sigma$ by Theorem 5.8. But $K \cup \sim K = \Sigma^* \in \mathfrak{D}_\Sigma$.

(c) **$\mathfrak{N}_\Sigma$ is not closed under concatenation.** Let $E = \{\boldsymbol{a}^n\boldsymbol{b}^n \mid n \geq 1\}$ and define $L = \{\boldsymbol{a}, \boldsymbol{b}\}^* - E$. Since $E \in \mathfrak{N}_\Sigma$, Theorem 5.8 gives $L \in \mathfrak{N}_\Sigma$. Also $\lambda \in L$, and if $x \not\in nL$ then $x = \boldsymbol{a}^n\boldsymbol{b}^n = \boldsymbol{a} \cdot \boldsymbol{a}^{n-1}\boldsymbol{b}^n$ with both factors in L. Hence $L \cdot L = \{\boldsymbol{a}, \boldsymbol{b}\}^* \in \mathfrak{D}_\Sigma$.

(d) **$\mathfrak{N}_\Sigma$ is not closed under Kleene closure.** With the same language L, we have $\lambda \in L$ and $L \cdot L = \Sigma^*$, so $L^* = \Sigma^* \in \mathfrak{D}_\Sigma$.

(e) **If $L \in \mathfrak{N}_\Sigma$, then $L^+ \in \mathfrak{N}_\Sigma$: No.** For the same language L, because $\lambda \in L$ we have $L^+ = L^* = \Sigma^*$, which is FAD.

5.52. **Why require $S_1 \cap S_2 = \emptyset$ in Theorem 5.4?**

If $S_1 \cap S_2 \neq \emptyset$, the transition function $\delta^\bullet$ would be ambiguous: for a state $s \in S_1 \cap S_2$, which transitions should apply—those of A_1 or A_2? The definition of $\delta^\bullet$ would not be well-defined. Moreover, the proof that paths through the concatenation machine correctly simulate A_1 then A_2 relies on the fact that once the computation enters S_2, it stays in S_2 (and similarly for S_1). If the state sets overlap, a path might inadvertently "mix" transitions from A_1 and A_2.

5.53. $\mathfrak{D}_\Sigma$ **is closed under** $E(L) = \{z \mid \exists y \in \Sigma^+, \exists x \in L \ni z = yx\}$**.**

Here $E(L) = \Sigma^+ \cdot L$, the set of words obtained by prepending a nonempty prefix to a word of L.

(a) **Construction.** Let B be the two-state DFA for Σ^+: start state p_0 (nonfinal), final state p_1, and for every $\boldsymbol{a} \in \Sigma$, $\delta_B(p_0, \boldsymbol{a}) = p_1$ and $\delta_B(p_1, \boldsymbol{a}) = p_1$. If A accepts L, apply the concatenation construction of Theorem 5.4 to B and A.

(b) **Proof.** A word z is accepted by the new machine iff it can be written as $z = yx$ where y is accepted by B and x is accepted by A. That is equivalent to saying $y \in \Sigma^+$ and $x \in L$, so the new machine accepts exactly $E(L)$.

(c) Since $\Sigma^+ \in \mathfrak{D}_\Sigma$ and $\mathfrak{D}_\Sigma$ is closed under concatenation, $E(L) = \Sigma^+ \cdot L \in \mathfrak{D}_\Sigma$. □

5.54. $\mathfrak{D}_\Sigma$ **is closed under** $B(L) = \{z \mid \exists x \in L, \exists y \in \Sigma^* \ni z = xy\}$**.**

$B(L)$ is the set of all words that have a prefix in L.

(a) **Construction.** Given $A = \langle \Sigma, S, s_0, \delta, F\rangle$ for L, build

$$A' = \langle \Sigma, S \times \{0,1\}, \langle s_0, 0\rangle, \delta', S \times \{1\}\rangle,$$

where the second component records whether some prefix read so far belongs to L. Define

$$\delta'(\langle s, b\rangle, a) = \langle t, b'\rangle,$$

where $t = \delta(s, a)$ and

$$b' = \begin{cases} 1 & \text{if } b = 1 \text{ or } t \in F, \\ 0 & \text{otherwise.} \end{cases}$$

Thus once the flag becomes 1, it stays 1 forever.

(b) **Proof:** By induction on the length of the input, after reading a word z the machine A' is in state

$$\langle \overline{\delta}(s_0, z), 1\rangle$$

exactly when some prefix of z is in L, and otherwise in

$$\langle \overline{\delta}(s_0, z), 0\rangle.$$

Therefore A' accepts exactly those words having a prefix in L, that is, exactly the words in $B(L)$.

(c) □

5.55. $\mathfrak{D}_\Sigma$ **is closed under** $M(L) = \{z \mid \exists x \in L, \exists y \in \Sigma^+ \ni z = xy\}$**.**

$M(L) = \{z \mid z = xy, x \in L, y \in \Sigma^+\} = L \cdot \Sigma^+$. By closure under concatenation, $L \cdot \Sigma^+ \in \mathfrak{D}_\Sigma$.

(a) **Construction:** Concatenate A (for L) with a DFA for Σ^+ (which is a two-state machine: start nonfinal, self-loop on Σ to final, final self-loops on Σ).

(b) **Proof:** By Theorem 5.4, $M(L) = L(A^\bullet) \in \mathfrak{D}_\Sigma$.

(c) □

5.56. **Prove** $\overline{\delta^\cap}(\langle s, t\rangle, x) = \langle \overline{\delta}_1(s,x), \overline{\delta}_2(t,x)\rangle$**.**

Proof. By induction on $|x|$.

Base: $\overline{\delta^\cap}(\langle s,t\rangle, \lambda) = \langle s,t\rangle = \langle \overline{\delta}_1(s,\lambda), \overline{\delta}_2(t,\lambda)\rangle$.

Step: Assume the result for x. Then:

$$\begin{aligned}
\overline{\delta^\cap}(\langle s,t\rangle, x\boldsymbol{a}) &= \delta^\cap(\overline{\delta^\cap}(\langle s,t\rangle, x), \boldsymbol{a}) \\
&= \delta^\cap(\langle \overline{\delta}_1(s,x), \overline{\delta}_2(t,x)\rangle, \boldsymbol{a}) \\
&= \langle \delta_1(\overline{\delta}_1(s,x), \boldsymbol{a}), \delta_2(\overline{\delta}_2(t,x), \boldsymbol{a})\rangle \\
&= \langle \overline{\delta}_1(s, x\boldsymbol{a}), \overline{\delta}_2(t, x\boldsymbol{a})\rangle. \quad \square
\end{aligned}$$

□

5.57. **Prove** $L(A^\cap) = L_1 \cap L_2$**.**

Proof. By Exercise 5.56:

$$\begin{aligned}
x \in L(A^\cap) &\Leftrightarrow \overline{\delta^\cap}(\langle s_0^1, s_0^2\rangle, x) \in F^\cap \\
&\Leftrightarrow \langle \overline{\delta}_1(s_0^1, x), \overline{\delta}_2(s_0^2, x)\rangle \in F_1 \times F_2 \\
&\Leftrightarrow \overline{\delta}_1(s_0^1, x) \in F_1 \wedge \overline{\delta}_2(s_0^2, x) \in F_2 \\
&\Leftrightarrow x \in L_1 \wedge x \in L_2 \\
&\Leftrightarrow x \in L_1 \cap L_2. \quad \square
\end{aligned}$$

□

5.58. **Prove Theorem 5.6 (**$\mathfrak{D}_\Sigma$ **closed under** Z**).**

Proof. Let $A = \langle \Sigma, S, s_0, \delta, F\rangle$ be a DFA for L. Build a λ-NDFA A_Z by adding, for every ordinary transition $s \xrightarrow{\boldsymbol{a}} t$ of A, a matching λ-transition $s \xrightarrow{\lambda} t$. Then a path in A_Z may either read a letter of the input or skip a letter that was present in some word of L.

Thus x is accepted by A_Z iff there exists some $w \in L$ from which x can be obtained by deleting zero or more letters, i.e., iff $x \in Z(L)$. Hence $Z(L) \in \mathfrak{D}_\Sigma$. □

5.59. **Prove Theorem 5.7 (**$\mathfrak{D}_\Sigma$ **closed under** Y**).**

Proof. Let $A = \langle \Sigma, S, s_0, \delta, F\rangle$ be a DFA for L. Following the text after Theorem 5.7, build two copies of A. Being in the first copy means that no letter has yet been deleted; being in the second copy means that exactly one letter has been deleted. Ordinary transitions stay within each copy, and for every transition $s \xrightarrow{\boldsymbol{a}} t$ in the first copy add a λ-transition from the first-copy state s to the second-copy state t.

An accepting path therefore behaves like a path of A on some word of L, with exactly one letter skipped at the moment when the path moves from the first copy to the second. So the resulting machine accepts exactly $Y(L)$, and therefore $Y(L) \in \mathfrak{D}_\Sigma$. $\square$

5.60. (a) **Alternative construction for Theorem 5.7 without λ-moves.**

Let $A = \langle \Sigma, S, s_0, \delta, F\rangle$ be a DFA for L. Use state set $S \times \{0,1\}$, where the second component records whether the one deletion has already occurred. The start state is $\langle s_0, 0\rangle$. Final states are

$$\{\langle s,1\rangle \mid s \in F\} \cup \{\langle s,0\rangle \mid (\exists \boldsymbol{a} \in \Sigma)\ \delta(s,\boldsymbol{a}) \in F\}.$$

On input $\boldsymbol{a}$, define

$$\delta'(\langle s,0\rangle, \boldsymbol{a}) = \{\langle \delta(s,\boldsymbol{a}), 0\rangle\} \cup \{\langle \delta(\delta(s,\boldsymbol{b}),\boldsymbol{a}), 1\rangle \mid \boldsymbol{b} \in \Sigma\},$$

and

$$\delta'(\langle s,1\rangle, \boldsymbol{a}) = \{\langle \delta(s,\boldsymbol{a}), 1\rangle\}.$$

(b) **Proof:** In the 0-copy the machine either reads the current input symbol normally, or guesses that one letter $\boldsymbol{b}$ of the original word was deleted immediately before it and moves into the 1-copy. Once in the 1-copy, the rest of the input is processed exactly as in A. The extra final states in the 0-copy handle the case where the deleted letter was the last letter of the original word. Hence this NDFA accepts exactly $Y(L)$.

5.61. **$\mathfrak{D}_\Sigma$ is closed under $W(L)$ (delete one or more letters).**

(a) **Construction.** Given DFA A for L, build λ-NDFA A' by adding, for each transition $s \xrightarrow{\boldsymbol{a}} t$, an optional λ-transition $s \xrightarrow{\lambda} t$ (allowing deletion of $\boldsymbol{a}$). Require at least one deletion by adding a flag bit.

More precisely: A' has state set $S \times \{0,1\}$ where the second component tracks whether at least one letter has been deleted. Initial state: $\langle s_0, 0\rangle$. Final states: $\{\langle f,1\rangle \mid f \in F\}$. For each $c \in \Sigma$,

$$\delta'(\langle s,b\rangle, c) = \{\langle \delta(s,c), b\rangle\}, \qquad \delta'(\langle s,b\rangle, \lambda) = \{\langle \delta(s,c), 1\rangle \mid c \in \Sigma\}.$$

The ordinary transition reads the next input symbol, while the λ-transition simulates deleting one symbol c from the original word and moves into the “already deleted” copy.

(b) **Proof.** A word z is in $W(L)$ iff z can be obtained by deleting at least one letter from some $w \in L$. A path in A' simulates a path in A but uses λ-transitions to skip letters; the flag ensures at least one skip. Hence $L(A') = W(L)$.

(c) Since A' is a λ-NDFA, $W(L) \in \mathfrak{D}_\Sigma$. $\square$

5.62. **$\mathfrak{D}_\Sigma$ is closed under $V(L)$ (delete odd-positioned letters).**

(a) **Construction.** Let $A = \langle \Sigma, S, s_0, \delta, F\rangle$. Build the λ-NDFA

$$A' = \langle \Sigma, S \times \{0,1\}, \langle s_0, 0\rangle, \delta', F \times \{0,1\}\rangle,$$

where the second component records the parity of the next position in the original word: 0 means odd, 1 means even. For each $s \in S$ and $a \in \Sigma$,

$$\delta'(\langle s,1\rangle, a) = \{\langle \delta(s,a), 0\rangle\},$$

because even-positioned letters are kept, and

$$\delta'(\langle s,0\rangle, \lambda) = \{\langle \delta(s,c), 1\rangle \mid c \in \Sigma\},$$

because an odd-positioned letter is deleted and may be any symbol c.

(b) **Proof.** A computation of A' alternates between deleting one symbol of the original word at every odd position and reading one input symbol at every even position. Thus the visible input is exactly the word obtained by deleting the odd-positioned letters of some word accepted by A. Conversely, if a word z arises from deleting the odd-positioned letters of some $w \in L$, then A' can guess those deleted odd-positioned letters by λ-moves and read precisely the surviving even-positioned letters. So $L(A') = V(L)$.

(c) $L(A') \in \mathfrak{D}_\Sigma$. □

5.63. **$\mathfrak{D}_\Sigma$ is closed under $U(L)$ (delete even-positioned letters).**

Let

$$A = \langle \Sigma, S, s_0, \delta, F\rangle.$$

Build the λ-NDFA

$$A' = \langle \Sigma, S \times \{0,1\}, \langle s_0, 0\rangle, \delta', F \times \{0,1\}\rangle,$$

where again 0 means the next position in the original word is odd and 1 means even. This time odd-positioned letters are kept and even-positioned letters are deleted:

$$\delta'(\langle s,0\rangle, a) = \{\langle \delta(s,a), 1\rangle\} \qquad (a \in \Sigma),$$

$$\delta'(\langle s,1\rangle, \lambda) = \{\langle \delta(s,c), 0\rangle \mid c \in \Sigma\}.$$

Exactly the same reasoning as in 5.62 shows that A' accepts precisely those words obtained from words of L by deleting the even-positioned letters. Hence $U(L) \in \mathfrak{D}_\Sigma$. □

5.64. **$\mathfrak{D}_\Sigma$ is closed under $T(L)$ (delete every 3rd, 6th, ... letter).**

(a) **Construction.** Let $A = \langle \Sigma, S, s_0, \delta, F\rangle$. Build the λ-NDFA

$$A' = \langle \Sigma, S \times \{1,2,0\}, \langle s_0, 1\rangle, \delta', F \times \{1,2,0\}\rangle,$$

where the second component records the next position modulo 3. For each $s \in S$ and $a \in \Sigma$,

$$\delta'(\langle s,1\rangle, a) = \{\langle \delta(s,a), 2\rangle\}, \qquad \delta'(\langle s,2\rangle, a) = \{\langle \delta(s,a), 0\rangle\},$$

because positions 1 and 2 modulo 3 are kept, while

$$\delta'(\langle s,0\rangle, \lambda) = \{\langle \delta(s,c), 1\rangle \mid c \in \Sigma\},$$

because every third letter is deleted and may be any symbol c.

(b) **Proof.** The machine simulates A on an original word while the counter records which positions are first, second, and third in each block of three. At positions congruent to 1 or 2 modulo 3, it reads the next input symbol and processes it through δ; at positions congruent to 0 modulo 3, it guesses and deletes one original symbol by a λ-move. Thus the visible input is exactly the word obtained from the original one by deleting every 3rd, 6th, 9th, ... letter. Hence $L(A') = T(L)$.

(c) $L(A') \in \mathfrak{D}_\Sigma$. $\square$

5.65. $I(L) = L \cap P$ **where** $P = \{x \mid |x| \text{ is prime}\}$**.**

(a) $\mathfrak{D}_\Sigma$ **not closed under** I**.** Let $\Sigma = \{\boldsymbol{a}\}$ and $L = \Sigma^*$. Then

$$I(L) = P = \{\boldsymbol{a}^q \mid q \text{ prime}\}.$$

We show $P \notin \mathfrak{D}_\Sigma$ by the pumping lemma. Assume P were regular, and let p be a pumping length. Choose a prime $q > p$, and consider $\boldsymbol{a}^q \in P$. Write

$$\boldsymbol{a}^q = uvw$$

with $|uv| \le p$ and $|v| \ge 1$. Since the word is unary, let $|v| = m \ge 1$. Pump with $i = q + 1$. Then

$$|uv^{q+1}w| = q + qm = q(1 + m).$$

This number is composite, because $q > 1$ and $1 + m > 1$. Hence $uv^{q+1}w \notin P$, contradicting the pumping lemma. Therefore $P \notin \mathfrak{D}_\Sigma$.

(b) F_Σ **closed under** I**.** $I(L) = L \cap P \subseteq L$; if L is finite, so is $I(L)$.

(c) C_Σ **closed under** I**? No.** $L = \Sigma^* \setminus F$ for finite F. $I(L) = P \setminus F'$ for some finite F'. If P is cofinite: P is not cofinite (infinitely many non-primes), so $P \setminus F'$ is not cofinite. **Not closed.**

(d) B_Σ **closed under** I**? No.** $I(\Sigma^*) = P \notin B_\Sigma$ (not finite, not cofinite).

(e) I_Σ **closed under** I**? No.** Let $\Sigma = \{\boldsymbol{a}\}$ and $L = \{\boldsymbol{a}^{2n} \mid n \ge 1\}$. Then $L \in I_\Sigma$, but

$$I(L) = L \cap P = \{\boldsymbol{aa}\},$$

since 2 is the only even prime. Thus $I(L) \notin I_\Sigma$.

(f) J_Σ **closed under** I**?** $I(L) = L \cap P$. If $\sim L$ is infinite, $\sim (L \cap P) \supseteq \sim P$ which is infinite (non-primes). **Yes, closed.**

(g) $\boldsymbol{E}$ **closed under** I**?** If $\boldsymbol{abba} \in L$ and $|\boldsymbol{abba}| = 4$ is not prime, then $\boldsymbol{abba} \notin I(L)$. So $I(L)$ might not contain $\boldsymbol{abba}$. **Not closed.**

(h) $\mathfrak{N}_\Sigma$ **closed under** I**? No.** Over $\Sigma = \{\boldsymbol{a}\}$, let $P = \{\boldsymbol{a}^p \mid p \text{ prime}\}$ and $L = \{\boldsymbol{a}\}^* - P$. Since $P \notin \mathfrak{D}_\Sigma$, Theorem 5.8 implies $L \in \mathfrak{N}_\Sigma$. But

$$I(L) = L \cap P = \emptyset \in \mathfrak{D}_\Sigma.$$

Hence $\mathfrak{N}_\Sigma$ is not closed under I.

5.66. C: all languages over $\{\boldsymbol{a}, \boldsymbol{b}\}$ not containing λ.

(a) **Complementation: No.** $\sim L$ contains λ (since $\lambda \not inL$ but $\lambda \in \Sigma^*$), so $\sim L \not inC$.

(b) **Union: Yes.** If $\lambda \not inL_1$ and $\lambda \not inL_2$, then $\lambda \not inL_1 \cup L_2$.

(c) **Intersection: Yes.** $\lambda \not inL_1 \cap L_2 \subseteq L_1$.

(d) **Concatenation: Yes.** If $\lambda \not inL_1$ and $\lambda \not inL_2$: every word in $L_1 \cdot L_2$ is of the form xy with $x \in L_1, y \in L_2$, and $x \neq \lambda, y \neq \lambda$, so $xy \neq \lambda$.

(e) **Kleene closure: No.** $\lambda \in L^*$ always.

(f) **Relative complement: Yes.** $\lambda \not inL_1 \setminus L_2 \subseteq L_1$.

(g) L^+**: Yes.** $L^+ = \bigcup_{k\geq 1} L^k$. Each L^k contains words of positive length (since $\lambda \not inL$), so $\lambda \not inL^+$.

5.67. (a) $\mathfrak{D}_\Sigma$ **closed under finite union.**

(i) *By theorems and induction:* Base: $\{L\} \subseteq \mathfrak{D}_\Sigma$ trivially. Step: if $L_1 \cup L_2 \cup \cdots \cup L_k \in \mathfrak{D}_\Sigma$ and $L_{k+1} \in \mathfrak{D}_\Sigma$, then $(L_1 \cup \cdots \cup L_k) \cup L_{k+1} \in \mathfrak{D}_\Sigma$ by Theorem 5.2.

(ii) *By construction:* Build $A^{(k)}$ iteratively by the union construction of Exercise 5.14; each step is deterministic.

(b) $\mathfrak{D}_\Sigma$ **is NOT closed under infinite union.** Each singleton $\{a^n b^n\} \in \mathfrak{D}_\Sigma$ (finite languages are regular), but $\bigcup_{n\geq 0}\{a^n b^n\} = \{a^n b^n \mid n \geq 0\} \not in\mathfrak{D}_\Sigma$ (not regular, by pumping).

5.68. (a) **Three homomorphisms under which $\mathfrak{N}_\Sigma$ is not closed** (over $\Sigma = \{\boldsymbol{a}, \boldsymbol{b}\}$): Use the non-FAD language $K = \{x \in \{\boldsymbol{a}, \boldsymbol{b}\}^* \mid |x|_{\boldsymbol{a}} = |x|_{\boldsymbol{b}}\}$ from Lemma 5.4.

(i) $\xi_1(\boldsymbol{a}) = \boldsymbol{a}$, $\xi_1(\boldsymbol{b}) = \boldsymbol{a}$. By Lemma 5.4, $\overline{\xi_1}(K)$ is the regular language of even-length $\boldsymbol{a}$-strings.

(ii) $\xi_2(\boldsymbol{a}) = \lambda$, $\xi_2(\boldsymbol{b}) = \lambda$. Then $\overline{\xi_2}(K) = \{\lambda\} \in \mathfrak{D}_\Sigma$.

(iii) $\xi_3(\boldsymbol{a}) = \boldsymbol{a}$, $\xi_3(\boldsymbol{b}) = \lambda$. For $L = \{\boldsymbol{a}^n \boldsymbol{b}^n \mid n \geq 0\} \in \mathfrak{N}_\Sigma$, we get $\overline{\xi_3}(L) = \{\boldsymbol{a}^n \mid n \geq 0\} \in \mathfrak{D}_\Sigma$.

(b) **Three homomorphisms under which $\mathfrak{N}_\Sigma$ is closed:**

(i) The identity map: $\psi_1(\boldsymbol{a}) = \boldsymbol{a}$, $\psi_1(\boldsymbol{b}) = \boldsymbol{b}$.

(ii) The swap map: $\psi_2(\boldsymbol{a}) = \boldsymbol{b}$, $\psi_2(\boldsymbol{b}) = \boldsymbol{a}$.

(iii) The doubling map: $\psi_3(\boldsymbol{a}) = \boldsymbol{aa}$, $\psi_3(\boldsymbol{b}) = \boldsymbol{bb}$.

For ψ_1 and ψ_2, if $\overline{\psi_i}(L)$ were FAD then applying the inverse letter permutation would make L FAD. For ψ_3, the map is injective, so

$$L = \psi_3^{-1}(\overline{\psi_3}(L)).$$

Thus, if $\overline{\psi_3}(L)$ were FAD, then Theorem 5.10 (closure under inverse homomorphism) would imply that L is FAD as well.

5.69. Over $\Sigma = \{\boldsymbol{a}\}$ there is only one homomorphism under which $\mathfrak{N}_\Sigma$ is not closed, namely the erasing homomorphism $\psi(\boldsymbol{a}) = \lambda$. For any non-FAD unary language L, $\overline{\psi}(L) = \{\lambda\} \in \mathfrak{D}_\Sigma$.

For every other unary homomorphism, $\psi(\boldsymbol{a}) = \boldsymbol{a}^k$ with $k \geq 1$. Such a homomorphism is injective, so if $\overline{\psi}(L)$ were FAD, then

$$L = \overline{\psi^{-1}}(\overline{\psi}(L))$$

would be FAD by Theorem 5.10. Therefore $\mathfrak{N}_\Sigma$ is closed under every nonerasing unary homomorphism. So the answer is **no**: there are not two different unary homomorphisms under which $\mathfrak{N}_\Sigma$ is not closed.

5.70. (a) **Prove** $\overline{\delta^{\psi}}(s,x) = \overline{\delta}(s, \overline{\psi}(x))$**.**

Proof. By induction on $|x|$.
Base: $\overline{\delta^{\psi}}(s,\lambda) = s = \overline{\delta}(s,\lambda) = \overline{\delta}(s,\overline{\psi}(\lambda))$ (since $\psi(\lambda) = \lambda$).
Step: $\overline{\delta^{\psi}}(s, x\boldsymbol{b}) = \delta^{\psi}(\overline{\delta^{\psi}}(s,x), \boldsymbol{b}) = \overline{\delta}(\overline{\delta}(s,\overline{\psi}(x)), \psi(\boldsymbol{b})) = \overline{\delta}(s, \overline{\psi}(x)\psi(\boldsymbol{b})) = \overline{\delta}(s, \overline{\psi}(x\boldsymbol{b}))$. □ □

(b) **Proof of Theorem 5.10:** $y \in L(A^{\psi})$ iff $\overline{\delta^{\psi}}(s_0, y) \in F$ iff $\overline{\delta}(s_0, \psi(y)) \in F$ iff $\psi(y) \in L$ iff $y \in \psi^{-1}(L)$. □

5.71. (a) **Apparent contradiction resolved.** $\xi(L)$ is the set of even-length $\boldsymbol{a}$-strings; $\xi^{-1}(\xi(L))$ is the set of all strings x with $\xi(x) \in \xi(L)$, not L itself. In general, $\xi^{-1}(\xi(L)) \supseteq L$ and can be much larger; it is not necessarily L. The inverse image lands in $\mathfrak{D}_\Sigma$ by Theorem 5.10, but it's $\xi^{-1}(\xi(L))$, not L.

(b) **Example where** $\overline{\psi}(\psi^{-1}(L)) \neq L$**:** Take $\psi(\boldsymbol{a}) = \boldsymbol{aa}$, $L = \{\boldsymbol{a}\}$ (singleton). Then $\psi^{-1}(L) = \emptyset$ (no word maps to $\boldsymbol{a}$ under ψ, since ψ always produces even-length strings), and $\overline{\psi}(\emptyset) = \emptyset \neq L$.

(c) **Example where** $\psi^{-1}(\overline{\psi}(L)) \neq L$**:** Same: $\psi^{-1}(\overline{\psi}(\{\boldsymbol{a}\})) = \psi^{-1}(\{\boldsymbol{a}\}) = \emptyset \neq \{\boldsymbol{a}\} = L$.

(d) **Prove** $\overline{\psi}(\psi^{-1}(L)) \subseteq L$**:** Let $y \in \overline{\psi}(\psi^{-1}(L))$. Then $y = \psi(x)$ for some $x \in \psi^{-1}(L)$. Since $x \in \psi^{-1}(L)$, $\psi(x) \in L$, i.e., $y \in L$. □

(e) **Prove** $L \subseteq \psi^{-1}(\overline{\psi}(L))$**:** Let $x \in L$. Then $\psi(x) \in \overline{\psi}(L)$ (since $\psi(x)$ is the image of some element of L, namely x itself). Hence $x \in \psi^{-1}(\overline{\psi}(L))$. □

5.72. **$\mathfrak{D}_\Sigma$ is closed under L^e (last halves of even-length words in L).**
$L^e = \{x \mid \exists y \in \Sigma^* \ni yx \in L \wedge |x| = |y|\}$.

Proof. Observe that

$$x \in L^e \iff \exists y \in \Sigma^* \ (yx \in L \ \wedge \ |y| = |x|) \iff x^r \in (L^r)^b.$$

Indeed, $yx \in L$ with $|y| = |x|$ iff $x^r y^r \in L^r$ with $|x^r| = |y^r|$, which says exactly that x^r is a first half of some word in L^r.

Therefore

$$L^e = \big((L^r)^b\big)^r.$$

Exercise 5.20 shows that $\mathfrak{D}_\Sigma$ is closed under reversal, and Theorem 5.11 shows that it is closed under b. Hence $\mathfrak{D}_\Sigma$ is closed under e. □

5.73. **Show that redefining final states to merely "midway" states is incorrect.**

Example 5.25 gives a concrete counterexample. For the highlighted state q,

$$I(A,q) = \{(\boldsymbol{abc})^n \mid n \geq 1\} \qquad \text{and} \qquad T(A,q) = \{\boldsymbol{a}^{2n} \mid n \geq 1\}.$$

So a word can use q as a midpoint only when its first half has a length that occurs in both sets, namely a length divisible by 6. The proof of Theorem 5.11 therefore keeps only

$$I(A,q) \cap \overline{\psi^{-1}}(\overline{\psi}(I(A,q)) \cap \overline{\psi}(T(A,q))) = \{(\boldsymbol{abc})^{2n} \mid n \geq 1\}.$$

If we simply declared q to be a final state, the resulting automaton would accept all of $I(A,q)$, including $\boldsymbol{abc}$. But $\boldsymbol{abc}$ is not a first half of any word in L, because after reaching q there is no continuation of length 3 to a final state. Hence "make the midway states final" accepts too many words and is incorrect.

5.74. **Flaw in the argument that K is not FAD because M is not FAD.**

Theorem 5.9 states that $\mathfrak{D}_\Sigma$ is closed under homomorphism: if L is FAD and ψ is a homomorphism, then $\psi(L)$ is FAD. This does *not* mean that if L is not FAD, then $\psi(L)$ is not FAD. A homomorphism can collapse the structure that made M non-FAD, yielding a simpler FAD image $K = \psi(M)$. Hence the fact that $M \not in \mathfrak{D}_\Sigma$ says nothing by itself about whether $K = \psi(M)$ is FAD.

5.75. **Prove Theorem 5.9 ($\mathfrak{D}_\Sigma$ closed under homomorphism).**

Proof. Let $A = \langle \Sigma, S, s_0, \delta, F \rangle$ be a DFA for L, and let $\psi : \Sigma \to \Sigma^*$ be a homomorphism. For each transition $s \xrightarrow{\boldsymbol{a}} t$ of A, replace that edge by a path labeled $\psi(\boldsymbol{a})$. If $\psi(\boldsymbol{a}) = \lambda$, use a λ-transition from s to t; if $\psi(\boldsymbol{a}) = \boldsymbol{b}_1 \cdots \boldsymbol{b}_k$ with $k \geq 1$, insert $k-1$ new states so that the new path reads $\boldsymbol{b}_1, \boldsymbol{b}_2, \ldots, \boldsymbol{b}_k$.

The resulting finite automaton accepts exactly the homomorphic images of words in L, namely $\overline{\psi}(L)$. Hence $\overline{\psi}(L) \in \mathfrak{D}_\Sigma$, proving Theorem 5.9. □

5.76. **Homomorphism capitalizing lowercase ASCII letters.**

Define $\psi : \text{ASCII}^* \to \text{ASCII}^*$ by $\psi(c) = \begin{cases} \text{uppercase}(c) & \text{if } c \in \{a, \ldots, z\} \\ c & \text{otherwise} \end{cases}$ (extended letter-by-letter to words). This is a homomorphism since $\psi(xy) = \psi(x)\psi(y)$ follows from the letter-by-letter definition.

5.77. (a) **Prove $L(A^{'}) = L_1/L_2$.**
$L_1/L_2 = \{x \mid \exists y \in L_2 \ni xy \in L_1\}$. $F^{'} = \{t \in S_1 \mid \exists y \in \Sigma^* \ni y \in L_2 \wedge \overline{\delta}_1(t, y) \in F_1\}$.

Proof. $x \in L(A^{'})$ iff $\overline{\delta}_1(s_{0_1}, x) \in F^{'}$ iff $\exists y \in L_2 \ni \overline{\delta}_1(\overline{\delta}_1(s_{0_1}, x), y) \in F_1$ iff $\exists y \in L_2 \ni \overline{\delta}_1(s_{0_1}, xy) \in F_1$ iff $\exists y \in L_2 \ni xy \in L_1$ iff $x \in L_1/L_2$. □ □

(b) **Algorithm to compute $F^{'}$.**
For each state $t \in S_1$, determine whether $\exists y \in L_2 \ni \overline{\delta}_1(t, y) \in F_1$:

1. Build the product machine $A_1 \times A_2$: state set $S_1 \times S_2$, start at $\langle t, s_{0_2} \rangle$ for each t.
2. In this product, run A_2 on the input and simultaneously advance A_1.
3. State $t \in F^{'}$ iff some state $\langle f_1, f_2 \rangle$ with $f_1 \in F_1$ and $f_2 \in F_2$ is reachable from $\langle t, s_{0_2} \rangle$ in the product.

This is a reachability computation in a finite graph, computable in $O(|S_1| \cdot |S_2|)$ time.

5.78. **Extend DFA over Σ_1 to handle $\Sigma_1 \cup \Sigma_2$.**

(a) Define $A' = \langle \Sigma_1 \cup \Sigma_2, S \cup \{d\}, s_0, \delta', F \rangle$ where d is a new dead state and $\delta'(s, \boldsymbol{a}) = \begin{cases} \delta(s, \boldsymbol{a}) & \boldsymbol{a} \in \Sigma_1 \\ d & \boldsymbol{a} \in \Sigma_2 \setminus \Sigma_1 \end{cases}$
and $\delta'(d, \boldsymbol{a}) = d$ for all $\boldsymbol{a}$. Final states: F (unchanged).

(b) **Proof.** If $w \in (\Sigma_1 \cup \Sigma_2)^*$ and w contains any letter from $\Sigma_2 \setminus \Sigma_1$, then A' goes to d and stays there, rejecting w. If $w \in \Sigma_1^*$, then A' behaves identically to A. Hence $L(A') = L(A)$ (viewed as a subset of $(\Sigma_1 \cup \Sigma_2)^*$). □

5.79. **If $\mathscr{S}$ is closed under union, concatenation, Kleene closure, and is infinite, must every language in $\mathscr{S}$ be FAD?**

No. The collection I_Σ of all infinite languages is infinite, and Exercise 5.4 shows that it is closed under union, concatenation, and Kleene closure. But I_Σ contains languages in $\mathfrak{N}_\Sigma$, so not every language in such a collection must be FAD.

5.80. **If $\mathscr{S}$ is closed under union, concatenation, Kleene closure, and is finite, must every language be FAD?**

No. Let

$$K = \{x \in \{\boldsymbol{a}, \boldsymbol{b}\}^* \mid |x|_{\boldsymbol{a}} = |x|_{\boldsymbol{b}}\},$$

which is in $\mathfrak{N}_\Sigma$ by Lemma 5.4, and define

$$\mathscr{S} = \{\emptyset, \{\lambda\}, K\}.$$

This collection is finite. It is closed under union because $\{\lambda\} \subseteq K$, so every union is still one of the three listed languages. It is closed under concatenation because

$$\emptyset \cdot L = \emptyset, \qquad \{\lambda\} \cdot L = L \cdot \{\lambda\} = L, \qquad K \cdot K = K$$

(the numbers of $\boldsymbol{a}$s and $\boldsymbol{b}$s add, and $\lambda \in K$). It is also closed under Kleene closure:

$$\emptyset^* = \{\lambda\}, \qquad \{\lambda\}^* = \{\lambda\}, \qquad K^* = K.$$

So a finite collection with these closure properties need not consist entirely of FAD languages.

5.81. **If $u \circ u = \mathbf{id}$ and $\mathfrak{D}_\Sigma$ is closed under u, then $\mathfrak{N}_\Sigma$ is closed under u.**

Proof. Let $L \in \mathfrak{N}_\Sigma$. Suppose, toward a contradiction, that $u(L) \in \mathfrak{D}_\Sigma$. Because $\mathfrak{D}_\Sigma$ is closed under u, applying u again gives

$$u(u(L)) \in \mathfrak{D}_\Sigma.$$

But $u \circ u = \mathrm{id}$, so $u(u(L)) = L$. Hence $L \in \mathfrak{D}_\Sigma$, contradicting $L \in \mathfrak{N}_\Sigma$. Therefore $u(L) \in \mathfrak{N}_\Sigma$ whenever $L \in \mathfrak{N}_\Sigma$, so $\mathfrak{N}_\Sigma$ is closed under u. $\square$

Chapter 6

Regular Expressions — Solutions

6.1. Even-length words over $\{\boldsymbol{a},\boldsymbol{b}\}$:

$$((\boldsymbol{a}\cup\boldsymbol{b})(\boldsymbol{a}\cup\boldsymbol{b}))^*$$

6.2. Words of length ≥ 2 over $\{\boldsymbol{a},\boldsymbol{b}\}$:

$$(\boldsymbol{a}\cup\boldsymbol{b})(\boldsymbol{a}\cup\boldsymbol{b})(\boldsymbol{a}\cup\boldsymbol{b})^*$$

6.3. Words with $|x|_{\boldsymbol{a}} = |x|_{\boldsymbol{b}}$ over $\{\boldsymbol{a},\boldsymbol{b}\}$: This language is *not* regular (not FAD). By the pumping lemma, if it were FAD with n states, take $w = \boldsymbol{a}^n\boldsymbol{b}^n$; any pump within the first n characters (all $\boldsymbol{a}$s) would unbalance the counts. Hence no regular expression exists.

6.4. Odd-length words over $\{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}\}$ not ending in $\boldsymbol{b}$:

A word of odd length not ending in $\boldsymbol{b}$ has an even-length prefix (possibly empty) followed by a non-$\boldsymbol{b}$ letter:

$$((\boldsymbol{a}\cup\boldsymbol{b}\cup\boldsymbol{c})(\boldsymbol{a}\cup\boldsymbol{b}\cup\boldsymbol{c}))^*(\boldsymbol{a}\cup\boldsymbol{c})$$

(When the Kleene star produces k pairs, the total length is $2k+1$, which is odd.)

6.5. Words over $\{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}\}$ not containing $\boldsymbol{cc}$: Words with no two consecutive $\boldsymbol{c}$s are those where every $\boldsymbol{c}$ is followed by a non-$\boldsymbol{c}$ (or is at the end). Let $N = \boldsymbol{a}\cup\boldsymbol{b}$ (non-$\boldsymbol{c}$ letters).

$$(N\cup\boldsymbol{c}N)^*(N\cup\boldsymbol{c}\cup\epsilon)$$

6.6. Words over $\{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}\}$ containing $\boldsymbol{cc}$:

$$(\boldsymbol{a}\cup\boldsymbol{b}\cup\boldsymbol{c})^*\boldsymbol{cc}(\boldsymbol{a}\cup\boldsymbol{b}\cup\boldsymbol{c})^*$$

6.7. Words over $\{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}\}$ with no $\boldsymbol{c}$s:

$$(\boldsymbol{a}\cup\boldsymbol{b})^*$$

6.8. Let $\Sigma = \{\mathbf{0},\mathbf{1}\}$.

(a) $L_1 = \{x \mid |x| \bmod 3 = 2\}$: words whose length is congruent to 2 modulo 3:

$$(\mathbf{0}\cup\mathbf{1})(\mathbf{0}\cup\mathbf{1})((\mathbf{0}\cup\mathbf{1})(\mathbf{0}\cup\mathbf{1})(\mathbf{0}\cup\mathbf{1}))^*$$

(b) $L_2 = \Sigma^* \setminus \{w \mid w \text{ ends in } \mathbf{1}\}$: words that do *not* end in $\mathbf{1}$, i.e., words that end in $\mathbf{0}$ or equal λ:

$$\epsilon \cup (\mathbf{0} \cup \mathbf{1})^* \mathbf{0}$$

(c) $L_3 = \{y \mid |y|_{\mathbf{0}} > |y|_{\mathbf{1}}\}$: this language is *not* regular. If it were regular, then its intersection with the regular language $\mathbf{0}^*\mathbf{1}^*$ would also be regular. But

$$L_3 \cap \mathbf{0}^*\mathbf{1}^* = \{\mathbf{0}^m\mathbf{1}^n \mid m > n\},$$

and this language is not regular by the pumping lemma: for pumping length p, the word $\mathbf{0}^{p+1}\mathbf{1}^p$ belongs to it, but pumping down any nonempty part from the initial $\mathbf{0}$-block yields a word with at most as many $\mathbf{0}$s as $\mathbf{1}$s. Hence no regular expression exists.

6.9. Let $\Sigma = \{\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}\}$. Let $\Sigma^* = (\boldsymbol{a} \cup \boldsymbol{b} \cup \boldsymbol{c})^*$.

(a) $L_1 = \{x \mid |x|_{\boldsymbol{a}} \text{ odd} \wedge |x|_{\boldsymbol{b}} \text{ even}\}$:
Let $C = \boldsymbol{c}^*$, $A = \boldsymbol{a}C$, and $B = \boldsymbol{b}C$. Then the non-$\boldsymbol{c}$ letters must contribute odd $\boldsymbol{a}$-parity and even $\boldsymbol{b}$-parity. One exact regular expression is:

$$C\big(AA \cup BB \cup (AB \cup BA)(AA \cup BB)^*(AB \cup BA)\big)^*\big(A \cup (AB \cup BA)(AA \cup BB)^* B\big)$$

(b) $L_2 = \{y \mid |y|_{\boldsymbol{c}} \text{ even} \vee |y|_{\boldsymbol{b}} \text{ odd}\}$:

$$\big((\boldsymbol{a} \cup \boldsymbol{b})^* \boldsymbol{c} (\boldsymbol{a} \cup \boldsymbol{b})^* \boldsymbol{c}\big)^* (\boldsymbol{a} \cup \boldsymbol{b})^* \cup \big((\boldsymbol{a} \cup \boldsymbol{c})^* \boldsymbol{b} (\boldsymbol{a} \cup \boldsymbol{c})^* \boldsymbol{b}\big)^* (\boldsymbol{a} \cup \boldsymbol{c})^* \boldsymbol{b} (\boldsymbol{a} \cup \boldsymbol{c})^*$$

The first expression gives all words with an even number of $\boldsymbol{c}$s; the second gives all words with an odd number of $\boldsymbol{b}$s.

(c) $L_3 = \{z \mid |z|_{\boldsymbol{a}} \text{ even}\}$:

$$((\boldsymbol{b} \cup \boldsymbol{c})^* \boldsymbol{a} (\boldsymbol{b} \cup \boldsymbol{c})^* \boldsymbol{a})^* (\boldsymbol{b} \cup \boldsymbol{c})^*$$

(d) $L_4 = \{z \mid |z|_{\boldsymbol{c}} \text{ prime}\}$: Not regular. The set of primes is not eventually periodic, so the language $\{c^p \mid p \text{ prime}\} \subseteq L_4$ restricted to $\Sigma = \{\boldsymbol{c}\}$ is not FAD (pumping lemma). Hence L_4 has no regular expression.

(e) $L_5 = \{x \mid \boldsymbol{abc} \text{ is a substring}\}$:

$$(\boldsymbol{a} \cup \boldsymbol{b} \cup \boldsymbol{c})^* \boldsymbol{abc} (\boldsymbol{a} \cup \boldsymbol{b} \cup \boldsymbol{c})^*$$

(f) $L_6 = \{x \mid \boldsymbol{acaba} \text{ is a substring}\}$:

$$(\boldsymbol{a} \cup \boldsymbol{b} \cup \boldsymbol{c})^* \boldsymbol{acaba} (\boldsymbol{a} \cup \boldsymbol{b} \cup \boldsymbol{c})^*$$

(g) $L_7 = \{x \mid |x|_{\boldsymbol{a}} \equiv 0 \pmod 3\}$:

$$(\boldsymbol{b} \cup \boldsymbol{c})^* (\boldsymbol{a} (\boldsymbol{b} \cup \boldsymbol{c})^* \boldsymbol{a} (\boldsymbol{b} \cup \boldsymbol{c})^* \boldsymbol{a} (\boldsymbol{b} \cup \boldsymbol{c})^*)^*$$

6.10. $\Psi = \{x \mid x \text{ begins with } \boldsymbol{d} \vee x \text{ contains } \boldsymbol{bb}\}$ over $\{\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{d}\}$:

$$\boldsymbol{d}(\boldsymbol{a} \cup \boldsymbol{b} \cup \boldsymbol{d})^* \cup (\boldsymbol{a} \cup \boldsymbol{b} \cup \boldsymbol{d})^* \boldsymbol{bb} (\boldsymbol{a} \cup \boldsymbol{b} \cup \boldsymbol{d})^*$$

6.11. $\Phi = \{x \mid \text{every } \boldsymbol{b} \text{ in } x \text{ is immediately followed by } \boldsymbol{c}\}$ over $\{\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}\}$: Words with no isolated $\boldsymbol{b}$ (every $\boldsymbol{b}$ must be followed immediately by $\boldsymbol{c}$). Let $S = \boldsymbol{a} \cup \boldsymbol{c} \cup \boldsymbol{bc}$ (safe building blocks):

$$(\boldsymbol{a} \cup \boldsymbol{c} \cup \boldsymbol{bc})^*$$

6.12. Numbers divisible by 3 over $\{0, \ldots, 9\}$:

A number is divisible by 3 iff the sum of its digits is divisible by 3. Let

$$D_0 = (\mathbf{0} \cup \mathbf{3} \cup \mathbf{6} \cup \mathbf{9}), \quad D_1 = (\mathbf{1} \cup \mathbf{4} \cup \mathbf{7}), \quad D_2 = (\mathbf{2} \cup \mathbf{5} \cup \mathbf{8}).$$

Using the 3-state DFA that tracks the running digit sum modulo 3, the terminal-set equations are

$$\begin{aligned} X_0 &= \epsilon \cup D_0 X_0 \cup D_1 X_1 \cup D_2 X_2 \\ X_1 &= D_2 X_0 \cup D_0 X_1 \cup D_1 X_2 \\ X_2 &= D_1 X_0 \cup D_2 X_1 \cup D_0 X_2. \end{aligned}$$

Applying Theorems 6.2 and 6.1 yields the following exact regular expression for the language:

$$\begin{aligned} X_0 = {} & ((D_0)^* \cup (D_0)^* D_1 ((D_0 \cup D_2 (D_0)^* D_1))^* D_2 (D_0)^*) \\ & \cup ((D_0)^* D_2 \cup (D_0)^* D_1 ((D_0 \cup D_2 (D_0)^* D_1))^* (D_1 \cup D_2 (D_0)^* D_2)) \\ & \quad \cdot (((D_0 \cup D_1 (D_0)^* D_2) \cup (D_2 \cup D_1 (D_0)^* D_1)((D_0 \cup D_2 (D_0)^* D_1))^* (D_1 \cup D_2 (D_0)^* D_2)))^* \\ & \quad \cdot (D_1 (D_0)^* \cup (D_2 \cup D_1 (D_0)^* D_1)((D_0 \cup D_2 (D_0)^* D_1))^* D_2 (D_0)^*)). \end{aligned}$$

6.13. $K = \{x \mid x \text{ is divisible by } 5\}$ over decimal digits:

A decimal number is divisible by 5 iff it ends in **0** or **5**.

$$K = (\mathbf{0} \cup \mathbf{1} \cup \cdots \cup \mathbf{9})^* (\mathbf{0} \cup \mathbf{5})$$

6.14. **Build NDFA for $\boldsymbol{b} \cup \boldsymbol{a}^* \boldsymbol{c}$ using exact constructs of Chapter 5.**

Step 1: $\boldsymbol{a}^*$ via Theorem 5.5 (Kleene closure of $\{\boldsymbol{a}\}$). Step 2: $\boldsymbol{a}^* \boldsymbol{c}$ via Theorem 5.4 (concatenation of $\boldsymbol{a}^*$ and $\{\boldsymbol{c}\}$). Step 3: $\boldsymbol{b} \cup \boldsymbol{a}^* \boldsymbol{c}$ via Theorem 5.2 (union of $\{\boldsymbol{b}\}$ and $\boldsymbol{a}^* \boldsymbol{c}$).

The NDFA:

- Machine for $\{\boldsymbol{a}\}$: one start state p_0, one final state p_1, $\delta(p_0, \boldsymbol{a}) = \{p_1\}$.
- Machine for $\boldsymbol{a}^*$ (Theorem 5.5): add start state q_0 (final); λ-transition $q_0 \to p_0$; λ-transition $p_1 \to q_0$.
- Machine for $\{\boldsymbol{c}\}$: start r_0, final r_1, $\delta(r_0, \boldsymbol{c}) = \{r_1\}$.
- Machine for $\boldsymbol{a}^* \boldsymbol{c}$ (Theorem 5.4): λ-transition $q_0 \to r_0$; final states $\{r_1\}$.
- Machine for $\{\boldsymbol{b}\}$: start s_0, final s_1, $\delta(s_0, \boldsymbol{b}) = \{s_1\}$.
- Union (Theorem 5.2): start state set $\{q_0, s_0\}$; final states $\{r_1, s_1\}$.

6.15. **Inequalities in Lemma 6.1:**

(a) $R_1 \cup \epsilon \neq R_1$: Let $R_1 = \boldsymbol{a}$. Then $R_1 \cup \epsilon = \{\lambda, \boldsymbol{a}\} \neq \{\boldsymbol{a}\} = R_1$.

(b) $R_1 \cdot R_2 \neq R_2 \cdot R_1$: Let $R_1 = \boldsymbol{a}$, $R_2 = \boldsymbol{b}$. $\boldsymbol{ab} \neq \boldsymbol{ba}$.

(c) $R_1 \cdot R_1 \neq R_1$: Let $R_1 = \boldsymbol{a}$. $\boldsymbol{aa} \neq \boldsymbol{a}$.

(d) $R_1 \cup (R_2 \cdot R_3) \neq (R_1 \cup R_2) \cdot (R_1 \cup R_3)$: Let $R_1 = \boldsymbol{a}$, $R_2 = \boldsymbol{b}$, $R_3 = \boldsymbol{c}$. Then

$$R_1 \cup (R_2 \cdot R_3) = \{\boldsymbol{a}, \boldsymbol{bc}\}, \qquad (R_1 \cup R_2) \cdot (R_1 \cup R_3) = \{\boldsymbol{a}, \boldsymbol{b}\} \cdot \{\boldsymbol{a}, \boldsymbol{c}\} = \{\boldsymbol{aa}, \boldsymbol{ac}, \boldsymbol{ba}, \boldsymbol{bc}\}.$$

These two sets are not equal.

(e) $(R_1 \cdot R_2)^* \neq (R_1^* \cdot R_2^*)^*$: Let $R_1 = \boldsymbol{a}$, $R_2 = \boldsymbol{b}$. $(ab)^* = \{\lambda, \boldsymbol{ab}, \boldsymbol{abab}, \ldots\}$. $(a^* b^*)^* = (\boldsymbol{a} \cup \boldsymbol{b})^*$. Not equal.

(f) $(R_1 \cdot R_2)^* \neq (R_1^* \cup R_2^*)^*$: Same example. $(R_1^* \cup R_2^*)^* = (a^* \cup b^*)^* = (a \cup b)^*$. Not equal to $(\boldsymbol{ab})^*$.

Examples where equality holds:

(g) $R_1 \cup \epsilon = R_1$: let $R_1 = \boldsymbol{a}^*$. Then $\boldsymbol{a}^* \cup \epsilon = \boldsymbol{a}^*$.

(h) $R_1 \cdot R_2 = R_2 \cdot R_1$: let $R_1 = \boldsymbol{a}$ and $R_2 = \boldsymbol{aa}$. Then $R_1 \neq R_2$, neither is ϵ, and both products equal $\boldsymbol{aaa}$.

(i) $R_1 \cdot R_1 = R_1$: let $R_1 = \Sigma^*$. Then $\Sigma^* \Sigma^* = \Sigma^*$.

(j) $R_1 \cup (R_2 \cdot R_3) = (R_1 \cup R_2) \cdot (R_1 \cup R_3)$: let $R_1 = (\boldsymbol{a} \cup \boldsymbol{b})^*$, $R_2 = \boldsymbol{a}$, and $R_3 = \boldsymbol{b}$. These three sets are distinct, and both sides equal $(\boldsymbol{a} \cup \boldsymbol{b})^*$.

(k) $(R_1 \cdot R_2)^* = (R_1^* \cdot R_2^*)^*$: let $R_1 = \boldsymbol{a}^*$ and $R_2 = \boldsymbol{aa}^*$. Then $R_1 \neq R_2$, neither is ϵ, and both sides equal $\boldsymbol{a}^*$.

(l) $(R_1 \cdot R_2)^* = (R_1^* \cup R_2^*)^*$: with the same choice $R_1 = \boldsymbol{a}^*$ and $R_2 = \boldsymbol{aa}^*$, both sides again equal $\boldsymbol{a}^*$.

6.16. **Prove the equalities of Lemma 6.1.**

Let $L_i = L(R_i)$. Each identity follows directly from the definitions.

(a) $R_1 \cup \emptyset = R_1$: $L_1 \cup \emptyset = L_1$.

(b) $R_1 \cdot \epsilon = R_1 = \epsilon \cdot R_1$: concatenating with $\{\lambda\}$ does not change a word.

(c) $R_1 \cdot \emptyset = \emptyset = \emptyset \cdot R_1$: there is no second factor to form a concatenation.

(d) $R_1 \cup R_2 = R_2 \cup R_1$: union is commutative.

(e) $R_1 \cup R_1 = R_1$: union is idempotent.

(f) $R_1 \cup (R_2 \cup R_3) = (R_1 \cup R_2) \cup R_3$: union is associative.

(g) $R_1 \cdot (R_2 \cdot R_3) = (R_1 \cdot R_2) \cdot R_3$: both sides are all words of the form xyz with $x \in L_1$, $y \in L_2$, $z \in L_3$.

(h) $R_1 \cdot (R_2 \cup R_3) = (R_1 \cdot R_2) \cup (R_1 \cdot R_3)$: a word on the left is xy with $x \in L_1$ and $y \in L_2 \cup L_3$, so $y \in L_2$ or $y \in L_3$.

(i) $\epsilon^* = \epsilon$: concatenating any number of empty words still gives λ.

(j) $\emptyset^* = \epsilon$: by definition, $L^* = \bigcup_{n \geq 0} L^n$, and $\emptyset^0 = \epsilon$ while $\emptyset^n = \emptyset$ for $n \geq 1$.

(k) $(R_1 \cup R_2)^* = (R_1^* \cup R_2^*)^*$: every factor from $L_1 \cup L_2$ already lies in $L_1^* \cup L_2^*$, so the left side is contained in the right; conversely, each factor from L_1^* or L_2^* is itself a finite concatenation of words from $L_1 \cup L_2$, so the right side is contained in the left.

(l) $(R_1 \cup R_2)^* = (R_1^* \cdot R_2^*)^*$: a block from $L_1^* L_2^*$ is a finite string of words from L_1 followed by a finite string of words from L_2, hence belongs to $(L_1 \cup L_2)^*$; the reverse inclusion holds because each factor from $L_1 \cup L_2$ can be viewed as a block with either the L_1^* or L_2^* part empty.

(m) $(R_1^*)^* = R_1^*$: L_1^* is already closed under finite concatenation.

(n) $R_1^* \cdot R_1^* = R_1^*$: concatenating two finite concatenations of words from L_1 still yields a finite concatenation of words from L_1, and every word of L_1^* is obtained by taking the second factor to be λ.

6.17. (a) **Non-uniqueness of $A^* E$ when $\lambda \in A$.**
Let $A = \{\lambda, \boldsymbol{a}\} = \epsilon \cup \boldsymbol{a}$ and $E = \{\boldsymbol{b}\}$. Then $A^* E = \{\boldsymbol{a}\}^* \{\boldsymbol{b}\} = \{\boldsymbol{a}^n \boldsymbol{b} \mid n \geq 0\}$. But $X = \Sigma^*$ also satisfies $X = A \cdot X \cup E$: indeed,

$$A \cdot \Sigma^* \cup \{\boldsymbol{b}\} = (\{\lambda, \boldsymbol{a}\} \cdot \Sigma^*) \cup \{\boldsymbol{b}\} = \Sigma^* \cup \{\boldsymbol{b}\} = \Sigma^*.$$

So Σ^* is also a solution, but $\Sigma^* \neq A^* E$. Hence the solution is not unique.

(b) **$A^* E$ as unique solution even if $\lambda \in A$.**
Let the alphabet be $\{\boldsymbol{a}\}$, let $A = \{\lambda, \boldsymbol{a}\}$, and let $E = \boldsymbol{a}^*$. Then

$$A^* E = (\epsilon \cup \boldsymbol{a})^* \boldsymbol{a}^* = \boldsymbol{a}^*.$$

The equation becomes

$$X = E \cup A \cdot X = \boldsymbol{a}^* \cup (\epsilon \cup \boldsymbol{a}) X.$$

Any solution must contain $E = \boldsymbol{a}^*$, but $\boldsymbol{a}^*$ is already the whole universe over this alphabet. Therefore the only possible solution is $X = \boldsymbol{a}^* = A^* E$.

6.18. **Solve the system of Exercise 6.18.**

The equations are

$$X_0 = (\mathbf{0} \cup \mathbf{1}) X_1, \qquad X_1 = \epsilon \cup \mathbf{1} X_0 \cup \mathbf{0} X_1.$$

Substitute the first equation into the second:

$$X_1 = \epsilon \cup \mathbf{1}(\mathbf{0} \cup \mathbf{1}) X_1 \cup \mathbf{0} X_1 = \epsilon \cup (\mathbf{10} \cup \mathbf{11} \cup \mathbf{0}) X_1.$$

Since $\lambda \not{i} n (\mathbf{10} \cup \mathbf{11} \cup \mathbf{0})$, Theorem 6.1 gives

$$X_1 = (\mathbf{10} \cup \mathbf{11} \cup \mathbf{0})^*.$$

Therefore

$$X_0 = (\mathbf{0} \cup \mathbf{1}) X_1 = (\mathbf{0} \cup \mathbf{1})(\mathbf{10} \cup \mathbf{11} \cup \mathbf{0})^*.$$

These are the terminal sets for the corresponding two-state DFA. In fact, this is the same state-equation system one obtains from the DFA in Example 3.4, where X_0 and X_1 are the two terminal sets.

6.19. **Solve for X_1, X_2, X_3 from the equations.**

Simplifying (noting $\emptyset \cdot X_i = \emptyset$):

$$X_1 = (\mathbf{0} \cup \mathbf{1}) X_2$$
$$X_2 = \epsilon \cup \mathbf{0} X_1 \cup \mathbf{1} X_2$$
$$X_3 = (\mathbf{0} \cup \mathbf{1}) X_2$$

Note $X_3 = X_1$.

(a) **Eliminate X_3, then X_2:**
Since $X_3 = X_1$, eliminate X_3. From $X_2 = \epsilon \cup \mathbf{0}X_1 \cup \mathbf{1}X_2$: by Theorem 6.1 ($\lambda \not\in \{\mathbf{1}\}$): $X_2 = \mathbf{1}^*(\epsilon \cup \mathbf{0}X_1) = \mathbf{1}^* \cup \mathbf{1}^*\mathbf{0}X_1$. Substitute into $X_1 = (\mathbf{0}\cup\mathbf{1})X_2 = (\mathbf{0}\cup\mathbf{1})(\mathbf{1}^*\cup\mathbf{1}^*\mathbf{0}X_1) = (\mathbf{0}\cup\mathbf{1})\mathbf{1}^*\cup(\mathbf{0}\cup\mathbf{1})\mathbf{1}^*\mathbf{0}X_1$. By Theorem 6.1: $X_1 = ((\mathbf{0}\cup\mathbf{1})\mathbf{1}^*\mathbf{0})^*(\mathbf{0}\cup\mathbf{1})\mathbf{1}^*$. Then

$$X_2 = \mathbf{1}^* \cup \mathbf{1}^*\mathbf{0}((\mathbf{0}\cup\mathbf{1})\mathbf{1}^*\mathbf{0})^*(\mathbf{0}\cup\mathbf{1})\mathbf{1}^*, \qquad X_3 = X_1.$$

(b) **Eliminate X_3, then X_1:**
Same $X_3 = X_1$. From $X_1 = (\mathbf{0}\cup\mathbf{1})X_2$, substitute into X_2: $X_2 = \epsilon \cup \mathbf{0}(\mathbf{0}\cup\mathbf{1})X_2 \cup \mathbf{1}X_2 = \epsilon \cup (\mathbf{0}(\mathbf{0}\cup\mathbf{1})\cup\mathbf{1})X_2$. Let $A = \mathbf{0}(\mathbf{0}\cup\mathbf{1})\cup\mathbf{1} = \mathbf{00}\cup\mathbf{01}\cup\mathbf{1}$; $\lambda \not\in A$. $X_2 = A^*$. Then $X_1 = (\mathbf{0}\cup\mathbf{1})A^*$ and $X_3 = X_1$.

(c) **Comparison:** Both solutions are equivalent: $(\mathbf{0}\cup\mathbf{1})\mathbf{1}^*(\mathbf{0}(\mathbf{0}\cup\mathbf{1})\mathbf{1}^*)^* = (\mathbf{0}\cup\mathbf{1})(\mathbf{00}\cup\mathbf{01}\cup\mathbf{1})^*$. The second form is more concise.

6.20. **Prove Lemma 6.2: every regular expression describes a FAD language.**

Proof. Let $P(m)$: "Every regular expression with $\le m$ operators represents a FAD language."

Base ($m = 0$): No operators: the expression is $\emptyset$, ϵ, or a single letter $\boldsymbol{a} \in \Sigma$. These represent $\emptyset$, $\{\lambda\}$, and $\{\boldsymbol{a}\}$ respectively—all finite, hence FAD.

Inductive step: Assume $P(m)$. Any regular expression with $m+1$ operators is either $R_1 \cup R_2$, $R_1 \cdot R_2$, or R_1^*, where R_1 and R_2 each have $\le m$ operators. By $P(m)$, $L(R_1), L(R_2) \in \mathfrak{D}_\Sigma$. Then:

- $L(R_1 \cup R_2) = L(R_1) \cup L(R_2) \in \mathfrak{D}_\Sigma$ (Theorem 5.2).
- $L(R_1 \cdot R_2) = L(R_1) \cdot L(R_2) \in \mathfrak{D}_\Sigma$ (Theorem 5.4).
- $L(R_1^*) = L(R_1)^* \in \mathfrak{D}_\Sigma$ (Theorem 5.5).

So $P(m+1)$ holds. By induction, every regular expression represents a FAD language. □ □

6.21. **All solutions to $X = X \cup \{\boldsymbol{b}\}$ over $\{\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}\}^*$.**

$X = X \cup \{\boldsymbol{b}\}$ means exactly that $\{\boldsymbol{b}\} \subseteq X$. Hence the solutions are precisely the languages X with $\boldsymbol{b} \in X$. Every such language is a solution, and the smallest one is $\{\boldsymbol{b}\}$.

6.22. **Prove $A^*E = E \cup A \cdot (A^*E)$ for any languages A and E.**

Proof. $A^*E = (\epsilon \cup A \cup A^2 \cup \cdots)E = \epsilon E \cup AE \cup A^2E \cup \cdots = E \cup A(E \cup AE \cup A^2E \cup \cdots) = E \cup A(A^*E)$. □ □

6.23. **Regular expression for $(\boldsymbol{ab} \cup \boldsymbol{b})^*\boldsymbol{a} \cap (\boldsymbol{ba} \cup \boldsymbol{a})^*$.**

Both are FAD; their intersection is FAD (Lemma 5.1). We use the product DFA approach.

$L_1 = (\boldsymbol{ab} \cup \boldsymbol{b})^*\boldsymbol{a}$: strings over $\{\boldsymbol{a}, \boldsymbol{b}\}$ of the form $w\boldsymbol{a}$ where $w \in (\boldsymbol{ab} \cup \boldsymbol{b})^*$. $L_2 = (\boldsymbol{ba} \cup \boldsymbol{a})^*$: strings over $\{\boldsymbol{a}, \boldsymbol{b}\}$ that are concatenations of $\boldsymbol{ba}$ or $\boldsymbol{a}$.

L_1 ends in $\boldsymbol{a}$. $L_2 = (\boldsymbol{ba} \cup \boldsymbol{a})^*$ consists of all strings that decompose as concatenations of $\boldsymbol{a}$ and $\boldsymbol{ba}$.

$L_2 = (\boldsymbol{ba} \cup \boldsymbol{a})^*$ means every $\boldsymbol{b}$ is immediately followed by $\boldsymbol{a}$. A word in $L_1 = (\boldsymbol{ab} \cup \boldsymbol{b})^*\boldsymbol{a}$ has prefix in $(\boldsymbol{ab} \cup \boldsymbol{b})^*$, so every $\boldsymbol{a}$ in the prefix is immediately followed by $\boldsymbol{b}$. Combining both constraints on the prefix atoms: each atom (whether $\boldsymbol{ab}$ or $\boldsymbol{b}$) must be immediately followed by something starting with $\boldsymbol{a}$ (since the atom ends in $\boldsymbol{b}$, and every $\boldsymbol{b}$ must be followed by $\boldsymbol{a}$). Consequently, $\boldsymbol{b}$-atoms can appear only once, at the start (any $\boldsymbol{b}$-atom followed by another $\boldsymbol{b}$-atom would put $\boldsymbol{b}$ before $\boldsymbol{b}$, violating L_2); and $\boldsymbol{ab}$-atoms must be followed by $\boldsymbol{ab}$-atoms or the final $\boldsymbol{a}$. The intersection is therefore:

$$L_1 \cap L_2 = (\epsilon \cup \boldsymbol{b})(\boldsymbol{ab})^*\boldsymbol{a}$$

Verification: $(\epsilon \cup \boldsymbol{b})(\boldsymbol{ab})^k\boldsymbol{a}$ gives words $\boldsymbol{a}, \boldsymbol{aba}, \boldsymbol{ababa}, \ldots$ and $\boldsymbol{ba}, \boldsymbol{baba}, \boldsymbol{bababa}, \ldots$, all of which belong to both L_1 and L_2; and any word in $L_1 \cap L_2$ has this form by the above analysis.

6.24. **Algorithm for intersection of two regular expressions.**

Given regular expressions R_1 and R_2 (each describing a FAD language):

1. Convert R_1 to an NDFA A_1 using the constructions of Chapter 5.
2. Convert R_2 to an NDFA A_2 similarly.
3. Apply the subset construction to get DFAs A_1^d and A_2^d.
4. Build the product DFA $A^\cap$ (Lemma 5.1) with $F^\cap = F_1 \times F_2$.
5. Use the state-equation method of Theorems 6.3 and 6.2 to convert $A^\cap$ to a regular expression for $L(A_1) \cap L(A_2)$.

6.25. **Verify that $X_1 = (\boldsymbol{a} \cup \boldsymbol{bb})^*$ and $X_2 = \boldsymbol{b} \cdot (\boldsymbol{a} \cup \boldsymbol{bb})^*$ satisfy:**

$$\begin{aligned} X_1 &= \epsilon \cup \boldsymbol{a} \cdot X_1 \cup \boldsymbol{b} \cdot X_2 \\ X_2 &= \boldsymbol{b} \cdot X_1 \end{aligned}$$

Verification of $X_2 = \boldsymbol{b} \cdot X_1$:

$$\boldsymbol{b} \cdot X_1 = \boldsymbol{b} \cdot (\boldsymbol{a} \cup \boldsymbol{bb})^* = X_2. \quad \checkmark$$

Verification of $X_1 = \epsilon \cup \boldsymbol{a} \cdot X_1 \cup \boldsymbol{b} \cdot X_2$:

$$\begin{aligned} \epsilon \cup \boldsymbol{a} \cdot X_1 \cup \boldsymbol{b} \cdot X_2 &= \epsilon \cup \boldsymbol{a}(\boldsymbol{a} \cup \boldsymbol{bb})^* \cup \boldsymbol{b} \cdot \boldsymbol{b}(\boldsymbol{a} \cup \boldsymbol{bb})^* \\ &= \epsilon \cup (\boldsymbol{a} \cup \boldsymbol{bb}) \cdot (\boldsymbol{a} \cup \boldsymbol{bb})^* \\ &= (\boldsymbol{a} \cup \boldsymbol{bb})^* = X_1. \quad \checkmark \end{aligned}$$

6.26. (a) $L(D)$ **for Example 6.10:** Example 6.10 gives $L(B) = X_1 = (\boldsymbol{a} \cup \boldsymbol{bb})^*$ and $L(C) = X_2 = \boldsymbol{b}(\boldsymbol{a} \cup \boldsymbol{bb})^*$. Since D has both corresponding start states, its language is

$$L(D) = X_1 \cup X_2 = (\epsilon \cup \boldsymbol{b})(\boldsymbol{a} \cup \boldsymbol{bb})^*.$$

(b) **General formula:** For a machine A with start states $s_{i_1}, \ldots, s_{i_m}$:

$$L(A) = X_{i_1} \cup X_{i_2} \cup \cdots \cup X_{i_m}$$

where each X_j is the solution to the state equation for state s_j.

6.27. **Regular expression for the complement of $(\boldsymbol{ab} \cup \boldsymbol{b})^*\boldsymbol{a}$.**

The language $(\boldsymbol{ab} \cup \boldsymbol{b})^*\boldsymbol{a}$ is exactly the set of words over $\{\boldsymbol{a}, \boldsymbol{b}\}$ that end in $\boldsymbol{a}$ and contain no $\boldsymbol{aa}$ as a substring. Therefore its complement is the set of words that either end in $\boldsymbol{b}$ (or are λ) or contain $\boldsymbol{aa}$. One regular expression is

$$\epsilon \cup (\boldsymbol{a} \cup \boldsymbol{b})^*\boldsymbol{b} \cup (\boldsymbol{a} \cup \boldsymbol{b})^*\boldsymbol{aa}(\boldsymbol{a} \cup \boldsymbol{b})^*.$$

6.28. **Algorithm to complement a regular expression.**

Given regular expression R:

1. Convert R to a DFA A (via NDFA → subset construction).
2. Complement A by swapping F and $S \setminus F$.
3. Use the state-equation method of Theorems 6.3 and 6.2 to convert the complemented DFA back to a regular expression.

6.29. **$\mathfrak{R}_\Sigma$ is closed under $E(L)$ using regular expressions.**

Here

$$E(L) = \{z \mid (\exists y \in \Sigma^+)(\exists x \in L)\ z = yx\} = \Sigma^+ \cdot L.$$

If $L = L(R)$ and $\Sigma = \{\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}\}$, then

$$E(L) = L\Big((\boldsymbol{a} \cup \boldsymbol{b} \cup \boldsymbol{c})(\boldsymbol{a} \cup \boldsymbol{b} \cup \boldsymbol{c})^* R\Big),$$

which is again a regular expression. Hence $\mathfrak{R}_\Sigma$ is closed under E.

6.30. **$\mathfrak{R}_\Sigma$ is closed under $B(L)$.**

$B(L) = L{\cdot}\Sigma^*$ (words starting with a word from L). If $L = L(R)$, then $B(L) = L(R{\cdot}\Sigma^*) = L(R{\cdot}(\boldsymbol{a}{\cup}\boldsymbol{b}{\cup}\boldsymbol{c})^*)$. This is a regular expression, so $B(L) \in \mathfrak{R}_\Sigma$.

6.31. **$\mathfrak{R}_\Sigma$ is closed under $M(L)$.**

$M(L) = L \cdot \Sigma^+$. If $L = L(R)$, then $M(L) = L(R \cdot (\boldsymbol{a} \cup \boldsymbol{b} \cup \boldsymbol{c})^+) = L(R \cdot (\boldsymbol{a} \cup \boldsymbol{b} \cup \boldsymbol{c}) \cdot (\boldsymbol{a} \cup \boldsymbol{b} \cup \boldsymbol{c})^*)$. Regular expression, hence $M(L) \in \mathfrak{R}_\Sigma$.

6.32. (a) **No unique solution to the given system:**

The equations: $X_1 = \boldsymbol{b} \cup \epsilon \cdot X_1 \cup \boldsymbol{a} \cdot X_2 = \boldsymbol{b} \cup X_1 \cup \boldsymbol{a}X_2$. Since $X_1 \subseteq X_1 \cup \boldsymbol{b} \cup \boldsymbol{a}X_2$, the equation $X_1 = \boldsymbol{b} \cup X_1 \cup \boldsymbol{a}X_2$ says $X_1 \supseteq \boldsymbol{b} \cup \boldsymbol{a}X_2$. Any X_1 that contains $\boldsymbol{b}$ and $\boldsymbol{a}X_2$ is a solution; there are infinitely many such X_1.

$X_2 = \boldsymbol{c} \cup \emptyset \cdot X_1 \cup \epsilon \cdot X_2 = \boldsymbol{c} \cup X_2$. Similarly, any $X_2 \supseteq \{\boldsymbol{c}\}$ is a solution.

(b) **No contradiction with Theorem 6.2:** Theorem 6.2 assumes $\lambda \not\! in A_{ij}$ for *every* coefficient in the system. Here $A_{11} = \epsilon$ and $A_{22} = \epsilon$, so the hypothesis fails. Therefore Theorem 6.2 does not apply, and there is no contradiction.

6.33. **Solve for X_0 and X_1.**

$$X_0 = \mathbf{0}^*\mathbf{1} \cup (\mathbf{10})^* X_0 \cup \mathbf{0}(\mathbf{0} \cup \mathbf{1}) X_1$$
$$X_1 = \epsilon \cup \mathbf{1}^*\mathbf{01} X_0 \cup \mathbf{0} X_1$$

From the X_1 equation (by Theorem 6.1, $\lambda \not\! in\{\mathbf{0}\}$): $X_1 = \mathbf{0}^*(\epsilon \cup \mathbf{1}^*\mathbf{01}X_0) = \mathbf{0}^* \cup \mathbf{0}^*\mathbf{1}^*\mathbf{01}X_0$.

Substitute into X_0: $X_0 = \mathbf{0}^*\mathbf{1} \cup (\mathbf{10})^* X_0 \cup \mathbf{0}(\mathbf{0} \cup \mathbf{1})(\mathbf{0}^* \cup \mathbf{0}^*\mathbf{1}^*\mathbf{01}X_0)$.

Let $C = \mathbf{0}(\mathbf{0} \cup \mathbf{1})\mathbf{0}^*\mathbf{1}^*\mathbf{01}$. Then: $X_0 = ((\mathbf{10})^* \cup C)X_0 \cup (\mathbf{0}^*\mathbf{1} \cup \mathbf{0}(\mathbf{0} \cup \mathbf{1})\mathbf{0}^*)$.

Since $(\mathbf{10})^*$ contains λ, Theorem 6.1 does not guarantee a unique solution here. Following the chapter's discussion of ambiguous equations, the least solution is

$$X_0 = ((\mathbf{10})^* \cup C)^* (\mathbf{0}^*\mathbf{1} \cup \mathbf{0}(\mathbf{0} \cup \mathbf{1})\mathbf{0}^*).$$

Back-substitution then gives

$$X_1 = \mathbf{0}^* \cup \mathbf{0}^*\mathbf{1}^*\mathbf{01}((\mathbf{10})^* \cup C)^* (\mathbf{0}^*\mathbf{1} \cup \mathbf{0}(\mathbf{0} \cup \mathbf{1})\mathbf{0}^*).$$

6.34. (a) Words ending with $\boldsymbol{c}$ with no $\boldsymbol{aa}$, $\boldsymbol{bb}$, $\boldsymbol{cc}$ substrings:

Build the DFA with states q_0 (start), q_a (last letter $\boldsymbol{a}$), q_b (last letter $\boldsymbol{b}$), q_c (last letter $\boldsymbol{c}$, accepting), and dead state q_d; transitions $q_x \xrightarrow{x} q_d$ and $q_x \xrightarrow{y} q_y$ for $y \neq x$. Write state equations for strings reaching q_c:

$$\begin{aligned} X_a &= \boldsymbol{b}X_b \cup \boldsymbol{c}X_c \\ X_b &= \boldsymbol{a}X_a \cup \boldsymbol{c}X_c \\ X_c &= \epsilon \cup \boldsymbol{a}X_a \cup \boldsymbol{b}X_b \end{aligned}$$

Eliminate X_a: substituting $X_b = \boldsymbol{a}X_a \cup \boldsymbol{c}X_c$ into X_a gives $X_a = \boldsymbol{ba}X_a \cup (\boldsymbol{bc} \cup \boldsymbol{c})X_c$, so $X_a = (\boldsymbol{ba})^*(\boldsymbol{b} \cup \epsilon)\boldsymbol{c}X_c$. Then $X_b = \boldsymbol{a}(\boldsymbol{ba})^*(\boldsymbol{b} \cup \epsilon)\boldsymbol{c}X_c \cup \boldsymbol{c}X_c$. Substituting into X_c:

$$X_c = \epsilon \cup \big((\boldsymbol{a} \cup \boldsymbol{ba})(\boldsymbol{ba})^*(\boldsymbol{b} \cup \epsilon)\boldsymbol{c} \cup \boldsymbol{bc}\big)X_c$$

Since λ is absent from the coefficient, $X_c = A^*$ where

$$A = (\boldsymbol{a} \cup \boldsymbol{ba})(\boldsymbol{ba})^*(\boldsymbol{b} \cup \epsilon)\boldsymbol{c} \cup \boldsymbol{bc}.$$

The language is $L = \boldsymbol{a}X_a \cup \boldsymbol{b}X_b \cup \boldsymbol{c}X_c = (\boldsymbol{a}(\boldsymbol{ba})^*(\boldsymbol{b} \cup \epsilon)\boldsymbol{c} \cup \boldsymbol{b}(\boldsymbol{a}(\boldsymbol{ba})^*(\boldsymbol{b} \cup \epsilon)\boldsymbol{c} \cup \boldsymbol{c}) \cup \boldsymbol{c})A^*$.

(b) Words beginning with $\boldsymbol{c}$ with no $\boldsymbol{aa}$, $\boldsymbol{bb}$, $\boldsymbol{cc}$ substrings: Let X_c be the set of suffixes that may follow an initial $\boldsymbol{c}$ while preserving the "no equal consecutive letters" condition. Solving the corresponding state equations gives

$$X_c = \big(((\boldsymbol{a} \cup \boldsymbol{ba})(\boldsymbol{ba})^*(\epsilon \cup \boldsymbol{b}) \cup \boldsymbol{b})\boldsymbol{c}\big)^*(\epsilon \cup \boldsymbol{a})(\boldsymbol{ba})^*(\epsilon \cup \boldsymbol{b}).$$

Therefore one regular expression for the language is

$$\boldsymbol{c}\big(((\boldsymbol{a} \cup \boldsymbol{ba})(\boldsymbol{ba})^*(\epsilon \cup \boldsymbol{b}) \cup \boldsymbol{b})\boldsymbol{c}\big)^*(\epsilon \cup \boldsymbol{a})(\boldsymbol{ba})^*(\epsilon \cup \boldsymbol{b}).$$

6.35. (a) Words with at most two $\boldsymbol{c}$s:

$$(\boldsymbol{a} \cup \boldsymbol{b})^* \cup (\boldsymbol{a} \cup \boldsymbol{b})^*\boldsymbol{c}(\boldsymbol{a} \cup \boldsymbol{b})^* \cup (\boldsymbol{a} \cup \boldsymbol{b})^*\boldsymbol{c}(\boldsymbol{a} \cup \boldsymbol{b})^*\boldsymbol{c}(\boldsymbol{a} \cup \boldsymbol{b})^*$$

(b) Words that do *not* have exactly one $\boldsymbol{c}$ (i.e., zero or ≥ 2 $\boldsymbol{c}$s):

$$(\boldsymbol{a} \cup \boldsymbol{b})^* \cup (\boldsymbol{a} \cup \boldsymbol{b})^*\boldsymbol{c}(\boldsymbol{a} \cup \boldsymbol{b})^*\boldsymbol{c}(\boldsymbol{a} \cup \boldsymbol{b} \cup \boldsymbol{c})^*$$

6.36. **Reversal of regular expressions.**

(a) Equivalences:
- $(R_1 \cup R_2)^r = R_1^r \cup R_2^r$
- $(R_1 \cdot R_2)^r = R_2^r \cdot R_1^r$
- $(R^*)^r = (R^r)^*$
- $\emptyset^r = \emptyset$
- $\epsilon^r = \epsilon$
- $\boldsymbol{a}^r = \boldsymbol{a}$ for each $\boldsymbol{a} \in \Sigma$

(b) To generate a regular expression for L^r from R: apply the reversal operator recursively to R using the rules above.

(c) **Proof by induction on the number of operators m in R:**
Base ($m = 0$): $R \in \{\emptyset, \epsilon, \boldsymbol{a}\}$. $L(R)^r = L(R^r)$ by direct check.
Step: Assume the result for all expressions with $< m$ operators. If $R = R_1 \cup R_2$: $(L(R_1) \cup L(R_2))^r = L(R_1)^r \cup L(R_2)^r = L(R_1^r) \cup L(R_2^r) = L(R_1^r \cup R_2^r) = L(R^r)$. Similarly for $\cdot$ and $*$. □

(d) Since every FAD language has a regular expression (Corollary 6.2), and the reversal of any regular expression (via the rules above) is also a regular expression, $\mathfrak{R}_\Sigma$ is closed under r.

6.37. **Complete details of Theorem 6.4.**

Theorem 6.4 is the left-linear analog of Theorem 6.2.

Proof. Focus on the mth equation:

$$Y_m = I_m \cup Y_1 B_{m1} \cup \cdots \cup Y_{m-1} B_{m(m-1)} \cup Y_m B_{mm} \cup Y_{m+1} B_{m(m+1)} \cup \cdots \cup Y_n B_{mn}.$$

Treating every term except $Y_m B_{mm}$ as a constant, the mirror-image of the argument in Theorem 6.1 gives

$$Y_m = (I_m \cup Y_1 B_{m1} \cup \cdots \cup Y_{m-1} B_{m(m-1)} \cup Y_{m+1} B_{m(m+1)} \cup \cdots \cup Y_n B_{mn}) B_{mm}^*.$$

This is part (b).

Substitute this expression for Y_m into each remaining equation

$$Y_k = I_k \cup Y_1 B_{k1} \cup \cdots \cup Y_m B_{km} \cup \cdots \cup Y_n B_{kn} \qquad (k \neq m),$$

and distribute B_{km} across the substituted expression. Collecting the constant term and the coefficients of each Y_j yields

$$\hat{I}_k = I_k \cup (I_m B_{mm}^* B_{km})$$

and

$$\hat{B}_{kj} = B_{kj} \cup (B_{mj} B_{mm}^* B_{km}) \qquad (j \neq m),$$

which is exactly the reduced system in part (a).

For a single left-linear equation

$$Y_1 = I_1 \cup Y_1 B_{11},$$

the same mirror-image argument gives the unique solution

$$Y_1 = I_1 B_{11}^*.$$

This is part (c). Hence Theorem 6.4 follows. □ □

6.38. (a) Words with no $\boldsymbol{b}$ immediately preceded by $\boldsymbol{a}$ over $\{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}\}$:

This means exactly that $\boldsymbol{ab}$ never occurs. Every maximal block of $\boldsymbol{a}$s must therefore be followed by $\boldsymbol{c}$ or occur at the end of the word. Hence one regular expression is

$$(\boldsymbol{b}\cup\boldsymbol{c}\cup\boldsymbol{aa}^*\boldsymbol{c})^*(\epsilon\cup\boldsymbol{aa}^*).$$

(b) Words with exactly two $\boldsymbol{c}$s and no $\boldsymbol{ab}$:

Over $\{\boldsymbol{a},\boldsymbol{b}\}$, avoiding $\boldsymbol{ab}$ means that once an $\boldsymbol{a}$ appears, only $\boldsymbol{a}$s may follow; hence the $\boldsymbol{a}/\boldsymbol{b}$ portions are exactly $\boldsymbol{b}^*\boldsymbol{a}^*$. Therefore the desired language is

$$\boldsymbol{b}^*\boldsymbol{a}^*\boldsymbol{c}\boldsymbol{b}^*\boldsymbol{a}^*\boldsymbol{c}\boldsymbol{b}^*\boldsymbol{a}^*.$$

6.39. (a) Words with no $\boldsymbol{b}$ immediately preceded by $\boldsymbol{c}$ (no $\boldsymbol{cb}$ substring):

This means exactly that every maximal block of $\boldsymbol{c}$s is followed by $\boldsymbol{a}$ or occurs at the end of the word. Hence one regular expression is

$$(\boldsymbol{a}\cup\boldsymbol{b}\cup\boldsymbol{cc}^*\boldsymbol{a})^*(\epsilon\cup\boldsymbol{cc}^*)$$

(b) Words with exactly one $\boldsymbol{c}$ and no $\boldsymbol{cb}$:

Exactly one $\boldsymbol{c}$, and that $\boldsymbol{c}$ is not immediately followed by $\boldsymbol{b}$:

$$(\boldsymbol{a}\cup\boldsymbol{b})^*\boldsymbol{c}(\boldsymbol{a}\cup\epsilon)(\boldsymbol{a}\cup\boldsymbol{b})^* \quad \text{where the } \boldsymbol{c} \text{ is followed by } \boldsymbol{a} \text{ or end}$$

More precisely:

$$(\boldsymbol{a}\cup\boldsymbol{b})^*\cdot\boldsymbol{c}\cdot((\boldsymbol{a}(\boldsymbol{a}\cup\boldsymbol{b})^*)\cup\epsilon)$$

6.40. (a) For Figure 6.11 the right-linear equations are

$$X_1 = \emptyset\cup\boldsymbol{a}X_1\cup\boldsymbol{a}X_2,$$
$$X_2 = \epsilon\cup(\boldsymbol{a}\cup\boldsymbol{b})X_1.$$

(b) Substitute the second equation into the first:

$$X_1 = \boldsymbol{a}X_1\cup\boldsymbol{a}(\epsilon\cup(\boldsymbol{a}\cup\boldsymbol{b})X_1) = \boldsymbol{a}\cup(\boldsymbol{a}\cup\boldsymbol{aa}\cup\boldsymbol{ab})X_1.$$

Hence

$$X_1 = (\boldsymbol{a}\cup\boldsymbol{aa}\cup\boldsymbol{ab})^*\boldsymbol{a}, \qquad X_2 = \epsilon\cup(\boldsymbol{a}\cup\boldsymbol{b})(\boldsymbol{a}\cup\boldsymbol{aa}\cup\boldsymbol{ab})^*\boldsymbol{a}.$$

(c) Therefore the NDFA accepts

$$L = (\boldsymbol{a}\cup\boldsymbol{aa}\cup\boldsymbol{ab})^*\boldsymbol{a}.$$

(d) The left-linear equations are

$$Y_1 = \epsilon\cup Y_1\boldsymbol{a}\cup Y_2(\boldsymbol{a}\cup\boldsymbol{b}),$$
$$Y_2 = Y_1\boldsymbol{a}.$$

Substituting gives

$$Y_1 = \epsilon\cup Y_1(\boldsymbol{a}\cup\boldsymbol{aa}\cup\boldsymbol{ab}),$$

so

$$Y_1 = (\boldsymbol{a}\cup\boldsymbol{aa}\cup\boldsymbol{ab})^*, \qquad Y_2 = (\boldsymbol{a}\cup\boldsymbol{aa}\cup\boldsymbol{ab})^*\boldsymbol{a}.$$

6.41. (a) For Figure 6.12 the right-linear equations are

$$\begin{aligned}
X_0 &= \boldsymbol{a}X_1,\\
X_1 &= \boldsymbol{a}X_1 \cup \boldsymbol{a}X_2,\\
X_2 &= \epsilon \cup \boldsymbol{b}X_2 \cup \boldsymbol{a}X_3,\\
X_3 &= \boldsymbol{b}X_0 \cup (\boldsymbol{a}\cup\boldsymbol{b})X_3.
\end{aligned}$$

(b) Solving from the end:

$$X_3 = (\boldsymbol{a}\cup\boldsymbol{b})^*\boldsymbol{b}X_0,$$

$$X_2 = \boldsymbol{b}^*(\epsilon\cup\boldsymbol{a}X_3) = \boldsymbol{b}^* \cup \boldsymbol{b}^*\boldsymbol{a}(\boldsymbol{a}\cup\boldsymbol{b})^*\boldsymbol{b}X_0,$$

$$X_1 = \boldsymbol{a}^*\boldsymbol{a}X_2, \qquad X_0 = \boldsymbol{aa}^*\boldsymbol{a}X_2.$$

Therefore

$$X_0 = \boldsymbol{aa}^*\boldsymbol{ab}^* \cup \boldsymbol{aa}^*\boldsymbol{ab}^*\boldsymbol{a}(\boldsymbol{a}\cup\boldsymbol{b})^*\boldsymbol{b}X_0,$$

so

$$X_0 = \big(\boldsymbol{aa}^*\boldsymbol{ab}^*\boldsymbol{a}(\boldsymbol{a}\cup\boldsymbol{b})^*\boldsymbol{b}\big)^*\boldsymbol{aa}^*\boldsymbol{ab}^*.$$

(c) Hence the language of the automaton is

$$L = \big(\boldsymbol{aa}^*\boldsymbol{ab}^*\boldsymbol{a}(\boldsymbol{a}\cup\boldsymbol{b})^*\boldsymbol{b}\big)^*\boldsymbol{aa}^*\boldsymbol{ab}^*.$$

(d) The left-linear equations are

$$\begin{aligned}
Y_0 &= \epsilon \cup Y_3\boldsymbol{b},\\
Y_1 &= Y_0\boldsymbol{a} \cup Y_1\boldsymbol{a},\\
Y_2 &= Y_1\boldsymbol{a} \cup Y_2\boldsymbol{b},\\
Y_3 &= Y_2\boldsymbol{a} \cup Y_3(\boldsymbol{a}\cup\boldsymbol{b}).
\end{aligned}$$

Solving gives

$$Y_0 = \big(\boldsymbol{aa}^*\boldsymbol{ab}^*\boldsymbol{a}(\boldsymbol{a}\cup\boldsymbol{b})^*\boldsymbol{b}\big)^*,$$

$$Y_1 = Y_0\boldsymbol{aa}^*, \qquad Y_2 = Y_0\boldsymbol{aa}^*\boldsymbol{ab}^*, \qquad Y_3 = Y_0\boldsymbol{aa}^*\boldsymbol{ab}^*\boldsymbol{a}(\boldsymbol{a}\cup\boldsymbol{b})^*.$$

6.42. (a) For Figure 6.13 the right-linear equations are

$$\begin{aligned}
X_0 &= \mathbf{0}X_0 \cup \mathbf{1}X_0 \cup \mathbf{0}X_1,\\
X_1 &= \mathbf{1}X_2,\\
X_2 &= \mathbf{0}X_3,\\
X_3 &= \mathbf{0}X_4,\\
X_4 &= \mathbf{1}X_5,\\
X_5 &= \mathbf{0}X_6,\\
X_6 &= \epsilon \cup \mathbf{0}X_6 \cup \mathbf{1}X_6.
\end{aligned}$$

(b) These equations collapse immediately:

$$X_6 = (\mathbf{0} \cup \mathbf{1})^*, \quad X_5 = \mathbf{0}(\mathbf{0} \cup \mathbf{1})^*, \quad X_4 = \mathbf{10}(\mathbf{0} \cup \mathbf{1})^*,$$

$$X_3 = \mathbf{010}(\mathbf{0} \cup \mathbf{1})^*, \quad X_2 = \mathbf{0010}(\mathbf{0} \cup \mathbf{1})^*, \quad X_1 = \mathbf{10010}(\mathbf{0} \cup \mathbf{1})^*.$$

Hence

$$X_0 = (\mathbf{0} \cup \mathbf{1})X_0 \cup \mathbf{010010}(\mathbf{0} \cup \mathbf{1})^*,$$

so

$$X_0 = (\mathbf{0} \cup \mathbf{1})^*\mathbf{010010}(\mathbf{0} \cup \mathbf{1})^*.$$

(c) Therefore the language is all binary strings containing **010010** as a substring:

$$L = (\mathbf{0} \cup \mathbf{1})^*\mathbf{010010}(\mathbf{0} \cup \mathbf{1})^*.$$

(d) The left-linear equations are

$$\begin{aligned}
Y_0 &= \epsilon \cup Y_0\mathbf{0} \cup Y_0\mathbf{1},\\
Y_1 &= Y_0\mathbf{0},\\
Y_2 &= Y_1\mathbf{1},\\
Y_3 &= Y_2\mathbf{0},\\
Y_4 &= Y_3\mathbf{0},\\
Y_5 &= Y_4\mathbf{1},\\
Y_6 &= Y_5\mathbf{0} \cup Y_6\mathbf{0} \cup Y_6\mathbf{1}.
\end{aligned}$$

Solving gives

$$Y_0 = (\mathbf{0} \cup \mathbf{1})^*, \quad Y_1 = (\mathbf{0} \cup \mathbf{1})^*\mathbf{0}, \quad Y_2 = (\mathbf{0} \cup \mathbf{1})^*\mathbf{01},$$

$$Y_3 = (\mathbf{0} \cup \mathbf{1})^*\mathbf{010}, \quad Y_4 = (\mathbf{0} \cup \mathbf{1})^*\mathbf{0100}, \quad Y_5 = (\mathbf{0} \cup \mathbf{1})^*\mathbf{01001},$$

and

$$Y_6 = (\mathbf{0} \cup \mathbf{1})^*\mathbf{010010}(\mathbf{0} \cup \mathbf{1})^*.$$

6.43. **Prove: for any A, E, Y, if $E \subseteq Y$ then $A \cdot E \subseteq A \cdot Y$.**

Proof. Let $w \in A \cdot E$. Then $w = xy$ with $x \in A$ and $y \in E$. Since $E \subseteq Y$, $y \in Y$. Hence $w = xy \in A \cdot Y$. □ □

6.44. (a) **Image of $\mathfrak{R}_\Sigma$ under substitution $\bar{s}$ is in $\mathfrak{R}_\Gamma$.**

Proof. Substitution $s : \Sigma \to \wp(\Gamma^*)$ maps each letter to a regular language. For a regular expression R over Σ, define $\bar{s}(R)$ by: $\bar{s}(\boldsymbol{a}) = s(\boldsymbol{a}) \in \mathfrak{R}_\Gamma$ (by assumption), $\bar{s}(\emptyset) = \emptyset$, $\bar{s}(\epsilon) = \{\lambda\}$, $\bar{s}(R_1 \cup R_2) = \bar{s}(R_1) \cup \bar{s}(R_2) \in \mathfrak{R}_\Gamma$ (closed under union), $\bar{s}(R_1 \cdot R_2) = \bar{s}(R_1) \cdot \bar{s}(R_2) \in \mathfrak{R}_\Gamma$ (closed under concatenation), $\bar{s}(R^*) = \bar{s}(R)^* \in \mathfrak{R}_\Gamma$ (closed under Kleene closure). By induction on the structure of R, $\bar{s}(R) \in \mathfrak{R}_\Gamma$. □ □

(b) **Image of $\mathfrak{N}_\Sigma$ under $\bar{s}$ need not be in $\mathfrak{N}_\Gamma$.**

Let

$$L = \{x \in \{\boldsymbol{a}, \boldsymbol{b}\}^* \mid |x|_{\boldsymbol{a}} = |x|_{\boldsymbol{b}}\} \in \mathfrak{N}_\Sigma.$$

Define the substitution by $s(\boldsymbol{a}) = (\boldsymbol{a} \cup \boldsymbol{b})$ and $s(\boldsymbol{b}) = (\boldsymbol{a} \cup \boldsymbol{b})$. Then $\overline{s}(L)$ is the set of all even-length words over $\{\boldsymbol{a}, \boldsymbol{b}\}$, which is regular. So the image of a nonregular language under a substitution need not remain in $\mathfrak{N}_\Gamma$.

6.45. **Proof of Lemma 6.3.**

Lemma 6.3 is the version of Theorem 6.2 that eliminates the mth unknown instead of necessarily the last unknown.

Proof. Proof. Renumber the unknowns so that X_m becomes the last unknown X_n, without changing the actual system except for notation. After this relabeling, Theorem 6.2 applies directly. It gives the elimination formulas

$$\hat{E}_i = E_i \cup (A_{im} A_{mm}^* E_m) \qquad \text{and} \qquad \hat{A}_{ij} = A_{ij} \cup (A_{im} A_{mm}^* A_{mj}),$$

for every index other than m, together with the back-substitution formula

$$X_m = A_{mm}^* \big(E_m \cup A_{m1} X_1 \cup \cdots \cup A_{m(m-1)} X_{m-1} \cup A_{m(m+1)} X_{m+1} \cup \cdots \cup A_{mn} X_n\big).$$

Renaming the variables back proves Lemma 6.3. □ □

6.46. $\Xi = \{x \in \{\boldsymbol{a}, \boldsymbol{b}\}^* \mid x$ contains at least two consecutive $\boldsymbol{b}$s $\wedge$ x contains no two consecutive $\boldsymbol{a}$s$\}$:

Words over $\{\boldsymbol{a}, \boldsymbol{b}\}$ avoiding $\boldsymbol{aa}$ are exactly $(\boldsymbol{b} \cup \boldsymbol{ab})^* (\boldsymbol{a} \cup \epsilon)$. Requiring a $\boldsymbol{bb}$ substring gives:

$$(\boldsymbol{b} \cup \boldsymbol{ab})^* (\boldsymbol{a} \cup \epsilon) \boldsymbol{bb} (\boldsymbol{b} \cup \boldsymbol{ab})^* (\boldsymbol{a} \cup \epsilon)$$

6.47. (a) Every $\boldsymbol{b}$ eventually followed by $\boldsymbol{c}$ (not necessarily immediately): A word is in the language iff either it has no $\boldsymbol{b}$ at all after its last $\boldsymbol{c}$, or it has no $\boldsymbol{c}$ and then must contain only $\boldsymbol{a}$s. Thus the suffix after the last $\boldsymbol{c}$ must lie in $\boldsymbol{a}^*$:

$$\boldsymbol{a}^* \cup (\boldsymbol{a} \cup \boldsymbol{b} \cup \boldsymbol{c})^* \boldsymbol{c} \boldsymbol{a}^*$$

Any $\boldsymbol{b}$ before the last $\boldsymbol{c}$ is discharged by that final $\boldsymbol{c}$, and after the last $\boldsymbol{c}$ only $\boldsymbol{a}$s may remain.

(b) Every $\boldsymbol{b}$ immediately followed by $\boldsymbol{c}$: (same as Exercise 6.11)

$$(\boldsymbol{a} \cup \boldsymbol{c} \cup \boldsymbol{bc})^*$$

6.48. Over $\Sigma = \{\boldsymbol{a}, \boldsymbol{b}\}$:

(a) No consecutive $\boldsymbol{a}$s and no consecutive $\boldsymbol{b}$s (strictly alternating):

$$(\boldsymbol{ab})^* (\boldsymbol{a} \cup \epsilon) \cup (\boldsymbol{ba})^* (\boldsymbol{b} \cup \epsilon)$$

(Alternating starting with $\boldsymbol{a}$ or $\boldsymbol{b}$, or empty word.)

(b) Words beginning and ending with different letters:

$$\boldsymbol{a}(\boldsymbol{a}\cup\boldsymbol{b})^*\boldsymbol{b}\cup\boldsymbol{b}(\boldsymbol{a}\cup\boldsymbol{b})^*\boldsymbol{a}$$

(c) Words for which the last two letters match ($\boldsymbol{aa}$ or $\boldsymbol{bb}$ at the end):

$$(\boldsymbol{a}\cup\boldsymbol{b})^*\boldsymbol{aa}\cup(\boldsymbol{a}\cup\boldsymbol{b})^*\boldsymbol{bb}$$

(d) Words for which the first two letters match ($\boldsymbol{aa}$ or $\boldsymbol{bb}$ at the start):

$$\boldsymbol{aa}(\boldsymbol{a}\cup\boldsymbol{b})^*\cup\boldsymbol{bb}(\boldsymbol{a}\cup\boldsymbol{b})^*$$

(Also include words of length < 2: λ and $\boldsymbol{a}$ and $\boldsymbol{b}$ trivially satisfy the vacuous condition; but "first two letters match" requires length ≥ 2.)

(e) Words for which the first and last letters match:

$$\epsilon\cup\boldsymbol{a}\cup\boldsymbol{b}\cup\boldsymbol{a}(\boldsymbol{a}\cup\boldsymbol{b})^*\boldsymbol{a}\cup\boldsymbol{b}(\boldsymbol{a}\cup\boldsymbol{b})^*\boldsymbol{b}$$

6.49. **The language of all valid regular expressions over $\{\boldsymbol{a},\boldsymbol{b}\}$ is not FAD.**

Proof. Let V be the language of all valid regular expressions over $\{\boldsymbol{a},\boldsymbol{b}\}$. Intersect V with the regular language

$$R=(\boldsymbol{(})^*\boldsymbol{a}\boldsymbol{)})^*.$$

The only valid regular expressions in R are the strings

$$\boldsymbol{(}^n\boldsymbol{a}\boldsymbol{)}^n \qquad (n\geq 0),$$

because the parentheses must match and simply enclose the one-letter expression $\boldsymbol{a}$. Therefore

$$V\cap R=\{\boldsymbol{(}^n\boldsymbol{a}\boldsymbol{)}^n \mid n\geq 0\},$$

which is not FAD by the pumping lemma. But the intersection of two FAD languages would be FAD. Hence V is not FAD. □

6.50. Apply Theorem 6.3 (state equations) and Theorem 6.2 (Arden's lemma) to each NDFA in Figure 6.14 to produce a regular expression.

(a) $(\boldsymbol{a}\cup\boldsymbol{c})^*$

(b) $(\boldsymbol{b}\cup\boldsymbol{ac}^*\boldsymbol{b})^*$

(c) $(\boldsymbol{a}\cup(\boldsymbol{a}\cup\boldsymbol{c})\boldsymbol{b})^*$

(d) $\boldsymbol{a}(\boldsymbol{bb})^*\boldsymbol{a}$

(e) $(\boldsymbol{a}\cup\boldsymbol{aa}\cup\boldsymbol{ab})^*\boldsymbol{a}$

(f) $\boldsymbol{aa}^*\boldsymbol{bc}(\boldsymbol{a}^*\boldsymbol{bc})^*$

(g) $\big(\boldsymbol{a}(\boldsymbol{bb}\cup\boldsymbol{cc})^*\boldsymbol{a}\big)^*$

6.51. **Complete details of Theorem 6.6.**

Theorem 6.6 proves closure of $\mathfrak{R}_\Sigma$ under substitution. The text supplies the basis step; the missing details are the three operator cases.

Proof. Assume inductively that whenever R_1 and R_2 have at most m operators, the expressions obtained by replacing each letter $\boldsymbol{a}$ with $s(\boldsymbol{a})$ represent $\overline{s}(R_1)$ and $\overline{s}(R_2)$.

If $R = R_1 \cdot R_2$, then by Definition 6.5,

$$\overline{s}(R_1 \cdot R_2) = \overline{s}(R_1) \cdot \overline{s}(R_2).$$

So the substituted expression $R_1' R_2'$ represents $\overline{s}(R)$.

If $R = R_1 \cup R_2$, then

$$\overline{s}(R_1 \cup R_2) = \overline{s}(R_1) \cup \overline{s}(R_2),$$

so the substituted expression $R_1' \cup R_2'$ represents $\overline{s}(R)$.

If $R = (R_1)^*$, then

$$\overline{s}((R_1)^*) = (\overline{s}(R_1))^*,$$

so $(R_1')^*$ represents $\overline{s}(R)$.

Thus substitution preserves each of the three operators used to build regular expressions. By induction on the number of operators in R, the expression obtained by substituting $s(\boldsymbol{a})$ for each letter $\boldsymbol{a}$ represents exactly $\overline{s}(R)$. Therefore $\mathfrak{R}_\Sigma$ is closed under substitution. □ □

6.52. **Prove Lemma 6.4.**

Proof. Let

$$A = \langle \Sigma, \{s_1, s_2, \dots, s_n\}, S_0, \delta, F \rangle$$

be an NDFA, and for each i let

$$Y_i = I(A, s_i) = \{x \mid \overline{\delta}(S_0, x) \ni s_i\}.$$

Define

$$I_i = \begin{cases} \epsilon & \text{if } s_i \in S_0, \\ \emptyset & \text{if } s_i \not{i} n S_0, \end{cases}$$

and

$$B_{ij} = \bigcup_{\boldsymbol{a} \in \Sigma \wedge s_i \in \delta(s_j, \boldsymbol{a})} \boldsymbol{a}.$$

We show that

$$Y_k = I_k \cup Y_1 B_{k1} \cup Y_2 B_{k2} \cup \cdots \cup Y_n B_{kn}.$$

If $x \in Y_k$, then either $x = \lambda$ and $s_k \in S_0$, so $x \in I_k$, or $x = za$ where z takes some start state to some s_j and a takes s_j to s_k. Then $z \in Y_j$ and $a \in B_{kj}$, so $x \in Y_j B_{kj}$. Hence $Y_k \subseteq I_k \cup Y_1 B_{k1} \cup \cdots \cup Y_n B_{kn}$.

Conversely, if $x \in I_k$, then $x = \lambda$ and $s_k \in S_0$, so $x \in Y_k$. If $x \in Y_j B_{kj}$, then $x = za$ for some $z \in Y_j$ and some letter $a \in B_{kj}$. Thus z reaches s_j from a start state, and then a takes s_j to s_k, so $x \in Y_k$. Therefore $I_k \cup Y_1 B_{k1} \cup \cdots \cup Y_n B_{kn} \subseteq Y_k$ as well.

So the required equations hold for every k. □ □

6.53. **Theorems 5.2, 5.4, 5.5 as corollaries of Theorem 6.6.**

By Theorem 6.5, every FAD language has a regular expression. Let $L_1 = L(R_1)$ and $L_2 = L(R_2)$.

For union, work over the alphabet $\{\boldsymbol{a}, \boldsymbol{b}\}$ with the regular expression

$$Q = \boldsymbol{a} \cup \boldsymbol{b}.$$

Define a substitution by

$$s(\boldsymbol{a}) = R_1, \qquad s(\boldsymbol{b}) = R_2.$$

Then the substituted expression is

$$\overline{s}(Q) = R_1 \cup R_2,$$

whose language is $L_1 \cup L_2$. So $L_1 \cup L_2$ is regular by Theorem 6.6.

For concatenation, use the regular expression

$$Q = \boldsymbol{ab}$$

with the same substitution. Then

$$\overline{s}(Q) = R_1 R_2,$$

whose language is $L_1 \cdot L_2$. So $L_1 \cdot L_2$ is regular.

For Kleene closure, work over the alphabet $\{\boldsymbol{a}\}$ and use the regular expression

$$Q = \boldsymbol{a}^*.$$

Define

$$s(\boldsymbol{a}) = R_1.$$

Then

$$\overline{s}(Q) = (R_1)^*,$$

whose language is L_1^*. So L_1^* is regular.

Thus Theorems 5.2, 5.4, and 5.5 follow as corollaries of Theorem 6.6.

6.54. **Class $\boldsymbol{F}$ generated by the five rules:**

The five rules generate exactly the class of *finite* languages over $\{\boldsymbol{a}, \boldsymbol{b}\}$:

(i) Singletons $\{\boldsymbol{a}\}$ and $\{\boldsymbol{b}\}$ are in $\boldsymbol{F}$.

(ii) $\emptyset \in \boldsymbol{F}$.

(iii) $\{\lambda\} \in \boldsymbol{F}$.

(iv) Concatenation: $F_1 \cdot F_2$ is finite if F_1 and F_2 are.

(v) Union: $F_1 \cup F_2$ is finite if F_1 and F_2 are.

No Kleene closure is allowed, so $\boldsymbol{F}$ consists only of finite languages. Every finite language over $\{\boldsymbol{a}, \boldsymbol{b}\}$ is in $\boldsymbol{F}$ (can be expressed as a finite union of finite concatenations of singletons), and $\boldsymbol{F} \subseteq F_{\{\boldsymbol{a},\boldsymbol{b}\}}$ (the finite languages). Thus $\boldsymbol{F} = F_{\{\boldsymbol{a},\boldsymbol{b}\}}$, the class of all finite languages over $\{\boldsymbol{a}, \boldsymbol{b}\}$.

Chapter 7

Finite State Transducers — Solutions

7.1. (a) **Prove** $\overline{\omega}(t, yx) = \overline{\omega}(t, y)\cdot\overline{\omega}(\overline{\delta}(t, y), x)$ for any Mealy machine.

Proof. By induction on $|x| = k$.
Base ($k = 0$): $\overline{\omega}(t, y\lambda) = \overline{\omega}(t, y) = \overline{\omega}(t, y)\cdot\lambda = \overline{\omega}(t, y)\cdot\overline{\omega}(\overline{\delta}(t, y), \lambda)$ since $\overline{\omega}(\cdot, \lambda) = \lambda$.
Step: Let $x = x'\boldsymbol{a}$ with $|x'| = k$. Then:

$$\begin{aligned}\overline{\omega}(t, yx'\boldsymbol{a}) &= \overline{\omega}(t, yx')\cdot\omega(\overline{\delta}(t, yx'), \boldsymbol{a})\\ &= \overline{\omega}(t, y)\cdot\overline{\omega}(\overline{\delta}(t, y), x')\cdot\omega(\overline{\delta}(\overline{\delta}(t, y), x'), \boldsymbol{a})\\ &\quad\text{(inductive hypothesis applied to } yx')\\ &= \overline{\omega}(t, y)\cdot\overline{\omega}(\overline{\delta}(t, y), x'\boldsymbol{a})\\ &= \overline{\omega}(t, y)\cdot\overline{\omega}(\overline{\delta}(t, y), x). \quad\square\end{aligned}$$

□

(b) **Prove** $\overline{\delta}(s, yx) = \overline{\delta}(\overline{\delta}(s, y), x)$.

Proof. By induction on $|x|$.
Base: $\overline{\delta}(s, y\lambda) = \overline{\delta}(s, y) = \overline{\delta}(\overline{\delta}(s, y), \lambda)$.
Step: $\overline{\delta}(s, yx'\boldsymbol{a}) = \delta(\overline{\delta}(s, yx'), \boldsymbol{a}) = \delta(\overline{\delta}(\overline{\delta}(s, y), x'), \boldsymbol{a}) = \overline{\delta}(\overline{\delta}(s, y), x'\boldsymbol{a})$. □ □

7.2. Let $\mu : S_A \to S_B$ be a homomorphism between Mealy machines.

(a) **Prove** $\mu(\overline{\delta}_A(s, x)) = \overline{\delta}_B(\mu(s), x)$.

Proof. By induction on $|x|$.
Base: $\mu(\overline{\delta}_A(s, \lambda)) = \mu(s) = \overline{\delta}_B(\mu(s), \lambda)$.
Step: $\mu(\overline{\delta}_A(s, x\boldsymbol{a})) = \mu(\delta_A(\overline{\delta}_A(s, x), \boldsymbol{a})) = \delta_B(\mu(\overline{\delta}_A(s, x)), \boldsymbol{a}) = \delta_B(\overline{\delta}_B(\mu(s), x), \boldsymbol{a}) = \overline{\delta}_B(\mu(s), x\boldsymbol{a})$.
(Used: $\mu(\delta_A(t, \boldsymbol{a})) = \delta_B(\mu(t), \boldsymbol{a})$ for all $t \in S_A$—homomorphism property.) □ □

(b) **Prove** $\overline{\omega}_A(s, x) = \overline{\omega}_B(\mu(s), x)$.

Proof. By induction on $|x|$.
Base: $\overline{\omega}_A(s, \lambda) = \lambda = \overline{\omega}_B(\mu(s), \lambda)$.

Step: $\overline{\omega}_A(s, x\boldsymbol{a}) = \overline{\omega}_A(s,x)\cdot\omega_A(\overline{\delta}_A(s,x),\boldsymbol{a})$. By induction: $= \overline{\omega}_B(\mu(s),x)\cdot\omega_A(\overline{\delta}_A(s,x),\boldsymbol{a})$. By homomorphism: $\omega_A(t,\boldsymbol{a}) = \omega_B(\mu(t),\boldsymbol{a})$ for all t; so: $= \overline{\omega}_B(\mu(s),x)\cdot\omega_B(\mu(\overline{\delta}_A(s,x)),\boldsymbol{a}) = \overline{\omega}_B(\mu(s),x)\cdot\omega_B(\overline{\delta}_B(\mu(s),x),\boldsymbol{a}) = \overline{\omega}_B(\mu(s),x\boldsymbol{a})$. □ □

7.3. **Prove Corollary 7.3.**

If there is a homomorphism $\mu : A \to B$ between connected Mealy machines, then A and B are equivalent.

Proof. By Exercise 7.2(b), $\overline{\omega}_A(s_0,x) = \overline{\omega}_B(\mu(s_0),x)$ for all $x \in \Sigma^*$. Since $\mu(s_{0_A}) = s_{0_B}$ (by definition of homomorphism), $\overline{\omega}_A(s_{0_A},x) = \overline{\omega}_B(s_{0_B},x)$, i.e., $f_A(x) = f_B(x)$ for all x. Hence $A \equiv B$. □ □

7.4. **Prove Corollary 7.4: M is minimal iff M is reduced and connected.**

Proof. (⇒) If M is minimal, then M must be connected by Corollary 7.2. Also, M must be reduced by Corollary 7.3. Hence M is both reduced and connected.

(⇐) Assume that M is reduced and connected. Let A be any minimal FST equivalent to M. By Corollaries 7.2 and 7.3, A is also reduced and connected. Therefore Theorem 7.5 applies, so $A \cong M$. Isomorphic machines have the same number of states, hence $\|S_A\| = \|S_M\|$. Since A was chosen minimal, no equivalent machine has fewer states than A, and therefore no equivalent machine has fewer states than M. Thus M is minimal. □ □

7.5. **Pumping lemma for FTD functions.**

(a) **Statement:** Let $f : \Sigma^* \to \Gamma^*$ be FTD, implemented by a Mealy machine M with n states. Then for any $x \in \Sigma^*$ with $|x| \ge n$, there exist strings u, v, w with $x = uvw$, $|uv| \le n$, $|v| \ge 1$, such that for all $k \ge 0$:

$$f(uv^k w) = \overline{\omega}(s_0,u)\cdot\overline{\omega}(\overline{\delta}(s_0,u),v)^k\cdot\overline{\omega}(\overline{\delta}(s_0,uv),w).$$

That is, the output of $uv^k w$ consists of the output while processing u, followed by k repetitions of the output while processing v (each time starting from state $\overline{\delta}(s_0,u)$), followed by the output while processing w from state $\overline{\delta}(s_0,uv) = \overline{\delta}(s_0,u)$.

(b) **Proof:** Since $|x| \ge n$ and M has n states, by the pigeonhole principle the sequence of states $s_0, \overline{\delta}(s_0,u), \overline{\delta}(s_0,uv), \ldots$ must repeat a state within the first $n+1$ steps. Let u, v, w be such that $uvw = x$, v causes a cycle: $\overline{\delta}(s_0,u) = \overline{\delta}(s_0,uv)$. Then for any $k \ge 0$, $\overline{\delta}(s_0,uv^k) = \overline{\delta}(s_0,u)$, so the machine processes w from the same state regardless of k. Thus $f(uv^k w) = \overline{\omega}(s_0,u)\cdot\overline{\omega}(\overline{\delta}(s_0,u),v)^k\cdot\overline{\omega}(\overline{\delta}(s_0,uv),w)$. □

7.6. **Prove Lemma 7.3 parts a–e.**

(a) **Part (a):** $E_{m+1,M}$ is a refinement of E_{mM}.

Proof. If $sE_{m+1,M}t$, then by definition the states s and t agree on all input strings of length at most $m+1$. In particular, they agree on all strings of length at most m. Hence $sE_{mM}t$. □ □

(b) **Part (b):** E_M is a refinement of E_{mM}.

Proof. If $sE_M t$, then $\overline{\omega}(s,x) = \overline{\omega}(t,x)$ for every $x \in \Sigma^*$. In particular, the two states agree on all strings of length at most m, so $sE_{mM} t$. Therefore $E_M \subseteq E_{mM}$. □ □

(c) **Part (c):** If $E_{mM} = E_{m+1,M}$, then $E_{m+k,M} = E_{mM}$ for every $k \in \mathbb{N}$.

Proof. We use induction on k. The case $k = 0$ is trivial, and the case $k = 1$ is the hypothesis. Assume $E_{m+k,M} = E_{mM}$ for some $k \geq 1$. By part (a), $E_{m+k+1,M} \subseteq E_{m+k,M} = E_{mM}$, so only the reverse inclusion needs proof.

Let $sE_{mM} t$. Since $E_{mM} = E_{m+1,M}$, we also have $sE_{m+1,M} t$. Therefore, for every $\boldsymbol{a} \in \Sigma$, the successor states $\delta(s,\boldsymbol{a})$ and $\delta(t,\boldsymbol{a})$ agree on all strings of length at most m, so

$$\delta(s,\boldsymbol{a}) E_{mM} \delta(t,\boldsymbol{a}).$$

By the induction hypothesis this implies

$$\delta(s,\boldsymbol{a}) E_{m+k,M} \delta(t,\boldsymbol{a})$$

for every $\boldsymbol{a}$. Hence s and t agree on all strings of length at most $m+k+1$, so $sE_{m+k+1,M} t$. Thus $E_{mM} \subseteq E_{m+k+1,M}$, and therefore $E_{m+k+1,M} = E_{mM}$. □ □

(d) **Part (d):** There exists $m \leq \|S\|$ such that $E_{mM} = E_{m+1,M}$.

Proof. By part (a), the relations

$$E_{0M}, E_{1M}, E_{2M}, \ldots$$

form a descending chain of refinements. Every proper refinement increases the number of equivalence classes by at least one. Since there are only $\|S\|$ states, there can be at most $\|S\|$ equivalence classes. Therefore the chain cannot refine properly more than $\|S\| - 1$ times, and for some $m \leq \|S\|$ we must have $E_{mM} = E_{m+1,M}$. □ □

(e) **Part (e):** If $E_{mM} = E_{m+1,M}$, then $E_{mM} = E_M$.

Proof. By part (b), $E_M \subseteq E_{mM}$. For the reverse inclusion, let $sE_{mM} t$ and let $x \in \Sigma^*$ be arbitrary, say $|x| = k$. By part (c),

$$E_{m+k,M} = E_{mM},$$

so $sE_{m+k,M} t$. By definition of $E_{m+k,M}$, the states s and t agree on every input string of length at most $m+k$, and in particular on the string x. Thus $\overline{\omega}(s,x) = \overline{\omega}(t,x)$. Since x was arbitrary, $sE_M t$. Hence $E_{mM} \subseteq E_M$, and therefore $E_{mM} = E_M$. □ □

7.7. **Prove parts of Lemma 7.4 for FST M.**

(a) E_{0M} **has one class consisting of all of** S**:**

E_{0M} is defined by $sE_{0M} t$ always (no conditions). Hence all states are in one class: $[s]_{E_{0M}} = S$ for all s.

(b) E_{1M} **iff** $\forall \boldsymbol{a} \in \Sigma, \omega(s,\boldsymbol{a}) = \omega(t,\boldsymbol{a})$**:**

E_{1M} refines E_{0M} by requiring that s and t agree on single-letter outputs. $sE_{1M} t$ iff $sE_{0M} t$ (trivially) and $\forall \boldsymbol{a} \in \Sigma, \omega(s,\boldsymbol{a}) = \omega(t,\boldsymbol{a})$.

(c) $sE_{i+1,M} t$ **iff** $sE_{iM} t$ **and** $\forall \boldsymbol{a} \in \Sigma, \delta(s,\boldsymbol{a}) E_{iM} \delta(t,\boldsymbol{a})$**:**

Proof. By definition, $E_{i+1,M}$ refines E_{iM} by additionally requiring that the successors are E_{iM}-equivalent. This is the standard iterative refinement.

($\Rightarrow$) If $sE_{i+1,M}t$, then s and t agree on all strings of length $\le i+1$. In particular, they agree on strings of length $\le i$ (so $sE_{iM}t$), and for each $\boldsymbol{a}$, $\delta(s,\boldsymbol{a})$ and $\delta(t,\boldsymbol{a})$ agree on all strings of length $\le i$ (so $\delta(s,\boldsymbol{a})E_{iM}\delta(t,\boldsymbol{a})$).

($\Leftarrow$) If $sE_{iM}t$ and $\delta(s,\boldsymbol{a})E_{iM}\delta(t,\boldsymbol{a})$ for all $\boldsymbol{a}$, then for any $w=\boldsymbol{a}w'$ with $|w|\le i+1$: $\overline{\omega}(s,\boldsymbol{a}w') = \omega(s,\boldsymbol{a})\cdot\overline{\omega}(\delta(s,\boldsymbol{a}),w')$. Since $sE_{1M}t$ (implied by $sE_{iM}t$, $i\ge 1$): $\omega(s,\boldsymbol{a})=\omega(t,\boldsymbol{a})$. And $\delta(s,\boldsymbol{a})E_{iM}\delta(t,\boldsymbol{a})$ gives $\overline{\omega}(\delta(s,\boldsymbol{a}),w') = \overline{\omega}(\delta(t,\boldsymbol{a}),w')$ for $|w'|\le i$. So $\overline{\omega}(s,w)=\overline{\omega}(t,w)$. □ □

7.8. **Prove Theorem 7.6 for Moore machines.**

For a Moore machine $A=\langle\Sigma,\Gamma,S,s_0,\delta,\omega\rangle$:

Proof. We prove that

$$(\forall x\in\Sigma^*)(\forall y\in\Sigma^*)(\forall t\in S)\overline{\omega}(t,yx)=\overline{\omega}(t,y)\cdot\overline{\omega}(\overline{\delta}(t,y),x)$$

by induction on $|y|$.

Base ($|y|=0$): If $y=\lambda$, then

$$\overline{\omega}(t,\lambda x)=\overline{\omega}(t,x)=\lambda\cdot\overline{\omega}(t,x)=\overline{\omega}(t,\lambda)\cdot\overline{\omega}(\overline{\delta}(t,\lambda),x),$$

since $\overline{\omega}(t,\lambda)=\lambda$ and $\overline{\delta}(t,\lambda)=t$.

Step: Assume the identity holds for a word y. Let $\boldsymbol{a}\in\Sigma$. Then, by Definition 7.17,

$$\overline{\omega}(t,\boldsymbol{a}yx)=\omega(\delta(t,\boldsymbol{a}))\cdot\overline{\omega}(\delta(t,\boldsymbol{a}),yx).$$

Apply the inductive hypothesis with start state $\delta(t,\boldsymbol{a})$:

$$\overline{\omega}(t,\boldsymbol{a}yx)=\omega(\delta(t,\boldsymbol{a}))\cdot\overline{\omega}(\delta(t,\boldsymbol{a}),y)\cdot\overline{\omega}(\overline{\delta}(\delta(t,\boldsymbol{a}),y),x).$$

Again by Definition 7.17, the first two factors are exactly $\overline{\omega}(t,\boldsymbol{a}y)$, and by Exercise 7.1(b), $\overline{\delta}(\delta(t,\boldsymbol{a}),y)=\overline{\delta}(t,\boldsymbol{a}y)$. Hence

$$\overline{\omega}(t,\boldsymbol{a}yx)=\overline{\omega}(t,\boldsymbol{a}y)\cdot\overline{\omega}(\overline{\delta}(t,\boldsymbol{a}y),x).$$

This proves the result. □ □

7.9. **Prove Theorem 7.7.**

Proof. Let

$$A=\langle\Sigma,\Gamma,S,s_0,\delta,\omega\rangle$$

be a Moore machine, and let

$$M=\langle\Sigma,\Gamma,S,s_0,\delta,\omega'\rangle$$

be the corresponding Mealy machine from Definition 7.19, where $\omega'(s,\boldsymbol{a})=\omega(\delta(s,\boldsymbol{a}))$.

We prove by induction on $|x|$ that

$$(\forall s\in S)(\forall x\in\Sigma^*)\overline{\omega}_M(s,x)=\overline{\omega}_A(s,x).$$

Base: If $x = \lambda$, then both sides equal λ.

Step: Let $x = \boldsymbol{a}y$. Then

$$\overline{\omega}_M(s, \boldsymbol{a}y) = \omega'(s, \boldsymbol{a}) \cdot \overline{\omega}_M(\delta(s, \boldsymbol{a}), y) = \omega(\delta(s, \boldsymbol{a})) \cdot \overline{\omega}_M(\delta(s, \boldsymbol{a}), y).$$

By the inductive hypothesis applied to the state $\delta(s, \boldsymbol{a})$,

$$\overline{\omega}_M(s, \boldsymbol{a}y) = \omega(\delta(s, \boldsymbol{a})) \cdot \overline{\omega}_A(\delta(s, \boldsymbol{a}), y) = \overline{\omega}_A(s, \boldsymbol{a}y),$$

by Definition 7.17. Therefore $\overline{\omega}_M(s, x) = \overline{\omega}_A(s, x)$ for all s and x. In particular, $f_M(x) = \overline{\omega}_M(s_0, x) = \overline{\omega}_A(s_0, x) = f_A(x)$, so $M \equiv A$. □ □

7.10. **Prove Theorem 7.8.**

Proof. Let

$$M = \langle \Sigma, \Gamma, S, s_0, \delta, \omega \rangle$$

be a Mealy machine, and let

$$A = \langle \Sigma, \Gamma, S \times \Gamma, \langle s_0, \alpha \rangle, \delta', \omega' \rangle$$

be the corresponding Moore machine from Definition 7.20.

We prove the stronger statement

$$(\forall s \in S)(\forall \boldsymbol{b} \in \Gamma)(\forall x \in \Sigma^*) \overline{\omega}_A(\langle s, \boldsymbol{b} \rangle, x) = \overline{\omega}_M(s, x).$$

Base: If $x = \lambda$, then both sides are λ.

Step: Let $x = \boldsymbol{a}y$. By Definition 7.17 for Moore machines,

$$\overline{\omega}_A(\langle s, \boldsymbol{b} \rangle, \boldsymbol{a}y) = \omega'(\delta'(\langle s, \boldsymbol{b} \rangle, \boldsymbol{a})) \cdot \overline{\omega}_A(\delta'(\langle s, \boldsymbol{b} \rangle, \boldsymbol{a}), y).$$

By Definition 7.20,

$$\delta'(\langle s, \boldsymbol{b} \rangle, \boldsymbol{a}) = \langle \delta(s, \boldsymbol{a}), \omega(s, \boldsymbol{a}) \rangle$$

and

$$\omega'(\langle \delta(s, \boldsymbol{a}), \omega(s, \boldsymbol{a}) \rangle) = \omega(s, \boldsymbol{a}).$$

Hence

$$\overline{\omega}_A(\langle s, \boldsymbol{b} \rangle, \boldsymbol{a}y) = \omega(s, \boldsymbol{a}) \cdot \overline{\omega}_A(\langle \delta(s, \boldsymbol{a}), \omega(s, \boldsymbol{a}) \rangle, y).$$

Apply the inductive hypothesis:

$$\overline{\omega}_A(\langle s, \boldsymbol{b} \rangle, \boldsymbol{a}y) = \omega(s, \boldsymbol{a}) \cdot \overline{\omega}_M(\delta(s, \boldsymbol{a}), y) = \overline{\omega}_M(s, \boldsymbol{a}y).$$

Therefore $\overline{\omega}_A(\langle s, \boldsymbol{b} \rangle, x) = \overline{\omega}_M(s, x)$ for all states and inputs. In particular,

$$f_A(x) = \overline{\omega}_A(\langle s_0, \alpha \rangle, x) = \overline{\omega}_M(s_0, x) = f_M(x),$$

so $A \equiv M$. □ □

7.11. **Use Lemma 7.6 to find E_C in Example 7.10.**

For the Moore machine C of Example 7.10,

$$\omega(r_0) = \omega(r_2) = \mathbf{0}, \qquad \omega(r_1) = \omega(r_3) = \mathbf{1}.$$

Hence

$$E_{0C} = \{\{r_0, r_2\}, \{r_1, r_3\}\}.$$

Now apply Lemma 7.6(b). The states r_0 and r_2 are separated at the next stage because

$$\delta(r_0, \boldsymbol{a}) = r_0, \qquad \delta(r_2, \boldsymbol{a}) = r_1,$$

and r_0 and r_1 are in different E_{0C}-classes. Likewise, r_1 and r_3 are separated because

$$\delta(r_1, \boldsymbol{a}) = r_0, \qquad \delta(r_3, \boldsymbol{a}) = r_1,$$

again landing in different E_{0C}-classes. Therefore

$$E_{1C} = \{\{r_0\}, \{r_1\}, \{r_2\}, \{r_3\}\}.$$

Since E_{1C} is already the identity relation, the process has stabilized and

$$E_C = E_{1C}.$$

7.12. **Homomorphism from machine M (Example 7.11) to machine B (Example 7.2).**

Let the states of M be $\{r_0, r_1, r_2, r_3\}$, as in Example 7.11, and let the states of B be $\{s_0, s_1\}$. Define

$$\mu(r_0) = \mu(r_1) = s_0, \qquad \mu(r_2) = \mu(r_3) = s_1.$$

This is the natural choice because in Example 7.11 the states r_0, r_1 print only $\mathbf{0}$, while r_2, r_3 print only $\mathbf{1}$, exactly as s_0 and s_1 do in Example 7.2.

We check the homomorphism conditions.

(i) $\mu(r_0) = s_0$, so start states are preserved.

(ii) For $\boldsymbol{a}$, Example 7.11 gives

$$\delta_M(r_0, \boldsymbol{a}) = \delta_M(r_1, \boldsymbol{a}) = r_0, \qquad \delta_M(r_2, \boldsymbol{a}) = \delta_M(r_3, \boldsymbol{a}) = r_1.$$

Therefore

$$\mu(\delta_M(r_i, \boldsymbol{a})) = s_0 = \delta_B(\mu(r_i), \boldsymbol{a})$$

for every i, since $\delta_B(s_0, \boldsymbol{a}) = \delta_B(s_1, \boldsymbol{a}) = s_0$ in Example 7.2.

For $\boldsymbol{b}$, Example 7.11 gives

$$\delta_M(r_0, \boldsymbol{b}) = \delta_M(r_1, \boldsymbol{b}) = r_2, \qquad \delta_M(r_2, \boldsymbol{b}) = \delta_M(r_3, \boldsymbol{b}) = r_3.$$

Hence

$$\mu(\delta_M(r_i, \boldsymbol{b})) = s_1 = \delta_B(\mu(r_i), \boldsymbol{b})$$

for every i, since $\delta_B(s_0, \boldsymbol{b}) = \delta_B(s_1, \boldsymbol{b}) = s_1$.

(iii) The output table of Example 7.11 gives

$$\omega_M(r_0, \boldsymbol{a}) = \omega_M(r_0, \boldsymbol{b}) = \omega_M(r_1, \boldsymbol{a}) = \omega_M(r_1, \boldsymbol{b}) = \mathbf{0},$$

and

$$\omega_M(r_2, \boldsymbol{a}) = \omega_M(r_2, \boldsymbol{b}) = \omega_M(r_3, \boldsymbol{a}) = \omega_M(r_3, \boldsymbol{b}) = \mathbf{1}.$$

These agree exactly with the outputs of B at s_0 and s_1.

Thus μ is a homomorphism from M onto B.

7.13. **Prove $\overline{\omega}(t, \boldsymbol{a}) = \omega(t, \boldsymbol{a})$ for any FST.**

Proof. By definition of $\overline{\omega}$ for a single letter: $\overline{\omega}(t, \boldsymbol{a}) = \overline{\omega}(t, \lambda) \cdot \omega(t, \boldsymbol{a}) = \lambda \cdot \omega(t, \boldsymbol{a}) = \omega(t, \boldsymbol{a})$. (Since $\overline{\omega}(t, \lambda) = \lambda$ and concatenation with λ is identity.) □ □

7.14. **Modified vending machine with coin return.**

Add to the machine:

- New input $\boldsymbol{r}$ (coin return).
- New output $\boldsymbol{a}$ (release coins).
- From each state s_k (representing k cents in holding), the transition on $\boldsymbol{r}$ goes to s_0 (reset to zero) and outputs $\boldsymbol{a}$.
- The output $\boldsymbol{a}$ could encode how many coins to return (one per 5 cents, etc.), but if $\boldsymbol{a}$ is a single symbol meaning "release all," then it's a single output.

Formally: add $\delta(s_k, \boldsymbol{r}) = s_0$ and $\omega(s_k, \boldsymbol{r}) = \boldsymbol{a}$ for each state s_k in the vending machine.

7.15. **δ_{E_M} is well defined.**

Proof. $\delta_{E_M}([s]_{E_M}, \boldsymbol{a}) = [\delta(s, \boldsymbol{a})]_{E_M}$. Suppose $[s]_{E_M} = [s']_{E_M}$, i.e., sE_Ms'. By Lemma 7.3c, $\delta(s, \boldsymbol{a})E_M\delta(s', \boldsymbol{a})$. Hence $[\delta(s, \boldsymbol{a})]_{E_M} = [\delta(s', \boldsymbol{a})]_{E_M}$, so δ_{E_M} gives the same result regardless of the representative chosen. □ □

7.16. **ω_{E_M} is well defined.**

Proof. $\omega_{E_M}([s]_{E_M}, \boldsymbol{a}) = \omega(s, \boldsymbol{a})$. Suppose sE_Ms'. By the definition of E_M (states are equivalent iff they produce the same output on all inputs), in particular on the single letter $\boldsymbol{a}$: $\omega(s, \boldsymbol{a}) = \omega(s', \boldsymbol{a})$. Hence ω_{E_M} is independent of the choice of representative. □ □

7.17. **Reduced is not sufficient for minimality.**

Example. A reduced machine may have unreachable states. Let M have states $\{s_0, s_1\}$ where s_1 is unreachable from s_0. If s_0 and s_1 have distinct output behaviors, M is reduced (no two states are equivalent). But s_1 is useless; removing it gives an equivalent machine with fewer states. Hence M is reduced but not minimal.

7.18. **μ in Theorem 7.5 is well defined.**

Proof. μ is defined by $\mu(s) = \overline{\delta}_2(s_{0_2}, x)$ where x is any word with $\overline{\delta}_1(s_{0_1}, x) = s$ (such x exists because M_1 is connected). Well-definedness requires: if $\overline{\delta}_1(s_{0_1}, x) = \overline{\delta}_1(s_{0_1}, x') = s$, then $\overline{\delta}_2(s_{0_2}, x) = \overline{\delta}_2(s_{0_2}, x')$.

Since M_1 and M_2 are equivalent, $\overline{\omega}_1(s_{0_1}, wx) = \overline{\omega}_2(s_{0_2}, wx)$ and $\overline{\omega}_1(s_{0_1}, wx') = \overline{\omega}_2(s_{0_2}, wx')$ for all w. In particular, $\overline{\omega}_1(s_{0_1}, xy) = \overline{\omega}_2(s_{0_2}, xy)$ implies $\overline{\omega}_1(s, y) = \overline{\omega}_2(\overline{\delta}_2(s_{0_2}, x), y)$ for all y. Similarly with x'. Since M_2 is reduced, the state $\overline{\delta}_2(s_{0_2}, x)$ is uniquely determined by its output behavior, which equals that of s. Hence $\overline{\delta}_2(s_{0_2}, x) = \overline{\delta}_2(s_{0_2}, x')$, and μ is well defined. □ □

7.19. **μ is an isomorphism.**

(a) $\mu(s_{0_1}) = \overline{\delta}_2(s_{0_2}, \lambda) = s_{0_2}$.

(b) $\mu(\delta_1(s, \boldsymbol{a})) = \overline{\delta}_2(s_{0_2}, x\boldsymbol{a}) = \delta_2(\overline{\delta}_2(s_{0_2}, x), \boldsymbol{a}) = \delta_2(\mu(s), \boldsymbol{a})$.

(c) $\omega_1(s, \boldsymbol{a}) = \overline{\omega}_1(s_{0_1}, x\boldsymbol{a})|_{\text{last output}} = \overline{\omega}_2(s_{0_2}, x\boldsymbol{a})|_{\text{last}} = \omega_2(\mu(s), \boldsymbol{a})$. (The output functions agree since the machines are equivalent and μ commutes with transitions.)

(d) **Injective:** If $\mu(s) = \mu(t)$, then $\overline{\delta}_2(s_{0_2}, x_s) = \overline{\delta}_2(s_{0_2}, x_t)$, so s and t have the same output behavior from s_{0_1}; by M_1 being reduced, $s = t$.

(e) **Surjective:** Since M_2 is connected, every $t \in S_2$ is reachable: $t = \overline{\delta}_2(s_{0_2}, y)$ for some y. Let $s = \overline{\delta}_1(s_{0_1}, y)$; then $\mu(s) = t$.

7.20. **One-unit delay Mealy and Moore machines.**

$\overline{\omega}(x_1 x_2 \cdots x_m) = \boldsymbol{x} \cdot x_1 \cdots x_{m-1}$ (output is $\boldsymbol{x}$ followed by the input shifted by one).

(a) **Mealy sextuple:** $M = \langle \{\boldsymbol{a}, \boldsymbol{b}\}, \{\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{x}\}, \{s_a, s_b, s_0\}, s_0, \delta, \omega \rangle$ where s_0 is the initial state, s_a means "last input was $\boldsymbol{a}$", s_b means "last input was $\boldsymbol{b}$". $\delta(s_0, \boldsymbol{a}) = s_a$, $\delta(s_0, \boldsymbol{b}) = s_b$, $\delta(s_a, \boldsymbol{a}) = s_a$, $\delta(s_a, \boldsymbol{b}) = s_b$, $\delta(s_b, \boldsymbol{a}) = s_a$, $\delta(s_b, \boldsymbol{b}) = s_b$. $\omega(s_0, \boldsymbol{a}) = \boldsymbol{x}$, $\omega(s_0, \boldsymbol{b}) = \boldsymbol{x}$, $\omega(s_a, \boldsymbol{a}) = \boldsymbol{a}$, $\omega(s_a, \boldsymbol{b}) = \boldsymbol{a}$, $\omega(s_b, \boldsymbol{a}) = \boldsymbol{b}$, $\omega(s_b, \boldsymbol{b}) = \boldsymbol{b}$.

(b) **Mealy machine diagram:** Three states; s_0 outputs $\boldsymbol{x}$ on both inputs; s_a outputs $\boldsymbol{a}$ on both inputs; s_b outputs $\boldsymbol{b}$ on both inputs.

(c) **Moore sextuple:** Same states but output is determined by state: $\omega(s_0) = \boldsymbol{x}$, $\omega(s_a) = \boldsymbol{a}$, $\omega(s_b) = \boldsymbol{b}$. (The Moore machine outputs at the current state, not the transition.)

(d) **Moore machine diagram:** Three states with state-output labels; same transition structure.

7.21. **Circuit for vending machine (Example 7.1).**

(a) **EOS:** No. The machine operates continuously, and the candy bar is dispensed on the transition where the button is pushed.

(b) **SOS:** Yes. As the text points out, a vending machine should reset to its zero-credit state after power is restored.

(c) **Input encoding:** We want codes for $\{\boldsymbol{b}, \boldsymbol{d}, \boldsymbol{n}, \boldsymbol{q}, \text{<}SOS\text{>}\}$, so three bits suffice. One convenient assignment is

$$\boldsymbol{b} = 000, \qquad \boldsymbol{d} = 001, \qquad \boldsymbol{n} = 010, \qquad \boldsymbol{q} = 011, \qquad \text{<}SOS\text{>} = 111,$$

with 100, 101, 110 unused.

(d) **Output encoding:** The output alphabet is

$$\{\boldsymbol{c}_0, \boldsymbol{c}_1, \boldsymbol{c}_2, \boldsymbol{c}_3, \boldsymbol{c}_4, \boldsymbol{d}', \boldsymbol{n}', \boldsymbol{q}', \boldsymbol{\varphi}\},$$

so four bits are needed. Let

$\boldsymbol{c}_0 = 0000$, $\boldsymbol{c}_1 = 0001$, $\boldsymbol{c}_2 = 0010$, $\boldsymbol{c}_3 = 0011$, $\boldsymbol{c}_4 = 0100$, $\boldsymbol{d}' = 0101$, $\boldsymbol{n}' = 0110$, $\boldsymbol{q}' = 0111$, $\boldsymbol{\varphi} = 1000$.

(e) **State encoding:** Write s_i for the state representing $5i$¢ for $0 \le i \le 10$. Then 11 states require four bits; for example

$$s_0 = 0000,\ s_1 = 0001,\ \ldots,\ s_{10} = 1010,$$

with $1011, 1100, 1101, 1110, 1111$ unused.

(f) t_2 **truth table and Boolean function:**

	$\boldsymbol{b}$	$\boldsymbol{d}$	$\boldsymbol{n}$	$\boldsymbol{q}$	<SOS>
s_0	0	1	0	0	0
s_1	0	1	1	1	0
s_2	1	0	1	1	0
s_3	1	0	0	0	0
s_4	0	1	0	0	0
s_5	0	1	1	1	0
s_6	0	1	1	1	0
s_7	0	1	1	1	0
s_8	0	0	0	0	0
s_9	0	0	0	0	0
s_{10}	0	1	1	1	0

and one minimized sum-of-products is

$$\begin{aligned} t_2' = {} & (\neg t_4 \wedge \neg t_2 \wedge t_1 \wedge \neg a_3 \wedge a_2) \vee (t_3 \wedge t_2 \wedge \neg a_3 \wedge a_2) \vee \\ & (\neg t_4 \wedge \neg t_3 \wedge t_2 \wedge \neg a_2 \wedge \neg a_1) \vee (t_4 \wedge t_2 \wedge \neg a_3 \wedge a_1) \vee \\ & (t_2 \wedge \neg t_1 \wedge \neg a_3 \wedge a_2) \vee (t_3 \wedge \neg a_2 \wedge a_1) \vee (\neg t_4 \wedge \neg t_2 \wedge \neg a_2 \wedge a_1). \end{aligned}$$

(g) w_2 **truth table and Boolean function:**

	$\boldsymbol{b}$	$\boldsymbol{d}$	$\boldsymbol{n}$	$\boldsymbol{q}$	<SOS>
s_0	0	0	0	0	0
s_1	0	0	0	0	0
s_2	0	0	0	0	0
s_3	0	0	0	0	0
s_4	0	0	0	0	0
s_5	0	0	0	0	0
s_6	0	0	1	1	0
s_7	0	0	1	1	0
s_8	1	0	1	1	0
s_9	1	0	1	1	0
s_{10}	0	0	1	1	0

and a minimized form is

$$w_2 = (t_4 \wedge \neg t_2 \wedge \neg a_1) \vee (t_3 \wedge t_2 \wedge \neg a_3 \wedge a_2) \vee (t_4 \wedge \neg a_3 \wedge a_2).$$

(h) The complete circuit uses four D flip-flops for the state bits t_1, t_2, t_3, t_4, three input lines a_1, a_2, a_3, and four output networks w_1, w_2, w_3, w_4. The full transition/output table is exactly the one read from Figure 7.2: for $0 \le i \le 5$, pushing $\boldsymbol{b}$ leaves s_i fixed with output $\boldsymbol{\varphi}$, while $\boldsymbol{n}, \boldsymbol{d}, \boldsymbol{q}$ send s_i to $s_{i+1}, s_{i+2}, s_{i+5}$ with output $\boldsymbol{\varphi}$; for $6 \le i \le 10$, pushing $\boldsymbol{b}$ sends s_i to s_0 with output $\boldsymbol{c}_{i-6}$, while $\boldsymbol{n}, \boldsymbol{d}, \boldsymbol{q}$ self-loop with outputs $\boldsymbol{n'}, \boldsymbol{d'}, \boldsymbol{q'}$; and <*SOS*> sends every real state to s_0 with output $\boldsymbol{\varphi}$. The other next-state bits and output bits are obtained from this same table in exactly the same way.

7.22. **Circuit for modified vending machine (Exercise 7.14).**

(a) **EOS:** No.

(b) **Input encoding:** We again keep a reset code, so use three bits for $\{\boldsymbol{b}, \boldsymbol{d}, \boldsymbol{n}, \boldsymbol{q}, \boldsymbol{r}, \text{<}SOS\text{>}\}$:

$$\boldsymbol{b} = 000, \quad \boldsymbol{d} = 001, \quad \boldsymbol{n} = 010, \quad \boldsymbol{q} = 011, \quad \boldsymbol{r} = 100, \quad \text{<}SOS\text{>} = 111.$$

(c) **Output encoding:** There are now ten outputs $\{\boldsymbol{a}, \boldsymbol{c}_0, \boldsymbol{c}_1, \boldsymbol{c}_2, \boldsymbol{c}_3, \boldsymbol{c}_4, \boldsymbol{d'}, \boldsymbol{n'}, \boldsymbol{q'}, \boldsymbol{\varphi}\}$, so four bits suffice. One convenient code is

$$\boldsymbol{a} = 0000,\ \boldsymbol{c}_0 = 0001,\ \boldsymbol{c}_1 = 0010,\ \boldsymbol{c}_2 = 0011,\ \boldsymbol{c}_3 = 0100,\ \boldsymbol{c}_4 = 0101,\ \boldsymbol{d'} = 0110,\ \boldsymbol{n'} = 0111,\ \boldsymbol{q'} = 1000,\ \boldsymbol{\varphi} = 1001.$$

(d) **State encoding:** Keep the state encoding from Exercise 7.21: s_i means $5i$¢ for $0 \le i \le 10$, with $s_0 = 0000, \ldots, s_{10} = 1010$.

(e) t_3 **truth table and Boolean function:**

	$\boldsymbol{b}$	$\boldsymbol{d}$	$\boldsymbol{n}$	$\boldsymbol{q}$	$\boldsymbol{r}$	<*SOS*>
s_0	0	0	0	1	0	0
s_1	0	0	0	1	0	0
s_2	0	1	0	1	0	0
s_3	0	1	1	0	0	0
s_4	1	1	1	0	0	0
s_5	1	1	1	0	0	0
s_6	0	1	1	1	0	0
s_7	0	1	1	1	0	0
s_8	0	0	0	0	0	0
s_9	0	0	0	0	0	0
s_{10}	0	0	0	0	0	0

and one minimized form is

$$\begin{aligned} t_3' = {} & (\neg t_4 \wedge t_2 \wedge \neg t_1 \wedge \neg a_3 \wedge a_1) \vee (t_2 \wedge t_1 \wedge a_2 \wedge \neg a_1) \vee \\ & (t_3 \wedge a_2 \wedge \neg a_1) \vee (\neg t_4 \wedge t_2 \wedge \neg a_2 \wedge a_1) \vee (t_3 \wedge t_2 \wedge \neg a_3 \wedge a_2) \vee \\ & (t_3 \wedge \neg t_2 \wedge \neg a_3 \wedge \neg a_2) \vee (\neg t_4 \wedge \neg t_3 \wedge \neg t_2 \wedge \neg a_3 \wedge a_2 \wedge a_1). \end{aligned}$$

(f) w_3 **truth table and Boolean function:**

	$\boldsymbol{b}$	$\boldsymbol{d}$	$\boldsymbol{n}$	$\boldsymbol{q}$	$\boldsymbol{r}$	<SOS>
s_0	0	0	0	0	0	0
s_1	0	0	0	0	0	0
s_2	0	0	0	0	0	0
s_3	0	0	0	0	0	0
s_4	0	0	0	0	0	0
s_5	0	0	0	0	0	0
s_6	0	1	1	0	0	0
s_7	0	1	1	0	0	0
s_8	0	1	1	0	0	0
s_9	1	1	1	0	0	0
s_{10}	1	1	1	0	0	0

and a minimized form is

$$w_3 = (t_4 \wedge \neg a_2 \wedge a_1) \vee (t_4 \wedge t_1 \wedge \neg a_3 \wedge \neg a_1) \vee (t_3 \wedge t_2 \wedge \neg a_2 \wedge a_1) \vee$$
$$(t_3 \wedge t_2 \wedge a_2 \wedge \neg a_1) \vee (t_4 \wedge a_2 \wedge \neg a_1) \vee (t_4 \wedge t_2 \wedge \neg a_3 \wedge \neg a_1).$$

(g) The complete circuit is obtained from Exercise 7.21 by adding one new input column for $\boldsymbol{r}$ and changing the output table accordingly. Every real state goes to s_0 on $\boldsymbol{r}$ and produces $\boldsymbol{a}$; every real state goes to s_0 on <SOS> and produces $\boldsymbol{\varphi}$; and all $\boldsymbol{b}, \boldsymbol{d}, \boldsymbol{n}, \boldsymbol{q}$ entries are unchanged from Exercise 7.21. The other state and output bits are obtained from this same combined table.

7.23. **Circuit for Example 7.2.**

Let $a_1 = 0$ code $\boldsymbol{a}$ and $a_1 = 1$ code $\boldsymbol{b}$; let $t_1 = 0$ code s_0 and $t_1 = 1$ code s_1; and let $w_1 = 0, 1$ code the output symbols $\mathbf{0}, \mathbf{1}$. Then

t_1	a_1	t_1'
0	0	0
0	1	1
1	0	0
1	1	1

t_1	a_1	w_1
0	0	0
0	1	0
1	0	1
1	1	1

so the minimized equations are

$$t_1' = a_1, \qquad w_1 = t_1.$$

Thus the circuit uses one D flip-flop holding t_1, the input line a_1 feeding the D input, and the present-state line t_1 feeding the output line w_1.

7.24. **Circuit for Example 7.6.**

Encode the four states of Figure 7.4 by

$$t_0 = 00, \qquad t_1 = 01, \qquad t_2 = 10, \qquad t_3 = 11,$$

let $a_1 = 0$ code $\boldsymbol{a}$ and $a_1 = 1$ code $\boldsymbol{b}$, and let $w_1 = 0, 1$ code the output symbols $\mathbf{0}, \mathbf{1}$. To avoid confusion with the state names, write the two state bits as $u_2 u_1$.

The truth tables are

u_2	u_1	a_1	u_2'	u_1'
0	0	0	0	1
0	0	1	0	0
0	1	0	1	0
0	1	1	0	0
1	0	0	1	0
1	0	1	1	1
1	1	0	0	1
1	1	1	0	0

u_2	u_1	a_1	w_1
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	0

and the minimized equations are

$$u_2' = (u_2 \wedge \neg u_1) \ \vee \ (\neg u_2 \wedge u_1 \wedge \neg a_1),$$

$$u_1' = (\neg u_2 \wedge \neg u_1 \wedge \neg a_1) \ \vee \ (u_2 \wedge \neg u_1 \wedge a_1) \ \vee \ (u_2 \wedge u_1 \wedge \neg a_1),$$

$$w_1 = u_2 \wedge \neg u_1 \wedge a_1.$$

So the circuit has two D flip-flops holding u_2, u_1, a combinational network for u_2', u_1', and one output gate computing w_1.

7.25. **Circuit for FST D (Example 7.8).**

Encode the states of Figure 7.5 by

$$q_0 = 00, \qquad q_1 = 01, \qquad q_2 = 10,$$

leaving 11 as a don't-care code. Let $a_1 = 0$ code $\boldsymbol{a}$ and $a_1 = 1$ code $\boldsymbol{b}$, and let $u_2 u_1$ again denote the state bits. Then

u_2	u_1	a_1	u_2'	u_1'
0	0	0	0	1
0	0	1	0	0
0	1	0	1	0
0	1	1	0	0
1	0	0	1	0
1	0	1	0	0

u_2	u_1	a_1	w_1
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1

with minimized equations

$$u_2' = \neg a_1 \wedge (u_1 \vee u_2), \qquad u_1' = \neg u_2 \wedge \neg u_1 \wedge \neg a_1, \qquad w_1 = u_2 \wedge a_1.$$

Thus the circuit again has two D flip-flops and one output gate.

7.26. **Connected is not sufficient for minimality.**

Example. Let M have states $\{s_0, s_1\}$, $\Sigma = \{\boldsymbol{a}\}$, $\Gamma = \{\mathbf{0}, \mathbf{1}\}$: $\delta(s_0, \boldsymbol{a}) = s_1$, $\delta(s_1, \boldsymbol{a}) = s_0$; $\omega(s_0, \boldsymbol{a}) = \mathbf{0}$, $\omega(s_1, \boldsymbol{a}) = \mathbf{0}$. Both states output $\mathbf{0}$ on $\boldsymbol{a}$ and alternate, so $s_0 E_M s_1$. M is connected (both states reachable from s_0) but not reduced (two equivalent states). Hence it is not minimal.

7.27. **Two-unit delay Mealy and Moore machines.**

$\overline{\omega}(x_1 \cdots x_m) = \boldsymbol{x}\boldsymbol{x} x_1 \cdots x_{m-2}$.

(a) **Mealy sextuple:**
Let

$$M = \langle\{\boldsymbol{a},\boldsymbol{b}\},\{\boldsymbol{a},\boldsymbol{b},\boldsymbol{x}\},S,s_{xx},\delta,\omega\rangle,$$

where

$$S = \{s_{xx}, s_{xa}, s_{xb}, s_{aa}, s_{ab}, s_{ba}, s_{bb}\}.$$

The state remembers the last two input symbols, using $\boldsymbol{x}$ as a padding symbol before enough input has been seen. The transition function is

$$\delta(s_{uv},\sigma) = s_{v\sigma},$$

where $u, v \in \{\boldsymbol{x},\boldsymbol{a},\boldsymbol{b}\}$ and the only states of this form that occur are the seven listed above. The output function is

$$\omega(s_{uv},\sigma) = u.$$

Thus the machine prints the symbol seen two steps earlier; from the start state s_{xx} the first two outputs are $\boldsymbol{x}$.

(b) **Mealy machine diagram:**
The state transition diagram has the seven states above. Every arrow into $s_{v\sigma}$ is labeled σ/u, where u is the first component of the source state s_{uv}. Concretely,

$$s_{xx} \xrightarrow{\boldsymbol{a}/\boldsymbol{x}} s_{xa}, \qquad s_{xx} \xrightarrow{\boldsymbol{b}/\boldsymbol{x}} s_{xb},$$

$$s_{xa} \xrightarrow{\boldsymbol{a}/\boldsymbol{x}} s_{aa}, \qquad s_{xa} \xrightarrow{\boldsymbol{b}/\boldsymbol{x}} s_{ab},$$

$$s_{xb} \xrightarrow{\boldsymbol{a}/\boldsymbol{x}} s_{ba}, \qquad s_{xb} \xrightarrow{\boldsymbol{b}/\boldsymbol{x}} s_{bb},$$

and from each state s_{uv} with $u, v \in \{\boldsymbol{a},\boldsymbol{b}\}$,

$$s_{uv} \xrightarrow{\boldsymbol{a}/u} s_{va}, \qquad s_{uv} \xrightarrow{\boldsymbol{b}/u} s_{vb}.$$

(c) **Moore sextuple:**
Let

$$A = \langle\{\boldsymbol{a},\boldsymbol{b}\},\{\boldsymbol{a},\boldsymbol{b},\boldsymbol{x}\},S,s_{xx},\delta,\eta\rangle,$$

where S and δ are exactly as above, and the Moore output function is

$$\eta(s_{uv}) = u.$$

Since the destination state records the two most recent input symbols, its first component is precisely the letter seen two steps earlier.

(d) **Moore machine diagram:**
Use the same transition graph as in part (b), but label each state by its Moore output:

$$s_{xx}/\boldsymbol{x}, \qquad s_{xa}/\boldsymbol{x}, \qquad s_{xb}/\boldsymbol{x},$$

$$s_{aa}/\boldsymbol{a}, \qquad s_{ab}/\boldsymbol{a}, \qquad s_{ba}/\boldsymbol{b}, \qquad s_{bb}/\boldsymbol{b}.$$

7.28. (a) Take $M_1 = C$ from Example 7.6 and $M_2 = D$ from Example 7.8. These two transducers are equivalent, but C has four states while D has only three, so they cannot be isomorphic. Thus the conclusion of Theorem 7.5 fails if M_1 is not reduced.

(b) The missing property is that the proposed map μ need not be one-to-one.

7.29. (a) Let $M_2 = D$ from Example 7.8. Let M_1 be obtained from D by adding one extra unreachable state r with the same outgoing arrows as q_0:

$$\delta(r, \boldsymbol{a}) = q_1, \qquad \delta(r, \boldsymbol{b}) = r, \qquad \omega(r, \boldsymbol{a}) = \omega(r, \boldsymbol{b}) = \mathbf{0}.$$

Then M_1 and M_2 are equivalent, but they are not isomorphic because M_1 has an extra unreachable state.

(b) The missing property is that μ need not even be well defined on all of S_1, since the definition of $\mu(s)$ requires a string reaching s from the start state.

7.30. (a) Again take $M_1 = D$ from Example 7.8 and $M_2 = C$ from Example 7.6. They are equivalent, but C is not reduced and has one extra state, so the machines are not isomorphic.

(b) The missing property is that μ need not be onto S_2.

7.31. (a) Let $M_1 = D$ from Example 7.8, and let M_2 be obtained from D by adjoining one extra unreachable state r with

$$\delta(r, \boldsymbol{a}) = q_1, \qquad \delta(r, \boldsymbol{b}) = r, \qquad \omega(r, \boldsymbol{a}) = \omega(r, \boldsymbol{b}) = \mathbf{0}.$$

Then $M_1 \equiv M_2$, but the two machines are not isomorphic because M_2 has an unreachable state that M_1 does not have.

(b) The missing property is surjectivity of μ onto S_2.

7.32. (a) Let $A = C$ from Example 7.6. The machine C is connected, so $A^c = A$. But C is not reduced, since Example 7.8 shows that it has an equivalent three-state quotient. Hence A is not reduced and A^c is not reduced.

(b) Let A be obtained from D in Example 7.8 by adjoining one extra unreachable state r satisfying

$$\delta(r, \boldsymbol{a}) = q_1, \qquad \delta(r, \boldsymbol{b}) = r, \qquad \omega(r, \boldsymbol{a}) = \omega(r, \boldsymbol{b}) = \mathbf{0}.$$

Then r is equivalent to q_0, so A is not reduced. However, the connected part A^c throws away r and is exactly D, which is reduced.

7.33. **Complete proof of Theorem 7.4.**

(a) $\overline{\omega}(t, y) = \overline{\omega}_{E_M}([t]_{E_M}, y)$:

Proof. By induction on $|y|$.
Base: If $y = \lambda$, then both sides equal λ.
Step: Let $y = \boldsymbol{a}x$. Then

$$\overline{\omega}_{E_M}([t]_{E_M}, \boldsymbol{a}x) = \omega_{E_M}([t]_{E_M}, \boldsymbol{a}) \cdot \overline{\omega}_{E_M}(\delta_{E_M}([t]_{E_M}, \boldsymbol{a}), x).$$

By the definitions of ω_{E_M} and δ_{E_M}, this is

$$\omega(t, \boldsymbol{a}) \cdot \overline{\omega}_{E_M}([\delta(t, \boldsymbol{a})]_{E_M}, x).$$

Apply the inductive hypothesis to the state $\delta(t,\boldsymbol{a})$:

$$\overline{\omega}_{E_M}([t]_{E_M},\boldsymbol{a}x) = \omega(t,\boldsymbol{a})\cdot\overline{\omega}(\delta(t,\boldsymbol{a}),x) = \overline{\omega}(t,\boldsymbol{a}x).$$

□

(b) $M/_{E_M} \equiv M$:

Proof. Part (a) with $t = s_0$ gives

$$\overline{\omega}_{E_M}([s_0]_{E_M}, y) = \overline{\omega}(s_0, y)$$

for every input string y. Since $[s_0]_{E_M}$ is the start state of $M/_{E_M}$, the two machines define the same translation. □

(c) $M/_{E_M}$ is reduced:

Proof. Suppose $[s]_{E_M}$ and $[t]_{E_M}$ are equivalent states of $M/_{E_M}$. Then for every $y \in \Sigma^*$,

$$\overline{\omega}_{E_M}([s]_{E_M}, y) = \overline{\omega}_{E_M}([t]_{E_M}, y).$$

By part (a), this implies

$$\overline{\omega}(s,y) = \overline{\omega}(t,y)$$

for every y. Hence $sE_M t$, so $[s]_{E_M} = [t]_{E_M}$. Therefore $M/_{E_M}$ is reduced. □

(d) If M connected, $M/_{E_M}$ connected:

Proof. Let $[t]_{E_M}$ be any state of the quotient. Since M is connected, there exists $x \in \Sigma^*$ with $\overline{\delta}(s_0, x) = t$. We claim that

$$\overline{\delta}_{E_M}([s_0]_{E_M}, x) = [t]_{E_M}.$$

This follows by induction on $|x|$ from the definition of δ_{E_M}. Hence every quotient state is reachable from the quotient start state, so the quotient is connected. □

7.34. (a) f_1 **is FTD:** Define Mealy machine M with states {read1, read0}, initial state start (extra state for first letter): From start: on $\mathbf{1}$, go to read1 with output $\boldsymbol{y}$; on $\mathbf{0}$, go to read0 with output $\boldsymbol{n}$. From read1: on any input, stay in read1, output $\boldsymbol{y}$. From read0: on any input, stay in read0, output $\boldsymbol{n}$. Then $\overline{\omega}(s_{\text{start}}, \boldsymbol{a}_1\cdots\boldsymbol{a}_m) = \boldsymbol{y}^m$ if $\boldsymbol{a}_1 = \mathbf{1}$, $\boldsymbol{n}^m$ otherwise. Hence f_1 is FTD.

(b) f_2 **is not FTD:** By the pumping lemma for FTD functions (Exercise 7.5): for any n and $x = \mathbf{0}^n\mathbf{1}$ (length $n+1 > n$), if f_2 were FTD with n states, we pump the $\mathbf{0}$ block. $f_2(\mathbf{0}^n\mathbf{1}) = \boldsymbol{nn}\cdots\boldsymbol{ny}$ (n $\boldsymbol{n}$s and one $\boldsymbol{y}$); but $f_2(\mathbf{0}^{n+k}\mathbf{1}) = \boldsymbol{n}^{n+k}\boldsymbol{y}$ for any k. The last letter of the output changes from $\boldsymbol{n}$ to $\boldsymbol{y}$ depending on the last input letter, which requires the machine to "remember" the entire input—impossible with finite memory. Formally: any two words differing only in the last letter (one ending in $\mathbf{0}$, one in $\mathbf{1}$) must have different output strings, but for any n-state machine, words of length $> n$ that agree on a prefix must produce the same output on that prefix; however, $f_2(\mathbf{0}^m\mathbf{1})$ and $f_2(\mathbf{0}^m\mathbf{0})$ disagree on the last output character. This means the machine must distinguish $m+1$ different lengths of leading zeros, requiring unbounded state space. □

7.35. f_3 **is not FTD.**

f_3 maps each input to the first $|x|$ characters of the fixed infinite sequence **010010001...**. The number of **0**s between successive **1**s grows without bound. For any n-state Mealy machine, the output sequence must eventually be periodic with period at most n. But the gaps between **1**s in the infinite sequence grow arbitrarily, so f_3 is not periodic—hence not implementable by a finite machine. More precisely: if f_3 were FTD, by the pumping lemma, processing a word of length $> n$ would pump a cycle; but cycling would repeat a finite gap between **1**s, preventing the ever-growing gaps. Contradiction. $\square$

7.36. **Prove: for FTD f, the first n letters of $f(xy)$ equal those of $f(xz)$.**

Proof. Let M be a Mealy machine for f with states S. For any $x \in \Sigma^n$: $\overline{\delta}(s_0, x)$ is some state s. Then $f(xy) = \overline{\omega}(s_0, xy) = \overline{\omega}(s_0, x) \cdot \overline{\omega}(s, y)$ and $f(xz) = \overline{\omega}(s_0, xz) = \overline{\omega}(s_0, x) \cdot \overline{\omega}(s, z)$. The first n letters of both are $\overline{\omega}(s_0, x)$ (since $|x| = n$ and the first n output letters come from processing x). Hence they agree on the first n letters. $\square$ $\square$

7.37. **Elevator transducer.**

(a) **Input alphabet:** Since several buttons may be active simultaneously, use

$$\Sigma = \{0,1\}^4,$$

with an input written $\langle u, d, c_1, c_2 \rangle$. Here u is the hall call "up from floor 1," d is the hall call "down from floor 2," and c_1, c_2 are the car buttons for floors 1 and 2.

(b) **Output alphabet:**

$$\Gamma = \{\text{close}, \text{open}, \text{goto1}, \text{goto2}\}.$$

(c) **Mealy sextuple:** Let

$$M = \langle \Sigma, \Gamma, S, (1, \text{closed}, \varnothing), \delta, \omega \rangle,$$

where

$$S = \{(f, m, P) \mid f \in \{1,2\},\ m \in \{\text{open}, \text{closed}\},\ P \subseteq \{1,2\}\}.$$

Here f is the current floor, m is the door status, and P is the set of pending requests. For an input $x = \langle u, d, c_1, c_2 \rangle$, define

$$R(x) = \{1 \mid u = 1 \text{ or } c_1 = 1\} \cup \{2 \mid d = 1 \text{ or } c_2 = 1\},$$

and let $P^* = P \cup R(x)$. Then:

(i) if $m = \text{open}$, then

$$\delta((f, \text{open}, P), x) = (f, \text{closed}, P^*), \qquad \omega((f, \text{open}, P), x) = \text{close};$$

(ii) if $m = \text{closed}$ and $f \in P^*$, then

$$\delta((f, \text{closed}, P), x) = (f, \text{open}, P^* \setminus \{f\}), \qquad \omega((f, \text{closed}, P), x) = \text{open};$$

(iii) if $m = \text{closed}$, $f = 1$, and $2 \in P^*$, then

$$\delta((1, \text{closed}, P), x) = (2, \text{closed}, P^* \setminus \{2\}), \qquad \omega((1, \text{closed}, P), x) = \text{goto2};$$

(iv) if $m = \text{closed}$, $f = 2$, and $1 \in P^*$, then

$$\delta((2, \text{closed}, P), x) = (1, \text{closed}, P^* \setminus \{1\}), \qquad \omega((2, \text{closed}, P), x) = \text{goto1};$$

(v) if $m = \text{closed}$ and $P^* = \emptyset$, then

$$\delta((f, \text{closed}, \emptyset), x) = (f, \text{closed}, \emptyset), \qquad \omega((f, \text{closed}, \emptyset), x) = \text{close}.$$

(d) **Mealy machine diagram:** The diagram is exactly the directed graph of the sextuple in part (c): each state is a triple (f, m, P), and each edge is labeled by the corresponding input/output pair from the five cases above.

(e) **Moore sextuple:** A Moore version can use

$$S' = \{C_{f,P}, O_{f,P} \mid f \in \{1,2\},\ P \subseteq \{1,2\}\} \cup \{U_P, D_P \mid P \subseteq \{1,2\}\},$$

where $C_{f,P}$ means "at floor f, doors closed, pending set P," $O_{f,P}$ means "doors open," U_P means "travelling to floor 2," and D_P means "travelling to floor 1." The output function is

$$\omega(C_{f,P}) = \text{close}, \qquad \omega(O_{f,P}) = \text{open}, \qquad \omega(U_P) = \text{goto2}, \qquad \omega(D_P) = \text{goto1}.$$

The transition function is the Moore analogue of part (c):

$$\delta(O_{f,P}, x) = C_{f,P \cup R(x)},$$

$$\delta(C_{1,P}, x) = \begin{cases} O_{1,P^* \setminus \{1\}} & \text{if } 1 \in P^*, \\ U_{P^*} & \text{if } 1 \not{\imath}nP^* \text{ and } 2 \in P^*, \\ C_{1,P^*} & \text{if } P^* = \emptyset, \end{cases}$$

$$\delta(C_{2,P}, x) = \begin{cases} O_{2,P^* \setminus \{2\}} & \text{if } 2 \in P^*, \\ D_{P^*} & \text{if } 2 \not{\imath}nP^* \text{ and } 1 \in P^*, \\ C_{2,P^*} & \text{if } P^* = \emptyset, \end{cases}$$

$$\delta(U_P, x) = C_{2,(P \cup R(x)) \setminus \{2\}}, \qquad \delta(D_P, x) = C_{1,(P \cup R(x)) \setminus \{1\}}.$$

(f) **Moore machine diagram:** The diagram is the directed graph of the Moore sextuple in part (e).

(g) **Circuit:** The Mealy machine in part (c) has 16 states, so four state bits suffice. The input is already given as four bits $\langle u, d, c_1, c_2 \rangle$, and the outputs can be encoded by two bits, for example

$$\text{close} = 00, \qquad \text{open} = 01, \qquad \text{goto1} = 10, \qquad \text{goto2} = 11.$$

The required circuit is therefore the standard four-flip-flop sequential network implementing the transition and output rules from part (c).

7.38. **Mealy machine for traffic signal (Example 7.16).**

Let

$$\Sigma = \{\langle 0,0 \rangle, \langle 0,1 \rangle, \langle 1,0 \rangle, \langle 1,1 \rangle\}$$

on the sensor pair $\langle \alpha, \beta \rangle$. A correct Mealy machine is obtained from the Moore machine of Figure 7.15 by placing the output of the destination state on each edge. Use the state set $\{s_0, s_1, s_2, s_3, s_4, s_5, s_6, s_7\}$ with

$$\begin{aligned} o(s_0) &= \langle G, R, G, R \rangle, & o(s_1) &= \langle Y, R, Y, R \rangle, \\ o(s_2) &= \langle R, R, R, G \rangle, & o(s_3) &= \langle G, R, Y, R \rangle, \\ o(s_4) &= \langle R, R, R, Y \rangle, & o(s_5) &= \langle G, Y, R, R \rangle, \\ o(s_6) &= \langle G, G, R, R \rangle, & o(s_7) &= \langle Y, Y, R, R \rangle. \end{aligned}$$

The transition table is

	$\langle 0,0 \rangle$	$\langle 0,1 \rangle$	$\langle 1,0 \rangle$	$\langle 1,1 \rangle$
s_0	s_0	s_1	s_3	s_1
s_1	s_2	s_2	s_2	s_2
s_2	s_4	s_2	s_4	s_2
s_3	s_6	s_6	s_6	s_6
s_4	s_0	s_0	s_6	s_6
s_5	s_0	s_0	s_0	s_0
s_6	s_5	s_7	s_6	s_6
s_7	s_2	s_2	s_2	s_2

and the Mealy output on any transition (s, x) is

$$\omega(s, x) = o(\delta(s, x)).$$

7.39. **Traffic signal with walk signals.**

(a) The new alphabets are

$$\Sigma = \{0, 1\}^3$$

on $\langle \alpha, \beta, \gamma \rangle$, and

$$\Gamma = \{R, Y, G\}^4 \times \{D, W\}.$$

(b) **Moore machine:** Use the states

$$s_0,\ s_1^D,\ s_1^W,\ s_2^D,\ s_2^W,\ s_3,\ s_4,\ s_5,\ s_6,\ s_7,$$

with outputs

$$\begin{aligned} \omega(s_0) &= \langle G, R, G, R, D \rangle, & \omega(s_1^D) &= \langle Y, R, Y, R, D \rangle, \\ \omega(s_1^W) &= \langle Y, R, Y, R, D \rangle, & \omega(s_2^D) &= \langle R, R, R, G, D \rangle, \\ \omega(s_2^W) &= \langle R, R, R, G, W \rangle, & \omega(s_3) &= \langle G, R, Y, R, D \rangle, \\ \omega(s_4) &= \langle R, R, R, Y, D \rangle, & \omega(s_5) &= \langle G, Y, R, R, D \rangle, \\ \omega(s_6) &= \langle G, G, R, R, D \rangle, & \omega(s_7) &= \langle Y, Y, R, R, D \rangle. \end{aligned}$$

The transition policy is:

(i) from s_0, stay in s_0 on 000; go to s_3 on 100; and whenever $\beta = 1$ or $\gamma = 1$, go to s_1^W if $\gamma = 1$ and to s_1^D otherwise;

(ii) s_1^D goes to s_2^D on every input, and s_1^W goes to s_2^W on every input;

(iii) s_2^W goes to s_2^D on every input, so the walk signal reverts to D before the side-street light turns yellow;

(iv) s_2^D stays in s_2^D while $\beta = 1$ and goes to s_4 when $\beta = 0$;

(v) s_4 goes to s_6 if $\alpha = 1$ and to s_0 otherwise;

(vi) s_3 goes to s_6 on every input;

(vii) s_6 stays in s_6 while $\alpha = 1$; if $\alpha = 0$ and either $\beta = 1$ or $\gamma = 1$, it goes to s_7; otherwise it goes to s_5;

(viii) s_5 goes to s_0 on every input, and s_7 goes to s_2^W when $\gamma = 1$ and to s_2^D otherwise.

(c) **Mealy machine:** Use the same directed graph as in part (b), but place on each edge the output of the destination state. If $o(t)$ denotes the five-component output attached to the Moore state t, then

$$\omega(s,x) = o(\delta(s,x)).$$

7.40. **Intersection with left-turn signals.**

Let

$$\Sigma = \{0,1\}^4$$

on $\langle \alpha, \beta, \gamma, \delta \rangle$, and let

$$\Gamma = \{R, Y, G\}^8$$

with coordinates ordered by signals 1, 2, 3, 4, 5, 6, 7, 8 in Figure 7.20.

A convenient Moore controller uses the twelve phase states

$$EW_G,\ EW_Y,\ NS_G,\ NS_Y,\ WB_G,\ WB_Y,\ NB_G,\ NB_Y,\ EB_G,\ EB_Y,\ SB_G,\ SB_Y,$$

with outputs

$$\begin{aligned}
EW_G &= \langle G,R,G,R,R,R,R,R\rangle, & EW_Y &= \langle Y,R,Y,R,R,R,R,R\rangle,\\
NS_G &= \langle R,G,R,G,R,R,R,R\rangle, & NS_Y &= \langle R,Y,R,Y,R,R,R,R\rangle,\\
WB_G &= \langle G,R,R,R,G,R,R,R\rangle, & WB_Y &= \langle Y,R,R,R,Y,R,R,R\rangle,\\
NB_G &= \langle R,G,R,R,R,G,R,R\rangle, & NB_Y &= \langle R,Y,R,R,R,Y,R,R\rangle,\\
EB_G &= \langle R,R,G,R,R,R,G,R\rangle, & EB_Y &= \langle R,R,Y,R,R,R,Y,R\rangle,\\
SB_G &= \langle R,R,R,G,R,R,R,G\rangle, & SB_Y &= \langle R,R,R,Y,R,R,R,Y\rangle.
\end{aligned}$$

One explicit transition policy is:

(i) start in EW_G; remain there while no sensor is active, and move to EW_Y as soon as some request appears;

(ii) from EW_Y, give priority to the two sensor-controlled lanes on the north-south side: go to NB_G if $\beta = 1$, else to SB_G if $\delta = 1$, else to NS_G;

(iii) NB_G stays in NB_G while $\beta = 1$, then goes to NB_Y, and NB_Y goes to SB_G if $\delta = 1$ and otherwise to NS_G;

(iv) SB_G stays in SB_G while $\delta = 1$, then goes to SB_Y, and SB_Y goes to NS_G;

(v) NS_G remains in NS_G while $\alpha = \gamma = 0$, and moves to NS_Y once some east-west left-turn request appears;

(vi) from NS_Y, give priority first to α and then to γ: go to WB_G if $\alpha = 1$, else to EB_G if $\gamma = 1$, else to EW_G;

(vii) WB_G stays in WB_G while $\alpha = 1$, then goes to WB_Y, and WB_Y goes to EB_G if $\gamma = 1$ and otherwise to EW_G;

(viii) EB_G stays in EB_G while $\gamma = 1$, then goes to EB_Y, and EB_Y goes to EW_G.

This gives a complete Moore machine. The corresponding Mealy machine uses the same graph and places the output of the destination phase on each transition.

7.41. **Base-3 adder.**

(a) **Input alphabet:**
$$\Sigma = \{0,1,2\} \times \{0,1,2\}.$$

(b) **Output alphabet:**
$$\Gamma = \{0,1,2\}.$$

(c) **Mealy machine:** Let
$$M = \langle \Sigma, \Gamma, \{C_0, C_1\}, C_0, \delta, \omega \rangle,$$
where C_0 means "no carry" and C_1 means "carry 1." For $\langle a, b\rangle \in \Sigma$, define
$$\delta(C_i, \langle a, b\rangle) = \begin{cases} C_1 & \text{if } a+b+i \geq 3, \\ C_0 & \text{if } a+b+i \leq 2, \end{cases}$$
and
$$\omega(C_i, \langle a, b\rangle) \equiv a+b+i \pmod 3.$$
This is exactly the base-3 analogue of the binary adder in Example 7.15.

(d) **Moore machine:** A corresponding Moore machine is
$$A = \langle \Sigma, \Gamma, S, q_*, \delta', \eta \rangle,$$
where
$$S = \{q_*\} \cup \{q_{ij} \mid i \in \{0,1\},\ j \in \{0,1,2\}\},$$
the output function is
$$\eta(q_*) = 0, \qquad \eta(q_{ij}) = j,$$
and the transition function is
$$\delta'(q_*, \langle a, b\rangle) = q_{c,s},$$
$$\delta'(q_{ij}, \langle a, b\rangle) = q_{c',s'},$$
where
$$s \equiv a+b \pmod 3, \qquad c = \begin{cases} 1 & \text{if } a+b \geq 3, \\ 0 & \text{if } a+b \leq 2, \end{cases}$$
and
$$s' \equiv a+b+i \pmod 3, \qquad c' = \begin{cases} 1 & \text{if } a+b+i \geq 3, \\ 0 & \text{if } a+b+i \leq 2. \end{cases}$$
Thus the Moore state records both the current carry and the digit that is to be printed.

(e) **Circuit with <EOS> and <SOS>:** Encode each trit by two bits, for example 0 = 00, 1 = 01, 2 = 10, and encode a column $\langle a, b\rangle$ by the four bits of the two trits. Reserve extra input codes for $<SOS>$ and $<EOS>$. One flip-flop stores the carry bit c. The $<SOS>$ input forces $c = 0$. On an ordinary input column $\langle a, b\rangle$, the combinational network computes

$$s \equiv a + b + c \pmod 3, \qquad c' = \begin{cases} 1 & \text{if } a + b + c \geq 3, \\ 0 & \text{if } a + b + c \leq 2, \end{cases}$$

and sends the two-bit code of s to the output device. On $<EOS>$, the circuit outputs the final carry digit: print 1 if $c = 1$ and 0 if $c = 0$, then reset the flip-flop to 0. This is the required circuit implementation of the Mealy machine in part (c).

7.42. **Base-10 adder.**

(a) **Input and output alphabets:**

$$\Sigma = \{0, 1, \ldots, 9\} \times \{0, 1, \ldots, 9\}, \qquad \Gamma = \{0, 1, \ldots, 9\}.$$

As in Example 7.15, one input symbol is a column of two digits, and the transducer reads the columns from right to left.

(b) **Mealy sextuple:**
Let

$$M = \langle \Sigma, \Gamma, \{N, C\}, N, \delta, \omega\rangle,$$

where N means "no carry" and C means "carry." For $(i, j) \in \Sigma$, define

$$\omega(N, (i, j)) = (i + j) \bmod 10, \qquad \delta(N, (i, j)) = \begin{cases} C & \text{if } i + j \geq 10, \\ N & \text{if } i + j < 10, \end{cases}$$

and

$$\omega(C, (i, j)) = (i + j + 1) \bmod 10, \qquad \delta(C, (i, j)) = \begin{cases} C & \text{if } i + j + 1 \geq 10, \\ N & \text{if } i + j + 1 < 10. \end{cases}$$

(c) **Moore sextuple:**
A convenient Moore version uses one start state s together with states

$$q_{c,d} \qquad (c \in \{0, 1\},\ d \in \{0, 1, \ldots, 9\}),$$

where c records the carry to be used on the next column and d is the most recently produced output digit. Let

$$A = \langle \Sigma, \Gamma, \{s\} \cup \{q_{c,d} \mid c \in \{0, 1\}, 0 \leq d \leq 9\}, s, \delta', \eta\rangle.$$

The output function is

$$\eta(q_{c,d}) = d,$$

while the value of $\eta(s)$ is irrelevant because the start state's output is not used on empty input. For $(i, j) \in \Sigma$, define

$$\delta'(s, (i, j)) = q_{c',d'},$$

where

$$d' = (i + j) \bmod 10, \qquad c' = \begin{cases} 1 & \text{if } i + j \geq 10, \\ 0 & \text{if } i + j < 10, \end{cases}$$

and in general

$$\delta'(q_{c,d}, (i, j)) = q_{c',d'},$$

where

$$d' = (i + j + c) \bmod 10, \qquad c' = \begin{cases} 1 & \text{if } i + j + c \geq 10, \\ 0 & \text{if } i + j + c < 10. \end{cases}$$

(d) **Circuit:**

Using <EOS> and <SOS>, encode each decimal digit in BCD and use one extra carry flip-flop. The combinational part computes

$$s = i + j + c,$$

outputs the BCD code for $s \bmod 10$, and feeds back the carry bit indicating whether $s \geq 10$. The <SOS> input initializes the carry flip-flop to 0, and the <EOS> input can be used to report whether a final overflow carry is present.

7.43. **Left-to-right addition is not FTD.**

Assume for contradiction that the left-to-right addition function f_4 is FTD. Then Exercise 7.36 applies: for every n, whenever two inputs agree on their first n symbols, the first n output symbols must also agree.

Fix any $n \geq 1$. Consider the following two length-$(n+1)$ input strings of column pairs:

$$x = \langle 0,0\rangle\langle 0,1\rangle^{n-1}\langle 1,1\rangle \qquad \text{and} \qquad y = \langle 0,0\rangle\langle 0,1\rangle^{n-1}\langle 0,1\rangle.$$

These two inputs agree on their first n symbols. However, their first output symbols are different when the addition is carried out correctly from right to left:

- For x, the last column $\langle 1,1\rangle$ generates a carry. Each middle column $\langle 0,1\rangle$ propagates that carry, so a carry reaches the first column $\langle 0,0\rangle$. Hence the first output bit is **1**.
- For y, the last column $\langle 0,1\rangle$ generates no carry, so no carry reaches the first column. Hence the first output bit is **0**.

Thus the first output symbols of $f_4(x)$ and $f_4(y)$ differ, even though the first n input symbols of x and y are identical. This contradicts Exercise 7.36. Therefore f_4 is not FTD.

7.44. **Prove Theorem 7.9 for MSM.**

(a) $M/_{E_M} \equiv M$:

Proof. As in the Mealy case, one first proves by induction on $|y|$ that for every state t of M,

$$\overline{\omega}_M(t, y) = \overline{\omega}_{E_M}([t]_{E_M}, y).$$

The base case is immediate since both sides are λ on empty input. For the inductive step, use the Moore-machine definition of $\overline{\omega}$, together with the facts that

$$\delta_{E_M}([s]_{E_M}, \boldsymbol{a}) = [\delta(s, \boldsymbol{a})]_{E_M} \quad \text{and} \quad \omega_{E_M}([s]_{E_M}) = \omega(s).$$

Taking $t = s_0$ yields $f_{M/_{E_M}} = f_M$, so the quotient is equivalent to M. $\square$

(b) $M/_{E_M}$ is reduced:

Proof. Suppose $[s]_{E_M}$ and $[t]_{E_M}$ are equivalent states of the quotient. Then

$$\overline{\omega}_{E_M}([s]_{E_M}, y) = \overline{\omega}_{E_M}([t]_{E_M}, y)$$

for all $y \in \Sigma^*$. By part (a), this implies $\overline{\omega}_M(s, y) = \overline{\omega}_M(t, y)$ for all y, so $sE_M t$. Therefore $[s]_{E_M} = [t]_{E_M}$, and the quotient is reduced. □

(c) If M connected, $M/_{E_M}$ connected:

Proof. Let $[t]_{E_M}$ be any state of the quotient. Since M is connected, there exists $x \in \Sigma^*$ with $\overline{\delta}_M(s_0, x) = t$. By induction on $|x|$,

$$\overline{\delta}_{E_M}([s_0]_{E_M}, x) = [\overline{\delta}_M(s_0, x)]_{E_M} = [t]_{E_M}.$$

Thus every quotient state is reachable, so the quotient is connected. □

7.45. **Minimal Moore machine is reduced.**

Proof. Suppose the minimal Moore machine M for f is not reduced. Then two states $s \neq t$ in M have $sE_M t$. Form M' by merging $[s] = [t]$ in the quotient $M/_{E_M}$; this gives an equivalent machine with fewer states. But M was defined as minimal—contradiction. Hence M is reduced. □ □

7.46. **Theorem 7.10 for MSM.**

(a) M minimal iff M is reduced and connected:

Proof. If M is minimal, then it must be connected, since unreachable states can be deleted without changing the translation. It must also be reduced by Theorem 7.9(d). Hence minimality implies reduced and connected.

Conversely, assume M is reduced and connected. Let B be any Moore machine equivalent to M. Replace B by $B^c/_{E_{B^c}}$ if necessary. By Theorem 7.9(a)–(c), this replacement is equivalent to B, reduced, and connected, and it has no more states than B. So it is enough to prove that every reduced connected Moore machine equivalent to M has at least as many states as M.

Let B now be reduced, connected, and equivalent to M. Since M is connected, for each state $s \in S_M$ choose a string $x_s \in \Sigma^*$ such that

$$\overline{\delta}_M(s_{0_M}, x_s) = s.$$

Define

$$\mu(s) = \overline{\delta}_B(s_{0_B}, x_s).$$

We show that μ is well defined and one-to-one.

First, suppose x_s and y_s both reach the same state s in M. For any $z \in \Sigma^*$, Theorem 7.6 gives

$$\overline{\omega}_M(s, z) = \overline{\omega}_B(\overline{\delta}_B(s_{0_B}, x_s), z)$$

and also

$$\overline{\omega}_M(s, z) = \overline{\omega}_B(\overline{\delta}_B(s_{0_B}, y_s), z),$$

because M and B are equivalent and both starts have processed the same prefixes x_s and y_s. Thus the two states of B reached by x_s and y_s are equivalent. Since B is reduced, they are equal. Hence μ is well defined.
Next, suppose $\mu(s) = \mu(t)$. Then for every $z \in \Sigma^*$,

$$\overline{\omega}_M(s,z) = \overline{\omega}_B(\mu(s),z) = \overline{\omega}_B(\mu(t),z) = \overline{\omega}_M(t,z),$$

so $sE_M t$. Because M is reduced, $s = t$. Therefore μ is injective.
Since μ is an injective map from S_M into S_B,

$$\|S_M\| \le \|S_B\|.$$

Hence no equivalent Moore machine can have fewer states than M, and M is minimal. □

(b) $M^c/_{E_{M^c}}$ is minimal:

Proof. By Theorem 7.9, the quotient $M^c/_{E_{M^c}}$ is equivalent to M^c, reduced, and connected. Since M^c is equivalent to M, the quotient is also equivalent to M. Part (a) now implies that $M^c/_{E_{M^c}}$ is minimal. □

7.47. **Prove Lemma 7.5 and Corollary 7.9 for MSMs.**

(a) $(\forall s \in S_A)(\forall x \in \Sigma^*)(\mu(\overline{\delta}_A(s,x)) = \overline{\delta}_B(\mu(s),x))$.

Proof. By induction on $|x|$.
Base: If $x = \lambda$, then $\mu(\overline{\delta}_A(s,\lambda)) = \mu(s) = \overline{\delta}_B(\mu(s),\lambda)$.
Step: Let $x = y\boldsymbol{a}$. Then

$$\mu(\overline{\delta}_A(s,y\boldsymbol{a})) = \mu(\delta_A(\overline{\delta}_A(s,y),\boldsymbol{a})) = \delta_B(\mu(\overline{\delta}_A(s,y)),\boldsymbol{a}),$$

by the homomorphism property. Apply the inductive hypothesis to obtain

$$\mu(\overline{\delta}_A(s,y\boldsymbol{a})) = \delta_B(\overline{\delta}_B(\mu(s),y),\boldsymbol{a}) = \overline{\delta}_B(\mu(s),y\boldsymbol{a}).$$

□

(b) $(\forall s \in S_A)(\forall x \in \Sigma^*)(\overline{\omega}_A(s,x) = \overline{\omega}_B(\mu(s),x))$.

Proof. By induction on $|x|$.
Base: If $x = \lambda$, then both sides equal λ.
Step: Let $x = \boldsymbol{a}y$. By Definition 7.17,

$$\overline{\omega}_A(s,\boldsymbol{a}y) = \omega_A(\delta_A(s,\boldsymbol{a})) \cdot \overline{\omega}_A(\delta_A(s,\boldsymbol{a}),y).$$

Since μ is a Moore homomorphism,

$$\omega_A(\delta_A(s,\boldsymbol{a})) = \omega_B(\mu(\delta_A(s,\boldsymbol{a}))) = \omega_B(\delta_B(\mu(s),\boldsymbol{a})).$$

Applying the inductive hypothesis to the state $\delta_A(s,\boldsymbol{a})$ gives

$$\overline{\omega}_A(s,\boldsymbol{a}y) = \omega_B(\delta_B(\mu(s),\boldsymbol{a})) \cdot \overline{\omega}_B(\delta_B(\mu(s),\boldsymbol{a}),y) = \overline{\omega}_B(\mu(s),\boldsymbol{a}y).$$

□

(c) $A \equiv B$:

Proof. Taking $s = s_{0_A}$ in part (b), and using $\mu(s_{0_A}) = s_{0_B}$, we obtain

$$\overline{\omega}_A(s_{0_A}, x) = \overline{\omega}_B(s_{0_B}, x)$$

for every $x \in \Sigma^*$. Hence $f_A = f_B$. □

7.48. **Prove Corollary 7.10.**

Proof. Start with E_{0M}, which groups together exactly those states that print the same output symbol. By Lemma 7.6(b), compute successively $E_{1M}, E_{2M}, \ldots$ from the transition function. By Lemma 7.3(d), some equality $E_{mM} = E_{m+1,M}$ must occur after finitely many steps, and by Lemma 7.3(e), that stabilized relation is exactly E_M. Thus there is an algorithm for computing E_M. □ □

7.49. **Prove Corollary 7.11.**

Proof. Compute the connected part M^c by the usual reachability algorithm. Then use Corollary 7.10 to compute E_{M^c}. Finally, form the quotient $M^c/_{E_{M^c}}$. By Theorem 7.10(b), this quotient is the minimal Moore machine equivalent to M. Hence there is an algorithm for computing the minimal machine equivalent to M. □ □

7.50. δ_{E_M} **and** ω_{E_M} **well-defined for FST quotient.**

(a) δ_{E_M} is well-defined:

Proof. Define

$$\delta_{E_M}([s]_{E_M}, \boldsymbol{a}) = [\delta(s, \boldsymbol{a})]_{E_M}.$$

If $[s]_{E_M} = [t]_{E_M}$, then $sE_M t$. By Lemma 7.3(c),

$$\delta(s, \boldsymbol{a}) E_M \delta(t, \boldsymbol{a}),$$

so

$$[\delta(s, \boldsymbol{a})]_{E_M} = [\delta(t, \boldsymbol{a})]_{E_M}.$$

Therefore δ_{E_M} does not depend on the chosen representative. □

(b) ω_{E_M} is well-defined:

Proof. Define

$$\omega_{E_M}([s]_{E_M}, \boldsymbol{a}) = \omega(s, \boldsymbol{a}).$$

If $[s]_{E_M} = [t]_{E_M}$, then $sE_M t$, so the states s and t agree on every one-letter input. In particular,

$$\omega(s, \boldsymbol{a}) = \omega(t, \boldsymbol{a}).$$

Hence ω_{E_M} is independent of the representative. □

7.51. δ_{E_M} **and** ω_{E_M} **well-defined for MSM quotient.**

(a) δ_{E_M} is well-defined:

Proof. Define

$$\delta_{E_M}([s]_{E_M}, \boldsymbol{a}) = [\delta(s, \boldsymbol{a})]_{E_M}.$$

Suppose $[s]_{E_M} = [t]_{E_M}$, so $sE_M t$. Then s and t produce the same output on every continuation. In particular, for every $x \in \Sigma^*$,

$$\overline{\omega}(s, \boldsymbol{a}x) = \overline{\omega}(t, \boldsymbol{a}x).$$

By the Moore-machine definition of $\overline{\omega}$, this implies

$$\omega(\delta(s, \boldsymbol{a})) \cdot \overline{\omega}(\delta(s, \boldsymbol{a}), x) = \omega(\delta(t, \boldsymbol{a})) \cdot \overline{\omega}(\delta(t, \boldsymbol{a}), x).$$

Hence $\delta(s, \boldsymbol{a}) E_M \delta(t, \boldsymbol{a})$, and so

$$[\delta(s, \boldsymbol{a})]_{E_M} = [\delta(t, \boldsymbol{a})]_{E_M}.$$

Therefore δ_{E_M} does not depend on the representative chosen. □

(b) ω_{E_M} is well-defined:

Proof. Define

$$\omega_{E_M}([s]_{E_M}) = \omega(s).$$

If $[s]_{E_M} = [t]_{E_M}$, then $sE_M t$. By Lemma 7.3(b), the full state equivalence relation E_M is a refinement of E_{0M}, so $sE_{0M} t$. Lemma 7.6(a) says that $sE_{0M} t$ is exactly the statement that the two states have the same Moore output. Hence

$$\omega(s) = \omega(t).$$

Thus ω_{E_M} is independent of the chosen representative. □

7.52. **Isomorphism from A to B and A connected $\Rightarrow$ B connected.**

(a) **Mealy machines:** Let $\psi : S_A \to S_B$ be an isomorphism. Every $t \in S_B$ has a preimage $s = \psi^{-1}(t)$. Since A is connected, $s = \overline{\delta}_A(s_{0_A}, x)$ for some x. Then $t = \psi(s) = \psi(\overline{\delta}_A(s_{0_A}, x)) = \overline{\delta}_B(\psi(s_{0_A}), x) = \overline{\delta}_B(s_{0_B}, x)$. So t is reachable in B.

(b) **Moore machines:** Let $\psi : S_A \to S_B$ be a Moore isomorphism. For any state $t \in S_B$, choose $s = \psi^{-1}(t)$. Since A is connected, there exists $x \in \Sigma^*$ with

$$s = \overline{\delta}_A(s_{0_A}, x).$$

Because an isomorphism commutes with the transition function,

$$t = \psi(s) = \psi(\overline{\delta}_A(s_{0_A}, x)) = \overline{\delta}_B(\psi(s_{0_A}), x) = \overline{\delta}_B(s_{0_B}, x).$$

Thus every state of B is reachable, so B is connected.

7.53. **Isomorphism from A to B and B connected $\Rightarrow$ A connected.**

Proof. If $\psi : A \to B$ is an isomorphism, then by Exercise 7.65(a) its inverse

$$\psi^{-1} : B \to A$$

is also an isomorphism. Since B is connected, Exercise 7.52 applied to ψ^{-1} shows that A is connected. □ □

7.54. **Homomorphism from A to B and A connected does NOT imply B connected.**

(a) **Mealy counterexample:** Let A have state $\{s_0\}$ (connected), $\Sigma = \{\boldsymbol{a}\}$, $\delta_A(s_0, \boldsymbol{a}) = s_0$, $\omega_A(s_0, \boldsymbol{a}) = \mathbf{0}$. Let B have states $\{t_0, t_1\}$ where t_1 is unreachable from t_0; $\delta_B(t_0, \boldsymbol{a}) = t_0$, $\omega_B(t_0, \boldsymbol{a}) = \mathbf{0}$, $\delta_B(t_1, \boldsymbol{a}) = t_1$, $\omega_B(t_1, \boldsymbol{a}) = \mathbf{0}$. Map $\mu(s_0) = t_0$. Then μ is a homomorphism, A is connected, but B is not connected (t_1 unreachable).

(b) **Moore machines:** Use the same state graph and the same map $\mu(s_0) = t_0$, but interpret outputs as state outputs rather than transition outputs. Then μ is again a homomorphism from a connected machine onto a machine with an unreachable state.

7.55. **Homomorphism from A to B and B connected does NOT imply A connected.**

(a) **Counterexample:** Let A have states $\{s_0, s_1\}$ where s_1 is unreachable; B has state $\{t_0\}$ (trivially connected). Map $\mu(s_0) = \mu(s_1) = t_0$. μ is a homomorphism, B is connected, but A is not connected.

(b) **Moore machines:** Use the same two-state domain and one-state codomain, but interpret outputs as state outputs. The map sending both states of A to the single state of B is again a homomorphism, while A remains disconnected.

7.56. **If $\psi : A \to B$ and $\mu : B \to A$ are isomorphisms of connected FSTs, then $\psi = \mu^{-1}$.**

Proof. $\mu \circ \psi : S_A \to S_A$ is an isomorphism (composition of isomorphisms). For any $s \in S_A$ reachable via x ($s = \overline{\delta}_A(s_{0_A}, x)$): $(\mu \circ \psi)(s) = \mu(\psi(\overline{\delta}_A(s_{0_A}, x))) = \mu(\overline{\delta}_B(s_{0_B}, x)) = \overline{\delta}_A(s_{0_A}, x) = s$. So $\mu \circ \psi = \mathrm{id}_{S_A}$, i.e., $\mu = \psi^{-1}$. □ □

7.57. **If A, B are FSTs (possibly not connected), $\psi = \mu^{-1}$ may fail.**

Example: Let $A = B$ have states $\{s_0, u, v\}$, start state s_0, alphabet $\{\boldsymbol{a}\}$, and output alphabet $\{\mathbf{0}\}$, with

$$\delta(s_0, \boldsymbol{a}) = s_0, \qquad \delta(u, \boldsymbol{a}) = u, \qquad \delta(v, \boldsymbol{a}) = v,$$

and every transition printing $\mathbf{0}$. The states u and v are both unreachable from the start state, so the machines are not connected. Now

$$\psi = \mathrm{id}_{\{s_0, u, v\}}$$

is an isomorphism from A to B, and so is

$$\mu(s_0) = s_0, \qquad \mu(u) = v, \qquad \mu(v) = u.$$

But $\psi^{-1} = \psi \neq \mu$. Thus the conclusion of Exercise 7.56 fails without connectedness.

7.58. **Three-state MSM with E_{0A} having one class; can $E_{0A} \neq E_{1A}$?**

Yes, such a three-state Moore machine exists: take

$$A = \langle \{\boldsymbol{a}\}, \{\mathbf{0}\}, \{r_0, r_1, r_2\}, r_0, \delta, \omega \rangle,$$

where

$$\omega(r_0) = \omega(r_1) = \omega(r_2) = \mathbf{0}.$$

and

$$\delta(r_0, \boldsymbol{a}) = r_1, \qquad \delta(r_1, \boldsymbol{a}) = r_2, \qquad \delta(r_2, \boldsymbol{a}) = r_2.$$

Then Lemma 7.6(a) gives

$$E_{0A} = \{(r_i, r_j) \mid 0 \le i, j \le 2\},$$

so E_{0A} has exactly one equivalence class.

However, in this situation it is *not* possible for E_{0A} to differ from E_{1A}. By Lemma 7.6(b),

$$sE_{1A}t \iff sE_{0A}t \ \wedge\ (\forall \boldsymbol{a} \in \Sigma)(\delta(s, \boldsymbol{a}) E_{0A} \delta(t, \boldsymbol{a})).$$

If E_{0A} has only one equivalence class, then every pair of successor states is automatically E_{0A}-related. Hence every pair of states is also E_{1A}-related, so in fact

$$E_{1A} = E_{0A}.$$

7.59. (a) Mealy M not connected and $M/_{E_M}$ not connected: take

$$M = \langle \{\boldsymbol{a}\}, \{\mathbf{0}, \mathbf{1}\}, \{s_0, s_1, s_2\}, s_0, \delta, \omega \rangle$$

where

$$\delta(s_0, \boldsymbol{a}) = s_1, \qquad \delta(s_1, \boldsymbol{a}) = s_1, \qquad \delta(s_2, \boldsymbol{a}) = s_2,$$

and

$$\omega(s_0, \boldsymbol{a}) = \mathbf{0}, \qquad \omega(s_1, \boldsymbol{a}) = \mathbf{0}, \qquad \omega(s_2, \boldsymbol{a}) = \mathbf{1}.$$

The state s_2 is unreachable, so M is not connected. Also $s_2 \not\!E_M s_0$ and $s_2 \not\!E_M s_1$, because on the one-letter input $\boldsymbol{a}$ it outputs $\mathbf{1}$ while the reachable states output $\mathbf{0}$. Hence $[s_2]$ remains a separate unreachable state in the quotient, so $M/_{E_M}$ is not connected.

(b) Mealy M not connected but $M/_{E_M}$ connected: take

$$M = \langle \{\boldsymbol{a}\}, \{\mathbf{0}\}, \{t_0, t_1, t_2\}, t_0, \delta, \omega \rangle$$

where every state loops on $\boldsymbol{a}$ and always outputs $\mathbf{0}$:

$$\delta(t_j, \boldsymbol{a}) = t_j, \qquad \omega(t_j, \boldsymbol{a}) = \mathbf{0} \qquad (j = 0, 1, 2).$$

The states t_1 and t_2 are unreachable from t_0, so M is not connected. But all three states are equivalent, since every input word produces the same output word $\mathbf{0}^{|x|}$ no matter where we start. Therefore

$$[t_0] = [t_1] = [t_2],$$

so $M/_{E_M}$ has just one state and is therefore connected.

7.60. (a) Moore M not connected and $M/_{E_M}$ not connected: take a machine with states $\{r_0, r_1, r_2\}$, start state r_0, alphabet $\{\boldsymbol{a}\}$, outputs

$$\omega(r_0) = \mathbf{0}, \qquad \omega(r_1) = \mathbf{1}, \qquad \omega(r_2) = \mathbf{0},$$

and transitions

$$\delta(r_0, \boldsymbol{a}) = r_1, \qquad \delta(r_1, \boldsymbol{a}) = r_1, \qquad \delta(r_2, \boldsymbol{a}) = r_2.$$

The state r_2 is unreachable, so M is not connected. Also, r_2 is not equivalent to either reachable state: it outputs $\mathbf{0}$ forever, while r_0 leads to $\mathbf{1}$ after one step and r_1 already outputs $\mathbf{1}$. Hence the quotient still has an unreachable class, so $M/_{E_M}$ is not connected.

(b) Moore M not connected but $M/_{E_M}$ connected: take states $\{r_0, r_1\}$ with start state r_0, alphabet $\{\boldsymbol{a}\}$, outputs

$$\omega(r_0) = \omega(r_1) = \mathbf{0},$$

and transitions

$$\delta(r_0, \boldsymbol{a}) = r_0, \qquad \delta(r_1, \boldsymbol{a}) = r_1.$$

The state r_1 is unreachable, so M is not connected. But r_0 and r_1 are equivalent, since both produce only $\mathbf{0}$ on every input. Therefore the quotient merges them into one state, and that one-state quotient is connected.

7.61. **Prove $\mu(s) E_B \mu(t) \Leftrightarrow s E_A t$ for Mealy homomorphism.**

Proof. ($\Rightarrow$) If $\mu(s) E_B \mu(t)$: $\overline{\omega}_B(\mu(s), x) = \overline{\omega}_B(\mu(t), x)$ for all x. By Exercise 7.2(b), $\overline{\omega}_A(s, x) = \overline{\omega}_B(\mu(s), x) = \overline{\omega}_B(\mu(t), x) = \overline{\omega}_A(t, x)$, so $s E_A t$.

($\Leftarrow$) If $s E_A t$: $\overline{\omega}_A(s, x) = \overline{\omega}_A(t, x)$ for all x. So $\overline{\omega}_B(\mu(s), x) = \overline{\omega}_A(s, x) = \overline{\omega}_A(t, x) = \overline{\omega}_B(\mu(t), x)$, giving $\mu(s) E_B \mu(t)$. □ □

7.62. **Prove $\mu(s) E_B \mu(t) \Leftrightarrow s E_A t$ for Moore homomorphism.**

Proof. ($\Rightarrow$) Suppose $\mu(s) E_B \mu(t)$. Then

$$\overline{\omega}_B(\mu(s), x) = \overline{\omega}_B(\mu(t), x)$$

for every $x \in \Sigma^*$. By Exercise 7.47(b),

$$\overline{\omega}_A(s, x) = \overline{\omega}_B(\mu(s), x) \quad \text{and} \quad \overline{\omega}_A(t, x) = \overline{\omega}_B(\mu(t), x),$$

so $\overline{\omega}_A(s, x) = \overline{\omega}_A(t, x)$ for all x. Hence $s E_A t$.

($\Leftarrow$) Suppose $s E_A t$. Then

$$\overline{\omega}_A(s, x) = \overline{\omega}_A(t, x)$$

for every $x \in \Sigma^*$. Again using Exercise 7.47(b),

$$\overline{\omega}_B(\mu(s), x) = \overline{\omega}_A(s, x) = \overline{\omega}_A(t, x) = \overline{\omega}_B(\mu(t), x).$$

Therefore $\mu(s) E_B \mu(t)$. □ □

7.63. (a) FST A not reduced and A^c not reduced:

Let

$$A = \langle\{\boldsymbol{a}\}, \{\mathbf{0}\}, \{s_0, s_1, s_2\}, s_0, \delta, \omega\rangle,$$

where

$$\delta(s_0, \boldsymbol{a}) = s_1, \qquad \delta(s_1, \boldsymbol{a}) = s_1, \qquad \delta(s_2, \boldsymbol{a}) = s_2,$$

and every transition prints $\mathbf{0}$. The states s_1 and s_2 are equivalent, so A is not reduced. Both s_0 and s_1 are reachable from s_0, and s_0 and s_1 are also equivalent, so the connected part A^c still has equivalent distinct states. Hence A^c is not reduced.

(b) FST A not reduced and A^c reduced:

Let

$$A = \langle\{\boldsymbol{a}\}, \{\mathbf{0}, \mathbf{1}\}, \{s_0, s_1, s_2\}, s_0, \delta, \omega\rangle,$$

where

$$\delta(s_0, \boldsymbol{a}) = s_1, \qquad \delta(s_1, \boldsymbol{a}) = s_1, \qquad \delta(s_2, \boldsymbol{a}) = s_2,$$

and

$$\omega(s_0, \boldsymbol{a}) = \mathbf{0}, \qquad \omega(s_1, \boldsymbol{a}) = \omega(s_2, \boldsymbol{a}) = \mathbf{1}.$$

Then $s_1 E_A s_2$, so A is not reduced. But s_2 is unreachable, while the connected part $A^c = \{s_0, s_1\}$ is reduced because s_0 and s_1 have different output behavior.

7.64. (a) Moore machine A not reduced and A^c not reduced:

Let the states be $\{r_0, r_1, r_2\}$ with start state r_0, alphabet $\{\boldsymbol{a}\}$, transitions

$$\delta(r_0, \boldsymbol{a}) = r_1, \qquad \delta(r_1, \boldsymbol{a}) = r_1, \qquad \delta(r_2, \boldsymbol{a}) = r_2,$$

and outputs

$$\omega(r_0) = \omega(r_1) = \omega(r_2) = \mathbf{0}.$$

Then all states are equivalent, so A is not reduced. The connected part A^c consists of r_0 and r_1, which are still equivalent, so A^c is not reduced.

(b) Moore machine A not reduced and A^c reduced:

Let the states be $\{r_0, r_1, r_2\}$ with start state r_0, alphabet $\{\boldsymbol{a}\}$, transitions

$$\delta(r_0, \boldsymbol{a}) = r_1, \qquad \delta(r_1, \boldsymbol{a}) = r_1, \qquad \delta(r_2, \boldsymbol{a}) = r_2,$$

and outputs

$$\omega(r_0) = \mathbf{0}, \qquad \omega(r_1) = \omega(r_2) = \mathbf{1}.$$

Then $r_1 E_A r_2$, so A is not reduced. But r_2 is unreachable, while the connected part $A^c = \{r_0, r_1\}$ is reduced because r_0 and r_1 have different output behavior.

7.65. **Isomorphism $\cong$ is an equivalence relation on Mealy machines.**

(a) **Symmetry:** If $\psi : A \to B$ is an isomorphism, then ψ is bijective and satisfies

$$\psi(s_{0_A}) = s_{0_B}, \qquad \psi(\delta_A(s, \boldsymbol{a})) = \delta_B(\psi(s), \boldsymbol{a}), \qquad \omega_A(s, \boldsymbol{a}) = \omega_B(\psi(s), \boldsymbol{a}).$$

Applying ψ^{-1} to these identities shows that

$$\psi^{-1}(s_{0_B}) = s_{0_A}, \qquad \psi^{-1}(\delta_B(t, \boldsymbol{a})) = \delta_A(\psi^{-1}(t), \boldsymbol{a}),$$

and

$$\omega_B(t, \boldsymbol{a}) = \omega_A(\psi^{-1}(t), \boldsymbol{a}).$$

Hence $\psi^{-1} : B \to A$ is also an isomorphism.

(b) **Reflexivity:** The identity map id_{S_A} preserves the start state, transition function, and output function, so it is an isomorphism from A to itself.

(c) **Composition:** If $\psi : A \to B$ and $\varphi : B \to C$ are isomorphisms, then $\varphi \circ \psi$ is bijective. Also,

$$(\varphi \circ \psi)(s_{0_A}) = \varphi(s_{0_B}) = s_{0_C},$$

and for every state s and input $\boldsymbol{a}$,

$$(\varphi \circ \psi)(\delta_A(s, \boldsymbol{a})) = \varphi(\delta_B(\psi(s), \boldsymbol{a})) = \delta_C((\varphi \circ \psi)(s), \boldsymbol{a}),$$

while

$$\omega_A(s, \boldsymbol{a}) = \omega_B(\psi(s), \boldsymbol{a}) = \omega_C((\varphi \circ \psi)(s), \boldsymbol{a}).$$

Thus $\varphi \circ \psi$ is an isomorphism.

(d) **Equivalence:** Parts (a), (b), and (c) show that $\cong$ is symmetric, reflexive, and transitive. Therefore it is an equivalence relation.

(e) **Homomorphism is not an equivalence relation:** Homomorphism need not be symmetric. Let A have two distinct states s_0, s_1 with the same transition and output behavior, and let B be the one-state machine obtained by identifying them. Then there is a homomorphism $A \to B$, but no inverse map $B \to A$ exists, so homomorphism is not symmetric.

7.66. **Isomorphism $\cong$ is an equivalence relation on Moore machines; homomorphism is not.**

(a) **Symmetry:** If $\psi : A \to B$ is a Moore isomorphism, then ψ^{-1} preserves the start state, transition function, and state-output function, so $\psi^{-1} : B \to A$ is also an isomorphism.

(b) **Reflexivity:** The identity map on the state set of a Moore machine is an isomorphism.

(c) **Composition:** The composition of two Moore isomorphisms again preserves start state, transitions, and outputs, hence is an isomorphism.

(d) **Equivalence:** Therefore Moore-machine isomorphism is reflexive, symmetric, and transitive.

(e) **Homomorphism is not an equivalence relation:** It is not symmetric. Let

$$A = \langle \{\boldsymbol{a}\}, \{\mathbf{0}\}, \{r_0, r_1\}, r_0, \delta_A, \omega_A \rangle$$

with

$$\omega_A(r_0) = \omega_A(r_1) = \mathbf{0}, \qquad \delta_A(r_0, \boldsymbol{a}) = r_1, \qquad \delta_A(r_1, \boldsymbol{a}) = r_1,$$

and let

$$B = \langle \{\boldsymbol{a}\}, \{\mathbf{0}\}, \{s_0\}, s_0, \delta_B, \omega_B \rangle$$

with

$$\omega_B(s_0) = \mathbf{0}, \qquad \delta_B(s_0, \boldsymbol{a}) = s_0.$$

The map $\mu(r_0) = \mu(r_1) = s_0$ is a homomorphism $A \to B$. But there is no homomorphism $B \to A$: any such map must send s_0 to the start state r_0, and then

$$\mu(\delta_B(s_0, \boldsymbol{a})) = \mu(s_0) = r_0$$

would have to equal

$$\delta_A(r_0, \boldsymbol{a}) = r_1,$$

which is impossible. So homomorphism is not symmetric, and therefore not an equivalence relation.

7.67. **Prove homomorphism $\mu : M \to M/_{E_M}$ exists.**

Proof. Define $\mu : S \to S_{E_M}$ by $\mu(s) = [s]_{E_M}$.

We verify the homomorphism properties.

(i) $\mu(s_0) = [s_0]_{E_M}$, which is the start state of the quotient.

(ii) For every state s and input $\boldsymbol{a}$,

$$\mu(\delta(s, \boldsymbol{a})) = [\delta(s, \boldsymbol{a})]_{E_M} = \delta_{E_M}([s]_{E_M}, \boldsymbol{a}) = \delta_{E_M}(\mu(s), \boldsymbol{a}).$$

(iii) For every state s and input $\boldsymbol{a}$,

$$\omega_{E_M}(\mu(s), \boldsymbol{a}) = \omega_{E_M}([s]_{E_M}, \boldsymbol{a}) = \omega(s, \boldsymbol{a}).$$

Thus μ is a homomorphism from M onto $M/_{E_M}$. □ □

7.68. **Prove homomorphism $\mu : M \to M/_{E_M}$ exists for Moore machines.**

Proof. Define $\mu(s) = [s]_{E_M}$ from the state set of M onto the state set of the quotient.

We verify the homomorphism properties.

(i) $\mu(s_0) = [s_0]_{E_M}$ is the start state of the quotient.

(ii) For every state s and input $\boldsymbol{a}$,

$$\mu(\delta(s, \boldsymbol{a})) = [\delta(s, \boldsymbol{a})]_{E_M} = \delta_{E_M}([s]_{E_M}, \boldsymbol{a}) = \delta_{E_M}(\mu(s), \boldsymbol{a}).$$

(iii) For every state s,

$$\omega_{E_M}(\mu(s)) = \omega_{E_M}([s]_{E_M}) = \omega(s).$$

Hence μ is a Moore homomorphism from M onto $M/_{E_M}$. □ □

7.69. **Limit time in state s_2 to 3 clock cycles.**

(a) Replace s_2 by three states s_{2a}, s_{2b}, s_{2c}, all with output $\langle R,R,R,G\rangle$. The old transition $s_1 \to s_2$ becomes $s_1 \to s_{2a}$. Then define

	$\langle 0,0\rangle$	$\langle 0,1\rangle$	$\langle 1,0\rangle$	$\langle 1,1\rangle$
s_{2a}	s_4	s_{2b}	s_4	s_{2b}
s_{2b}	s_4	s_{2c}	s_4	s_{2c}
s_{2c}	s_4	s_4	s_4	s_4

so the configuration $\langle R,R,R,G\rangle$ can last for one, two, or three cycles, but never four.

(b) Replace s_6 by three states s_{6a}, s_{6b}, s_{6c}, all with output $\langle G,G,R,R\rangle$. Every old edge entering s_6 now enters s_{6a}. Then define

	$\langle 0,0\rangle$	$\langle 0,1\rangle$	$\langle 1,0\rangle$	$\langle 1,1\rangle$
s_{6a}	s_5	s_7	s_{6b}	s_{6b}
s_{6b}	s_5	s_7	s_{6c}	s_{6c}
s_{6c}	s_5	s_7	s_5	s_7

which bounds the left-turn green in the same way.

(c) To guarantee a *minimum* of two cycles and a *maximum* of four cycles for the left-turn green, replace s_6 by four states $s_{6a}, s_{6b}, s_{6c}, s_{6d}$, all outputting $\langle G,G,R,R\rangle$, and let

	$\langle 0,0\rangle$	$\langle 0,1\rangle$	$\langle 1,0\rangle$	$\langle 1,1\rangle$
s_{6a}	s_{6b}	s_{6b}	s_{6b}	s_{6b}
s_{6b}	s_5	s_7	s_{6c}	s_{6c}
s_{6c}	s_5	s_7	s_{6d}	s_{6d}
s_{6d}	s_5	s_7	s_5	s_7

so the first transition out of s_{6a} is forced, the machine may remain in the green left-turn configuration for two, three, or four cycles, and the fourth cycle is the last possible one.

7.70. **Ensure east-west traffic gets green occasionally.**

(a) Without adding states, force every excursion away from s_0 to return to s_0 after one service. Concretely, remove the self-loops on s_2 and s_6, and redefine

$$\delta(s_2, \langle i,j\rangle) = s_4 \qquad \text{for all } i,j \in \{0,1\},$$

$$\delta(s_6,\langle 0,0\rangle) = s_5, \quad \delta(s_6,\langle 1,0\rangle) = s_5, \quad \delta(s_6,\langle 0,1\rangle) = s_7, \quad \delta(s_6,\langle 1,1\rangle) = s_7,$$

and then force the yellow states back to east-west green:

$$\delta(s_4,\langle i,j\rangle) = \delta(s_5,\langle i,j\rangle) = \delta(s_7,\langle i,j\rangle) = s_0$$

for all i,j. Then the machine must revisit $\langle G,R,G,R\rangle$ regularly.

(b) To obtain a genuine fairness controller, pair each phase state with a memory set $H \subseteq \{A,B\}$, where A means "the α left-turn lane has already had green" and B means "the β side-street lane has already had green" since the last visit to s_0. Use states (s,H) with $s \in \{s_0,\ldots,s_7\}$ and start state $(s_0,\emptyset)$. Whenever the machine enters the old state s_2, replace H by $H \cup \{B\}$; whenever it enters the old state s_6, replace H by $H \cup \{A\}$; whenever it returns to s_0, reset the memory to $\emptyset$. The yellow states now dispatch according to the set of sensor-driven lanes that are still owed service:

(i) from a yellow state that follows B-service, if α is active and $A \not\!\! A n H$, go next to the A-branch instead of returning to s_0; otherwise return to s_0;

(ii) from a yellow state that follows A-service, if β is active and $B \not\!\! A n H$, go next to the B-branch instead of returning to s_0; otherwise return to s_0.

Thus no sensor-controlled lane can receive a second green before every other active sensor-controlled lane has received a first one.

7.71. **Prove: if two Moore machines are homomorphic, they are equivalent.**

Proof. If $\mu : A \to B$ is a homomorphism, by Exercise 7.47(c), $f_A = f_B$. Hence $A \equiv B$. □ □

7.72. **Prove $\mathfrak{D}_\Sigma$ is closed under any FTD function $f : \Sigma^* \to \Sigma^*$.**

Proof. Let $L \in \mathfrak{D}_\Sigma$ and f be FTD, implemented by Mealy machine M. We want to show $f^{-1}(L) \in \mathfrak{D}_\Sigma$ (inverse image) and $f(L) \in \mathfrak{D}_\Sigma$ (image).

Closure under inverse: Given DFA A for L and Mealy machine M for f, build a product machine P whose states are $S_A \times S_M$. On input $\boldsymbol{a}$: P advances M to compute the output $f(\boldsymbol{a})$, then advances A on that output. If the output of M is a single letter per input (Mealy), then P directly simulates A on the translation. The result is a DFA for $f^{-1}(L) = \{x \mid f(x) \in L\} \in \mathfrak{D}_\Sigma$.

Closure under image: $f(L) = \{f(x) \mid x \in L\}$. Build an NDFA that reads the output of M while M's input was accepted by A: nondeterministically guess the input $x \in L$ and simulate M on x to generate output. Since $L \in \mathfrak{D}_\Sigma$ and M is a finite transducer, $f(L) \in \mathfrak{D}_\Sigma$. (More precisely: the composition of a DFA with an FST yields an NDFA, and by Theorem 4.1, this is equivalent to a DFA.) □ □

Chapter 8

Regular and Context-Free Grammars — Solutions

8.1. **Can strings like $\boldsymbol{ab}BA\boldsymbol{d}B\boldsymbol{c}$ be derived from the start symbol in a right-linear grammar?**

No. In a right-linear grammar, every production has the form $A \to w$ or $A \to wB$ where $w \in \Sigma^*$ and B is a nonterminal. Thus any sentential form derived from the start symbol has the structure w or wB where all terminal symbols precede the single nonterminal (if any). A string like $\boldsymbol{ab}BA\boldsymbol{d}B\boldsymbol{c}$ contains nonterminals interspersed with and following terminals, which cannot occur in any right-linear grammar derivation.

8.2. **Prove $P(n)$: for all $x \in \Sigma^n$, if $t_0 \stackrel{*}{\Rightarrow} x t_j$ then $\overline{\delta}_A(t_0, x) = t_j$.**

Base case ($n = 0$): The only string of length 0 is λ. The derivation $t_0 \stackrel{*}{\Rightarrow} \lambda t_0$ uses zero steps, and $\overline{\delta}_A(t_0, \lambda) = t_0$ by definition, so $P(0)$ holds.

Inductive step: Assume $P(n)$ holds for all strings of length n. Let $x = ya$ where $y \in \Sigma^n$ and $a \in \Sigma$. If $t_0 \stackrel{*}{\Rightarrow} ya \cdot t_j$, then the derivation must pass through some intermediate state: $t_0 \stackrel{*}{\Rightarrow} y \cdot t_k \Rightarrow ya \cdot t_j$ for some t_k. By the inductive hypothesis applied to y, $\overline{\delta}_A(t_0, y) = t_k$. The final step $t_k \Rightarrow a \cdot t_j$ corresponds to a production $t_k \to \boldsymbol{a} t_j$ in the grammar (by construction of G_A), which corresponds to a transition $\delta(t_k, a) = t_j$ in A. Therefore $\overline{\delta}_A(t_0, ya) = \delta(\overline{\delta}_A(t_0, y), a) = \delta(t_k, a) = t_j$.

8.3. **Regular expressions for the languages of:**

(a) G_4: Tracing the grammar, we get

$$\begin{aligned} &S = \boldsymbol{ab}A \cup \boldsymbol{bb}B \cup \boldsymbol{cc}V, \\ &A = \boldsymbol{b}C \cup \boldsymbol{c}X, \qquad B = \boldsymbol{ab}, \qquad C = \lambda \cup \boldsymbol{c}S, \\ &V = \boldsymbol{a}V \cup \boldsymbol{c}X, \qquad X = \boldsymbol{b}V \cup \boldsymbol{aa}X. \end{aligned}$$

From

$$V = \boldsymbol{a}V \cup \boldsymbol{c}X,$$

Arden's lemma gives

$$V = \boldsymbol{a}^* \boldsymbol{c}X.$$

Substituting into the equation for X gives

$$X = \boldsymbol{ba}^*\boldsymbol{c}X \cup \boldsymbol{aa}X = (\boldsymbol{ba}^*\boldsymbol{c} \cup \boldsymbol{aa})X.$$

There is no constant term, so the only solution is

$$X = \emptyset,$$

and therefore also $V = \emptyset$.

Thus only the S, A, B, C portion contributes:

$$S = \boldsymbol{ab}A \cup \boldsymbol{bbab}, \qquad A = \boldsymbol{b}C \cup \boldsymbol{c}\emptyset = \boldsymbol{b}C, \qquad C = \lambda \cup \boldsymbol{c}S.$$

Hence

$$A = \boldsymbol{b} \cup \boldsymbol{bc}S,$$

so

$$S = \boldsymbol{abb} \cup \boldsymbol{abbc}S \cup \boldsymbol{bbab}.$$

By Arden's lemma,

$$L(G_4) = (\boldsymbol{abbc})^*(\boldsymbol{abb} \cup \boldsymbol{bbab}).$$

(b) G_5 generates strings with an even number of **1**s (binary parity even):

$$L(G_5) = (\mathbf{0}^*\mathbf{1}\mathbf{0}^*\mathbf{1})^*\mathbf{0}^* = (\mathbf{0}^*(\mathbf{1}\mathbf{0}^*\mathbf{1})^*\mathbf{0}^*).$$

The regular expression is $(\mathbf{0} \cup \mathbf{1}\mathbf{0}^*\mathbf{1})^*$.

(c) G_6: From the productions $Z \to \boldsymbol{a}Z|\boldsymbol{b}T$, $T \to \boldsymbol{a}Z$. Language equations: $Z = \boldsymbol{a}Z \cup \boldsymbol{b}T$ and $T = \boldsymbol{a}Z$. Substituting: $Z = \boldsymbol{a}Z \cup \boldsymbol{ba}Z = (\boldsymbol{a} \cup \boldsymbol{ba})Z$. Since there is no terminal production (no production $Z \to w$ or $T \to w$ for $w \in \Sigma^*$), $L(G_6) = \emptyset$.

Every derivation cycles: $Z \Rightarrow \boldsymbol{b}T \Rightarrow \boldsymbol{ba}Z \Rightarrow \cdots$, and neither Z nor T has a terminal production, confirming $L(G_6) = \emptyset$.

(d) G_7: $S \to \boldsymbol{a}S|\boldsymbol{ab}B|\boldsymbol{c}C$, $B \to \boldsymbol{ab}B|\lambda$, $C \to \boldsymbol{c}C|\boldsymbol{ca}$.

- From B: $B = \boldsymbol{ab}B \cup \lambda$, so $L(B) = (\boldsymbol{ab})^*$.
- From C: $C = \boldsymbol{c}C \cup \boldsymbol{ca}$, so $L(C) = \boldsymbol{c}^*\boldsymbol{ca} = \boldsymbol{c}^+\boldsymbol{a}$.
- From S: $S = \boldsymbol{a}S \cup \boldsymbol{ab}B \cup \boldsymbol{c}C = \boldsymbol{a}S \cup \boldsymbol{ab}(\boldsymbol{ab})^* \cup \boldsymbol{c}^+\boldsymbol{a}$. $S = \boldsymbol{a}S \cup \boldsymbol{a}(\boldsymbol{ab})^+ \cup \boldsymbol{c}^+\boldsymbol{a}$, so $L(S) = \boldsymbol{a}^*(\boldsymbol{a}(\boldsymbol{ab})^+ \cup \boldsymbol{c}^+\boldsymbol{a})$.

The regular expression is $\boldsymbol{a}^+(\boldsymbol{ab})^+ \cup \boldsymbol{a}^*\boldsymbol{c}^+\boldsymbol{a}$.

8.4. **Use Exercise 8.2 to formally prove Lemma 8.1.**

Lemma 8.1 states that for the grammar G_A constructed from DFA A, $L(G_A) = L(A)$.

By the inductive result of Exercise 8.2, for all $x \in \Sigma^*$ and all states t_i, t_j: $t_i \stackrel{*}{\Rightarrow} xt_j$ in G_A iff $\overline{\delta}_A(t_i, x) = t_j$. Taking $t_i = t_0$ (start state):

$x \in L(G_A)$ iff $t_0 \stackrel{*}{\Rightarrow} x$ in G_A iff $(\exists t_j \in F)$ $t_0 \stackrel{*}{\Rightarrow} xt_j \Rightarrow x$ iff $(\exists t_j \in F)$ $\overline{\delta}_A(t_0, x) = t_j$ iff $\overline{\delta}_A(t_0, x) \in F$ iff $x \in L(A)$. $\square$

8.5. **Draw the automata for the grammars in Exercise 8.3.**

Use the construction of Lemma 8.2: states are the suffix-strings that arise from right-hand sides, the start state is $\langle S\rangle$, and the only final state is $\langle\lambda\rangle$.

(a) For G_4, the state set is

$\{\langle S\rangle, \langle A\rangle, \langle B\rangle, \langle C\rangle, \langle V\rangle, \langle W\rangle, \langle X\rangle, \langle \boldsymbol{ab}A\rangle, \langle \boldsymbol{b}A\rangle, \langle \boldsymbol{bb}B\rangle, \langle \boldsymbol{b}B\rangle, \langle \boldsymbol{cc}V\rangle, \langle \boldsymbol{c}V\rangle, \langle \boldsymbol{b}C\rangle, \langle \boldsymbol{c}X\rangle, \langle \boldsymbol{ab}\rangle, \langle \boldsymbol{b}\rangle, \langle \boldsymbol{c}S\rangle, \langle \boldsymbol{a}V\rangle, \langle \boldsymbol{aa}\rangle, \langle \boldsymbol{a}$

The nonempty λ-transitions are

$$\delta(\langle S\rangle, \lambda) = \{\langle \boldsymbol{ab}A\rangle, \langle \boldsymbol{bb}B\rangle, \langle \boldsymbol{cc}V\rangle\},$$

$$\delta(\langle A\rangle, \lambda) = \{\langle \boldsymbol{b}C\rangle, \langle \boldsymbol{c}X\rangle\}, \quad \delta(\langle B\rangle, \lambda) = \{\langle \boldsymbol{ab}\rangle\},$$

$$\delta(\langle C\rangle, \lambda) = \{\langle\lambda\rangle, \langle \boldsymbol{c}S\rangle\}, \quad \delta(\langle V\rangle, \lambda) = \{\langle \boldsymbol{a}V\rangle, \langle \boldsymbol{c}X\rangle\},$$

$$\delta(\langle W\rangle, \lambda) = \{\langle \boldsymbol{aa}\rangle, \langle \boldsymbol{a}W\rangle\}, \quad \delta(\langle X\rangle, \lambda) = \{\langle \boldsymbol{b}V\rangle, \langle \boldsymbol{aa}X\rangle\}.$$

The nonempty letter-transitions are the suffix moves, for example

$$\delta(\langle \boldsymbol{ab}A\rangle, \boldsymbol{a}) = \{\langle \boldsymbol{b}A\rangle\}, \qquad \delta(\langle \boldsymbol{b}A\rangle, \boldsymbol{b}) = \{\langle A\rangle\},$$

$$\delta(\langle \boldsymbol{ab}\rangle, \boldsymbol{a}) = \{\langle \boldsymbol{b}\rangle\}, \qquad \delta(\langle \boldsymbol{b}\rangle, \boldsymbol{b}) = \{\langle\lambda\rangle\},$$

and similarly for the other string-states.

(b) For G_5, the corresponding automaton is simply the three-state DFA

$$\langle\{\mathbf{0}, \mathbf{1}\}, \{S_0, S_1, S_2\}, S_0, \delta, \{S_0\}\rangle,$$

with

$$\delta(S_0, \mathbf{0}) = S_2, \quad \delta(S_0, \mathbf{1}) = S_1, \quad \delta(S_1, \mathbf{0}) = S_1, \quad \delta(S_1, \mathbf{1}) = S_2,$$

$$\delta(S_2, \mathbf{0}) = S_2, \quad \delta(S_2, \mathbf{1}) = S_0.$$

(c) For G_6, the state set is

$$\{\langle Z\rangle, \langle T\rangle, \langle \boldsymbol{a}Z\rangle, \langle \boldsymbol{b}T\rangle, \langle\lambda\rangle\},$$

with start state $\langle Z\rangle$, final state $\langle\lambda\rangle$, and nonempty transitions

$$\delta(\langle Z\rangle, \lambda) = \{\langle \boldsymbol{a}Z\rangle, \langle \boldsymbol{b}T\rangle\}, \qquad \delta(\langle T\rangle, \lambda) = \{\langle \boldsymbol{a}Z\rangle\},$$

$$\delta(\langle \boldsymbol{a}Z\rangle, \boldsymbol{a}) = \{\langle Z\rangle\}, \qquad \delta(\langle \boldsymbol{b}T\rangle, \boldsymbol{b}) = \{\langle T\rangle\}.$$

There is no path to $\langle\lambda\rangle$, so the language is empty.

(d) For G_7, the state set is

$$\{\langle S\rangle, \langle B\rangle, \langle C\rangle, \langle \boldsymbol{a}S\rangle, \langle \boldsymbol{ab}B\rangle, \langle \boldsymbol{b}B\rangle, \langle \boldsymbol{c}C\rangle, \langle \boldsymbol{ab}\rangle, \langle \boldsymbol{b}\rangle, \langle \boldsymbol{ca}\rangle, \langle \boldsymbol{a}\rangle, \langle\lambda\rangle\},$$

with start state $\langle S\rangle$, final state $\langle\lambda\rangle$, and nonempty λ-transitions

$$\delta(\langle S\rangle, \lambda) = \{\langle \boldsymbol{a}S\rangle, \langle \boldsymbol{ab}B\rangle, \langle \boldsymbol{c}C\rangle\},$$

$$\delta(\langle B\rangle, \lambda) = \{\langle \boldsymbol{ab}B\rangle, \langle\lambda\rangle\}, \qquad \delta(\langle C\rangle, \lambda) = \{\langle \boldsymbol{c}C\rangle, \langle \boldsymbol{ca}\rangle\},$$

and nonempty letter-transitions

$$\delta(\langle \boldsymbol{a}S\rangle, \boldsymbol{a}) = \{\langle S\rangle\}, \quad \delta(\langle \boldsymbol{ab}B\rangle, \boldsymbol{a}) = \{\langle \boldsymbol{b}B\rangle\}, \quad \delta(\langle \boldsymbol{b}B\rangle, \boldsymbol{b}) = \{\langle B\rangle\},$$

$$\delta(\langle \boldsymbol{c}C\rangle, \boldsymbol{c}) = \{\langle C\rangle\}, \quad \delta(\langle \boldsymbol{ab}\rangle, \boldsymbol{a}) = \{\langle \boldsymbol{b}\rangle\}, \quad \delta(\langle \boldsymbol{b}\rangle, \boldsymbol{b}) = \{\langle\lambda\rangle\},$$

$$\delta(\langle \boldsymbol{ca}\rangle, \boldsymbol{c}) = \{\langle \boldsymbol{a}\rangle\}, \qquad \delta(\langle \boldsymbol{a}\rangle, \boldsymbol{a}) = \{\langle\lambda\rangle\}.$$

8.6. **Right-linear grammars for the given languages.**

(a) All words in $\{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}\}^*$ *not* containing two consecutive $\boldsymbol{b}$s:

$$G = \langle \{S, A\}, \{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}\}, S, \{S \to \boldsymbol{a}S|\boldsymbol{b}A|\boldsymbol{c}S|\lambda, A \to \boldsymbol{a}S|\boldsymbol{c}S|\lambda\}\rangle$$

State S = "last symbol was not $\boldsymbol{b}$" (or start); state A = "last symbol was $\boldsymbol{b}$". From A, reading another $\boldsymbol{b}$ is disallowed.

(b) All words *containing* two consecutive $\boldsymbol{b}$s:

$$G = \langle \{S, A, B\}, \{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}\}, S, \{S \to \boldsymbol{a}S|\boldsymbol{c}S|\boldsymbol{b}A, A \to \boldsymbol{b}B|\boldsymbol{a}S|\boldsymbol{c}S, B \to \boldsymbol{a}B|\boldsymbol{b}B|\boldsymbol{c}B|\lambda\}\rangle$$

(c) All words with $|w|_{\boldsymbol{a}} = |w|_{\boldsymbol{b}}$: This language is not regular (it includes $\{\boldsymbol{a}^n\boldsymbol{b}^n\}$), so no right-linear grammar exists.

(d) All words with an even number of $\boldsymbol{a}$s:

$$G = \langle \{S, A\}, \{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}\}, S, \{S \to \boldsymbol{b}S|\boldsymbol{c}S|\boldsymbol{a}A|\lambda, A \to \boldsymbol{b}A|\boldsymbol{c}A|\boldsymbol{a}S\}\rangle$$

(e) All words *not* ending in $\boldsymbol{b}$:

$$G = \langle \{S, A\}, \{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}\}, S, \{S \to \boldsymbol{a}S|\boldsymbol{b}A|\boldsymbol{c}S|\boldsymbol{a}|\boldsymbol{c}|\lambda, A \to \boldsymbol{a}S|\boldsymbol{b}A|\boldsymbol{c}S|\boldsymbol{a}|\boldsymbol{c}\}\rangle$$

(Equivalently, $G = \langle \{S, A\}, \ldots, \{S \to (\boldsymbol{a}|\boldsymbol{b}|\boldsymbol{c})^*(\boldsymbol{a}|\boldsymbol{c})|\lambda\}\rangle$.)

(f) All words containing no $\boldsymbol{c}$s:

$$G = \langle \{S\}, \{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}\}, S, \{S \to \boldsymbol{a}S|\boldsymbol{b}S|\lambda\}\rangle$$

8.7. **Left-linear grammars for the languages in Exercise 8.6.**

A left-linear grammar has productions of the form $A \to w$ or $A \to Bw$.

(a) No two consecutive $\boldsymbol{b}$s:

$$G = \langle \{S, A\}, \{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}\}, S, \{S \to S\boldsymbol{a}|S\boldsymbol{c}|A\boldsymbol{b}|\lambda, A \to S\boldsymbol{a}|S\boldsymbol{c}|A\boldsymbol{b}\cdot\text{(blocked)}\}\rangle$$

More carefully: S = words not ending in $\boldsymbol{b}$; A = words ending in exactly one $\boldsymbol{b}$ (last char is $\boldsymbol{b}$, but second-to-last is not $\boldsymbol{b}$). Productions: $S \to S\boldsymbol{a}|S\boldsymbol{c}|A\boldsymbol{a}|A\boldsymbol{c}|\lambda$, $A \to S\boldsymbol{b}$.

(b) Containing two consecutive $\boldsymbol{b}$s: $S \to S\boldsymbol{a}|S\boldsymbol{b}|S\boldsymbol{c}|A\boldsymbol{b}$, $A \to B\boldsymbol{b}$, $B \to B\boldsymbol{a}|B\boldsymbol{b}|B\boldsymbol{c}|\lambda$.

(c) Even number of $\boldsymbol{a}$s: $S \to S\boldsymbol{b}|S\boldsymbol{c}|A\boldsymbol{a}|\lambda$, $A \to S\boldsymbol{a}|A\boldsymbol{b}|A\boldsymbol{c}$.

(d) Not ending in $\boldsymbol{b}$: $S \to S\boldsymbol{a}|S\boldsymbol{c}|A\boldsymbol{a}|A\boldsymbol{c}|\lambda$, $A \to S\boldsymbol{b}|A\boldsymbol{b}$.

(e) No $\boldsymbol{c}$s: $S \to S\boldsymbol{a}|S\boldsymbol{b}|\lambda$.

8.8. **Complete the inductive portion of the proof of Theorem 8.8.**

Let $G = \langle \Omega, \Sigma, S, P\rangle$ already be in the normal form of Theorem 8.7, and let $G^0 = \langle \Omega \cup \{Z\}, \Sigma, Z, P^0\rangle$ be the grammar constructed in Theorem 8.8.

The needed induction is on the number of productions used after the initial step $Z \to S$.

Claim: For every $A \in \Omega$ and every $x \in \Sigma^+$,

$$A \overset{*}{\Rightarrow}_{G^0} x \quad \text{if and only if} \quad A \overset{*}{\Rightarrow}_G x.$$

We prove the claim by induction on the number of productions in the derivation.

Base: A one-step derivation in G^0 must use a production $A \to \boldsymbol{a}$. By construction of P^0, there is some $B \in \Omega$ such that $A \to \boldsymbol{a}B \in P$ and $B \to \lambda \in P$. Hence in G we have $A \Rightarrow \boldsymbol{a}B \Rightarrow \boldsymbol{a}$. Conversely, if $A \overset{*}{\Rightarrow}_G x$ in two steps, then necessarily $x = \boldsymbol{a}$, with productions $A \to \boldsymbol{a}B$, $B \to \lambda$, so $A \to \boldsymbol{a}$ is in P^0 and $A \overset{*}{\Rightarrow}_{G^0} \boldsymbol{a}$.

Step: Assume the claim holds for derivations using at most n productions, and suppose $A \overset{*}{\Rightarrow}_{G^0} x$ uses $n+1$ productions.

If the first production is $A \to \boldsymbol{a}B$, then

$$A \Rightarrow_{G^0} \boldsymbol{a}B \overset{*}{\Rightarrow}_{G^0} x,$$

so $x = \boldsymbol{a}y$ and $B \overset{*}{\Rightarrow}_{G^0} y$ uses n productions. By the induction hypothesis, $B \overset{*}{\Rightarrow}_G y$, and therefore $A \Rightarrow_G \boldsymbol{a}B \overset{*}{\Rightarrow}_G x$.

The other possibility is that the first production is $A \to \boldsymbol{a}$; then $x = \boldsymbol{a}$ and this is already the base case.

For the converse direction, suppose $A \overset{*}{\Rightarrow}_G x$ and the derivation uses $n+2$ productions. The first production cannot be $A \to \lambda$, since $x \in \Sigma^+$. If the first production is $A \to \boldsymbol{a}B$, and the remainder is $B \overset{*}{\Rightarrow}_G y$ with $x = \boldsymbol{a}y$, then either $y \in \Sigma^+$, in which case the induction hypothesis gives $B \overset{*}{\Rightarrow}_{G^0} y$ and hence $A \Rightarrow_{G^0} \boldsymbol{a}B \overset{*}{\Rightarrow}_{G^0} x$, or else $y = \lambda$, in which case $B \to \lambda \in P$ and therefore $A \to \boldsymbol{a} \in P^0$, so $A \overset{*}{\Rightarrow}_{G^0} \boldsymbol{a} = x$.

Now apply the claim with $A = S$. Any terminal derivation in G^0 either is $Z \Rightarrow \lambda$ (which occurs exactly when $S \to \lambda \in P$) or begins $Z \Rightarrow S$ and then follows a derivation covered by the claim. Hence $L(G^0) = L(G)$. □

8.9. **Complete the inductive portion of the proof of Theorem 8.5.**

Let $G_* = \langle \Omega \cup \{Z\}, \Sigma, Z, P_* \rangle$ be the grammar of Theorem 8.5. The theorem says that derivations using exactly n productions of the form $A \to xZ$ generate exactly the words in $L(G)^n$.

Claim: For every $n \in \mathbb{N}$ and every $w \in \Sigma^*$,

$$Z \overset{*}{\Rightarrow}_{G_*} w \text{ using exactly } n \text{ productions of the form } A \to xZ$$

if and only if $w \in L(G)^n$.

We prove this by induction on n.

Base ($n = 0$): The only terminal derivation from Z that uses no production of the form $A \to xZ$ is $Z \Rightarrow \lambda$. Hence the words generated are exactly $L(G)^0 = \{\lambda\}$.

Step: Assume the claim holds for n, and consider a derivation using exactly $n+1$ productions of the form $A \to xZ$. Since the start symbol Z appears only in the productions $Z \to \lambda$ and $Z \to S$, such a derivation must begin

$$Z \Rightarrow S \overset{*}{\Rightarrow}_{G_*} uZ \overset{*}{\Rightarrow}_{G_*} uv,$$

where the portion $S \overset{*}{\Rightarrow}_{G_*} uZ$ uses exactly one production $A \to xZ$. Replacing that final production by the corresponding production $A \to x$ from G shows that $u \in L(G)$. The remainder $Z \overset{*}{\Rightarrow}_{G_*} v$ uses exactly n productions of the form $A \to xZ$, so by the induction hypothesis $v \in L(G)^n$. Therefore $w = uv \in L(G) \cdot L(G)^n = L(G)^{n+1}$.

Conversely, let $w \in L(G)^{n+1}$. Then $w = uv$ for some $u \in L(G)$ and $v \in L(G)^n$. Choose a derivation $S \overset{*}{\Rightarrow}_G u$ in G. Its last step uses some production $A \to x$ in P; in G_* this production has been replaced by $A \to xZ$, so the same derivation yields $S \overset{*}{\Rightarrow}_{G_*} uZ$. By the induction hypothesis, $Z \overset{*}{\Rightarrow}_{G_*} v$ using exactly n productions of the form $A \to xZ$. Thus

$$Z \Rightarrow S \overset{*}{\Rightarrow}_{G_*} uZ \overset{*}{\Rightarrow}_{G_*} uv = w$$

uses exactly $n+1$ such productions. Hence the words generated in this way are exactly the words in $L(G)^{n+1}$. □

8.10. **Use the efficient algorithm of Theorem 8.3 to find regular expressions for $L(G_5)$, $L(G_6)$, $L(G_7)$ in Exercise 8.3.**

The efficient algorithm eliminates one variable at a time using Arden's lemma.

- G_5: Equations: $S_0 = \mathbf{0}S_2 \cup \mathbf{1}S_1 \cup \lambda$, $S_1 = \mathbf{0}S_1 \cup \mathbf{1}S_2$, $S_2 = \mathbf{0}S_2 \cup \mathbf{1}S_0$. Solve for S_2 from third equation: $S_2 = \mathbf{0}^*\mathbf{1}S_0$. Substitute into S_1: $S_1 = \mathbf{0}S_1 \cup \mathbf{10}^*\mathbf{1}S_0$, so $S_1 = \mathbf{0}^*\mathbf{10}^*\mathbf{1}S_0$. Substitute S_1 and S_2 into S_0: $S_0 = \mathbf{0}\cdot\mathbf{0}^*\mathbf{1}S_0 \cup \mathbf{1}\cdot\mathbf{0}^*\mathbf{10}^*\mathbf{1}S_0 \cup \lambda = (\mathbf{0}^+\mathbf{1} \cup \mathbf{10}^*\mathbf{10}^*\mathbf{1})S_0 \cup \lambda$. By Arden's lemma: $L(G_5) = (\mathbf{0}^+\mathbf{1} \cup \mathbf{10}^*\mathbf{10}^*\mathbf{1})^*$. Simplified: $(\mathbf{0} \cup \mathbf{10}^*\mathbf{1})^*$.
- G_6: Equations: $Z = \boldsymbol{a}Z \cup \boldsymbol{b}T$, $T = \boldsymbol{a}Z$. Substitute: $Z = \boldsymbol{a}Z \cup \boldsymbol{ba}Z = (\boldsymbol{a} \cup \boldsymbol{ba})Z \cup \emptyset$. By Arden's lemma: $Z = (\boldsymbol{a} \cup \boldsymbol{ba})^* \cdot \emptyset = \emptyset$. So $L(G_6) = \emptyset$.
- G_7: Equations: $S = \boldsymbol{a}S \cup \boldsymbol{ab}B \cup \boldsymbol{c}C$, $B = \boldsymbol{ab}B \cup \lambda$, $C = \boldsymbol{c}C \cup \boldsymbol{ca}$. Solve $B = (\boldsymbol{ab})^*$. Solve $C = \boldsymbol{c}^*\boldsymbol{ca}$. Substitute: $S = \boldsymbol{a}S \cup \boldsymbol{ab}(\boldsymbol{ab})^* \cup \boldsymbol{c}\cdot\boldsymbol{c}^*\boldsymbol{ca} = \boldsymbol{a}S \cup \boldsymbol{a}(\boldsymbol{ab})^+ \cup \boldsymbol{c}^+\boldsymbol{a}$. By Arden's: $L(G_7) = \boldsymbol{a}^*(\boldsymbol{a}(\boldsymbol{ab})^+ \cup \boldsymbol{c}^+\boldsymbol{a}) = \boldsymbol{a}^+(\boldsymbol{ab})^+ \cup \boldsymbol{a}^*\boldsymbol{c}^+\boldsymbol{a}$.

8.11. **Left-linear version of Theorem 8.3 and Lemmas 8.4–8.5.**

(a) For left-linear grammars, the language equations take the form $A = BA \cup w$ (variables appear on the right of concatenations). The restatement: if a language equation system for a left-linear grammar is $\{X_i = \bigcup_j X_j\alpha_{ji} \cup \beta_i\}$, then solve by right-side substitution.

(b) Lemma 8.4 restated for left-linear: If $X = XA \cup B$ (where A, B do not contain X), then $X = BA^*$. Lemma 8.5 restated: If $X = XA \cup Y$ and X does not appear in A or Y, then $X = YA^*$.

(c) For G_2 in Example 8.12, the language equations are

$$S = A\boldsymbol{baa}, \qquad A = A\boldsymbol{a} \cup \lambda.$$

By the left-linear version of Arden's lemma,

$$A = \boldsymbol{a}^*.$$

Therefore

$$L(G_2) = \boldsymbol{a}^*\boldsymbol{baa}.$$

8.12. **Grammar Q for binary numbers with decimal point.**

(a) A correct NDFA is

$$A = \langle\{\mathbf{0},\mathbf{1},\mathbf{.}\},\{I,D,F\},I,\delta,\{F\}\rangle,$$

where

$$\delta(I,\mathbf{0}) = \{I,D\}, \qquad \delta(I,\mathbf{1}) = \{I,D\},$$

$$\delta(D,\mathbf{.}) = \{F\}, \qquad \delta(F,\mathbf{0}) = \{F\}, \qquad \delta(F,\mathbf{1}) = \{F\},$$

and all other transitions are empty. The state D means "the last digit before the decimal point has just been read." This accepts strings such as **011.** and **101.11**, but not strings beginning with a dot.

(b) Right-linear equations:

$$I = \mathbf{0}I \cup \mathbf{1}I \cup \mathbf{0.}F \cup \mathbf{1.}F$$
$$F = \lambda \cup \mathbf{0}F \cup \mathbf{1}F$$

(c) Solving: $F = (\mathbf{0} \cup \mathbf{1})^*$ (by Arden's on $F = \mathbf{0}F \cup \mathbf{1}F \cup \lambda$). Substituting into I: $I = (\mathbf{0} \cup \mathbf{1})I \cup (\mathbf{0} \cup \mathbf{1})\mathbf{.}(\mathbf{0} \cup \mathbf{1})^*$. By Arden's: $I = (\mathbf{0} \cup \mathbf{1})^+\mathbf{.}(\mathbf{0} \cup \mathbf{1})^*$.

8.13. **Right-linear grammars for the given regular expressions.**

(a) $(\boldsymbol{a} \cup \boldsymbol{b})\boldsymbol{c}^*(\boldsymbol{d} \cup (\boldsymbol{ab})^*)$:

$$G = \langle\{S,A,B,C\},\{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c},\boldsymbol{d}\},S,P\rangle$$
$$P = \{S \to \boldsymbol{a}A|\boldsymbol{b}A,\ A \to \boldsymbol{c}A|\boldsymbol{d}|\boldsymbol{a}B|\lambda,\ B \to \boldsymbol{b}C,\ C \to \boldsymbol{a}B|\lambda\}$$

State A handles $\boldsymbol{c}^*$ and the choice of $\boldsymbol{d}$ or $(\boldsymbol{ab})^*$; state B after seeing $\boldsymbol{a}$ in the $(\boldsymbol{ab})^*$ part; state C after $\boldsymbol{ab}$.

(b) $(\boldsymbol{a} \cup \boldsymbol{b})^*\boldsymbol{a}(\boldsymbol{a} \cup \boldsymbol{b})^*$ (strings containing at least one $\boldsymbol{a}$):

$$G = \langle\{S,A\},\{\boldsymbol{a},\boldsymbol{b}\},S,\{S \to \boldsymbol{a}S|\boldsymbol{b}S|\boldsymbol{a}A,\ A \to \boldsymbol{a}A|\boldsymbol{b}A|\lambda\}\rangle$$

8.14. **Left-linear grammars for the same expressions.**

(a) $(\boldsymbol{a} \cup \boldsymbol{b})\boldsymbol{c}^*(\boldsymbol{d} \cup (\boldsymbol{ab})^*)$: Reverse the order of derivation.

$$G = \langle\{S,A,B\},\{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c},\boldsymbol{d}\},S,\{S \to A\boldsymbol{d}|B,\ A \to A\boldsymbol{c}|\boldsymbol{a}|\boldsymbol{b},\ B \to C\boldsymbol{b}|A,\ C \to B\boldsymbol{a}\}\rangle$$

Here A generates $(\boldsymbol{a} \cup \boldsymbol{b})\boldsymbol{c}^*$, so the production $S \to A\boldsymbol{d}$ gives the $\boldsymbol{d}$ branch. The other branch starts with $S \to B$, then $B \to A$ gives the $(\boldsymbol{ab})^0$ case, while the cycle $B \to C\boldsymbol{b} \to B\boldsymbol{ab}$ adds one more copy of $(\boldsymbol{ab})$ on the right.

(b) $(\boldsymbol{a} \cup \boldsymbol{b})^* \boldsymbol{a} (\boldsymbol{a} \cup \boldsymbol{b})^*$:

$$G = \langle \{S, A\}, \{\boldsymbol{a}, \boldsymbol{b}\}, S, \{S \to S\boldsymbol{a} | S\boldsymbol{b} | A\boldsymbol{a},\ A \to A\boldsymbol{a} | A\boldsymbol{b} | \lambda\} \rangle$$

8.15. **Algorithm to convert right-linear to left-linear grammar.**

(a) **Algorithm**: 1. From the right-linear grammar G, construct the NDFA A_G of Lemma 8.2.

2. Reverse all arrows in A_G and interchange initial and final states. The resulting machine accepts $L(G)^r$.

3. Apply the subset construction from Chapter 4 to obtain a DFA for $L(G)^r$.

4. Apply Lemma 8.1 to that DFA to get a right-linear grammar H for $L(G)^r$.

5. Reverse every production in H. By Lemma 8.3, the resulting grammar is left-linear and generates

$$L(H)^r = (L(G)^r)^r = L(G).$$

(b) Applying this to G_4: first construct A_{G_4} from Lemma 8.2, then reverse it, determinize, and apply Lemma 8.1. The resulting right-linear grammar for $L(G_4)^r$ is

$$H = \langle \{R, U, V, W, X, L, M, N, P\}, \Sigma, R, P_H \rangle,$$

where

$$\begin{aligned}
R &\to \boldsymbol{b}U \\
U &\to \boldsymbol{b}V \mid \boldsymbol{a}W \\
V &\to \boldsymbol{a}L \\
W &\to \boldsymbol{b}X \\
X &\to \boldsymbol{b}L \\
L &\to \lambda \mid \boldsymbol{c}M \\
M &\to \boldsymbol{b}N \\
N &\to \boldsymbol{b}P \\
P &\to \boldsymbol{a}L.
\end{aligned}$$

Reversing every production now yields a left-linear grammar for $L(G_4)$:

$$\begin{aligned}
R &\to U\boldsymbol{b} \\
U &\to V\boldsymbol{b} \mid W\boldsymbol{a} \\
V &\to L\boldsymbol{a} \\
W &\to X\boldsymbol{b} \\
X &\to L\boldsymbol{b} \\
L &\to \lambda \mid M\boldsymbol{c} \\
M &\to N\boldsymbol{b} \\
N &\to P\boldsymbol{b} \\
P &\to L\boldsymbol{a}.
\end{aligned}$$

8.16. **Algorithm to convert a regular grammar to one generating the complement.**

(a) Construct the DFA A from grammar G (via Lemma 8.1 or 8.2).

(b) Compute the complement DFA A^c by swapping final and nonfinal states (after making A complete/total).

(c) Construct a right-linear grammar G^c from A^c. Then $L(G^c) = \overline{L(G)}$.

8.17. **Algorithm to determine whether $L(G) = \emptyset$ for a regular grammar G.**

(a) Build the automaton A_G from G.

(b) Compute the set of states reachable from the start state (BFS/DFS from start state).

(c) If no final state is reachable, $L(G) = \emptyset$; otherwise $L(G) \neq \emptyset$.

This terminates since the automaton has finitely many states.

8.18. **Algorithm to determine whether $L(G)$ is infinite for a regular grammar G.**

(a) Build the automaton A_G from G.

(b) Find all states reachable from the start state.

(c) Find all states from which a final state is reachable.

(d) Check whether the intersection of these two sets (the "useful" states) contains any cycle (e.g., by DFS looking for back edges).

(e) $L(G)$ is infinite iff a cycle exists among the useful states.

8.19. **Algorithm to determine whether two right-linear grammars G_1 and G_2 generate the same language.**

(a) Construct DFAs A_1 and A_2 from G_1 and G_2 respectively.

(b) Compute the minimized DFA for each.

(c) Check isomorphism of the two minimized DFAs (compare state by state).

(d) $L(G_1) = L(G_2)$ iff the minimal DFAs are isomorphic.

8.20. **Determine $L(H)$ for grammar H.**

The grammar is $H = \langle \{A, B, S\}, \{\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}\}, S, \{S \to \boldsymbol{a}SB\boldsymbol{c}, S \to \lambda, SB \to \boldsymbol{b}S, \boldsymbol{c}B \to B\boldsymbol{c}\}\rangle$.

This grammar generates λ, and for every $n \geq 2$ it generates $\boldsymbol{a}^n\boldsymbol{b}^n\boldsymbol{c}^n$. For example,

$$S \Rightarrow \boldsymbol{aa}SB\boldsymbol{c}B\boldsymbol{c} \Rightarrow \boldsymbol{aa}SBB\boldsymbol{cc} \Rightarrow \boldsymbol{aab}SB\boldsymbol{cc} \Rightarrow \boldsymbol{aabb}S\boldsymbol{cc} \Rightarrow \boldsymbol{aabbcc}.$$

In general, after n uses of $S \to \boldsymbol{a}SB\boldsymbol{c}$ we have

$$\boldsymbol{a}^n SB^n \boldsymbol{c}^n.$$

Repeatedly use $\boldsymbol{c}B \to B\boldsymbol{c}$ to move one B left until it is adjacent to S, then apply $SB \to \boldsymbol{b}S$. Doing this n times yields

$$\boldsymbol{a}^n\boldsymbol{b}^n S\boldsymbol{c}^n,$$

and then $S \to \lambda$ gives $\boldsymbol{a}^n\boldsymbol{b}^n\boldsymbol{c}^n$.

However, $\boldsymbol{abc}$ is *not* generated: after one use of $S \to \boldsymbol{a}SB\boldsymbol{c}$ and then $S \to \lambda$ we obtain $\boldsymbol{a}B\boldsymbol{c}$, and there is no rule that can eliminate that lone B. Thus the nonempty terminal strings begin at length 6.

Therefore

$$L(H) = \{\lambda\} \cup \{\boldsymbol{a}^n \boldsymbol{b}^n \boldsymbol{c}^n \mid n \geq 2\}.$$

8.21. **What is wrong with the concatenation construction** $G^\bullet = \langle \Omega_1 \cup \Omega_2 \cup \{Z\}, \Sigma, Z, P_1 \cup P_2 \cup \{Z \to S_1 \cdot S_2\}\rangle$**?**

The production $Z \to S_1 S_2$ has two nonterminals on the right-hand side. In a right-linear grammar, all productions must have the form $A \to w$ or $A \to wB$ (a string of terminals followed by at most one nonterminal). The production $S_1 S_2$ violates this since S_1 is a nonterminal followed by S_2 (also nonterminal), with no terminal prefix. This is a context-free production, not a right-linear one, so the resulting grammar is not right-linear and cannot directly represent a language in $\mathfrak{G}_\Sigma$.

8.22. **Prove $\mathfrak{G}_\Sigma$ is closed under concatenation.**

(a) Construction: Let $G_1 = \langle \Omega_1, \Sigma, S_1, P_1\rangle$ and $G_2 = \langle \Omega_2, \Sigma, S_2, P_2\rangle$ be right-linear grammars with $\Omega_1 \cap \Omega_2 = \emptyset$. Define:

$$G^\bullet = \langle \Omega_1 \cup \Omega_2, \Sigma, S_1, P^\bullet\rangle$$

where $P^\bullet$ is obtained from P_1 by replacing each terminal production $A \to w$ (where $A \in \Omega_1$) with $A \to wS_2$, plus all productions from P_2 unchanged.

(b) Proof that $L(G^\bullet) = L(G_1) \cdot L(G_2)$: If $u \in L(G_1)$ and $v \in L(G_2)$, then $S_1 \overset{*}{\Rightarrow} u$ in G_1 via some derivation ending with $A \to u'$ (terminal production). In $G^\bullet$, this becomes $S_1 \overset{*}{\Rightarrow} u'S_2 \overset{*}{\Rightarrow} u'v = uv$. Conversely, any derivation in $G^\bullet$ must use rules from P_1 until reaching a production $A \to wS_2$, producing some prefix, then switch to P_2. $\square$

8.23. **Construct a λ-NDFA A' with one start and one final state from an NDFA A without λ-transitions.**

Let $A = \langle \Sigma, S, s_0, \delta, F\rangle$ be an NDFA without λ-transitions.

Add two new states i and f, where i will be the unique start state and f the unique final state. Define

$$A' = \langle \Sigma, S \cup \{i, f\}, i, \delta', \{f\}\rangle,$$

where

$$\delta'(s, \boldsymbol{a}) = \delta(s, \boldsymbol{a}) \qquad (s \in S,\ \boldsymbol{a} \in \Sigma),$$
$$\delta'(i, \lambda) = \{s_0\},$$

and for every old final state $p \in F$ we add the λ-transition

$$f \in \delta'(p, \lambda).$$

There are no other new transitions.

Every accepting path of A becomes an accepting path of A' by first taking $i \to s_0$, following the old transitions of A, and then taking a final λ-move into f. Conversely, any accepting path in A' must begin with $i \to s_0$, use only old transitions while reading the input, and end with a λ-move from an old final state into f. Hence

$$L(A') = L(A),$$

and A' has exactly one start state and exactly one final state.

8.24. **Complete the proof of Lemma 8.2.**

(a) Inductive statement: For all $n \geq 0$ and all $x \in \Sigma^n$ and all nonterminals A: $A \overset{*}{\Rightarrow} x$ in G iff $\overline{\delta}_A(A, x) \in F$.

(b) Proof by induction on n: **Base** ($n = 0$): $A \overset{*}{\Rightarrow} \lambda$ iff $A \to \lambda \in P$ iff $A \in F$ in the NDFA iff $\overline{\delta}(A, \lambda) = A \in F$. **Step**: Assume the result holds for length n. For $x = ay$ ($|y| = n$): $A \overset{*}{\Rightarrow} ay$ iff $(\exists B)$ $A \to \boldsymbol{a}B \in P$ and $B \overset{*}{\Rightarrow} y$ iff $(\exists B)$ $B \in \delta(A, \boldsymbol{a})$ and (by IH) $\overline{\delta}(B, y) \in F$ iff $\overline{\delta}(A, ay) \in F$. □

8.25. **Complete the proof of Lemma 8.3.**

(a) Inductive statement: for every $n \geq 0$, every pair of nonterminals A, B, and every $x \in \Sigma^*$,

$$A \overset{*\,n}{\Rightarrow}_G Bx \qquad \Longleftrightarrow \qquad A \overset{*\,n}{\Rightarrow}_{G^r} x^r B.$$

(b) Proof by induction on n.

Base ($n = 0$): A zero-step derivation has the form $A \overset{*\,0}{\Rightarrow} A$, so $B = A$ and $x = \lambda$. Then also

$$A \overset{*\,0}{\Rightarrow}_{G^r} \lambda^r A = A.$$

Step: Assume the statement true for n productions. Suppose

$$A \overset{*\,n+1}{\Rightarrow}_G By.$$

Then the first production used in G has the form

$$A \to Cz$$

for some $C \in \Omega \cup \{\lambda\}$ and some $z \in \Sigma^*$, and after that

$$C \overset{*\,n}{\Rightarrow}_G Bx$$

with $y = zx$. By the induction hypothesis,

$$C \overset{*\,n}{\Rightarrow}_{G^r} x^r B.$$

Since G^r contains the reversed production

$$A \to z^r C,$$

we obtain

$$A \Rightarrow_{G^r} z^r C \overset{*\,n}{\Rightarrow}_{G^r} z^r x^r B = (xz)^r B = y^r B.$$

The converse implication is identical with the roles of G and G^r reversed. Therefore the statement holds for all n. In particular, terminal derivations are reversed, so

$$L(G^r) = L(G)^r.$$

□

8.26. **Reasons for the assertions in the second half of the proof of Theorem 8.2.**

The second half of Theorem 8.2 begins with a language L generated by a right-linear grammar.

(a) Since L is generated by a right-linear grammar, Theorem 8.1 implies that $L \in \mathfrak{D}_\Sigma$.

(b) Exercise 5.20 (or 6.36) shows that $\mathfrak{D}_\Sigma$ is closed under reverse, so $L^r \in \mathfrak{D}_\Sigma$.

(c) Because $\mathfrak{D}_\Sigma = \mathfrak{G}_\Sigma$, there is a right-linear grammar G such that $L(G) = L^r$.

(d) Lemma 8.3 says that G^r is a left-linear grammar and that

$$L(G^r) = L(G)^r.$$

Since $L(G) = L^r$, this becomes

$$L(G^r) = (L^r)^r = L.$$

(e) Therefore G^r is a left-linear grammar that generates L. This is the "corresponding left-linear grammar" claimed in the theorem.

8.27. **Formal inductive proofs.**

(a) $L(G_1) = \{\boldsymbol{a}^n\boldsymbol{baa} \mid n \in \mathbb{N}\}$ (Example 8.7):

Claim: For all $n \geq 0$, $S \overset{*}{\Rightarrow} \boldsymbol{a}^n\boldsymbol{baa}$ in G_1. **Base** ($n = 0$): $S \Rightarrow \boldsymbol{baa}$ (production $S \to \boldsymbol{baa}$). **Step**: If $S \overset{*}{\Rightarrow} \boldsymbol{a}^n\boldsymbol{baa}$, then $S \Rightarrow \boldsymbol{a}S \overset{*}{\Rightarrow} \boldsymbol{a} \cdot \boldsymbol{a}^n\boldsymbol{baa} = \boldsymbol{a}^{n+1}\boldsymbol{baa}$. Conversely, any derivation from S either uses $S \to \boldsymbol{baa}$ (giving $\lambda \cdot \boldsymbol{baa}$) or $S \to \boldsymbol{a}S$ (giving a longer string by IH). So $L(G_1) = \{\boldsymbol{a}^n\boldsymbol{baa}\}$. □

(b) $L(G_8) = \{\boldsymbol{a}^n\boldsymbol{baa} \mid n \in \mathbb{N}\}$ (Example 8.9): The proof is analogous, using the inductive structure of the left-linear grammar G_8. □**Claim**: For every $n \geq 0$, $Z \overset{*}{\Rightarrow} \boldsymbol{a}^n\boldsymbol{baa}$ in G_8.

Base ($n = 0$): $Z \Rightarrow \boldsymbol{baa}$.

Step: If $Z \overset{*}{\Rightarrow} \boldsymbol{a}^n\boldsymbol{baa}$, then $Z \Rightarrow \boldsymbol{a}Z \overset{*}{\Rightarrow} \boldsymbol{a}^{n+1}\boldsymbol{baa}$.

Conversely, any derivation from Z either uses $Z \to \boldsymbol{baa}$ immediately or begins with $Z \to \boldsymbol{a}Z$. Repeating the second choice n times and then using $Z \to \boldsymbol{baa}$ yields exactly the string $\boldsymbol{a}^n\boldsymbol{baa}$, and there is no other way to terminate a derivation. Therefore $L(G_8) = \{\boldsymbol{a}^n\boldsymbol{baa} \mid n \in \mathbb{N}\}$. □

8.28. **Mixed grammar C with both right-linear and left-linear productions.**

$C = \langle\{S, A, B\}, \{\mathbf{0}, \mathbf{1}\}, S, \{S \to \mathbf{0}A|\mathbf{1}B|\mathbf{0}|\mathbf{1}|\lambda, A \to S\mathbf{0}, B \to S\mathbf{1}\}\rangle$.

(a) $L(C) = \{w \in \{\mathbf{0}, \mathbf{1}\}^* \mid w = w^r\}$ — the set of all palindromes over $\{\mathbf{0}, \mathbf{1}\}$.

Derivation: $S \Rightarrow \mathbf{0}A \Rightarrow \mathbf{0}S\mathbf{0}$. Each step wraps the current sentential form between matching symbols. The grammar generates all even-length and odd-length palindromes.

(b) Yes, $L(C) = \{w \mid w = w^r\}$ is not regular (it requires unbounded memory to match the first and last symbols). So $L(C)$ is FAD only for the trivial case; in general, $L(C)$ is not FAD.

(c) The definition of regular grammars should *not* be expanded to include such mixed grammars, because the resulting class of languages would no longer coincide with the regular languages. Mixed grammars generate languages (like palindromes) that are context-free but not regular.

8.29. **Importance of $\Omega_1 \cap \Omega_2 = \emptyset$ in Theorem 8.4.**

(a) **Why important**: If $\Omega_1 \cap \Omega_2 \neq \emptyset$, then a shared nonterminal A would have its productions from both P_1 and P_2 merged. Derivations starting from S_1 could use P_2 productions for A and vice versa, producing strings not in $L(G_1) \cup L(G_2)$. **Example**: Let G_1 have $\{S_1 \to \boldsymbol{a}A, A \to \boldsymbol{b}\}$ and G_2 have $\{S_2 \to \boldsymbol{c}A, A \to \boldsymbol{d}\}$. If A is shared, the union grammar with $Z \to S_1|S_2$ would derive $\boldsymbol{a}A$ and then use $A \to \boldsymbol{d}$ from G_2, producing $\boldsymbol{ad}\ \not{i}nL(G_1) \cup L(G_2)$.

(b) **Why possible**: We can always rename nonterminals in G_2 (choosing fresh names) to make $\Omega_1 \cap \Omega_2 = \emptyset$ without changing the language $L(G_2)$. This is a standard renaming/relabeling argument.

8.30. **If A_G is disconnected, what does this say about G?**

If A_G (the NDFA constructed from G) is disconnected, it means some states (nonterminals) are not reachable from the start state. This means the corresponding nonterminals in G are *useless*: they cannot appear in any derivation from the start symbol. The portions of G defining those nonterminals contribute nothing to $L(G)$ and can be removed without affecting the language generated.

8.31. **Apply Lemma 8.1 to the automata in Figure 8.4.**

Applying Lemma 8.1 means: use one nonterminal per state, use the start state as start symbol, and add one production for each labeled transition; final states also get λ-productions.

(a) Figure 8.4(a):

$$\begin{aligned}
r_0 &\to \boldsymbol{a}r_3 \mid \boldsymbol{b}r_1 \mid \lambda \\
r_1 &\to \boldsymbol{a}r_0 \mid \boldsymbol{b}r_2 \\
r_2 &\to \boldsymbol{a}r_1 \mid \boldsymbol{b}r_3 \\
r_3 &\to \boldsymbol{a}r_2 \mid \boldsymbol{b}r_0.
\end{aligned}$$

(b) Figure 8.4(b):

$$\begin{aligned}
q_0 &\to \boldsymbol{a}q_1 \mid \boldsymbol{b}q_0 \mid \boldsymbol{c}q_0 \mid \lambda \\
q_1 &\to \boldsymbol{a}q_0 \mid \boldsymbol{b}q_0 \mid \boldsymbol{c}q_1.
\end{aligned}$$

(c) Figure 8.4(c):

$$\begin{aligned}
S_0 &\to \boldsymbol{a}S_1 \mid \boldsymbol{b}S_2 \\
S_1 &\to \boldsymbol{a}S_3 \mid \boldsymbol{b}S_2 \\
S_2 &\to \boldsymbol{a}S_2 \mid \boldsymbol{b}S_2 \mid \lambda \\
S_3 &\to \boldsymbol{a}S_0 \mid \boldsymbol{b}S_1.
\end{aligned}$$

(d) Figure 8.4(d):

$$\begin{aligned}
t_0 &\to \boldsymbol{a}t_1 \mid \boldsymbol{b}t_2 \\
t_1 &\to \boldsymbol{a}t_0 \mid \boldsymbol{b}t_2 \mid \lambda \\
t_2 &\to \boldsymbol{a}t_3 \mid \boldsymbol{b}t_3 \\
t_3 &\to \boldsymbol{a}t_1 \mid \boldsymbol{b}t_1 \mid \lambda.
\end{aligned}$$

(e) Figure 8.4(e):

$$\begin{aligned}
S &\to \boldsymbol{a}r \mid \boldsymbol{b}r \mid \boldsymbol{b}t \mid \boldsymbol{c}q \mid \lambda \\
r &\to \boldsymbol{a}S \mid \boldsymbol{b}t \mid \boldsymbol{c}q \\
q &\to \boldsymbol{a}r \mid \boldsymbol{b}r \mid \boldsymbol{c}q \mid \lambda \\
t &\to \boldsymbol{a}r \mid \boldsymbol{c}q.
\end{aligned}$$

8.32. **Restate Lemma 8.1 for NDFAs.**

(a) **Restated Lemma**: Given an NDFA

$$A = \langle \Sigma, S, s_0, \delta, F \rangle,$$

define

$$G_A = \langle S, \Sigma, s_0, P \rangle$$

by adding

- a production $s \to \boldsymbol{a}t$ for every $t \in \delta(s, \boldsymbol{a})$,
- a production $s \to t$ for every $t \in \delta(s, \lambda)$,
- a production $s \to \lambda$ for every final state $s \in F$.

Then $L(G_A) = L(A)$.

(b) **Proof**: Every production records one NDFA move: letter transitions become productions of the form $s \to \boldsymbol{a}t$, λ-transitions become unit productions $s \to t$, and final states get the production $s \to \lambda$. Thus a derivation in G_A corresponds exactly to a path in A, and the derivation ends in a terminal word precisely when the corresponding path ends in a final state. Therefore $L(G_A) = L(A)$.

(c) Applied to Figure 8.5:

i. Figure 8.5(a):

$$t_0 \to \boldsymbol{a}t_0 \mid \boldsymbol{c}t_0 \mid \lambda.$$

ii. Figure 8.5(b):

$$\begin{aligned} q_1 &\to \boldsymbol{a}q_0 \mid \boldsymbol{b}q_1 \mid \lambda \\ q_0 &\to \boldsymbol{b}q_1 \mid \boldsymbol{c}q_0. \end{aligned}$$

iii. Figure 8.5(c):

$$\begin{aligned} r_0 &\to \boldsymbol{a}r_0 \mid \boldsymbol{a}r_1 \mid \boldsymbol{c}r_1 \mid \lambda \\ r_1 &\to \boldsymbol{b}r_0. \end{aligned}$$

iv. Figure 8.5(d):

$$\begin{aligned} S_0 &\to \boldsymbol{a}S_1 \\ S_1 &\to \boldsymbol{a}S_2 \mid \boldsymbol{b}S_3 \\ S_2 &\to \lambda \\ S_3 &\to \boldsymbol{b}S_1. \end{aligned}$$

v. Figure 8.5(e):

$$\begin{aligned} S_1 &\to \boldsymbol{a}S_1 \mid \boldsymbol{a}S_2 \\ S_2 &\to \boldsymbol{a}S_1 \mid \boldsymbol{b}S_1 \mid \lambda. \end{aligned}$$

vi. Figure 8.5(f):

$$\begin{aligned} S_0 &\to \boldsymbol{a}S_1 \\ S_1 &\to S_0 \mid \boldsymbol{b}S_2 \\ S_2 &\to \boldsymbol{c}S_3 \\ S_3 &\to S_1 \mid \lambda. \end{aligned}$$

vii. Figure 8.5(g):

$$\begin{aligned}
S_0 &\to \boldsymbol{a}S_1 \mid \lambda \\
S_1 &\to \boldsymbol{a}S_0 \mid \boldsymbol{b}S_3 \mid \boldsymbol{c}S_2 \\
S_2 &\to \boldsymbol{c}S_1 \\
S_3 &\to \boldsymbol{b}S_1.
\end{aligned}$$

8.33. **Context-free grammars for languages over $\Sigma = \{\boldsymbol{a}, \boldsymbol{b}\}$.**

(a) L_1 = words where last letter matches first letter:

$$G_1 : S \to \boldsymbol{a}X\boldsymbol{a} \mid \boldsymbol{b}X\boldsymbol{b} \mid \boldsymbol{a} \mid \boldsymbol{b} \mid \lambda, \quad X \to \boldsymbol{a}X \mid \boldsymbol{b}X \mid \lambda$$

(b) L_2 = odd-length words where first letter matches center letter: Let $n = 2k+1$.

$$G_2 : S \to \boldsymbol{a} \mid \boldsymbol{b} \mid \boldsymbol{a}A\boldsymbol{a} \mid \boldsymbol{a}A\boldsymbol{b} \mid \boldsymbol{b}B\boldsymbol{a} \mid \boldsymbol{b}B\boldsymbol{b},$$

$$A \to \boldsymbol{a} \mid \boldsymbol{a}A\boldsymbol{a} \mid \boldsymbol{a}A\boldsymbol{b} \mid \boldsymbol{b}A\boldsymbol{a} \mid \boldsymbol{b}A\boldsymbol{b},$$

$$B \to \boldsymbol{b} \mid \boldsymbol{a}B\boldsymbol{a} \mid \boldsymbol{a}B\boldsymbol{b} \mid \boldsymbol{b}B\boldsymbol{a} \mid \boldsymbol{b}B\boldsymbol{b}$$

The nonterminal A generates all odd-length words whose center letter is $\boldsymbol{a}$, and B does the same for center letter $\boldsymbol{b}$. The productions for S force the first letter to match that center letter.

(c) L_3 = words where last letter matches none of the other letters:

$$G_3 : S \to X\boldsymbol{a} \mid X\boldsymbol{b}, \quad \text{where } X \text{ generates strings not ending in } \boldsymbol{a} \text{ (resp. } \boldsymbol{b}\text{)}$$

Since $\Sigma = \{\boldsymbol{a}, \boldsymbol{b}\}$, "last letter matches none of the others" means all other letters differ. This is regular: last letter $\boldsymbol{a}$, all others $\boldsymbol{b}$; or last letter $\boldsymbol{b}$, all others $\boldsymbol{a}$: $L_3 = \boldsymbol{b}^*\boldsymbol{a} \cup \boldsymbol{a}^*\boldsymbol{b}$. So $G_3 : S \to B\boldsymbol{a} \mid A\boldsymbol{b} \mid \boldsymbol{a} \mid \boldsymbol{b}$, $B \to B\boldsymbol{b} \mid \lambda$, $A \to A\boldsymbol{a} \mid \lambda$.

(d) L_4 = even-length words where two center letters match:

$$G_4 : S \to \boldsymbol{a}S\boldsymbol{a} \mid \boldsymbol{a}S\boldsymbol{b} \mid \boldsymbol{b}S\boldsymbol{a} \mid \boldsymbol{b}S\boldsymbol{b} \mid \boldsymbol{aa} \mid \boldsymbol{bb}$$

(e) L_5 = odd-length words where center letter matches none of the others: Similar to L_3; since $|\Sigma| = 2$, if center is $\boldsymbol{a}$ all others are $\boldsymbol{b}$, and vice versa. So $L_5 = \{\boldsymbol{b}^k\boldsymbol{a}\boldsymbol{b}^k\} \cup \{\boldsymbol{a}^k\boldsymbol{b}\boldsymbol{a}^k\}$, which is context-free but not regular:

$$G_5 : S \to \boldsymbol{b}S\boldsymbol{b} \mid \boldsymbol{a}T\boldsymbol{a} \mid \boldsymbol{a} \mid \boldsymbol{b}, \quad T \to \boldsymbol{a}T\boldsymbol{a} \mid \boldsymbol{b}$$

(f) Regular languages: L_1 is regular; L_2 is not regular; L_3 is regular ($\boldsymbol{b}^*\boldsymbol{a} \cup \boldsymbol{a}^*\boldsymbol{b}$); L_4 is not regular; L_5 is not regular.

8.34. **Context-free grammars for the given languages.**

(a) $L = \{x \in \{\boldsymbol{a}, \boldsymbol{b}\}^* \mid |x|_{\boldsymbol{a}} < |x|_{\boldsymbol{b}}\}$:

$$G : S \to E\boldsymbol{b}E \mid E\boldsymbol{b}S, \qquad E \to \boldsymbol{a}E\boldsymbol{b}E \mid \boldsymbol{b}E\boldsymbol{a}E \mid \lambda$$

Here E generates exactly the strings with equally many $\boldsymbol{a}$'s and $\boldsymbol{b}$'s. The nonterminal S inserts one or more extra $\boldsymbol{b}$'s between balanced pieces, so every derived string has strictly more $\boldsymbol{b}$'s than $\boldsymbol{a}$'s, and every such string can be decomposed in that way.

(b) $G = \{x \in \{\boldsymbol{a}, \boldsymbol{b}\}^* \mid |x|_{\boldsymbol{a}} \geq |x|_{\boldsymbol{b}}\}$:

$$\begin{aligned} &S \to E \mid A \\ &E \to \boldsymbol{a}E\boldsymbol{b}E \mid \boldsymbol{b}E\boldsymbol{a}E \mid \lambda \\ &A \to E\boldsymbol{a}E \mid E\boldsymbol{a}A. \end{aligned}$$

Again E generates the balanced strings. The nonterminal A inserts one or more extra $\boldsymbol{a}$'s between balanced pieces, so it generates exactly the strings with more $\boldsymbol{a}$'s than $\boldsymbol{b}$'s. Therefore S generates exactly the strings with $|x|_{\boldsymbol{a}} \geq |x|_{\boldsymbol{b}}$.

(c) $K = \{w \in \{\mathbf{0}, \mathbf{1}\}^* \mid w = w^r\}$ (palindromes): $S \to \mathbf{0}S\mathbf{0} \mid \mathbf{1}S\mathbf{1} \mid \mathbf{0} \mid \mathbf{1} \mid \lambda$.

(d) $\Phi = \{\boldsymbol{a}^j \boldsymbol{b}^k \boldsymbol{c}^m \mid j \geq 3, k = m\}$: $S \to \boldsymbol{aaa}AT$, $A \to \boldsymbol{a}A \mid \lambda$, $T \to \boldsymbol{b}T\boldsymbol{c} \mid \lambda$.

8.35. **More context-free grammars.**

(a) $L_1 = \{x \in \{\boldsymbol{a}, \boldsymbol{b}\}^* \mid |x|_{\boldsymbol{a}} = 2|x|_{\boldsymbol{b}}\}$: $S \to \boldsymbol{aa}S\boldsymbol{b} \mid \boldsymbol{a}S\boldsymbol{ab} \mid S\boldsymbol{aab} \mid \lambda$ (or use $S \to SS \mid \boldsymbol{aab} \mid \boldsymbol{aba} \mid \boldsymbol{baa} \mid \lambda$).

(b) $L_2 = \{x \in \{\boldsymbol{a}, \boldsymbol{b}\}^* \mid |x|_{\boldsymbol{a}} \neq |x|_{\boldsymbol{b}}\}$:

$$\begin{aligned} &S \to A \mid B \\ &E \to \boldsymbol{a}E\boldsymbol{b}E \mid \boldsymbol{b}E\boldsymbol{a}E \mid \lambda \\ &A \to E\boldsymbol{a}E \mid E\boldsymbol{a}A \\ &B \to E\boldsymbol{b}E \mid E\boldsymbol{b}B. \end{aligned}$$

Here E generates the words with equally many $\boldsymbol{a}$'s and $\boldsymbol{b}$'s. Then A inserts one or more extra $\boldsymbol{a}$'s between balanced pieces, so it generates exactly the words with more $\boldsymbol{a}$'s than $\boldsymbol{b}$'s; similarly B generates exactly the words with more $\boldsymbol{b}$'s than $\boldsymbol{a}$'s.

(c) Postfix expressions over $\{\boldsymbol{A}, \boldsymbol{B}, \mathbf{+}, \mathbf{-}\}$: $S \to \boldsymbol{A} \mid \boldsymbol{B} \mid SS\mathbf{+} \mid SS\mathbf{-}$.

(d) Parenthesized infix expressions over $\{\boldsymbol{A}, \boldsymbol{B}, \mathbf{+}, \mathbf{-}, \mathbf{(}, \mathbf{)}\}$: $S \to \boldsymbol{A} \mid \boldsymbol{B} \mid \mathbf{(}S\mathbf{+}S\mathbf{)} \mid \mathbf{(}S\mathbf{-}S\mathbf{)}$.

8.36. **Context-sensitive grammars for the given languages.**

(a) $\Gamma = \{w\mathbf{2}w \mid w \in \{\mathbf{0}, \mathbf{1}\}^*\}$: A pure context-sensitive grammar for Γ is

$$G_\Gamma = \langle \{S, A, B\}, \{\mathbf{0}, \mathbf{1}, \mathbf{2}\}, S, P \rangle,$$

where

$$\begin{aligned} P = \{&S \to \mathbf{2},\ S \to \mathbf{0}AS,\ S \to \mathbf{1}BS, \\ &A\mathbf{0} \to \mathbf{0}A,\ A\mathbf{1} \to \mathbf{1}A,\ B\mathbf{0} \to \mathbf{0}B,\ B\mathbf{1} \to \mathbf{1}B, \\ &A\mathbf{2} \to \mathbf{20},\ B\mathbf{2} \to \mathbf{21}\}. \end{aligned}$$

The productions $S \to \mathbf{0}AS$ and $S \to \mathbf{1}BS$ generate a sentential form $M\mathbf{2}w$, where the marker string M records the symbols of w in the same left-to-right order. The productions $A\mathbf{2} \to \mathbf{20}$ and $B\mathbf{2} \to \mathbf{21}$ convert the rightmost remaining marker into the next symbol of the left copy, while the swapping rules move the other markers right until they too meet the $\mathbf{2}$. For example,

$$S \Rightarrow \mathbf{0}AS \Rightarrow \mathbf{01}ABS \Rightarrow \mathbf{01}AB\mathbf{2} \Rightarrow \mathbf{01}A\mathbf{21} \Rightarrow \mathbf{0}A\mathbf{121} \Rightarrow \mathbf{01201}.$$

Thus $L(G_\Gamma) = \Gamma$.

(b) $\Phi = \{\boldsymbol{b}^{2^j} \mid j \in \mathbb{N}\} = \{\boldsymbol{b}, \boldsymbol{bb}, \boldsymbol{bbbb}, \ldots\}$: A pure context-sensitive grammar for Φ is

$$G_\Phi = \langle \{S, L, D, R\}, \{\boldsymbol{b}\}, S, P\rangle,$$

where

$$P = \{S \to \boldsymbol{b},\ S \to LR,\ L \to LD,\ L \to D,\ D\boldsymbol{b} \to \boldsymbol{bb}D,\ DR \to \boldsymbol{b}R,\ DR \to \boldsymbol{bb}\}.$$

Starting from $S \to LR$, repeated use of $L \to LD$ followed by $L \to D$ produces $D^j R$. The rightmost D changes R into one $\boldsymbol{b}$ by $DR \to \boldsymbol{b}R$ (or into $\boldsymbol{bb}$ if it is the last D). Each remaining D then moves rightward through the current block of $\boldsymbol{b}$'s by means of $D\boldsymbol{b} \to \boldsymbol{bb}D$, doubling the length of that block. Hence each D multiplies the number of $\boldsymbol{b}$'s by 2, and a sentential form with j copies of D yields exactly $\boldsymbol{b}^{2^j}$. The first derivations are

$$S \Rightarrow \boldsymbol{b}, \qquad S \Rightarrow LR \Rightarrow DR \Rightarrow \boldsymbol{bb},$$

and

$$S \Rightarrow LR \Rightarrow LDR \Rightarrow DDR \Rightarrow D\boldsymbol{b}R \Rightarrow \boldsymbol{bb}DR \Rightarrow \boldsymbol{bbbb}.$$

Therefore $L(G_\Phi) = \Phi$.

8.37. **Show G (context-sensitive) is not equivalent to G'' in Example 8.3.**

The two grammars do *not* generate the same language.

In G'' from Example 8.3 we have

$$S \Rightarrow \boldsymbol{a}SB\boldsymbol{c} \Rightarrow \boldsymbol{a}TB\boldsymbol{c} \Rightarrow \boldsymbol{ab}T\boldsymbol{c} \Rightarrow \boldsymbol{abc},$$

so

$$\boldsymbol{abc} \in L(G'').$$

But $\boldsymbol{abc}\ \not{i}nL(G)$. In the grammar G, after one use of $S \to \boldsymbol{a}SB\boldsymbol{c}$ the only way to remove the remaining S is $S \to \lambda$, which gives

$$\boldsymbol{a}B\boldsymbol{c}.$$

That sentential form is stuck: there is no occurrence of SB to which $SB \to \boldsymbol{b}S$ can be applied, and there is no occurrence of $\boldsymbol{c}B$ to which $\boldsymbol{c}B \to B\boldsymbol{c}$ can be applied. Hence no derivation in G produces $\boldsymbol{abc}$.

Therefore

$$\boldsymbol{abc} \in L(G'') \setminus L(G),$$

so G is not equivalent to G''.

8.38. **Context-free grammars for the given languages.**

(a) $L_1 = \boldsymbol{a}^*(\boldsymbol{b} \cup \boldsymbol{c})^* \cap \{x \mid |x|_{\boldsymbol{a}} = |x|_{\boldsymbol{b}} + |x|_{\boldsymbol{c}}\}$: Strings in L_1 are of the form $\boldsymbol{a}^n\boldsymbol{b}^j\boldsymbol{c}^k$ where $n = j + k$:

$$G_1 : S \to \boldsymbol{a}S\boldsymbol{b} \mid \boldsymbol{a}S\boldsymbol{c} \mid \lambda$$

Each step adds one $\boldsymbol{a}$ at the left and one $\boldsymbol{b}$ or $\boldsymbol{c}$ at the right.

(b) $L_2 = \{\boldsymbol{a}^i\boldsymbol{b}^j\boldsymbol{c}^k \mid i + j = k\}$:

$$G_2 : S \to \boldsymbol{a}S\boldsymbol{c} \mid \boldsymbol{b}T\boldsymbol{c} \mid \lambda, \quad T \to \boldsymbol{b}T\boldsymbol{c} \mid \lambda$$

(c) $L_3 = \{x \in \{\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}\}^* \mid |x|_{\boldsymbol{a}} + |x|_{\boldsymbol{b}} = |x|_{\boldsymbol{c}}\}$:

$$G_3 : S \to \boldsymbol{a}S\boldsymbol{c} \mid \boldsymbol{b}S\boldsymbol{c} \mid SS \mid \lambda$$

8.39. **Prove** $L(G') = L(G) \cup \{\lambda\}$ **(Definition 8.4).**

Definition 8.4 constructs G' from G by adding a new start symbol Z with $Z \to S \mid \lambda$ (where S was the original start). Then: $x \in L(G')$ iff $Z \overset{*}{\Rightarrow} x$ in G'. $Z \to \lambda$ gives $\lambda \in L(G')$. $Z \to S \overset{*}{\Rightarrow} x$ gives $x \in L(G) \subseteq L(G')$. Conversely, any derivation from Z begins with either $Z \to \lambda$ or $Z \to S$, so $L(G') \subseteq L(G) \cup \{\lambda\}$. Hence $L(G') = L(G) \cup \{\lambda\}$. □

8.40. **Prove** $L(G') = L(G) \cup \{\lambda\}$ **(Definition 8.6).**

Definition 8.6 constructs

$$G' = \langle \Omega \cup \{Z\}, \Sigma, Z, P \cup \{Z \to \lambda,\ Z \to S\}\rangle$$

from the pure context-free grammar $G = \langle \Omega, \Sigma, S, P\rangle$.

If $x \in L(G)$, then $S \overset{*}{\Rightarrow}_G x$, so in G' we have

$$Z \Rightarrow S \overset{*}{\Rightarrow} x,$$

hence $x \in L(G')$. Also $Z \Rightarrow \lambda$, so $\lambda \in L(G')$. Therefore

$$L(G) \cup \{\lambda\} \subseteq L(G').$$

Conversely, let $x \in L(G')$. Any derivation from the new start symbol Z must begin with either $Z \to \lambda$ or $Z \to S$, because these are the only productions having Z on the left. If the first step is $Z \to \lambda$, then $x = \lambda$. If the first step is $Z \to S$, the rest of the derivation uses only productions from P, so $S \overset{*}{\Rightarrow}_G x$ and therefore $x \in L(G)$. Thus

$$L(G') \subseteq L(G) \cup \{\lambda\}.$$

Hence

$$L(G') = L(G) \cup \{\lambda\}.$$

□

8.41. **Properties of the normal form-preserving construction.**

(a) If G is right-linear (Theorem 8.8), the naive star construction $Z \to S \mid ZZ \mid \lambda$ does *not* preserve right-linearity: $Z \to ZZ$ places a nonterminal in a non-final position, violating the form $A \to wB$ or $A \to w$. A right-linear Kleene-star construction avoids ZZ by folding the restart into the accepting productions of G: whenever $A \to w$ is a terminal production of G, add $A \to wS$ (to restart) and $A \to w$ (to halt). This keeps all productions right-linear.

(b) Example:

$$G_1 = \langle\{S_1\}, \{\boldsymbol{a}, \boldsymbol{b}\}, S_1, \{S_1 \to \boldsymbol{a}\}\rangle, \qquad G_2 = \langle\{S_2\}, \{\boldsymbol{a}, \boldsymbol{b}\}, S_2, \{S_2 \to \boldsymbol{b}\}\rangle.$$

Both grammars are in the Theorem 8.8 normal form: every production is of the form $A \to \boldsymbol{a}$ or $A \to \boldsymbol{b}$, and there are no contracting productions. But the union construction of Theorem 8.4 gives

$$G^{\cup} = \langle\{Z, S_1, S_2\}, \{\boldsymbol{a}, \boldsymbol{b}\}, Z, \{Z \to S_1,\ Z \to S_2,\ S_1 \to \boldsymbol{a},\ S_2 \to \boldsymbol{b}\}\rangle.$$

The productions $Z \to S_1$ and $Z \to S_2$ are unit productions, not of the form $A \to \boldsymbol{a}B$ or $A \to \boldsymbol{a}$, so $G^{\cup}$ is not in the normal form of Theorem 8.8.

(c) The $G^{\bullet}$ construction for concatenation in Example 8.18 is not normal form-preserving in general, since it requires replacing terminal productions $A \to w$ with $A \to wS_2$, which is a right-linear production but may violate other normal form constraints.

8.42. **Left-linear grammars for union, concatenation, and Kleene closure.**

Given left-linear grammars $G_1 = \langle \Omega_1, \Sigma, S_1, P_1 \rangle$ and $G_2 = \langle \Omega_2, \Sigma, S_2, P_2 \rangle$ with $\Omega_1 \cap \Omega_2 = \emptyset$:

(a) **Union**: $G^{\cup} = \langle \Omega_1 \cup \Omega_2 \cup \{Z\}, \Sigma, Z, P_1 \cup P_2 \cup \{Z \to S_1 \mid S_2\} \rangle$. The production $Z \to S_1$ is left-linear (since $Z \to S_1$ has a nonterminal on the right with no symbols; this is a unit production, which is allowed). $L(G^{\cup}) = L(G_1) \cup L(G_2)$.

(b) **Concatenation**: For left-linear grammars, to concatenate $L(G_1) \cdot L(G_2)$: replace each "initial" production $A \to w$ in G_1 (where A generates a terminal string) with $A \to S_2 w$ (prepend S_2 which generates $L(G_2)$ on the left). More precisely, for each production $S_1 \to w$ in G_1, replace with $S_1 \to S_2 w$ (left-linear). The resulting grammar generates $L(G_1) \cdot L(G_2)$.

(c) **Kleene closure**: Add new start Z with $Z \to ZS_1 \mid \lambda$. This is left-linear. $L(G_*) = L(G_1)^*$.

Chapter 9

Normal Forms and Pumping Lemmas — Solutions

9.1. **Characterize the parse trees of left-linear grammars.**

In a left-linear grammar, every production has the form $A \to w$ (terminal string) or $A \to Bw$ (nonterminal followed by a terminal string). Consequently, in any parse tree for a derivation in a left-linear grammar:

- The tree has a "left-spine": the leftmost child of any internal node is always a nonterminal, and the remaining children are terminal symbols.
- All nonterminals appear in the leftmost branch of the parse tree.
- The tree is maximally left-branching; at every level, the left subtree contains all nonterminal structure while the right siblings are terminals.

This is the mirror image of right-linear parse trees (which are maximally right-branching). The depth of the parse tree equals the number of nonterminals in the derivation.

9.2. **Context-free grammars for the following languages.**

(a) $\{\boldsymbol{a}^n\boldsymbol{b}^n\boldsymbol{c}^m\boldsymbol{d}^m \mid n, m \in \mathbb{N}\}$:

$$S \to AB, \quad A \to \boldsymbol{a}A\boldsymbol{b} \mid \lambda, \quad B \to \boldsymbol{c}B\boldsymbol{d} \mid \lambda.$$

(b) $\{\boldsymbol{a}^i\boldsymbol{b}^j\boldsymbol{c}^j\boldsymbol{d}^i \mid i, j \in \mathbb{N}\}$:

$$S \to \boldsymbol{a}S\boldsymbol{d} \mid T \mid \lambda, \quad T \to \boldsymbol{b}T\boldsymbol{c} \mid \lambda.$$

(c) $\{\boldsymbol{a}^n\boldsymbol{b}^n\boldsymbol{c}^m\boldsymbol{d}^m\} \cup \{\boldsymbol{a}^i\boldsymbol{b}^j\boldsymbol{c}^j\boldsymbol{d}^i\}$:

$$S \to S_1 \mid S_2$$

where S_1 generates the first language (as in part (a)) and S_2 generates the second (as in part (b)).

9.3. **Unambiguous context-free grammars.**

(a) i. For $\{\boldsymbol{a}^n\boldsymbol{b}^n\boldsymbol{c}^m\boldsymbol{d}^m\}$: The grammar $S \to AB$, $A \to \boldsymbol{a}A\boldsymbol{b} \mid \lambda$, $B \to \boldsymbol{c}B\boldsymbol{d} \mid \lambda$ is already unambiguous (every string has a unique parse).

ii. For $\{\boldsymbol{a}^i\boldsymbol{b}^j\boldsymbol{c}^j\boldsymbol{d}^i\}$: The grammar $S \to \boldsymbol{a}S\boldsymbol{d} \mid T \mid \lambda$, $T \to \boldsymbol{b}T\boldsymbol{c} \mid \lambda$ is unambiguous.

iii. For the union $L_1 \cup L_2$, it *is* possible to make the grammar unambiguous by splitting the union into disjoint pieces:

$$L_1 \cup L_2 = L_1 \cup \{a^i b^j c^j d^i \mid i < j\} \cup \{a^i b^j c^j d^i \mid i > j\}.$$

An unambiguous grammar is

$$S \to S_1 \mid S_< \mid S_>,$$

where S_1 is the unambiguous grammar from part (i), and

$$S_< \to \boldsymbol{ab}S_<\boldsymbol{cd} \mid U, \qquad U \to \boldsymbol{b}U\boldsymbol{c} \mid \boldsymbol{bc},$$

$$S_> \to \boldsymbol{a}S_>\boldsymbol{d} \mid \boldsymbol{a}V\boldsymbol{d}, \qquad V \to \boldsymbol{ab}V\boldsymbol{cd} \mid \lambda.$$

The $S_<$ branch generates exactly the case $j > i$, the $S_>$ branch generates exactly the case $i > j$, and the S_1 branch generates L_1. These three subsets are disjoint, so the grammar is unambiguous.

(b) The statement is *false*. The example from part (i)(iii) already shows that two unambiguous context-free languages can have a union that is still unambiguous. On the other hand, Theorem 9.1 shows that inherently ambiguous context-free languages do exist, and the standard example

$$\{a^i b^j c^k \mid i = j\} \cup \{a^i b^j c^k \mid j = k\}$$

shows that the union of two unambiguous context-free languages can fail to be unambiguous.

(c) No. Since there are unambiguous context-free languages whose union is not unambiguous, $\mathfrak{U}_\Sigma$ is not closed under union.

9.4. **State and prove the inductive result needed in Theorem 9.2.**

Let $G' = G_{(A_G^{\sigma d})}$ be the grammar built in Theorem 9.2 from the deterministic automaton $A_G^{\sigma d}$.

Inductive statement: For every state q of $A_G^{\sigma d}$ and every word $x \in \Sigma^n$, there is at most one derivation in G' from the nonterminal q that yields x.

Proof: By induction on n.

Base ($n = 0$): The only possible word is λ. In the grammar obtained from a deterministic automaton, the only way to derive λ from q is via the production $q \to \lambda$, which exists exactly when q is a final state. Hence there is at most one such derivation.

Step: Assume the statement true for words of length n, and let $x = \boldsymbol{a}y$ with $|y| = n$. Any derivation of x from q must begin with a production of the form

$$q \to \boldsymbol{a}r,$$

where $r = \delta(q, \boldsymbol{a})$. Since the machine is deterministic, there is at most one such state r, hence at most one possible first production. After that first step, the rest of the derivation must derive y from r, and by the induction hypothesis there is at most one way to do so. Therefore there is at most one derivation of x from q.

Applying this to the start state shows that every word in $L(G')$ has a unique derivation tree, so G' is unambiguous. $\square$

9.5. **Algorithm to compute $B^u = \{C \mid B \overset{*}{\Rightarrow} C\}$ for each nonterminal B in P^u.**

Consider the unit-production graph: nodes are nonterminals, directed edge $B \to C$ for each unit production $B \to C$ in P^u. Then B^u is the set of nonterminals reachable from B in this graph (including B itself).

Algorithm: For each nonterminal B, compute B^u by BFS or DFS from B in the unit-production graph:

(a) Initialize $B^u = \{B\}$.

(b) While there exists $C \in B^u$ with a unit production $C \to D \in P^u$ and $D \not{i}nB^u$: add D to B^u.

(c) Repeat until no new nonterminals are added.

This is a standard graph reachability computation and terminates since $|\Omega|$ is finite.

9.6. **Closure of $\mathfrak{C}_\Sigma$ under concatenation.**

(a) **Problem with naive construction**: The production $Z \to S_1 \cdot S_2$ is fine as a context-free production, but the issue is with lambda productions. If $\lambda \in L(G_1)$ (i.e., $S_1 \overset{*}{\Rightarrow} \lambda$), then $Z \to S_1 S_2 \overset{*}{\Rightarrow} S_2$, so $L(G^*) \supseteq L(G_2)$, not just $L(G_1) \cdot L(G_2)$ minus λ-contributions. Similarly, if $\lambda \in L(G_2)$, then $Z \overset{*}{\Rightarrow} S_1$, giving $L(G^*) \supseteq L(G_1)$. The issue: when both G_1 and G_2 derive λ, the simple construction may include unintended strings or exclude others due to improper lambda-production treatment.

(b) **Fix**: First transform G_1 and G_2 so that λ is not derivable from their respective start symbols (or handle it explicitly). If $\lambda \in L(G_i)$, add explicit productions:

$$Z \to S_1 S_2 \mid S_1 \mid S_2 \mid \lambda \quad \text{(appropriate subset based on whether } \lambda \in L(G_i))$$

Specifically: add $Z \to S_2$ if $\lambda \in L(G_1)$; add $Z \to S_1$ if $\lambda \in L(G_2)$; add $Z \to \lambda$ if $\lambda \in L(G_1) \cap L(G_2)$.

(c) **Proof**: $w \in L(G^*_{\text{corrected}})$ iff $Z \overset{*}{\Rightarrow} w$ iff $w = uv$ with $u \in L(G_1)$ and $v \in L(G_2)$ (considering the λ cases via the added productions) iff $w \in L(G_1) \cdot L(G_2)$. □

9.7. $\{\boldsymbol{a}^i \boldsymbol{b}^j \boldsymbol{c}^k \mid i, j, k \in \mathbb{N},\ i + j = k\}$ **is context free.**

Grammar: $S \to \boldsymbol{a}S\boldsymbol{c} \mid T$, $T \to \boldsymbol{b}T\boldsymbol{c} \mid \lambda$.

Proof: Any derivation produces $\boldsymbol{a}^i \cdot \boldsymbol{b}^j \boldsymbol{c}^j \cdot \boldsymbol{c}^i = \boldsymbol{a}^i \boldsymbol{b}^j \boldsymbol{c}^{i+j}$. Taking $k = i + j$, we get exactly the language. □

9.8. **Ambiguous right-linear grammar and its disambiguation.**

(a) $G = \langle \{S, A, B\}, \{\boldsymbol{a}\}, S, \{S \to A, S \to B, A \to \boldsymbol{aa}A, A \to \lambda, B \to \boldsymbol{aaa}B, B \to \lambda\}\rangle$. The string $\boldsymbol{aaaaaa} = \boldsymbol{a}^6$ can be derived as $S \Rightarrow A \overset{*}{\Rightarrow} \boldsymbol{aa}A \overset{*}{\Rightarrow} \boldsymbol{aaaa}A \overset{*}{\Rightarrow} \boldsymbol{aaaaaa}$ (via A: three $\boldsymbol{aa}$ steps) or as $S \Rightarrow B \overset{*}{\Rightarrow} \boldsymbol{aaa}B \overset{*}{\Rightarrow} \boldsymbol{aaaaaa}$ (via B: two $\boldsymbol{aaa}$ steps). So G is ambiguous.

(b) $L(G) = \{w \mid |w| \equiv 0 \pmod 2\} \cup \{w \mid |w| \equiv 0 \pmod 3\} = \{w \mid 2 \mid |w| \text{ or } 3 \mid |w|\}$. An unambiguous right-linear grammar is obtained from the DFA whose states record $|w| \bmod 6$. Let the nonterminals be $S_0, S_1, S_2, S_3, S_4, S_5$, where S_i corresponds to remainder i. The start symbol is S_0, and the accepting remainders are $0,2,3,4$. The grammar is

$$\begin{aligned}
&S_0 \to \lambda \mid \boldsymbol{a}S_1,\\
&S_1 \to \boldsymbol{a}S_2,\\
&S_2 \to \lambda \mid \boldsymbol{a}S_3,\\
&S_3 \to \lambda \mid \boldsymbol{a}S_4,\\
&S_4 \to \lambda \mid \boldsymbol{a}S_5,\\
&S_5 \to \boldsymbol{a}S_0.
\end{aligned}$$

This grammar is unambiguous because at each nonterminal there is at most one production beginning with $\boldsymbol{a}$, so each word determines a unique path through the six remainders. A derivation ends with λ exactly in the states whose remainders are divisible by 2 or 3, so the grammar generates precisely $L(G)$.

9.9. **Grammar for regular expressions, modified.**

(a) Grammar that strips outermost parentheses:

$$R \to \boldsymbol{a} \mid \boldsymbol{b} \mid \boldsymbol{c} \mid \boldsymbol{\epsilon} \mid \boldsymbol{\emptyset} \mid E \boldsymbol{\cdot} E \mid E \boldsymbol{\cup} E \mid R^{\boldsymbol{*}}$$

where $E \to \boldsymbol{(}R\boldsymbol{)}$ or E is any atomic expression.

(b) Grammar allowing extraneous parentheses:

$$R \to \boldsymbol{(}R\boldsymbol{)} \mid \boldsymbol{a} \mid \boldsymbol{b} \mid \boldsymbol{c} \mid \boldsymbol{\epsilon} \mid \boldsymbol{\emptyset} \mid \boldsymbol{(}R \boldsymbol{\cdot} R\boldsymbol{)} \mid \boldsymbol{(}R \boldsymbol{\cup} R\boldsymbol{)} \mid R^{\boldsymbol{*}}$$

9.10. **Leftmost and rightmost derivations for $\boldsymbol{((a^* \cdot b) \cup (c \cdot d)^*)}$.**

(a) **Leftmost derivation** (always expand leftmost nonterminal):

$$\begin{aligned}
R &\Rightarrow \boldsymbol{(}R \boldsymbol{\cup} R\boldsymbol{)}\\
&\Rightarrow \boldsymbol{((}R \boldsymbol{\cdot} R\boldsymbol{)} \boldsymbol{\cup} R\boldsymbol{)}\\
&\Rightarrow \boldsymbol{((}R^{\boldsymbol{*}} \boldsymbol{\cdot} R\boldsymbol{)} \boldsymbol{\cup} R\boldsymbol{)}\\
&\Rightarrow \boldsymbol{((a^*} \boldsymbol{\cdot} R\boldsymbol{)} \boldsymbol{\cup} R\boldsymbol{)}\\
&\Rightarrow \boldsymbol{((a^*\cdot b)} \boldsymbol{\cup} R\boldsymbol{)}\\
&\Rightarrow \boldsymbol{((a^* \cdot b) \cup (}R \boldsymbol{\cdot} R\boldsymbol{)^*)}\\
&\Rightarrow \boldsymbol{((a^* \cdot b) \cup (c \cdot} R\boldsymbol{)^*)}\\
&\Rightarrow \boldsymbol{((a^* \cdot b) \cup (c\cdot d)^*)}
\end{aligned}$$

(b) **Rightmost derivation** (always expand rightmost nonterminal):

$$\begin{aligned}
R &\Rightarrow \boldsymbol{(}R \boldsymbol{\cup} R\boldsymbol{)}\\
&\Rightarrow \boldsymbol{(}R \boldsymbol{\cup (}R \boldsymbol{\cdot} R\boldsymbol{)^*)}\\
&\Rightarrow \boldsymbol{(}R \boldsymbol{\cup (}R \boldsymbol{\cdot d)^*)}\\
&\Rightarrow \boldsymbol{(}R \boldsymbol{\cup (c\cdot d)^*)}\\
&\Rightarrow \boldsymbol{((}R \boldsymbol{\cdot} R\boldsymbol{) \cup (c \cdot d)^*)}\\
&\Rightarrow \boldsymbol{((}R \boldsymbol{\cdot b) \cup (c \cdot d)^*)}\\
&\Rightarrow \boldsymbol{((}R^{\boldsymbol{*}} \boldsymbol{\cdot b) \cup (c \cdot d)^*)}\\
&\Rightarrow \boldsymbol{((a^* \cdot b) \cup (c \cdot d)^*)}
\end{aligned}$$

9.11. **Prove $L(G) = L(G')$ (Theorem 9.5) by induction on derivation steps.**

For every production $A \to \alpha_1\alpha_2\cdots\alpha_n$ of G, the grammar G' contains a replacement set that begins with $A \to \alpha_1 Y_{k1}$ and ends with $Y_{k,n-2} \to \alpha_{n-1}\alpha_n$ (with terminal symbols replaced by the new symbols $X_{\boldsymbol{a}}$ when necessary). Those replacement rules together produce exactly the same strings as the original rule.

Claim: If $S \overset{*\ m}{\Rightarrow}_G x$ with $x \in \Sigma^*$, then $S \overset{*}{\Rightarrow}_{G'} x$.

Proof: By induction on m.

Base ($m = 1$): Then $S \to x$ is a single production of G. If it is already in CNF form, the same production is in G'. Otherwise, the replacement set for that production derives x in G'.

Step: Assume the claim true for derivations of length m. Suppose

$$S \Rightarrow_G \alpha \overset{*\ m}{\Rightarrow}_G x.$$

Replace the first production $S \to \alpha$ by its corresponding replacement set in G'. This produces the same sentential form α, except possibly that some terminal symbols have been replaced by the new nonterminals $X_{\boldsymbol{a}}$, each of which immediately derives the corresponding terminal. After these finitely many extra steps, the remainder of the original derivation can be copied by the induction hypothesis. Hence $S \overset{*}{\Rightarrow}_{G'} x$.

Conversely, every derivation in G' can be compressed to one in G: each new symbol Y_{ki} occurs in only one replacement set, so any maximal block of steps using Y_{ki} symbols is simply the expanded form of one original production of G. Replacing each such block by the corresponding production of G yields a derivation of the same terminal string in G.

Therefore $L(G) = L(G')$. □

9.12. **For CNF grammar G and $x \in L(G)$, $x \neq \lambda$: number of productions in any derivation of x is $2|x|$.**

Proof: In the chapter's definition of CNF, a nonempty derivation begins with the start step $Z \to S$. After that initial step, every production is either of the form $A \to BC$ or $A \to \boldsymbol{a}$.

If x has length n, then exactly n terminal productions are needed, one for each terminal symbol in x. Before those terminal productions can be applied, the derivation must create exactly n nonterminals ready to become those terminals. Starting from the single nonterminal S, each production $A \to BC$ increases the number of nonterminals by 1, so exactly $n - 1$ productions of the form $A \to BC$ are required.

Thus the total number of productions is

$$1 + (n-1) + n = 2n = 2|x|,$$

where the initial 1 is the production $Z \to S$. □

9.13. **Bounds on parse tree depth for CNF grammars.**

(a) **Lower bound**: For $x \in L(G)$ with $|x| = n > 0$, the parse tree has n leaf nodes (one per terminal). Since each internal node has exactly 2 children (CNF), a complete binary tree with n leaves has depth at least $\lceil \log_2 n \rceil$. Thus the depth of the parse tree is at least $\lceil \log_2 n \rceil$.

(b) **Upper bound**: The parse tree can be maximally skewed (a path): at each level, one child is a leaf (terminal) and the other is the next internal node. This gives depth $n-1$ from the root to the last nonterminal, plus 1 for the final terminal, giving depth n. (Or $2n-1$ if we count from a different convention.) Thus the depth is at most $n = |x|$.
Summary: $\lceil \log_2 |x| \rceil \le \text{depth} \le |x|$.

9.14. **Convert grammars to Chomsky Normal Form.**

(a) $\langle \{S, B, C\}, \{\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}\}, S, \{S \to \boldsymbol{a}B, S \to \boldsymbol{ab}C, B \to \boldsymbol{bc}, C \to \boldsymbol{c}\} \rangle$:
Introduce terminal nonterminals: $T_a \to \boldsymbol{a}$, $T_b \to \boldsymbol{b}$, $T_c \to \boldsymbol{c}$. - $S \to \boldsymbol{a}B$ becomes $S \to T_a B$ (already binary CNF). - $S \to \boldsymbol{ab}C$ becomes $S \to T_a D_1$ where $D_1 \to T_b C$. - $B \to \boldsymbol{bc}$ becomes $B \to T_b T_c$. - $C \to \boldsymbol{c}$ becomes $C \to T_c$ — but $C \to T_c$ is a unit production! Replace: wherever C appears as a target, substitute. Since $C \to \boldsymbol{c}$ and C does not appear in the right side of B or S (only in $S \to \boldsymbol{ab}C$), we can just keep $C \to T_c$ and note it will be eliminated: replace $S \to T_a D_1$ (where $D_1 \to T_b C$) by substituting C: $D_1 \to T_b T_c$.
Final CNF:

$$S \to T_a B \mid T_a D_1, \quad D_1 \to T_b T_c, \quad B \to T_b T_c, \quad T_a \to \boldsymbol{a}, \quad T_b \to \boldsymbol{b}, \quad T_c \to \boldsymbol{c}.$$

(b) $\langle \{S, A, B\}, \{\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}\}, S, \{S \to \boldsymbol{c}BA, S \to B, A \to \boldsymbol{c}B, A \to A\boldsymbol{bb}S, B \to \boldsymbol{aaa}\} \rangle$:
Step 1: Remove unit production $S \to B$: add all B-rules to S: $S \to \boldsymbol{aaa}$. Step 2: Introduce terminal symbols: $T_a \to \boldsymbol{a}$, $T_b \to \boldsymbol{b}$, $T_c \to \boldsymbol{c}$. Step 3: Break long right-hand sides: - $S \to \boldsymbol{c}BA$: $S \to T_c E_1$, $E_1 \to BA$. - $S \to \boldsymbol{aaa}$: $S \to T_a E_2$, $E_2 \to T_a T_a$. - $A \to \boldsymbol{c}B$: $A \to T_c B$. - $A \to A\boldsymbol{bb}S$: $A \to AE_3$, $E_3 \to T_b E_4$, $E_4 \to T_b S$. - $B \to \boldsymbol{aaa}$: $B \to T_a E_2$ (reuse E_2).
Final CNF:

$$\begin{aligned} &S \to T_c E_1 \mid T_a E_2 \\ &E_1 \to BA, \quad E_2 \to T_a T_a \\ &A \to T_c B \mid AE_3 \\ &E_3 \to T_b E_4, \quad E_4 \to T_b S \\ &B \to T_a E_2 \\ &T_a \to \boldsymbol{a}, \quad T_b \to \boldsymbol{b}, \quad T_c \to \boldsymbol{c} \end{aligned}$$

(c) $\langle \{R\}, \{\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}, \boldsymbol{(}, \boldsymbol{)}, \boldsymbol{\epsilon}, \boldsymbol{\emptyset}, \cup, \cdot, ^{*}\}, R, \{R \to \boldsymbol{a}|\boldsymbol{b}|\boldsymbol{c}|\boldsymbol{\epsilon}|\boldsymbol{\emptyset}|(R \cdot R)|(R \cup R)|R^{*}\} \rangle$:
Introduce terminal nonterminals for each terminal symbol: $T_a, T_b, T_c, T_\epsilon, T_\emptyset, T_(, T_), T_\cup, T_\cdot, T_*$.
$R \to \boldsymbol{a}|\boldsymbol{b}|\boldsymbol{c}|\boldsymbol{\epsilon}|\boldsymbol{\emptyset}$: Each of these is a single-terminal production $R \to a$ (with $|a| = 1$), which is already in CNF form.
$R \to (R \cdot R)$: Break as $R \to T_(D_1$, $D_1 \to RD_2$, $D_2 \to T_\cdot D_3$, $D_3 \to RT_)$. $R \to (R \cup R)$: Similarly $R \to T_(D_4$, $D_4 \to RD_5$, $D_5 \to T_\cup D_6$, $D_6 \to RT_)$. $R \to R^{*}$: $R \to RD_7$, $D_7 \to T_*$—but $D_7 \to T_*$ is a unit production; substitute: $R \to RT_*$.
Final CNF:

$$\begin{aligned} &R \to \boldsymbol{a}|\boldsymbol{b}|\boldsymbol{c}|\boldsymbol{\epsilon}|\boldsymbol{\emptyset} \mid T_(D_1 \mid T_(D_4 \mid RT_* \\ &D_1 \to RD_2, \quad D_2 \to T_\cdot D_3, \quad D_3 \to RT_) \\ &D_4 \to RD_5, \quad D_5 \to T_\cup D_6, \quad D_6 \to RT_) \\ &T_(\to \boldsymbol{(}, \quad T_) \to \boldsymbol{)}, \quad T_\cdot \to \cdot, \quad T_\cup \to \cup, \quad T_* \to ^{*} \end{aligned}$$

(d) $\langle\{T\},\{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c},\boldsymbol{d},\boldsymbol{-},\boldsymbol{+}\},T,\{T\to\boldsymbol{a}|\boldsymbol{b}|\boldsymbol{c}|\boldsymbol{d}|T\boldsymbol{-}T|T\boldsymbol{+}T\}\rangle$:
$T\to\boldsymbol{a}|\boldsymbol{b}|\boldsymbol{c}|\boldsymbol{d}$: Already in CNF (terminal productions). $T\to T\boldsymbol{-}T$: $T\to TD_-$ where $D_-\to T_-T$ and $T_-\to\boldsymbol{-}$. $T\to T\boldsymbol{+}T$: $T\to TD_+$ where $D_+\to T_+T$ and $T_+\to\boldsymbol{+}$.
Final CNF:

$$T\to\boldsymbol{a}|\boldsymbol{b}|\boldsymbol{c}|\boldsymbol{d}\mid TD_-\mid TD_+,\quad D_-\to T_-T,\quad D_+\to T_+T,\quad T_-\to\boldsymbol{-},\quad T_+\to\boldsymbol{+}.$$

9.15. **Convert grammars to Greibach Normal Form.**

(a) $\langle\{S_1,S_2\},\{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c},\boldsymbol{d},\boldsymbol{e}\},S_1,\{S_1\to S_2S_1\boldsymbol{e},S_1\to S_2\boldsymbol{b},S_2\to S_1S_2,S_2\to\boldsymbol{c}\}\rangle$:
One equivalent GNF grammar is:

$$\begin{aligned}
S_1 &\to \boldsymbol{c}S_1\boldsymbol{e}\mid\boldsymbol{cb}\mid\boldsymbol{c}S_1\boldsymbol{e}R\mid\boldsymbol{cb}R,\\
S_2 &\to \boldsymbol{c}\mid\boldsymbol{c}S_1\boldsymbol{e}S_2\mid\boldsymbol{cb}S_2\mid\boldsymbol{c}S_1\boldsymbol{e}RS_2\mid\boldsymbol{cb}RS_2,\\
R &\to \boldsymbol{c}S_1\boldsymbol{e}\mid\boldsymbol{cb}\mid\boldsymbol{c}S_1\boldsymbol{e}R\mid\boldsymbol{cb}R\\
&\qquad\mid\boldsymbol{c}S_1\boldsymbol{e}S_2S_1\boldsymbol{e}\mid\boldsymbol{cb}S_2S_1\boldsymbol{e}\mid\boldsymbol{c}S_1\boldsymbol{e}RS_2S_1\boldsymbol{e}\mid\boldsymbol{cb}RS_2S_1\boldsymbol{e}\\
&\qquad\mid\boldsymbol{c}S_1\boldsymbol{e}S_2\boldsymbol{b}\mid\boldsymbol{cb}S_2\boldsymbol{b}\mid\boldsymbol{c}S_1\boldsymbol{e}RS_2\boldsymbol{b}\mid\boldsymbol{cb}RS_2\boldsymbol{b}.
\end{aligned}$$

Every production now begins with the terminal $\boldsymbol{c}$, so this grammar is in GNF.

(b) $\langle\{S_1,S_2,S_3\},\ldots,\{S_1\to S_3S_1,S_1\to S_2\boldsymbol{a},S_2\to\boldsymbol{be},S_3\to S_2\boldsymbol{c}\}\rangle$:
$S_2\to\boldsymbol{be}$ (already in GNF). $S_3\to S_2\boldsymbol{c}\to\boldsymbol{bec}$ (substitute): $S_3\to\boldsymbol{bec}$. $S_1\to S_3S_1\to\boldsymbol{bec}S_1$ and $S_1\to S_2\boldsymbol{a}\to\boldsymbol{bea}=\boldsymbol{bea}$. GNF: $S_1\to\boldsymbol{bec}S_1\mid\boldsymbol{bea}$, $S_2\to\boldsymbol{be}$, $S_3\to\boldsymbol{bec}$.

(c) $\langle\{S_1,S_2,S_3\},\ldots,\{S_1\to S_1S_2\boldsymbol{c},S_1\to\boldsymbol{d}S_3,S_2\to S_1S_1,S_2\to\boldsymbol{a},S_3\to S_3\boldsymbol{e}\}\rangle$:
As printed, this grammar generates no terminal strings at all. The nonterminal S_3 has only the production $S_3\to S_3\boldsymbol{e}$, so S_3 is unproductive. Hence the production $S_1\to\boldsymbol{d}S_3$ cannot lead to a terminal word. The other production for S_1 is $S_1\to S_1S_2\boldsymbol{c}$, which still contains S_1 on the left, so it cannot terminate either. Therefore

$$L(G)=\emptyset.$$

An equivalent grammar in pure Greibach normal form is simply

$$\langle\{S_1\},\{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c},\boldsymbol{d},\boldsymbol{e}\},S_1,\emptyset\rangle.$$

9.16. **Prove there exists a CFG G'' equivalent to G' (with added $A\to\lambda$ rules).**

G' is obtained from G by adding rules $A\to\lambda$ for various nonterminals A. We want to show that there is an equivalent grammar G'' in which all of those lambda productions have been removed, except possibly the special start rule $Z\to\lambda$.

Proof: Apply the standard lambda-elimination construction to G'. First identify all *nullable* nonterminals, that is, all A for which $A\overset{*}{\Rightarrow}{}^{*}\ \lambda$ in G'. For each production in which nullable nonterminals occur on the right, add all productions obtained by deleting any chosen subset of those nullable occurrences, except that we do not create an empty right-hand side unless it is the special start production $Z\to\lambda$. After all such replacement productions have been added, delete the old productions $A\to\lambda$.

Every derivation in G' can now be simulated in the new grammar G'': whenever G' uses a nullable nonterminal and later erases it by $A \to \lambda$, the grammar G'' simply chooses the corresponding shortened production from the start. Conversely, every shortened production of G'' represents a derivation in G' in which the omitted nullable nonterminals are first produced and then erased. Thus G'' generates exactly the same language as G', except that the only remaining lambda production, if needed, is the allowable special rule $Z \to \lambda$.

Therefore an equivalent grammar G'' always exists. □

9.17. **Generalization of Lemma 9.1.**

Lemma: If $G = \langle \Omega, \Sigma, S, P \rangle$, $X \to \gamma B \alpha \in P$, all B-rules are $\{B \to \beta_i\}$, and G' replaces $X \to \gamma B \alpha$ with $\{X_B \to \gamma \beta_i \alpha\}$, then $L(G) = L(G')$.

Proof: By induction on derivation length.

($L(G) \subseteq L(G')$): Any derivation in G that uses $X \to \gamma B \alpha$ followed later by $B \to \beta_i$ can be replaced in G' by the single production $X_B \to \gamma \beta_i \alpha$. All other derivations are unchanged.

($L(G') \subseteq L(G)$): Any production $X_B \to \gamma \beta_i \alpha$ in G' can be simulated in G by $X \to \gamma B \alpha$ followed by $B \to \beta_i$. □

9.18. $P = \{d^p \mid p$ **prime**$\}$ **is not context free.**

(a) **Pumping theorem proof**: Assume P is context free with pumping constant n. Take $s = \boldsymbol{d}^p$ where $p > n$ is prime. By the pumping lemma, $s = uvwxy$ with $|vwx| \le n$, $|vx| \ge 1$, and $uv^i wx^i y \in P$ for all $i \ge 0$.
Let $|vx| = m$ where $1 \le m \le n$. Then $uv^i wx^i y = \boldsymbol{d}^{p+(i-1)m}$. For $i = 0$: $\boldsymbol{d}^{p-m}$. For $i = 2$: $\boldsymbol{d}^{p+m}$.
Choose $i = p+1$: the resulting string has length $p + pm = p(1+m)$. Since $p \ge 2$ and $m \ge 1$, $p(1+m)$ is composite (product of two integers each ≥ 2). So $\boldsymbol{d}^{p(1+m)} \not{i}nP$, contradiction. □

(b) **Using nonregularity**: The language P is a language over the single-letter alphabet $\{\boldsymbol{d}\}$. If P were context free, then Theorem 9.15 would imply that P is regular, because every context-free language over a one-letter alphabet is regular. But P is known to be nonregular. This contradiction shows that P is not context free. □

9.19. $\Gamma = \{w\mathbf{2}w \mid w \in \{\mathbf{0}, \mathbf{1}\}^*\}$ **is not context free.**

Assume Γ is context free with pumping constant n. Choose $s = \mathbf{0}^n\mathbf{1}^n\mathbf{2}\mathbf{0}^n\mathbf{1}^n \in \Gamma$ (take $w = \mathbf{0}^n\mathbf{1}^n$, $|s| = 4n+1$).

By the pumping lemma: $s = uvwxy$, $|vwx| \le n$, $|vx| \ge 1$, all pumped strings in Γ.

Since $|vwx| \le n$, the substring vwx can meet at most one boundary of the five-block word

$$\mathbf{0}^n \mid \mathbf{1}^n \mid \mathbf{2} \mid \mathbf{0}^n \mid \mathbf{1}^n.$$

We consider the possibilities.

- If vwx does *not* contain the separator $\mathbf{2}$, then it lies entirely in the left half $\mathbf{0}^n\mathbf{1}^n$ or entirely in the right half $\mathbf{0}^n\mathbf{1}^n$, possibly crossing the internal boundary between $\mathbf{0}$'s and $\mathbf{1}$'s. Pumping changes one side of the separator but leaves the other side unchanged. Therefore the resulting word cannot be of the form $w'\mathbf{2}w'$.

- If vwx does contain the separator **2**, then there is only one occurrence of **2** in s. If **2** appears in vx, then pumping with $i = 0$ or $i = 2$ changes the number of **2**'s, so the pumped word is not in Γ.
- The only remaining possibility is that **2** lies in w but not in vx. Then pumping changes symbols on one side of **2** while leaving the other side and the separator itself unchanged. Again the two halves around the separator are no longer identical, so the pumped word is not in Γ.

In every case some pumped word lies outside Γ, contradicting the pumping lemma. Therefore Γ is not context free. □

9.20. $\Psi = \{ww \mid w \in \{\mathbf{0},\mathbf{1}\}^*\}$ **is not context free.**

Assume Ψ is context free with pumping constant n. Choose $s = \mathbf{0}^n\mathbf{1}^n\mathbf{0}^n\mathbf{1}^n \in \Psi$ (take $w = \mathbf{0}^n\mathbf{1}^n$). $|s| = 4n$.

By the pumping lemma: $s = uvwxy$, $|vwx| \le n$, $|vx| \ge 1$, all pumped strings in Ψ.

Since $|vwx| \le n$, vwx is contained within a span of n consecutive characters. The string s has the pattern $0^n1^n0^n1^n$. The middle two blocks meet at position $2n$; the window vwx cannot span from the first 0^n-block to the second 0^n-block.

In all cases, pumping (or de-pumping with $i = 0$) changes the length or composition of one half of s without matching change in the other half, so the result cannot have the form $w'w'$. Contradiction. □

9.21. $\Xi = \{\boldsymbol{b}^{2^j} \mid j \in \mathbb{N}\}$ **is not context free.**

Assume Ξ is context free with pumping constant n. Choose j such that $2^j > n$, and let $s = \boldsymbol{b}^{2^j}$.

By the pumping lemma: $s = uvwxy$, $|vwx| \le n$, $|vx| = m \ge 1$. Then $|uv^iwx^iy| = 2^j + (i-1)m$ for each $i \ge 0$.

For $i = 2$: length $= 2^j + m$. We need $2^j + m$ to be a power of 2. But $2^j < 2^j + m \le 2^j + n < 2^{j+1}$ (since $m \le n < 2^j$). So $2^j + m$ is strictly between 2^j and 2^{j+1}, which means it is not a power of 2. Contradiction. □

9.22. $\Phi = \{\boldsymbol{a}^{j^2} \mid j \in \mathbb{N}\}$ **is not context free; application to Φ'.**

(a) Φ **is not context free**: Assume Φ is context free with pumping constant n. Let $j > n$; take $s = \boldsymbol{a}^{j^2}$.

By the pumping lemma: $s = uvwxy$, $|vwx| \le n$, $|vx| = m \ge 1$. Then pumped strings have length $j^2 + (i-1)m$. For $i = 2$: length $j^2 + m$. We need $j^2 + m$ to be a perfect square. Choose $j > n/2$ so that $n < 2j$; then $j^2 < j^2 + m \le j^2 + n < j^2 + 2j + 1 = (j+1)^2$. So $j^2 + m$ is strictly between two consecutive perfect squares, hence not a perfect square. Contradiction. □

(b) Φ' **not context free via homomorphism**: Define $h : \{\boldsymbol{b},\boldsymbol{c},\boldsymbol{d}\}^* \to \{\boldsymbol{a}\}^*$ by $h(\boldsymbol{c}) = \boldsymbol{a}$ and $h(\boldsymbol{b}) = h(\boldsymbol{d}) = \lambda$. If Φ' were a CFL, then $h(\Phi') = \Phi$ would also be a CFL (CFLs are closed under homomorphism), contradicting part (a). Hence $\Phi' \not{i}n\mathfrak{C}_\Sigma$.

(c) **Direct Ogden's lemma proof**: Assume Φ' is context free, and let p be the constant from Ogden's lemma. Choose the word

$$s = \boldsymbol{b}\boldsymbol{c}^{p^2}\boldsymbol{d}^p \in \Phi',$$

and mark all p^2 occurrences of $\boldsymbol{c}$. Ogden's lemma gives a decomposition $s = uvwxy$ such that vx contains at least one marked position, and vwx contains at most p marked positions.

Let $\alpha = |vx|_{\boldsymbol{c}}$ and $\beta = |vx|_{\boldsymbol{d}}$. Because vx contains a marked position, $\alpha \geq 1$. Also $\alpha \leq p$ and $\beta \leq p$. Pump once upward, to uv^2wx^2y. Its number of $\boldsymbol{c}$'s is $p^2 + \alpha$, while its number of $\boldsymbol{d}$'s is $p + \beta$.

If $\beta = 0$, then the number of $\boldsymbol{d}$'s stays equal to p while the number of $\boldsymbol{c}$'s changes, so the relation $|x|_{\boldsymbol{c}} = (|x|_{\boldsymbol{d}})^2$ fails.

If $\beta \geq 1$, then

$$(p+\beta)^2 - p^2 = 2p\beta + \beta^2 > p \geq \alpha,$$

so

$$p^2 + \alpha < (p+\beta)^2.$$

Again the relation $|x|_{\boldsymbol{c}} = (|x|_{\boldsymbol{d}})^2$ fails. In either case, $uv^2wx^2y \not{i}n\Phi'$, contradicting Ogden's lemma. Therefore Φ' is not context free.

(d) **Pumping theorem**: *Yes.* The ordinary pumping theorem also works. Assume Φ' is context free, and let n be its pumping constant. Take

$$s = \boldsymbol{b}\boldsymbol{c}^{n^2}\boldsymbol{d}^n \in \Phi'.$$

Write $s = uvwxy$ with $|vwx| \leq n$ and $|vx| \geq 1$.

If vx contains the initial $\boldsymbol{b}$, then pumping down ($i = 0$) removes that only $\boldsymbol{b}$, so $uv^0wx^0y \not{i}n\Phi'$.

Otherwise, vx lies entirely in the $\boldsymbol{c}^*\boldsymbol{d}^*$ portion of the word. Let $\alpha = |vx|_{\boldsymbol{c}}$ and $\beta = |vx|_{\boldsymbol{d}}$. Then $\alpha + \beta \geq 1$ and $\alpha + \beta \leq n$.

If $\alpha = 0$, pumping up changes the number of $\boldsymbol{d}$'s while leaving the number of $\boldsymbol{c}$'s fixed, so the quadratic relation fails.

If $\beta = 0$, pumping up changes the number of $\boldsymbol{c}$'s while leaving the number of $\boldsymbol{d}$'s fixed, so the quadratic relation fails.

If both α and β are positive, then pumping up gives $n^2 + \alpha$ occurrences of $\boldsymbol{c}$ and $n + \beta$ occurrences of $\boldsymbol{d}$. But

$$(n+\beta)^2 - n^2 = 2n\beta + \beta^2 > n \geq \alpha,$$

so $n^2 + \alpha \neq (n+\beta)^2$. Thus the pumped word is not in Φ'.

In every case some pumped word lies outside Φ', contradicting the pumping theorem. So it is possible to prove directly from the pumping theorem that Φ' is not context free.

9.23. $L = \{y \in \{\mathbf{0},\mathbf{1}\}^* \mid |y|_{\mathbf{0}} = |y|_{\mathbf{1}}\}$ **is context free.**

Grammar: $S \to \mathbf{0}S\mathbf{1} \mid \mathbf{1}S\mathbf{0} \mid SS \mid \lambda$.

Proof: By induction, any string with equal numbers of **0**s and **1**s is derivable, and all derivations yield strings with equal counts. So $L(G) = L$. □

9.24. **Formal recursive definition of the path in Theorem 9.9 (pumping lemma).**

(a) Let the parse tree have root r corresponding to start symbol S. Define a *leftmost longest path* recursively: - **Boundary**: If r is a leaf, the path is $\{r\}$. - **Rule**: From a node v with children $c_1, c_2, \ldots, c_k$, choose c_i to be the child whose subtree has the greatest number of leaves. (Break ties by choosing the leftmost.) Append c_i to the path, and recurse from c_i.

(b) The depth of this path is at least $\log_2$ (number of leaves) $= \log_2 |x|$ (by the binary branching property of CNF). If $|x| \geq b^{h+1}$ where b is the maximum branching factor and $h = |\Omega|$, then the path has length $> h$, so some nonterminal repeats on the path. Let A be the repeated nonterminal at positions $i < j$ on the path. Then the subtree at position j generates w, and the subtree at position i generates vwx (with v, x possibly empty but $|vx| \geq 1$ by the choice that $vwx \neq w$). Pumping follows from substituting the subtree at j for the subtree at i. $\square$

9.25. **$\mathfrak{C}_\Sigma$ is closed under $\cup$ via direct grammar construction.**

Let $G_1 = \langle \Omega_1, \Sigma, S_1, P_1 \rangle$ and $G_2 = \langle \Omega_2, \Sigma, S_2, P_2 \rangle$ with $\Omega_1 \cap \Omega_2 = \emptyset$. Choose $Z \not{\in} n \Omega_1 \cup \Omega_2$ and define:

$$G^\cup = \langle \Omega_1 \cup \Omega_2 \cup \{Z\}, \Sigma, Z, P_1 \cup P_2 \cup \{Z \to S_1, Z \to S_2\} \rangle.$$

Then $w \in L(G^\cup)$ iff $Z \overset{*}{\Rightarrow} w$ iff $Z \Rightarrow S_i \overset{*}{\Rightarrow} w$ for $i = 1$ or $i = 2$ iff $w \in L(G_1) \cup L(G_2)$. $\square$

9.26. **Closure properties of $\mathfrak{X}_\Sigma$ (languages that are NOT context free).**

(a) **Union**: Not closed. Let

$$A = \{\boldsymbol{a}^{2^j} \mid j \in \mathbb{N}\}.$$

Exercise 9.21 shows that A is not context free. Let $B = \boldsymbol{a}^* - A$. If B were context free, then by Theorem 9.15 it would be regular, and therefore $A = \boldsymbol{a}^* - B$ would also be regular, contrary to Exercise 9.21. Hence $B \in \mathfrak{X}_\Sigma$ as well. But

$$A \cup B = \boldsymbol{a}^*,$$

which is regular, hence context free. Therefore $\mathfrak{X}_\Sigma$ is not closed under union.

(b) **Complement**: Not closed. Lemma 9.4 gives a context-free language L whose complement is not context free. Let $X = \overline{L}$. Then $X \in \mathfrak{X}_\Sigma$, but $\overline{X} = L \in \mathfrak{C}_\Sigma$. So $\mathfrak{X}_\Sigma$ is not closed under complement.

(c) **Intersection**: Not closed. Using the same unary languages A and $B = \boldsymbol{a}^* - A$ from part (a), both A and B belong to $\mathfrak{X}_\Sigma$, but

$$A \cap B = \emptyset,$$

which is context free. Hence $\mathfrak{X}_\Sigma$ is not closed under intersection.

(d) **Kleene closure**: Not closed. For the same language $A = \{\boldsymbol{a}^{2^j} \mid j \in \mathbb{N}\}$, we have $A \in \mathfrak{X}_\Sigma$. Since $\boldsymbol{a} = \boldsymbol{a}^{2^0} \in A$, every unary word belongs to A^*, and therefore

$$A^* = \boldsymbol{a}^*.$$

Thus A^* is regular, so $\mathfrak{X}_\Sigma$ is not closed under Kleene closure.

(e) **Concatenation**: Not closed. Let

$$A' = A \cup \{\lambda\}, \qquad B' = B \cup \{\lambda\},$$

where A and B are as in part (a). If A' were context free, then by Theorem 9.15 it would be regular; removing the single word λ would show that A is regular, contradiction. So $A' \in \mathfrak{X}_\Sigma$, and similarly $B' \in \mathfrak{X}_\Sigma$.

Now every word $\boldsymbol{a}^n$ belongs to $A'B'$: if $\boldsymbol{a}^n \in A$, then $\boldsymbol{a}^n = \boldsymbol{a}^n \lambda$ with $\boldsymbol{a}^n \in A'$ and $\lambda \in B'$; if $\boldsymbol{a}^n \not{\in} n A$, then $\boldsymbol{a}^n \in B$ and $\boldsymbol{a}^n = \lambda \boldsymbol{a}^n$ with $\lambda \in A'$ and $\boldsymbol{a}^n \in B'$. Hence $A'B' = \boldsymbol{a}^*$, which is context free. Therefore $\mathfrak{X}_\Sigma$ is not closed under concatenation.

9.27. $\mathfrak{C}_\Sigma$ **is closed under reversal.**

Let $G = \langle \Omega, \Sigma, S, P\rangle$ generate L. Define $G^r = \langle \Omega, \Sigma, S, P^r\rangle$ where $P^r = \{A \to \alpha^r \mid A \to \alpha \in P\}$ (reverse the right-hand side of each production).

Claim: $L(G^r) = L^r$.

Proof: We prove a stronger statement: for every nonterminal $A \in \Omega$ and every terminal string $x \in \Sigma^*$,

$$A \overset{*\ *}{\Rightarrow}_G x \quad \Longleftrightarrow \quad A \overset{*\ *}{\Rightarrow}_{G^r} x^r.$$

We prove the forward implication by induction on the derivation length of $A \overset{*\ *}{\Rightarrow}_G x$.

Base: If the derivation has one step, then $A \to x$ is a production of G. By definition, $A \to x^r$ is a production of G^r, so $A \overset{*}{\Rightarrow}_{G^r} x^r$.

Step: Suppose the first production is

$$A \to X_1 X_2 \cdots X_k$$

in G, where each X_i is either a terminal or a nonterminal, and suppose the rest of the derivation yields

$$X_1 X_2 \cdots X_k \overset{*\ *}{\Rightarrow}_G x_1 x_2 \cdots x_k = x,$$

where each x_i is the terminal string ultimately produced from X_i (possibly a single terminal if $X_i \in \Sigma$). By the induction hypothesis, each nonterminal X_i derives x_i^r in G^r, while terminals already equal their own one-letter reversals. Since G^r contains the production

$$A \to X_k X_{k-1} \cdots X_1,$$

we obtain

$$A \Rightarrow_{G^r} X_k X_{k-1} \cdots X_1 \overset{*\ *}{\Rightarrow}_{G^r} x_k^r x_{k-1}^r \cdots x_1^r = (x_1 x_2 \cdots x_k)^r = x^r.$$

The reverse implication follows by symmetry, since $(G^r)^r = G$. Taking $A = S$, we conclude that

$$x \in L(G) \quad \Longleftrightarrow \quad x^r \in L(G^r),$$

so $L(G^r) = L^r$. □

9.28. $\mathfrak{C}_\Sigma$ **is closed under operator** T **(**$T(L) = \{w^r w \mid w \in L\}$**).**

Claim: T does not preserve $\mathfrak{C}_\Sigma$ in general; i.e., there exists $L \in \mathfrak{C}_\Sigma$ such that $T(L) \not{\imath}n\mathfrak{C}_\Sigma$.

Counterexample: Let

$$L = \{\boldsymbol{a}^n \boldsymbol{b} \boldsymbol{a}^n \mid n \ge 0\}.$$

This language is context free; for example, it is generated by $S \to \boldsymbol{a}S\boldsymbol{a} \mid \boldsymbol{b}$. Since every word of L is a palindrome, we have

$$T(L) = \{\boldsymbol{a}^n \boldsymbol{b} \boldsymbol{a}^{2n} \boldsymbol{b} \boldsymbol{a}^n \mid n \ge 0\}.$$

We claim $T(L) \not{\imath}n\mathfrak{C}_\Sigma$. Suppose instead that $T(L)$ is context free, and let N be its pumping constant. Choose

$$s = \boldsymbol{a}^N \boldsymbol{b} \boldsymbol{a}^{2N} \boldsymbol{b} \boldsymbol{a}^N \in T(L).$$

Write $s = uvwxy$ with $|vwx| \le N$ and $|vx| \ge 1$.

Because $|vwx| \le N$, the substring vwx can contain at most one of the two $\boldsymbol{b}$'s. Thus it lies entirely within one block, or else it crosses just one boundary between adjacent blocks.

- If vwx lies entirely within the leftmost $\boldsymbol{a}^N$ block, then pumping changes only that block. The rightmost $\boldsymbol{a}$-block remains of length N, so the pumped word cannot have the form $\boldsymbol{a}^n\boldsymbol{b}\boldsymbol{a}^{2n}\boldsymbol{b}\boldsymbol{a}^n$.
- If vwx lies entirely within the middle $\boldsymbol{a}^{2N}$ block, then pumping changes only the middle block. The outer blocks still both have length N, so the middle block should still have length $2N$, but it does not.
- If vwx lies entirely within the rightmost $\boldsymbol{a}^N$ block, the argument is symmetric to the first case.
- If vwx contains one of the $\boldsymbol{b}$'s, then pumping with $i = 0$ or $i = 2$ either changes the number of $\boldsymbol{b}$'s or changes one adjacent $\boldsymbol{a}$-block while leaving the other outer $\boldsymbol{a}$-block unchanged. In either event, the pumped word is not of the form $\boldsymbol{a}^n\boldsymbol{b}\boldsymbol{a}^{2n}\boldsymbol{b}\boldsymbol{a}^n$.

Every possible placement of vwx leads to a contradiction. Therefore $T(L)$ is not context free, and $\mathfrak{C}_\Sigma$ is not closed under T. □

9.29. $\mathfrak{C}_{\{\boldsymbol{a},\boldsymbol{b}\}}$ **is not closed under relative complement.**

Disproof: Let $L_1 = \{x \in \{\boldsymbol{a},\boldsymbol{b}\}^* \mid |x|_{\boldsymbol{a}} = |x|_{\boldsymbol{b}}\}$ (CFL) and $L_2 = \{\boldsymbol{a}\}^*\{\boldsymbol{b}\}^*$ (regular, hence CFL). Then:

$$L_1 - L_2 = \{x \mid |x|_{\boldsymbol{a}} = |x|_{\boldsymbol{b}}\} - \{\boldsymbol{a}^n\boldsymbol{b}^n\} = \{x \mid |x|_{\boldsymbol{a}} = |x|_{\boldsymbol{b}}, x \not{in} \boldsymbol{a}^*\boldsymbol{b}^*\}.$$

This language (balanced strings that are not in $\boldsymbol{a}^*\boldsymbol{b}^*$) is context free (construct a grammar). So this does not give a counterexample.

A better approach: use the fact that if $\mathfrak{C}_\Sigma$ were closed under relative complement and we know $\mathfrak{C}_\Sigma$ is closed under union, then $\mathfrak{C}_\Sigma$ would be closed under complement (since $\overline{L_1} = \Sigma^* - L_1$, and Σ^* is CFL). But Lemma 9.4 shows that $\mathfrak{C}_\Sigma$ is not closed under complement. Hence $\mathfrak{C}_\Sigma$ is *not* closed under relative complement. □

9.30. $\mathfrak{C}_\Sigma$ **is not closed under intersection or complement.**

(a) Let $L_1 = \{\boldsymbol{a}^i\boldsymbol{b}^i\boldsymbol{c}^j \mid i,j \in \mathbb{N}\}$ and $L_2 = \{\boldsymbol{a}^i\boldsymbol{b}^j\boldsymbol{c}^j \mid i,j \in \mathbb{N}\}$. Both are context free. Their intersection:

$$L_1 \cap L_2 = \{\boldsymbol{a}^n\boldsymbol{b}^n\boldsymbol{c}^n \mid n \in \mathbb{N}\},$$

which is not context free (by the pumping lemma). So $\mathfrak{C}_\Sigma$ is not closed under intersection.

If $\mathfrak{C}_\Sigma$ were closed under complement, then $\overline{L_1}$ and $\overline{L_2}$ would be context free, and $\overline{\overline{L_1} \cup \overline{L_2}} = L_1 \cap L_2$ would be context free (by De Morgan and closure under union and complement). But $L_1 \cap L_2$ is not CFL. Contradiction. So $\mathfrak{C}_\Sigma$ is not closed under complement.

(b) For $|\Sigma| > 1$, choose two distinct symbols of Σ and call them $\boldsymbol{a}$ and $\boldsymbol{b}$. Consider the languages

$$L_1 = \{\boldsymbol{a}^i\boldsymbol{b}^j\boldsymbol{a}^j \mid i,j \in \mathbb{N}\}, \qquad L_2 = \{\boldsymbol{a}^i\boldsymbol{b}^i\boldsymbol{a}^j \mid i,j \in \mathbb{N}\}.$$

Both are context free: L_1 is generated by $S \to AB$, $A \to \boldsymbol{a}A \mid \lambda$, $B \to \boldsymbol{b}B\boldsymbol{a} \mid \lambda$, and L_2 is generated by $S \to CD$, $C \to \boldsymbol{a}C\boldsymbol{b} \mid \lambda$, $D \to \boldsymbol{a}D \mid \lambda$.

But

$$L_1 \cap L_2 = \{\boldsymbol{a}^n\boldsymbol{b}^n\boldsymbol{a}^n \mid n \in \mathbb{N}\},$$

which is not context free by the pumping lemma. Therefore $\mathfrak{C}_\Sigma$ is not closed under intersection whenever $|\Sigma| > 1$.

If $\mathfrak{C}_\Sigma$ were closed under complement, then by De Morgan's law it would also be closed under intersection, since it is already closed under union. This contradiction shows that $\mathfrak{C}_\Sigma$ is not closed under complement either. □

9.31. **Automaton-based method for Theorem 9.3 (epsilon elimination).**

Consider a pushdown automaton (PDA) with states labeled by subsets $T \subseteq \Omega$ (the power set of nonterminals). Start state: $\{S\}$. Transitions: from state T, for each nonterminal $A \in T$ and production $A \to \gamma$, compute the set of nonterminals derivable in one step from γ, and union these into the new state.

This is similar to the subset construction for NFAs, adapted for CFGs. The set T represents all nonterminals reachable in derivations. The nullable nonterminals (N_λ) are those states from which λ is reachable.

9.32. **Grammars in GNF with at most two symbols on the right.**

(a) This is not a normal form for all CFLs: The language $\{\boldsymbol{a}^n \mid n \geq 1\}$ can be generated (e.g., $S \to \boldsymbol{a}S \mid \boldsymbol{a}$), but more complex CFLs like $\{ww^r\}$ require productions of the form $S \to \boldsymbol{a}S\boldsymbol{a}$ in GNF which has 3 symbols on the right (more than 2). Converting to this restricted GNF may require more nonterminals and fail to represent the language if we strictly limit to 2 symbols. In fact $\boldsymbol{a}Xa$ (where a is terminal and X is nonterminal) has 3 symbols; with at most 2 on the right in GNF (first terminal, at most one nonterminal), we can only write $\boldsymbol{a}X$ (dropping the trailing $\boldsymbol{a}$). So $\{ww^r \mid w \in \{\boldsymbol{a},\boldsymbol{b}\}^*\}$ cannot be generated.

(b) Languages generated by this restricted form: GNF with at most two RHS symbols means productions are $A \to \boldsymbol{a}$ or $A \to \boldsymbol{a}B$. These are exactly the *right-linear grammars*, which generate precisely the *regular languages* ($\mathfrak{D}_\Sigma = \mathfrak{G}_\Sigma = \mathfrak{R}_\Sigma$). □

9.33. **Prove L^* must be regular for *any* collection L over Σ.**

The statement is *false* as written.

Take

$$L = \{\boldsymbol{a}^n\boldsymbol{b}^n \mid n \geq 1\}.$$

Then

$$L^* = \{x_1x_2\cdots x_k \mid k \geq 0,\ x_i \in L\}.$$

Assume for contradiction that L^* is regular. The language $\boldsymbol{a}^*\boldsymbol{b}^*$ is regular, so

$$L^* \cap \boldsymbol{a}^*\boldsymbol{b}^*$$

would also be regular.

But

$$L^* \cap \boldsymbol{a}^*\boldsymbol{b}^* = L.$$

Indeed, any concatenation of two or more nonempty words from L has at least two alternations between $\boldsymbol{a}$'s and $\boldsymbol{b}$'s, so it cannot lie in $\boldsymbol{a}^*\boldsymbol{b}^*$. Thus the only words of L^* that belong to $\boldsymbol{a}^*\boldsymbol{b}^*$ are the single-block words $\boldsymbol{a}^n\boldsymbol{b}^n$.

Therefore L would be regular, contrary to the standard result that $\{\boldsymbol{a}^n\boldsymbol{b}^n \mid n \geq 1\}$ is not regular. So L^* need not be regular.

9.34. **If $|\Sigma| = 1$, is $\mathfrak{C}_\Sigma$ closed under complementation?**

Yes. If $|\Sigma| = 1$, say $\Sigma = \{\boldsymbol{a}\}$, and $L \in \mathfrak{C}_\Sigma$, then Theorem 9.15 implies that L is regular. The regular languages are closed under complementation, so $\overline{L}$ is also regular. Every regular language is context free, hence $\overline{L} \in \mathfrak{C}_\Sigma$.

Therefore $\mathfrak{C}_\Sigma$ is closed under complementation when $|\Sigma| = 1$. □

9.35. $\{\boldsymbol{a}^n\boldsymbol{b}^n\boldsymbol{c}^m \mid n, m \in \mathbb{N}\}$ **is context free.**

Grammar: $S \to AB$, $A \to \boldsymbol{a}A\boldsymbol{b} \mid \lambda$, $B \to \boldsymbol{c}B \mid \lambda$.

A generates $\{\boldsymbol{a}^n\boldsymbol{b}^n\}$ and B generates $\boldsymbol{c}^*$. So $L(G) = \{\boldsymbol{a}^n\boldsymbol{b}^n\boldsymbol{c}^m\}$. □

9.36. **Use Ogden's lemma to prove** $\{\boldsymbol{a}^k\boldsymbol{b}^n\boldsymbol{c}^m \mid k \neq n \wedge n \neq m\}$ **is not context free.**

Assume $L = \{\boldsymbol{a}^k\boldsymbol{b}^n\boldsymbol{c}^m \mid k \neq n, n \neq m\}$ is context free with Ogden constant N.

Choose $p > N$ and let $s = \boldsymbol{a}^{p!+p}\boldsymbol{b}^p\boldsymbol{c}^{p!+p} \in L$ (since $p! + p \neq p$ and $p \neq p! + p$). Mark all p occurrences of $\boldsymbol{b}$ in s.

By Ogden's lemma: $s = uvwxy$ where vwx contains at most $N \leq p$ marked positions, vx contains at least 1 marked position, and all pumped strings $uv^i wx^i y \in L$.

Let $m = |vx|_{\boldsymbol{b}} \geq 1$ (the number of $\boldsymbol{b}$s in vx; at least 1 since vx contains a marked symbol). Choose $i - 1 = p!/m$, which is a positive integer since $m \leq N < p$ implies $m \leq p - 1$ and hence $m \mid p!$. The pumped string has $\boldsymbol{b}$-count equal to $p + (i-1)m = p + p! = p! + p$.

Since $|vwx| \leq N < p$ and the $\boldsymbol{b}$-block has exactly p symbols, the window vwx cannot reach both the $\boldsymbol{a}$-block on the left and the $\boldsymbol{c}$-block on the right simultaneously. Let $s_a = |vx|_{\boldsymbol{a}}$ and $s_c = |vx|_{\boldsymbol{c}}$. Then either $s_a = 0$ (window does not touch the $\boldsymbol{a}$-block) or $s_c = 0$ (window does not touch the $\boldsymbol{c}$-block).

- If $s_a = 0$: the $\boldsymbol{a}$-count remains $p! + p = \boldsymbol{b}$-count, so $k = n = p! + p$, violating $k \neq n$. Contradiction.
- If $s_c = 0$: the $\boldsymbol{c}$-count remains $p! + p = \boldsymbol{b}$-count, so $n = m = p! + p$, violating $n \neq m$. Contradiction.

In every case the pumped string lies outside L, contradicting Ogden's lemma. □

Chapter 10

Pushdown Automata — Solutions

10.1. **Show by induction:** $\overline{\delta}(s_0, x) = t \Leftrightarrow \langle s_0, x, \text{¢}\rangle \overset{*}{\vdash} \langle t, \lambda, \text{¢}\rangle$**.**

Claim ($P(n)$): For all $x \in \Sigma^n$, $\overline{\delta}(s_0, x) = t$ iff $\langle s_0, x, \text{¢}\rangle \overset{*}{\vdash} \langle t, \lambda, \text{¢}\rangle$.

Base ($n = 0$): $\overline{\delta}(s_0, \lambda) = s_0$ and $\langle s_0, \lambda, \text{¢}\rangle \overset{*}{\vdash} \langle s_0, \lambda, \text{¢}\rangle$ (zero moves). Both sides give $t = s_0$. ✓

Step: Assume $P(n)$. For $x = ya$ ($|y| = n$, $a \in \Sigma$):

($\Rightarrow$): $\overline{\delta}(s_0, ya) = \delta(\overline{\delta}(s_0, y), a) = t$. Let $q = \overline{\delta}(s_0, y)$. By IH, $\langle s_0, y, \text{¢}\rangle \overset{*}{\vdash} \langle q, \lambda, \text{¢}\rangle$. By Theorem 10.1, $\langle q, a, \text{¢}\rangle \vdash \langle t, \lambda, \text{¢}\rangle$ (single step reading a). So $\langle s_0, ya, \text{¢}\rangle \overset{*}{\vdash} \langle t, \lambda, \text{¢}\rangle$.

($\Leftarrow$): If $\langle s_0, ya, \text{¢}\rangle \overset{*}{\vdash} \langle t, \lambda, \text{¢}\rangle$, then the computation passes through some state q after reading y: $\langle s_0, y, \text{¢}\rangle \overset{*}{\vdash} \langle q, \lambda, \text{¢}\rangle$. By IH, $\overline{\delta}(s_0, y) = q$. The last step reads a from state q to reach t, so $\delta(q, a) = t$, hence $\overline{\delta}(s_0, ya) = t$. □

10.2. **Define a one-state DPDA P'_1 accepting $\{\boldsymbol{a}^n\boldsymbol{b}^n \mid n \geq 0\}$ via final state.**

A one-state DPDA accepting via final state always accepts λ: the unique state must be both the initial state and the only final state, so the machine is in a final state at the very start with empty input. Hence the best a one-state final-state DPDA can achieve is $\{\boldsymbol{a}^n\boldsymbol{b}^n \mid n \geq 0\}$ (which includes λ).

Let $P'_1 = \langle\{\boldsymbol{a}, \boldsymbol{b}\}, \{Z, A\}, \{t\}, t, \delta, Z, \{t\}\rangle$ where:

$$\delta(t, \boldsymbol{a}, Z) = \langle t, AZ\rangle \quad \text{(push } A \text{ on reading first } \mathbf{a}\text{)}$$
$$\delta(t, \boldsymbol{a}, A) = \langle t, AA\rangle \quad \text{(push another } A \text{ for each subsequent } \mathbf{a}\text{)}$$
$$\delta(t, \boldsymbol{b}, A) = \langle t, \lambda\rangle \quad \text{(pop } A \text{ on reading } \mathbf{b}\text{)}$$

The machine pushes one A for each $\boldsymbol{a}$ read, then pops one A for each $\boldsymbol{b}$ read. It accepts when the input is exhausted in state t: this occurs precisely when the $\boldsymbol{a}$s and $\boldsymbol{b}$s are balanced (stack returned to Z) or when the input is empty. Note: $\delta(t, \boldsymbol{b}, Z)$ is undefined, so a string with more $\boldsymbol{b}$s than $\boldsymbol{a}$s causes the machine to get stuck (reject). Hence $\Lambda(P'_1) = \{\boldsymbol{a}^n\boldsymbol{b}^n \mid n \geq 0\}$.

10.3. **PDA P'_2: proofs and language.**

$P'_2 = \langle\{\boldsymbol{a}, \boldsymbol{b}\}, \{S, C\}, \{t\}, t, \delta, S, \{t\}\rangle$ with $\delta(t, \boldsymbol{a}, S) = \{\langle t, SC\rangle, \langle t, C\rangle\}$, $\delta(t, \boldsymbol{b}, C) = \{\langle t, \lambda\rangle\}$, others empty.

(a) **Inductive proof**: $(\forall i \in \mathbb{N})$: if $\langle t, \boldsymbol{a}^i, S\rangle \overset{*}{\vdash} \langle t, \lambda, \alpha\rangle$ then $\alpha = SC^i$ or $\alpha = C^i$.

Base ($i = 0$): $\langle t, \lambda, S\rangle \overset{*}{\vdash} \langle t, \lambda, S\rangle$; here $\alpha = S = SC^0$. ✓

Step: Assume the result for i. Reading $\boldsymbol{a}^{i+1}$ from S: first move on $(\boldsymbol{a}, S)$ either pushes SC or C.

Case 1: Push SC. Then $\langle t, \boldsymbol{a}^{i+1}, S\rangle \vdash \langle t, \boldsymbol{a}^i, SC\rangle$. Now process $\boldsymbol{a}^i$ from S: by IH, $\langle t, \boldsymbol{a}^i, S\rangle \overset{*}{\vdash} \langle t, \lambda, \beta\rangle$ where $\beta \in \{SC^i, C^i\}$. So $\langle t, \boldsymbol{a}^i, SC\rangle \overset{*}{\vdash} \langle t, \lambda, \beta C\rangle \in \{SC^{i+1}, C^{i+1}\}$.

Case 2: Push C. Then $\langle t, \boldsymbol{a}^{i+1}, S\rangle \vdash \langle t, \boldsymbol{a}^i, C\rangle$. Processing $\boldsymbol{a}^i$ from C: there are no moves (neither $\delta(t, \boldsymbol{a}, C)$ nor $\delta(t, \lambda, C)$ are defined for reading $\boldsymbol{a}$), so this branch is a dead end unless $i = 0$.

So $\alpha \in \{SC^{i+1}, C^{i+1}\}$. □

(b) **Inductive proof**: $(\forall i \in \mathbb{N})$: if $\langle t, x, C^i\rangle \overset{*}{\vdash} \langle t, \lambda, \beta\rangle$ then $x = \boldsymbol{b}^i$.

Base ($i = 0$): $\langle t, x, \lambda\rangle$ can only have $x = \lambda$ (stack empty, no moves possible). So $x = \lambda = \boldsymbol{b}^0$. ✓

Step: Assume for i. From $\langle t, x, C^{i+1}\rangle$: the top of the stack is C. The only move consuming C reads $\boldsymbol{b}$: $\delta(t, \boldsymbol{b}, C) = \langle t, \lambda\rangle$. So x must start with $\boldsymbol{b}$: $x = \boldsymbol{b}x'$ and $\langle t, \boldsymbol{b}x', C^{i+1}\rangle \vdash \langle t, x', C^i\rangle$. By IH, $x' = \boldsymbol{b}^i$. Hence $x = \boldsymbol{b}^{i+1}$. □

(c) P_2' accepts by empty stack (the final-state component $\{t\}$ is not used for acceptance here; a string is accepted iff the machine empties its stack while in state t with empty input). From (a), processing $\boldsymbol{a}^n$ from start symbol S can yield stack C^n (by choosing $\delta(t, \boldsymbol{a}, S) = \langle t, SC\rangle$ at each of the first $n-1$ steps and then $\langle t, C\rangle$ at the last, so the final stack is C^n). From (b), the stack C^n is emptied by reading exactly $\boldsymbol{b}^n$. Thus:

$$L(P_2') = \{\boldsymbol{a}^n\boldsymbol{b}^n \mid n \geq 1\}.$$

For $n = 0$: the initial configuration $\langle t, \lambda, S\rangle$ has S still on the stack with no moves available, so the stack is never emptied and $\lambda \not\!\!\;\; nL(P_2')$. Hence $L(P_2') = \{\boldsymbol{a}^n\boldsymbol{b}^n \mid n \geq 1\}$.

10.4. $L = \{\boldsymbol{a}^i\boldsymbol{b}^j\boldsymbol{c}^k \mid i, j, k \in \mathbb{N},\ i + j = k\}$**.**

(a) **PDA via final state**: Use a single counting symbol A: push one A for each $\boldsymbol{a}$ or $\boldsymbol{b}$, then pop one A for each $\boldsymbol{c}$. A machine is

$$P = \langle\{\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}\}, \{Z, A\}, \{q_0, q_1, q_2, q_f\}, q_0, \delta, Z, \{q_0, q_f\}\rangle,$$

where the nonempty transitions are

$$\begin{array}{ll}
\delta(q_0, \boldsymbol{a}, Z) = \langle q_0, AZ\rangle, & \delta(q_0, \boldsymbol{a}, A) = \langle q_0, AA\rangle, \\
\delta(q_0, \boldsymbol{b}, Z) = \langle q_1, AZ\rangle, & \delta(q_0, \boldsymbol{b}, A) = \langle q_1, AA\rangle, \\
\delta(q_0, \boldsymbol{c}, A) = \langle q_2, \lambda\rangle, & \delta(q_1, \boldsymbol{b}, A) = \langle q_1, AA\rangle, \\
\delta(q_1, \boldsymbol{c}, A) = \langle q_2, \lambda\rangle, & \delta(q_2, \boldsymbol{c}, A) = \langle q_2, \lambda\rangle, \\
\delta(q_2, \lambda, Z) = \langle q_f, Z\rangle. &
\end{array}$$

State q_0 handles the $\boldsymbol{a}$-block, q_1 the $\boldsymbol{b}$-block, and q_2 the $\boldsymbol{c}$-block. The string λ is accepted because q_0 is final; otherwise, acceptance occurs exactly when all pushed A's have been matched by $\boldsymbol{c}$'s.

(b) **PDA via empty stack**: Use the same three reading states and replace the last move by $\delta(q_2, \lambda, Z) = \langle q_2, \lambda\rangle$. To include λ, also allow $\delta(q_0, \lambda, Z) = \langle q_0, \lambda\rangle$.

(c) **Counting automaton**: Yes. In fact the PDA in part (a) already uses only one genuine counting symbol A, so it is a counting automaton.

(d) **DPDA**: Yes. At each stage there is only one possible action: push during the $\boldsymbol{a}$- and $\boldsymbol{b}$-blocks, then pop during the $\boldsymbol{c}$-block. Thus the machine above can be made deterministic.

(e) **Grammar from PDA** (using Definition 10.7): Applying Definition 10.7 to part (a) and deleting useless variables yields the simpler equivalent grammar

$$S \to \boldsymbol{a}S\boldsymbol{c} \mid T, \qquad T \to \boldsymbol{b}T\boldsymbol{c} \mid \lambda.$$

This generates exactly the strings $\boldsymbol{a}^i\boldsymbol{b}^j\boldsymbol{c}^{i+j}$.

10.5. $L = \{x \in \{\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}\}^* \mid |x|_{\boldsymbol{a}} + |x|_{\boldsymbol{b}} = |x|_{\boldsymbol{c}}\}$.

(a) **PDA via final state**: Use two stack symbols A (surplus of $\boldsymbol{a}/\boldsymbol{b}$) and B (surplus of $\boldsymbol{c}$) to handle symbols in any order.

P: single reading state q, final state f, bottom marker Z:

$$\begin{array}{lll}
\delta(q, \boldsymbol{a}, Z) = \langle q, AZ\rangle, & \delta(q, \boldsymbol{a}, A) = \langle q, AA\rangle, & \delta(q, \boldsymbol{a}, B) = \langle q, \lambda\rangle, \\
\delta(q, \boldsymbol{b}, Z) = \langle q, AZ\rangle, & \delta(q, \boldsymbol{b}, A) = \langle q, AA\rangle, & \delta(q, \boldsymbol{b}, B) = \langle q, \lambda\rangle, \\
\delta(q, \boldsymbol{c}, Z) = \langle q, BZ\rangle, & \delta(q, \boldsymbol{c}, B) = \langle q, BB\rangle, & \delta(q, \boldsymbol{c}, A) = \langle q, \lambda\rangle.
\end{array}$$

When $\boldsymbol{a}$ or $\boldsymbol{b}$ is read with B on top, the pending $\boldsymbol{c}$-surplus is cancelled, and vice versa. Accept by final state: $\delta(q, \lambda, Z) = \langle f, Z\rangle$ (fires when input exhausted and stack balanced).

(b) **PDA via empty stack**: Pop Z as well. Add $\delta(q, \lambda, Z) = \langle q, \lambda\rangle$ (the stack becomes empty when balanced).

(c) **Counting automaton**: Yes. One counter is enough to store the current difference between $|x|_{\boldsymbol{a}} + |x|_{\boldsymbol{b}}$ and $|x|_{\boldsymbol{c}}$, while the finite control records which side currently has the surplus.

(d) **DPDA**: Yes. The two-symbol construction in part (a) is deterministic: for each (q, σ, X) triple there is at most one transition, since every input symbol and every stack top symbol determine a unique action. The only potential conflict is the λ-transition $\delta(q, \lambda, Z) = \langle f, Z\rangle$, which must not coexist with other q-transitions on Z. This is resolved by moving the λ-acceptance to a separate final state reached only when input is exhausted; in practice, the construction is deterministic for all reachable configurations. Hence a DPDA for L exists.

(e) **Grammar from PDA**: Applying Definition 10.7 to part (a) and simplifying gives an equivalent grammar such as

$$S \to \boldsymbol{a}S\boldsymbol{c} \mid \boldsymbol{c}S\boldsymbol{a} \mid \boldsymbol{b}S\boldsymbol{c} \mid \boldsymbol{c}S\boldsymbol{b} \mid SS \mid \lambda.$$

Each production preserves the invariant $|x|_{\boldsymbol{a}}+|x|_{\boldsymbol{b}} = |x|_{\boldsymbol{c}}$, and every nonempty balanced word can be split at some point into one balanced word or into a balanced word bracketed by one of the four possible outer pairs.

10.6. **Closure of $\mathfrak{P}_\Sigma$ and $\mathscr{A}_\Sigma$ under inverse homomorphism.**

(a) $\mathfrak{P}_\Sigma$ **is closed under inverse homomorphism**: Given a context-free language L over Σ and a homomorphism $h : \Delta^* \to \Sigma^*$, we want $h^{-1}(L) = \{w \in \Delta^* \mid h(w) \in L\}$ to be a CFL.

Proof: Construct a PDA for $h^{-1}(L)$ from the PDA P for L. The new PDA P' uses a buffer: for each input symbol $a \in \Delta$, it replaces a with $h(a)$ in a buffer and then simulates P on $h(a)$ before reading the next input symbol. The new PDA processes $h(a)$ letter-by-letter from the buffer. Since h maps to a finite string, this is a finite process per input symbol. Thus P' accepts exactly $h^{-1}(L)$, which is a CFL.

(b) $\mathscr{A}_\Sigma$ **(DCFL) is closed under inverse homomorphism**: Given a DCFL L and a homomorphism h, $h^{-1}(L)$ is also a DCFL.

Proof: The same buffering construction applied to a DPDA for L yields a DPDA for $h^{-1}(L)$, since the DPDA is deterministic and the buffering of $h(a)$ (a fixed string per input symbol a) preserves determinism. □

10.7. **A finite language not recognized by any one-state PDA via final state.**

Example: $L = \{\boldsymbol{ab}\}$.

Proof: Every one-state PDA P with state set $\{t\}$, start state t, and final states $\{t\}$ trivially accepts λ: the initial configuration has P already in the (unique) final state with empty input, so λ is accepted regardless of the stack content. Since $\lambda \not\in L = \{\boldsymbol{ab}\}$, no one-state PDA via final state can recognize L.

More generally, any finite language that does not contain λ is not recognizable by any one-state PDA via final state, since such a machine always accepts λ. □

10.8. $L = \{\boldsymbol{a}^n \boldsymbol{b}^n \boldsymbol{c}^m \boldsymbol{d}^m \mid n, m \in \mathbb{N}\}$**.**

(a) **PDA via final state**: Push A on $\boldsymbol{a}$; pop A on $\boldsymbol{b}$; push B on $\boldsymbol{c}$; pop B on $\boldsymbol{d}$; accept when balanced. States: $\{q_0, q_1, q_2, q_f\}$. Initial state q_0, final state q_f. Stack alphabet $\{Z, A, B\}$. $\delta(q_0, \boldsymbol{a}, Z) = \langle q_0, AZ\rangle$, $\delta(q_0, \boldsymbol{a}, A) = \langle q_0, AA\rangle$, $\delta(q_0, \boldsymbol{b}, A) = \langle q_1, \lambda\rangle$, $\delta(q_1, \boldsymbol{b}, A) = \langle q_1, \lambda\rangle$, After reading $\boldsymbol{b}^n$, the machine is in state q_1 with Z on top of the stack. Then: $\delta(q_1, \boldsymbol{c}, Z) = \langle q_2, BZ\rangle$, $\delta(q_2, \boldsymbol{c}, B) = \langle q_2, BB\rangle$, $\delta(q_2, \boldsymbol{d}, B) = \langle q_2, \lambda\rangle$, $\delta(q_2, \lambda, Z) = \langle q_f, Z\rangle$. Also $\delta(q_0, \lambda, Z) = \langle q_1, Z\rangle$ to handle $n = 0$, and $\delta(q_1, \lambda, Z) = \langle q_2, Z\rangle$ for $m = 0$ (with appropriate final states).

(b) **PDA via empty stack**: Similar; pop Z at the end.

(c) **DPDA**: Yes. The DPDA deterministically pushes As, then pops As, then pushes Bs, then pops Bs. Each step is deterministic.

(d) **Counting automaton**: Yes. A single stack symbol suffices: use it first to count the $\boldsymbol{a}$'s against the $\boldsymbol{b}$'s and, after the stack returns to Z, reuse the same symbol to count the $\boldsymbol{c}$'s against the $\boldsymbol{d}$'s.

(e) **Grammar from PDA** (Definition 10.7): After applying Definition 10.7 to part (b) and deleting useless variables, we get the equivalent grammar

$$S \to AB, \qquad A \to \boldsymbol{a}A\boldsymbol{b} \mid \lambda, \qquad B \to \boldsymbol{c}B\boldsymbol{d} \mid \lambda.$$

10.9. **Prove by induction (Theorem 10.2):** $\langle s, x, S\rangle \overset{*}{\vdash} \langle s, \lambda, \beta\rangle \Leftrightarrow S \overset{*}{\Rightarrow} x\beta$ **as a leftmost derivation.**

Claim ($P(n)$): For all $n \geq 0$ and strings x, β: $\langle s, x, S\rangle \overset{*}{\vdash} \langle s, \lambda, \beta\rangle$ in n moves iff $S \overset{*}{\Rightarrow} x\beta$ in n leftmost-derivation steps.

Base ($n=0$): $\langle s,\lambda,S\rangle \overset{*}{\vdash} \langle s,\lambda,S\rangle$ (zero moves) iff $x=\lambda, \beta=S$, and $S \overset{*}{\Rightarrow}{}^{0}\ \lambda\cdot S$ (zero steps). ✓

Step: Assume $P(n)$ for all shorter computations. A computation of $n+1$ steps: $\langle s,ay,S\rangle \vdash \langle s,y,\gamma\rangle \overset{*}{\vdash} \langle s,\lambda,\beta\rangle$ (first move reads a, pushes γ for top of S). By the construction of the PDA (Theorem 10.2), this first step corresponds to applying a grammar production $S \to a\gamma$ (GNF production, starting with terminal a). By $P(n)$, $\langle s,y,\gamma\rangle \overset{*}{\vdash} \langle s,\lambda,\beta\rangle$ iff $\gamma \overset{*}{\Rightarrow} y\beta$ as leftmost derivation. So the full computation corresponds to $S \Rightarrow a\gamma \overset{*}{\Rightarrow} ay\beta = x\beta$. □

10.10. **Grammar for regular expressions: convert to GNF, find corresponding PDA.**

Grammar: $R \to \boldsymbol{a}|\boldsymbol{b}|\boldsymbol{c}|\boldsymbol{\epsilon}|\boldsymbol{\emptyset}|(R\cdot R)|(R\cup R)|R^{\boldsymbol{*}}$.

(a) **GNF conversion**: Since pure Greibach normal form only requires the first symbol on the right to be terminal, the only problem is the left-recursive rule $R \to R^{\boldsymbol{*}}$. Introduce a new nonterminal Y generating one or more copies of $^{\boldsymbol{*}}$:

$$Y \to {}^{\boldsymbol{*}} \mid {}^{\boldsymbol{*}}Y.$$

Then an equivalent PGNF grammar is

$$\begin{aligned} R \to\, & \boldsymbol{a} \mid \boldsymbol{b} \mid \boldsymbol{c} \mid \boldsymbol{\epsilon} \mid \boldsymbol{\emptyset} \mid \boldsymbol{(}R\cdot R\boldsymbol{)} \mid \boldsymbol{(}R\cup R\boldsymbol{)} \\ & \mid \boldsymbol{a}Y \mid \boldsymbol{b}Y \mid \boldsymbol{c}Y \mid \boldsymbol{\epsilon}Y \mid \boldsymbol{\emptyset}Y \mid \boldsymbol{(}R\cdot R\boldsymbol{)}Y \mid \boldsymbol{(}R\cup R\boldsymbol{)}Y. \end{aligned}$$

This simply replaces the old rule $R \to R^{\boldsymbol{*}}$ by attaching a nonempty string of stars through Y.

(b) **PDA from GNF grammar** (Definition 10.6): Let

$$P = \langle \Sigma, \Gamma, \{s\}, s, \delta, R, \emptyset \rangle,$$

where

$$\Sigma = \{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c},\boldsymbol{(},\boldsymbol{)},\boldsymbol{\epsilon},\boldsymbol{\emptyset},\cup,\cdot,{}^{\boldsymbol{*}}\}$$

and $\Gamma = \Sigma \cup \{R, Y\}$. The nonempty transitions are exactly those dictated by Definition 10.6:

$$\begin{aligned} \delta(s,\boldsymbol{a},R) &= \{\langle s,\lambda\rangle, \langle s,Y\rangle\}, \\ \delta(s,\boldsymbol{b},R) &= \{\langle s,\lambda\rangle, \langle s,Y\rangle\}, \\ \delta(s,\boldsymbol{c},R) &= \{\langle s,\lambda\rangle, \langle s,Y\rangle\}, \\ \delta(s,\boldsymbol{\epsilon},R) &= \{\langle s,\lambda\rangle, \langle s,Y\rangle\}, \\ \delta(s,\boldsymbol{\emptyset},R) &= \{\langle s,\lambda\rangle, \langle s,Y\rangle\}, \\ \delta(s,\boldsymbol{(},R) &= \{\langle s,R\cdot R\boldsymbol{)}\rangle, \langle s,R\cup R\boldsymbol{)}\rangle, \langle s,R\cdot R\boldsymbol{)}Y\rangle, \langle s,R\cup R\boldsymbol{)}Y\rangle\}, \\ \delta(s,{}^{\boldsymbol{*}},Y) &= \{\langle s,\lambda\rangle, \langle s,Y\rangle\}, \end{aligned}$$

and for each terminal $\tau \in \Sigma$,

$$\delta(s,\tau,\tau) = \{\langle s,\lambda\rangle\}.$$

This PDA accepts via empty stack.

(c) **PDA accepting via final state** (Theorem 10.4): Apply Theorem 10.4 to the machine in part (b). Using fresh bottom markers U and V, we obtain

$$P_f = \langle \Sigma, \Gamma \cup \{U, V\}, \{s, f\}, s, \delta_f, V, \{f\} \rangle,$$

where δ_f agrees with δ on all old stack symbols, and the additional nonempty transitions are

$$\delta_f(s, \lambda, V) = \{\langle s, RU \rangle\}, \qquad \delta_f(s, \lambda, U) = \{\langle f, U \rangle\}.$$

There are no other moves involving U or V, and no moves out of f. By Theorem 10.4, $L(P_f) = \Lambda(P)$.

10.11. $L = \{\boldsymbol{a}^i \boldsymbol{b}^j \boldsymbol{c}^j \boldsymbol{d}^i \mid i, j \in \mathbb{N}\}$.

(a) **PDA via final state**: Push A on $\boldsymbol{a}$; push B on $\boldsymbol{b}$; pop B on $\boldsymbol{c}$; pop A on $\boldsymbol{d}$; accept when stack returns to Z. States $\{q_0, q_1, q_2, q_3, q_f\}$, alphabet $\{Z, A, B\}$: $\delta(q_0, \boldsymbol{a}, Z) = \langle q_0, AZ \rangle$, $\delta(q_0, \boldsymbol{a}, A) = \langle q_0, AA \rangle$, $\delta(q_0, \boldsymbol{b}, A) = \langle q_1, BA \rangle$, $\delta(q_0, \boldsymbol{b}, Z) = \langle q_1, BZ \rangle$ (for $i = 0$), $\delta(q_1, \boldsymbol{b}, B) = \langle q_1, BB \rangle$, $\delta(q_1, \boldsymbol{c}, B) = \langle q_2, \lambda \rangle$, $\delta(q_2, \boldsymbol{c}, B) = \langle q_2, \lambda \rangle$, $\delta(q_2, \boldsymbol{d}, A) = \langle q_3, \lambda \rangle$, $\delta(q_3, \boldsymbol{d}, A) = \langle q_3, \lambda \rangle$, $\delta(q_2, \lambda, Z) = \langle q_f, Z \rangle$, $\delta(q_3, \lambda, Z) = \langle q_f, Z \rangle$.

(b) **PDA via empty stack**: Pop Z as well; add $\delta(\cdot, \lambda, Z) = \langle q_f, \lambda \rangle$ with appropriate state.

(c) **DPDA**: Yes. The language has a fixed block structure: $\boldsymbol{a}^i, \boldsymbol{b}^j, \boldsymbol{c}^j, \boldsymbol{d}^i$, so the DPDA deterministically processes each block. The stack remembers i (via As) and j (via Bs overlaid on top). This is deterministic.

(d) **Counting automaton**: No; both i and j are unbounded.

(e) **Grammar from PDA** (Definition 10.7): Applying Definition 10.7 to part (b) and simplifying yields the equivalent grammar

$$S \to \boldsymbol{a} S \boldsymbol{d} \mid T, \qquad T \to \boldsymbol{b} T \boldsymbol{c} \mid \lambda.$$

10.12. **Find G_{P_3} from Example 10.5 using Definition 10.7.**

Since Definition 10.7 applies to PDAs that accept by empty stack, first convert P_3 to an equivalent empty-stack PDA by Theorem 10.5. Applying Definition 10.7 to that machine and deleting useless variables yields an equivalent grammar for the language of Example 10.5:

$$Z \to X \mid Y,$$

$$X \to \boldsymbol{a} X \boldsymbol{b} \mid \boldsymbol{ab}, \qquad Y \to \boldsymbol{aa} Y \boldsymbol{b} \mid \boldsymbol{aab}.$$

Here X generates $\{\boldsymbol{a}^n \boldsymbol{b}^n \mid n \geq 1\}$ and Y generates $\{\boldsymbol{a}^{2m} \boldsymbol{b}^m \mid m \geq 1\}$, so Z generates exactly the language recognized by P_3.

10.13. **Prove by induction (Theorem 10.3):** $A^{st} \stackrel{*}{\Rightarrow} x \Leftrightarrow \langle s, x, A \rangle \stackrel{*}{\vdash} \langle t, \lambda, \lambda \rangle$**.**

Claim ($P(n)$): For all n, for all states s, t, stack symbol A, string x: $A^{st} \stackrel{*}{\Rightarrow} x$ in n steps iff $\langle s, x, A \rangle \stackrel{*}{\vdash} \langle t, \lambda, \lambda \rangle$ in n moves.

Base ($n = 1$): $A^{st} \to a$ (terminal production from grammar of Definition 10.7) corresponds to $\delta(s, a, A) \ni \langle t, \lambda \rangle$, giving $\langle s, a, A \rangle \vdash \langle t, \lambda, \lambda \rangle$. ✓

Step: Assume $P(k)$ for all $k < n$. A derivation $A^{st} \overset{*}{\Rightarrow} x$ of length n begins with $A^{st} \Rightarrow aB_1^{s_0 s_1} B_2^{s_1 s_2} \cdots B_m^{s_{m-1} t}$ (using a production from $\delta(s, a, A) \ni \langle s_0, B_1 \cdots B_m \rangle$). Then each $B_i^{s_{i-1} s_i}$ independently derives x_i in fewer than n steps. By IH, $\langle s_{i-1}, x_i, B_i \rangle \overset{*}{\vdash} \langle s_i, \lambda, \lambda \rangle$. Chaining: $\langle s, ax_1 \cdots x_m, A \rangle \vdash \langle s_0, x_1 \cdots x_m, B_1 \cdots B_m \rangle \overset{*}{\vdash} \langle t, \lambda, \lambda \rangle$. So $x = ax_1 \cdots x_m$ and $\langle s, x, A \rangle \overset{*}{\vdash} \langle t, \lambda, \lambda \rangle$. □

10.14. $L = \{\boldsymbol{a}^n \boldsymbol{b}^n \boldsymbol{c}^m \boldsymbol{d}^m\} \cup \{\boldsymbol{a}^i \boldsymbol{b}^j \boldsymbol{c}^j \boldsymbol{d}^i\}$**.**

(a) **PDA via final state**: Use nondeterminism to choose which language the input belongs to. P: from initial state, nondeterministically branch to handle either $\boldsymbol{a}^n \boldsymbol{b}^n \boldsymbol{c}^m \boldsymbol{d}^m$ (Exercise 10.8) or $\boldsymbol{a}^i \boldsymbol{b}^j \boldsymbol{c}^j \boldsymbol{d}^i$ (Exercise 10.11).

(b) **PDA via empty stack**: Start with a new initial state and make a λ-choice to the empty-stack PDA from Exercise 10.8 or the one from Exercise 10.11. This is the standard PDA construction for union.

(c) **DPDA**: No. This is the classic ambiguous union

$$\{\boldsymbol{a}^n \boldsymbol{b}^n \boldsymbol{c}^m \boldsymbol{d}^m \mid n, m \geq 0\} \cup \{\boldsymbol{a}^i \boldsymbol{b}^j \boldsymbol{c}^j \boldsymbol{d}^i \mid i, j \geq 0\}.$$

Every word $\boldsymbol{a}^n \boldsymbol{b}^n \boldsymbol{c}^n \boldsymbol{d}^n$ belongs to both halves, so the language has two incompatible structural readings on an infinite family of words. Since deterministic context-free languages are unambiguous, this language cannot be a DCFL.

(d) **Counting automaton**: No; the string lengths are unbounded.

(e) **Grammar from PDA**: An equivalent grammar is

$$S \to AB \mid C,$$

$$A \to \boldsymbol{a}A\boldsymbol{b} \mid \lambda, \qquad B \to \boldsymbol{c}B\boldsymbol{d} \mid \lambda,$$
$$C \to \boldsymbol{a}C\boldsymbol{d} \mid T, \qquad T \to \boldsymbol{b}T\boldsymbol{c} \mid \lambda.$$

10.15. **Find G_{P_G} from Example 10.6 using Definition 10.7.**

Since P_G has only one state s, the nonterminals are Z, S^{ss}, $\boldsymbol{a}^{ss}$, and $\boldsymbol{b}^{ss}$. Definition 10.7 gives

$$Z \to S^{ss},$$
$$S^{ss} \to \boldsymbol{a} S^{ss} \boldsymbol{b}^{ss} \mid \boldsymbol{a}\boldsymbol{b}^{ss},$$
$$\boldsymbol{b}^{ss} \to \boldsymbol{b}.$$

The variable $\boldsymbol{a}^{ss}$ is useless and may be deleted. Renaming S^{ss} as S and $\boldsymbol{b}^{ss}$ as $\boldsymbol{b}$ gives the familiar grammar

$$S \to \boldsymbol{a}S\boldsymbol{b} \mid \boldsymbol{a}\boldsymbol{b}.$$

10.16. **Prove (Theorem 10.4): $\langle s, xy, \alpha \rangle \overset{*}{\vdash} \langle s, y, \beta \rangle$ in P iff $\langle s, xy, \alpha Y \rangle \overset{*}{\vdash} \langle s, y, \beta Y \rangle$ in P_f.**

Proof by induction on the number of moves:

Base (0 moves): Both sides are trivially equivalent (same configuration).

Step: If $\langle s, xy, \alpha \rangle \vdash \langle s', x'y, \alpha' \rangle \overset{*}{\vdash} \langle s, y, \beta \rangle$ in P: The first move in P corresponds to $\delta_P(s, a, A) \ni \langle s', \gamma \rangle$ (reading a or λ, popping A, pushing γ). In P_f, the same move is available: δ_{P_f} includes all moves of δ_P plus new moves for the marker Y. The marker Y sits below α and is not affected by moves on α. By IH on the remaining $k-1$ moves, the result follows for P_f with Y appended. □

10.17. **Prove (Theorem 10.5): same correspondence for P_λ.**

Proof by induction: Symmetric to Exercise 10.16; the marker Y at the bottom of the stack (representing "no more stack to pop") is unaffected by all moves above it, so the computation in P with stack α corresponds exactly to the computation in P_λ with stack αY. □

10.18. $\{x \in \{\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}\}^* \mid |x|_{\boldsymbol{a}} = |x|_{\boldsymbol{b}} \wedge |x|_{\boldsymbol{b}} > |x|_{\boldsymbol{c}}\}$ **is not context free.**

Proof using closure properties:

Let $K = \{x \mid |x|_{\boldsymbol{a}} = |x|_{\boldsymbol{b}} \wedge |x|_{\boldsymbol{b}} > |x|_{\boldsymbol{c}}\}$.

Suppose $K \in \mathfrak{P}_\Sigma$. Consider the regular language $R = \boldsymbol{a}^* \boldsymbol{b}^* \boldsymbol{c}^*$ (strings of the form $\boldsymbol{a}^i \boldsymbol{b}^j \boldsymbol{c}^k$).

$K \cap R = \{\boldsymbol{a}^n \boldsymbol{b}^n \boldsymbol{c}^k \mid n > k\}$.

If K were a CFL, then $K \cap R$ would be a CFL (by Theorem 10.6: CFL $\cap$ regular = CFL). But $\{\boldsymbol{a}^n \boldsymbol{b}^n \boldsymbol{c}^k \mid n > k\}$ is not context free (one can apply the pumping lemma to strings $\boldsymbol{a}^p \boldsymbol{b}^p \boldsymbol{c}^{p-1}$; any pumping either changes the $\boldsymbol{a}$-count differently from the $\boldsymbol{b}$-count, or changes the $\boldsymbol{c}$-count, in each case violating the constraints).

So K is not a CFL. □

10.19. **Two-tape automaton.**

(a) **Transition function**: For a two-tape automaton with tapes over Σ, the state transition function δ has domain $S \times (\Sigma \cup \{\boldsymbol{B}\}) \times (\Sigma \cup \{\boldsymbol{B}\})$ (current state, symbol read from tape 1, symbol read from tape 2) and range $S \times (\Sigma \cup \{\boldsymbol{B}\}) \times \{L, R, S\} \times (\Sigma \cup \{\boldsymbol{B}\}) \times \{L, R, S\}$ (new state, write to tape 1, move head 1, write to tape 2, move head 2).

(b) **Two-tape automaton for** $\{\boldsymbol{a}^n \boldsymbol{b}^n \boldsymbol{c}^n \mid n \geq 1\}$: Let

$$M = \langle \{\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}\}, \Gamma, \{q_0, q_1, q_2, q_3, q_4, q_f, q_r\}, q_0, \delta, q_f \rangle,$$

where tape 1 initially holds the input and tape 2 is blank, and $\Gamma = \{\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}, X, \boldsymbol{B}\}$. The intended behavior is:

1. In q_0, scan the $\boldsymbol{a}$-block on tape 1. For each $\boldsymbol{a}$ read on tape 1, write an X on tape 2 and move both heads right. If no $\boldsymbol{a}$ is seen, reject. On the first $\boldsymbol{b}$, leave tape 1 where it is, move tape 2 left into state q_1, and rewind tape 2 to its leftmost X.
2. In q_2, compare the $\boldsymbol{b}$-block with the stored X's: on $(\boldsymbol{b}, X)$ move right on both tapes. If the tape-1 symbol changes to $\boldsymbol{c}$ exactly when tape 2 reaches blank, go to q_3; otherwise reject.
3. In q_3, hold tape 1 on the first $\boldsymbol{c}$ and rewind tape 2 left to its leftmost X again.
4. In q_4, compare the $\boldsymbol{c}$-block with the same X's: on $(\boldsymbol{c}, X)$ move right on both tapes. Accept exactly when both heads simultaneously reach blank; any other mismatch or extra symbol causes rejection.

Thus tape 2 stores the number n once, and the machine checks both the $\boldsymbol{b}$-block and the $\boldsymbol{c}$-block against that same count.

10.20. $\{\boldsymbol{a}^n \boldsymbol{b}^n \boldsymbol{c}^n\}$ **is not context free;** $\{x \mid |x|_{\boldsymbol{a}} = |x|_{\boldsymbol{b}}\}$ **is not context free.**

(a) $\{\boldsymbol{a}^n\boldsymbol{b}^n\boldsymbol{c}^n\}$ **is not CFL**: Assume it is, with pumping constant N. Take $s = \boldsymbol{a}^N\boldsymbol{b}^N\boldsymbol{c}^N$. By the pumping lemma, $s = uvwxy$ with $|vwx| \le N$ and $|vx| \ge 1$.

Since $|vwx| \le N$, the substring vwx cannot span all three blocks. So v and x together involve at most two of $\{\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}\}$. Pumping changes the count of at most two symbols while leaving the third unchanged. But all three must be equal, so the pumped string fails $|u|_{\boldsymbol{a}} = |u|_{\boldsymbol{b}} = |u|_{\boldsymbol{c}}$ for some $i \ne 1$. Contradiction. □

(b) **Disproof**: the printed statement is false, because

$$L = \{x \mid |x|_{\boldsymbol{a}} = |x|_{\boldsymbol{b}}\}$$

is context free. For example, the grammar

$$S \to \boldsymbol{c}S \mid SS \mid \boldsymbol{a}S\boldsymbol{b} \mid \boldsymbol{b}S\boldsymbol{a} \mid \lambda$$

generates exactly the strings over $\{\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}\}$ having equally many $\boldsymbol{a}$'s and $\boldsymbol{b}$'s. Therefore the claim to be proved is false.

10.21. **DPDA constructions and non-DCFL example.**

(a) DPDA for $\{\boldsymbol{c}^n\boldsymbol{b}^m \mid n \ge 1, n = m\} \cup \{\boldsymbol{a}^n\boldsymbol{b}^m \mid n \ge 1, n = 2m\}$:

Let

$$P = \langle\{\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}\}, \{Z, A\}, \{q_0, q_c, q_{a,1}, q_{a,0}, q_b, q_f\}, q_0, \delta, Z, \{q_f\}\rangle.$$

The first symbol chooses the branch deterministically.

For the $\boldsymbol{c}^n\boldsymbol{b}^n$ branch, use the standard one-counter routine:

$$\begin{aligned}
\delta(q_0, \boldsymbol{c}, Z) &= \langle q_c, AZ\rangle,\\
\delta(q_c, \boldsymbol{c}, A) &= \langle q_c, AA\rangle,\\
\delta(q_c, \boldsymbol{b}, A) &= \langle q_b, \lambda\rangle,\\
\delta(q_b, \boldsymbol{b}, A) &= \langle q_b, \lambda\rangle,\\
\delta(q_b, \lambda, Z) &= \langle q_f, Z\rangle.
\end{aligned}$$

For the $\boldsymbol{a}^n\boldsymbol{b}^m$ branch with $n = 2m$, the machine records the parity of the number of $\boldsymbol{a}$'s: state $q_{a,1}$ means an odd number of $\boldsymbol{a}$'s has been seen since the last push, and state $q_{a,0}$ means an even number.

$$\begin{aligned}
\delta(q_0, \boldsymbol{a}, Z) &= \langle q_{a,1}, Z\rangle,\\
\delta(q_{a,1}, \boldsymbol{a}, Z) &= \langle q_{a,0}, AZ\rangle,\\
\delta(q_{a,1}, \boldsymbol{a}, A) &= \langle q_{a,0}, AA\rangle,\\
\delta(q_{a,0}, \boldsymbol{a}, A) &= \langle q_{a,1}, A\rangle,\\
\delta(q_{a,0}, \boldsymbol{b}, A) &= \langle q_b, \lambda\rangle.
\end{aligned}$$

Every second $\boldsymbol{a}$ pushes one A, so after reading $\boldsymbol{a}^n$ the stack contains exactly $n/2$ copies of A. The $\boldsymbol{b}$-phase then pops one A per $\boldsymbol{b}$. Acceptance occurs only when the input ends and the stack returns to Z, so this branch accepts exactly $\{\boldsymbol{a}^{2m}\boldsymbol{b}^m \mid m \ge 1\}$.

(b) Define $h : \{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}\}^* \to \{\boldsymbol{a},\boldsymbol{b}\}^*$ by $h(\boldsymbol{a}) = \boldsymbol{a}$, $h(\boldsymbol{b}) = \boldsymbol{b}$, $h(\boldsymbol{c}) = \boldsymbol{a}$. Then: $h(\{\boldsymbol{c}^n\boldsymbol{b}^m \mid n = m\}) = \{\boldsymbol{a}^n\boldsymbol{b}^n \mid n \geq 1\}$ and $h(\{\boldsymbol{a}^n\boldsymbol{b}^m \mid n = 2m\}) = \{\boldsymbol{a}^n\boldsymbol{b}^m \mid n = 2m\}$. $h(L) = \{\boldsymbol{a}^n\boldsymbol{b}^n\} \cup \{\boldsymbol{a}^n\boldsymbol{b}^m \mid n = 2m\}$. The discussion immediately before Theorem 10.9 identifies this union as a context-free language that is *not* deterministic context free. Hence this homomorphism takes a DCFL to a non-DCFL, as required.

10.22. **Ogden's lemma: $\{\boldsymbol{a}^k\boldsymbol{b}^n\boldsymbol{c}^m \mid k \neq n \wedge n \neq m\}$ is not CFL.**

Assume $L = \{\boldsymbol{a}^k\boldsymbol{b}^n\boldsymbol{c}^m \mid k \neq n, n \neq m\}$ is CFL with Ogden constant N.

Choose $p > N$. Let $s = \boldsymbol{a}^{p!+p}\boldsymbol{b}^p\boldsymbol{c}^{p!+p} \in L$ (since $p! + p \neq p$). Mark all $\boldsymbol{b}$-symbols.

By Ogden's lemma: $s = uvwxy$ with vwx containing at most N marked positions, vx has at least one marked position, all pumped strings in L.

Since all marks are on $\boldsymbol{b}$s, and $|vwx| \leq N \leq p$, vwx lies within the $\boldsymbol{b}$-block (possibly with some $\boldsymbol{a}$s or $\boldsymbol{c}$s on the boundary). Since vx has at least one $\boldsymbol{b}$, $|vx|_{\boldsymbol{b}} \geq 1$; let $|vx|_{\boldsymbol{b}} = j$ $(1 \leq j \leq N)$.

For the pumped string uv^iwx^iy: the count of $\boldsymbol{b}$s is $p + (i-1)j$. The counts of $\boldsymbol{a}$s and $\boldsymbol{c}$s may also change if v or x extend into the $\boldsymbol{a}$ or $\boldsymbol{c}$ blocks.

Case 1: vx contains only $\boldsymbol{b}$s. Then $k = p! + p$ and $m = p! + p$ for all i. Choose i such that $p + (i-1)j = p! + p$: then $(i-1)j = p!$, so $i - 1 = p!/j$. Since $j \leq N \leq p-1$, $j \mid p!$, so i is an integer. The pumped string has $|\boldsymbol{b}| = p! + p = k = m$, violating $k \neq n$ or $n \neq m$. Contradiction.

Case 2: vx spans the $\boldsymbol{a}/\boldsymbol{b}$ boundary. Let $|vx|_{\boldsymbol{a}} = \ell \geq 0$ and $|vx|_{\boldsymbol{b}} = j \geq 1$ (at least one marked $\boldsymbol{b}$). Choose $i - 1 = p!/j$ (an integer since $j \leq N < p$). Then the pumped string has $n = p + p! = p! + p$ and $m = p! + p$ (the $\boldsymbol{c}$-block is untouched), so $n = m$, violating $n \neq m$. Contradiction.

Case 3: vx spans the $\boldsymbol{b}/\boldsymbol{c}$ boundary. Let $|vx|_{\boldsymbol{b}} = j \geq 1$ and $|vx|_{\boldsymbol{c}} = r \geq 0$. Choose $i - 1 = p!/j$. The pumped string has $n = p! + p$ and $k = p! + p$ (the $\boldsymbol{a}$-block is untouched), so $k = n$, violating $k \neq n$. Contradiction. □

10.23. **Prove (Theorem 10.6) by induction.**

Theorem 10.6 states that if P_1 is a PDA and A_2 is a DFA (or regular language), then their "intersection" machine $P^\cap$ (Product machine) satisfies the correspondence stated.

Proof by induction on the number of moves:

Base (0 moves): $\langle\langle s_1, s_2\rangle, y, \alpha\rangle \overset{*}{\vdash} \langle\langle s_1, s_2\rangle, y, \alpha\rangle$ (trivially). Here $t_1 = s_1$, $t_2 = s_2$, $x = \lambda$, so $\overline{\delta}_2(s_2, \lambda) = s_2 = t_2$. ✓

Step: If the product machine takes $k+1$ moves from $\langle\langle s_1, s_2\rangle, xy, \alpha\rangle$: the first move reads some a (or λ) and moves to $\langle\langle s_1', s_2'\rangle, \cdot, \cdot\rangle$. In the product machine, this corresponds to a joint transition: P_1 moves from s_1 and A_2 reads a to move from s_2. By IH on k moves, the remaining computation in the product machine corresponds to $P^\cap$ and $\overline{\delta}_2$. The A_2 component is precisely $\overline{\delta}_2(s_2, x)$ after reading x. □

10.24. **Prove: DPDA P has equivalent DPDA P' with total transitions, no empty stack, scans full input.**

(a) **Never empty stack**: First place a permanent bottom marker **\$** under the original start symbol. Introduce a new start state s_0 with the single move $\delta(s_0, \lambda, \$) = \langle s, Z\$\rangle$, where s is the old start state and Z the old initial stack symbol. Whenever the old machine would have popped its last stack symbol, the new machine instead reaches a designated state whose stack top is **\$**. Thus the stack is never empty.

(b) **Always scans full input**: Add an endmarker **¢** to the input alphabet. Replace every old accepting halt by a move into an accepting scan state q_{acc}, and every rejecting halt by a move into a rejecting scan state q_{rej}. In both scan states, on every ordinary input symbol the machine simply moves right and leaves the stack unchanged; on **¢** it stops. Hence every computation reads the entire input before it halts.

(c) **Total transitions**: Now fill in whatever cases remain undefined, but in a way that respects the DPDA rules. If for some pair (q, X) there is already a λ-move, then no input move may be added there; the existing λ-move already supplies the unique next step. If there is no λ-move for (q, X), then for each missing input symbol add a transition to q_{rej} that preserves the stack. If neither input moves nor a λ-move existed for (q, X), add the single λ-move to q_{rej}. Thus every reachable configuration has a next move, and no input/λ conflict is introduced.

These three repairs preserve the accepted language and keep the machine deterministic, so the resulting DPDA has all the required properties. □

10.25. $\mathscr{A}_\Sigma$ **(DCFL) is closed under complementation.**

Using Exercise 10.24: For a DPDA P (accepting via final state), P can be made total (all transitions defined), never emptying stack, and always scanning full input. Under these conditions:

Every computation on every input string either ends in a final state (accept) or a nonfinal state (reject). There are no infinite loops or premature halts.

Construct P' by swapping final and nonfinal states of P: a string is accepted by P' iff it is rejected by P. The main issue (mentioned in the hint) is λ-moves cycling through both final and nonfinal states: since Exercise 10.24 ensures the DPDA is well-behaved (complete, non-emptying, full-input scanning), after processing any input string, the computation ends deterministically in exactly one state. Swapping final/nonfinal gives the complement. □

10.26. **Example:** $\mathscr{A}_\Sigma$ **not closed under concatenation.**

Let

$$L_1 = \{\mathbf{0}^i\mathbf{1}^i\mathbf{2}^k \mid i,k \ge 0\} \qquad \text{and} \qquad L_2 = \{\mathbf{0}^i\mathbf{1}^k\mathbf{2}^k \mid i,k \ge 0\}.$$

Both are DCFLs, but $L_1 \cup L_2$ is not. Now define

$$L_3 = \boldsymbol{a}L_1 \cup L_2.$$

L_3 is a DCFL, because the initial $\boldsymbol{a}$ tells the DPDA whether to use the L_1-routine or the L_2-routine. The regular language $\boldsymbol{a}^*$ is certainly a DCFL. Suppose $\mathscr{A}_\Sigma$ were closed under concatenation. Then $\boldsymbol{a}^* L_3$ would be a DCFL, and therefore so would

$$L_4 = \boldsymbol{a}^* L_3 \cap \boldsymbol{a}\mathbf{0}^*\mathbf{1}^*\mathbf{2}^*.$$

But

$$L_4 = \boldsymbol{a}L_1 \cup \boldsymbol{a}L_2.$$

If a DPDA recognized L_4, we could recognize $L_1 \cup L_2$ by simulating that DPDA after an imaginary initial $\boldsymbol{a}$. Since $L_1 \cup L_2$ is not a DCFL, L_4 cannot be a DCFL, and so $\mathscr{A}_\Sigma$ is not closed under concatenation.

10.27. **Example: $\mathscr{A}_\Sigma$ not closed under Kleene closure.**

With L_3 as in Exercise 10.26, let

$$L_5 = \{\boldsymbol{a}\} \cup L_3.$$

L_5 is a DCFL. Suppose $\mathscr{A}_\Sigma$ were closed under Kleene closure. Then L_5^* would be a DCFL, and hence so would

$$L_5^* \cap \boldsymbol{a}\mathbf{0}^*\mathbf{1}^*\mathbf{2}^*.$$

The only strings of L_5^* that have the form $\boldsymbol{a}\mathbf{0}^*\mathbf{1}^*\mathbf{2}^*$ are

$$\{\boldsymbol{a}\} \cup \boldsymbol{a}L_1 \cup \boldsymbol{a}L_2.$$

Intersecting further with the regular complement of $\{\boldsymbol{a}\}$ would leave $\boldsymbol{a}L_1 \cup \boldsymbol{a}L_2$, which is not a DCFL by the proof of Exercise 10.26. Therefore L_5^* cannot be a DCFL, and $\mathscr{A}_\Sigma$ is not closed under Kleene closure.

10.28. $\{\boldsymbol{c}\boldsymbol{a}^n\boldsymbol{b}^m \mid n = m\} \cup \{\boldsymbol{a}^n\boldsymbol{b}^m \mid n = 2m\}$ **is a DCFL.**

A DPDA can distinguish the two cases by the first symbol. On the $\boldsymbol{c}$-branch, it first consumes the initial $\boldsymbol{c}$ and then uses the standard DPDA for $\{\boldsymbol{a}^n\boldsymbol{b}^n\}$: push one A per $\boldsymbol{a}$ and pop one per $\boldsymbol{b}$. On the $\boldsymbol{a}$-branch, it uses the parity-counting construction from Exercise 10.20(a): every second $\boldsymbol{a}$ pushes one A, and each $\boldsymbol{b}$ pops one A. Thus the second branch accepts exactly $\{\boldsymbol{a}^{2m}\boldsymbol{b}^m \mid m \geq 1\}$. Since the first symbol determines the branch uniquely, the whole machine is deterministic. □

10.29. $L_1 - R_2$ **is context free if** L_1 **is CFL and** R_2 **is regular.**

(a) **Via product machine**: Extend the intersection proof (Theorem 10.6). Construct a product machine where the PDA simulates L_1 and the DFA simulates R_2; modify final states to accept when the PDA is in a final state and the DFA is in a *nonfinal* state. This gives $L_1 \cap \overline{R_2} = L_1 - R_2$.

(b) **Via closure**: $L_1 - R_2 = L_1 \cap \overline{R_2}$. Since regular languages are closed under complement, $\overline{R_2}$ is regular. Since CFLs are closed under intersection with regular languages (Theorem 10.6), $L_1 \cap \overline{R_2}$ is a CFL. □

10.30. $L_1 \cup R_2$ **is context free if** L_1 **is CFL and** R_2 **is regular.**

(a) **Via modified product machine**: Accept when the PDA is in a final state (for L_1) OR the DFA is in a final state (for R_2). This is a product machine with final states $F_1 \times S_2 \cup S_1 \times F_2$.

(b) **Via closure**: $L_1 \cup R_2$: since R_2 is regular, it is also a CFL. CFLs are closed under union: $L_1 \cup R_2 \in \mathfrak{P}_\Sigma$. □

10.31. **If** L_1 **is DCFL and** R_2 **is regular,** $R_2 - L_1$ **is DCFL.**

$R_2 - L_1 = R_2 \cap \overline{L_1}$. Since L_1 is DCFL, $\overline{L_1}$ is DCFL (Exercise 10.25). Since R_2 is regular, and the discussion immediately after Theorem 10.6 notes that the product construction remains deterministic when the PDA factor is deterministic, $R_2 \cap \overline{L_1}$ is a DCFL. □

10.32. $\{w\mathbf{2}w^r\}$ **is DCFL;** $\{ww^r\}$ **is not DCFL.**

(a) $\{w\mathbf{2}w^r\}$ **is DCFL**: The DPDA pushes all symbols before $\mathbf{2}$ onto the stack, then pops matching symbols for each character after $\mathbf{2}$ (reversed). Since $\mathbf{2}$ marks the center deterministically, the DPDA knows exactly when to switch from push to pop mode. □

(b) $\{ww^r\}$ **is not DCFL**: Let

$$E = \{ww^r \mid w \in \{\mathbf{0},\mathbf{1}\}^*\}.$$

Assume E were a DCFL. By Exercise 10.13, DCFLs are closed under inverse homomorphism. Define a homomorphism

$$h(\boldsymbol{a}) = \mathbf{01}, \qquad h(\boldsymbol{b}) = \mathbf{1010}, \qquad h(\boldsymbol{c}) = \mathbf{0101},$$

and let $R = \boldsymbol{c}^+\boldsymbol{b}^+ \cup \boldsymbol{a}^+\boldsymbol{b}^+$. For $x = \boldsymbol{c}^n\boldsymbol{b}^m$ we have

$$h(x) = (\mathbf{01})^{2n}(\mathbf{10})^{2m},$$

which is an even palindrome iff $n = m$. For $x = \boldsymbol{a}^n\boldsymbol{b}^m$ we have

$$h(x) = (\mathbf{01})^{n}(\mathbf{10})^{2m},$$

which is an even palindrome iff $n = 2m$. Therefore

$$R \cap h^{-1}(E) = \{\boldsymbol{c}^n\boldsymbol{b}^n \mid n \geq 1\} \cup \{\boldsymbol{a}^n\boldsymbol{b}^m \mid n \geq 1,\ n = 2m\}.$$

But Exercise 10.20 and the discussion immediately before Theorem 10.9 identify this language as not deterministic context free. This contradiction shows that E is not a DCFL. $\square$

10.33. **Non-closure of $\mathscr{A}_\Sigma$ under various operations.**

(a) $L_1 \cdot L_2$ need not be DCFL: Example from Exercise 10.26.

(b) $L_1 - L_2$ need not be DCFL: Let

$$L_1 = \{\boldsymbol{a}^n\boldsymbol{b}^n\boldsymbol{c}^m \mid n,m \geq 0\} \qquad \text{and} \qquad K = \{\boldsymbol{a}^m\boldsymbol{b}^n\boldsymbol{c}^n \mid m,n \geq 0\}.$$

Both L_1 and K are DCFLs. By Exercise 10.25, $\sim K$ is also a DCFL. Now

$$L_1 - (\sim K) = L_1 \cap K = \{\boldsymbol{a}^n\boldsymbol{b}^n\boldsymbol{c}^n \mid n \geq 0\},$$

which is not even context free. So difference does not preserve $\mathscr{A}_\Sigma$.

(c) L_1^* need not be DCFL: Exercise 10.27.

(d) L_1^r need not be DCFL: let

$$L = \{\boldsymbol{a}\mathbf{2}^j\mathbf{1}^i\mathbf{0}^i \mid i,j \geq 0\} \cup \{\boldsymbol{b}\mathbf{2}^j\mathbf{1}^j\mathbf{0}^i \mid i,j \geq 0\}.$$

This is a DCFL, because the first symbol determines which deterministic routine to use. But

$$L^r = \{\mathbf{0}^i\mathbf{1}^i\mathbf{2}^j\boldsymbol{a} \mid i,j \geq 0\} \cup \{\mathbf{0}^i\mathbf{1}^j\mathbf{2}^j\boldsymbol{b} \mid i,j \geq 0\}.$$

If L^r were a DCFL, then the same one-letter-lookahead argument as in Exercise 10.30 would turn it into a DPDA for

$$\{\mathbf{0}^i\mathbf{1}^i\mathbf{2}^j \mid i,j \geq 0\} \cup \{\mathbf{0}^i\mathbf{1}^j\mathbf{2}^j \mid i,j \geq 0\},$$

which is the same standard non-DCFL union pattern discussed before Theorem 10.9, now with an extra trailing block. So reversal does not preserve $\mathscr{A}_\Sigma$.

10.34. **Closure of $\mathfrak{P}_\Sigma$ and $\mathscr{A}_\Sigma$ under quotient.**

(a) $\mathfrak{P}_\Sigma$ **is not closed under quotient**: Let

$$L = \{\boldsymbol{a}^n\boldsymbol{b}^n\boldsymbol{c}^{n+m}\boldsymbol{d}^m \mid n, m \geq 0\} \qquad \text{and} \qquad K = \{\boldsymbol{c}^m\boldsymbol{d}^m \mid m \geq 0\}.$$

The language K is context free. So is L: a PDA pushes one marker for each $\boldsymbol{a}$, and on each $\boldsymbol{b}$ replaces one such marker by a marker that must later be matched by a $\boldsymbol{c}$. After those forced $\boldsymbol{c}$'s have been read, any additional $\boldsymbol{c}$'s are counted against the final $\boldsymbol{d}$'s. Now

$$L/K = \{\boldsymbol{a}^n\boldsymbol{b}^n\boldsymbol{c}^n \mid n \geq 0\}.$$

Indeed, every $\boldsymbol{a}^n\boldsymbol{b}^n\boldsymbol{c}^n$ is in the quotient by taking the suffix $\lambda \in K$. Conversely, if $x \in L/K$, then $xy \in L$ for some $y = \boldsymbol{c}^m\boldsymbol{d}^m \in K$. Since every word of L has the form $\boldsymbol{a}^n\boldsymbol{b}^n\boldsymbol{c}^{n+r}\boldsymbol{d}^r$, the suffix $\boldsymbol{c}^m\boldsymbol{d}^m$ can only begin after exactly the first n $\boldsymbol{c}$'s; otherwise the number of $\boldsymbol{c}$'s preceding the $\boldsymbol{d}$-block would exceed the number of trailing $\boldsymbol{d}$'s. Hence $x = \boldsymbol{a}^n\boldsymbol{b}^n\boldsymbol{c}^n$. But $\{\boldsymbol{a}^n\boldsymbol{b}^n\boldsymbol{c}^n \mid n \geq 0\}$ is not context free, so $\mathfrak{P}_\Sigma$ is not closed under quotient.

(b) $\mathscr{A}_\Sigma$ **is not closed under quotient**: The same counterexample works for deterministic context-free languages, because both L and K are accepted by DPDAs. For K, use the standard DPDA for $\{\boldsymbol{c}^m\boldsymbol{d}^m\}$. For L, use the deterministic phase structure described above:

$$\boldsymbol{a}\text{-phase} \to \boldsymbol{b}\text{-phase} \to \text{forced } \boldsymbol{c}\text{-phase} \to \text{extra } \boldsymbol{c}\text{-phase} \to \boldsymbol{d}\text{-phase}.$$

Thus $L, K \in \mathscr{A}_\Sigma$, but

$$L/K = \{\boldsymbol{a}^n\boldsymbol{b}^n\boldsymbol{c}^n \mid n \geq 0\}$$

is not even context free. Therefore $\mathscr{A}_\Sigma$ is not closed under quotient.

10.35. **Closure under operator b (Theorem 5.11).**

(a) $\mathfrak{P}_\Sigma$ is **not** closed under b. Let

$$A = \{\boldsymbol{a}^m\boldsymbol{b}^m\boldsymbol{c}^n\boldsymbol{\#\#}\boldsymbol{d}^{3n} \mid m, n \geq 0\}.$$

A is context free; for example,

$$S \to TU, \qquad T \to \boldsymbol{a}T\boldsymbol{b} \mid \lambda, \qquad U \to \boldsymbol{c}U\boldsymbol{ddd} \mid \boldsymbol{\#\#}.$$

Now intersect A^b with the regular language $\boldsymbol{a}^*\boldsymbol{b}^*\boldsymbol{c}^*\boldsymbol{\#}$. If $x = \boldsymbol{a}^i\boldsymbol{b}^j\boldsymbol{c}^k\boldsymbol{\#}$ belongs to this intersection, then some y of the same length satisfies $xy \in A$. The only possible choice is

$$xy = \boldsymbol{a}^n\boldsymbol{b}^n\boldsymbol{c}^n\boldsymbol{\#\#}\boldsymbol{d}^{3n}$$

for some n, so $i = j = k = n$. Conversely, every $\boldsymbol{a}^n\boldsymbol{b}^n\boldsymbol{c}^n\boldsymbol{\#}$ does arise this way. Hence

$$A^b \cap \boldsymbol{a}^*\boldsymbol{b}^*\boldsymbol{c}^*\boldsymbol{\#} = \{\boldsymbol{a}^n\boldsymbol{b}^n\boldsymbol{c}^n\boldsymbol{\#} \mid n \geq 0\},$$

which is not context free. Therefore $\mathfrak{P}_\Sigma$ is not closed under b.

(b) $\mathscr{A}_\Sigma$ is **not** closed under b. The same language A is accepted by a DPDA: deterministically match the $\boldsymbol{a}$'s against the $\boldsymbol{b}$'s, then push three markers for each $\boldsymbol{c}$, read the fixed delimiter ***##***, and pop one marker for each $\boldsymbol{d}$. Thus $A \in \mathscr{A}_\Sigma$, but A^b is not even context free by part (a). So $\mathscr{A}_\Sigma$ is not closed under b.

10.36. **Closure under operator Y (Theorem 5.7).**

(a) $\mathfrak{P}_\Sigma$ is closed under Y. If P is a PDA for L, build a new PDA with two copies of each state. In the first copy it simulates P normally, but once during the computation it may read one input letter and move to the second copy without changing the simulated stack. That single move represents the deleted letter. After that point the second copy continues to simulate P exactly. The new PDA therefore accepts precisely those strings obtained by deleting exactly one letter from a string of L.

(b) $\mathscr{A}_\Sigma$ is **not** closed under Y. Let
$$L = \{w\mathbf{2}w^r \mid w \in \{\mathbf{0},\mathbf{1}\}^*\}.$$
By Exercise 10.30(a), L is a DCFL. If we delete exactly one symbol from a word of L, the resulting string has no $\mathbf{2}$ exactly when the deleted symbol was the middle $\mathbf{2}$. Hence
$$Y(L) \cap \{\mathbf{0},\mathbf{1}\}^* = \{ww^r \mid w \in \{\mathbf{0},\mathbf{1}\}^*\}.$$
By Exercise 10.30(b), the language on the right is not a DCFL. Since $\mathscr{A}_\Sigma$ is closed under intersection with regular sets, $Y(L)$ cannot be a DCFL. So $\mathscr{A}_\Sigma$ is not closed under Y.

10.37. **Closure under prefix operator P (Exercise 5.16).**

(a) $\mathfrak{P}_\Sigma$ **is closed under** P: Let M be a PDA for L. Build a new PDA M' with two copies of each state of M. In the first copy, M' simulates M on the actual input. At any moment, M' may take a λ-move to the second copy. In that second copy, every transition of M that originally read a symbol of Σ is replaced by a λ-transition with the same stack effect, and the original λ-moves are kept as λ-moves. Thus, once the real input has been exhausted, M' may nondeterministically *guess* a continuation and simulate M on that guessed suffix. Therefore M' accepts exactly those x for which some y satisfies $xy \in L$, that is, exactly $P(L)$.

(b) $\mathscr{A}_\Sigma$ **is closed under** P: Let P be a DPDA normalized as in Exercise 10.24. For each stack symbol X, use two tagged copies X^{live} and X^{dead}. While simulating P, the new machine maintains on the stack not just the current stack symbols but also, for each stack position, whether the subconfiguration beginning at that position can still be extended to an accepting computation. Because the original machine is deterministic, this tag is forced by the current state and the tagged symbols that replace X in the next move. The machine therefore updates the tags deterministically whenever it pushes or pops. After the input ends, the simulated configuration is live exactly when the topmost tag is live, and that is exactly the condition $x \in P(L)$. Hence a DPDA can accept the prefix language.

10.38. **Closure under operator F (Exercise 5.19).**

(a) $\mathfrak{P}_\Sigma$ **is not closed under** F. Let
$$K = \{\boldsymbol{a}^i\boldsymbol{b}^j\boldsymbol{c}^k \mid i = j\} \cup \{\boldsymbol{a}^i\boldsymbol{b}^j\boldsymbol{c}^k \mid j = k\}.$$

K is context free, but by Exercise 10.23 its complement over $\{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}\}^*$ is not context free. Introduce a new symbol **$** and let

$$B = \{x\$ \mid x \in \{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}\}^*\} \cup \{x\$\$ \mid x \in K\}.$$

B is context free. A word $x\$$ is in $F(B)$ exactly when $x \not inK$, because it fails to be maximal precisely when it can be extended to $x\$\$$. Therefore

$$F(B) \cap \{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}\}^*\$ = \{x\$ \mid x \not inK\}.$$

If $F(B)$ were context free, then intersecting with the regular language $\{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}\}^*\$$ and deleting the final **$** would give $\sim K$ as a CFL, contradiction.

(b) $\mathscr{A}_\Sigma$ **is closed under** F: Start with a DPDA normalized as in Exercise 10.24. For each input word x, let C_x be the unique configuration reached after all of x has been scanned. Then $x \in F(L)$ exactly when two conditions hold: C_x is accepting, and C_x is not *prolific*, meaning that no nonempty continuation from C_x can still lead to acceptance. This is the same configuration-liveness idea used in part (b) for the prefix operator, with "live" replaced by "accepting but not prolific." A DPDA can therefore simulate the original machine and accept exactly the words in $F(L)$, so $\mathscr{A}_\Sigma$ is closed under F.

Chapter 11

Turing Machines — Solutions

11.1. **How is a Turing machine like an elevator?**

Analogy:

- **States** correspond to *floors* (the elevator's current position/state).
- **Input** corresponds to *button presses* (the sequence of requested floors).
- The elevator transitions from floor to floor based on button requests, just as the Turing machine transitions from state to state based on input symbols.

Essential missing component: An elevator has only *finite, bounded memory* — it can remember which buttons have been pressed (a finite set), but it cannot write new symbols to an unbounded tape. The critical component preventing an elevator from modeling a general computing device is the *unbounded read/write tape*. Without it, the elevator can only implement a finite state machine (regular language computation), not a Turing machine. The ability to store and retrieve arbitrarily large amounts of information is essential for general computation.

11.2. **Turing machine for palindromes** $L = \{w \mid w = w^r\}$ **over** $\Sigma = \{\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}\}$**.**

(a) **One-tape, one-head TM**:
Strategy: Repeatedly match the leftmost and rightmost unprocessed symbols.

1. Mark the leftmost symbol a (replace with $\overline{a}$); move right to the rightmost unmarked symbol.
2. If the rightmost symbol matches a, mark it and move left; continue.
3. If the rightmost symbol does not match, reject.
4. When all symbols are marked (or only the center symbol for odd-length), accept.

States: $\{q_{\text{start}}, q_a, q_b, q_c, q_{\text{right}}, q_{\text{check},a}, q_{\text{check},b}, q_{\text{check},c}, q_{\text{return}}, q_{\text{accept}}, q_{\text{reject}}\}$.
Transitions: $\delta(q_{\text{start}}, \boldsymbol{a}) = (q_a, \overline{\boldsymbol{a}}, R)$ [mark $\boldsymbol{a}$, go right]; etc.

(b) **Linear Bounded Automaton (LBA)**: Since the tape head never needs to go beyond the input boundaries (matching leftmost/rightmost within the input), the TM above is already an LBA. The computation uses only the n tape cells containing the input.

(c) **Nondeterminism/multiple tapes**: A two-tape TM copies the input to tape 2, reverses tape 2, and compares tape 1 with tape 2 symbol by symbol — $O(n)$ time. A nondeterministic TM can guess the center and verify matching from both ends simultaneously.

11.3. **TM recognizing $\{a^n \mid n$ is a perfect square$\}$ over $\Sigma = \{a\}$.**

Strategy: The k-th perfect square is k^2. Note $(k+1)^2 = k^2 + (2k+1)$: successive perfect squares differ by successive odd numbers $1, 3, 5, 7, \ldots$

Algorithm:

1. If input is λ: accept (since $0 = 0^2$).
2. Mark the first $\boldsymbol{a}$ and subtract 1; this corresponds to $1^2 = 1$.
3. Then subtract 3; if exhausted, accept (that was $2^2 = 4$). If not enough, reject.
4. Then subtract 5; if exhausted accept ($3^2 = 9$). Etc.
5. Subtract $2k+1$ at each step for $k = 0, 1, 2, \ldots$; accept if the tape is exactly emptied, reject if running short.

The TM maintains the current step k (in unary, using a separate tape region or using a scratch area) and repeats the subtraction of $2k+1$. It accepts when exactly k^2 symbols have been processed.

11.4. **TM for languages over $\Sigma = \{a, b, c\}$.**

(a) $\{\boldsymbol{a}^k \boldsymbol{b}^n \boldsymbol{c}^m \mid k \neq n \wedge n \neq m\}$:
Strategy: Count $|x|_a$, $|x|_b$, $|x|_c$ separately and verify $|x|_a \neq |x|_b$ and $|x|_b \neq |x|_c$.
A practical approach:
   1. Verify the input has the form $\boldsymbol{a}^k \boldsymbol{b}^n \boldsymbol{c}^m$ (check symbol ordering); reject if not.
   2. Pair off $\boldsymbol{a}$s and $\boldsymbol{b}$s: erase one $\boldsymbol{a}$ and one $\boldsymbol{b}$ in each pass. If they run out simultaneously, $k = n$; reject.
   3. Pair off $\boldsymbol{b}$s and $\boldsymbol{c}$s similarly. If they run out simultaneously, $n = m$; reject.
   4. If both checks show inequality, accept.

(b) $\{x \mid |x|_a \neq |x|_b \wedge |x|_b \neq |x|_c\}$ (arbitrary mixing):
Strategy: Count occurrences by marking symbols.
   1. In pass 1: pair $\boldsymbol{a}$s with $\boldsymbol{b}$s by crossing out one of each per scan. If the pairing runs out on one side before the other, $|x|_a \neq |x|_b$ confirmed. If they balance, $|x|_a = |x|_b$; reject.
   2. In pass 2: pair $\boldsymbol{b}$s with $\boldsymbol{c}$s. Similarly check $|x|_b \neq |x|_c$.
   3. Accept iff both checks confirm inequality.

11.5. **Machine M with $L(M) = L_1(M) = L_2(M) = L_3(M) = \{x \in \{a, b, c\}^* \mid |x|_a = |x|_b = |x|_c\}$.**

(a) **Construction**: First let M deterministically test whether the input has equally many $\boldsymbol{a}$'s, $\boldsymbol{b}$'s, and $\boldsymbol{c}$'s by the usual crossing-off routine. Then:
   1. If the test succeeds, erase the whole tape, write a single $\boldsymbol{Y}$, position the head on that $\boldsymbol{Y}$, and halt in a final state.
   2. If the test fails, do *not* halt: enter a simple two-state loop that moves right and left forever on blanks.

   For a yes-instance, the machine halts in a final state, with $\boldsymbol{Y}$ somewhere on the tape and in fact as the rightmost nonblank symbol. Thus the input belongs to all four sets $L(M), L_1(M), L_2(M), L_3(M)$. For a no-instance, the machine never halts, so it belongs to none of them. Therefore all four acceptance notions define exactly the required language.

(b) **Is this always possible?**: Yes, for any Turing-acceptable language L. Proof: Given any TM A accepting L, by the $L_1/L_2/L_3$ equivalence theorems (Exercise 11.19), there is a halting TM A' accepting L in any of the four modes. If we require all four modes to agree, we need A to halt on *all* inputs (not just those in L). This is possible iff L is *recursive* (decidable). For Turing-acceptable but not recursive languages, $L(M) = L$ but M may not halt on all inputs, preventing $L_1(M) = L$. So the answer is: **yes if and only if L is recursive**. For every recursive language, we can build M with all four modes agreeing.

11.6. $L = \{ww \mid w \in \{\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}\}^*\}$**.**

(a) **One-tape TM**: Strategy:

1. If input has odd length, reject (since $|ww| = 2|w|$ is even).
2. Mark the midpoint by counting: move right counting n symbols, then left counting n symbols to find the center (binary search or two-pass count).
3. From the start, match symbol i with symbol $n+i$ (for each i from 1 to n), crossing off matched pairs.
4. Accept if all pairs match; reject on any mismatch.

(b) **LBA**: Since the computation uses only the $2n$ cells of the input, the machine is already an LBA.

(c) **Multiple tapes**: A two-tape TM copies the first half of the input to tape 2, then compares tape 2 with the second half of tape 1 symbol by symbol. A nondeterministic TM can guess the midpoint and verify.

11.7. **Lexicographic ordering Turing machines.**

Given $\Sigma = \{\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n\}$ with lexicographic order based on base-n numbering.

(a) **Successor TM**: Machine M_{succ} maps x to its lexicographic successor.
Algorithm: Process x from right to left. Find the rightmost symbol $\boldsymbol{a}_i$ with $i < n$; replace it with $\boldsymbol{a}_{i+1}$ and replace all symbols to its right with $\boldsymbol{a}_1$. If all symbols are $\boldsymbol{a}_n$ (e.g., $x = \boldsymbol{a}_n^k$), the successor is $\boldsymbol{a}_1^{k+1}$: write $k+1$ copies of $\boldsymbol{a}_1$ (extending the tape by one cell).

(b) **Generator TM**: Start with blank tape. Write $\boldsymbol{a}_1$ (the first word). Apply M_{succ} to get the next word; erase the current word, write the successor. Repeat indefinitely. This generates words of Σ^* in lexicographic order, one at a time.

(c) **Enumerator TM**: Start with blank tape. Apply M_{succ} repeatedly, but instead of erasing the previous word, write the next word to the right (separated by a blank). This enumerates Σ^* by writing all words in sequence on the tape.

(d) **Deterministic simulation of NDTM**: Given a nondeterministic TM N, simulate it deterministically using the lexicographic ordering of computation paths. At step k, enumerate all possible k-step computation paths (sequences of transitions) using the lexicographic order, and simulate each one. If any path reaches an accepting state, accept. This gives a deterministic simulation (though potentially much slower). M_{succ} generates the next computation path to try.

11.8. **Semi-infinite tape Turing machine.**

A *semi-infinite tape* has a left boundary (leftmost cell = cell 1) and extends indefinitely to the right.

(a) **Two-track simulation**: Given a TM M with a bi-infinite tape, simulate it on a two-track semi-infinite tape: - Track 1 (top): cells $0,1,2,3,\ldots$ of the original tape. - Track 2 (bottom): cells $-1,-2,-3,\ldots$ of the original tape (stored in reverse). - Cell k of the two-track tape contains (c_k, c_{-k}) for $k \geq 0$ (where c_i is the content of cell i on the original tape), with a special marker at cell 0 to indicate the boundary.

The head of the two-track TM simulates both directions: moving right on Track 1 corresponds to moving right; moving left (if at cell 0) switches to Track 2 and moves right.

(b) **Equivalence proof**: By induction on the number of TM steps: - Each step of M corresponds to a step (or at most two steps) of the two-track machine. - The content of the two tracks correctly represents the bi-infinite tape of M. - Acceptance/rejection conditions are preserved. □

11.9. **TM recognizing $\{\boldsymbol{a}^n \mid n$ is a power of 2$\}$.**

Strategy: Repeatedly check if n is divisible by 2 and halve it until reaching 1.

Algorithm:

1. If input is $\boldsymbol{a}$ ($n = 1 = 2^0$), accept.
2. If input has odd length > 1, reject.
3. Sweep left to right, marking every other $\boldsymbol{a}$ with $\overline{\boldsymbol{a}}$. If n is odd (and > 1) an unmarked $\boldsymbol{a}$ remains at the end — reject. Otherwise, replace all unmarked $\boldsymbol{a}$s with blanks; the $\overline{\boldsymbol{a}}$s form the halved string. Repeat.
4. Repeat until one symbol remains; accept.

11.10. **Three-head TM for $\{x \mid |x|_a = |x|_b = |x|_c\}$.**

Design: All three heads start at the leftmost character.

Strategy: - Head 1 scans for $\boldsymbol{a}$-symbols (moving right, skipping $\boldsymbol{b}$ and $\boldsymbol{c}$). - Head 2 scans for $\boldsymbol{b}$-symbols. - Head 3 scans for $\boldsymbol{c}$-symbols.

In each synchronization step: advance all three heads to their next unmatched symbol of the respective types. If at any point one head reaches the end while others haven't (or vice versa), reject. If all three heads reach the end simultaneously, accept.

No head ever needs to move left: Each head only moves right (it scans past symbols of other types, looking for its own). Since the heads never backtrack, none needs to move left. □

11.11. **Every CFL is Turing-acceptable.**

Proof (Lemma 11.1): Let L be a CFL with PDA $P = \langle \Sigma, \Gamma, S, s_0, \delta, Z_0, F \rangle$.

Construct a TM M that simulates P: - Tape 1: input string x. - Tape 2: stack of P (top of stack at left). - M simulates each PDA move: reads next input symbol (or λ), looks up δ, pops the stack top, pushes the new stack string (nondeterministically if $|\delta| > 1$). - If the PDA reaches a final state with empty input, M accepts.

Since the PDA is nondeterministic, M is a nondeterministic TM. Every NTM can be simulated by a deterministic TM (by enumerating computation paths as in Exercise 11.7(d)).

Thus $L = L(P) = L(M) \in \mathfrak{T}_\Sigma$. □

11.12. **Every type 1 language is a LBL.**

Proof (Theorem 11.3): Let $G = \langle \Sigma, \Omega, Z, P \rangle$ be a context-sensitive grammar (type 1: every production $\alpha \to \beta$ has $|\beta| \geq |\alpha|$, except possibly $Z \to \lambda$).

Construct a nondeterministic LBA M that accepts $L(G)$: - Tape: input x of length n. - M nondeterministically simulates derivations of G in *reverse*: it starts with the candidate string x and applies productions in reverse ($\beta \to \alpha$, which decreases string length since $|\alpha| \leq |\beta|$), trying to reduce back to Z. - Since productions only decrease (or maintain) length in reverse, the tape never needs to grow beyond the original input length n. - If M successfully reduces to Z, accept; otherwise reject.

Since all computations stay within the original n cells, M is a linear bounded automaton. □

11.13. **Convert LBA M to three-track strict LBA.**

(a) **New alphabets**: - The three-track tape has symbols $(a, b, c) \in (\Sigma \cup \{\boldsymbol{B}\}) \times \{L, R, \boldsymbol{B}\} \times \{L, R, \boldsymbol{B}\}$, where: - Track 1: original tape content. - Track 2: left-boundary marker (L in the leftmost cell, $\boldsymbol{B}$ elsewhere). - Track 3: right-boundary marker (R in the rightmost cell of input, $\boldsymbol{B}$ elsewhere).

(b) **New transitions**: For any old transition $\delta(s, a) = (s', a', d)$ in M: - The new TM reads the three-track symbol at the current head position. - It writes the new three-track symbol (updating track 1, preserving tracks 2 and 3). - If the head would move left from the leftmost cell (track 2 $= L$) or right from the rightmost (track 3 $= R$), the new TM instead enters a unique "boundary error" state and rejects. - This ensures the head never leaves the input region.

(c) **Proof of equivalence** (for $|x| \geq 2$): By induction on the number of steps: - The strict LBA M' exactly simulates M as long as the head stays within boundaries. - Since M is a linear bounded automaton, it never leaves the input region (this is the definition of LBA). - Therefore the boundary checks never trigger, and M' accepts exactly when M does. □

11.14. **Handle strings of length 1; strict LBA for type 1 languages.**

(a) **Strings of length 1**: Add a special symbol $\underline{\overline{\boldsymbol{b}}}$ that represents "left boundary AND right boundary" (single-cell input). The modified machine enters a special mode when it sees $\underline{\overline{\boldsymbol{b}}}$ and handles single-character inputs by appropriate transitions that do not require movement.

(b) **Strict LBA accepting λ**: The empty string requires the initial scan of a blank cell (since the tape head must look at something). Define: if the initial tape cell is blank, the strict LBA interprets this as λ and immediately decides acceptance based on whether $\lambda \in L$. For type 1 grammars that allow $Z \to \lambda$ (the only way $\lambda \in L$), the strict LBA accepts the blank tape. All other computations proceed as before. □

11.15. **Prove (Theorem 11.4) by induction: TM computation iff LBA grammar derivation.**

Statement: $\alpha s \beta \overset{*}{\vdash} \gamma t \omega$ in M (Turing machine) iff $[\#^i \alpha s \beta \#^j] \overset{*}{\Rightarrow} [\#^m \gamma t \omega \#^n]$ (in the grammar constructed for Theorem 11.4), for some $i, j, m, n \in \mathbb{N}$.

Base (0 transitions): $\alpha s \beta \overset{*}{\vdash} \alpha s \beta$, and $[\#^i \alpha s \beta \#^j] \overset{*}{\Rightarrow} [\#^i \alpha s \beta \#^j]$ (no derivation steps). ✓

Step: If $\alpha s \beta \vdash \gamma t \omega$ by one TM transition, the grammar applies the corresponding production (which is a context-sensitive rule encoding the TM transition). The # boundary markers are adjusted as needed (if the head moves to a new blank cell, a # is converted). By induction, chaining k steps: k TM transitions correspond to k grammar derivation steps. □

11.16. **General induction statement for Theorem 11.5.**

Theorem 11.5 establishes that every LBL is a type 1 (context-sensitive) language.

Inductive statement: For all $k \geq 0$, for all LBA configurations $C = (s, \alpha, i)$ (state s, tape α, head at position i) reachable from the initial configuration on input x in at most k steps:

C is reachable in k LBA steps from the initial configuration iff the sentential form ϕ_C (encoding C as a string of nonterminals in the grammar for $L(M)$) is derivable in k grammar steps from the start symbol.

Proof by induction on k: - Base ($k = 0$): The initial configuration corresponds to the start of the grammar derivation. ✓ - Step: Each LBA transition corresponds to a grammar production. By the IH, all k-step reachable configurations correspond to k-step derivations; adding one more LBA step extends to $k+1$ on both sides. □

11.17. **Type 0 language Kleene closure.**

(a) **Flaw in** $G_* = \langle \Sigma, \Omega \cup \{W\}, W, P \cup \{W \to \lambda, W \to WW, W \to Z\}\rangle$:

The production $W \to WW$ in combination with $W \to Z \to \ldots$ may generate strings with nonterminals from different "copies" of the grammar interfering with each other. Specifically, type 0 grammars can have productions $\alpha\beta \to \gamma$ (context-sensitive or unrestricted) where α and β come from different parts of the derivation. With two independent copies of G (from $W \to WW$) sharing nonterminals, the productions of one copy may interact with nonterminals of the other copy in unintended ways.

(b) **Example**: Let $G = \langle \{A, B, C\}, \{a, b\}, Z, \{Z \to AB, AB \to BA, A \to a, B \to b\}\rangle$. Then $L(G) = \{ab, ba\}$. With G_*: $W \to WW \to ZZ \to ABAB$. The productions of G might apply to the combined string $ABAB$ in unintended ways: $AB \to BA$ could apply to either the first AB or the second AB or the internal pair BA (B from copy 1 and A from copy 2), producing incorrect strings.

(c) **Correct** G_*: To avoid cross-copy interference, rename nonterminals: for each copy of G used in the derivation, use a fresh copy of all nonterminals (with indices). Alternatively, use the Turing machine characterization: since $\mathfrak{T}_\Sigma$ is closed under concatenation and Kleene closure can be simulated by nondeterministically splitting the input and recognizing each piece.

Formally: Let $G_* = \langle \Sigma, \Omega \cup \{W, W'\}, W, P'\rangle$ where P' contains a *fresh* copy P_i of P for each "level" of recursion, with nonterminals marked to prevent interaction. The construction ensures each derivation of a word from $L(G)$ uses an isolated copy of G.

(d) **Proof**: $w \in L(G_*)$ iff w can be written as $w = w_1 w_2 \cdots w_k$ with each $w_i \in L(G)$ iff $W \overset{*}{\Rightarrow} w$ in G_* (by the isolated copy construction). □

11.18. $\mathfrak{T}_\Sigma$ **is closed under various operations.**

(a) **Homomorphism**: Given $L \in \mathfrak{T}_\Sigma$ (TM M accepting L) and homomorphism h, construct TM M' that accepts $h(L)$: M' nondeterministically guesses a string w and verifies $h(w) = x$ (the input) and $w \in L(M)$. Alternatively, directly: for $h(L)$, the TM M' on input y nondeterministically guesses an x with $h(x) = y$ and simulates M on x.

(b) **Inverse homomorphism**: M' on input x computes $h(x)$ and simulates M on $h(x)$. Since h is a computable function (homomorphisms are computable), this works. So $h^{-1}(L) \in \mathfrak{T}_\Sigma$.

(c) **Concatenation**: Given TMs M_1 and M_2 for L_1 and L_2, a nondeterministic TM M for $L_1 \cdot L_2$ guesses a split point $x = uv$, simulates M_1 on u and M_2 on v, accepts if both accept.

(d) **Substitution**: For each symbol $a \in \Sigma$, let $L_a \in \mathfrak{T}_\Sigma$. The substitution $L[a \mapsto L_a]$ replaces each a in words of L with words from L_a. A nondeterministic TM handles this by guessing the substitution. □

11.19. **Equivalence of acceptance modes L_1 and L_2.**

(a) $L(M) = L_1(M) \Rightarrow L(M) = L_2(M)$: Given TM A_1 with $L = L_1(A_1)$ (accept = halt), construct A_2 with $L = L_2(A_2)$ (accept = halt with $\boldsymbol{Y}$ on tape).

Modification: When A_1 enters the halt state: - If A_1 was in a final/accepting configuration: A_2 erases the tape and writes $\boldsymbol{Y}$. - If A_1 was in a rejecting halt: A_2 writes $\boldsymbol{N}$.

For non-halting A_1 computations: same behavior. Since A_1 halts on inputs in L (by L_1), A_2 also halts on those inputs and writes $\boldsymbol{Y}$. For inputs not in L: A_1 may not halt; A_2 has the same non-halting behavior, so $L_2(A_2) = L_1(A_1) = L$. □

(b) $L(M) = L_2(M) \Rightarrow L(M) = L_3(M)$: Given TM A_2 with $L = L_2(A_2)$ (accept = halt with $\boldsymbol{Y}$), construct A_3 with $L = L_3(A_3)$ (accept = halt with blank tape).

Modification: When A_2 halts with $\boldsymbol{Y}$: A_3 erases the entire tape. When A_2 halts with $\boldsymbol{N}$: A_3 writes some non-blank symbol. Then $L_3(A_3)$ = strings for which A_3 halts on blank tape = strings for which A_2 wrote $\boldsymbol{Y} = L_2(A_2) = L$. □

11.20. **$\mathfrak{H}_\Sigma$ (halting TM languages) is closed under the same operations.**

(a) **Homomorphism**: Given a halting TM M for $L \in \mathfrak{H}_\Sigma$, construct a halting TM for $h(L)$: on input y, nondeterministically enumerate all x with $h(x) = y$ in lexicographic order. For each x, simulate M on x (which halts since M is a halting TM). If M accepts some x, accept y; if all x are rejected, reject y. Since h is a homomorphism, the set of x with $h(x) = y$ is finite and computable. The TM halts on all inputs.

(b) **Inverse homomorphism**: On input x, compute $h(x)$ (finite computation), simulate halting TM M on $h(x)$; since M halts on all inputs, the whole machine halts.

(c) **Concatenation**: Nondeterministically split input $x = uv$ (finite choices), simulate halting TMs M_1 on u and M_2 on v (both halt). Accept if both accept. The constructed TM halts on all inputs.

(d) **Substitution**: For each a, the substituted language $L_a \in \mathfrak{H}_\Sigma$ has a halting TM. The substitution machine nondeterministically applies substitutions and simulates all relevant halting TMs, which all halt. Hence the combined machine halts. □

Chapter 12

Decidability — Solutions

12.1. **Verify assertions in proof of Theorem 12.1 concerning Theorem 2.7.**

Theorem 2.7 (Nerode's theorem) states that a language is FAD (finite automaton definable) iff its Nerode right congruence R_L has finite index.

Theorem 12.1 (decidability of $L(A) = \emptyset$) uses the following from Theorem 2.7: - The minimal DFA A_{min} has states in bijection with equivalence classes of R_L. - $L(A) = \emptyset$ iff no final state is reachable from s_0 iff no equivalence class of R_L containing an accepted string is reachable. - Reachability is decidable by BFS/DFS on the finite graph of A.

The assertion to verify: the equivalence classes of R_L for a finite-index R_L correspond exactly to the states of the minimal DFA. This follows directly from Theorem 2.7: each class $[x]_{R_L}$ maps to the state $\overline{\delta}(s_0, x)$, and this mapping is a bijection between classes and states. □

12.2. **Prove Lemma 12.1.**

Lemma 12.1 (typically states): For a DFA $A = \langle \Sigma, S, s_0, \delta, F\rangle$ with $|S| = n$, $L(A) \neq \emptyset$ iff $L(A)$ contains a string of length less than n.

Proof: ($\Leftarrow$) Obvious.

($\Rightarrow$): If $L(A) \neq \emptyset$, let $x \in L(A)$. If $|x| < n$, done. If $|x| \geq n$, then the sequence of states $s_0, \overline{\delta}(s_0, x_1), \overline{\delta}(s_0, x_1x_2), \dots, \overline{\delta}(s_0, x)$ has $|x|+1 > n$ states, so by the pigeonhole principle, some state repeats. Say $\overline{\delta}(s_0, u) = \overline{\delta}(s_0, uv) = q$ where $|v| \geq 1$. Then $\overline{\delta}(s_0, u) = q = \overline{\delta}(s_0, uv)$, and since $x = uvw \in L(A)$, $\overline{\delta}(s_0, uw) = \overline{\delta}(s_0, uvw) \in F$. So $uw \in L(A)$ and $|uw| < |uvw| = |x|$. Repeat until length $< n$. □

12.3. **Number of nonfinal states in minimal DFA $\leq$ nonfinal states in any machine B with $L(B) = L$.**

Proof: Let A_{min} be the minimal DFA for L with n_{min} nonfinal states and B any machine with $L(B) = L$ having n_B nonfinal states.

Each nonfinal state q of A_{min} corresponds to a unique right-congruence class $[x]_{R_L}$ (by Nerode's theorem) with $\overline{\delta}_{min}(s_0, x) = q$ and $x \not\!\!\, nL$ (since q is nonfinal). In any machine B with $L(B) = L$, two strings in different R_L-classes must lead to different states (otherwise B would accept/reject some string incorrectly). Therefore B must have at least as many nonfinal states as A_{min}. More precisely, the number of nonfinal classes of R_L equals the number of nonfinal states in A_{min}, and B must have at least this many nonfinal states. □

12.4. **Decidability of $L(A_1) \subseteq L(A_2)$.**

Algorithm:

(a) Construct DFA A_2^c for $\overline{L(A_2)}$ (complement: swap final/nonfinal states of A_2's complete DFA).

(b) Construct DFA $A^\cap$ for $L(A_1) \cap \overline{L(A_2)}$ (product machine).

(c) $L(A_1) \subseteq L(A_2)$ iff $L(A_1) \cap \overline{L(A_2)} = \emptyset$ iff $L(A^\cap) = \emptyset$ (decidable by Theorem 12.1). □

12.5. **Decidability of** $L(A)$ **is cofinite.**

$L(A)$ is cofinite iff $\overline{L(A)}$ is finite.

Algorithm:

(a) Construct DFA A^c for $\overline{L(A)}$ (complement).

(b) $\overline{L(A)}$ is finite iff $L(A^c)$ is finite iff the DFA A^c (after removing unreachable states) has no cycle reachable from the start state that also leads to a final state (Theorem 12.2 analog).

(c) Check for cycles in A^c among useful states (BFS/DFS). Decidable. □

12.6. **Decidability: does** $L(A)$ **contain any string of length** > 1228**?**

Algorithm (using the pumping lemma and DFA structure): - By Theorem 12.2 (pumping lemma), $L(A)$ is infinite iff $L(A)$ contains a string of length $n \leq k < 2n$ (where $n = |S|$). - For strings of length > 1228: if $|S| \leq 1228$, enumerate all strings of length n to $2n-1$ and check if any is in $L(A)$ (finitely many checks). If $L(A)$ contains such a string, it also contains strings of length > 1228. - More directly: compute $\{\overline{\delta}(s_0, y) \mid |y| = 1229\}$ (reachable states at depth 1229) by BFS. If any final state is reachable at depth > 1228 (specifically, any path of length > 1228 reaches a final state), the answer is yes. BFS on the DFA graph answers this. □

12.7. **Decidability: does** A **accept any even-length strings?**

Algorithm:

(a) Construct DFA A_{even} that recognizes $\{x \mid |x| \text{ is even}\}$ (a simple 2-state DFA alternating between even/odd states).

(b) Construct product DFA $A^\cap$ for $L(A) \cap L(A_{even})$.

(c) A accepts an even-length string iff $L(A^\cap) \neq \emptyset$. Decidable by Theorem 12.1. □

12.8. **Decidability:** R_1 **and** R_2 **represent complementary languages.**

Algorithm:

(a) Convert regular expressions R_1 and R_2 to DFAs A_1 and A_2.

(b) Check $L(A_1) = \overline{L(A_2)}$: equivalently, check $L(A_1) \subseteq \overline{L(A_2)}$ and $\overline{L(A_2)} \subseteq L(A_1)$.

(c) By Exercise 12.4, each subset relation is decidable. Equivalently, check $L(A_1) \cup L(A_2) = \Sigma^*$ and $L(A_1) \cap L(A_2) = \emptyset$.

(d) Both conditions are decidable (using complement, intersection, and emptiness). □

12.9. **Decidability:** R_1 **and** R_2 **describe common strings.**

Algorithm: $L(R_1) \cap L(R_2) \neq \emptyset$?

(a) Convert R_1, R_2 to DFAs A_1, A_2.

(b) Compute the product DFA $A^\cap$ for $L(A_1) \cap L(A_2)$.

(c) Check $L(A^\cap) \neq \emptyset$ (decidable by Theorem 12.1). □

12.10. **Decidability: is there a DFA with** < 31 **states accepting** $L(R_1)$**?**

Algorithm:

(a) Convert R_1 to a DFA A (via NFA construction).

(b) Minimize A to get A_{min}.

(c) There is a DFA with < 31 states accepting $L(R_1)$ iff $|A_{min}| < 31$.

(d) Count the states of A_{min}: decidable. □

12.11. **Decidability: is there a DFA with** > 31 **states accepting** $L(R_1)$**?**

One-step algorithm: Yes, always. If A_{min} has k states, then we can construct a DFA with $k+\ell$ states for any $\ell > 0$ by adding dead (sink) states. If $k \leq 31$, add $32-k$ unreachable dead states. If $k > 31$, A_{min} itself already has > 31 states.

Therefore, for *any* regular language, there always exists a DFA with > 31 states accepting it. The answer is **always yes**. □

12.12. **Decidability:** $\exists$ λ**-NDFA with** ≤ 1 **final state accepting** R**.**

Algorithm:

(a) Convert R to a λ-NDFA N using the Kleene/Thompson construction. This construction produces NDFAs with exactly one final state. So the answer is always yes for any regular expression R. □

12.13. **Decidability:** $\exists$ **NDFA (without** λ**-moves) with** ≤ 1 **final state accepting** $L(A)$**.**

Algorithm:

(a) Minimize the DFA A to get A_{min}.

(b) A DFA (hence NDFA) with one final state accepting L exists iff no two distinct final states of A_{min} are distinguishable by the language continuation. More precisely: the language L can be accepted by a one-final-state NDFA iff the set F of final-state equivalence classes (in the Myhill-Nerode quotient) can be "collapsed" into one: this is possible iff all strings in L lead to the same right-congruence class (i.e., L has a single final Nerode class), or more generally, the structure allows merging.

(c) Check: if A_{min} has exactly 1 final state, yes. Otherwise, check if the language structure allows a 1-final-state NDFA: convert to minimal DFA, check how many final Nerode classes exist. Decidable. □

12.14. **Decidability of** $R_1 = R_2$ **(same language).**

Algorithm:

(a) Convert R_1, R_2 to DFAs A_1, A_2.

(b) Minimize both: A_1^{min}, A_2^{min}.

(c) $L(R_1) = L(R_2)$ iff $A_1^{min} \cong A_2^{min}$ (isomorphic as DFAs). Check isomorphism: decidable in polynomial time. $\square$

12.15. **Decidability: R_1 and R_2 generate the same right congruence ($R_{L_1} = R_{L_2}$).**

Algorithm: $R_{L_1} = R_{L_2}$ iff L_1 and L_2 have the same Nerode congruence iff the minimal DFAs for L_1 and L_2 are isomorphic (same state structure, same transitions, same final states). By Exercise 12.14, this reduces to checking $L(R_1) = L(R_2)$, which is decidable. $\square$

12.16. **Prove Theorem 12.5.**

Theorem 12.5: The equivalence problem for DFAs is decidable.

Proof: Given DFAs A_1 and A_2, $L(A_1) = L(A_2)$ iff $L(A_1) \triangle L(A_2) = \emptyset$ (symmetric difference). $L(A_1) \triangle L(A_2) = (L(A_1) - L(A_2)) \cup (L(A_2) - L(A_1)) = (L(A_1) \cap \overline{L(A_2)}) \cup (L(A_2) \cap \overline{L(A_1)})$.

Construct the corresponding DFA (product machine) for this symmetric difference. Check if it is empty (decidable by Theorem 12.1). $\square$

12.17. **Efficient algorithm for $\{\overline{\delta}(s_0, y) \mid y \in \Sigma^*,\ n \le |y| < 2n\}$.**

Algorithm: BFS on the DFA state graph, tracking string length.

(a) Initialize: at length 0, state is $\{s_0\}$.

(b) BFS layer by layer (one transition per layer, tracking the current layer as the string length).

(c) At layer n: record all states reached at length n (i.e., states reachable by strings of length n).

(d) At layers $n, n+1, \ldots, 2n-1$: continue BFS, recording states reached at each length.

(e) Return the union of all states reached at lengths n through $2n-1$.

Since the DFA has $|S|$ states, the BFS terminates in at most $2n$ steps, each taking $O(|S| \cdot |\Sigma|)$ work. Total: $O(n \cdot |S| \cdot |\Sigma|)$.

Justification for halting: The DFA has a finite number of states. BFS on the finite DFA graph terminates in at most $|S|$ layers. We run for $2n$ layers, which is finite. $\square$

12.18. **Would intersecting $\{\overline{\delta}(s_0, y) \mid 5n \le |y| < 6n\}$ with F always work?**

Answer: Yes, it would always produce the correct answer.

Justification: By the pumping lemma (Theorem 12.2), $L(A)$ is infinite iff $L(A)$ contains a string of length $n \le k < 2n$ (where $n = |S|$).

Claim: $L(A)$ is infinite iff $\{\overline{\delta}(s_0, y) \mid 5n \le |y| < 6n\} \cap F \ne \emptyset$.

($\Rightarrow$): If $L(A)$ is infinite, it contains strings of arbitrary length, so in particular strings of length between $5n$ and $6n-1$. Hence the intersection is nonempty.

($\Leftarrow$): If the intersection is nonempty, there exists a string of length $5n \le k < 6n$ in $L(A)$. Since $k \ge 5n > n = |S|$, by the pigeonhole principle some state repeats in the path of length k; thus a shorter string is also accepted, and by applying the pumping argument, $L(A)$ is infinite.

So yes, this alternative range would work correctly. $\square$

12.19. **Decidability of equivalence of two Mealy machines.**

Algorithm:

(a) Minimize both Mealy machines M_1 and M_2.

(b) Two Mealy machines are equivalent iff they produce identical outputs on all inputs.

(c) Construct a product machine $M^\times$ that tracks the states of M_1 and M_2 simultaneously and checks if outputs agree on every input.

(d) Use a reachability analysis: find all states (q_1, q_2) of the product machine reachable from the initial state where the outputs differ. If no such state is reachable, the machines are equivalent.

(e) Reachability is decidable (BFS/DFS on the finite product machine graph). □

12.20. **Decidability of equivalence of two Moore machines.**

Algorithm: Same as Exercise 12.19. Moore machines output based on state (not transition), so:

(a) Build the product machine tracking states of both Moore machines.

(b) Check if any reachable state pair (q_1, q_2) has $\omega_1(q_1) \neq \omega_2(q_2)$ (different outputs).

(c) If not, the machines are equivalent. Decidable by BFS. □

12.21. **Decide whether R represents strings of length > 28, without finite automata.**

Algorithm (direct on regular expressions):

By structural induction on the regular expression R: - $R = \emptyset$: max length $= -\infty$; answer: no. - $R = \lambda$ or $R = a$ ($a \in \Sigma$): max length ≤ 1; answer: no. - $R = R_1 \cup R_2$: yes if either R_1 or R_2 represents a string of length > 28. - $R = R_1 \cdot R_2$: yes if the sum of max lengths exceeds 28, or if either has unbounded length (due to *). - $R = R_1^*$: yes if R_1 represents any nonempty string (then R_1^{29} has length > 28); formally, R_1^* represents strings of length > 28 iff R_1 is not $\emptyset$ and R_1 is not $\{\lambda\}$ alone.

This induction on the structure of R terminates in finitely many steps. □

12.22. **Decide whether $L(G)$ contains a string of length > 28 for right-linear grammar G.**

Algorithm (without finite automata):

Solve the language equations for G: compute the set of nonterminals reachable from the start symbol, and check for cycles in the nonterminal dependency graph. If any cycle involves a nonterminal that generates nonempty strings, the language is infinite; in particular, it contains strings of length > 28. Otherwise, solve the equations to find the maximum derivable string length.

Alternatively: perform BFS in the derivation tree up to depth 29 (each step adds at most one terminal). If any terminal string of length > 28 is derivable, answer yes. This terminates since the grammar is finite. □

12.23. **Prove $Z_0 \subseteq Z_1 \subseteq \cdots \subseteq Z_n \subseteq \cdots \subseteq \Omega$.**

In the proof of Theorem 12.6 (decidability of $L(G) = \emptyset$ for CFG G), Z_i is the set of nonterminals reachable from the start symbol S in i or fewer derivation steps.

$Z_0 = \{S\}$ (only the start symbol). $Z_{i+1} = Z_i \cup \{B \mid \exists A \in Z_i, \exists \alpha : A \to \alpha \in P, B \in \alpha\}$ (all nonterminals appearing in right-hand sides of productions of nonterminals in Z_i).

Proof: $Z_0 \subseteq Z_1$ since $Z_1 = Z_0 \cup (\cdots) \supseteq Z_0$. By induction: if $Z_k \subseteq Z_{k+1}$, then $Z_{k+1} \subseteq Z_{k+2}$ since any nonterminal in Z_{k+1} is reachable in $k+1$ steps, hence reachable in $k+2$ steps as well. The chain is bounded above by Ω (which is finite). □

12.24. **If $Z_m = Z_{m+1}$, then Z_m is the set of all nonterminals reachable from S.**

Proof: If $Z_m = Z_{m+1}$, then no new nonterminals are added in step $m+1$. This means: every nonterminal appearing in the right-hand side of any production of any nonterminal in Z_m is already in Z_m.

Claim: Z_m is closed under reachability, i.e., every nonterminal reachable from S is in Z_m.

Suppose B is reachable from S (i.e., $S \overset{*}{\Rightarrow} \alpha B \beta$ for some α, β). The shortest derivation has some length k. By monotonicity ($Z_0 \subseteq Z_1 \subseteq \cdots$), $B \in Z_k$. Since $Z_m = Z_{m+1} = Z_{m+2} = \cdots$ (the sequence stabilizes), $B \in Z_m$ for all $k \leq m$ already. If $k > m$, then by the stabilization, every nonterminal reachable in k steps is already in Z_m. So Z_m = all reachable nonterminals. □

12.25. **Prove $\exists m \in \mathbb{N} : Z_m = Z_{m+1}$.**

Proof: The sequence $Z_0 \subseteq Z_1 \subseteq \cdots \subseteq \Omega$ is a non-decreasing chain of subsets of Ω. Since $|\Omega|$ is finite, the chain can increase at most $|\Omega|$ times. Therefore there exists $m \leq |\Omega|$ such that $Z_m = Z_{m+1}$ (the chain stabilizes). □

12.26. **Definition of W_i and monotonicity.**

(a) In the proof of Theorem 12.6, W_i is the set of nonterminals that can derive a terminal string in i or fewer steps: $W_0 = \emptyset$. $W_{i+1} = W_i \cup \{A \mid \exists (A \to \alpha) \in P : \text{every symbol in } \alpha \in \Sigma \cup W_i\}$. (A nonterminal A can derive a terminal string in $\leq i+1$ steps if some production of A has a right-hand side consisting only of terminals and nonterminals already known to derive terminal strings.)

(b) **Proof of monotonicity**: $W_0 = \emptyset \subseteq W_1$ (trivially). By induction: if $W_k \subseteq W_{k+1}$, then for any $A \in W_{k+1}$: A derives a terminal string in $\leq k+1$ steps, hence in $\leq k+2$ steps as well, so $A \in W_{k+2}$. Thus $W_{k+1} \subseteq W_{k+2}$. The chain is bounded by Ω. □

12.27. **If $W_m = W_{m+1}$, then W_m is the set of all nonterminals producing terminal strings.**

Proof: If $A \overset{*}{\Rightarrow}{}^{*} w \in \Sigma^*$ in k steps, then $A \in W_k$. Since $W_m = W_{m+k}$ for all $k \geq 0$ (by stabilization), $A \in W_m$. Conversely, every $A \in W_m$ derives a terminal string by definition. So W_m is precisely the set of productive nonterminals. □

12.28. **Prove $\exists m \in \mathbb{N} : W_m = W_{m+1}$.**

Proof: Same as Exercise 12.25: $W_0 \subseteq W_1 \subseteq \cdots \subseteq \Omega$ is a non-decreasing chain in the finite set Ω, so it stabilizes. □

12.29. **Decidability questions about NDFA A with λ-moves.**

(a) **Exists string with no complete path**: A "complete path" is one reading all input symbols. Check if there is any state (reachable from s_0 along some path) from which some transition is undefined. Convert A to DFA A^d (subset construction); check if the dead state (empty subset) is reachable in A^d. Decidable.

(b) **Exists string with a path to a nonfinal state**: BFS/DFS on A to find if any complete computation path ends in a nonfinal state. Decidable (finite automaton has finitely many configurations per string-length).

(c) **Exists accepted string with all paths leading to final states**: Find strings where every path from s_0 to an accepting state passes only through transitions that can also lead to final states. Check using the product of A with a tracking automaton. Decidable.

(d) **All accepted strings have all complete paths to final states**: More complex, but still decidable: check if the complement of this property is empty. Using DFA A^d (subset construction), acceptance corresponds to subsets containing at least one final state. Check if any reachable subset in A^d contains both a final and a nonfinal state of A. Decidable.

(e) **All strings have unique paths**: A string has a unique path iff the NDFA is deterministic on that string (no two applicable transitions from any state on any symbol along the path). Check: for each state q and symbol a, if $|\delta(q,a)| > 1$ or there are λ-move alternatives, the NDFA is nondeterministic on some strings. More carefully: check if any string leads to a state with multiple transition choices. Decidable (check transitions structure). □

12.30. **Decidability for two DFAs A_1 and A_2.**

(a) **Decidability of homomorphism existence**: A homomorphism $h: A_1 \to A_2$ (of DFAs) is a state map preserving start state, transitions, and taking final states to final states. Check all functions $h: S_1 \to S_2$: finitely many ($|S_2|^{|S_1|}$). For each, check the homomorphism conditions. Decidable (finite check).

(b) **Decidability of isomorphism existence**: An isomorphism requires h to be bijective (so $|S_1| = |S_2|$). Check all bijections: $(|S_1|)!$ many. For each, verify the isomorphism conditions. Decidable (finite check for finite S_1).

(c) **Decidability of > 3 isomorphisms**: Similarly, count the number of valid isomorphisms (at most $|S_1|!$ total). If the count exceeds 3, answer yes. Decidable. □

12.31. **Decidability: R_1 describes infinitely many strings (no DFA construction).**

Algorithm (structural induction on R_1): - $R_1 = \emptyset$: finite (empty). No. - $R_1 = \lambda$ or $R_1 = a$: finite. No. - $R_1 = R \cup R'$: infinite iff R or R' is infinite. - $R_1 = R \cdot R'$: infinite iff (R is infinite and $L(R')$ is nonempty) or ($L(R)$ is nonempty and R' is infinite). - $R_1 = R^*$: infinite iff $L(R)$ contains a nonempty string.

Each check is computable by structural recursion on R_1. Terminates since R_1 is finite. □

12.32. **Decidability of Mealy machine equivalent to Moore machine.**

Algorithm:

(a) Minimize the Mealy machine M and the Moore machine A.

(b) Convert one to the other's representation (Mealy to Moore or vice versa) using the standard transformation.

(c) Check equivalence (Exercise 12.19 or 12.20). Decidable. □

12.33. **Decidability: $L(R_2)$ properly contains $L(R_1)$ ($L(R_1) \subset L(R_2)$).**

Algorithm:

(a) Convert R_1, R_2 to DFAs.

(b) Check $L(R_1) \subseteq L(R_2)$ (decidable, Exercise 12.4).

(c) Check $L(R_1) \neq L(R_2)$ (decidable, Exercise 12.14 or 12.15).

(d) $L(R_1) \subset L(R_2)$ iff both conditions hold. □

12.34. **Efficient method for $L(A) = \emptyset$ for NDFA A, without DFA conversion.**

(a) **Method**: BFS/DFS directly on the NDFA A (viewed as a graph). Start from s_0, compute the λ-closure, then follow transitions. $L(A) = \emptyset$ iff no final state is reachable from s_0 (via any sequence of transitions and λ-moves). Reachability on the finite graph of A (without constructing the exponential subset DFA).

(b) **Method for infinite** $L(A)$: BFS on A to find all states reachable from s_0 and from which a final state is reachable. Among these "useful" states, check if there is a cycle (a state that appears twice on some accepting path). BFS/DFS to find a cycle among useful states. If yes, $L(A)$ is infinite; else finite. No DFA conversion needed. □

12.35. **Decidability: $L(M)$ is a regular set for DPDA M.**

Reference result: It is known (Stearns 1967) that it is decidable whether the language accepted by a DPDA is regular. The decision procedure involves analyzing the structure of the DPDA (looking for certain patterns of stack behavior called "bounded stack"), but the proof is complex.

Outline: A DPDA accepts a regular language iff its stack depth is bounded (the stack never grows beyond a fixed bound). This can be checked by analyzing the DPDA's transition structure for cycles that strictly increase stack depth. Such analysis is decidable since the DPDA has finite state and transition structure. □

12.36. **Algorithm for path-finding in graphs (Theorem 12.9).**

(a) **Appropriate algorithm**: BFS (breadth-first search) from the source node to find all reachable nodes and shortest paths. For directed graphs (as in the PDA product machine):

a. Initialize a queue with the start node; mark it visited.
b. At each step, dequeue a node, enqueue all unvisited neighbors, mark them visited.
c. Terminate when the queue is empty or the target is found.

Time complexity: $O(|V|+|E|)$ where V is vertices and E is edges.

(b) **More efficient recursive algorithm**: DFS with memoization. `Reach`(v, t): returns true if t is reachable from v. - If $v = t$: return true. - If v is marked: return false. - Mark v; for each neighbor u of v: if `Reach`(u, t): return true. - Return false.

This avoids reprocessing nodes and runs in $O(|V|+|E|)$ time. □

12.37. $\mathfrak{H}_\Sigma$ **(halting-TM languages) closed under union, intersection, concatenation, reversal.**

(a) **Union**: Given halting TMs M_1 (L_1) and M_2 (L_2), build M for $L_1 \cup L_2$: run M_1 and M_2 in parallel (interleaved simulation). Since both halt on all inputs, M also halts. M accepts iff M_1 or M_2 accepts. □

(b) **Intersection**: Same parallel simulation; accept iff both accept. Both halt, so M halts. □

(c) **Concatenation**: M for $L_1 \cdot L_2$ on input x: enumerate all splits $x = uv$, run M_1 on u and M_2 on v. Since both halt, each check halts. There are $|x|+1$ splits (finitely many), all decidable. Accept if any split works. □

(d) **Reversal**: M for L_1^r on input x: compute x^r (finitely computable), run M_1 on x^r. M_1 halts on x^r; so M halts. Accept iff M_1 accepts x^r. □

12.38. **Halting TM T' equivalent to $L_2(T)$ for halting TM T.**

$L = L_2(T)$: T halts on all inputs and L consists of strings for which T halts with $\boldsymbol{Y}$ on the tape.

Construct T':

1. Simulate T on input w.
2. T halts (since it halts on all inputs by assumption).
3. If T halted with $\boldsymbol{Y}$ somewhere on the tape: erase the tape, write w then $\boldsymbol{Y}$ after w on an otherwise blank tape; halt in the accepting state.
4. If T halted with no $\boldsymbol{Y}$: erase the tape, write w then $\boldsymbol{N}$ after w; halt in a rejecting state.

T' satisfies all three conditions: (1) halts on all input (since T does and the post-processing is finite); (2) writes $\boldsymbol{Y}$ after accepted words; (3) writes $\boldsymbol{N}$ after rejected words. And $L(T') = L_2(T) = L$. □

12.39. **Machine $M_{\mathfrak{M}}$ and application to Theorem 12.15.**

(a) **Proof**: Let X be the set of encodings of halting TMs (all in $\mathfrak{M}$). Suppose every encoding in X represents a halting TM. Assume $M_{\mathfrak{M}}$ decides membership in X.

By diagonalization: construct a TM D that on input $\langle T\rangle$ (encoding of TM T): - Simulates $M_{\mathfrak{M}}$ on $\langle T\rangle$. - If $M_{\mathfrak{M}}$ accepts (i.e., $T \in X$): loop forever (don't halt). - If $M_{\mathfrak{M}}$ rejects: accept and halt.

D is a halting TM on inputs not in X, but loops on inputs in X. What about D itself? If $\langle D\rangle \in X$, then D is a halting TM, but by construction D loops on $\langle D\rangle$ (contradiction). If $\langle D\rangle \not inX$, then D halts on $\langle D\rangle$ (accepting), so $\langle D\rangle$ represents a halting TM, contradicting $\langle D\rangle \not inX$.

So $M_{\mathfrak{M}}$ cannot exist: there is a halting TM D whose language is not in $\mathfrak{M}$. □

(b) **Application to Theorem 12.15**: Theorem 12.15 states that the halting problem is undecidable. Applying part (a) with $X = \{\langle T\rangle \mid T \text{ halts on all inputs}\}$ and $\mathfrak{M}$ = all Turing-acceptable languages: if there were an $M_{\mathfrak{M}}$ deciding whether T halts on all inputs, then by part (a), some halting TM D exists whose language is not Turing-acceptable — contradiction. Hence no such $M_{\mathfrak{M}}$ exists. □

12.40. **Encoding context-sensitive grammars.**

(a) **Encoding scheme**: Assume 2 terminal symbols $\{a, b\}$ and arbitrarily many nonterminals $N_1, N_2, N_3, \ldots$

Encoding: - Terminals: $a \mapsto \texttt{00}$, $b \mapsto \texttt{01}$. - Nonterminal $N_i \mapsto \texttt{1}^i\texttt{0}$ (unary encoding with terminator). - Productions: $\alpha \to \beta$ encoded as $\overline{\alpha}\#\#\overline{\beta}$ where $\overline{x}$ is the encoding of x symbol by symbol. - Grammar: list of productions separated by ###, preceded by the start symbol encoding.

This encoding handles an unbounded number of nonterminals via unary encoding and terminates on any finite grammar.

(b) **Algorithm to decide valid CSG encoding**:

i. Parse the encoding to extract the alphabet, nonterminal list, start symbol, and productions.

ii. Check that: (a) terminal symbols are from $\{a, b\}$; (b) nonterminals are distinct and properly encoded; (c) each production $\alpha \to \beta$ has $|\beta| \geq |\alpha|$ (context-sensitive condition), except possibly $Z \to \lambda$.
iii. Verify the syntactic structure of the encoding (valid separators, well-formed strings).
iv. The TM accepts valid encodings and rejects invalid ones. Terminates in polynomial time in the encoding length. □

12.41. **Undecidability of $L(X) = \emptyset$.**

(a) **Arbitrary Turing machines**: Undecidable. If it were decidable, we could solve the halting problem: reduce the question "does T halt on w?" to "does $L(T_w) = \emptyset$?" where T_w runs T on w and accepts iff T halts.

(b) **Halting Turing machines**: Still undecidable. By Rice's theorem applied to halting TMs: any nontrivial property of the language of a halting TM is undecidable.

(c) **Context-sensitive grammars**: Undecidable (same reduction: membership in the halting set reduces to emptiness of the language of a CSG constructed from the TM encoding).

(d) **Linear bounded automata**: Undecidable. The emptiness problem for LBAs is undecidable (since CSG = LBL and CSG emptiness is undecidable). □

12.42. **Undecidability of $L(X) = \Sigma^*$.**

(a) **Arbitrary TMs**: Undecidable (reduce from halting problem).

(b) **Halting TMs**: Undecidable (Rice's theorem for halting TMs).

(c) **Context-sensitive grammars**: Undecidable.

(d) **Linear bounded automata**: Undecidable.

(e) **Context-free grammars**: Undecidable. The universality problem for CFGs ("does G generate Σ^*?") is undecidable. This is a classical result: it can be shown by reducing from the Post Correspondence Problem.

(f) **Pushdown automata**: Undecidable (equivalent to CFG universality). □

12.43. **Set E of encodings of TMs halting on λ.**

(a) $E \in \mathfrak{T}_\Sigma$: Yes. The TM that recognizes E: on input $\langle T \rangle$, simulate T on λ. If T halts, accept (since that means T halts on λ, so $\langle T \rangle \in E$). If T doesn't halt, the TM doesn't halt either. So E is Turing-acceptable ($\mathfrak{T}_\Sigma$). But $E \not in \mathfrak{D}_\Sigma$ (not decidable): if it were, we could decide the halting problem.

(b) $E \in \mathfrak{H}_\Sigma$: No. E is not recursive (decidable). If $E \in \mathfrak{H}_\Sigma$ (halting TM language = recursive language), then E would be decidable. But the halting problem for T on λ is undecidable. So $E \not in \mathfrak{H}_\Sigma$.

(c) $E \in \mathfrak{D}_\Sigma$: No. $\mathfrak{D}_\Sigma = \mathfrak{H}_\Sigma$ (halting = recursive = decidable). As argued above, E is not decidable. $E \not in \mathfrak{D}_\Sigma$.

12.44. **Set N of encodings of TMs that do NOT halt on λ.**

(a) $N \in \mathfrak{T}_\Sigma$: No. $N = \overline{E}$ (complement of E from Exercise 12.41). Since $E \in \mathfrak{T}_\Sigma$ but $E \not in \mathfrak{D}_\Sigma$, the complement $N = \overline{E} \not in \mathfrak{T}_\Sigma$ (if $E \in \mathfrak{T}_\Sigma$ and $\overline{E} \in \mathfrak{T}_\Sigma$, then E would be recursive, contradicting $E \not in \mathfrak{D}_\Sigma$). So $N \not in \mathfrak{T}_\Sigma$.

(b) $N \in \mathfrak{H}_\Sigma$: No. Same argument: N would be recursive (decidable) if $N \in \mathfrak{H}_\Sigma$, contradicting undecidability of the non-halting problem.

(c) $N \in \mathfrak{D}_\Sigma$: No. $\mathfrak{D}_\Sigma = \mathfrak{H}_\Sigma$. □

12.45. **Decidability questions about equivalence classes of $R_{L(A)}$.**

(a) **At least one equivalence class is finite**: For a DFA A, the equivalence classes of $R_{L(A)}$ correspond to states of the minimal DFA A_{min}. An equivalence class $[x]$ is finite iff the state $q = \overline{\delta}_{min}(s_0, x)$ has no cycles reachable from itself that lead back to q. More precisely, $[x] = \{y \mid \overline{\delta}_{min}(s_0, y) = q\}$ is the set of strings leading to state q.

This set is finite iff q is not reachable from itself in A_{min} (i.e., there is no cycle involving q). Check: in the DFA graph, does state q have any path from itself back to itself? BFS/DFS in the state graph. Decidable.

(b) **All equivalence classes are finite**: All classes are finite iff the DFA graph is acyclic (no cycles, since any cycle would create an infinite class). Check if the DFA graph has any cycles: DFS looking for back edges. Decidable. □

12.46. **Decide $L(G_1) \subset L(G_2)$ for right-linear grammars G_1, G_2.**

Algorithm:

(a) Convert G_1, G_2 to DFAs A_1, A_2 (via Lemma 8.1).

(b) Check $L(A_1) \subseteq L(A_2)$ (decidable, Exercise 12.4).

(c) Check $L(A_1) \neq L(A_2)$ (decidable, Exercise 12.14).

(d) $L(G_1) \subset L(G_2)$ (proper subset) iff $L(A_1) \subseteq L(A_2)$ and $L(A_1) \neq L(A_2)$. □